

- Friedman's ANOVA compares several conditions when the data are related (usually because the same participants take part in each condition) and the resulting data have unusual cases or violate any assumption in Chapter 6.
- Look at the row labelled *Asymptotic Sig*. If the value is less than 0.05 then typically people conclude that the conditions are significantly different.
- You can follow up the main analysis with pairwise comparisons. These tests compare all possible pairs of conditions using a *p*-value that is adjusted such that the overall Type I error rate remains at 5%.
- Report the χ^2 statistic, the degrees of freedom and the significance value for the main analysis. For any follow-up tests, report an effect size, the corresponding *z* and the significance value.
- Report the medians and their ranges (or draw a boxplot).



7.8 Brian's attempt to woo Jane

'Jane is an anomaly,' Brian thought, 'an awkward, mysterious, complicated set of observations that don't conform to my conventional assumptions.' Maybe that was why he couldn't shake her from his mind. She had a brilliant mind that compelled her to him. He'd never felt that comfortable around girls. He'd had a few relationships, but they all ended the same way: he'd get dumped for some more exciting guy. His dad said he was too nice, and maybe he was – he'd certainly had his fair share of being a doormat. He'd never lost hope – he

dreamed of a Disney fairy tale, where the last person he expected turned out to be his soul mate. Maybe he was compensating for growing up without a mum. He probably needed to get real. Maybe with Jane the writing was on the wall. He should forget about her and knuckle down to his studies. The trouble was, he got excited every time he understood something new about statistics, and Jane was the person with whom he wanted to share his excitement.

'The campus guy is an anomaly,' Jane thought, 'an awkward, mysterious, complicated set of observations that don't conform to my conventional assumptions.' Maybe that was why she couldn't shake him from her mind. She was sure that he had the mind of an idiot, but his efforts to impress her were sweet. He certainly didn't fit into her mental model of men. She'd got into the habit of being curt to men; it was what they deserved. She'd liked a guy, Josh, at school. She was crazy about him, but she was a nerd and he was one of the popular kids. She was not on his radar. Not until he asked her to the cinema. She could hardly breathe when he asked. She bought clothes, spent hours getting ready, and became the stereotype that she'd fought against being. She arrived and waited. Most of her class were in the foyer. She hadn't expected that and it made her feel conspicuous and awkward. He was late and every minute increased her embarrassment and stress. She wanted to run. She should have run: it would have spared her. When he paraded confidently through the doors like a film star with Eliza Hamilton on his arm she knew instantly that she'd been set up. They turned to Jane, placed their fingers in an L on their foreheads, and mouthed 'loser', causing the foyer to erupt in laughter. Jane felt a rage that she hadn't known she was capable of, but she didn't run. She put her headphones in, calmly walked past Josh and smiled. It was a smile that wiped his own one from his face.

Figure 7.17 What Brian learnt from this chapter



The funny thing was that Josh wasn't popular for long. First a bunch of emails leaked in which he tried to sell information about the school football team, then he posted highly personal information about Eliza all over social media. He became an outcast. He lost his place in the team, his grades plummeted and no one spoke to him. He became a total loser. He protested his innocence, but only a genius could have hacked his online life so comprehensively. Jane smiled at the memory. Campus guy seemed different. He was sweet … but not enough to let her guard down.

7.9 What next?



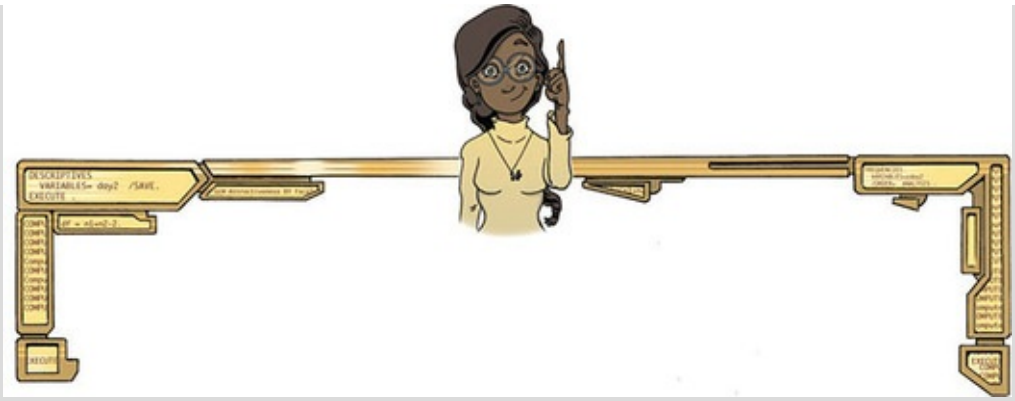
'You promised us swans,' I hear you cry, 'and all we got was Kruskal this and Wilcoxon that. Where were the bloody swans?!' Well, the Queen owns them all so I wasn't allowed to have them. Nevertheless, this chapter did negotiate Dante's eighth circle of hell (Malebolge), where data of deliberate and knowing

evil dwell. That is, data don't always behave themselves. Unlike the data in this chapter, my formative years at school were spent being very well behaved and uninteresting. However, a mischievous and rebellious streak was growing inside. Perhaps the earliest signs were my taste in music. Even from about the age of 3 music was my real passion: one of my earliest memories is of listening to my dad's rock and soul records (back in the days of vinyl) while waiting for my older brother to come home from school. I still have a nostalgic obsession with vinyl. The first record I asked my parents to buy me was 'Take on the world' by Judas Priest, which I'd heard on *Top of the Pops* (a now defunct UK TV show) and liked. Watching the Priest on *Top of the Pops* is a very vivid memory – it had a huge impact. This record came out in 1978 when I was 5. Some people think that this sort of music corrupts young minds. Let's see if it did …

7.10 Key terms that I've discovered

Cochran's *Q*
Friedman's ANOVA
Jonckheere–Terpstra test
Kendall's *W*
Kolmogorov–Smirnov *Z*
Kruskal–Wallis test
Mann–Whitney test
McNemar's test
Median test
Monte Carlo method
Moses extreme reactions
Nonparametric tests
Pairwise comparisons
Ranking
Sign test
Wald–Wolfowitz runs
Wilcoxon rank-sum test
Wilcoxon signed-rank test

Smart Alex's tasks



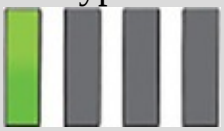
- **Task 1**: A psychologist was interested in the cross-species differences between men and dogs. She observed a group of dogs and a group of men in a naturalistic setting (20 of each). She classified several behaviours as being dog-like (urinating against trees and lampposts, attempts to copulate with anything that moved, and attempts to lick their own genitals). For each man and dog she counted the number of dog-like behaviours displayed in a 24-hour period. It was hypothesized that dogs would display more dog-like behaviours than men. Analyze the data in **MenLikeDogs.sav** with a Mann–Whitney test.



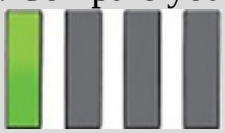
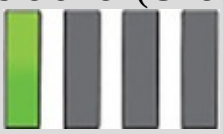
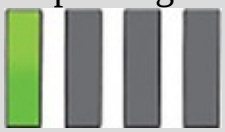
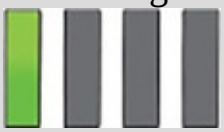
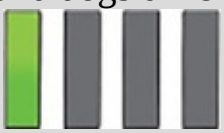
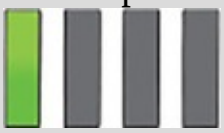
- **Task 2**: Both Ozzy Osbourne and Judas Priest have been accused of putting backward masked messages on their albums that subliminally influence poor unsuspecting teenagers into doing things like blowing their heads off with shotguns. A psychologist was interested in whether backward masked messages could have an effect. He created a version of Britney Spears's 'Baby one more time' that contained the masked message 'deliver your soul to the dark lord' repeated in the chorus. He took this version, and the original, and played one version (randomly) to a group of 32 people. Six months later he played them whatever version they hadn't heard the time before. So each person heard both the original and the version with the masked message, but at different points in time. The psychologist measured the number of goats that were sacrificed in the week after listening to each version. Test the hypothesis that the backward message would lead to more goats being sacrificed using a Wilcoxon signed-rank test (**DarkLord.sav**).



- **Task 3**: A media researcher was interested in the effect of television programmes on domestic life. She hypothesized that through 'learning by watching', certain programmes encourage people to behave like the characters within them. She exposed 54 couples to three popular TV shows, after which the couple were left alone in the room for an hour. The experimenter measured the number of times the couple argued. Each couple viewed all TV shows but at different points in time (a week apart) and in a counterbalanced order. The TV shows were *EastEnders* (which portrays the lives of extremely miserable, argumentative, London folk who spend their lives assaulting each other, lying and cheating), *Friends* (which portrays unrealistically considerate and nice people who love each other oh so very much – but I love it anyway), and a National Geographic programme about whales (this was a control). Test the hypothesis with Friedman's ANOVA (**Eastenders.sav**).



- **Task 4**: A researcher was interested in preventing coulrophobia (fear of clowns) in children. She did an experiment in which different groups of children (15 in each) were exposed to positive information about clowns. The first group watched adverts in which Ronald McDonald is seen cavorting with children and singing about how they should love their mums. A second group was told a story about a clown who helped some children when they got lost in a forest (what a clown was doing in a forest remains a mystery). A third group was entertained by a real clown, who made balloon animals for the children. A final, control, group had nothing done to them at all. Children rated how much they liked clowns from 0 (not scared of clowns at all) to 5 (very scared of clowns). Use a Kruskal–Wallis test to see whether the interventions were successful (**coulrophobia.sav**). 

- **Task 5**: Thinking back to Labcoat Leni's Real Research 4.1, test whether the number of offers was significantly different in people listening to Bon Scott compared to those listening to Brian Johnson (**Oxoby (2008) Offers.sav**). Compare your results to those reported by Oxoby (2008). 
- **Task 6**: Repeat the analysis above, but using the minimum acceptable offer (**Oxoby (2008) MAO.sav**) – see Chapter 4, Task 3. 
- **Task 7**: Using the data in **Shopping Exercise.sav** (Chapter 4, Task 4), test whether men and women spent significantly different amounts of time shopping. 
- **Task 8**: Using the same data, test whether men and women walked significantly different distances while shopping. 
- **Task 9**: Using the data in **Goat or Dog.sav** (Chapter 4, Task 5), test whether people married to goats and dogs differed significantly in their life satisfaction. 
- **Task 10**: Use the **SPSSExam.sav** (Chapter 6, Task 2) data to test whether students at the Universities of Sussex and Duncetown differed significantly in their SPSS exam scores, their numeracy, their computer literacy, and the number of lectures attended. 
- **Task 11**: Use the **DownloadFestival.sav** data from Chapter 6 to test whether hygiene levels changed significantly over the three days of the festival. 

Answers & additional resources are available on the book's website at **https://edge.sagepub.com/field5e**



8 Correlation

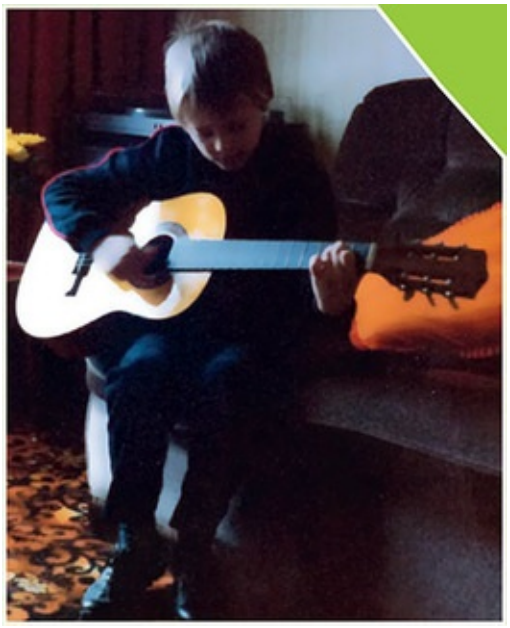
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8.1 What will this chapter tell me?

When I was 8 years old, my parents bought me a guitar for Christmas. Even then, I'd desperately wanted to play the guitar for years. I could not contain my excitement at getting this gift (had it been an *electric* guitar I think I would actually have exploded with excitement). The guitar came with a 'learn to play' book, and after some time trying to play what was on page 1 of this book, I readied myself to unleash a riff of universe-crushing power on the world (well, 'Skip to my Lou'). But, I couldn't do it. I burst into tears and ran upstairs to hide.[1] My dad sat with me and said something like 'Don't worry, Andy, everything is hard to begin with, but the more you practise the easier it gets.' With his comforting words, my dad was inadvertently teaching me about the relationship, or correlation, between two variables. These two variables could be related in three ways: (1) *positively related*, meaning that the more I practised my guitar, the better a guitar player I would become (i.e., my dad was telling me the truth); (2) *not related* at all, meaning that as I practised the guitar my playing ability would remain completely constant (i.e., my dad had fathered a cretin); or (3) *negatively related*, which would mean that the more I practised the guitar the worse a guitar player I would become (i.e., my dad had fathered an indescribably strange child). This chapter looks first at how we can express the relationships between variables statistically by looking at two measures: *covariance* and the *correlation coefficient*. We discover how to carry out and interpret correlations in SPSS Statistics. The chapter ends by looking at more complex measures of relationships; in doing so, it acts as a precursor to the chapter on the linear model.

1 This is not a dissimilar reaction to the one I have when publishers ask me for new editions of statistics textbooks.

Figure 8.1 I don't have a photo from Christmas 1981, but this was taken about that time at my grandparents' house. I'm trying to play an E by the looks of it, no doubt because it's in 'Take on the world'.



8.2 Modelling relationships

In Chapter 5 I stressed the importance of looking at your data graphically before fitting models to them. Our starting point with a correlation analysis is, therefore, to look at scatterplots of the variables we have measured. I am not going to repeat how to produce these graphs, but I urge you (if you haven't done so already) to read Section 5.8 before embarking on the rest of this chapter.



Way back in Chapter 2 we started talking about fitting models to your data that represent the hypothesis you're trying to test. In the previous chapter we looked at this process using a specific set of models that are useful when the data contain unusual cases or fail to meet the assumptions we discussed in Chapter 6. However, when these assumptions are met we can use the general linear model, which is an incredibly versatile and simple model that we've already encountered. I mentioned in Section 2.3 that everything in statistics boils down to one simple idea (expressed in equation (2.1)):

$$\text{outcome}_i = (\text{model}) + \text{error}_i \quad (8.1)$$

To recap, this equation means that the data we observe can be predicted from the model we choose to fit to the data plus some error in prediction. The 'model' in the equation varies depending on the design of your study, the type of data you

have and what you're trying to achieve. If we want to model a relationship between variables then we're trying to predict an outcome variable from a predictor variable. Therefore, we need to include the predictor variable in the model. As we saw in equation (2.3), we denote predictor variables with the letter X, so our model will be:

$$\text{outcome}_i = (b_1 X_i) + \text{error}_i \quad (8.2)$$

This equation means 'the outcome for an entity is predicted from their score on the predictor variable plus some error'. The model is described by a parameter, b_1, which in this context represents the relationship between the predictor variable (X) and the outcome. If we work with the raw data then to make predictions we need to know where the outcome is centred: in other words, what the value of the outcome is when the predictor is absent from the model (i.e., it is zero). This gives us a starting point for our prediction (i.e., if there were no predictor variable, what value of the outcome would we expect?). We add this value into the model as a constant, b_0, known as the intercept (we discuss this in detail in the next chapter). If we work with standardized scores (i.e., z-scores) then both the predictor and outcome have a mean of 0, so we already know the average value of the outcome when the predictor isn't in the model: it's 0. In other words, the intercept drops out of the model, leaving us with b_1:

$$z(\text{outcome})_i = b_1 z(X_i) + \text{error}_i \quad (8.3)$$

This equation means that the outcome expressed as z-scores can be predicted from the predictor variable (also expressed as z-scores) multiplied by b_1. When working with standardized scores like this, b_1 is called the Pearson product-moment correlation coefficient, and when we're not formally expressing the model as in equation (8.3), it is denoted by the letter r. Remember that we use the sample to estimate a value for b_1 (i.e., r) in the population, and its value quantifies the strength and direction of relationship between the predictor and outcome. How do we estimate this parameter? Like a quest for fire, we could search across the land … or we could use maths.

8.2.1 A detour into the murky world of covariance



The simplest way to look at whether two variables are associated is to look at whether they *covary*. To understand what covariance is, we first need to think back to the concept of variance that we met in Chapter 1. Remember that the variance of a single variable represents the average amount that the data vary from the mean. Numerically, it is described by:

$$\text{variance}\left(s^2\right) = \frac{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2}{N-1} = \frac{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)\left(x_i - \bar{x}\right)}{N-1} \quad (8.4)$$

The mean of the sample is represented by $\bar{x}$, x_i is the data point in question and N is the number of observations. If two variables are related, then changes in one variable should be met with similar changes in the other variable. Therefore, when one variable deviates from its mean we would expect the other variable to deviate from its mean in a similar way.

To illustrate what I mean, imagine we took five people and subjected them to a certain number of advertisements promoting toffee sweets, and then measured how many packets of those sweets each person bought during the next week. The data are in Table 8.1, as well as the mean and standard deviation (s) of each variable.

If there were a relationship between these two variables, then as one variable deviates from its mean, the other variable should deviate from its mean in the same or the directly opposite way. Figure 8.2 shows the data for each participant (orange circles represent the number of packets bought and blue circles represent the number of adverts watched); the orange line is the average number of packets bought and the blue line is the average number of adverts watched. The vertical lines represent the differences (remember that these differences are called *deviations* or *residuals*) between the observed values and the mean of the relevant variable. The first thing to notice about Figure 8.2 is that there is a similar pattern of deviations for both variables. For the first three participants the observed values are below the mean for both variables, for the last two people the observed values are above the mean for both variables. This pattern is indicative of a potential relationship between the two variables (because it seems that if a person's score is below the mean for one variable then their score for the other will also be below the mean).

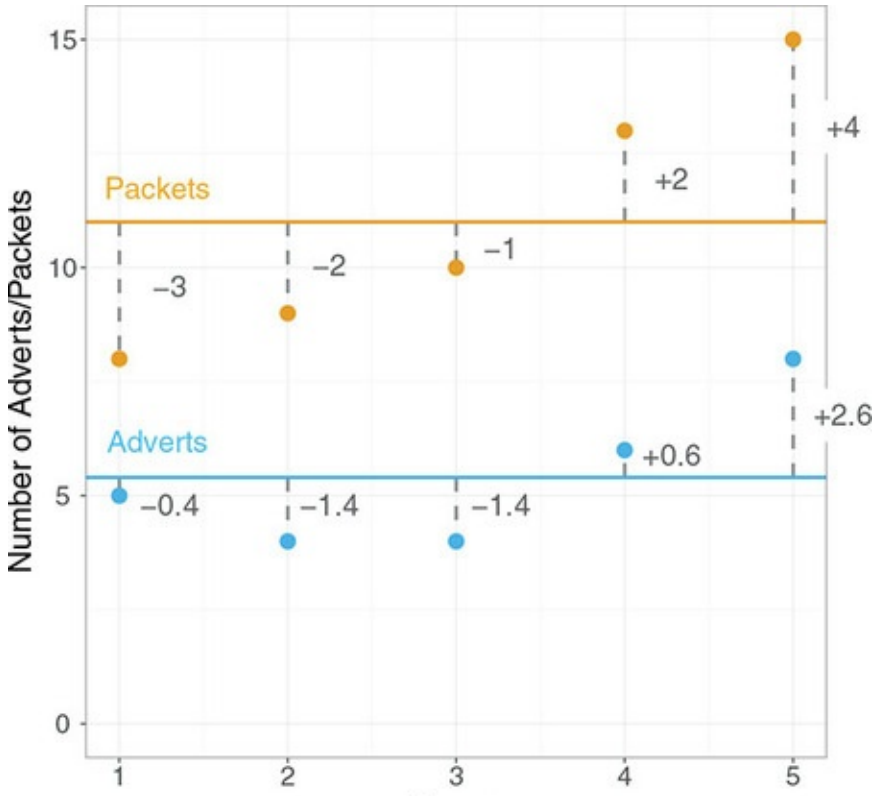
So, how do we calculate the exact similarity between the patterns of differences of the two variables displayed in Figure 8.2? One possibility is to calculate the total amount of deviation, but we would have the same problem as in the single-variable case: the positive and negative deviations would cancel out (see Section 1.8.5). Also, by adding the deviations, we would gain little insight into the *relationship* between the variables. In the single-variable case, we squared the deviations to eliminate the problem of positive and negative deviations cancelling each other out. When there are two variables, rather than squaring each deviation, we can multiply the deviation for one variable by the corresponding deviation for the second variable. If both deviations are positive or negative then this will give us a positive value (indicative of the deviations

being in the same direction), but if one deviation is positive and one negative then the resulting product will be negative (indicative of the deviations being opposite in direction). When we multiply the deviations of one variable by the corresponding deviations of a second variable, we get the **cross-product deviations**. As with the variance, if we want an average value of the combined deviations for the two variables, we divide by the number of observations (we actually divide by $N - 1$ for reasons explained in Jane Superbrain Box 2.2). This averaged sum of combined deviations is known as the **covariance**. We can write the covariance in equation form:

Table 8.1 Some data about toffees and adverts

Participant:	1	2	3	4	5	Mean	s
Adverts Watched	5	4	4	6	8	5.4	1.673
Packets Bought	8	9	10	13	15	11.0	2.915

Figure 8.2 Graphical display of the differences between the observed data and the means of two variables



$$\text{covariance}(x, y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{N - 1} \quad (8.5)$$

Notice that the equation is the same as the equation for variance (equation (1.7)), except that instead of squaring the deviances, we multiply them by the corresponding deviance of the second variable.

For the data in Table 8.1 and Figure 8.2 we get a value of 4.25:

$$\begin{aligned}\text{covariance}(x,y) &= \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{N-1} \\ &= \frac{(-0.4)(-3)+(-1.4)(-2)+(-1.4)(-1)+(0.6)(2)+(2.6)(4)}{N-1} \\ &= \frac{1.2+2.8+1.4+1.2+10.4}{4} = \frac{17}{4} = 4.25\end{aligned} \tag{8.6}$$

A positive covariance indicates that as one variable deviates from the mean, the other variable deviates in the same direction. On the other hand, a negative covariance indicates that as one variable deviates from the mean (e.g., increases), the other deviates from the mean in the opposite direction (e.g., decreases). However, the covariance depends upon the scales of measurement used: it is not a standardized measure. For example, if we use the data above and assume that they represented two variables measured in miles then the covariance is 4.25 square miles (as calculated above). If we convert these data into kilometres (by multiplying all values by 1.609) and calculate the covariance then we find that it increases to 11 square kilometres. This dependence on the scale of measurement is a problem because we cannot compare covariances in an objective way – we cannot say whether a covariance is particularly large or small relative to another data set unless both data sets were measured in the same units.

8.2.2 Standardization and the correlation coefficient



To overcome the problem of dependence on the measurement scale, we need to convert the covariance into a standard set of units. This process is known as **standardization**. We need a unit of measurement into which any variable can be converted, and typically we use the *standard deviation*. We came across this measure in Section 1.8.5 and saw that, like the variance, it is a measure of the average deviation from the mean. If we divide any distance from the mean by the standard deviation, it gives us that distance in standard deviation units. For example, for the data in Table 8.1, the standard deviation for the number of packets bought is approximately 3.0 (the exact value is 2.92). In Figure 8.2 we can see that the observed value for participant 1 was 3 packets less than the mean (so there was an error of −3 packets of sweets). If we divide this deviation, −3, by the standard deviation, which is approximately 3, then we get a value of −1. This tells us that the difference between participant 1's score and the mean

was −1 standard deviation. In this way we can express the deviation from the mean for a participant in standard units by dividing the observed deviation by the standard deviation.

It follows from this logic that if we want to express the covariance in a standard unit of measurement we can divide by the standard deviation. However, there are two variables and, hence, two standard deviations. When we calculate the covariance we calculate two deviations (one for each variable) and multiply them. We do the same for the standard deviations: we multiply them and divide the covariance by the product of this multiplication. The standardized covariance is known as a *correlation coefficient* and is defined as follows:

$$r = \frac{\text{cov}_{xy}}{s_x s_y} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(N-1)s_x s_y} \quad (8.7)$$

in which s_x is the standard deviation of the first variable and s_y is the standard deviation of the second variable (all other letters are the same as in the equation defining covariance). This coefficient, the *Pearson product-moment correlation coefficient* or **Pearson's correlation coefficient**, r, was invented by Karl Pearson with Florence Nightingale David[2] doing a lot of the hard maths to derive distributions for it (see Figure 8.3 and Jane Superbrain Box 8.1).[3] If we look back at Table 8.1 we see that the standard deviation for the number of adverts watched (s_x) was 1.673, and for the number of packets of crisps bought (s_y) was 2.915. If we multiply these together we get 1.673 × 2.915 = 4.877. Now all we need to do is take the covariance, which we calculated a few pages ago as being 4.250, and divide by these multiplied standard deviations. This gives us r = 4.250/4.877 = 0.871.

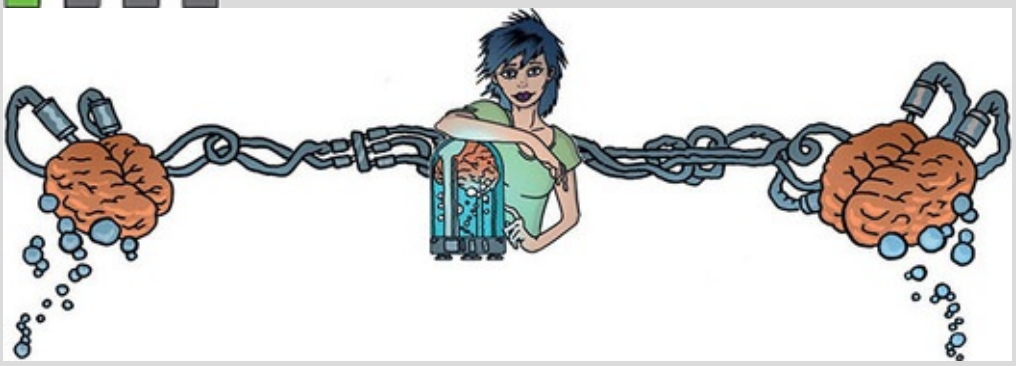
2 Not to be confused with the Florence Nightingale in Chapter 5 who she was named after.

3 Pearson's product-moment correlation coefficient is denoted by r but, just to confuse us, when we square r (as in Section 8.4.2.2) an upper-case R is typically used.

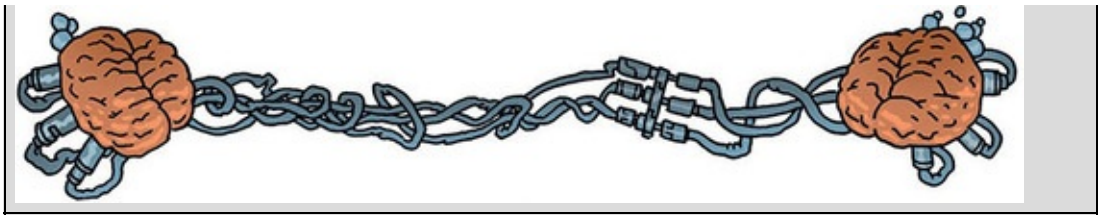
Figure 8.3 Karl Pearson and Florence Nightingale David



Jane Superbrain 8.1 Who said statistics was dull? Part 2



We saw in Jane Superbrain Box 2.3 that Fisher and Neyman had a feud over their different views of hypothesis testing. Fisher seemed to believe that if you're going to feud with one of your prominent peers, you may as well feud with them all, and he didn't care much for Karl Pearson either. This wasn't a great career move on Fisher's part because the more senior Pearson wielded huge influence through his journal *Biometrika*. The feud began when Pearson published a paper of Fisher's in his journal but belittled it in his editorial. Two years later, Pearson's group published work following on from Fisher's paper without consulting him. Fisher turned down a job in Pearson's group and published 'improvements' on Pearson's ideas that were greeted by Pearson like a fish would greet a cat as she entered his tank. For his part, Pearson wrote in his own journal about apparent errors made by Fisher (Barnard, 1963; Field, 2005d; Savage, 1976). Statistics was never dull back then.



By standardizing the covariance we end up with a value that has to lie between −1 and +1 (if you find a correlation coefficient less than −1 or more than +1 you can be sure that something has gone hideously wrong). We saw in Section 3.7.2 that a coefficient of +1 indicates that the two variables are perfectly positively correlated: as one variable increases, the other increases by a proportionate amount. This does not mean that the change in one variable *causes* the other to change, only that their changes coincide (Misconception Mutt 8.1). Conversely, a coefficient of −1 indicates a perfect negative relationship: if one variable increases, the other decreases by a proportionate amount. A coefficient of zero indicates no linear relationship at all and so as one variable changes, the other stays the same. We also saw that because the correlation coefficient is a standardized measure of an observed effect, it is a commonly used effect size measure and that values of ±0.1 represent a small effect, ±0.3 is a medium effect and ±0.5 is a large effect (although we should interpret the effect size within the context of the research literature and not use these canned effect sizes).

We have just described a **bivariate correlation**, which is a correlation between two variables. Later in the chapter we'll look at variations on the correlation that adjust for one or more additional variables.

8.2.3 The significance of the correlation coefficient



Although we can interpret the size of a correlation coefficient directly (Section 3.7.2), we have seen in Chapter 2 that scientists like to test hypotheses using probabilities. In the case of a correlation coefficient we can test the hypothesis that the correlation is different from zero (i.e., different from 'no relationship'). If we find that our observed coefficient was very unlikely to be (at least) as big as it would be if there were no effect in the population then we might gain confidence that the relationship that we have observed is statistically meaningful.

There are two ways that we can go about testing this hypothesis. The first is to use the trusty *z*-scores that keep cropping up in this book. As we have seen, *z*-scores are useful because we know the probability of a given value of *z* occurring, if the distribution from which it comes is normal. There is one

problem with Pearson's *r*, which is that it is known to have a sampling distribution that is not normally distributed. This would be a bit of a nuisance, except that thanks to our friend Fisher (1921) we can adjust *r* so that its sampling distribution *is* normal:

$$z_r = \frac{1}{2}\log_e\left(\frac{1+r}{1-r}\right) \quad (8.8)$$

The resulting z_r has a standard error given by:

$$SE_{z_r} = \frac{1}{\sqrt{N-3}} \quad (8.9)$$

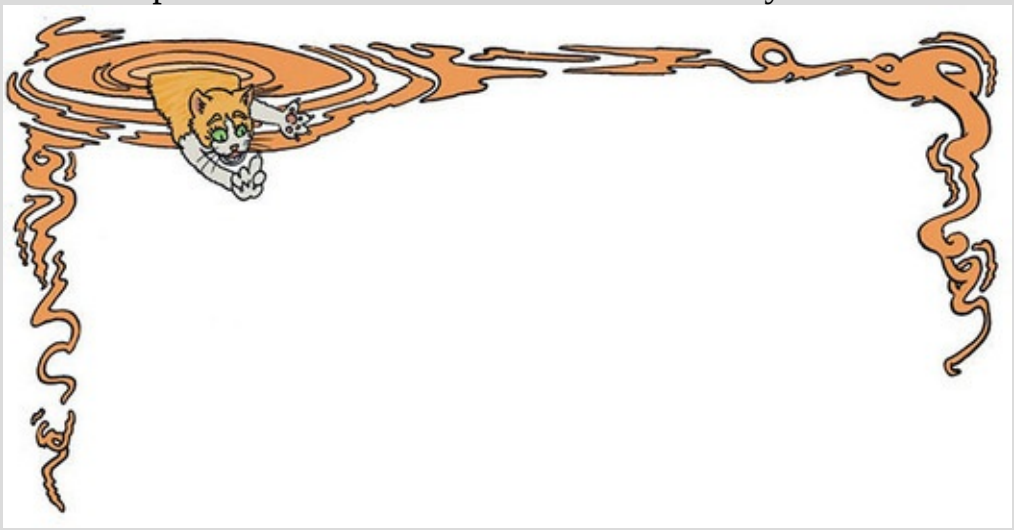
For our advert example, our $r = 0.871$ becomes 1.337 with a standard error of 0.707.

We can then transform this adjusted *r* into a *z*-score in the usual way. If we want a *z*-score that represents the size of the correlation relative to a particular value, then we compute a *z*-score by subtracting the value that we want to test against and then dividing by the standard error. Normally we want to see whether the correlation is different from 0, in which case we subtract 0 from the observed value of z_r and divide by the standard error, which is the same as simply dividing z_r by its standard error:

$$z = \frac{z_r}{SE_{z_r}} \quad (8.10)$$

For our advert data this gives us 1.337/0.707 = 1.891. We can look up this value of *z* (1.89) in the table for the normal distribution in the Appendix and get the one-tailed probability from the column labelled 'Smaller Portion' (think back to Section 1.8.6). In this case the value is 0.02938. To get the two-tailed probability we multiply this value by 2, which gives us 0.05876. As such the correlation is not significant, because $p > 0.05$.

Misconception Mutt 8.1 Correlations and causality



The Misconception Mutt had just come out of a lecture on correlation with his owner. It had made him think about the relationships between certain important variables in his life. There seemed to be a correlation between him getting under his owner's feet in the kitchen and being fed, staring lovingly at his owner seemed to get him strokes, and walking nicely and not chasing squirrels got him treats. As his mind wandered, he thought about how much he liked the campus. It was full of trees, grass and squirrels that he really wanted to chase if only it didn't stop the supply of treats. 'I really like treats,' he thought, 'and I really like chasing squirrels. But chasing squirrels is negatively correlated with treats, so if I chase the squirrel it will cause the number of treats to go down.'

He was hungry, so he started walking extra nicely. A squirrel darted up to him and jumped on his head as if trying to goad him into a chase. It started to grow, becoming heavier and more ginger until the grinning Correcting Cat stared down into the mutt's eyes from on top of his head.

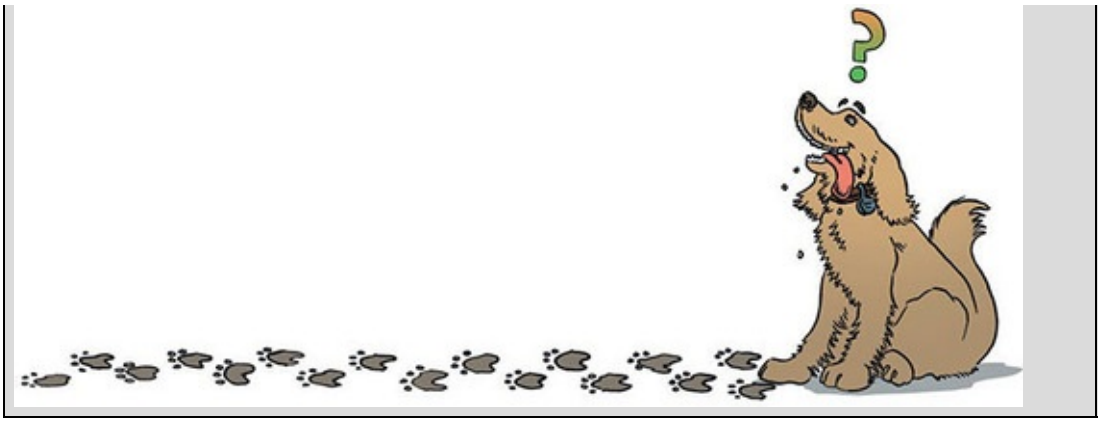
'Correlation coefficients give no indication of the direction of causality,' he said to the dog. 'You might think that chasing squirrels causes fewer treats, but statistically speaking there's no reason why fewer treats isn't causing you to chase squirrels.'

'That's ridiculous,' replied the mutt.

'It may be less intuitive to think of fewer treats causing you to chase squirrels more, but statistically speaking the correlation between those variables provides no information about cause: it is purely a measure of the degree to which variables covary. Think back to the lecture: the correlation measures whether differences between scores on one variable and its mean correspond to differences between scores on a second variable and its mean. Causality does not feature in the computation.'

'Another issue,' the cat continued, 'is that there could be other measured or unmeasured variables affecting the two things that correlate. This is known as the third-variable problem or the *tertium quid* (Section 1.7.2). Perhaps the time of day affects both how many treats you get and how many squirrels you chase.'

Annoyingly, the cat had a point, so the mutt shook him from his head and, after watching his feline form shrink back to that of a squirrel, chased him across campus.



In fact, the hypothesis that the correlation coefficient is different from 0 is usually (SPSS, for example, does this) tested not using a *z*-score, but using a different test statistic called a *t*-statistic with $N - 2$ degrees of freedom. This statistic can be obtained directly from *r*:

$$t_r = \frac{r\sqrt{N-2}}{\sqrt{1-r^2}} \tag{8.11}$$

So you might wonder why I told you about *z*-scores. Partly it was to keep the discussion framed in concepts with which you are already familiar (we don't encounter the *t*-statistic properly for a few chapters), but also it is useful background information for the next section.

8.2.4 Confidence intervals for r

We saw in Chapter 2 that 95% confidence intervals tell us about the likely value (in this case of the correlation) in the population (assuming that your sample is one of the 95% for which the confidence interval contains the true value). To compute confidence intervals for *r*, we take advantage of what we learnt in the previous section about converting *r* to z_r (to make the sampling distribution normal), and use the associated standard errors. By using z_r we can construct a confidence interval in the usual way. For example, a 95% confidence interval is calculated (see Eq. 2.15) as:

$$\begin{aligned}\text{lower boundary of confidence interval} &= \bar{X} - (1.96 \times SE)\\ \text{upper boundary of confidence interval} &= \bar{X} + (1.96 \times SE)\end{aligned} \tag{8.12}$$

In the case of our transformed correlation coefficients, these equations become:

$$\begin{aligned}\text{lower boundary of confidence interval} &= z_r - \left(1.96 \times SE_{z_r}\right)\\ \text{upper boundary of confidence interval} &= z_r + \left(1.96 \times SE_{z_r}\right)\end{aligned} \tag{8.13}$$

For our advert data we get 1.337 − (1.96 × 0.707) = −0.049, and 1.337 + (1.96 × 0.707) = 2.723. Remember that these values are in the z_r metric, but we can

convert back to a correlation coefficient using:

$$r = \frac{e^{2z_r} - 1}{e^{2z_r} + 1} \quad (8.14)$$

This gives us an upper bound of $r = 0.991$ and a lower bound of −0.049 (because this value is so close to zero the transformation to z has no impact).

I was moaning earlier on about how SPSS doesn't make tea for you. Another thing that it doesn't do is compute these confidence intervals for you, although there is a good macro available by Weaver & Koopman (2014). However, SPSS Statistics does something even better (than computing confidence intervals, not than making tea): it computes a bootstrap confidence interval. We learnt about the percentile bootstrap confidence interval in Section 6.12.3: it is a confidence interval that is derived from the actual data and, therefore, we know it will be accurate even when the sampling distribution of r is not normal. This is very good news indeed.

8.3 Data entry for correlation analysis

When looking at relationships between variables, each variable is entered in a separate column in the data editor. So, for each variable you have measured, create a variable with an appropriate name, and enter a participant's scores across one row. If you have categorical variables (such as the participant's sex) these can also be entered in a column (remember to define appropriate value labels). For example, if we wanted to calculate the correlation between the two variables in Table 8.1 we would enter these data as in Figure 8.4; each variable is entered in a separate column, and each row represents a single individual's data (so the first consumer saw 5 adverts and bought 8 packets).

Cramming Sam's Tips Correlation

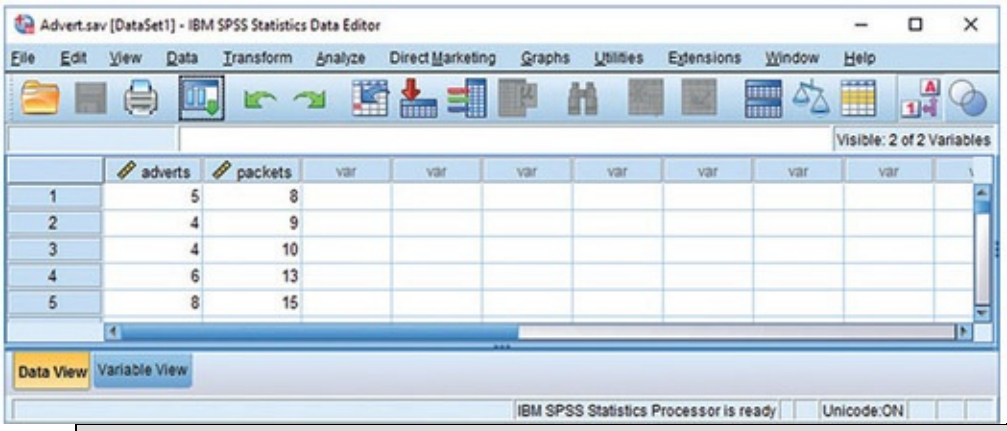


- A crude measure of the relationship between variables is the covariance.
- If we standardize this value we get Pearson's correlation coefficient, r.

- The correlation coefficient has to lie between −1 and +1.
- A coefficient of +1 indicates a perfect positive relationship, a coefficient of −1 indicates a perfect negative relationship, and a coefficient of 0 indicates no linear relationship.
- The correlation coefficient is a commonly used measure of the size of an effect: values of ±0.1 represent a small effect, ±0.3 is a medium effect and ±0.5 is a large effect. However, interpret the size of correlation within the context of the research you've done rather than blindly following these benchmarks.



Figure 8.4 Data entry for correlation





Enter the advert data and use the chart editor to produce a scatterplot (number of packets bought on the *y*-axis, and adverts watched on the *x*-axis) of the data.

8.4 Bivariate correlation

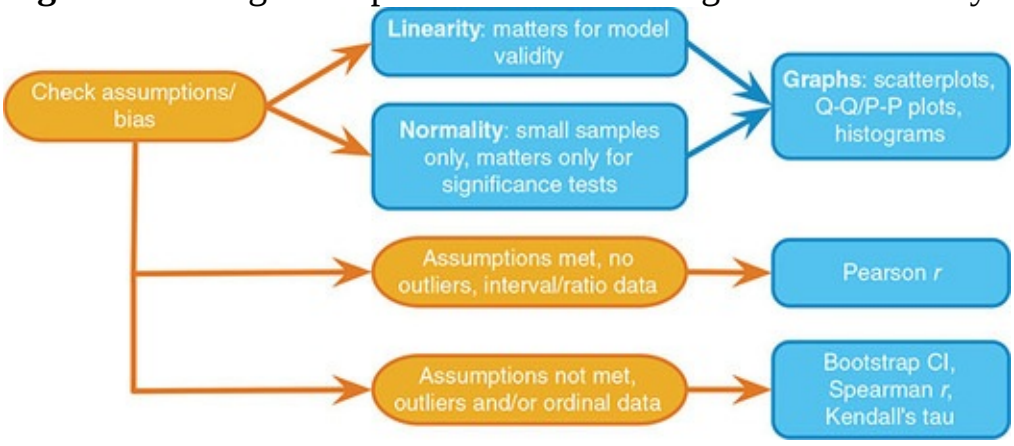
Figure 8.5 shows a general procedure to follow when computing a bivariate correlation coefficient. First, check for sources of bias as outlined in Chapter 6. The two most important ones in this context are linearity and normality. Remember that we're fitting a linear model to the data, so if the relationship

between variables is not linear then this model is invalid (a transformation might help to make the relationship linear). To meet this requirement, the outcome variable needs to be measured at the interval or ratio level (see Section 1.6.2), as does the predictor variable (one exception is that a predictor variable can be a categorical variable with only two categories – we'll get onto this in Section 8.4.5). As far as normality is concerned, we care about this only if we want confidence intervals or significance tests and if the sample size is small (Section 6.6.1).

If the data have outliers, are not normal (and the sample is small) or your variables are measured at the ordinal level then you can use Spearman's rho (Section 8.4.3) or Kendall's tau (Section 8.4.4), which are versions of the correlation coefficient applied to ranked data (just like the tests in the previous chapter). Ranking the data reduces the impact of outliers. Furthermore, given that normality matters only for inferring significance and computing confidence intervals, we could use a bootstrap to compute the confidence interval, then we don't need to worry about the distribution.

In Chapter 5 we looked at an example relating to exam anxiety: a psychologist was interested in the effects of exam stress and revision on exam performance. She had devised and validated a questionnaire to assess state anxiety relating to exams (called the Exam Anxiety Questionnaire, or EAQ). This scale produced a measure of anxiety scored out of 100. Anxiety was measured before an exam, and the percentage mark of each student on the exam was used to assess the exam performance. She also measured the number of hours spent revising. These data are in **Exam Anxiety.sav**. We have already created scatterplots for these data (Section 5.8) so we don't need to do that again; however, we will look at the distributions of the three main variables.

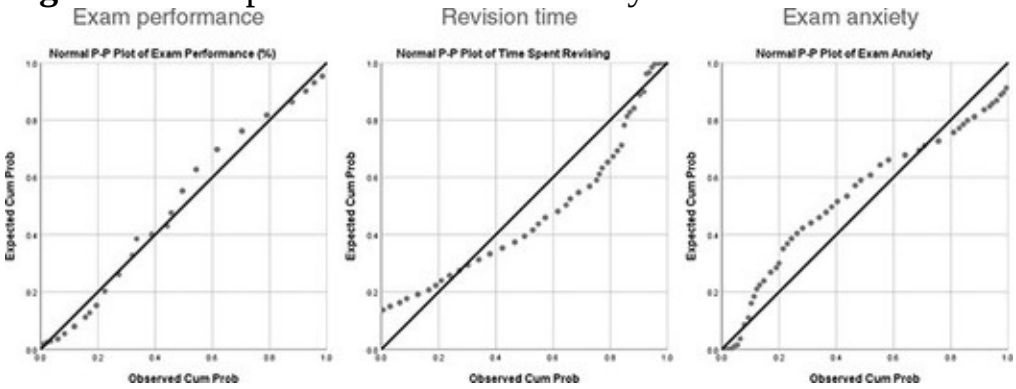
Figure 8.5 The general process for conducting correlation analysis





Create P-P plots of the variables **Revise, Exam**, and **Anxiety**.

Figure 8.6 P-P plots for the exam anxiety variables



It's clear from the P-P plots in Figure 8.6 that exam performance is the most normally distributed of the variables (the dots hover close to the line), but for exam revision and exam anxiety there is evidence of skew (the dots snake around the diagonal line). This skew is a problem if we want to do significance tests or look at confidence intervals. The sample contains 103 observations, which is reasonably large, and possibly large enough for the central limit theorem to relieve us of concerns about normality. However, it would be advisable to use a bootstrap to get robust confidence intervals. We might also consider using a rank-based method to compute the correlation coefficient itself.

Oditi's Lantern Correlations



'I, Oditi, understand the importance of relationships. Being leader of the cult of undiscovered numerical truths, no one wants a relationship with me. This truth makes me sad. I need my cult to help me better understand relationships so that I might have one and leave my empty and soulless existence behind. To this end, we must look within the data and quantify all relationships we find. Stare into my

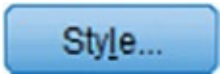
lantern and discover how…. Stare too long and possibly you'll never have another relationship.'



8.4.1 General procedure for correlations using SPSS Statistics

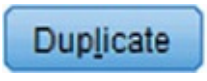
You conduct bivariate correlation using the dialog box accessed through *Analyze* *Correlate* Bivariate... . In this dialog box (Figure 8.7) the variables in the data editor are listed on the left-hand side and there is an empty box labelled *Variables* on the right-hand side. You can select any variables from the list using the mouse and transfer them to the *Variables* box by dragging them there or clicking .SPSS creates a table (called a *correlation matrix*) of correlation coefficients for all combinations of specified variables. For our current example, select the variables **Exam performance, Exam anxiety** and **Time spent revising** and transfer them to the *Variables* box. Having selected the variables of interest, you can choose between three correlation coefficients: the default is Pearson's product-moment correlation coefficient (Pearson), but you can also select Spearman's rho (Spearman) and Kendall's tau (Kendall's tau-b) – we'll explore the differences in due course. You can choose more than one of these correlation coefficients if you like.

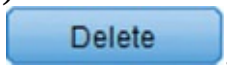
You can also specify whether the test is one-or two-tailed. In Section 2.9.5 I advised against one-tailed tests, so I would leave the default of Two-tailed, but if you don't like my advice then you might select One-tailed if your hypothesis is directional (e.g., 'The more anxious someone is about an exam, the worse their mark will be') and Two-tailed if it is non-directional (i.e., 'I'm not sure whether exam anxiety will improve or reduce exam marks').

Clicking 

opens the dialog box in Figure 8.8, which allows you to apply formatting to the table of correlations in the output. The drop-down menu in the column called *Value* is for selecting types of cells in the output: you could select the cells containing the correlation coefficients (as I have in the figure), the cells containing the sample sizes (*N*), the means, the significance values, or all the cells in the table. Clicking in the column labelled *Condition* opens a dialog box for setting a condition for formatting. In the figure I have set a condition of formatting cells that contain an absolute value greater than or equal to 0.5. Doing so will apply the formatting only to cells containing correlation coefficients greater than 0.5 or smaller than −0.5. Clicking in the column labelled *Format* opens a dialog box for defining what formatting you wish to apply. In the figure I have chosen to change the background colour of the cells to yellow, but you can do other things such as change the text colour, or make it bold or italic and so on. By using the settings in Figure 8.8, I have (1) chosen to format only the cells containing the correlation coefficients; and (2) asked that cells containing a value greater than 0.5 or smaller than −0.5 have a yellow background. The effect will be that these cells are highlighted in the output: I will very quickly be able to see which variables were strongly correlated because their cells will be highlighted. Not that I recommend this (because it's the sort of thing that will encourage you to blindly apply the $p < 0.05$ criterion), but I imagine that some people use this facility to highlight cells with significance values less than 0.05. You know better than to do that though.

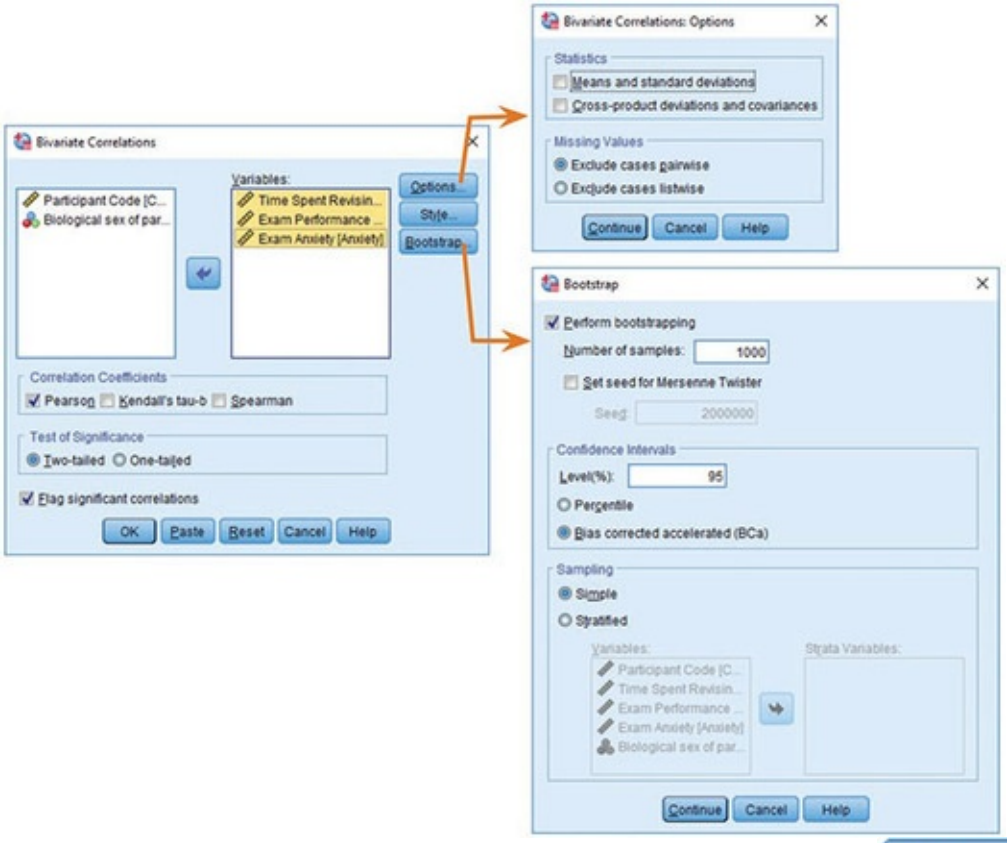
You can set several formatting rules by clicking 

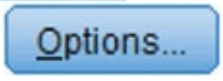
to create a new rule (it will appear as a new row in the dialog box) and then editing the rule. For example, we could set up four rules to use different background colours for cells containing correlations based on whether they are tiny (absolute values between 0 and 0.1), small to medium (absolute values between 0.1 and 0.3), medium to large (absolute values between 0.3 and 0.5) or large (absolute values greater than 0.5). These four rules would be quite similar so once we have set the first rule we could save time by selecting it and clicking 

to create a new rule that duplicates its settings (which we could then tweak). You can delete a rule by selecting the row representing the rule and clicking 

.

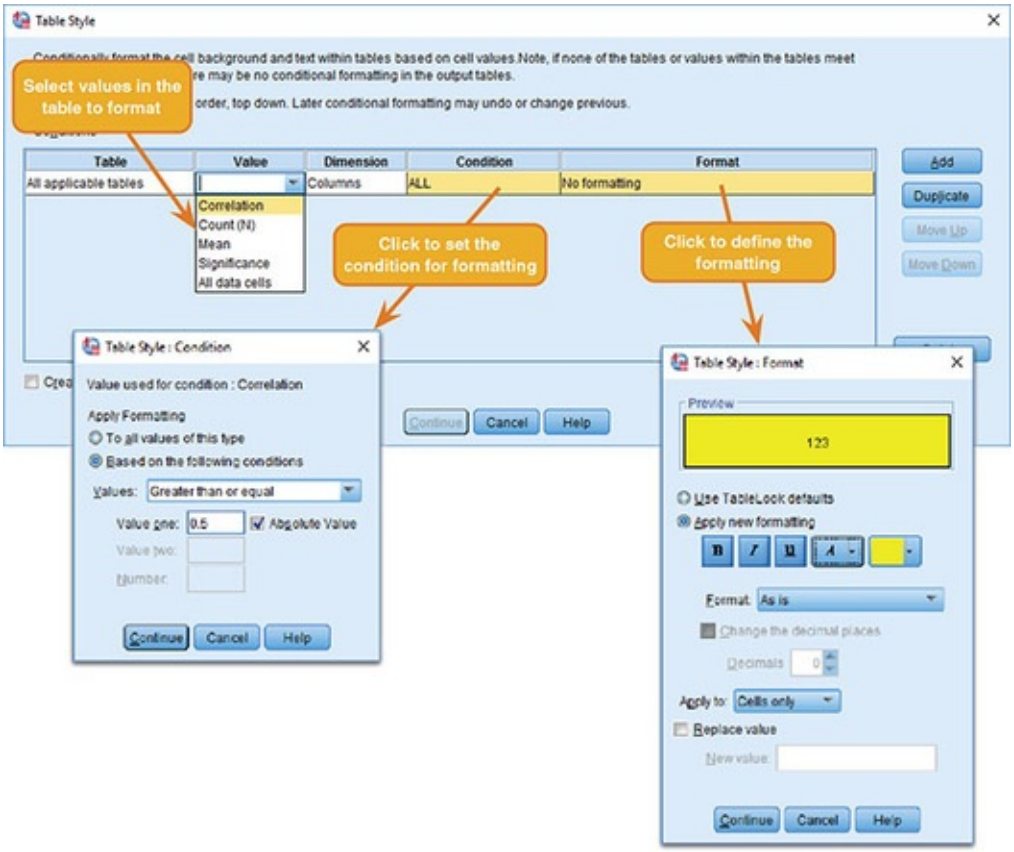
Figure 8.7 Dialog box for conducting a bivariate correlation



Going back to the main dialog box (Figure 8.7), clicking 

opens a dialog box with two *Statistics* options and two options for missing values. The *Statistics* options are enabled only when Pearson's correlation is selected (otherwise they appear 'greyed out'). These two options are meaningful only for interval data, which the Pearson correlation requires, so it makes sense that these options are deactivated if the Pearson correlation has not been selected. Selecting the tick-box labelled *Means and standard deviations* produces the mean and standard deviation of the variables selected for analysis. Selecting the tick-box labelled *Cross-product deviations and covariances* produces the values of these statistics for the variables in the analysis. The cross-product deviations are the values of the numerator (top half) of equation (8.5). The covariances between variables are what you would get from applying equation (8.5) to your variables. In other words, the covariance values are the cross-product deviations divided by $N - 1$ and represent the unstandardized correlation coefficient. In most instances you won't need these options, but they occasionally come in handy (see Oliver Twisted). At this point we need to decide how to deal with missing values (look back to SPSS Tip 6.1).

Figure 8.8 The *Table Style* dialog box



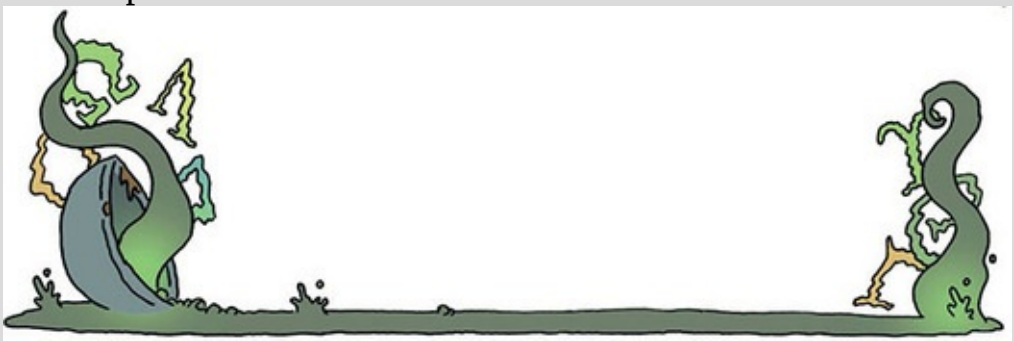
Finally, to get bootstrapped confidence intervals for the correlation coefficient click Bootstrap... . We discussed this dialog box in Section 6.12.3; to recap, select ☑ Perform bootstrapping to activate bootstrapping for the correlation coefficient, and to get a 95% confidence interval click ◉ Percentile or ◉ Bias corrected accelerated (BCa). For this analysis, let's ask for a bias corrected (BCa) confidence interval.

Oliver Twisted Please, Sir, can I have some more … options?



Oliver is so excited to get onto analysing his data that he doesn't want me to spend pages waffling on about options that you will probably never use. 'Stop writing, you waffling fool,' he says. 'I want to analyse my data.' Well, he's got a point. If you want to find out more about what the Options... do in correlation, then look at

the companion website.

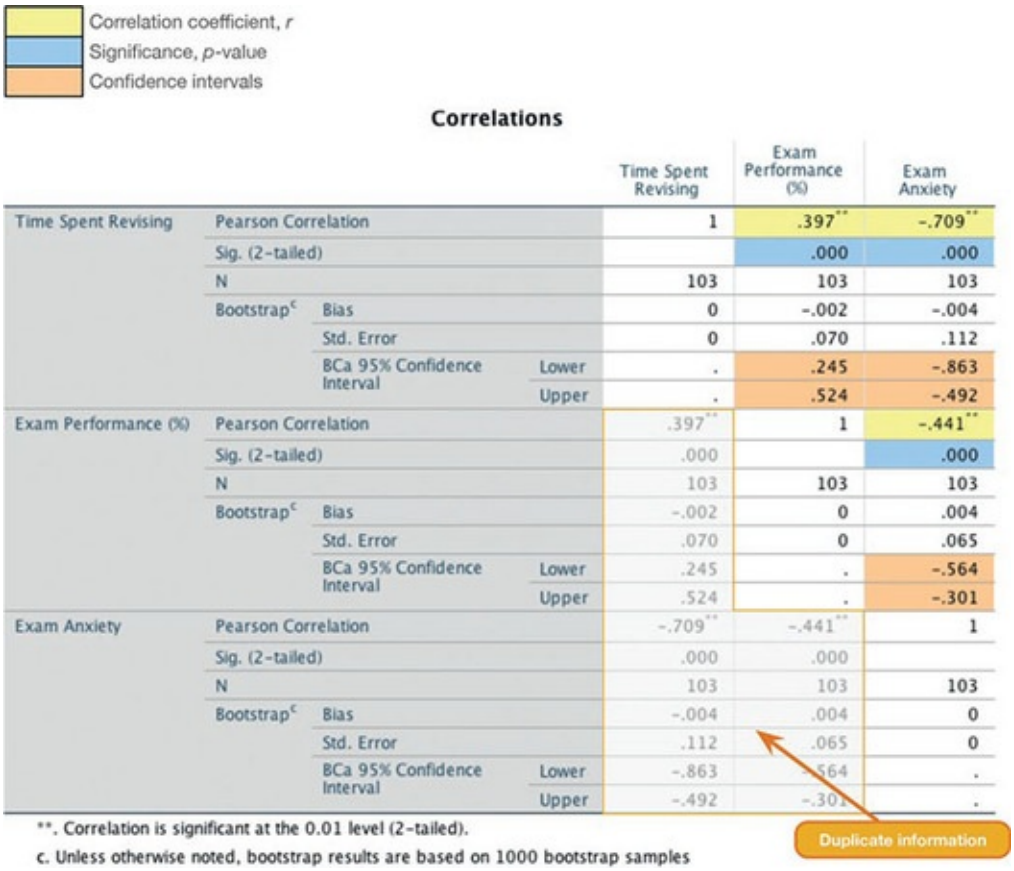


8.4.2 Pearson's correlation coefficient using SPSS

Statistics

To obtain Pearson's correlation coefficient, run through the general procedure that we've just looked at (Section 8.4.1), selecting Pearson (the default) and then clicking OK (Figure 8.7). In the viewer we'll see a matrix of results (Output 8.1), which is not as bewildering as it looks. For one thing, the information in the top part of the table is the same as in the bottom half (which I have faded out), so we can ignore half of the table. The first row tells us about time spent revising. This row is subdivided into other rows, the first of which contains the correlation coefficients with the exam performance ($r = 0.397$) and exam anxiety ($r = -0.709$). The second major row in the table tells us about exam performance, and from this part of the table we get the correlation coefficient for its relationship with exam anxiety, $r = -0.441$. Directly underneath each correlation coefficient we're told the significance value of the correlation and the sample size (N) on which it is based. The significance values are all less than 0.001 (as indicated by the double asterisk after the coefficient). This significance value tells us that the probability of getting a correlation coefficient at least this big in a sample of 103 people if the null hypothesis were true (there was no relationship between these variables) is very low (close to zero in fact). All the significance values are below the standard criterion of 0.05, indicating a 'statistically significant' relationship.

Output 8.1



Given the lack of normality in some of the variables, we should be more concerned with the bootstrapped confidence intervals than the significance *per se*: this is because the bootstrap confidence intervals will be unaffected by the distribution of scores, whereas the significance value might be. These confidence intervals are labelled *BCa 95% Confidence Interval*, and you're given two values: the upper boundary and the lower boundary. For the relationship between revision time and exam performance the interval is 0.245 to 0.524, for revision time and exam anxiety it is −0.863 to −0.492, and for exam anxiety and exam performance it is −0.564 to −0.301.

There are two important points here. First, because the confidence intervals are derived empirically using a random sampling procedure (i.e., bootstrapping) the results will be slightly different each time you run the analysis. The confidence intervals you get won't be the same as in Output 8.1, and that's normal and nothing to worry about. Second, think about what a correlation of zero represents: it is no effect whatsoever. A confidence interval is the boundary between which the population value falls in 95% of samples. If we assume that our sample is one of the 95% of samples that produces a confidence interval containing the true value of the correlation (and be aware that this assumption might be wrong), then if the interval crosses zero it means that (1) the population

value could be zero (i.e., no effect at all), and (2) we can't be sure if the true relationship is positive or negative because the population value could plausibly be a negative or positive value. None of our confidence intervals cross zero, therefore we might take this information to mean that there is a genuine effect in the population. In psychological terms, this means that there is a complex interrelationship between the three variables: (1) as anxiety about an exam increases, the percentage mark obtained in that exam significantly decreases; (2) as the amount of time revising increases, the percentage obtained in the exam significantly increases; and (3) as revision time increases, the student's anxiety about the exam significantly decreases.

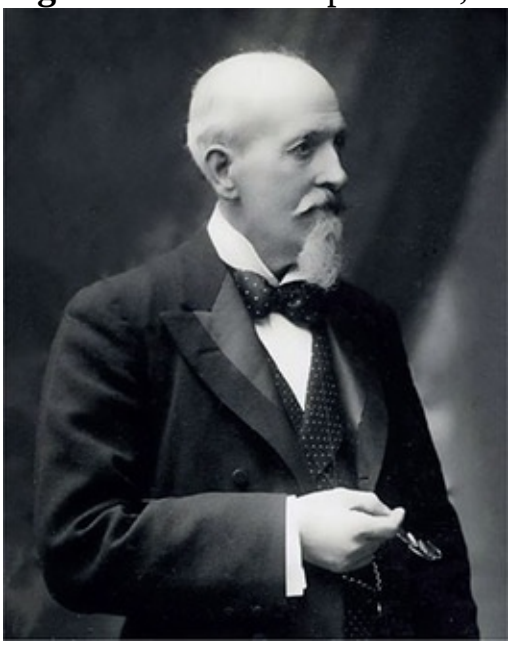
Although we cannot make direct conclusions about causality from a correlation coefficient (Misconception Mutt 8.1), we can take it a step further by squaring it. The correlation coefficient squared (known as the **coefficient of determination**, R^2) is a measure of the amount of variability in one variable that is shared by the other. Exam performance in our data varied, reflecting the fact that people's scores were not identical; they will have varied for any number of reasons (different ability, different levels of preparation and so on). If we add up this individual variability we get an estimate of how much variability in exam performances there was in total (this is computed with equation (1.7) from Section 1.8.5). It turns out that the variance is about 672 for the exam performance scores. Imagine that this value is the surface area of a biscuit. The size of the biscuit represents 100% of the variance in exam scores. Imagine that exam anxiety comes along and takes a bite out of the biscuit. R^2 tells us how much of this exam performance 'biscuit' has been eaten by exam anxiety. In other words, it tells us the proportion of variance (what proportion of the 672 units of variation) overlaps with exam anxiety. These two variables had a correlation of −0.4410 and so the value of R^2 is $-0.4410^2 = 0.194$, which means that 0.194 of the variability in exam performance is shared by exam anxiety. It's a bit easier to think of this value as a percentage rather than a proportion, which we can obtain by multiplying by 100. In this example, then, exam anxiety shares 19.4% of the variability in exam performance. To put this value into perspective, this leaves 80.6% of the variability unexplained.



You'll often see people write things about R^2 that imply causality: they might

write ‘the variance in *y accounted for* by *x*’, or ‘the variation in one variable *explained* by the other’. Although R^2 is a useful measure of the substantive importance of an effect, it cannot be used to infer causal relationships. Exam anxiety might well share 19.4% of the variation in exam scores, but it does not necessarily cause this variation.

Figure 8.9 Charles Spearman, ranking furiously



8.4.3 Spearman’s correlation coefficient

Spearman’s correlation coefficient, denoted by r_s (Figure 8.9), is a non-parametric statistic that is useful to minimize the effects of extreme scores or the effects of violations of the assumptions discussed in Chapter 6. You’ll sometimes hear the test referred to as Spearman’s rho (pronounced ‘row’, as in ‘row your boat gently down the stream’). Spearman’s test works by first ranking the data (see Section 7.4.1), and then applying Pearson’s equation (equation (8.7)) to those ranks (Spearman, 1910).



I was born in England, which has some bizarre traditions. One such oddity is the World’s Biggest Liar competition held annually at the Santon Bridge Inn in Wasdale (in the Lake District). The contest honours a local publican, ‘Auld Will

Ritson', who in the nineteenth century was famous in the area for his far-fetched stories (one such tale being that Wasdale turnips were big enough to be hollowed out and used as garden sheds). Each year locals are encouraged to attempt to tell the biggest lie in the world (lawyers and politicians are apparently banned from the competition). Over the years there have been tales of mermaid farms, giant moles, and farting sheep blowing holes in the ozone layer. (I am thinking of entering next year and reading out some sections of this book.)

Imagine I wanted to test a theory that more creative people will be able to create taller tales. I gathered together 68 past contestants from this competition and noted where they were placed in the competition (first, second, third, etc.); I also gave them a creativity questionnaire (maximum score 60). The position in the competition is an ordinal variable (see Section 1.6.2) because the places are categories but have a meaningful order (first place is better than second place and so on). Therefore, Spearman's correlation coefficient should be used (Pearson's *r* requires interval or ratio data). The data for this study are in the file **The Biggest Liar.sav**. The data are in two columns: one labelled **Creativity** and one labelled **Position** (there's a third variable in there, but we will ignore it for the time being). For the **Position** variable, each of the categories described above has been coded with a numerical value. First place has been coded with the value 1, with other positions being labelled 2, 3 and so on. Note that for each numeric code I have provided a value label (just like we did for coding variables). I have also set the *Measure* property of this variable to  Ordinal.

Output 8.2

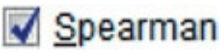
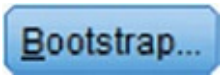
Correlations

					Creativity	Position in Biggest Liar Competition
Spearman's rho	Creativity	Correlation Coefficient			1.000	-.373**
		Sig. (2-tailed)			.	.002
		N			68	68
		Bootstrap[c]	Bias		.000	.001
			Std. Error		.000	.125
			BCa 95% Confidence Interval	Lower	.	-.602
				Upper	.	-.122
	Position in Biggest Liar Competition	Correlation Coefficient			-.373**	1.000
		Sig. (2-tailed)			.002	.
		N			68	68
		Bootstrap[c]	Bias		.001	.000
			Std. Error		.125	.000
			BCa 95% Confidence Interval	Lower	-.602	.
				Upper	-.122	.

**. Correlation is significant at the 0.01 level (2-tailed).

c. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

The procedure for doing a Spearman correlation is the same as for a Pearson correlation, except that in the *Bivariate Correlations* dialog box (Figure 8.7) we

need to select  and deselect the option for a Pearson correlation. As with the Pearson correlation, we should use the  option to get some robust confidence intervals.

The output for a Spearman correlation (Output 8.2) is like that of the Pearson correlation, giving the correlation coefficient between the two variables (−0.373), the significance value of this coefficient (0.002) and the sample size (68).[4] We also have the BCa 95% confidence interval, ranging from −0.602 to −0.122.[5] The fact that the confidence interval does not cross zero (and the significance is less than 0.05) tells us that there is a significant negative relationship between creativity scores and how well someone did in the World's Biggest Liar competition: as creativity increased, position decreased. This might seem contrary to what we predicted until you remember that a low number means that you did well in the competition (a low number such as 1 means you came first, and a high number like 4 means you came fourth). Therefore, our hypothesis is supported: as creativity increased, so did success in the competition.

4 It is good to check that the value of *N* corresponds to the number of observations that were made. If it doesn't then data may have been excluded for some reason.

5 Remember that these confidence intervals are based on a random sampling procedure so the values you get will differ slightly from mine, and will change if you rerun the analysis.



Did creativity cause success in the World's Biggest Liar competition?

8.4.4 Kendall's tau (non-parametric)

Kendall's tau, denoted by τ, is another non-parametric correlation and it should be used rather than Spearman's coefficient when you have a small data set with a large number of tied ranks. This means that if you rank the scores and many scores have the same rank, then Kendall's tau should be used. Although Spearman's statistic is the more popular of the two coefficients, there is much to suggest that Kendall's statistic is a better estimate of the correlation in the population (see Howell, 2012). As such, we can draw more accurate

generalizations from Kendall's statistic than from Spearman's. To carry out Kendall's correlation on the World's Biggest Liar data follow the same steps as for Pearson and Spearman correlations but select ☑ Kendall's tau-b and deselect the Pearson and Spearman options.

The output is much the same as for Spearman's correlation (Output 8.3). Notice that the value of the correlation coefficient is closer to zero than the Spearman correlation (it has changed from −0.373 to −0.300). Despite the difference in the correlation coefficients, we can still interpret this result as being a highly significant relationship because the significance value of 0.001 is less than 0.05 and the robust confidence interval does not cross zero (−0.473 to −0.117). However, Kendall's value is likely a more accurate gauge of what the correlation in the population would be. As with any correlation, we cannot assume that creativity caused success in the World's Biggest Liar competition.



Conduct a Pearson correlation analysis of the advert data from the beginning of the chapter.

Output 8.3

Correlations

					Creativity	Position in Biggest Liar Competition
Kendall's tau_b	Creativity	Correlation Coefficient			1.000	-.300**
		Sig. (2-tailed)			.	.001
		N			68	68
		Bootstrap[c]	Bias		.000	.001
			Std. Error		.000	.094
			BCa 95% Confidence Interval	Lower	.	-.473
				Upper	.	-.117
	Position in Biggest Liar Competition	Correlation Coefficient			-.300**	1.000
		Sig. (2-tailed)			.001	.
		N			68	68
		Bootstrap[c]	Bias		.001	.000
			Std. Error		.094	.000
			BCa 95% Confidence Interval	Lower	-.473	.
				Upper	-.117	.

**. Correlation is significant at the 0.01 level (2-tailed).

c. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

8.4.5 Biserial and point-biserial correlations

Often it is necessary to investigate relationships between two variables when one of the variables is dichotomous (i.e., it is categorical with only two categories). An example of a dichotomous variable is being pregnant, because a woman can

be either pregnant or not (she cannot be ‘a bit pregnant’). The biserial and point-biserial correlation coefficients should be used in these situations. These correlations are distinguished only by a conceptual difference, but their statistical calculation is quite different. The difference between the use of biserial and point-biserial correlations depends on whether the dichotomous variable is discrete or continuous. This difference is very subtle. A discrete, or true, dichotomy is one for which there is no underlying continuum between the categories. An example of this is whether someone is dead or alive: a person can be only dead or alive, they can’t be ‘a bit dead’. Although you might describe a person as being ‘half-dead’ – especially after a heavy drinking session – there is a definition of clinically dead and if you don’t meet that definition then you are alive. As such, there is no continuum between the two categories. However, it is possible to have a dichotomy for which a continuum does exist. An example is passing or failing a statistics test: some people will only just fail while others will fail by a large margin; likewise some people will scrape a pass while others will excel. Although participants fall into only two categories (pass or fail) there is an underlying continuum along which people lie.

The **point-biserial correlation** coefficient (r_{pb}) is used when one variable is a discrete dichotomy (e.g., pregnancy), whereas the **biserial correlation** coefficient (r_b) is used when one variable is a continuous dichotomy (e.g., passing or failing an exam). The biserial correlation coefficient cannot be calculated directly in SPSS Statistics; first you must calculate the point-biserial correlation coefficient and then adjust the value. Let’s look at an example. Imagine that I was interested in the relationship between the sex of a cat and how much time it spent away from home (I love cats, so these things interest me). I had heard that male cats disappeared for substantial amounts of time on long-distance roams around the neighbourhood (something about hormones driving them to find mates) whereas female cats tended to be more homebound. I used this as a purr-fect (sorry!) excuse to go and visit lots of my friends and their cats. I took a note of the sex of the cat (**Sex**) and then asked the owners to note down the number of hours that their cat was absent from home over a week (**Time**). The time spent away from home is measured at a ratio level – and let’s assume it meets the other assumptions of parametric data – while the sex of the cat is a discrete dichotomy. The data are in the file **Roaming Cats.sav**.

We want to calculate a point-biserial correlation, and this is simplicity itself: it is a Pearson correlation when the dichotomous variable is coded with 0 for one category and 1 for the other (in practice, you can use any values because SPSS changes the lower one to 0 and the higher one to 1 when it does the

calculations). In the saved data I coded the **Sex** variable 1 for male and 0 for female. The **Time** variable contains the roaming time over the week in hours.



Using the **Roaming Cats.sav** file, compute a Pearson correlation between **Sex** and **Time**.

Congratulations: if you did the self-test task then you have conducted your first point-biserial correlation. Despite the horrible name, it's really quite easy to do. You should find that you have the same output as Output 8.4, which shows the correlation matrix of **Time** and **Sex**. The point-biserial correlation coefficient is $r_{pb} = 0.378$, which has a significance value of 0.003. The significance test for this correlation is the same as performing an independent-samples *t*-test on the data (see Chapter 10). The sign of the correlation (i.e., whether the relationship was positive or negative) depends entirely on which way round the dichotomous variable was coded. To prove this point, the data file has an extra variable called **Recode** which is the same as the variable **Sex** except that the coding is reversed (1 = female, 0 = male). If you repeat the Pearson correlation using **Recode** instead of **Sex** you will find that the correlation coefficient becomes −0.378. The sign of the coefficient is completely dependent on which category you assign to which code, and so we must ignore all information about the direction of the relationship. However, we can still interpret R^2 as before. In this example, $R^2 = 0.378^2 = 0.143$, which equates to the sex of the cat sharing 14.3% of the variability in the time spent away from home.

Output 8.4

Correlations

				Time away from home (hours)	Sex of cat
Time away from home (hours)	Pearson Correlation			1	.378**
	Sig. (2-tailed)				.003
	N			60	60
	Bootstrap[c]	Bias		0	-.005
		Std. Error		0	.113
		BCa 95% Confidence Interval	Lower	.	.153
			Upper	.	.588
Sex of cat	Pearson Correlation			.378**	1
	Sig. (2-tailed)			.003	
	N			60	60
	Bootstrap[c]	Bias		-.005	0
		Std. Error		.113	0
		BCa 95% Confidence Interval	Lower	.153	.
			Upper	.588	.

**. Correlation is significant at the 0.01 level (2-tailed).

c. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

8.5 Partial and semi-partial correlation



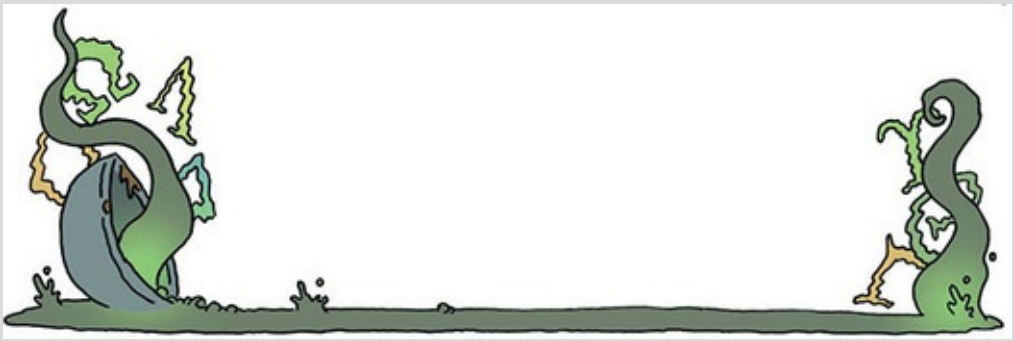
8.5.1 Semi-partial (or part) correlation

I mentioned earlier that there is a type of correlation that can be done that allows you to look at the relationship between two variables, accounting for the effect of a third variable. For example, in the exam anxiety data (in the file **Exam Anxiety.sav**) exam performance (EP) was negatively related to exam anxiety (EA), but positively related to revision time (RT), and revision time itself was negatively related to exam anxiety. If revision time is related to both exam anxiety and exam performance, then to get a measure of the unique relationship between exam anxiety and exam performance we need to account for revision time.

Oliver Twisted Please, Sir, can I have some more … biserial correlation?



‘Some of the male cats were neutered and so there might be a continuum of maleness that underlies the sex variable, you nugget brain,’ Oliver hurls at me. ‘We need to convert the point-biserial correlation into the biserial correlation coefficient (r_b). I think you’re Fagin your knowledge of how to do this.’ Oliver, if you go to the companion website you’ll find that I am not artfully dodging how to do the conversion.



Cramming Sam’s Tips Correlations



- Spearman’s correlation coefficient, r_s, is a non-parametric statistic and requires only ordinal data for both variables.
- Kendall’s correlation coefficient, τ, is like Spearman’s r_s but probably better for small samples.
- The point-biserial correlation coefficient, r_{pb}, quantifies the relationship between a continuous variable and a variable that is a discrete dichotomy (e.g., there is no continuum underlying the two categories, such as dead or alive).
- The biserial correlation coefficient, r_b, quantifies the relationship between a continuous variable and a variable that is a continuous dichotomy (e.g., there is a continuum underlying the two categories, such as passing or failing an exam).



Let's begin by transforming the correlation coefficients from Output 8.1 into proportions of variance by squaring them:

$$\text{Proportion of variance shared by EP and EA} = r^2_{\text{EP-EA}} = -0.441^2 = 0.194$$
$$\text{Proportion of variance shared by EP and RT} = r^2_{\text{EP-RT}} = 0.397^2 = 0.158 \quad (8.15)$$
$$\text{Proportion of variance shared by EA and RT} = r^2_{\text{EA-RT}} = -0.709^2 = 0.503$$

If we multiply the resulting proportions by 100 to turn them into percentages (which people typically find easier to understand) we see that exam performance shares 19.4% of its variance with exam anxiety, and 15.8% with revision time. Revision time shares about half of its variance (50.3%) with exam anxiety.

Figure 8.10 depicts the variance of each variable as a shape: exam performance and anxiety are blue and red squares respectively, and revision time is a sort of weird yellow spacecraft or spark plug. The area of each shape is 100% of the variance of the variable it represents. The figure shows the overlap in the variances of these variables. The depicted overlaps are proportional (approximately) to the actual data: if two variables share 10% of variance then about 10% of the surface areas of their shapes overlap in the diagram. The left-hand side shows only the overlap of shapes/variances and the right duplicates the image but with specific areas marked out and labelled.

To see what the overlaps mean, let's look at exam performance and exam anxiety. Notice that the bottom of the square for exam performance overlaps with the top of the square for exam anxiety. For both variables (squares) this overlap is about 19.4% of their surface area, which corresponds to the amount of variance they share (equation (8.15)). This overlap is (I hope) clear on the left-hand side, but on the right I have marked out these areas as A and C. What distinguishes area A from area C is that area C also overlaps with revision time, whereas area A does not. Together (A + C) these areas are the overlap between exam performance and exam anxiety (19.4%), but this overlap can be decomposed into variance in exam performance that is unique to exam anxiety (A) and variance in exam performance that is not unique to exam anxiety because it is also shared with revision time (C).

Similarly, look at the overlap between exam performance and revision time by focusing on the area where the yellow shape overlaps with the blue square. This overlap is represented by a sort of L-shape that in total represents 15.8% of the

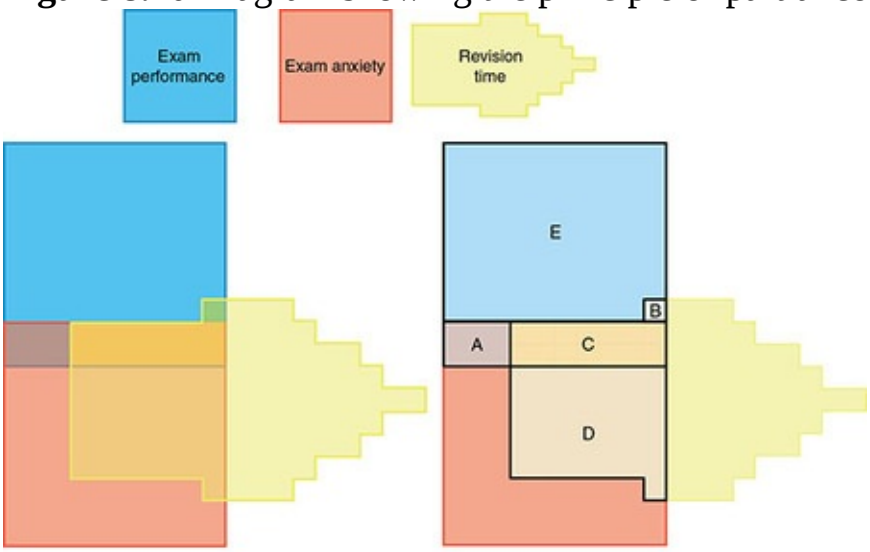
surface area of both shapes (i.e., 15.8% of shared variance). I have marked out the overlap between the blue square and yellow shape on the right with areas labelled B and C. Like before, these areas combine (B + C) to represent the overlap between exam performance and revision time (15.8%), but this overlap can be decomposed into variance in exam performance that is unique to revision time (B) and variance in exam performance that is not unique to revision time because it is also shared with exam anxiety (C).

We know from equation (8.15) that revision time and exam anxiety share 50.3% of their variance. This is represented in Figure 8.10 by the overlap between the yellow shape and red square, which is represented by areas C and D. Like before, these areas combine (C + D) to represent the total overlap between exam anxiety and revision time (50.3%), but this shared variance can be decomposed into variance in exam anxiety that is unique to revision time (D) and variance in exam anxiety that is not unique to revision time because it is also shared with exam performance (C).

Figure 8.10 contains the size of the areas A, B, C and D as well as some of their sums. It also explains what areas represent in isolation and in combination; for example, area B is the variance in exam performance that is *uniquely shared* with revision time (1.5%), but in combination with area C it is the *total* variance in exam performance that is shared with revision time (15.8%). Think about that. The correlation between exam performance and revision time tells us that they share 15.8% of the variance, but in reality, only 1.5% is unique to revision time whereas the remaining 14.3% (area C) is also shared with exam anxiety. This issue gets to the heart of **semi-partial correlation** (also referred to as a **part correlation**), which we'll return to in the next chapter. The areas of unique variance in Figure 8.10 represent the semi-partial correlation. For example, area A is the unique variance in exam performance shared with exam anxiety expressed as a proportion of the variance in exam performance. It is 5.1%. This means that 5.1% of the variance in exam performance is shared uniquely with exam anxiety (and no other variable that we're currently using to predict exam performance). As a proportion this value is 0.051, and remember that to get these proportions we square the correlation coefficient, so to get back to r we would take the square root of 0.051, which is 0.226 (because r can be negative the value could also be −0.226). This is the semi-partial correlation between exam performance and exam anxiety. It is correlation between exam performance and exam anxiety ignoring the portion of that relationship that is also shared by revision time. Likewise, we can look at area B which represents the unique variance in exam performance shared with revision time (1.5%). As a proportion this value is 0.015, and to get this proportion back to a correlation coefficient,

we take the square root, which gives us 0.122 (or −0.122). This value is the semi-partial correlation between exam performance and revision time. It is the correlation between exam performance and revision time ignoring the portion of that relationship that is also shared by exam anxiety.

Figure 8.10 Diagram showing the principle of partial correlation



A = variance in exam performance uniquely shared with exam anxiety (5.1%)
B = variance in exam performance uniquely shared with revision time (1.5%)
C = variance in exam performance shared by both exam anxiety and revision time (14.3%)
D = variance shared by exam anxiety and revision time but not exam performance (36%)
E = variance in exam performance not shared by any measured variable (79.1%)
A + C = variance shared by exam performance and exam anxiety (19.4%)
C + B = variance shared by exam performance and revision time (15.8%)
C + D = variance shared by revision time and exam anxiety (50.3%)
A + B + C = variance in exam performance accounted for by revision time and exam anxiety (20.9%)

As such, the semi-partial correlation expresses the unique relationship between two variables as a function of their total variance. In general terms, imagine we want to look at the relationship between two variables *X* and *Y*, adjusting for the effect of a third variable *Z*. Again it's easier to think in terms of proportions (R^2) and not *r* itself; the semi-partial correlation squared is the uniquely shared variance between *X* and *Y*, expressed as a proportion of the *total variance in Y*. We can show this in terms of the areas in Figure 8.10:

$$sr^2_{\text{EP-EA}} = \frac{\text{A}}{\text{A}+\text{B}+\text{C}+\text{E}} = \frac{5.1}{100} = 0.051$$
$$sr_{\text{EP-EA}} = \sqrt{sr^2_{\text{EP-EA}}} = \sqrt{0.051} = \pm 0.226 \quad (8.16)$$

The semi-partial correlation squared (sR^2) between exam performance and exam anxiety is the uniquely overlapping area (A) expressed as a function of the total area of exam performance (A + B + C + E). The values of each area are underneath Figure 8.10 if you want to plug the numbers into equation (8.16) as I have. As such a semi-partial correlation is the relationship between *X* and *Y*

accounting for the overlap in *X* and *Z*, but not the overlap in *Y* and *Z*. In our specific example, the semi-partial correlation for exam performance and exam anxiety (area A) quantifies their relationship accounting for the overlap between exam anxiety and revision time (area C) *but not* the overlap between exam performance and revision time (area B).

8.5.2 Partial correlation 

Another way to express the unique relationship between two variables (i.e., the relationship accounting for other variables) is the **partial correlation**. Remember that the semi-partial correlation expresses the unique relationship between two variables, *X* and *Y*, as a function of the *total variance* in *Y*. We can instead express this unique variance in terms of the *variance* in *Y left over when other variables have been considered*. The following equation shows this in terms of the areas in Figure 8.10:

$$pr^2_{EP-EA} = \frac{A}{A+E} = \frac{5.1}{5.1+79.1} = 0.061$$

$$pr_{EP-EA} = \sqrt{sr^2_{EP-EA}} = \sqrt{0.066} = \pm 0.247 \quad (8.17)$$

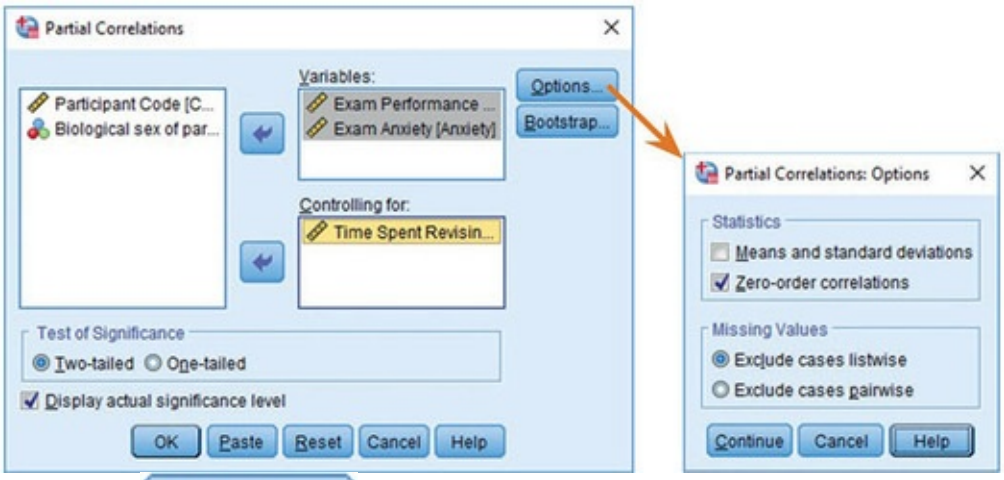
Compare this equation with equation (8.16) and notice that the denominator has changed from the total area for exam performance (A + B + C + E) to the area for exam performance that is left over once we have considered revision time (A + E). In other words, the areas in exam performance that overlap with revision time (areas B and C) are removed from the denominator. Hopefully this makes clear the point that the partial correlation expresses the unique relationship between *X* and *Y* as a function of the *variance* in *Y left over when other variables have been considered*. Returning to our exam example, partial correlation squared (pR^2) between exam performance and exam anxiety is the uniquely overlapping area (A) expressed as a function of the area of exam performance that does *not* overlap with revision time (A + E). Again you can use the values in Figure 8.10 to follow equation (8.17).

By ignoring the variance in *Y* that overlaps with *Z*, a partial correlation adjusts for both the overlap that *X* and *Z* have *and* the overlap in *Y* and *Z*, whereas a semi-partial correlation adjusts only for the overlap that *X* and *Z* have. In our specific example, the partial correlation for exam performance and exam anxiety (area A) adjusts for both the overlap in exam anxiety and revision time (area C) *and* the overlap in exam performance and revision time (area B).

8.5.3 Partial correlation using SPSS Statistics

Reload the **Exam Anxiety.sav** file so that we can conduct a partial correlation between exam anxiety and exam performance while adjusting for the effect of revision time. Access the *Partial Correlations* dialog box (Figure 8.11) using the *Analyze* *Correlate* Partial... menu. Your variables will be listed in the left-hand box, and on the right the box labelled *Variables* is for specifying the variables that you want to correlate and the box labelled *Controlling for* is for declaring the variables for which you want to adjust. If we want to look at the unique effect of exam anxiety on exam performance we'd correlate the variables **exam** and **anxiety** while adjusting for **revise** as shown in Figure 8.11. In this instance we are going to adjust for one variable, which is known as a *first-order partial correlation*. It is possible to control for the effects of two variables (a *second-order partial correlation*), three variables (a *third-order partial correlation*) and so on by dragging more variables to the *Controlling for* box.

Figure 8.11 Main dialog box for conducting a partial correlation



Clicking Options... accesses options such as those in bivariate correlation. Within this dialog box you can select *Zero-order correlations*, which are the Pearson correlation coefficients without adjusting for other variables. If we select this option SPSS will produce a correlation matrix of **anxiety, exam** and **revise**, which might be useful if you haven't already looked at the raw (or zero-order) correlations between variables, but we have (Output 8.1), so don't tick this option for this example. As we have done throughout, use the Bootstrap... option to get some robust confidence intervals.

Output 8.5 shows the partial correlation of exam anxiety and exam performance, controlling for revision time. Note that the top and bottom of the table contain identical values, so we can ignore one half of the table. The partial correlation between exam performance and exam anxiety is −0.247, which is the same value as we computed in equation (8.17). This value is considerably less than when we didn't adjust for the effect of revision time ($r = -0.441$). Although this correlation is still statistically significant (its p-value is still below 0.05) and the confidence interval [−0.430, −0.030] still doesn't contain zero, the relationship is diminished. In terms of variance, the value of R^2 for the partial correlation is 0.061, which means that exam anxiety shares only 6.1% of the variance in exam performance that is left over by revision time (compared to 19.4% when revision time was not factored in). Running this analysis has shown us that exam anxiety alone does explain some of the variation in exam scores, but there is a complex relationship between anxiety, revision and exam performance that might otherwise have been ignored. Although causality is still not certain, because relevant variables are being included, the third-variable problem is, at least, being addressed to some degree.

Partial correlations can be done when variables are dichotomous (including the 'third' variable). For example, we could look at the relationship between bladder relaxation (did the person wet themselves or not?) and the number of large tarantulas crawling up the person's leg, adjusting for fear of spiders (the first variable is dichotomous, but the second variable and 'control' variable are both continuous). Similarly, to use an earlier example, we could examine the relationship between creativity and success in the World's Biggest Liar competition, adjusting for whether someone had previous experience in the competition (and therefore had some idea of the type of tale that would win). In this case the 'control' variable is dichotomous.[6]

6 Both these examples are, in fact, simple cases of hierarchical regression (see Chapter 9), and the first example is also an example of analysis of covariance. This may not mean much yet, but it illustrates what I have repeatedly said about all statistical models being variations of the same linear model.

Output 8.5 Output from a partial correlation

Correlations

Control Variables					Exam Performance (%)	Exam Anxiety
Time Spent Revising	Exam Performance (%)	Correlation			1.000	-.247
		Significance (2-tailed)			.	.012
		df			0	100
		Bootstrap[a]	Bias		.000	.007
			Std. Error		.000	.100
			BCa 95% Confidence Interval	Lower	.	-.430
				Upper	.	-.030
	Exam Anxiety	Correlation			-.247	1.000
		Significance (2-tailed)			.012	.
		df			100	0
		Bootstrap[a]	Bias		.007	.000
			Std. Error		.100	.000
			BCa 95% Confidence Interval	Lower	-.430	.
				Upper	-.030	.

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

Cramming Sam's Tips Partial and semi-partial correlations



- A partial correlation quantifies the relationship between two variables while accounting for the effects of a third variable on both variables in the original correlation.
- A semi-partial correlation quantifies the relationship between two variables while accounting for the effects of a third variable on only one of the variables in the original correlation.



8.6 Comparing correlations

8.6.1 Comparing independent rs

Sometimes we want to know whether one correlation coefficient is bigger than

another. For example, when we looked at the effect of exam anxiety on exam performance, we might have been interested to know whether this correlation was different in men and women. We can compute the correlation in these two samples, but how do we assess whether the difference is meaningful?



Use the *split file* command to compute the correlation coefficient between exam anxiety and exam performance in men and women.

If you do the self-test you'll find that the correlations are $r_{\text{Male}} = -0.506$ and $r_{\text{Female}} = -0.381$. These two samples are independent (they contain different entities). To compare these correlations we can use what we discovered in Section 8.2.3 to convert these coefficients to z_r (because it makes the sampling distribution normal). Do the conversion and you should obtain z_r (males) = −0.557 and z_r (females) = −0.401. We can calculate a *z*-score of the differences between these correlations using:



$$z_{\text{Difference}} = \frac{z_{r_1} - z_{r_2}}{\sqrt{\frac{1}{N_1 - 3} + \frac{1}{N_2 - 3}}} \tag{8.18}$$

We had 52 men and 51 women, so the resulting *z* is:

$$z_{\text{Difference}} = \frac{-0.557 - (-0.401)}{\sqrt{\frac{1}{49} + \frac{1}{48}}} = \frac{-0.156}{0.203} = -0.768 \tag{8.19}$$

We can look up this value of *z* (0.768, we can ignore the minus sign) in the table for the normal distribution in the Appendix and get the one-tailed probability from the column labelled 'Smaller Portion'. The value is 0.221, which we need to double to get the two-tailed probability of 0.442. This value is less than the 0.05 criterion *z*-score of 1.96, so the correlation between exam anxiety and exam performance is not significantly different in men and women.

8.6.2 Comparing dependent rs

If you want to compare correlation coefficients that come from the same entities

then things are a little more complicated. You can use a *t*-statistic to test whether a difference between two dependent correlations is significant. For example, in our exam anxiety data we might want to see whether the relationship between exam anxiety (*x*) and exam performance (*y*) is stronger than the relationship between revision (*z*) and exam performance. To address this question, we need the three *r*s that quantify the relationships between these variables: r_{xy}, the relationship between exam anxiety and exam performance (−0.441); r_{zy}, the relationship between revision and exam performance (0.397); and r_{xz}, the relationship between exam anxiety and revision (−0.709). The *t*-statistic is computed as follows (Chen & Popovich, 2002):

$$t_{\text{Difference}} = \left(r_{xy} - r_{zy}\right)\sqrt{\frac{(n-3)(1+r_{xz})}{2\left(1-r_{xy}^2-r_{xz}^2-r_{zy}^2+2r_{xy}r_{xz}r_{zy}\right)}} \tag{8.20}$$

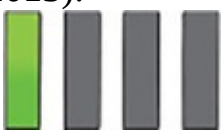
Admittedly that equation looks hideous, but really it's not too bad when you remember that it just uses the three correlation coefficients and the sample size *N*. Place the numbers from the exam anxiety example in (*N* was 103) and you should end up with:

$$t_{\text{Difference}} = (-0.838)\sqrt{\frac{29.1}{2(1-0.194-0.503-0.158+0.248)}} = -5.09 \tag{8.21}$$

This value can be checked against the appropriate critical value for *t* with $N - 3$ degrees of freedom (in this case 100). The critical values in the table (see the Appendix) are 1.98 ($p < 0.05$) and 2.63 ($p < 0.01$), two-tailed. As such we can say that the correlation between exam anxiety and exam performance was significantly higher than the correlation between revision time and exam performance (this isn't a massive surprise, given that these relationships went in the opposite directions from each other).

8.6.3 Comparing rs using SPSS Statistics 

You can't compare correlations through the dialog boxes in SPSS, but there is a macro available by Weaver and Wuensch (2013).

8.7 Calculating the effect size 

Calculating effect sizes for correlation coefficients couldn't be easier because, as we saw earlier in the book, correlation coefficients *are* effect sizes. So, no calculations (other than those you have already done) are necessary. However, although the Spearman and Kendall correlations are comparable to Pearson's *r* in many respects (their power, for example, is similar under parametric

conditions), there are important differences (Strahan, 1982).



First, we can square the value of Pearson's r to get the proportion of shared variance, R^2. For Spearman's r_s we can do this too because it uses the same equation as Pearson's r. However, the resulting R^2_s is the proportion of variance in the *ranks* that two variables share. Having said this, R^2_s is usually a good approximation of R^2 (especially in conditions of near-normal distributions). Kendall's τ is not numerically similar to either r or r_s, and so τ^2 does not tell us about the proportion of variance shared by two variables (or the ranks of those two variables).

The second difference relates to a more general point that when using correlations as effect sizes (both when reporting your own analysis and when interpreting others) be mindful that the choice of correlation coefficient can make a substantial difference to the apparent size of the effect. For example, Kendall's τ is 66–75% smaller than both Spearman's r_s and Pearson's r, but r and r_s are generally similar in size (Strahan, 1982). Therefore, if τ is used as an effect size it is not comparable to r and r_s. The point-biserial and biserial correlations differ in size too, so you should think very carefully about whether your dichotomous variable has an underlying continuum, or whether it is a truly discrete variable.

8.8 How to report correlation coefficents 

Reporting correlation coefficients is pretty easy: you report how big they are, their confidence intervals and significance value (the significance value is probably the least important because the correlation coefficient is an effect size). Some general points (see Sections 1.9.3 and 3.8) are as follows: (1) if you use APA style there should be no zero before the decimal point for the correlation coefficient or the probability value (because neither can exceed 1); (2) coefficients are usually reported to 2 or 3 decimal places because this is a reasonable level of precision; (3) generally people report 95% confidence intervals; (4) each correlation coefficient is represented by a different letter (and some of them are Greek); and (5) report exact p-values. Let's take a few

examples from this chapter:

✓ There was not a significant relationship between the number of adverts watched and the number of packets of sweets purchased, $r = 0.87$, $p = 0.054$.

✓ Bias corrected and accelerated bootstrap 95% CIs are reported in square brackets. Exam performance was significantly correlated with exam anxiety, $r = -0.44$ [−0.56, −0.30], and time spent revising, $r = 0.40$ [0.25, 0.52]; the time spent revising was also correlated with exam anxiety, $r = -0.71$ [−0.86, −0.49] (all ps < 0.001).

✓ Creativity was significantly related to how well people did in the World's Biggest Liar competition, $r_s = -0.37$, 95% BCa CI [−0.60, −0.12], $p = 0.002$.

✓ Creativity was significantly related to a person's placing in the World's Biggest Liar competition, $\tau = -0.30$, 95% BCa CI [−0.47, −0.12], $p = 0.001$. (Note that I've quoted Kendall's τ.)

✓ The sex of the cat was significantly related to the time the cat spent away from home, $r_{pb} = 0.38$, 95% BCa CI [0.15, 0.59], $p = 0.003$.

✓ The sex of the cat was significantly related to the time the cat spent away from home, $r_b = 0.48$, $p = 0.003$.

A table is a good way to report lots of correlations. Our exam anxiety correlations could be reported as in Table 8.2. Note that above the diagonal I have reported the correlation coefficients and used symbols to represent different levels of significance. The confidence intervals are reported underneath. Under the table there is a legend to tell readers what symbols represent. (None of the correlations were non-significant or had p bigger than 0.001 so most of the table footer is there to give you a template – you would normally include only symbols that you had actually used in the table.) Finally, in the lower part of the table I have reported the sample sizes. These are all the same (103), but when you have missing data it is useful to report the sample sizes in this way because different values of the correlation will be based on different sample sizes. You could alternatively use the bottom part of the table to report exact p-values.

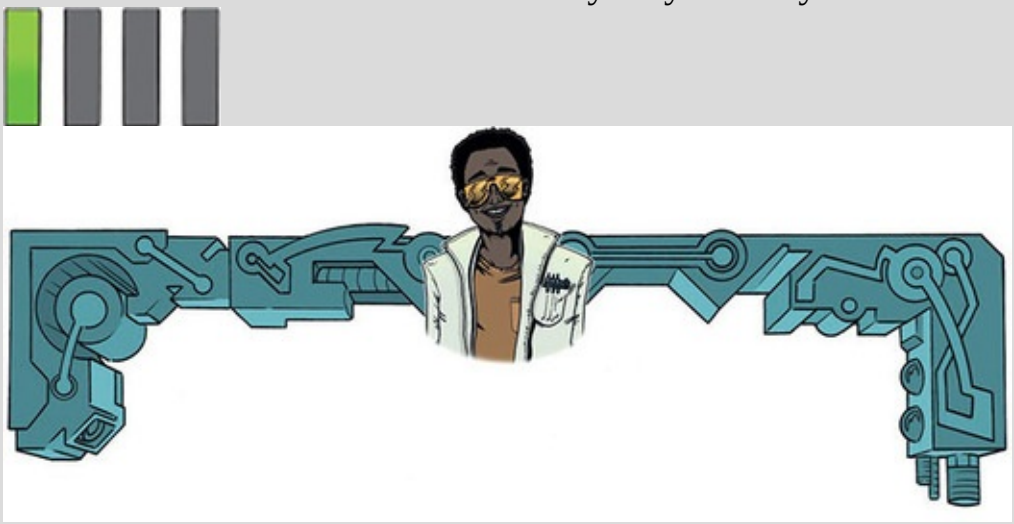
Table 8.2 An example of reporting a table of correlations

	Exam Performance	Exam Anxiety	Revision Time
Exam Performance	1	−0.44*** [−0.56, −0.30]	0.40*** [0.25, 0.52]
Exam Anxiety	103	1	−0.71*** [−0.86, −0.49]
Revision Time	103	103	1

ns = not significant ($p > 0.05$), * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. BCa bootstrap 95% CIs reported in brackets.

ns = not significant ($p > 0.05$), * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. BCa bootstrap 95% CIs reported in brackets.

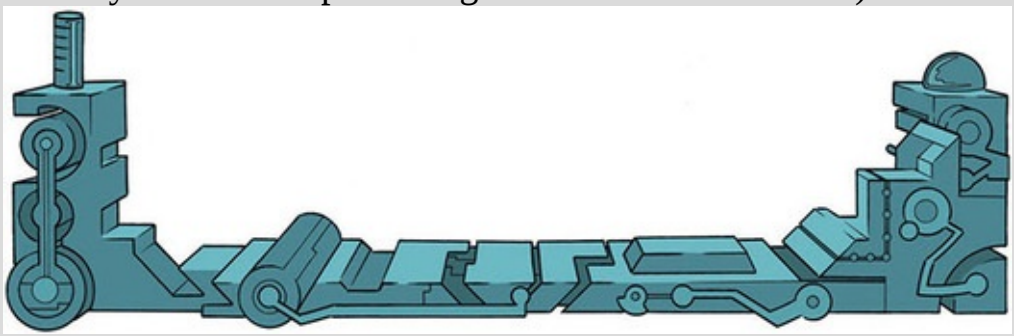
Labcoat Leni's Real Research 8.1 Why do you like your lecturers?



Chamorro-Premuzic, T., *et al.* (2008). *Personality and Individual Differences*, *44*, 965–976.

As students you probably have to rate your lecturers at the end of the course. There will be some lecturers you like and others you don't. As a lecturer I find this process horribly depressing (although this has a lot to do with the fact that I tend to focus on negative feedback and ignore the good stuff). There is some evidence that students tend to pick courses of lecturers they perceive to be enthusiastic and good communicators. In a fascinating study, Tomas Chamorro-Premuzic and his colleagues (Chamorro-Premuzic, Furnham, Christopher, Garwood, & Martin, 2008) tested the hypothesis that students tend to like lecturers who are like themselves. (This hypothesis will have the students on my course who like my lectures screaming in horror.) The authors measured students' own personalities using a very well-established measure (the NEO-FFI) which measures five fundamental personality traits: neuroticism, extroversion, openness to experience, agreeableness and conscientiousness. Students also completed a questionnaire in which they were given descriptions (e.g., 'warm: friendly, warm, sociable, cheerful, affectionate, outgoing') and asked to rate how much they wanted to see this in a lecturer from −5 (I don't want this characteristic at all) through 0 (the characteristic is not important) to +5 (I really want this characteristic in my lecturer). The characteristics were the same as those measured by the NEO-FFI.

As such, the authors had a measure of how much a student had each of the five core personality characteristics, but also a measure of how much they wanted to see those same characteristics in their lecturer. Tomas and his colleagues could then test whether, for instance, extroverted students want extroverted lecturers. The data from this study are in the file **Chamorro-Premuzic.sav**. Run Pearson correlations on these variables to see if students with certain personality characteristics want to see those characteristics in their lecturers. What conclusions can you draw? Answers are on the companion website (or look at Table 3 in the original article, which shows you how to report a large number of correlations).



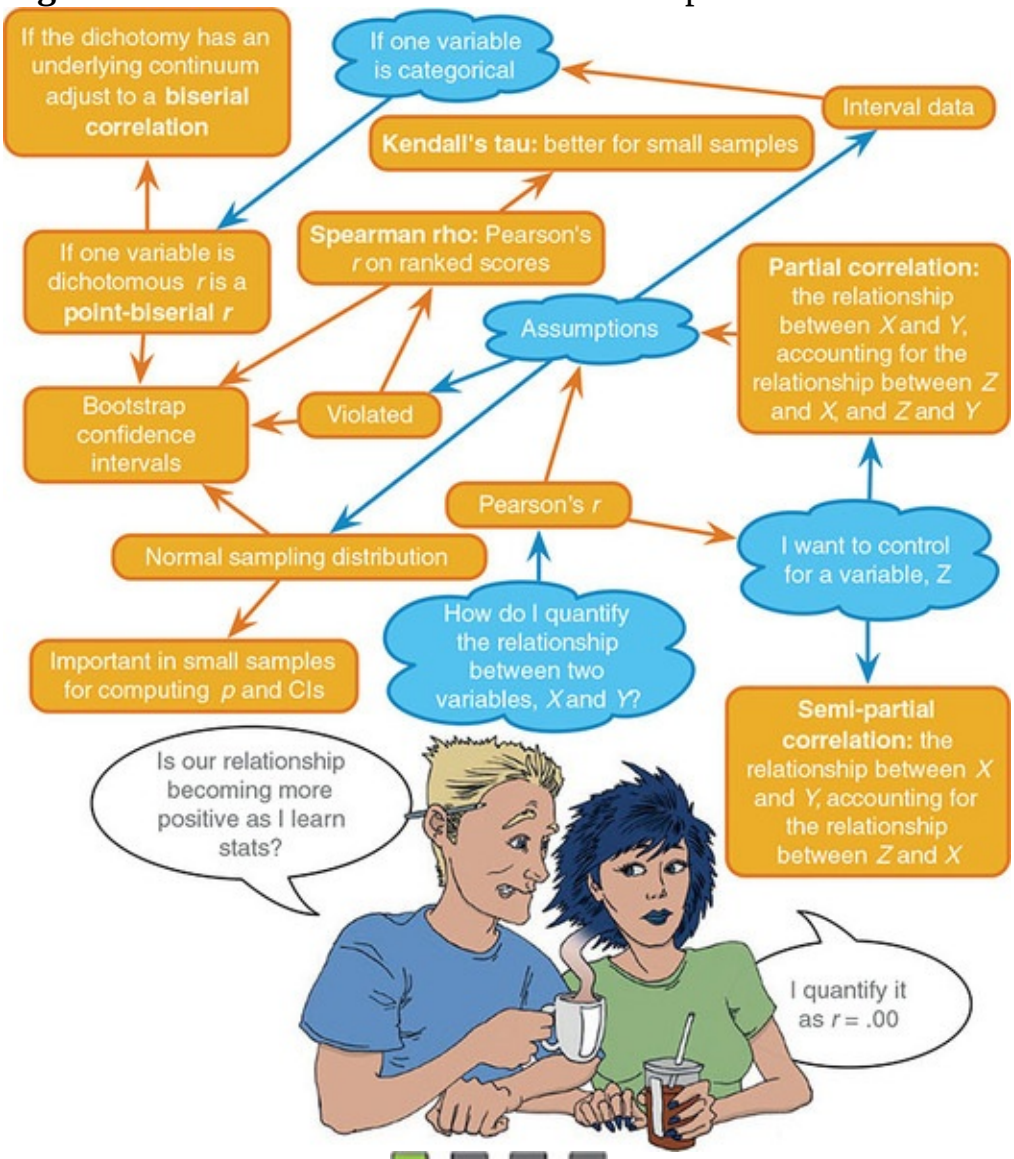
8.9 Brian's attempt to woo Jane

Brian wandered out of his lecture about correlations. An hour learning about relationships had made him feel flat. He knew it was stupid: the lecturer wasn't talking about those kinds of relationships, but the word kept buzzing around his head, distracting him. Jane was complex. On the one hand she always gave him the brush-off, on the other he sensed a playful humour in her responses to him. Was he kidding himself, or was that her idea of flirting? Should he be trying to get involved with someone who thought *that* was how to flirt? It was confusing. He wanted to get in her head. Did she think they were developing a positive relationship? Was it negative? Non-existent? He decided to be direct.

Jane's morning had been strange. She'd worked late into the night, but woke early in a mental haze. She was acting on autopilot. She ate breakfast, spent more time on her make-up than she normally would, and put on some of her favourite clothes. She didn't remember doing any of it. It didn't register as odd to her. Neither did it seem unusual to her that she passed by the entrance of the Leviathan lecture hall at exactly 10:50, the time that Brian's statistics lecture kicked out. Seeing Brian dispersed the fog in her mind. She panicked as she realized that her unconscious mind had brought her to this place at this exact

time. She felt overdressed and awkward. He seemed sad, so she asked him if he was OK. He told her about the lecture he had just had, and the familiarity of correlations calmed her.

Figure 8.12 What Brian learnt from this chapter



8.10 What next?

At the age of 8 my dad taught me a valuable lesson, which is that if you really want something then you need to work at it, and the harder you work at it the more likely you are to get what you want. I did practise my guitar and before long the tears gave way to a competent version of ‘Skip to my Lou’. My dad had also aspired to be a musician when he was young and encouraged my new passion.[7] He found me a guitar teacher and found the money for lessons. These

lessons illustrate how being a good student often depends on finding the right teacher. Ken Steers, despite his best efforts, was on a completely different wavelength than me. I wanted to learn some crushing metal riffs, and he wanted me to work through Bert Weedon's *Play in a Day* and learn trad jazz classics. As an adult, I wish I had paid more attention to Ken because I'd have been a better guitar player than I am. I was a terrible student and adopted a strategy of selective practice: I'd practise if I wanted to do something but not if I thought it was 'boring'. Perhaps this is why I am so obsessed with trying not to be a boring teacher. Nevertheless, my dad and Ken did get me going and soon enough, like my favourite record of the time, I was ready to 'Take on the world'. Well, Wales at any rate …
7 My dad, like me, never made it in his band, but, unlike me, did sing on the UK TV show *Stars in Their Eyes*, which made us all pretty proud.

8.11 Key terms that I've discovered

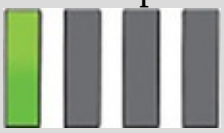
Biserial correlation
Bivariate correlation
Coefficient of determination
Covariance
Cross-product deviations
Kendall's tau
Part correlation
Pearson correlation coefficient
Point-biserial correlation
Semi-partial correlation
Spearman's correlation coefficient
Standardization

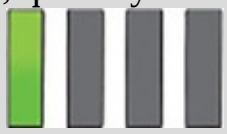
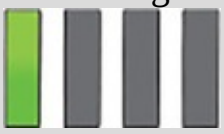
Smart Alex's tasks

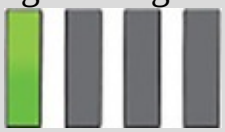


- **Task 1**: A student was interested in whether there was a positive relationship between the time spent doing an essay and the mark

received. He got 45 of his friends and timed how long they spent writing an essay (**hours**) and the percentage they got in the essay (**essay**). He also translated these grades into their degree classifications (**grade**): in the UK, a student can get a first-class mark (the best), an upper-second-class mark, a lower second, a third, a pass or a fail (the worst). Using the data in the file **EssayMarks.sav**, find out what the relationship was between the time spent doing an essay and the eventual mark in terms of percentage and degree class (draw a scatterplot too).



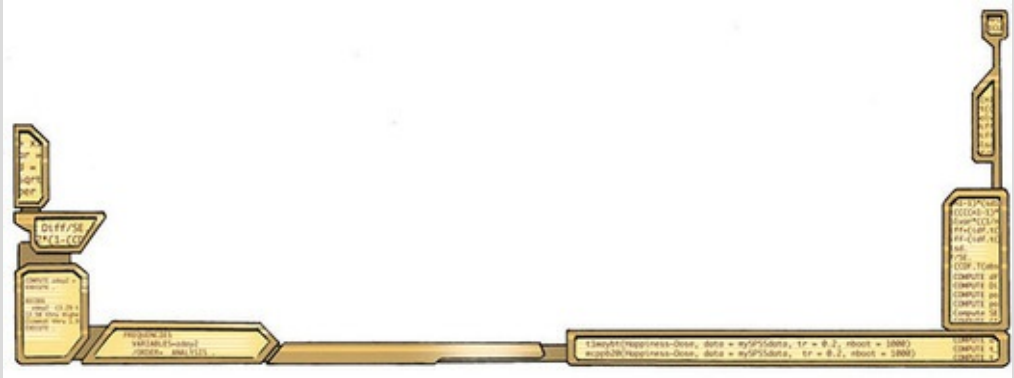
- **Task 2**: Using the **Notebook.sav** data from Chapter 3, find out the size of relationship between the participant's sex and arousal. 
- **Task 3**: Using the notebook data again, quantify the relationship between the film watched and arousal. 
- **Task 4**: As a statistics lecturer I am interested in the factors that determine whether a student will do well on a statistics course. Imagine I took 25 students and looked at their grades for my statistics course at the end of their first year at university: first, upper second, lower second and third class (see Task 1). I also asked these students what grade they got in their high school maths exams. In the UK, GCSEs are school exams taken at age 16 that are graded A, B, C, D, E or F (an A grade is the best). The data for this study are in the file **grades.sav**. To what degree does GCSE maths grade correlate with first-year statistics grade? 
- **Task 5**: In Figure 2.3 we saw some data relating to people's ratings of dishonest acts and the likeableness of the perpetrator (for a full description see Jane Superbrain Box 2.1). Compute the Spearman correlation between ratings of dishonesty and likeableness of the perpetrator. The data are in **HonestyLab.sav**.

- **Task 6**: In Chapter 4 (Task 6) we looked at data from people who had been forced to marry goats and dogs and measured their life satisfaction and, also, how much they like animals (**Goat or Dog.sav)**. Is there a significant correlation between life satisfaction and the type of animal to which a person was married? 
- **Task 7**: Repeat the analysis above, taking account of animal liking when computing the correlation between life satisfaction and the animal to which a person was married. 
- **Task 8**: In Chapter 4 (Task 7) we looked at data based on findings that the number of cups of tea drunk was related to cognitive functioning (Feng et al., 2010). The data are in the file **Tea Makes You Brainy 15.sav**. What is the correlation between tea drinking and cognitive functioning? Is there a significant effect? 
- **Task 9**: The research in the previous task was replicated but in a larger sample (N = 716), which is the same as the sample size in Feng et al.'s research (**Tea Makes You Brainy 716.sav**). Conduct a correlation between tea drinking and cognitive functioning. Compare the correlation coefficient and significance in this large sample with the previous task. What statistical point do the results illustrate? 
- **Task 10**: In Chapter 6 we looked at hygiene scores over three days of a rock music festival (**Download Festival.sav**). Using Spearman's correlation, were hygiene scores on day 1 of the festival significantly correlated with those on day 3? 
- **Task 11**: Using the data in **Shopping Exercise.sav** (Chapter 4, Task 5), find out if there is a significant relationship between the

time spent shopping and the distance covered. 

- **Task 12**: What effect does accounting for the participant's sex have on the relationship between the time spent shopping and the distance covered? 

Answers & additional resources are available on the book's website at **https://edge.sagepub.com/field5e**



9 The Linear Model (Regression)

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9.1 What will this chapter tell me?

Although none of us can know the future, predicting it is so important that organisms are hard-wired to learn about predictable events in their environment. We saw in the previous chapter that I received a guitar for Christmas when I was eight. My first foray into public performance was a weekly talent show at a holiday camp called 'Holimarine' in Wales (it doesn't exist any more because I am old and this was 1981). I sang a Chuck Berry song called 'My ding-a-ling'[1] and to my absolute amazement I won the competition.[2] Suddenly other 8-year-olds across the land (well, a ballroom in Wales) worshipped me (I made lots of friends after the competition). I had tasted success, it tasted like praline chocolate, and so I wanted to enter the competition in the second week of our holiday. To ensure success, I needed to know why I had won in the first week. One way to do this would have been to collect data and to use these data to predict people's evaluations of children's performances in the contest from certain variables: the age of the performer, what type of performance they gave (singing, telling a joke, magic tricks), and perhaps how cute they looked. Obviously actual talent wouldn't be a factor. A linear model (regression) fitted to these data would enable us to predict the future (success in next week's competition) based on values of the variables we'd measured. If, for example, singing was an important factor in getting a good audience evaluation, I could sing again the following week; but if jokers tended to do better then I might switch to a comedy routine. When I was eight I wasn't the pathetic nerd that I am today, so I didn't know about linear models (nor did I wish to); however, my dad thought that success was due to the winning combination of a cherub-looking 8-year-old singing songs that can be interpreted in a filthy way. He wrote a song for me to sing about the keyboard player in the Holimarine Band 'messing about with his organ'. He said 'take this song, son, and steal the show' … and that's what I did: I came first again. There's no accounting for taste.

1 It appears that even then I had a passion for lowering the tone.

2 I have a very grainy video of this performance recorded by my dad's friend on a video camera the size of a medium-sized dog that had to be accompanied at all times by a 'battery pack' the size and weight of a tank (see Oditi's Lantern).

Figure 9.1 Me playing with my ding-a-ling in the Holimarine Talent Show. Note the groupies queuing up at the front



Oditi's Lantern Words that go unspoken, deeds that go undone



'I, Oditi, do not want my followers to get distracted by playing with their ding-a-lings. To warn you all of the dangers of such frivolity, I have uncovered a song, sung by an innocent child, that explains the risks. Stare into my lantern and shake your booty to the funky tune.'



9.2 An introduction to the linear model (regression)



9.2.1 The linear model with one predictor



In the previous chapter we started getting down to the nitty-gritty of the linear model that we've been discussing since way back in Chapter 2. We saw that if we wanted to look at the relationship between two variables we could use the model in equation (2.3):

$$\text{outcome}_i = (b_1 X_i) + \text{error}_i \quad (9.1)$$

I mentioned then that if we work with raw scores we must add information about where the outcome variable is centred. I wrote that we add a constant, b_0, known as the intercept to the model that represents the value of the outcome when the predictor is absent (i.e., it is zero). The resulting model is:

$$\text{outcome}_i = (b_0 + b_1 X_i) + \text{error}_i \quad (9.2)$$

$$Y_i = (b_0 + b_1 X_i) + \varepsilon_i$$

This equation keeps the fundamental idea that an outcome for a person can be predicted from a model (the stuff in parentheses) and some error associated with that prediction (ε_i). We still predict an outcome variable (Y_i) from a predictor variable (X_i) and a parameter, b_1, associated with the predictor variable that quantifies the relationship it has with the outcome variable. This model differs from that of a correlation only in that it uses an *unstandardized* measure of the relationship (b_1) and consequently we include a parameter, b_0, that tells us the value of the outcome when the predictor is zero.

As a quick diversion, let's imagine that instead of b_0 we use the letter c, and instead of b_1 we use the letter m. Let's also ignore the error term. We could predict our outcome as follows:

$$\text{outcome}_i = mx + c$$

Or if you're American, Canadian or Australian let's use the letter b instead of c:

$$\text{outcome}_i = mx + b$$

Perhaps you're French, Dutch or Brazilian, in which case let's use a instead of m:

$$\text{outcome}_i = ax + b$$

Do any of these equations look familiar? If not, there are two explanations: (1) you didn't pay enough attention at school; or (2) you're Latvian, Greek, Italian, Swedish, Romanian, Finnish, Russian or from some other country that has a different variant of the equation of a straight line. The different forms of the equation illustrate how the symbols or letters in an equation can be somewhat arbitrary choices.[3] Whether we write $mx + c$ or $b_1X + b_0$ doesn't really matter; what matters is what the symbols represent. So, what do the symbols represent?

[3] For example, you'll sometimes see equation (9.2) written as $Y_i = (\beta_0 + \beta_1 X_i) + \varepsilon_i$. The only difference is that this equation has got βs in it instead of bs. Both versions are the same thing, they just use different letters to represent the coefficients.

I have talked throughout this book about fitting 'linear models', and linear

simply means ‘straight line’. All the equations above are forms of the equation of a straight line. Any straight line can be defined by two things: (1) the slope (or gradient) of the line (usually denoted by b_1); and (2) the point at which the line crosses the vertical axis of the graph (known as the *intercept* of the line, b_0). These parameters b_1 and b_0 are known as the regression coefficients and will crop up throughout this book, where you see them referred to generally as b (without any subscript) or ***bi*** (meaning the b associated with variable i). Figure 9.2 (left) shows a set of lines that have the same intercept but different gradients. For these three models, b_0 is the same in each but b_1 is different for each line. Figure 9.2 (right) shows models that have the same gradients (b_1 is the same in each model) but different intercepts (b_0 is different in each model).

In Chapter 8 we saw how relationships can be either positive or negative (and I don’t mean whether you and your partner argue all the time). A model with a positive b_1 describes a positive relationship, whereas a line with a negative b_1 describes a negative relationship. Looking at Figure 9.2 (left), the orange line describes a positive relationship whereas the green line describes a negative relationship. As such, we can use a linear model (i.e., a straight line) to summarize the relationship between two variables: the gradient (b_1) tells us what the model looks like (its shape) and the intercept (b_0) locates the model in geometric space.

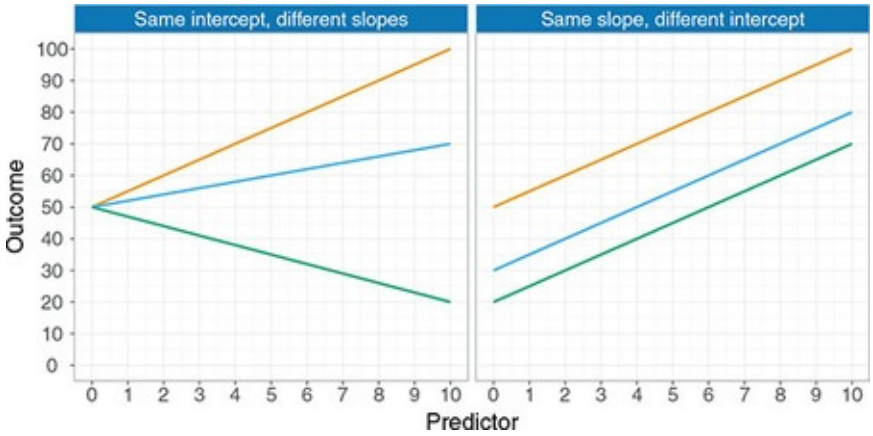
Let’s look at an example. Imagine that I was interested in predicting physical and downloaded album sales (outcome) from the amount of money spent advertising that album (predictor). We could adapt the linear model (equation (9.2)) by replacing the predictor and outcome with our variable names:

$$\begin{aligned} Y_i &= b_0 + b_1 X_i + \varepsilon_i \\ \text{album sales}_i &= b_0 + b_1 \text{advertising budget}_i + \varepsilon_i \end{aligned} \tag{9.3}$$

Once we have estimated the values of the bs we would be able to make a prediction about album sales by replacing ‘advertising’ with a number representing how much we wanted to spend advertising an album. For example, imagine that b_0 turned out to be 50 and b_1 turned out to be 100. Our model would be:

$$\text{album sales}_i = 50 + (100 \times \text{advertising budget}_i) + \varepsilon_i \tag{9.4}$$

Figure 9.2 Lines that share the same intercept but have different gradients, and lines with the same gradients but different intercepts



Note that I have replaced the *b*s with their numeric values. Now, we can make a prediction. Imagine we wanted to spend £5 on advertising. We can replace the variable 'advertising budget' with this value and solve the equation to discover how many album sales we will get:

$$\begin{aligned}\text{album sales}_i &= 50 + (100 \times 5) + \varepsilon_i \\ &= 550 + \varepsilon_i\end{aligned} \tag{9.5}$$

So, based on our model we can predict that if we spend £5 on advertising, we'll sell 550 albums. I've left the error term in there to remind you that this prediction will probably not be perfectly accurate. This value of 550 album sales is known as a **predicted value**.

9.2.2 The linear model with several predictors



Life is usually complicated and there will be numerous variables that might be related to the outcome that you want to predict. To take our album sales example, variables other than advertising are likely to affect sales. For example, how much someone hears songs from the album on the radio, or the 'look' of the band. One of the beautiful things about the linear model is that it expands to include as many predictors as you like. We hinted at this in Chapter 2 (equation (2.4)). An additional predictor can be placed in the model and given a *b* to estimate its relationship to the outcome:



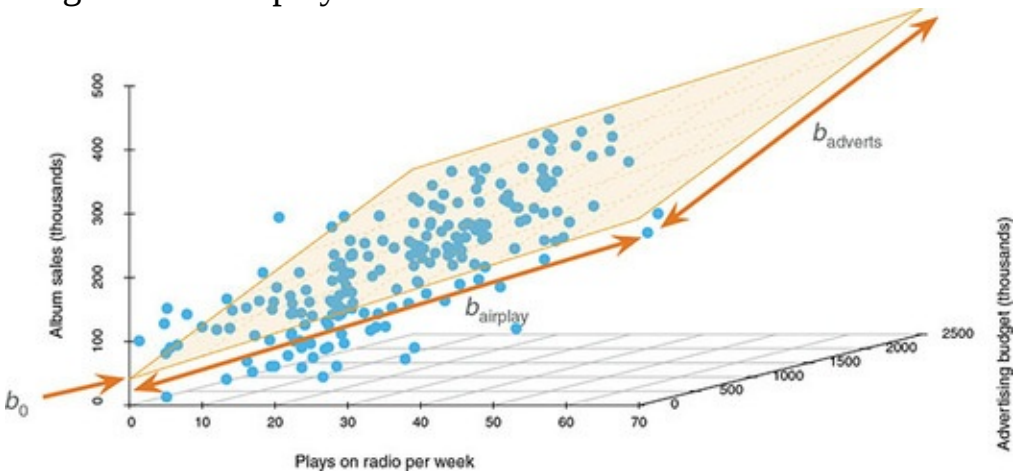
$$Y_i = (b_0 + b_1 X_{1i} + b_2 X_{2i}) + \varepsilon_i \tag{9.6}$$

All that has changed is the addition of a second predictor (X_2) and an associated

parameter (b_2). To make things more concrete, if we add the number of plays of the band on the radio per week (airplay) to the model in equation (9.3), we get:

$$\text{album sales}_i = b_0 + b_1\text{advertising budget}_i + b_2\text{airplay}_i + \varepsilon_i \quad (9.7)$$

Figure 9.3 Scatterplot of the relationship between album sales, advertising budget and radio play



The new model includes a *b*-value for both predictors (and, of course, the constant, b_0). By estimating the *b*-values, we can make predictions about album sales based not only on the amount spent on advertising but also on airplay. The resulting model is visualized in Figure 9.3. The tinted trapezium (the regression *plane*) is described by equation (9.7) and the dots represent the observed data points. Like a regression line, a regression plane aims to give the best prediction for the observed data. However, there are invariably differences between the model and the real-life data (this fact is evident because most of the dots do not lie exactly on the plane). The vertical distances between the plane and each data point are the errors or *residuals* in the model. The *b*-value for advertising describes the slope of the left and right sides of the plane, whereas the *b*-value for airplay describes the slope of the top and bottom of the plane. Just like with one predictor, these two slopes describe the shape of the model (what it looks like) and the intercept locates the model in space.

It is easy enough to visualize a linear model with two predictors, because it is possible to plot the plane using a 3-D scatterplot. However, with three, four or even more predictors you can't immediately visualize what the model looks like, or what the *b*-values represent, but you can apply the principles of these basic models to more complex scenarios. For example, in general, we can add as many predictors as we like, provided we give them a *b*, and the linear model expands accordingly:

$$Y_i = \left(b_0 + b_1X_{1i} + b_2X_{2i} + \cdots + b_nX_{ni}\right) + \varepsilon_i \quad (9.8)$$

Y is the outcome variable, b_1 is the coefficient of the first predictor (X_1), b_2 is the

coefficient of the second predictor (X_2), b_n is the coefficient of the nth predictor (X_{ni}), and ε_i is the error for the ith entity. (The parentheses aren't necessary, they're there to make the connection to equation (9.2).) This equation illustrates that we can add predictors to the model until we reach the final one (X_n), and each time we add one, we assign it a regression coefficient (b).

To sum up, regression analysis is a term for fitting a linear model to data and using it to predict values of an **outcome variable** (a.k.a. dependent variable) from one or more **predictor variables** (a.k.a. independent variables). With one predictor variable, the technique is sometimes referred to as **simple regression**, but with several predictors it is called **multiple regression**. Both are merely terms for the linear model.

9.2.3 Estimating the model 

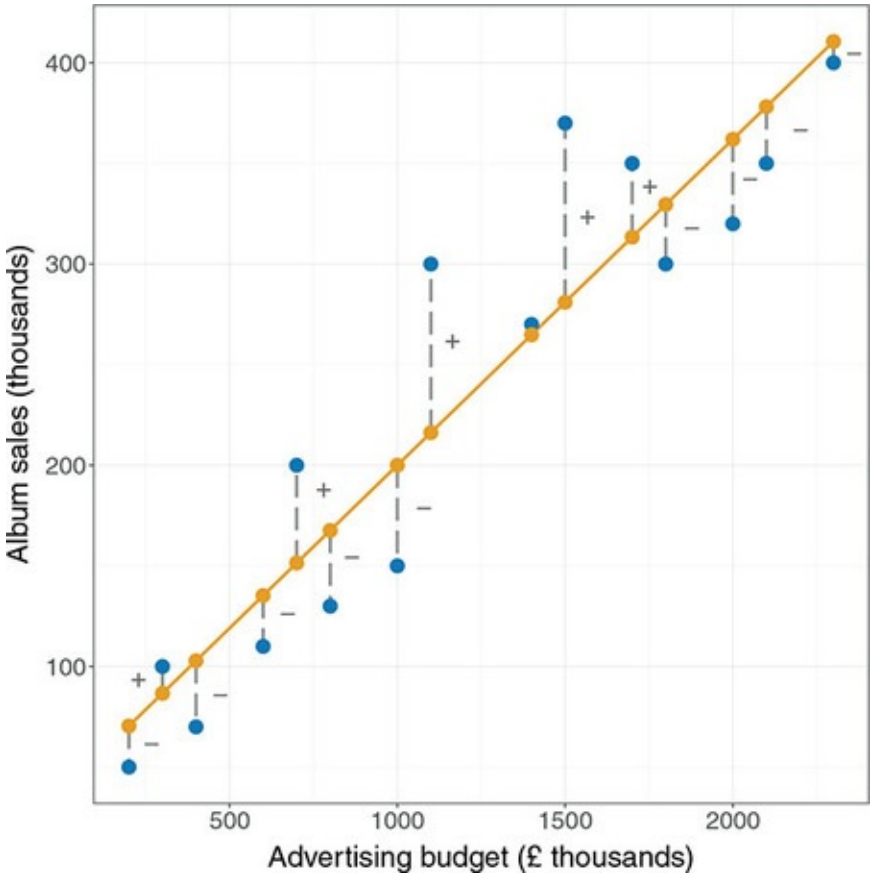
We have seen that the linear model is a versatile model for summarizing the relationship between one or more predictor variables and an outcome variable. No matter how many predictors we have, the model can be described entirely by a constant (b_0) and by parameters associated with each predictor (bs). You might wonder how we estimate these parameters, and the quick answer is that we typically use the method of least squares that was described in Section 2.6. We saw then that we could assess the fit of a model (the example we used was the mean) by looking at the deviations between the model and the data collected. These deviations were the vertical distances between what the model predicted and each data point that was observed. We can do the same to assess the fit of a regression line (or plane).



Figure 9.4 shows some data about advertising budget and album sales. A model has been fitted to these data (the straight line). The blue circles are the observed data. The line is the model. The orange dots on the line are the predicted values. We saw earlier that predicted values are the values of the outcome variable calculated from the model. In other words, if we estimated the values of b that define the model and put these values into the linear model (as we did in equation (9.4)), then insert different values for advertising budget, the predicted values are the resulting estimates of album sales. If we insert the observed values

of advertising budget into the model to get these predicted values, then we can gauge how well the model fits (i.e., makes accurate predictions). If the model is a perfect fit to the data then for a given value of the predictor(s) the model will predict the same value of the outcome as was observed. In terms of Figure 9.4 this would mean that the orange and blue dots fall in the same locations. They don't, because the model is not perfect (and it never will be): sometimes it overestimates the observed value of the outcome and sometimes it underestimates it. With the linear model the differences between what the model predicts and the observed data are usually called **residuals** (they are the same as *deviations* when we looked at the mean); they are the vertical dashed lines in Figure 9.4.

Figure 9.4 A scatterplot of some data with a line representing the general trend. The vertical lines (dotted) represent the differences (or residuals) between the line and the observed data



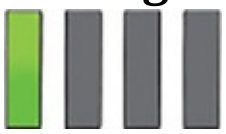
We saw in Chapter 2, equation (2.11), that to calculate the total error in a model we square the differences between the observed values of the outcome, and the predicted values that come from the model:

$$\text{total error} = \sum_{i=1}^{n} (\text{observed}_i - \text{model}_i)^2 \qquad (9.9)$$

Sometimes the predicted value of the outcome is less than the actual value and sometimes it is greater. Consequently, some residuals are positive but others are negative, and if we summed them they would cancel out. The solution is to square them before we add them up (this idea should be familiar from Section 2.5.2). Therefore, to assess the error in a linear model, just like when we assessed the fit of the mean using the variance, we use a sum of squared errors, and because we call these errors residuals, this total is called the *sum of squared residuals* or **residual sum of squares** (SS_R). The residual sum of squares is a gauge of how well a linear model fits the data: if the squared differences are large, the model is not representative of the data (there is a lot of error in prediction); if the squared differences are small, the line is representative.

Let's get back to how we estimate the *b*-values. If you were particularly bored, you could draw every possible straight line (linear model) through your data and calculate the residual sum of squares for each one. You could then compare these 'goodness-of-fit' measures and keep the line with the smallest SS_R because it would be the best-fitting model. We have better things to do, so like when we estimate the mean, we use the method of least squares to estimate the parameters (*b*) that define the regression model for which the sum of squared errors is the minimum it can be (given the data). This method is known as **ordinary least squares (OLS)** regression. How exactly the method of least squares does this is beyond me: it uses a mathematical technique for finding maxima and minima to find the *b*-values that describe the model that minimizes the sum of squared differences.

I don't know much more about it than that, to be honest, so with one predictor I tend to think of the process as a little bearded wizard called Nephwick the Line Finder who just magically finds lines of best fit. Yes, he lives inside your computer. For more complex models, Nephwick invites his brother Clungglewad the Beta Seeker for tea and cake and together they stare into the tea leaves in their cups until the optimal beta-values are revealed to them. Then they compare beard growth since their last meeting. I'm pretty sure that's how the method of least squares works.

9.2.4 Assessing the goodness of fit, sums of squares, R and R^2 

Once Nephwick and Clungglewad have found the values of *b* that define the model of best fit we assess how well this model fits the observed data (i.e., the **goodness of fit**). We do this because even though the model is the best one

available, it can still be a lousy fit (the best of a bad bunch). We saw above that the residual sum of squares measures how much error there is in the model: it quantifies the error in prediction, but it doesn't tell us whether using the model is better than nothing. We need to compare the model against a baseline to see whether it 'improves' how well we can predict the outcome. So, we fit a baseline model and use equation (9.9) to calculate the fit of this model. Then we fit the best model, and calculate the error, SS_R, within it using equation (9.9). If the best model is any good, it should have significantly less error within it than the baseline model.

What would be a good baseline model? Let's go back to our example of predicting album sales (Y) from the amount of money spent advertising that album (X). In my fictional world where I am a statistician employed by a record company or my favourite football team, my boss one day bursts into my office. He says, 'Andy, I know you wanted to be a rock star but have ended up working as my stats-monkey, but how many albums will we sell if we spend £100,000 on advertising?' If I didn't have an accurate model of the relationship between album sales and advertising, what would my best guess be? Probably the best answer I could give would be the mean number of album sales (say, 200,000) because – on average – that's how many albums we expect to sell. This response might satisfy a brainless record company executive (who didn't offer my band a record contract). The next day he bursts in again and demands to know how many albums we will sell if we spend £1 on advertising. In the absence of any better information, I'm again going to have to say the average number of sales (200,000). This is getting embarrassing for me: whatever amount of money is spent on advertising, I predict the same levels of sales. My boss will think I'm an idiot.



The mean of the outcome is a model of 'no relationship' between the variables: as one variable changes the prediction for the other remains constant (see Section 3.7.2). I hope this illustrates that the mean of the outcome is a good baseline of 'no relationship'. Using the mean of the outcome as a baseline model, we can calculate the difference between the observed values and the values predicted by the mean (equation (9.9)). We saw in Section 2.5.1 that we square these differences to give us the sum of squared differences. This sum of

squared differences is known as the **total sum of squares** (denoted by SS_T) and it represents how good the mean is as a model of the observed outcome scores (Figure 9.5, top left).

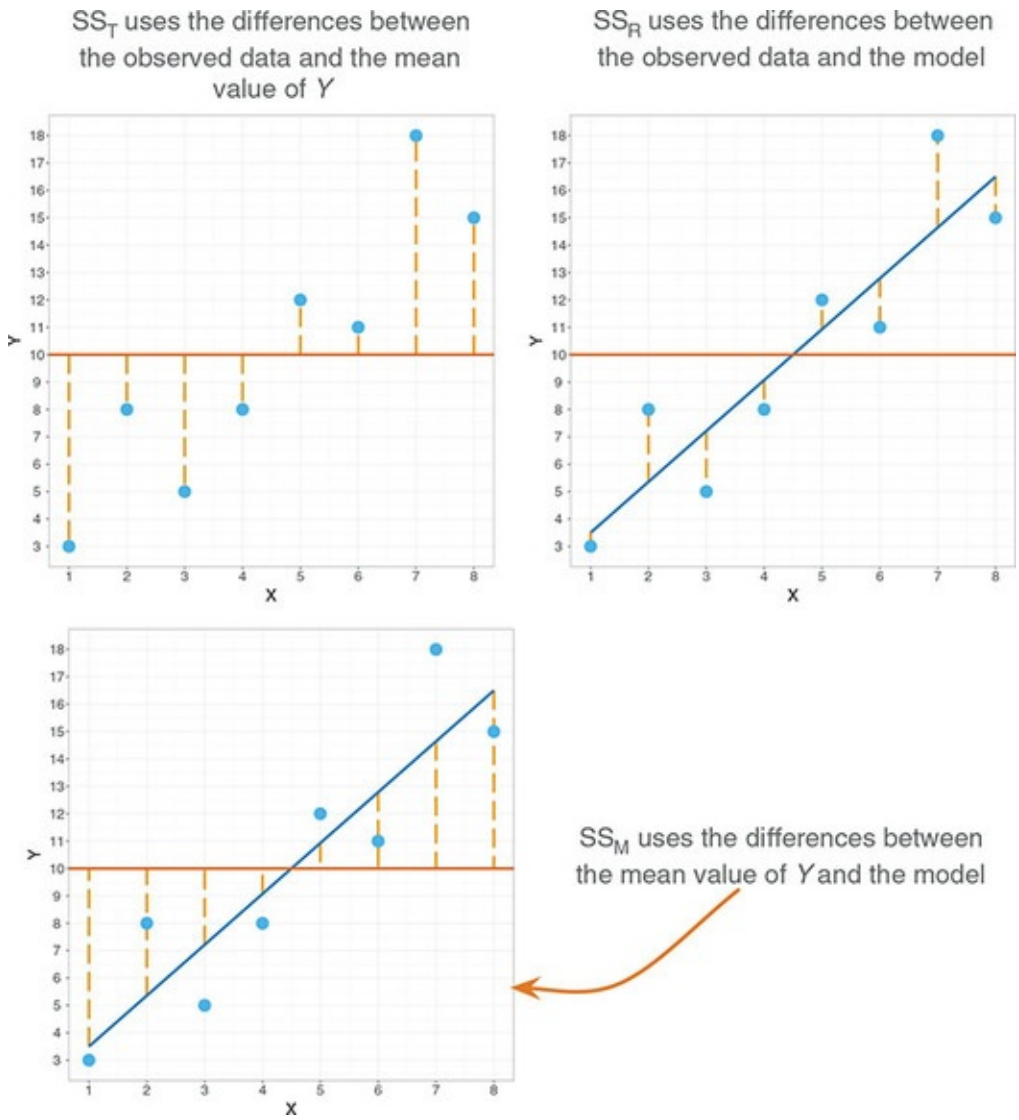
We then fit a more sophisticated model to the data, such as a linear model, and again work out the differences between what this new model predicts and the observed data (again using equation (9.9)). This value is the residual sum of squares (SS_R) discussed in the previous section. It represents the degree of inaccuracy when the best model is fitted to the data (Figure 9.5, top right).

We can use the values of SS_T and SS_R to calculate how much better the linear model is than the baseline model of 'no relationship'. The improvement in prediction resulting from using the linear model rather than the mean is calculated as the difference between SS_T and SS_R (Figure 9.5, bottom). This difference shows us the reduction in the inaccuracy of the model resulting from fitting the regression model to the data. This improvement is the **model sum of squares** (SS_M). Figure 9.5 shows each sum of squares graphically where the model is a line (i.e., one predictor) but the same principles apply with more than one predictor.

If the value of SS_M is large, the linear model is very different from using the mean to predict the outcome variable. This implies that the linear model has made a big improvement to predicting the outcome variable. If SS_M is small then using the linear model is little better than using the mean (i.e., the best model is no better than predicting from 'no relationship'). A useful measure arising from these sums of squares is the proportion of improvement due to the model. This is calculated by dividing the sum of squares for the model by the total sum of squares to give a quantity called R^2:

$$R^2 = \frac{SS_M}{SS_T} \tag{9.10}$$

Figure 9.5 Diagram showing from where the sums of squares derive



To express this value as a percentage multiply it by 100. R^2 represents the amount of variance in the outcome explained by the model (SS_M) relative to how much variation there was to explain in the first place (SS_T); it is the same as the R^2 we met in Section 8.4.2 and it is interpreted in the same way: it represents the proportion of the variation in the outcome that can be predicted from the model. We can take the square root of this value to obtain Pearson's correlation coefficient for the relationship between the values of the outcome predicted by the model and the observed values of the outcome.[4] As such, the correlation coefficient provides us with a good estimate of the overall fit of the regression model (i.e., the correspondence between predicted values of the outcome and the actual values), and R^2 provides us with a gauge of the substantive size of the model fit.[5]

4 This is the correlation between the orange and blue dots in Figure 9.4. With

only one predictor in the model this value will be the same as the Pearson correlation coefficient between the predictor and outcome variable.
5 When the model contains more than one predictor, people sometimes refer to R^2 as multiple R^2. This is another example of how people attempt to make statistics more confusing than it needs to be by referring to the same thing in different ways. The meaning and interpretation of R^2 are the same regardless of how many predictors you have in the model or whether you choose to call it multiple R^2: it is the squared correlation between values of the outcome predicted by the model and the values observed in the data.
A second use of the sums of squares in assessing the model is the *F*-test. I mentioned way back in Chapter 2 that test statistics (like *F*) are usually the amount of systematic variance divided by the amount of unsystematic variance, or, put another way, the model compared to the error in the model. This is true here: *F* is based upon the ratio of the improvement due to the model (SS_M) and the error in the model (SS_R). I say 'based upon' because the sums of squares depend on the number of differences that were added up, and so the average sums of squares (referred to as the **mean squares** or MS) are used to compute *F*. The mean sum of squares is the sum of squares divided by the associated degrees of freedom (this is comparable to calculating the variance from the sums of squares – see Section 2.5.2). For SS_M the degrees of freedom are the number of predictors in the model (k), and for SS_R they are the number of observations (N) minus the number of parameters being estimated (i.e., the number of b coefficients including the constant). We estimate a b for each predictor and the intercept (b_0), so the total number of bs estimated will be $k + 1$, giving us degrees of freedom of $N - (k + 1)$ or, more simply, $N - k - 1$. Thus

$$MS_M = \frac{SS_M}{k} \qquad MS_R = \frac{SS_R}{N-k-1} \tag{9.11}$$

There is more on mean squares in Chapter 12. The ***F*-statistic** computed from these mean squares,

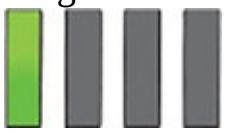
$$F = \frac{MS_M}{MS_R} \tag{9.12}$$

is a measure of how much the model has improved the prediction of the outcome compared to the level of inaccuracy of the model. If a model is good, then the improvement in prediction from using the model should be large (MS_M will be large) and the difference between the model and the observed data should be small (MS_R will be small). In short, for a good model the numerator in equation (9.12) will be bigger than the denominator, resulting in a large *F*-statistic (greater than 1 at least).

This F has an associated probability distribution from which a p-value can be derived to tell us the probability of getting an F at least as big as the one we have if the null hypothesis were true. The null hypothesis in this case is a flat model (predicted values of the outcome are the same regardless of the value of the predictors). If you want to go old school, you can compare the F-statistic against critical values for the corresponding degrees of freedom (as in the Appendix).
The F-statistic is also used to calculate the significance of R^2 using the following equation:

$$F = \frac{(N-k-1)R^2}{k\left(1-R^2\right)} \tag{9.13}$$

in which N is the number of cases or participants, and k is the number of predictors in the model. This F tests the null hypothesis that R^2 is zero (i.e., there is no improvement in the sum of squared error due to fitting the model).

9.2.5 Assessing individual predictors 

We've seen that any predictor in a linear model has a coefficient (b_1). The value of b represents the change in the outcome resulting from a unit change in a predictor. If a predictor was useless at predicting the outcome, then what might we expect the change in the outcome to be as values of the predictor change? If a predictor had 'no relationship' with the outcome then the change would be zero. Think back to Figure 9.5. In the panel representing SS_T we saw that the line representing 'no relationship' or 'mean of the outcome' is flat: as the predictor variable changes, the predicted value of the outcome does *not* change (it is a constant value). A 'flat' model, a model in which the same predicted value arises from all values of the predictor variables, will have b-values of 0 for the predictors.
A regression coefficient of 0 means: (1) a unit change in the predictor variable results in no change in the predicted value of the outcome (the predicted value of the outcome is constant); and (2) the linear model is 'flat' (the line or plane doesn't deviate from the horizontal). Therefore, logically, if a variable significantly predicts an outcome, it should have a b-value that is *different* from zero. This hypothesis is tested using a ***t*-statistic** that tests the null hypothesis that the value of b is 0. If the test is significant, we might interpret this information as supporting a hypothesis that the b-value is significantly different from 0 and that the predictor variable contributes significantly to our ability to estimate values of the outcome.
Like F, the t-statistic is based on the ratio of explained variance against

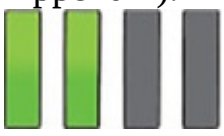
unexplained variance or error. What we're interested in here is not so much variance but whether the *b* we have is big compared to the amount of error in that estimate. Remember that the standard error for *b* tells us something about how different *b*-values would be across different samples (think back to Section 2.7). If the standard error is very small, then most samples are likely to have a *b*-value similar to the one in our sample (because there is little variation across samples). Therefore, the standard error is a good estimate of how much error there is likely to be in our *b*.

The following equation shows how the *t*-test is calculated:

$$t = \frac{b_{\text{observed}} - b_{\text{expected}}}{SE_b} = \frac{b_{\text{observed}}}{SE_b} \tag{9.14}$$

You'll find a general version of this equation in Section 10.5.1 (equation (10.5)). The b_{expected} is the value of *b* that we would expect to obtain if the null hypothesis were true. The null hypothesis is that *b* is 0, and so this value is replaced by 0 and drops out of the equation. The resulting *t* is the observed value of *b* divided by the standard error with which it is associated. The *t*, therefore, tells us whether the observed *b* is different from 0 relative to the variation in *b*s across samples. When the standard error is small even a small deviation from zero can reflect a significant difference because *b* is representative of the majority of possible samples.

The statistic *t* has a probability distribution that differs according to the degrees of freedom for the test. In this context, the degrees of freedom are $N - k - 1$, where *N* is the total sample size and *k* is the number of predictors. With only one predictor, this reduces to $N - 2$. Using the appropriate *t*-distribution, it's possible to calculate a *p*-value that indicates the probability of getting a *t* at least as large as the one we observed if the null hypothesis were true (i.e., if *b* was in fact 0 in the population). If this observed *p*-value is less than 0.05, then scientists tend to assume that *b* is significantly different from 0; put another way, the predictor makes a significant contribution to predicting the outcome. However, remember the potential pitfalls of blindly applying this 0.05 rule. If you want to pretend it's 1935 then instead of computing an exact *p*, you can compare your observed *t* against critical values in a table (in the Appendix).

9.3 Bias in linear models? 

In Chapter 6 we saw that statistical models can be biased by unusual cases or by failing to meet certain assumptions. Therefore, the next questions to ask are whether the model: (1) is influenced by a small number of cases; and (2) generalizes to other samples. These questions are, in some sense, hierarchical

because we wouldn't want to generalize a bad model. However, it is a mistake to think that because a model fits the observed data well we can draw conclusions beyond our sample. **Generalization** (Section 9.4) is a critical additional step, and if we find that our model is not generalizable, then we must restrict any conclusions to the sample used. First, let's look at bias. To answer the question of whether the model is influenced by a small number of cases, we can look for outliers and influential cases (the difference is explained in Jane Superbrain Box 9.1).



9.3.1 Outliers

An outlier is a case that differs substantially from the main trend in the data (see Section 6.3). Outliers can affect the estimates of the regression coefficients. For example, Figure 9.6 uses the same data as Figure 9.4 except that the score of one album has been changed to be an outlier (in this case an album that sold relatively few copies despite a very large advertising budget). The blue line shows the original model, and the orange line shows the model with the outlier included. The outlier makes the line flatter (i.e., b_1 gets smaller) and increases the intercept (b_0 gets larger). If outliers affect the estimates of the *b*s that define the model then it is important to detect them. But how?

An outlier, by its nature, is very different from the other scores. In which case, do you think that the model will predict an outlier's score very accurately? Probably not: in Figure 9.6 it's evident that even though the outlier has dragged the model towards it, the model still predicts it very badly (the line is a long way from the outlier). Therefore, if we compute the residuals (the differences between the observed values of the outcome and the values predicted by the model), outliers could be spotted because they'd have large values. In other words, we'd look for cases that the model predicts inaccurately.

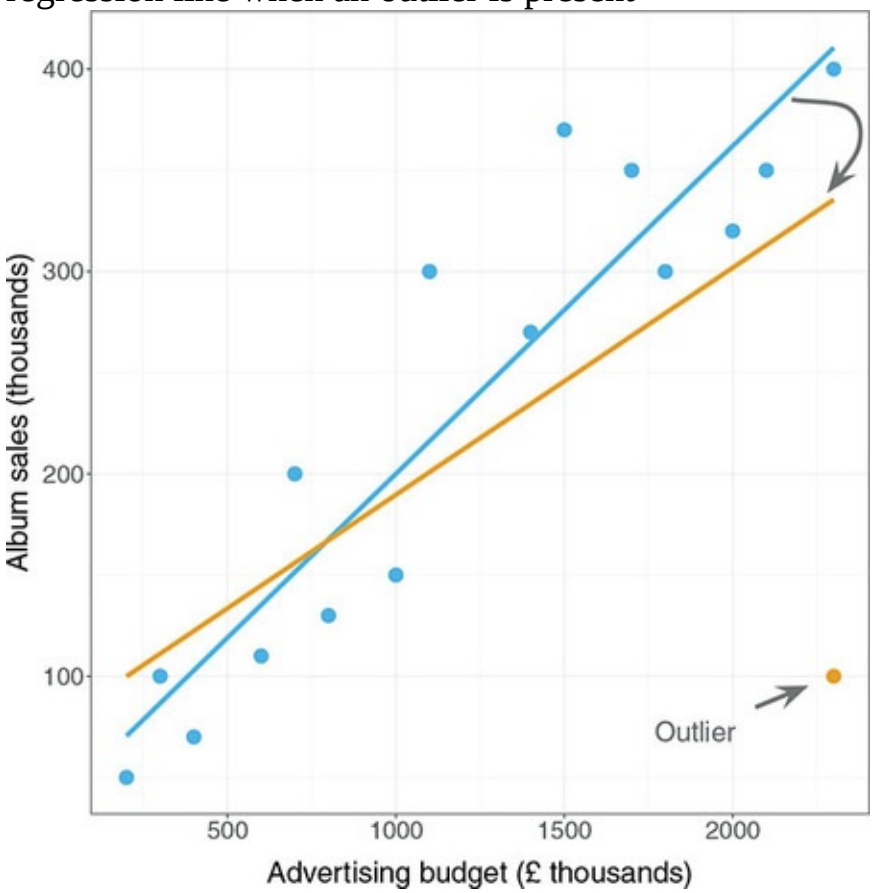


Residuals are used to compute which of the three sums of squares?

Remember that residuals represent the error present in the model. If a model fits

the sample data well then all residuals will be small (if the model was a perfect fit of the sample data – all data points fall on the regression line – then all residuals would be zero). If a model is a poor fit to the sample data then the residuals will be large. Up to now we have discussed *normal* or **unstandardized residuals**. These are the raw differences between predicted and observed values of the outcome variable. They are measured in the same units as the outcome variable, which makes it difficult to apply general rules (because what constitutes 'large' depends on the outcome variable). All we can do is to look for residuals that stand out as being particularly large.

Figure 9.6 Graph demonstrating the effect of an outlier. The blue line represents the original regression line for these data, whereas the orange line represents the regression line when an outlier is present



To overcome this problem, we can use **standardized residuals**, which are the residuals converted to *z*-scores (see Section 1.8.6) and so are expressed in standard deviation units. Regardless of the variables in your model, standardized residuals (like any *z*-scores) are distributed around a mean of 0 with a standard deviation of 1. Therefore, we can compare standardized residuals from different models and use what we know about *z*-scores to apply universal guidelines for what is expected. For example, in a normally distributed sample, 95% of *z*-

scores should lie between −1.96 and +1.96, 99% should lie between −2.58 and +2.58, and 99.9% (i.e., nearly all of them) should lie between −3.29 and +3.29 (see Chapter 1). Based on this: (1) standardized residuals with an absolute value greater than 3.29 (we can use 3 as an approximation) are cause for concern because in an average sample a value this high is unlikely to occur; (2) if more than 1% of our sample cases have standardized residuals with an absolute value greater than 2.58 (2.5 will do) there is evidence that the level of error within our model may be unacceptable; and (3) if more than 5% of cases have standardized residuals with an absolute value greater than 1.96 (2 for convenience) then the model may be a poor representation of the data.

A third form of residual is the **studentized residual**, which is the unstandardized residual divided by an estimate of its standard deviation that varies point by point. These residuals have the same properties as the standardized residuals but usually provide a more precise estimate of the error variance of a specific case.

9.3.2 Influential cases 

It is also possible to look at whether certain cases exert undue influence over the parameters of the model. In other words, if we were to delete a certain case, how different would the regression coefficients be? This analysis helps to determine whether the model is stable across the sample, or whether it is biased by a few influential cases. This process can also unveil outliers.

There are several statistics used to assess the influence of a case. The **adjusted predicted value** for a case is the predicted value of the outcome for that case from a model in which the case is excluded. In effect, you estimate the model parameters excluding a particular case and use this new model to predict the outcome for the case that was excluded. If a case does not exert a large influence over the model then the adjusted predicted value should be similar to the predicted value when the case is included. Put simply, if the model is stable then the predicted value of a case should be the same regardless of whether that case was used to estimate the model.

We can also look at the **deleted residual**, which is the difference between the adjusted predicted value and the original observed value. The deleted residual can be divided by the standard error to give a standardized value known as the **studentized deleted residual**. This residual can be compared across different regression analyses because it is measured in standard units.

The deleted residuals are very useful to assess the influence of a case on the ability of the model to predict that case. However, they do not provide any information about how a case influences the model as a whole (i.e., the impact

that a case has on the model's ability to predict *all* cases). **Cook's distance** is a measure of the overall influence of a case on the model, and Cook and Weisberg (1982) have suggested that values greater than 1 may be cause for concern.
The **leverage** (sometimes called **hat values**) gauges the influence of the observed value of the outcome variable over the predicted values. The average leverage value is defined as $(k + 1)/n$, in which k is the number of predictors in the model and n is the number of cases.[6] The maximum value for leverage is $(N – 1)/N$; however, IBM SPSS Statistics calculates a version of the leverage that has a maximum value of 1 (indicating that the case has complete influence over prediction).
6 You may come across the average leverage denoted as p/n, in which p is the number of parameters being estimated. In regression, we estimate parameters for each predictor and also for a constant, and so p is equivalent to the number of predictors plus one $(k + 1)$.

- If no cases exert undue influence over the model then all leverage values should be close to the average value $((k + 1)/n)$.
- We should investigate cases with values greater than twice the average, $2(k + 1)/n$ (Hoaglin & Welsch, 1978), or three times the average, $3(k + 1)/n$ (Stevens, 2002).

We will see how to use these cut-off points later. However, cases with large leverage values will not necessarily have a large influence on the regression coefficients because they are measured on the outcome variables, not the predictors.
Related to the leverage values are the **Mahalanobis distances**, which measure the distance of cases from the mean(s) of the predictor variable(s). Look for the cases with the highest values. These distances have a chi-square distribution, with degrees of freedom equal to the number of predictors (Tabachnick & Fidell, 2012). One way to establish a cut-off point is to find the critical value of chi-square for the desired alpha level (values for $p = 0.05$ and 0.01 are in the Appendix). For example, with three predictors, a distance greater than 7.81 ($p = 0.05$) or 11.34 ($p = 0.01$) would be cause for concern. As general context, based on Barnett and Lewis (1978), with large samples ($N = 500$) and five predictors, values above 25 are cause for concern. In smaller samples ($N = 100$) and fewer predictors (namely, three), values greater than 15 are problematic. In very small samples ($N = 30$) with only two predictors, values greater than 11 should be examined.
Another approach is to look at how the estimates of b in a model change as a result of excluding a case (i.e., compare the values of b estimated from the full data to those estimated from the data excluding the particular case). The change

in *bs* tells us how much influence a case has on the parameters of the model. To take a hypothetical example, imagine two variables that have a perfect negative relationship except for a single case (case 30). These data are in the file **DFBeta.sav.**

Figure 9.7 Prasanta Chandra Mahalanobis staring into his distances





Once you have read Section 9.7, fit a linear model first with all the cases included and then with case 30 deleted.

The results of these two models are summarized in Table 9.1, which shows: (1) the parameters for the regression model when the extreme case is included or excluded; (2) the resulting regression equations; and (3) the value of *Y* predicted from participant 30's score on the *X* variable (which is obtained by replacing the *X* in the regression equation with participant 30's score for *X*, which was 1). When case 30 is excluded, these data have a perfect negative relationship; hence the coefficient for the predictor (b_1) is −1, and the coefficient for the constant (the intercept, b_0) is 31. However, when case 30 is included, both parameters are reduced[7] and the difference between the parameters is also displayed. The difference between a parameter estimated using all cases and estimated when one case is excluded is known as the **DFBeta**. DFBeta is calculated for every case and for each of the parameters in the model. So, in our hypothetical example, the DFBeta for the constant is −2, and the DFBeta for the predictor variable is 0.1. The values of DFBeta help us to identify cases that have a large influence on the parameters of the model. The units of measurement used will affect these values, and so you can use **standardized DFBeta** to apply universal cut-off points. Standardized DFBetas with absolute values above 1 indicate cases that substantially influence the model parameters (although Stevens, 2002,

suggests looking at cases with absolute values greater than 2).
7 The value of b_1 is reduced because the variables no longer have a perfect linear relationship and so there is now variance that the predictor cannot explain.
A related statistic is the **DFFit**, which is the difference between the predicted values for a case when the model is estimated including or excluding that case: in this example the value is −1.90 (see Table 9.1). If a case has no influence then its DFFit should be zero – hence, we expect non-influential cases to have small DFFit values. As with DFBeta, this statistic depends on the units of measurement of the outcome, and so a DFFit of 0.5 will be very small if the outcome ranges from 1 to 100, but very large if the outcome varies from 0 to 1. To overcome this problem we can look at standardized versions of the DFFit values (**standardized DFFit**) which are expressed in standard deviation units. A final measure is the **covariance ratio (CVR)**, which quantifies the degree to which a case influences the variance of the regression parameters. A description of the computation of this statistic leaves me dazed and confused, so suffice to say that when this ratio is close to 1 the case is having very little influence on the variances of the model parameters. Belsey, Kuh, & Welsch (1980) recommend the following:

Table 9.1 The difference in the parameters of the regression model when one case is excluded

Parameter (*b*)	Case 30 Included	Case 30 Excluded	Difference
Constant (intercept)	29.00	31.00	−2.00
Predictor (gradient)	−0.90	−1.00	0.10
Model (regression line):	$Y = -0.9X + 29$	$Y = -1X + 31$	
Predicted *Y*	28.10	30.00	−1.90

- If $CVR_i > 1 + [3(k + 1)/n]$ then deleting the *i*th case will damage the precision of some of the model's parameters.
- If $CVR_i < 1 - [3(k + 1)/n]$ then deleting the *i*th case will improve the precision of some of the model's parameters.

In both inequalities, k is the number of predictors, CVR_i is the covariance ratio for the *i*th participant, and n is the sample size.

9.3.3 A final comment on diagnostic statistics



I'll conclude this section with a point made by Belsey *et al.* (1980): diagnostics are tools to see how well your model fits the sampled data and *not* a way of justifying the removal of data points to effect some desirable change in the regression parameters (e.g., deleting a case that changes a non-significant *b*-value into a significant one). Stevens (2002) similarly notes that if a case is a

significant outlier but is not having an influence (e.g., Cook's distance is less than 1, DFBetas and DFFit are small) there is no real need to worry about that point because it's not having a large impact on the model parameters. Nevertheless, you should still be interested in *why* the case didn't fit the model.

9.4 Generalizing the model 

The linear model produces an equation that is correct for the sample of observed values. However, we are usually interested in generalizing our findings beyond the sample. For a linear model to generalize the underlying assumptions must be met, and to test whether the model does generalize we can cross-validate it.

9.4.1 Assumptions of the linear model 

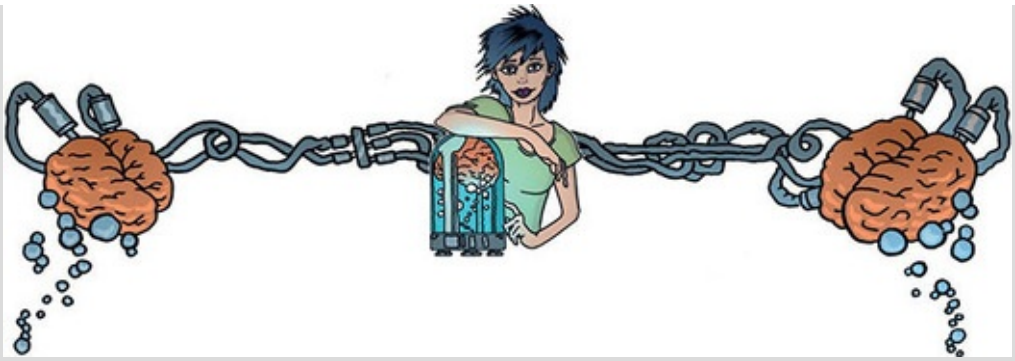
We have already looked at the main assumptions of the linear model and how to assess them in Chapter 6. The main ones in order of importance (Field & Wilcox, 2017; Gelman & Hill, 2007) are:

- *Additivity and linearity*: The outcome variable should, in reality, be linearly related to any predictors, and, with several predictors, their combined effect is best described by adding their effects together. In other words, the process we're trying to model can be described by the linear model. If this assumption isn't met then the model is invalid. You can sometimes transform variables to make their relationships linear (see Chapter 6).
- **Independent errors**: For any two observations the residual terms should be uncorrelated (i.e., independent). This eventuality is sometimes described as a lack of **autocorrelation**. If we violate this assumption then the model standard errors will be invalid, as will the confidence intervals and significance tests based upon them. In terms of the model parameters themselves, the estimates from the method of least squares will be valid but not optimal (see Section 6.8). This assumption can be tested with the **Durbin–Watson test**, which tests for serial correlations between errors. Specifically, it tests whether adjacent residuals are correlated. As such it is affected by the order of cases and only makes sense when your cases have a

meaningful order (which they don't in the album sales example). The test statistic varies between 0 and 4, with a value of 2 meaning that the residuals are uncorrelated. A value greater than 2 indicates a negative correlation between adjacent residuals, whereas a value below 2 indicates a positive correlation. The size of the Durbin–Watson statistic depends upon the number of predictors in the model and the number of observations. If this test is relevant to you, look up the critical values in Durbin and Watson (1951). As a very conservative rule of thumb, values less than 1 or greater than 3 are cause for concern.

- *Homoscedasticity* (see Section 6.7): At each level of the predictor variable(s), the variance of the residual terms should be constant. This assumption means that the residuals at each level of the predictor(s) should have the same variance (**homoscedasticity**); when the variances are very unequal there is said to be **heteroscedasticity**. Violating this assumption invalidates confidence intervals and significance tests; estimates of the model parameters (*b*) using the method of least squares are valid but not optimal. This problem is overcome using weighted least squares regression, in which each case is weighted by a function of its variance, or using robust regression.
- *Normally distributed errors* (see Section 6.6): It can be helpful if the residuals in the model are random, normally distributed variables with a mean of 0. This assumption means that the differences between the predicted and observed data are most frequently zero or very close to zero, and that differences much greater than zero happen only occasionally. Some people confuse this assumption with the idea that predictors have to be normally distributed, which they don't. In small samples a lack of normality invalidates confidence intervals and significance tests, whereas in large samples it will not because of the central limit theorem. If you are concerned only with estimating the model parameters (and not significance tests and confidence intervals) then this assumption barely matters. If you bootstrap confidence intervals then you can ignore this assumption.

Jane Superbrain 9.1 The difference between residuals and influence statistics 



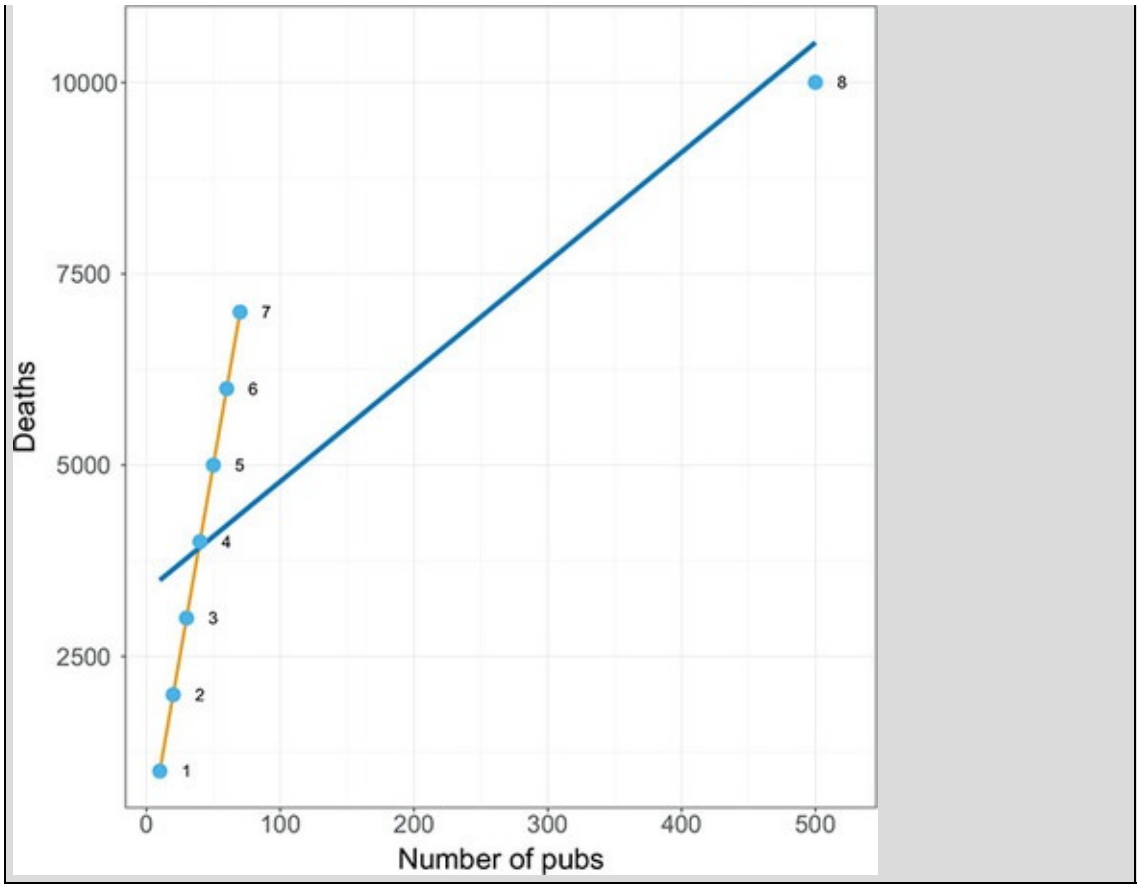
To illustrate how residuals and influence statistics differ, imagine that the Mayor of London in 1900 was interested in how drinking affected mortality. London is divided up into different regions called boroughs, and so he measured the number of pubs and the number of deaths over a period of time in eight of his boroughs. The data are in a file called **pubs.sav**.

The scatterplot of these data (Figure 9.8) reveals that without the last case there is a perfect linear relationship (the orange line). However, the presence of the last case (case 8) changes the line of best fit dramatically (although this line is still a significant fit to the data – fit the model and see for yourself).

The residuals and influence statistics are interesting (Output 9.1). The standardized residual for case 8 is the second *smallest*: it produces a very small residual (most of the non-outliers have larger residuals) because it sits very close to the line that has been fitted to the data. According to the residual it is not an outlier, but how is that possible when it is so different from the rest of the data? The answer lies in the influence statistics, which are all massive for case 8: it exerts a huge influence over the model – so huge that the model predicts that case very well.

When you see a statistical oddity like this, ask what's happening in the real world. District 8 is the City of London, a tiny area of only 1 square mile in the centre of London where very few people lived but where thousands of commuters (even then) came to work and needed pubs. Therefore, there was a massive number of pubs. (I'm very grateful to David Hitchin for this example, and he in turn got it from Dr Richard Roberts.)

Figure 9.8 Relationship between the number of pubs and the number of deaths in 8 London districts

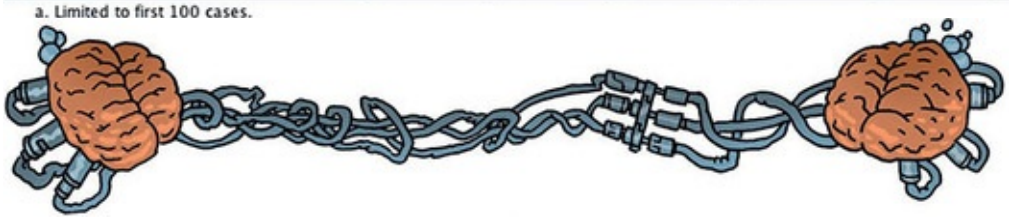


Output 9.1

Case Summaries[a]

	Standardized Residual	Mahalanobis Distance	Cook's Distance	Centered Leverage Value	DFFIT	DFBETA Intercept	DFBETA pubs
1	-1.33839	.28515	.21328	.04074	-495.72692	-509.65184	1.39249
2	-.87895	.22370	.08530	.03196	-305.09716	-321.12768	.80153
3	-.41950	.16969	.01814	.02424	-137.20167	-147.10661	.33016
4	.03995	.12314	.00015	.01759	12.38769	13.45081	-.02658
5	.49940	.08403	.02294	.01200	147.81622	161.44976	-.27267
6	.95885	.05237	.08092	.00748	273.00807	297.67748	-.41116
7	1.41830	.02817	.17107	.00402	391.72124	422.81664	-.44422
8	-.27966	6.03375	227.14286	.86196	-39478.585	3351.95531	-85.66108
Total N	8	8	8	8	8	8	8

a. Limited to first 100 cases.



There are other considerations that we haven't touched on (see Berry, 1993):

- *Predictors are uncorrelated with 'external variables'*: *External variables*

are variables that haven't been included in the model and that influence the outcome variable.[8] These variables are like the 'third variable' that we discussed in the correlation chapter. This assumption means that there should be no external variables that correlate with any of the variables included in the regression model. Obviously, if external variables do correlate with the predictors, then the conclusions we draw from the model become unreliable (because other variables exist that can predict the outcome just as well).

- *Variable types*: All predictor variables must be quantitative or categorical (with two categories), and the outcome variable must be quantitative, continuous and unbounded. By 'quantitative' I mean that they should be measured at the interval level and by 'unbounded' I mean that there should be no constraints on the variability of the outcome. If the outcome is a measure ranging from 1 to 10 yet the data collected vary between 3 and 7, then these data are constrained.
- *No perfect* **multicollinearity**: If your model has more than one predictor then there should be no perfect linear relationship between two or more of the predictors. So, the predictor variables should not correlate too highly (see Section 9.9.3).
- *Non-zero variance*: The predictors should have some variation in value (i.e., they should not have variances of 0). This is self-evident really.

8 Some authors refer to these external variables as part of an error term that includes any random factor in the way in which the outcome varies. However, to avoid confusion with the residual terms in the regression equations I have chosen the label 'external variables'. Although this term implicitly washes over any random factors, I acknowledge their presence.

As we saw in Chapter 6, violating these assumptions has implications mainly for significance tests and confidence intervals; the estimates of *b*s are not dependent on these assumptions (although least squares methods will be optimal when the assumptions are met). However, the 95% confidence interval for a *b* tells us the boundaries within which the population values of that *b* are likely to fall.[9] Therefore, if confidence intervals are inaccurate (as they are when these assumptions are broken) we cannot accurately estimate the likely population value. In other words, we can't generalize our model to the population. When the assumptions are met then *on average* the regression model from the sample is the same as the population model. However, you should be clear that even when the assumptions are met, it is possible that a model obtained from a sample is not

the same as the population model – but the likelihood of them being the same is increased.

9 Assuming your sample is one of the 95% that generates a confidence interval containing the population value. Yes, I do have to keep making this point – it's important.

9.4.2 Cross-validation of the model 

Even if we can't be confident that the model derived from our sample accurately represents the population, we can assess how well our model might predict the outcome in a different sample. Assessing the accuracy of a model across different samples is known as **cross-validation**. If a model can be generalized, then it must be capable of accurately predicting the same outcome variable from the same set of predictors in a different group of people. If the model is applied to a different sample and there is a severe drop in its predictive power, then the model does *not* generalize. First, we should collect enough data to obtain a reliable model (see the next section). Once we have a estimated the model there are two main methods of cross-validation:

- *Adjusted* R^2: Whereas R^2 tells us how much of the variance in Y overlaps with predicted values from the model in our sample, **adjusted *R*2** tells us how much variance in Y would be accounted for if the model had been derived from the population from which the sample was taken. Therefore, the adjusted value indicates the loss of predictive power or **shrinkage**. SPSS derives the adjusted R^2 using Wherry's equation. This equation has been criticized because it tells us nothing about how well the model would predict scores of a different sample of data from the same population. Stein's formula,

$$\text{adjusted } R^2 = 1 - \left[\left(\frac{n-1}{n-k-1}\right)\left(\frac{n-2}{n-k-2}\right)\left(\frac{n+1}{n}\right)\right]\left(1-R^2\right) \qquad (9.15)$$

- does tell us how well the model cross-validates (see Stevens, 2002), and the more mathematically minded of you might want to try using it instead of what SPSS chugs out. In Stein's formula, R^2 is the unadjusted value, n is the number of cases and k is the number of predictors in the model.

- *Data splitting*: This approach involves randomly splitting your sample data, estimating the model in both halves of the data and comparing the resulting models. When using stepwise methods (see Section 9.9.1), cross-validation is particularly important; you should run the stepwise regression on a random selection of about 80% of your cases. Then force this model on the remaining 20% of the data. By comparing values of R^2 and b in the two samples you can tell how well the original model generalizes (see Tabachnick & Fidell, 2012).

9.5 Sample size and the linear model 

In the previous section I said that it's important to collect enough data to obtain a reliable regression model. Also, larger samples enable us to assume that our bs have normal sampling distributions because of the central limit theorem (Section 6.6.1). Well, how much is enough?



You'll find a lot of rules of thumb floating about, the two most common being that you should have 10 cases of data for each predictor in the model, or 15 cases of data per predictor. These rules are very pervasive but they oversimplify the issue to the point of being useless. The sample size required depends on the size of effect that we're trying to detect (i.e., how strong the relationship is that we're trying to measure) and how much power we want to detect these effects. The simplest rule of thumb is that the bigger the sample size, the better: the estimate of R that we get from regression is dependent on the number of predictors, k, and the sample size, N. In fact, the expected R for random data is $k/(N - 1)$ and so with small sample sizes random data can appear to show a strong effect: for example, with six predictors and 21 cases of data, $R = 6/(21 - 1) = 0.3$ (a medium effect size by Cohen's criteria described in Section 3.7.2). Obviously for random data we'd want the expected R to be 0 (no effect), and for this to be true we need large samples (to take the previous example, if we had 100 cases rather than 21, then the expected R would be a more acceptable 0.06).

Figure 9.9 The sample size required to test the overall regression model depending on the number of predictors and the size of expected effect, R^2 = 0.02 (small), 0.13 (medium) and 0.26 (large)

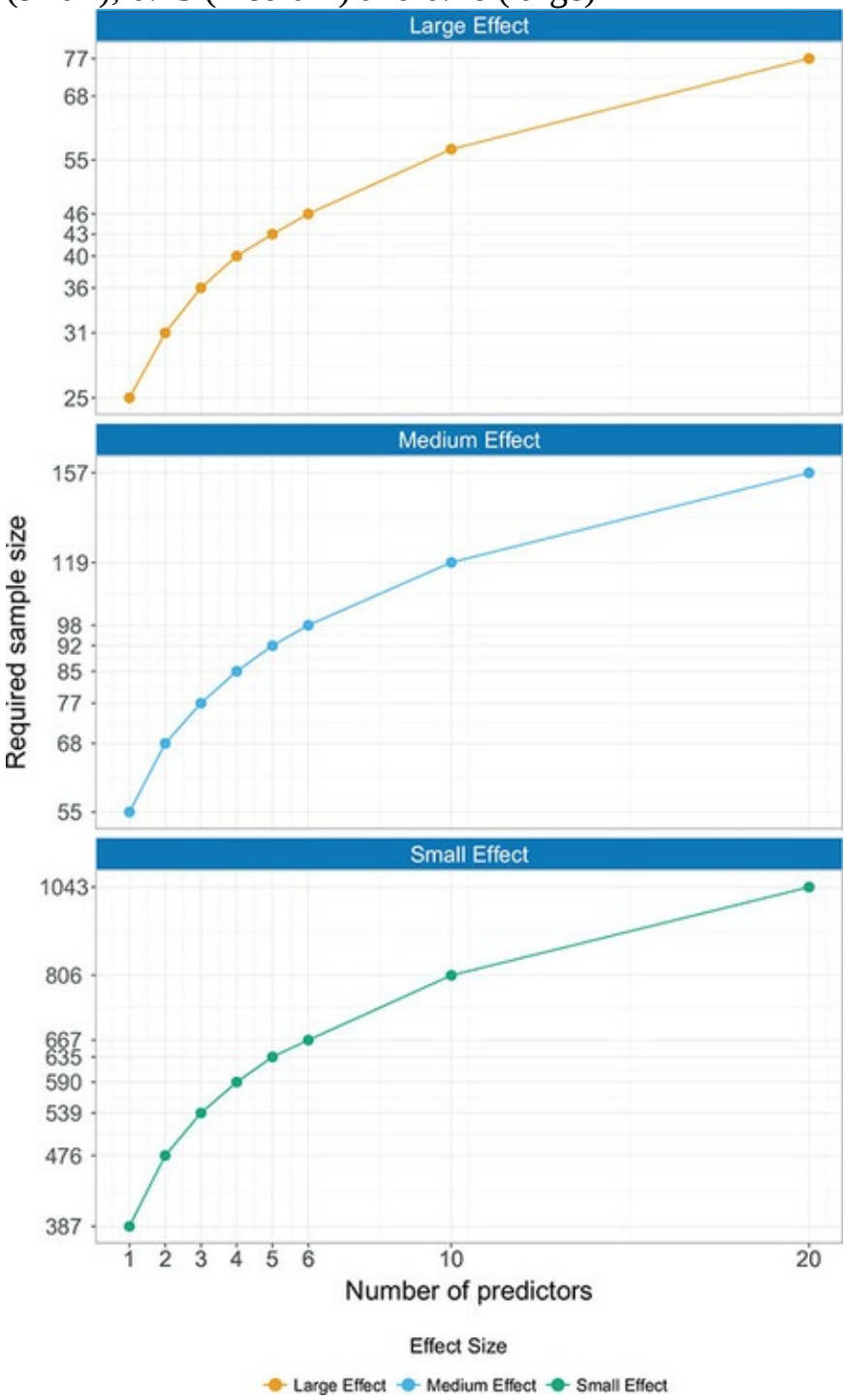


Figure 9.9 shows the sample size required[10] to achieve a high level of power (I've taken Cohen's, 1988, benchmark of 0.8) to test that the model is significant

overall (i.e., R^2 is not equal to zero). I've varied the number of predictors and the size of expected effect: I used R^2 = 0.02 (small), 0.13 (medium) and 0.26 (large), which correspond to benchmarks in Cohen (1988). Broadly speaking, if your aim is to test the overall fit of the model: (1) if you expect to find a large effect then a sample size of 77 will always suffice (with up to 20 predictors) and if there are fewer predictors then you can afford to have a smaller sample; (2) if you're expecting a medium effect, then a sample size of 160 will always suffice (with up to 20 predictors), you should always have a sample size above 55, and with six or fewer predictors you'll be fine with a sample of 100; and (3) if you're expecting a small effect size then just don't bother unless you have the time and resources to collect hundreds of cases of data. Miles and Shevlin (2001) produce more detailed graphs that are worth a look, but the take-home message is that if you're looking for medium to large effects sample sizes don't need to be massive, regardless of how many predictors you have.

10 I used the program G*power, mentioned in Section 2.9.8, to compute these values.

9.6 Fitting linear models: the general procedure

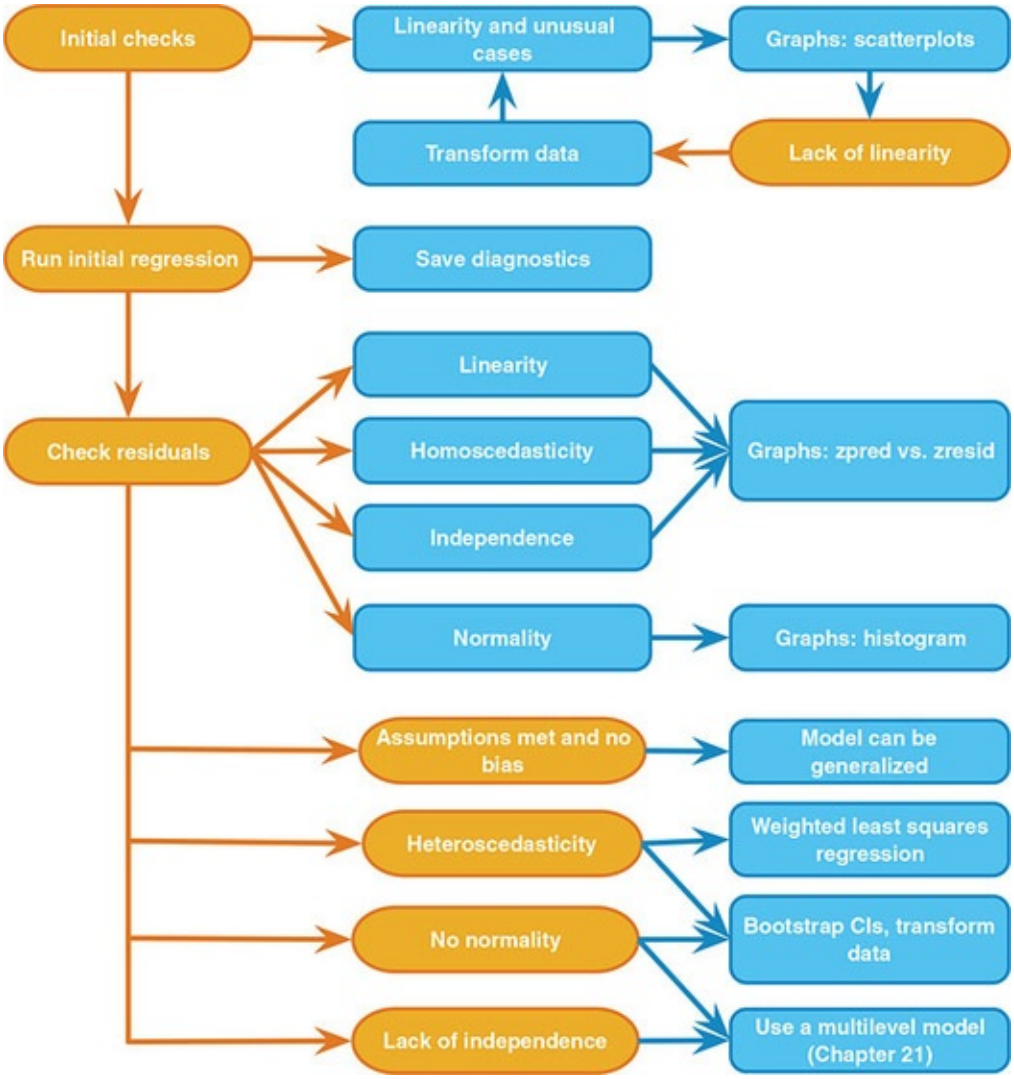


Figure 9.10 shows the general process of fitting linear models. First, we should produce scatterplots to get some idea of whether the assumption of linearity is met, and to look for outliers or obvious unusual cases. At this stage we might transform the data to correct problems. Having done this initial screen for problems, we fit a model and save the various diagnostic statistics that we discussed in Section 9.3. If we want to generalize our model beyond the sample, or we are interested in interpreting significance tests and confidence intervals, then we examine these residuals to check for homoscedasticity, normality, independence and linearity (although this will likely be fine, given our earlier screening). If we find problems then we take corrective action and re-estimate the model. This process might seem complex, but it's not as bad as it seems. Also, it's probably wise to use bootstrapped confidence intervals when we first estimate the model because then we can basically forget about things like normality.

9.7 Using SPSS Statistics to fit a linear model with one predictor

Earlier on I asked you to imagine that I worked for a record company and that my boss was interested in predicting album sales from advertising. There are data for this example in the file **Album Sales.sav**. This data file has 200 rows, each one representing a different album. There are also several columns, one of which contains the sales (in thousands) of each album in the week after release (**Sales**) and one containing the amount (in thousands of pounds) spent promoting the album before release (**Adverts**). The other columns represent how many times songs from the album were played on a prominent national radio station in the week before release (**Airplay**), and how attractive people found the band's image out of 10 (**Image**). Ignore these last two variables for now; we'll use them later. Note how the data are laid out (Figure 9.11): each variable is in a column and each row represents a different album. So, the first album had £10,260 spent advertising it, sold 330,000 copies, received 43 plays on radio the week before release, and was made by a band with a pretty sick image.

Figure 9.10 The process of fitting a regression model





Produce a scatterplot of sales (*y*-axis) against advertising budget (*x*-axis). Include the regression line.

Figure 9.12 shows that a positive relationship exists: the more money spent advertising the album, the more it sells. Of course there are some albums that sell well regardless of advertising (top left of scatterplot), but there are none that sell badly when advertising levels are high (bottom right of scatterplot). The scatterplot shows the line of best fit for these data: bearing in mind that the mean would be represented by a flat line at around the 200,000 sales mark, the regression line is noticeably different.

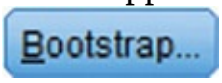
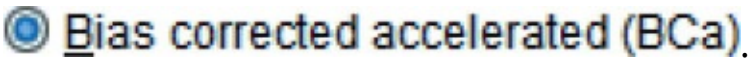
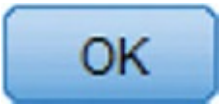
To fit the model, access the main dialog box by selecting *Analyze*  *Regression*  (Figure 9.13). First, we define the outcome variable (in this example **Sales**). Select **Sales** from the list on the left-hand side, and transfer it to the space labelled *Dependent* by dragging it or clicking  . In this model we're going to enter only one predictor (**Adverts**) so select it from the list and click  (or drag it) to transfer it to the box labelled *Independent(s)*. There are a tonne of options available, but we'll explore these when we build up the model in due course. For now, request bootstrapped confidence intervals for the regression coefficients by clicking  (see Section 6.12.3). Select  to activate bootstrapping, and to get a 95% confidence interval select  . Click  in the main dialog box to fit the model.

Figure 9.11 The data editor for fitting a linear model

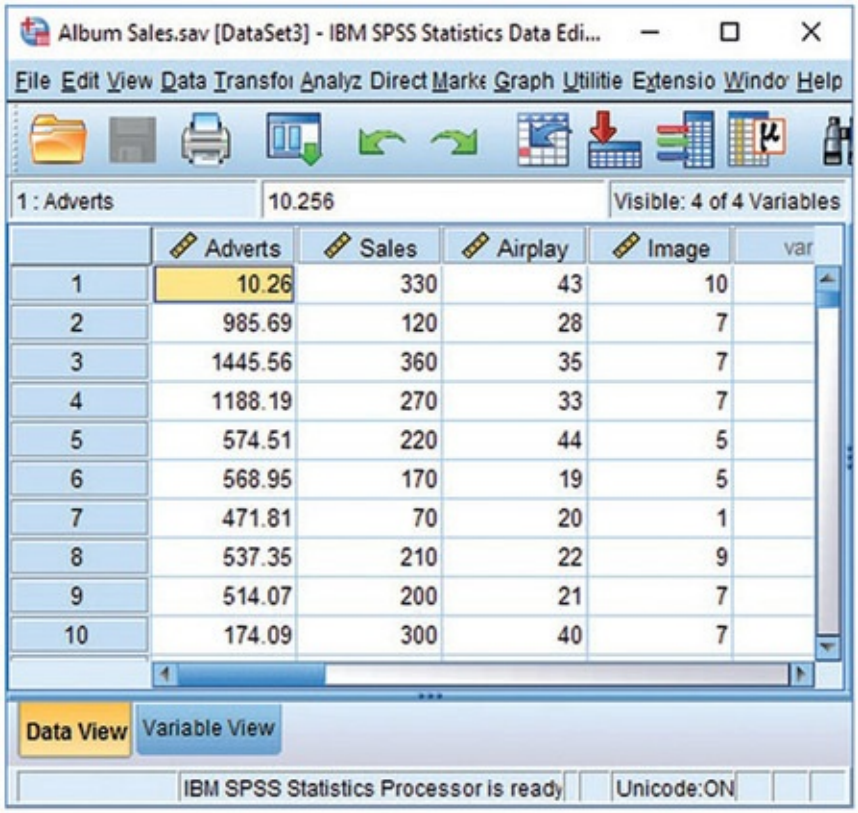


Figure 9.12 Scatterplot showing the relationship between album sales and the amount spent promoting the album

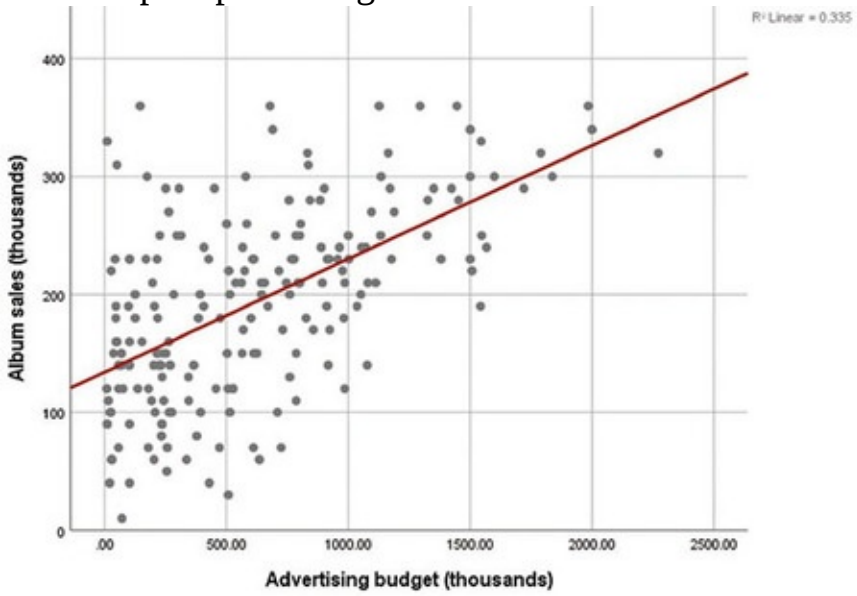
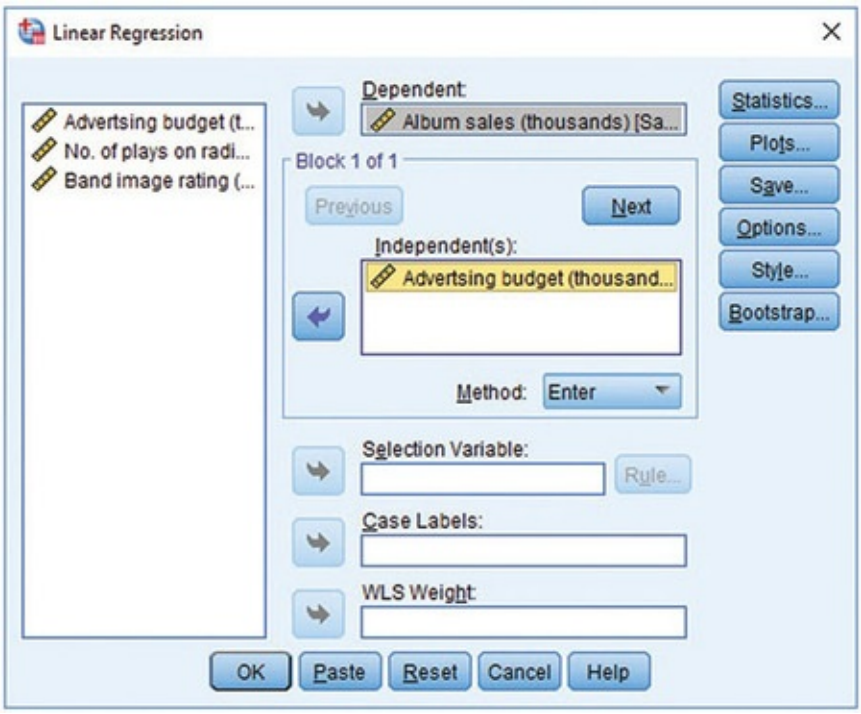


Figure 9.13 Main dialog box for regression



9.8 Interpreting a linear model with one predictor

9.8.1 Overall fit of the model

The first table is a summary of the model (Output 9.2). This summary table provides the value of R and R^2 for the model. For these data, R has a value of 0.578 and because there is only one predictor, this value is the correlation between advertising and album sales (you can confirm this by running a correlation using what you learnt in Chapter 8). The value of R^2 is 0.335, which tells us that advertising expenditure can account for 33.5% of the variation in album sales. This means that 66.5% of the variation in album sales remains unaccounted for: there might be other variables that have an influence also.

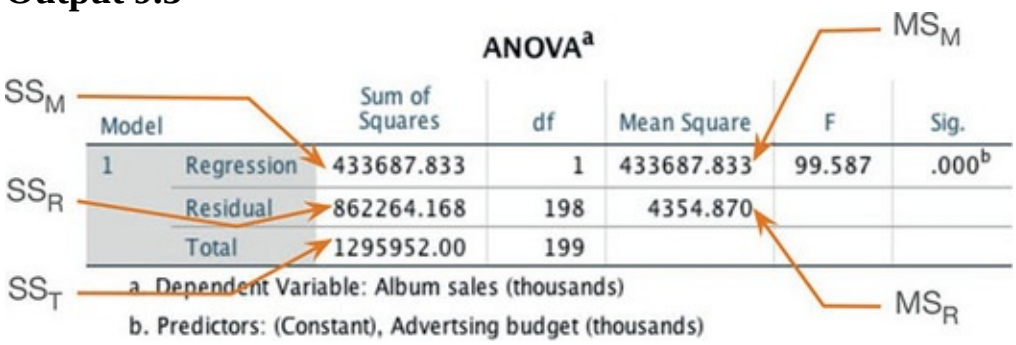
Output 9.2

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.578[a]	.335	.331	65.991

a. Predictors: (Constant), Advertsing budget (thousands)

Output 9.3



The next part of the output (Output 9.3) reports the various sums of squares described in Figure 9.5, the degrees of freedom associated with each and the resulting mean squares (equation (9.11)). The most important part of the table is the *F*-statistic (equation (9.12)) of 99.59 and its associated significance value of $p < 0.001$ (expressed this way because the value in the column labelled *Sig.* is less than 0.001). This *p*-value tells us that there is less than a 0.1% chance that an *F*-statistic at least this large would happen if the null hypothesis were true. Therefore, we could conclude that our model results in significantly better prediction of album sales than if we used the mean value of album sales. In short, the linear model overall predicts album sales significantly.

9.8.2 Model parameters



Output 9.4 provides estimates of the model parameters (the beta values) and the significance of these values. We saw in equation (9.2) that b_0 was the *Y* intercept, and this value is 134.14 (*B* for the constant in Output 9.4). This value can be interpreted as meaning that when no money is spent on advertising (when $X = 0$), the model predicts that 134,140 albums will be sold (remember that our unit of measurement is thousands of albums). We can also read off the value of b_1 from the table, which is 0.096. Although this value is the slope of the line for the model, it is more useful to think of this value as representing *the change in*

the outcome associated with a unit change in the predictor. In other words, if our predictor variable is increased by one unit (if the advertising budget is increased by 1), then our model predicts that 0.096 extra albums will be sold. Our units of measurement were thousands of pounds and thousands of albums sold, so we can say that for an increase in advertising of £1000 the model predicts 96 (0.096 × 1000 = 96) extra album sales. This investment is pretty useless for the record company: it invests £1000 and gets only 96 extra sales! Fortunately, as we already know, advertising accounts for only one-third of album sales.



We saw earlier that if a predictor is having a significant impact on our ability to predict the outcome then its *b* should be different from 0 (and large relative to its standard error). We also saw that the *t*-test and associated *p*-value tell us whether the *b*-value is significantly different from 0. The column *Sig.* contains the exact probability that a value of *t* at least as big as the one in the table would occur if the value of *b* in the population were zero. If this probability is less than 0.05, then people interpret that as the predictor being a 'significant' predictor of the outcome (see Chapter 2). For both *t*s, the probabilities are given as 0.000 (zero to 3 decimal places), and so we can say that the probability of these *t* values (or larger) occurring if the values of *b* in the population were zero is less than 0.001. In other words, the *b*s are significantly different from 0. In the case of the *b* for advertising budget this result means that the advertising budget makes a significant contribution ($p < 0.001$) to predicting album sales.

Output 9.4

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	134.140	7.537		17.799	.000
	Advertising budget (thousands)	.096	.010	.578	9.979	.000

a. Dependent Variable: Album sales (thousands)

Bootstrap for Coefficients

Model		B	Bootstrap[a] Bias	Std. Error	Sig. (2-tailed)	BCa 95% Confidence Interval Lower	Upper
1	(Constant)	134.140	-.049	8.087	.001	118.699	150.746
	Advertising budget (thousands)	.096	.000	.008	.001	.079	.113

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

If our sample is one of the 95% producing confidence intervals that contain the population value then the bootstrap confidence interval tells us that the population value of *b* for advertising budget is likely to fall between 0.079 and 0.113, and because this interval doesn't include zero we might conclude that there is a genuine positive relationship between advertising budget and album sales in the population. Also, the significance associated with this confidence interval is $p = 0.001$, which is highly significant. Note that the bootstrap process involves re-estimating the standard error (it changes from 0.01 in the original table to a bootstrap estimate of 0.008). This is a very small change. The bootstrap confidence intervals and significance values are useful to report and interpret because they do not rely on assumptions of normality or homoscedasticity.



How is the *t* in Output 9.4 calculated? Use the values in the table to see if you can get the same value as SPSS.

9.8.3 Using the model



We have discovered that we have a model that significantly improves our ability to predict album sales. The next stage is often to use that model to make predictions about the future. First we specify the model by replacing the *b*-values in equation (9.2) with the values from Output 9.4. We can also replace the *X* and *Y* with the variable names:

$$\begin{aligned}\text{album sales}_i &= b_0 + b_1\text{advertising budget}_i \\ &= 134.14 + (0.096 \times \text{advertising budget}_i)\end{aligned} \tag{9.16}$$

We can make a prediction about album sales by replacing the advertising budget with a value of interest. For example, if we spend £100,000 on advertising a new album, remembering that our units are already in thousands of pounds, we simply replace the advertising budget with 100. We discover that album sales should be around 144,000 for the first week of sales:

$$\begin{aligned}\text{album sales}_i &= 134.14 + (0.096 \times \text{advertising budget}_i) \\ &= 134.14 + (0.096 \times 100) \\ &= 143.74\end{aligned} \tag{9.17}$$

Cramming Sam's Tips Linear models



- A linear model (regression) is a way of predicting values of one variable from another based on a model that describes a straight line.
- This line is the line that best summarizes the pattern of the data.
- To assess how well the model fits the data use:
  - R^2, which tells us how much variance is explained by the model compared to how much variance there is to explain in the first place. It is the proportion of variance in the outcome variable that is shared by the predictor variable.
  - *F*, which tells us how much variability the model can explain relative to how much it can't explain (i.e., it's the ratio of how good the model is compared to how bad it is).
  - the *b*-value, which tells us the gradient of the regression line and the strength of the relationship between a predictor and the outcome variable. If it is significant (*Sig.* < 0.05 in the SPSS output) then the predictor variable significantly predicts the outcome variable.





How many albums would be sold if we spent £666,000 on advertising the latest album by Deafheaven?

9.9 The linear model with two or more predictors (multiple regression) 

Imagine that the record company executive wanted to extend the model of albums sales to incorporate other predictors. Before an album is released, the executive notes the amount spent on advertising, the number of times songs from the album are played on a prominent radio station the week before release (**Airplay**), and ratings of the band's image (**Image**). He or she does this for 200 albums (each by a different band). The credibility of the band's image was rated by a random sample of the target audience on a scale from 0 (dad dancing at a disco) to 10 (sicker than a dog that's eaten a bag of onions). The mode rating was used because the executive was interested in what most people thought, not the average opinion.

When we build a model with several predictors, everything we have discussed so far applies. However, there are some additional things to think about. The first is what variables to enter into the model. A great deal of care should be taken in selecting predictors for a model because the estimates of the regression coefficients depend upon the variables in the model (and the order in which they are entered). *Do not enter hundreds of predictors, just because you've measured them, and expect the resulting model to make sense*. SPSS Statistics will happily generate output based on any garbage you decide to feed it – it will not judge you, but others will. Select predictors based on a sound theoretical rationale or well-conducted past research that has demonstrated their importance.[11] In our example, it seems logical that the band's image and radio play ought to affect sales, so these are sensible predictors. It would not be sensible to measure how much the album cost to make because this won't affect sales directly: you would just add noise to the model. If predictors are being added that have never been looked at before (in your research context) then select these variables based on their substantive *theoretical* importance. The key point is that the most important thing when building a model is to use your brain – which is slightly worrying if your brain is as small as mine.

11 Preferably past research that is methodologically and statistically rigorous and yielded reliable, generalizable models.

9.9.1 Methods of entering predictors into the model



Having chosen predictors, you must decide the order to enter them into the model. When predictors are completely uncorrelated the order of variable entry has very little effect on the parameters estimated; however, we rarely have uncorrelated predictors, and so the method of variable entry has consequences and is, therefore, important.

Other things being equal, use **hierarchical regression**, in which you select predictors based on past work and decide in which order to enter them into the model. Generally speaking, you should enter known predictors (from other research) into the model first in order of their importance in predicting the outcome. After having entered known predictors, you can add new predictors into the model simultaneously, in a stepwise manner, or hierarchically (entering the new predictor suspected to be the most important first).

An alternative is forced entry (or *Enter* as it is known in SPSS), in which you force all predictors into the model simultaneously. Like hierarchical, this method relies on good theoretical reasons for including the chosen predictors, but unlike hierarchical, you make no decision about the order in which variables are entered. Some researchers believe that this method is the only appropriate method for theory testing (Studenmund & Cassidy, 1987), because stepwise techniques are influenced by random variation in the data and so seldom give replicable results if the model is retested.

The final option, **stepwise regression**, is generally frowned upon by statisticians. Nevertheless, SPSS Statistics makes it easy to do and actively encourages it in the *Automatic Linear Modeling* process (probably because this function is aimed at people who don't know better) – see Oditi's Lantern. I'm assuming that you wouldn't wade through 900 pages of my drivel unless you wanted to know better, so we'll give stepwise a wide berth. However, you probably ought to know what it does so you can understand why to avoid it. The stepwise method bases decisions about the order in which predictors enter the model on a purely mathematical criterion. In the *forward* method, an initial model is defined that contains only the constant (b_0). The computer then searches for the predictor (out of the ones available) that best predicts the

outcome variable – it does this by selecting the predictor that has the highest simple correlation with the outcome. If this predictor significantly improves the model's ability to predict the outcome then it is retained and the computer looks to add a second predictor from the available pool of variables. The next predictor the computer tries will be the one that has the largest semi-partial correlation with the outcome. Remember that the semi-partial correlation quantifies the unique overlap between two variables *X* and *Y*: it 'partials out' or accounts for the relationship that *X* has with other predictors. Therefore, the computer looks for the variable that has the largest *unique* overlap with the outcome. This variable is retained if it significantly improves the fit of the model, otherwise it is rejected and the process stops. If it is retained and there are still potential predictors left out of the model then these are reviewed and the one with the largest semi-partial correlation with the outcome is entered, evaluated and retained if it significantly improves the fit, and so on until there are no more potential predictors or none of the potential predictors significantly improves the model if it is entered.

Let's make this process a bit more concrete. In Section 8.5, we used an example of the relationships between exam performance, exam anxiety and revision time. Imagine our goal is to predict exam performance from the other two variables. Think back to Figure 8.10. If we build the model stepwise, the first step is to see which of exam anxiety and revision time overlaps more with exam performance. The area of overlap between exam performance and exam anxiety is area A + C (19.4%), whereas for revision time it is area B + C (15.8%). Therefore, exam anxiety will enter the model first and is retained only if it significantly improves the model's predictive power. If not, no predictor variables are entered.

In our exam performance example, there is only one other potential predictor (revision time), so this will be entered next. Remember that its unique overlap with exam performance is area B in Figure 8.10: we ignore the part of its overlap with exam performance that is shared with exam anxiety (area C), because that variable is already in the model. If area B is big enough to improve the fit of the model significantly then revision time will be retained. If not the final model will contain only exam anxiety.

In the case where we had another potential predictor (let's say we measured the difficulty of the exam) and exam anxiety had been entered first, then the unique overlap for revision time and exam performance (area B) would be compared to the equivalent area for exam difficulty. The variable with the bigger area would

be entered next, evaluated and retained only if its inclusion improved the fit of the model.

The *stepwise* method in SPSS Statistics is the same as the forward method, except that each time a predictor is added to the equation, a removal test is made of the least useful predictor. As such, the regression equation is constantly reassessed to see whether redundant predictors can be removed. The *backward* method is the opposite of the forward method in that the model initially contains all predictors and the contribution of each is evaluated with the *p*-value of its *t*-test. This significance value is compared against a removal criterion (which can be either an absolute value of the test statistic or a *p*-value). If a predictor meets the removal criterion (i.e., it is not making a statistically significant contribution to the model) it is removed and the model is re-estimated for the remaining predictors. The contribution of the remaining predictors is then reassessed.

Oditi's Lantern *Automatic Linear Modeling*



'I, Oditi, come with a warning. Your desparation to bring me answers to numerical truths so as to gain a privileged place within my heart may lead you into the temptation that is SPSS's *Automatic Linear Modeling*. This feature promises answers without thought, and like a cat who is promised a fresh salmon, you will drool and purr in anticipation. If you want to find out more then stare into my lantern, but be warned, sometimes what looks like a juicy salmon is a rotting pilchard in disguise.'



Which of these methods should you use? The short answer is 'not stepwise', because variables are selected based upon mathematical criteria. The issue is that these criteria (e.g., the semi-partial correlation) are at the mercy of sampling

variation. That is, a particular variable might have a large semi-partial correlation in your sample but a small one in a different sample. Therefore, models built using stepwise methods are less likely to generalize across samples because the selection of variables in the model is affected by the sampling process. Also, because the criterion for retaining variables is based on statistical significance, your sample size affects the model you get: in large samples significance tests are highly powered, resulting in predictors being retained that make trivial contributions to predicting the outcome, and in small samples where power is low, predictors that make a large contribution may get overlooked. Consequently, there is the danger of overfitting (having too many variables in the model that essentially make little contribution to predicting the outcome) and underfitting (leaving out important predictors) the model. Stepwise methods also take important methodological decisions out of the hands of the researcher.



The main problem with stepwise methods is that they assess the fit of a variable based on the other variables in the model. Jeremy Miles (who has worked with me on other books) illustrates this problem by imagining getting dressed using a stepwise method. You wake up one morning and on your dressing table (or floor, if you're me) you have underwear, some jeans, a T-shirt and jacket. Imagine these items are predictor variables. It's a cold day and you're trying to keep warm. A stepwise method will put your trousers on first because they fit your goal best. It then looks around and tries the other clothes (variables). It tries to put your underwear on you but it won't fit over your jeans. It decides they are 'a poor fit' and discards them. It tries a jacket – that fits, but your T-shirt doesn't go over the top and is discarded. You end up leaving the house in jeans and a jacket with nothing underneath. You are very cold. Later in the day during a university seminar you stand up and your trousers fall down (because your body has shrunk from the cold), exposing you to your year group. It's a mess. The problem is that the underwear was a poor fit only because when you tried to put it on you were already wearing jeans. In stepwise methods, variables might be considered bad predictors only because of what has already been put in the model.

For these reasons, stepwise methods are best avoided except for exploratory

model building. If you do decide to use a stepwise method then let the statistical blood be on your hands, not mine. Use the backward method rather than the forward method to minimize **suppressor effects**, which occur when a predictor has a significant effect only when another variable is held constant. Forward selection is more likely than backward elimination to exclude predictors involved in suppressor effects. As such, the forward method runs a higher risk of making a Type II error (i.e., missing a predictor that does in fact predict the outcome). It is also advisable to cross-validate your model by splitting the data (see Section 9.4.2).

9.9.2 Comparing models 

Hierarchical and (although obviously you'd never use them) stepwise methods involve adding predictors to the model in stages, and it is useful to assess the improvement to the model at each stage. Given that larger values of R^2 indicate better fit, a simple way to quantify the improvement when predictors are added is to compare the R^2 for the new model to that for the old model. We can assess the significance of the change in R^2 using equation (9.13), but because we're looking at the change in models we use the change in R^2 (R^2_{change}) and the change in the number of predictors (k_{change}), as well as the R^2 (R^2_{new}) and number of predictors (k_{new}) in the new model:

$$F_{\text{change}} = \frac{(N - k_{\text{new}} - 1)R^2_{\text{change}}}{k_{\text{change}}(1 - R^2_{\text{new}})} \tag{9.18}$$

We can compare models using this F-statistic. The problem with R^2 is that when you add more variables to the model, it always goes up. So, if you are deciding which of two models fits the data better, the model with more predictor variables in will always fit better. The **Akaike information criterion (AIC)**[12] is a measure of fit that penalizes the model for having more variables. If the AIC is bigger, the fit is worse; if the AIC is smaller, the fit is better. If you use the *Automatic Linear Model* function in SPSS Statistics then you can use the AIC to select models rather than the change in R^2. The AIC doesn't mean anything on its own: you cannot say that an AIC value of 10 is small, or that a value of 1000 is large. The only thing you do with the AIC is compare it to other models with the same outcome variable: if it's getting smaller, the fit of your model is

improving.

12 Hirotsugu Akaike (pronounced ‘A-ka-ee-kay’) was a Japanese statistician who gave his name to the AIC, which is used in a huge range of different places.

9.9.3 Multicollinearity 

A final consideration for models with more than one predictor is multicollinearity, which exists when there is a strong correlation between two or more predictors. **Perfect collinearity** exists when at least one predictor is a perfect linear combination of the others (the simplest example being two predictors that are perfectly correlated – they have a correlation coefficient of 1). If there is perfect collinearity between predictors it becomes impossible to obtain unique estimates of the regression coefficients because there are an infinite number of combinations of coefficients that would work equally well. Put simply, if we have two predictors that are perfectly correlated, then the values of *b* for each variable are interchangeable. The good news is that perfect collinearity is rare in real-life data. The bad news is that less than perfect collinearity is virtually unavoidable. Low levels of collinearity pose little threat to the model estimates, but as collinearity increases there are three problems that arise:

- **Untrustworthy *b*s**: As collinearity increases, so do the standard errors of the *b* coefficients. Big standard errors for *b* coefficients mean more variability in these *b*s across samples, and a greater chance of (1) predictor equations that are unstable across samples too; and (2) *b* coefficients in the sample that are unrepresentative of those in the population. Crudely put, multicollinearity leads to untrustworthy *b*-values. Don’t lend them money and don’t let them go for dinner with your boy-or girlfriend.
- **It limits the size of *R***: Remember that *R* is a measure of the correlation between the predicted values of the outcome and the observed values and that R^2 indicates the variance in the outcome for which the model accounts. Imagine a situation in which a single variable predicts the outcome variable with $R = 0.80$ and a second predictor variable is added to the model. This second variable might account for a lot of the variance in the outcome (which is why it is included in the model), but the variance it accounts for is the same variance accounted for by the first variable (the second variable

accounts for very little *unique* variance). Hence, the overall variance in the outcome accounted for by the two predictors is little more than when only one predictor is used (*R* might increase from 0.80 to 0.82). If, however, the two predictors are completely uncorrelated, then the second predictor is likely to account for *different* variance in the outcome than that accounted for by the first predictor. The second predictor might account for only a little of the variance in the outcome, but the variance it does account for is different from that of the other predictor (and so when both predictors are included, *R* is substantially larger, say 0.95).

- **Importance of predictors**: Multicollinearity between predictors makes it difficult to assess the individual importance of a predictor. If the predictors are highly correlated, and each accounts for similar variance in the outcome, then how can we know which of the two variables is important? We can't – the model could include either one, interchangeably.

A 'ball park' method of identifying multicollinearity (that will miss subtler forms) is to scan the correlation matrix for predictor variables that correlate very highly (values of *r* above 0.80 or 0.90). SPSS Statistics can compute the **variance inflation factor (VIF)**, which indicates whether a predictor has a strong linear relationship with the other predictor(s), and the **tolerance** statistic, which is its reciprocal (1/VIF). Some general guidelines have been suggested for interpreting the VIF:

- If the largest VIF is greater than 10 (or the tolerance is below 0.1) then this indicates a serious problem (Bowerman & O'Connell, 1990; Myers, 1990).
- If the average VIF is substantially greater than 1 then the regression may be biased (Bowerman & O'Connell, 1990).
- Tolerance below 0.2 indicates a potential problem (Menard, 1995).

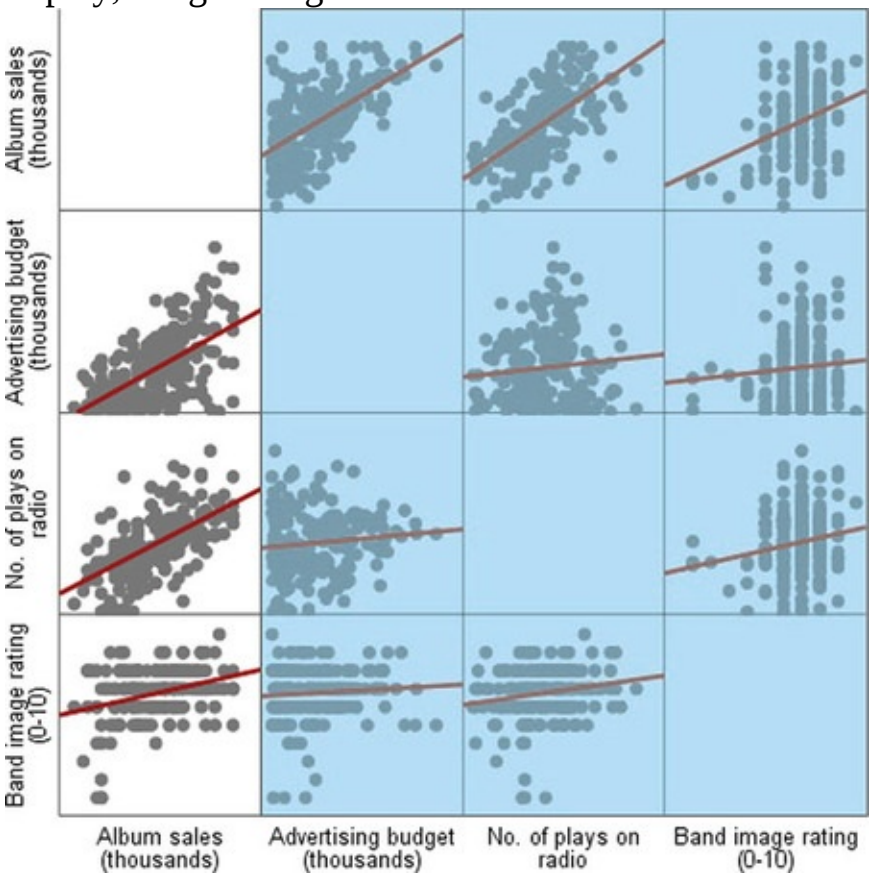
Other measures that are useful in discovering whether predictors are dependent are the *eigenvalues of the scaled, uncentred cross-products matrix*, the *condition indexes* and the *variance proportions*. These statistics will be covered as part of the interpretation of SPSS output (see Section 9.11.5). If none of this made sense, Hutcheson and Sofroniou (1999) explain multicollinearity very clearly.

9.10 Using SPSS Statistics to fit a linear model with

several predictors 

Remember the general procedure in Figure 9.10. First, we could look at scatterplots of the relationships between the outcome variable and the predictors. Figure 9.14 shows a matrix of scatterplots for our album sales data, but I have shaded all the scatterplots except the three related to the outcome, album sales. Although the data are messy, the three predictors have reasonably linear relationships with the album sales and there are no obvious outliers (except maybe in the bottom left of the scatterplot with band image).

Figure 9.14 Matrix scatterplot of the relationships between advertising budget, airplay, image rating and album sales



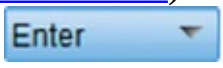


Produce a matrix scatterplot of **Sales, Adverts, Airplay** and **Image** including the regression

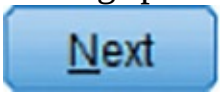
line.

9.10.1 Main options 

Past research shows that advertising budget is a significant predictor of album sales, and so we should include this variable in the model first, entering the new variables (**Airplay** and **Image**) afterwards. This method is hierarchical (we decide the order that variables are entered based on past research). To do a hierarchical regression we enter the predictors in blocks, with each block representing one step in the hierarchy. Access the main *Linear Regression* dialog box by selecting *Analyze*  *Regression* 

. We encountered this dialog box when we looked at a model with only one predictor (Figure 9.13). To set up the first block we do what we did before: drag **Sales** to the box labelled *Dependent* (or click ). We also need to specify the predictor variable for the first block, which we decided should be advertising budget. Drag this variable from the left-hand list to the box labelled *Independent(s)* (or click ). Underneath the *Independent(s)* box is a drop-down menu for specifying the *Method* of variable entry (see Section 9.9.1). You can select a different method for each block by clicking on 

. The default option is forced entry, and this is the option we want, but if you were carrying out more exploratory work, you might use a different method.

Having specified the first block, we can specify a second by clicking 

. This process clears the *Independent(s)* box so that you can enter the new predictors (note that it now reads *Block 2 of 2* above this box to indicate

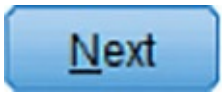
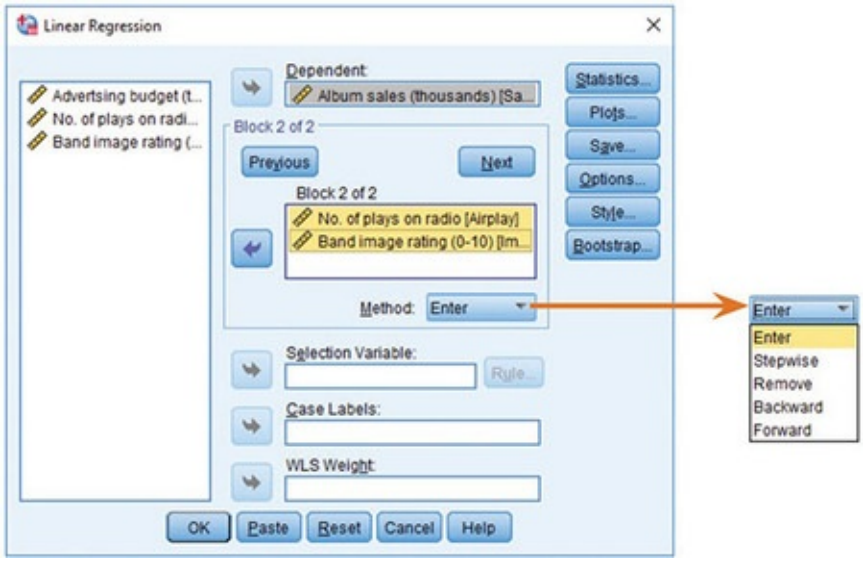
that you are in the second block of the two that you have so far specified). We decided that the second block would contain both of the new predictors, so select **Airplay** and **Image** and drag them to the *Independent(s)* box (or click ). The dialog box should look like Figure 9.15. To move between blocks use the  and  buttons (e.g., to move back to block 1, click ).

Figure 9.15 Main dialog box for block 2 of the multiple regression



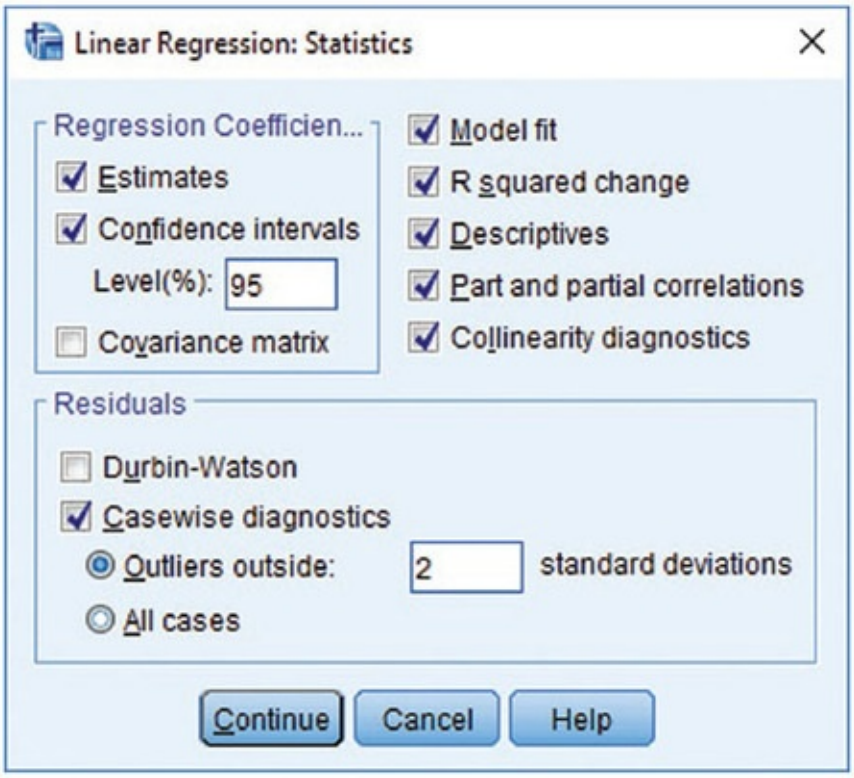
It is possible to select different methods of variable entry for different blocks. For example, having specified forced entry for the first block, we could now specify a stepwise method for the second. Given that we have no previous research regarding the effects of image and airplay on album sales, we might be justified in doing this. However, because of the problems with stepwise methods, I am going to stick with forced entry for both blocks.

9.10.2 Statistics 

In the main *Regression* dialog box click Statistics... to open the dialog box in Figure 9.16. Below is a run-down of the options available. Select the options you require and click Continue to return to the main dialog box.

- *Estimates*: This option is selected by default because it gives us the estimated *b*-values for the model as well as the associated *t*-test and *p*-value (see Section 9.2.5).
- *Confidence intervals*: This option produces confidence intervals for each *b*-value in the model. Remember that if the model assumptions are not met these confidence intervals will be inaccurate and bootstrap confidence intervals should be used instead.
- *Covariance matrix*: This option produces a matrix of the covariances, correlation coefficients and variances between the *b*-values for each variable in the model. A variance–covariance matrix displays variances along the diagonal and covariances as off-diagonal elements. Correlations are produced in a separate matrix.
- *Model fit*: This option produces the *F*-test, *R*, R^2 and the adjusted R^2 (described in Sections 9.2.4. and 9.4.2).
- *R squared change*: This option displays the change in R^2 resulting from the inclusion of a new predictor (or block of predictors) – see Section 9.9.2.
- *Descriptives*: This option displays a table of the mean, standard deviation and number of observations of the variables included in the model. A correlation matrix is produced too, which can be helpful for spotting multicollinearity.

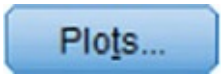
Figure 9.16 *Statistics* dialog box for regression analysis



- *Part and partial correlations*: This option produces the zero-order correlation (the Pearson correlation) between each predictor and the outcome variable. It also produces the semi-partial (part) and partial correlation between each predictor and the outcome (see Sections 8.5 and 9.9.1).
- *Collinearity diagnostics*: This option produces collinearity statistics such as the VIF, tolerance, eigenvalues of the scaled, uncentred cross-products matrix, condition indexes and variance proportions (see Section 9.9.3).
- *Durbin-Watson*: This option produces the Durbin–Watson test statistic, which tests the assumption of independent errors when cases have some meaningful sequence. In this case it isn't useful because our cases do not have a meaningful order.
- *Casewise diagnostics*: This option produces a table that lists the observed value of the outcome, the predicted value of the outcome, the difference between these values (the residual) and this difference standardized. You can choose to have this information for all cases, but that will result in a big table in large samples. The alternative option is to list only cases for which the standardized residual is greater than 3 (when the ± sign is ignored). I usually change this to 2 (so that I don't miss cases with standardized residuals not quite reaching the threshold of 3) A summary table of residual

statistics indicating the minimum, maximum, mean and standard deviation of both the values predicted by the model and the residuals is also produced (see Section 9.10.4).

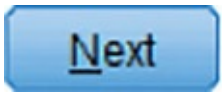
9.10.3 Regression plots 

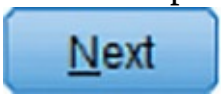
Once you are back in the main dialog box, click 

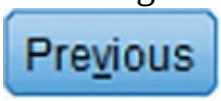
to activate the dialog box in Figure 9.17, which we can use to test some assumptions of the model. Most of these plots involve various *residual* values, which were described in Section 9.3. The left-hand side lists several variables:

- **DEPENDNT**: the outcome variable.
- ***ZPRED**: the standardized predicted values of the outcome based on the model. These values are standardized forms of the values predicted by the model.
- ***ZRESID**: the standardized residuals, or errors. These values are the standardized differences between the observed values of the outcome and those predicted by the model.
- ***DRESID**: the deleted residuals described in Section 9.3.2.
- ***ADJPRED**: the adjusted predicted values described in Section 9.3.2.
- ***SRESID**: the Studentized residual described in Section 9.3.1.
- ***SDRESID**: the Studentized deleted residual described in Section 9.3.2.

In Section 6.11.1 we saw that a plot of ***ZRESID** (*y*-axis) against ***ZPRED** (*x*-axis) is useful for testing the assumptions of independent errors, homoscedasticity, and linearity. A plot of ***SRESID** (*y*-axis) against ***ZPRED** (*x*-axis) will show up heteroscedasticity also. Although often these two plots are virtually identical, the latter is more sensitive on a case-by-case basis. To create these plots drag a variable from the list to the space labelled either *X* or *Y* (which refer to the axes), or select the variable and click  . When you have selected two variables for the first plot (as in Figure 9.17) you can specify a new

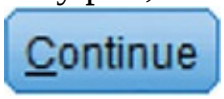
plot (up to nine different plots) by clicking on 

. This process clears the dialog box and you can specify a second plot. Click 

or 

to move between plots you have specified.

Ticking the box labelled *Produce all partial plots* will produce scatterplots of the residuals of the outcome variable and each of the predictors when both variables are regressed separately on the remaining predictors. Regardless of whether the previous sentence made any sense to you, these plots have important characteristics that make them worth inspecting. First, the gradient of the regression line between the two residual variables is equivalent to the coefficient of the predictor in the regression equation. As such, any obvious outliers on a partial plot represent cases that might have undue influence on a predictor's coefficient, *b*. Second, non-linear relationships between a predictor and the outcome variable are much more evident on these plots. Finally, they are useful for spotting collinearity.

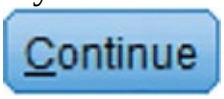
There are two other tick-boxes labelled *Standardized Residual Plots*. One produces a histogram of the standardized residuals and the other produces a normal probability plot, both of which are useful for checking for normality of errors). Click 

to return to the main dialog box.

9.10.4 Saving regression diagnostics 

Section 9.3 described numerous variables that we can use to diagnose outliers and influential cases. We can save these diagnostic variables for our model in the data editor (SPSS calculates them and places the values in new columns in the data editor) by clicking 

to access the dialog box in Figure 9.18. Most of the available options were explained in Section 9.3, and Figure 9.18 shows what I consider to be a reasonable set of diagnostic statistics. Standardized (and Studentized) versions of these diagnostics are generally easier to interpret, and so I tend to select them in preference to the unstandardized

versions. Once the model has been estimated, SPSS creates a column in your data editor for each statistic requested; it uses a standard set of variable names to describe each one. After the name, there will be a number that refers to the model from which they were generated. For example, for the first model fitted to a data set the variable names will be followed by a 1, if you estimate a second model it will create a new set of variables with names followed by a 2, and so on. For reference, the names used by SPSS are listed below. Selected the diagnostics you require and click 

to return to the main dialog box.

Figure 9.17 The *Plots* dialog box

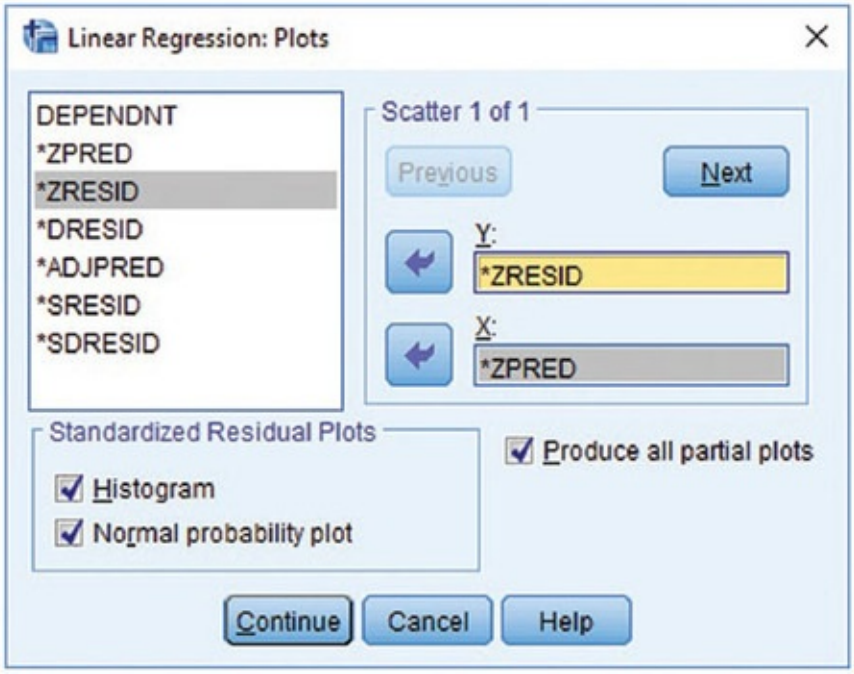
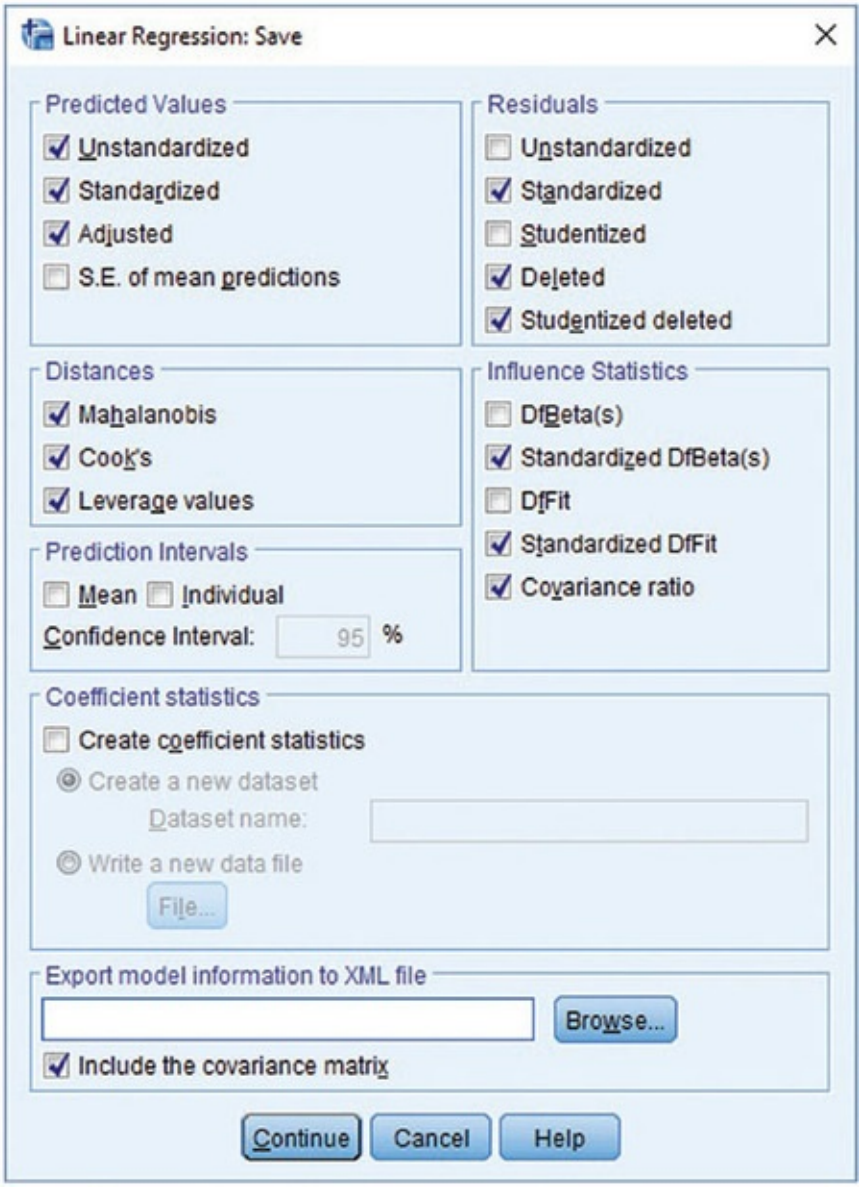


Figure 9.18 Dialog box for regression diagnostics



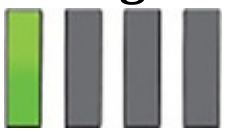
- **pre_1**: unstandardized predicted value;
- **zpr_1**: standardized predicted value;
- **adj_1**: adjusted predicted value;
- **sep_1**: standard error of predicted value;
- **res_1**: unstandardized residual;
- **zre_1**: standardized residual;
- **sre_1**: Studentized residual;
- **dre_1**: deleted residual;
- **sdr_1**: Studentized deleted residual;
- **mah_1**: Mahalanobis distance;
- **coo_1**: Cook's distance;

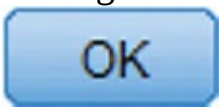
- **lev_1**: centred leverage value;
- **sdb0_1**: standardized DFBeta (intercept);
- **sdb1_1**: standardized DFBeta (predictor 1);
- **sdb2_1**: standardized DFBeta (predictor 2);
- **sdf_1**: standardized DFFit;
- **cov_1**: covariance ratio.

9.10.5 Further options 

Clicking 

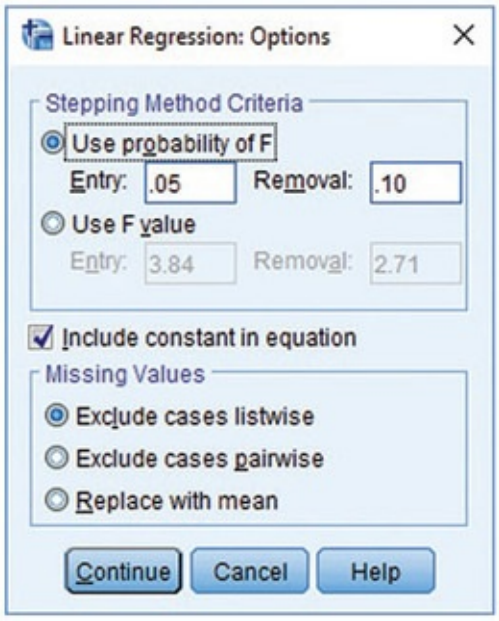
activates the dialog box in Figure 9.19. The first set of options allows you to change the criteria used for entering variables in a stepwise regression. If you insist on doing stepwise regression, then it's probably best that you leave the default criterion of 0.05 probability for entry alone. However, you can make this criterion more stringent (0.01). There is also the option to build a model that doesn't include a constant (i.e., has no *Y* intercept). This option should also be left alone. Finally, you can select a method for dealing with missing data points (see SPSS Tip 6.1 for a description). Just a hint, but leave the default of listwise alone because using pairwise can lead to absurdities such as R^2 that is negative or greater than 1.0.

9.11 Interpreting a linear model with several predictors 

Having selected the relevant options and returned to the main dialog box, click 

and watch in awe as SPSS Statistics spews forth quite terrifying amounts of output in the viewer window.

Figure 9.19 Options for linear regression



Oditi's Lantern The linear model



'I, Oditi, wish to predict when I can take over the world, and rule you pathetic mortals with a will of pure iron … erm .. ahem, I mean, I wish to predict how to save cute kittens from the jaws of rabid dogs, because I'm nice like that, and have no aspirations to take over the world. This chapter is so long that some of you will die before you reach the end, so ignore the author's bumbling drivel and stare instead into my lantern of wonderment.'



9.11.1 Descriptives

The output described in this section is produced using the options in Figure 9.16. If you selected the *Descriptives* option, you'll get Output 9.5, which tells us the mean and standard deviation of each variable in our model. This table is a useful summary of the variables. You'll also see a correlation matrix containing the Pearson correlation coefficient between every pair of variables, the one-tailed significance of each correlation, and the number of cases contributing to each correlation. Along the diagonal of the matrix the values for the correlation coefficients are all 1.00 (a perfect positive correlation) because they are the correlation of each variable with itself.

We can use the correlation matrix to get a sense of the relationships between predictors and the outcome, and for a preliminary look for multicollinearity. If there is no multicollinearity in the data then there should be no substantial correlations ($r > 0.9$) between predictors. If we look only at the predictors (ignore album sales) then the highest correlation is between the ratings of the band's image and the amount of airplay, which is significant at the 0.01 level ($r = 0.182$, $p = 0.005$). Despite the significance, the coefficient itself is small and so there is no collinearity to worry about. If we look at the outcome variable, then it's apparent that of the predictors airplay correlates best with the outcome ($r = 0.599$, $p < 0.001$).

9.11.2 Summary of the model 

Output 9.6 describes the overall fit of the model. There are two models in the table because we chose a hierarchical method with two blocks and the summary statistics are repeated for each model/block. Model 1 refers to the first stage in the hierarchy when only advertising budget is used as a predictor. Model 2 refers to when all three predictors are used. We can tell this from the footnotes under the table. If you selected the *R squared change* and *Durbin-Watson* options, then these values are included also (we didn't select Durbin–Watson so it is missing from Output 9.6).

Output 9.5

Descriptive Statistics

	Mean	Std. Deviation	N
Album sales (thousands)	193.20	80.699	200
Advertising budget (thousands)	614.41	485.65521	200
No. of plays on radio	27.50	12.270	200
Band image rating (0–10)	6.77	1.395	200

Correlations

		Album sales (thousands)	Advertising budget (thousands)	No. of plays on radio	Band image rating (0–10)
Pearson Correlation	Album sales (thousands)	1.000	.578	.599	.326
	Advertising budget (thousands)	.578	1.000	.102	.081
	No. of plays on radio	.599	.102	1.000	.182
	Band image rating (0–10)	.326	.081	.182	1.000
Sig. (1-tailed)	Album sales (thousands)	.	.000	.000	.000
	Advertising budget (thousands)	.000	.	.076	.128
	No. of plays on radio	.000	.076	.	.005
	Band image rating (0–10)	.000	.128	.005	.
N	Album sales (thousands)	200	200	200	200
	Advertising budget (thousands)	200	200	200	200
	No. of plays on radio	200	200	200	200
	Band image rating (0–10)	200	200	200	200

Cramming Sam's Tips Descriptive statistics



- Use the descriptive statistics to check the correlation matrix for multicollinearity; that is, predictors that correlate too highly with each other, $r > 0.9$.



The column labelled *R* contains the multiple correlation coefficient between the predictors and the outcome. When only advertising budget is used as a predictor, this is the simple correlation between advertising and album sales (0.578). In

fact, all of the statistics for model 1 are the same as the simple regression model earlier (see Section 9.8). The next column gives us a value of R^2, which we already know is a measure of how much of the variability in the outcome is accounted for by the predictors. For the first model its value is 0.335, which means that advertising budget accounts for 33.5% of the variation in album sales. However, when the other two predictors are included as well (model 2), this value increases to 0.665 or 66.5% of the variance in album sales. If advertising accounts for 33.5%, then image and airplay must account for an additional 33%.[13]

13 That is, 33% = 66.5% − 33.5% (this value is the *R Square Change* in the table).

Output 9.6

Model Summary[c]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.578[a]	.335	.331	65.991	.335	99.587	1	198	.000
2	.815[b]	.665	.660	47.087	.330	96.447	2	196	.000

a. Predictors: (Constant), Advertising budget (thousands)

b. Predictors: (Constant), Advertising budget (thousands), Band image rating (0–10), No. of plays on radio

c. Dependent Variable: Album sales (thousands)

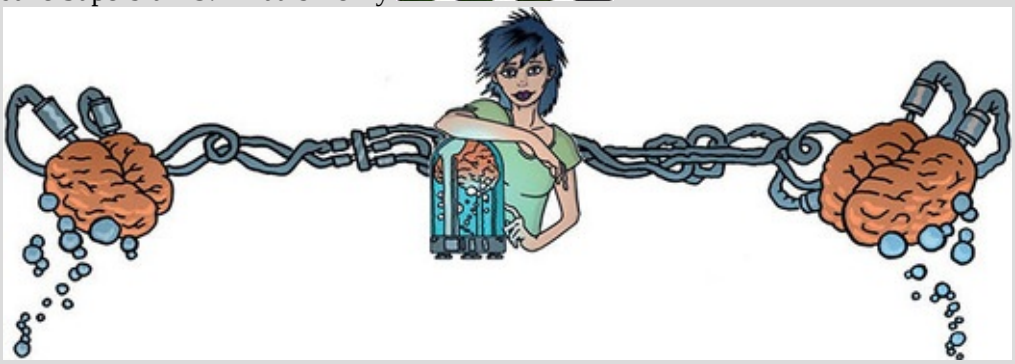
The adjusted R^2 gives us some idea of how well our model generalizes, and ideally we'd like its value to be the same as, or very close to, the value of R^2. In this example the difference for the final model is small (it is 0.665 − 0.660 = 0.005 or about 0.5%). This shrinkage means that if the model were derived from the population rather than a sample it would account for approximately 0.5% less variance in the outcome. If you apply Stein's formula (equation (9.15)) you'll get an adjusted value of 0.653 (Jane Superbrain Box 9.2), which is very close to the observed value of R^2 (0.665), indicating that the cross-validity of this model is very good.

The change statistics are provided only if requested and they tell us whether the change in R^2 is significant (i.e., how much does the model fit improve as predictors are added?). The change is reported for each block of the hierarchy: for model 1, R^2 changes from 0 to 0.335 and gives rise to an F-statistic of 99.59, which is significant with a probability less than 0.001. In model 2, in which image and airplay have been added as predictors, R^2 increases by 0.330, making the R^2 of the new model 0.665 with a significant ($p < 0.001$) F-statistic of 96.44

(Jane Superbrain Box 9.2).

Output 9.7 shows the *F*-test of whether the model is significantly better at predicting the outcome than using the mean outcome (i.e., no predictors). The *F*-statistic represents the ratio of the improvement in prediction that results from fitting the model, relative to the inaccuracy that still exists in the model (see Section 9.2.4). This table again reports the information for each model separately. The table contains the sum of squares for the model (the value of SS_M from Section 9.2.4), the residual sum of squares (the value of SS_R from Section 9.2.4) and their respective degrees of freedom. For SS_M the *df* are the number of predictors (1 for the first model and 3 for the second). For SS_R the *df* are the number of observations (200) minus the number of coefficients in the regression model. The first model has two coefficients (one each for the predictor and constant) whereas the second has four (the constant plus one for each of the three predictors). Therefore, model 1 has 198 residual degrees of freedom whereas model 2 has 196. Remember, the mean sum of squares (MS) is the SS divided by the *df* and the *F*-statistic is average improvement in prediction by the model (MS_M) divided by the average error in prediction (MS_R). The *p*-value tells us the probability of getting an *F* at least as large as the one we have if the null hypothesis were true (if we used the outcome mean to predict album sales). The *F*-statistic is 99.59, $p < 0.001$ for the initial model and 129.498, $p < 0.001$ for the second. We can interpret these results as meaning that both models significantly improved our ability to predict the outcome variable compared to not fitting the model.

Jane Superbrain 9.2 Maths frenzy 



We can look at how some of the values in the output are computed by thinking back to the

theory part of the chapter. For example, looking at the change in R^2 for the first model, we have only one predictor (so $k = 1$) and 200 cases ($N = 200$), so the F comes from equation (9.13):[14]

[14] To get the same values as SPSS we have to use the exact value of R^2, which is 0.3346480676231 (if you don't believe me, double-click in the table in the SPSS output that reports this value, then double-click the cell of the table containing the value of R^2 and you'll see that what was 0.335 now has a lot more decimal places).

$$F_{\text{Model 1}} = \frac{(200-1-1)0.334648}{1(1-0.334648)} = 99.59$$

In model 2 in Output 9.6, two predictors have been added (image and airplay), so the new model has 3 predictors (k_{new}) and the previous model had only 1, which is a change of 2 (k_{change}). The addition of these two predictors increases R^2 by 0.330 (R^2_{change}), making the R^2 of the new model 0.665 (R^2_{new}).[15] The F-statistic for this change comes from equation (9.18):

[15] The more precise value is 0.664668.

$$F_{\text{change}} = \frac{(N-3-1)0.33}{2(1-0.664668)} = 96.44$$

We can apply Stein's formula (equation (9.15)) to R^2 to get an idea of its likely value in different samples. We replace n with the sample size (200) and k with the number of predictors (3):

$$\begin{aligned}\text{adjusted } R^2 &= 1-\left[\left(\frac{n-1}{n-k-1}\right)\left(\frac{n-2}{n-k-2}\right)\left(\frac{n+1}{n}\right)\right]\left(1-R^2\right)\\ &= 1-\left[(1.015)(1.015)(1.005)\right](0.335)\\ &= 1-0.347\\ &= 0.653\end{aligned}$$



Output 9.7

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	433687.833	1	433687.833	99.587	.000[b]
	Residual	862264.168	198	4354.870		
	Total	1295952.00	199			
2	Regression	861377.418	3	287125.806	129.498	.000[c]
	Residual	434574.582	196	2217.217		
	Total	1295952.00	199			

a. Dependent Variable: Album sales (thousands)

b. Predictors: (Constant), Advertising budget (thousands)

c. Predictors: (Constant), Advertising budget (thousands), Band image rating (0–10), No. of plays on radio

Cramming Sam's Tips The model summary



- The fit of the linear model can be assessed using the *Model Summary* and *ANOVA* tables from SPSS.
- R^2 tells you the proportion of variance explained by the model.
- If you have done a hierarchical regression, assess the improvement of the model at each stage by looking at the change in R^2 and whether it is significant (values less than 0.05 in the column labelled *Sig. F Change*).
- The *F*-test tells us whether the model is a significant fit to the data overall (look for values less than 0.05 in the column labelled *Sig.*).



9.11.3 Model parameters

Output 9.8 shows the model parameters for both steps in the hierarchy. The first

step in our hierarchy was to include advertising budget, and so the parameters for this first model are identical to those we obtained earlier in this chapter in Output 9.4. Therefore, we will focus on the parameters for the final model (in which all predictors were included). The format of the table of coefficients depends on the options selected in Figure 9.16; for example, the confidence intervals *b*, collinearity diagnostics and the part and partial correlations will be present only if you checked those options.

Earlier in the chapter we saw that a linear model with several predictors takes the form of equation (9.8), which contains several unknown parameters (the *b*-values). The first column in Output 9.8 contains estimates for these *b*-values, which indicate the individual contribution of each predictor to the model. By replacing the *X*s in equation (9.8) with variables names and taking the *b*-values from Output 9.8 we can define our specific model as:

$$\begin{aligned}\text{sales}_i &= b_0 + b_1\text{advertising}_i + b_2\text{airplay}_i + b_3\text{image}_i \\ &= -26.61 + (0.085\ \text{advertising}_i) + (3.367\ \text{airplay}_i) + (11.086\ \text{image}_i)\end{aligned} \tag{9.19}$$

The *b*-values quantify the relationship between album sales and each predictor. The *direction* of the coefficient – positive or negative – corresponds to whether the relationship with the outcome is positive or negative. All three predictors have positive *b*-values, indicating positive relationships. So, as advertising budget, plays on the radio, and image rating increase so do album sales. The *size* of the *b* indicates the degree to which each predictor affects the outcome *if the effects of all other predictors are held constant*:

- **Advertising budget**: *b* = 0.085 indicates that as advertising budget increases by one unit, album sales increase by 0.085 units. Both variables were measured in thousands; therefore, for every £1000 more spent on advertising, an extra 0.085 thousand albums (85 albums) are sold. This interpretation is true only if the effects of band image and airplay are held constant.
- **Airplay**: *b* = 3.367 indicates that as the number of plays on radio in the week before release increases by one, album sales increase by 3.367 units. Every additional play of a song on radio (in the week before release) is associated with an extra 3.367 thousand albums (3367 albums) being sold. This interpretation is true only if the effects of the band's image and advertising budget are held constant.
- **Image**: *b* = 11.086 indicates that if a band can increase their image rating by 1 unit they can expect additional album sales of 11.086 units. Every unit

increase in the band's image rating is associated with an extra 11.086 thousand albums (11,086 albums) being sold. This interpretation is true only if the effects of airplay and advertising are held constant.

Output 9.8[16]

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			95.0% Confidence Interval for B	
		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound
1	(Constant)	134.140	7.537		17.799	.000	119.278	149.002
	Advertising budget (thousands)	.096	.010	.578	9.979	.000	.077	.115
2	(Constant)	-26.613	17.350		-1.534	.127	-60.830	7.604
	Advertising budget (thousands)	.085	.007	.511	12.261	.000	.071	.099
	No. of plays on radio	3.367	.278	.512	12.123	.000	2.820	3.915
	Band image rating (0–10)	11.086	2.438	.192	4.548	.000	6.279	15.894

a. Dependent Variable: Album sales (thousands)

Coefficients[a]

Model		Correlations			Collinearity Statistics	
		Zero-order	Partial	Part	Tolerance	VIF
1	Advertising budget (thousands)	.578	.578	.578	1.000	1.000
2	Advertising budget (thousands)	.578	.659	.507	.986	1.015
	No. of plays on radio	.599	.655	.501	.959	1.043
	Band image rating (0–10)	.326	.309	.188	.963	1.038

a. Dependent Variable: Album sales (thousands)

16 To spare your eyesight I have split this part of the output into two tables; however, it should appear as one long table.

Each of the beta values has an associated standard error indicating to what extent these values vary across different samples. The standard errors are also used to compute a *t*-statistic that tests whether the *b*-value is significantly different from 0 (Section 9.2.5). Remember that if a predictor's *b* is zero then its relationship to the outcome is zero also. By testing whether an observed *b* is significantly different from zero, we're testing whether the relationship between the predictor and outcome is different from zero. The *p*-value associated with a *b*'s *t*-statistic (in the column *Sig.*) is the probability of getting a *t* at least as big as the one we have if the population value of *b* was zero (i.e., if there was no relationship between that predictor and the outcome).

For this model, the advertising budget, $t(196) = 12.26$, $p < 0.001$, the amount of radio play prior to release, $t(196) = 12.12$, $p < 0.001$ and band image, $t(196) = 4.55$, $p < 0.001$, are all significant predictors of album sales.[17] Remember that these significance tests are accurate only if the assumptions discussed in Chapter 6 are met. From the magnitude of the *t*-statistics we can see that the advertising budget and radio play had a similar impact, whereas the band's image had less impact.

17 For all of these predictors I wrote $t(196)$. The number in brackets is the degrees of freedom. We saw in Section 9.2.5 that the degrees of freedom are $N - k - 1$, where N is the total sample size (in this case 200) and k is the number of predictors (in this case 3). For these data we get 200 – 3 – 1 = 196.

The standardized versions of the *b*-values are sometimes easier to interpret (because they are not dependent on the units of measurement of the variables). The standardized beta values (in the column labelled *Beta*, β_i) tell us the number of standard deviations that the outcome changes when the predictor changes by one standard deviation. Because the standardized beta values are measured in standard deviation units they are directly comparable: the values for airplay and advertising budget are virtually identical (0.512 and 0.511, respectively), suggesting that both variables have a comparably large effect, whereas image (standardized beta of 0.192) has a relatively smaller effect (this concurs with what the magnitude of the *t*-statistics told us). To interpret these values literally, we need to know the standard deviations of the variables, and these values can be found in Output 9.5.

- **Advertising budget**: Standardized $\beta = 0.511$ indicates that as advertising budget increases by one standard deviation (£485,655), album sales increase by 0.511 standard deviations. The standard deviation for album sales is 80,699, so this constitutes a change of 41,240 sales (0.511 × 80,699). Therefore, for every £485,655 more spent on advertising, an extra 41,240 albums are sold. This interpretation is true only if the effects of the band's image and airplay are held constant.
- **Airplay**: Standardized $\beta = 0.512$ indicates that as the number of plays on radio in the week before release increases by 1 standard deviation (12.27), album sales increase by 0.512 standard deviations. The standard deviation for album sales is 80,699, so this is a change of 41,320 sales (0.512 × 80,699). Basically, if the station plays the song an extra 12.27 times in the week before release, 41,320 extra album sales can be expected. This interpretation is true only if the effects of the band's image and advertising are held constant.
- **Image**: Standardized $\beta = 0.192$ indicates that a band rated one standard deviation (1.40 units) higher on the image scale can expect additional album sales of 0.192 standard deviations units. This is a change of 15,490 sales (0.192 × 80,699). A band with an image rating 1.40 higher than another band can expect 15,490 additional sales. This interpretation is true only if the effects of airplay and advertising are held constant.



Think back to what the confidence interval of the mean represented (Section 2.8). Can you work out what the confidence intervals for *b* represent?

Output 9.8 also contains the confidence intervals for the *b*s (again these are accurate only if the assumptions discussed in Chapter 6 are met). A bit of revision. Imagine that we collected 100 samples of data measuring the same variables as our current model. For each sample we estimate the same model that we have in this chapter, including confidence intervals for the unstandardized beta values. These boundaries are constructed such that in 95% of samples they contain the population value of *b* (see Section 2.8). Therefore, 95 of our 100 samples will yield confidence intervals for *b* that contain the population value. The trouble is that we don't know if our sample is one of the 95% with confidence intervals containing the population values or one of the 5% that misses.

The typical pragmatic solution to this problem is to assume that your sample is one of the 95% that hits the population value. If you assume this, then you can reasonably interpret the confidence interval as providing information about the population value of *b*. A narrow confidence interval suggests that all samples would yield estimates of *b* that are fairly close to the population value, whereas wide intervals suggest a lot of uncertainty about what the population value of *b* might be. If the interval contains zero then it suggests that the population value of *b* might be zero – in other words, no relationship between that predictor and the outcome – and could be positive but might be negative. All of these statements are reasonable if you're prepared to believe that your sample is one of the 95% for which the intervals contain the population value. Your belief will be wrong 5% of the time, though.

In our model of album sales, the two best predictors (advertising and airplay) have very tight confidence intervals, indicating that the estimates for the current model are likely to be representative of the true population values. The interval for the band's image is wider (but still does not cross zero), indicating that the parameter for this variable is less representative, but nevertheless significant.

If you asked for part and partial correlations, then they appear in separate

columns of the table. The zero-order correlations are the Pearson's correlation coefficients and correspond to the values in Output 9.5. Semi-partial (part) and partial correlations were described in Section 8.5; in effect, the part correlations quantify the unique relationship that each predictor has with the outcome. If you opted to do a stepwise regression, you would find that variable entry is based initially on the variable with the largest zero-order correlation and then on the part correlations of the remaining variables. Therefore, airplay would be entered first (because it has the largest zero-order correlation), then advertising budget (because its part correlation is bigger than that of image rating) and then finally the band's image rating – try running a forward stepwise regression on these data to see if I'm right. Finally, Output 9.8 contains collinearity statistics, but we'll discuss these in Section 9.11.5.

9.11.4 Excluded variables 

At each stage of fitting a linear model a summary is provided of predictors that are not yet in the model.

We had a two-block hierarchy with one predictor entered (and two excluded) in block 1, and three predictors entered (and none excluded) in block 2. Output 9.9 details excluded variables only for the first block of our hierarchy, because in the second block no predictors were excluded. The table includes an estimate of the *b*-value and associated *t*-statistic for each predictor *if* it entered the model at this point. Using a stepwise method, the predictor with the highest *t*-statistic will enter the model next, and predictors will continue to be entered until there are none left with *t*-statistics that have significance values less than 0.05. The partial correlation also indicates what contribution (if any) an excluded predictor would make if it entered the model.

Output 9.9

Excluded Variables[a]

Model		Beta In	t	Sig.	Partial Correlation	Collinearity Statistics: Tolerance	Collinearity Statistics: VIF	Collinearity Statistics: Minimum Tolerance
1	No. of plays on radio	.546[b]	12.513	.000	.665	.990	1.010	.990
	Band image rating (0–10)	.281[b]	5.136	.000	.344	.993	1.007	.993

a. Dependent Variable: Album sales (thousands)

b. Predictors in the Model: (Constant), Advertising budget (thousands)

Cramming Sam's Tips Coefficients



- The individual contribution of variables to the regression model can be found in the *Coefficients* table. If you have done a hierarchical regression then look at the values for the final model.
- You can see whether each predictor variable has made a significant contribution to predicting the outcome by looking at the column labelled *Sig.* (values less than 0.05 are significant).
- The standardized beta values tell you the importance of each predictor (bigger absolute value = more important).
- The tolerance and VIF values will also come in handy later, so make a note of them.



9.11.5 Assessing multicollinearity

I promised to come back to the measures of collinearity in Output 9.8, so here we go. The output contains the VIF and tolerance statistics (with tolerance being 1 divided by the VIF), and we need to apply the guidelines from Section 9.9.3. The VIF values are well below 10 and the tolerance statistics are well above 0.2. The average VIF, obtained by adding the VIF values for each predictor and dividing by the number of predictors (k), is also very close to 1:

$$\overline{\text{VIF}} = \frac{\sum_{i=1}^{k} \text{VIF}_i}{k} = \frac{1.015 + 1.043 + 1.038}{3} = 1.032 \tag{9.20}$$

It seems unlikely, therefore, that we need to worry about collinearity among predictors.

The other information we get about collinearity is a table of eigenvalues of the scaled, uncentred cross-products matrix, condition indexes and variance proportions. I discuss collinearity and variance proportions at length in Section 20.8.2, so here I'll just give you the headline: look for large variance proportions on the same *small* eigenvalues (Jane Superbrain Box 9.3). Therefore, in Output 9.10 inspect the bottom few rows of the table (these are the small eigenvalues) and look for variables that *both* have high variance proportions for that eigenvalue. The variance proportions vary between 0 and 1, and you'd like to see each predictor having a high proportion on a different eigenvalue to other predictors (in other words, the large proportions are distributed across different eigenvalues). For our model, each predictor has most of its variance loading onto a different dimension than other predictors (advertising has 96% of variance on dimension 2, airplay has 93% of variance on dimension 3 and image rating has 92% of variance on dimension 4). These data represent no multicollinearity. For an example of when collinearity exists in the data and some suggestions about what can be done, see Chapters 20 (Section 20.8.2) and 18 (Section 18.3.3).

Output 9.10

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Advertising budget (thousands)	No. of plays on radio	Band image rating (0–10)
1	1	1.785	1.000	.11	.11		
	2	.215	2.883	.89	.89		
2	1	3.562	1.000	.00	.02	.01	.00
	2	.308	3.401	.01	.96	.05	.01
	3	.109	5.704	.05	.02	.93	.07
	4	.020	13.219	.94	.00	.00	.92

a. Dependent Variable: Album sales (thousands)

Cramming Sam's Tips Multicollinearity

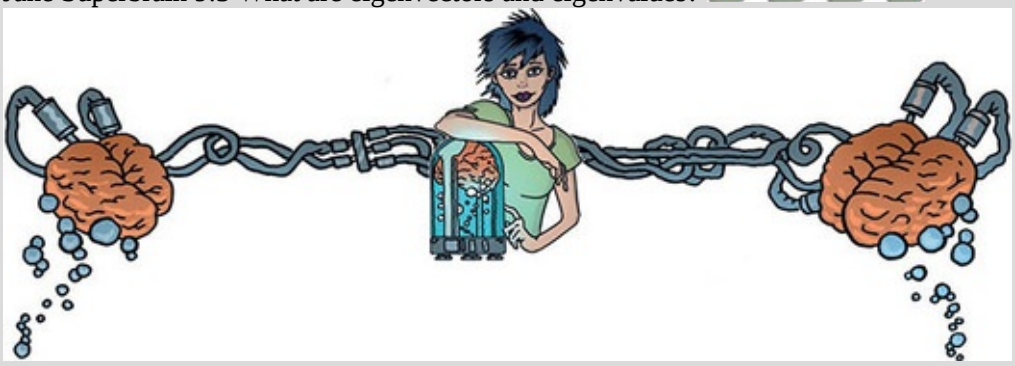


- To check for multicollinearity, use the VIF values from the table labelled *Coefficients*.
- If these values are less than 10 then that indicates there probably isn't cause for concern.
- If you take the average of VIF values, and it is not substantially greater than 1, then

there's also no cause for concern.



Jane Superbrain 9.3 What are eigenvectors and eigenvalues?

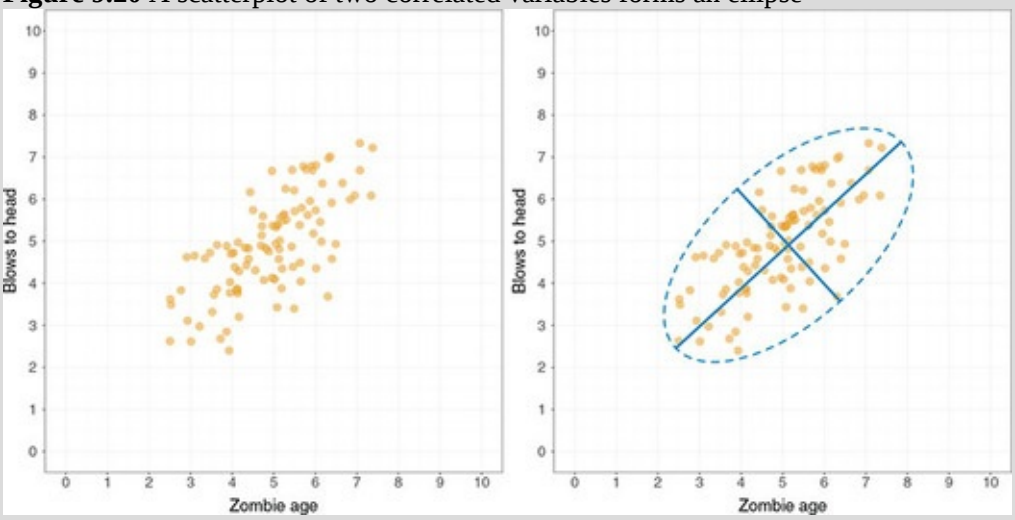


The definitions and mathematics of eigenvalues and eigenvectors are complicated and most of us need not worry about them (although they do crop up again in Chapters 17 and 18). Although the mathematics is hard, we can get a sense of what they represent visually. Imagine we have two variables: the age of a zombie (how long it has been a zombie), and how many blows to the head it takes to kill it.[18] These two variables are normally distributed and can be considered together as a bivariate normal distribution. If these variables are correlated their scatterplot forms an ellipse: if we draw a dashed line around the outer values of the scatterplot we get an oval shape (Figure 9.20). Imagine two lines to measure the length and height of this ellipse: these represent the *eigenvectors* of the correlation matrix for these two variables (a vector is a set of numbers that tells us the location of a line in geometric space). Note that the two straight lines in Figure 9.20 are at 90 degrees to each other, which means that they are independent of one another. So, with two variables, think of eigenvectors as lines measuring the length and height of the ellipse that surrounds the scatterplot of data for those variables. If we add a third variable (e.g., force of blow) our scatterplot gets a third dimension (depth), the ellipse turns into something shaped like a rugby ball (or American football), and we get an extra eigenvector to measure the extra dimension. If we add a fourth variable, a similar logic applies (although it's harder to visualize).

18 Assuming you can ever kill a zombie, that is.

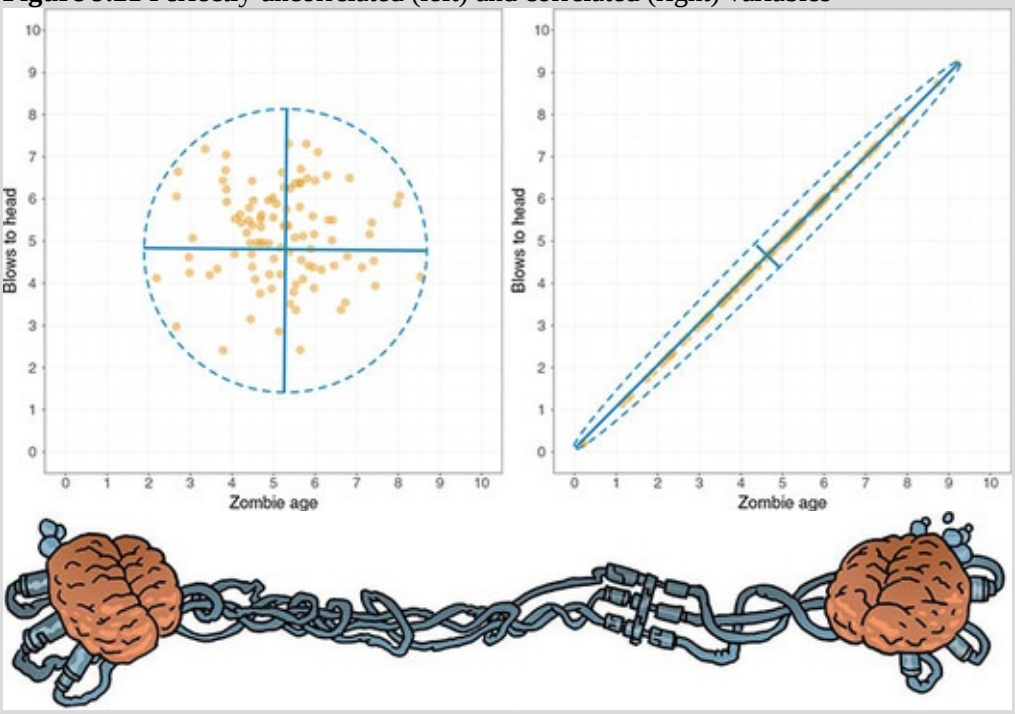
Each eigenvector has an *eigenvalue* that tells us its length (i.e., the distance from one end of the eigenvector to the other). By looking at the eigenvalues for a data set, we know the dimensions of the ellipse (length and height) or rugby ball (length, height, depth); more generally, we know the dimensions of the data. Therefore, the eigenvalues quantify how evenly (or otherwise) the variances of the matrix are distributed.

Figure 9.20 A scatterplot of two correlated variables forms an ellipse



In the case of two variables, the *condition* of the data is related to the ratio of the larger eigenvalue to the smaller. Figure 9.21 shows the two extremes: when there is no relationship at all between variables (left), and when there is a perfect relationship (right). When there is no relationship, the data cloud will be contained roughly within a circle (or a sphere if we had three variables). If we draw lines that measure the height and width of this circle they'll be the same length, which means they'll have the same eigenvalues. Consequently, when we divide the largest eigenvalue by the smallest we'll get a value of 1. When the variables are perfectly correlated (i.e., there is perfect collinearity) the data cloud (and ellipse around it) will collapse to a straight line. The height of the ellipse will be very small indeed (it will approach zero). Therefore, the largest eigenvalue divided by the smallest will tend to infinity (because the smallest eigenvalue is close to zero). An infinite condition index is a sign of deep trouble.

Figure 9.21 Perfectly uncorrelated (left) and correlated (right) variables



9.11.6 Bias in the model: casewise diagnostics



The final stage of the general procedure outlined in Figure 9.10 is to check the residuals for evidence of bias. The first step is to examine the casewise diagnostics. Output 9.11 shows any cases that have a standardized residual less than −2 or greater than 2 (remember that we changed the default criterion from 3 to 2 in Figure 9.16). In an ordinary sample we would expect 95% of cases to have standardized residuals within about ±2 (Jane Superbrain Box 6.4). We have a sample of 200, therefore it is reasonable to expect about 10 cases (5%) to have standardized residuals outside these limits. Output 9.11 shows that we have 12 cases (6%) that are outside of the limits: pretty close to what we would expect. In addition, 99% of cases should lie within ±2.5 and only 1% of cases should lie outside these limits. We have two cases that lie outside of the limits (cases 164 and 169), which is 1% and what we would expect. These diagnostics give us no cause for concern, except that case 169 has a standardized residual greater than 3, which is probably large enough for us to investigate this case further.

Output 9.11

Casewise Diagnostics[a]

Case Number	Std. Residual	Album sales (thousands)	Predicted Value	Residual
1	2.125	330	229.92	100.080
2	-2.314	120	228.95	-108.949
10	2.114	300	200.47	99.534
47	-2.442	40	154.97	-114.970
52	2.069	190	92.60	97.403
55	-2.424	190	304.12	-114.123
61	2.098	300	201.19	98.810
68	-2.345	70	180.42	-110.416
100	2.066	250	152.71	97.287
164	-2.577	120	241.32	-121.324
169	3.061	360	215.87	144.132
200	-2.064	110	207.21	-97.206

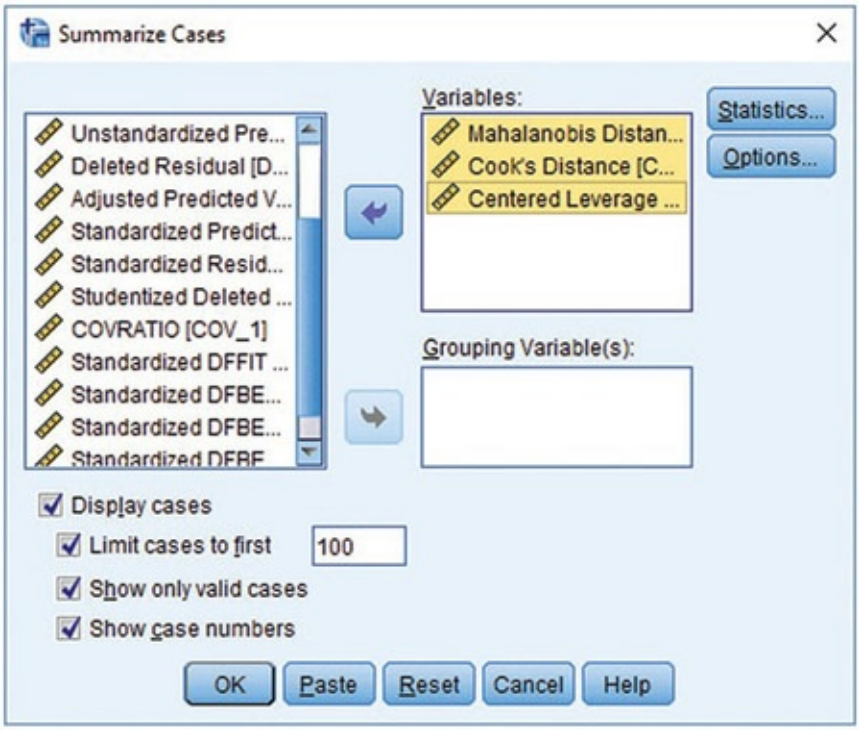
a. Dependent Variable: Album sales (thousands)

In Section 9.10.4 we opted to save various diagnostic statistics. You should find that the data editor contains columns for these variables. You can check these values in the data editor, or list values in the viewer window. To create a table of

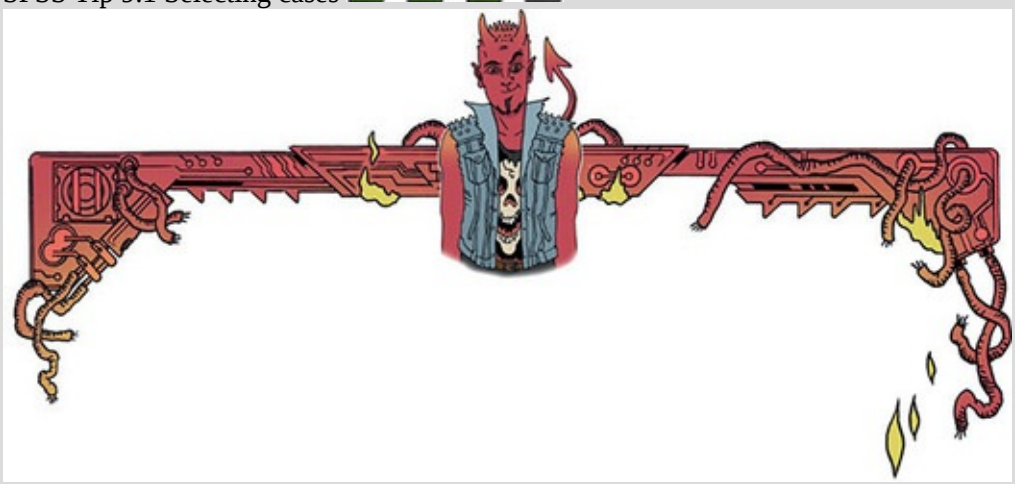
values in the viewer, select *Analyze*  *Reports*  to access the dialog box in Figure 9.22. Select and drag the variables that you want to list into the box labelled *Variables* (or click ). By default, the output is limited to the first 100 cases, but if you want to list all cases deselect this option (also see SPSS Tip 9.1). It is also useful to select *Show case numbers* to enable you to identify the case numbers of any problematic cases.

To save space, Output 9.12 shows the influence statistics for 12 cases that I selected. None of them has a Cook's distance greater than 1 (even case 169 is well below this criterion), and so no case appears to have an undue influence on the model. The average leverage can be calculated as $(k + 1)/n = 4/200 = 0.02$, and we should look for values either twice (0.04) or three times (0.06) this value (see Section 9.3.2). All cases are within the boundary of three times the average, and only case 1 is close to two times the average. For the Mahalanobis distances we saw earlier in the chapter that with a sample of 100 and three predictors, values greater than 15 are problematic. Also, with 3 predictors values greater than 7.81 are significant ($p < 0.05$). None of our cases comes close to exceeding the criterion of 15 although case 1 would be deemed 'significant'.

Figure 9.22 The *Summarize Cases* dialog box



SPSS Tip 9.1 Selecting cases

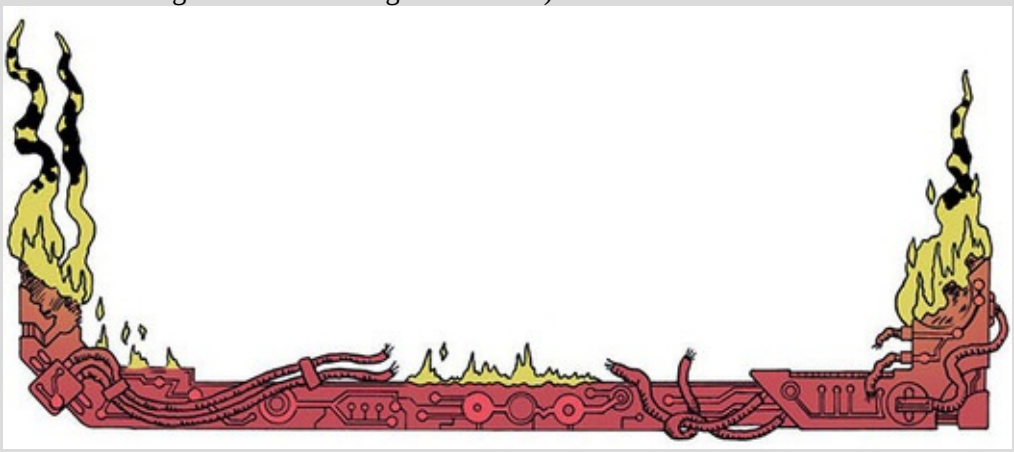


In large data sets, a useful strategy when summarizing cases is to use the *select cases* function (see Section 6.12.2) and set conditions that select problematic cases. For example, you could create a variable that selects cases with a Cook's distance greater than 1 by running this syntax:

```
USE ALL.
COMPUTE cook_problem=(COO_1 > 1).
VARIABLE LABELS cook_problem 'Cooks distance greater than 1'.
VALUE LABELS cook_problem 0 'Not Selected' 1 'Selected'.
FILTER BY cook_problem.
```

EXECUTE.

This syntax creates a variable called **cook_problem**, based on whether Cook's distance is greater than 1 (the *compute* command), it labels this variable as 'Cooks distance greater than 1' (the *variable labels* command), sets value labels to be 1 = include, 0 = exclude (the *value labels* command), and finally filters the data set by this new variable (the *filter* command). Having selected cases, you can use case summaries to see which cases meet the condition you set (in this case having Cook's distance greater than 1).



The DFBeta statistics tell us how much influence each case has on the model parameters. An absolute value greater than 1 is a problem, and all cases in Output 9.12 have values within ±1, which is good news.

Output 9.12

Case Summaries[a]

	Case Number	COVRATIO	Standardized DFFIT	Standardized DFBETA Intercept	Standardized DFBETA Adverts	Standardized DFBETA Airplay	Standardized DFBETA Image
1	1	.97127	.48929	-.31554	-.24235	.15774	.35329
2	2	.92018	-.21110	.01259	-.12637	.00942	-.01868
3	10	.94392	.26896	-.01256	-.15612	.16772	.00672
4	47	.91458	-.31469	.06645	.19602	.04829	-.17857
5	52	.95995	.36742	.35291	-.02881	-.13667	-.26965
6	55	.92486	-.40736	.17427	-.32649	-.02307	-.12435
7	61	.93654	.15562	.00082	-.01539	.02793	.02054
8	68	.92370	-.30216	-.00281	.21146	-.14766	-.01760
9	100	.95888	.35732	.06113	.14523	-.29984	.06766
10	164	.92037	-.54029	.17983	.28988	-.40088	-.11706
11	169	.85325	.46132	-.16819	-.25765	.25739	.16968
12	200	.95435	-.31985	.16633	-.04639	.14213	-.25907
Total N		12	12	12	12	12	12

a. Limited to first 100 cases.

	Mahalanobis Distance	Cook's Distance	Centered Leverage Value
1	8.39591	.05870	.04219
2	.59830	.01089	.00301
3	2.07154	.01776	.01041
4	2.12475	.02412	.01068
5	4.81841	.03316	.02421
6	4.19960	.04042	.02110
7	.06880	.00595	.00035
8	2.13106	.02229	.01071
9	4.53310	.03136	.02278
10	6.83538	.07077	.03435
11	3.14841	.05087	.01582
12	3.49043	.02513	.01754
Total N	12	12	12

a. Limited to first 100 cases.

For the covariance ratio we need to use the following criteria (Section 9.3.2):

- $CVR_i > 1 + [3(k + 1)/n] = 1 + [3(3 + 1)/200] = 1.06$
- $CVR_i < 1 - [3(k + 1)/n] = 1 - [3(3 + 1)/200] = 0.94$.

Therefore, we are looking for any cases that deviate substantially from these boundaries. Most of our 12 potential outliers have CVR values within or just outside these boundaries. The only case that causes concern is case 169 (again), whose CVR is some way below the bottom limit. However, given the Cook's distance for this case, there is probably little cause for alarm. You will have requested other diagnostic statistics and you can apply what we learnt earlier in the chapter when glancing over them.

From this minimal set of diagnostics there's nothing to suggest that there are influential cases (although we'd need to look at all 200 cases to confirm this conclusion); we appear to have a fairly reliable model that has not been unduly

influenced by any subset of cases.

Cramming Sam's Tips Residuals



- Look for cases that might be influencing the model.
- Look at standardized residuals and check that no more than 5% of cases have absolute values above 2, and that no more than about 1% have absolute values above 2.5. Any case with a value above about 3 could be an outlier.
- Look in the data editor for the values of Cook's distance: any value above 1 indicates a case that might be influencing the model.
- Calculate the average leverage and look for values greater than twice or three times this average value.
- For Mahalanobis distance, a crude check is to look for values above 25 in large samples (500) and values above15 in smaller samples (100). However, Barnett and Lewis (1978) should be consulted for more refined guidelines.
- Look for absolute values of DFBeta greater than 1.
- Calculate the upper and lower limit of acceptable values for the covariance ratio, CVR. Cases that have a CVR that fall outside these limits may be problematic.



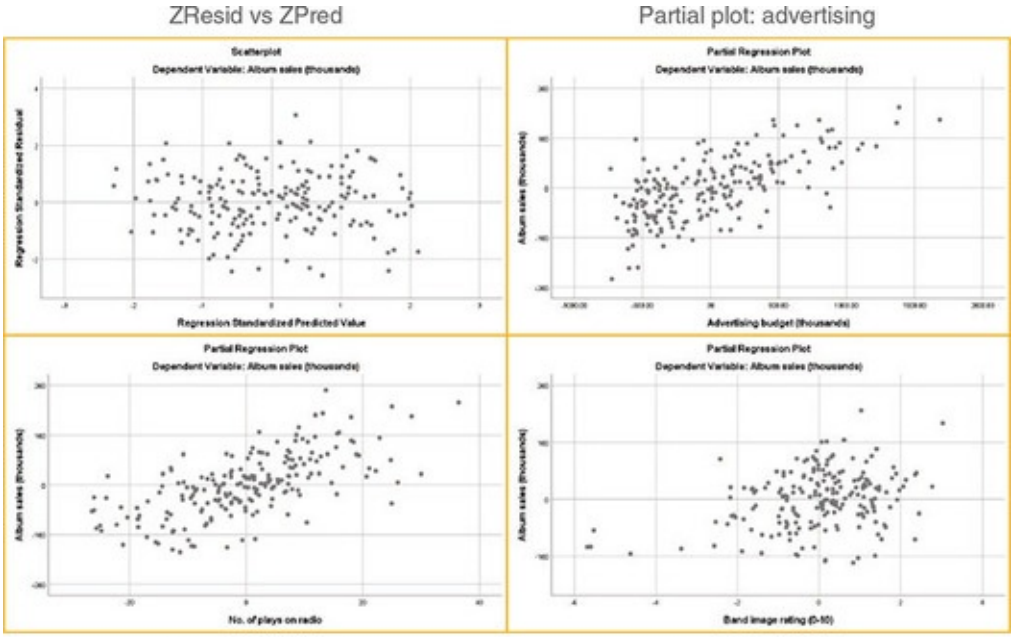
9.11.7 Bias in the model: assumptions

The general procedure outlined in Figure 9.10 suggests that having fitted a model, we need to look for evidence of bias, and the second stage of this process is to check the assumptions described in Chapter 6. We saw in Section 6.11.1 that we can look for heteroscedasticity and non-linearity using a plot of standardized predicted values against standardized residuals. We asked for this

plot in Section 9.10.3. If everything is OK then this graph should look like a random array of dots. Figure 9.23 (top left) shows the graph for our model. Note how the points are randomly and evenly dispersed throughout the plot. This pattern is indicative of a situation in which the assumptions of linearity and homoscedasticity have been met; compare it with the examples in Figure 6.19.

Figure 9.23 also shows the partial plots, which are scatterplots of the residuals of the outcome variable and each of the predictors when both variables are regressed separately on the remaining predictors. Obvious outliers on a partial plot represent cases that might have undue influence on a predictor's *b* coefficient. Non-linear relationships and heteroscedasticity can be detected using these plots as well. For advertising budget (Figure 9.23, top right) the partial plot shows the strong positive relationship to album sales. There are no obvious outliers and the cloud of dots is evenly spaced out around the line, indicating homoscedasticity. The plot for airplay (Figure 9.23, bottom left) also shows a strong positive relationship to album sales, there are no obvious outliers, and the cloud of dots is evenly spaced around the line, again indicating homoscedasticity. For image (Figure 9.23, bottom right) the plot again shows a positive relationship to album sales, but the dots show funnelling, indicating greater spread for bands with a high image rating. There are no obvious outliers on this plot, but the funnel-shaped cloud indicates a violation of the assumption of homoscedasticity.

Figure 9.23 Plot of standardized predicted values against standardized residuals (top left), and partial plots of album sales against advertising (top right), airplay (bottom left) and image of the band (bottom right)



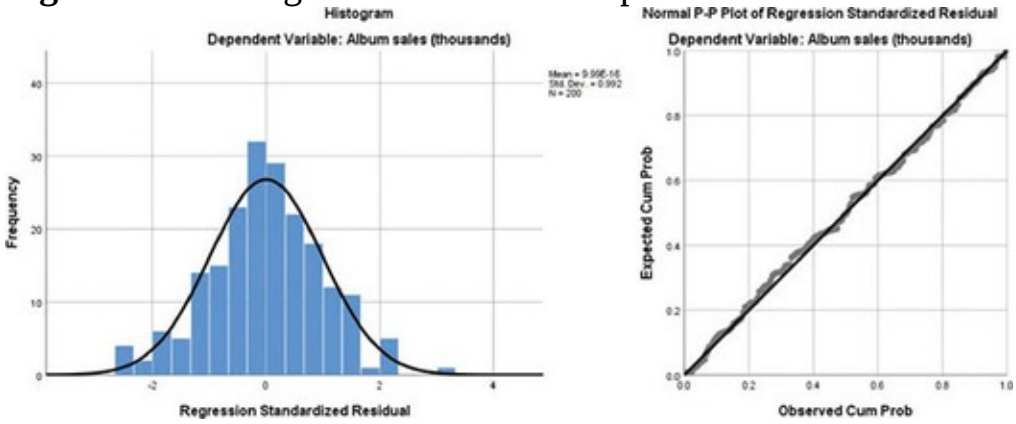
To test the normality of residuals, we look at the histogram and normal probability plot selected in Figure 9.17 and shown in Figure 9.24. Compare these plots to examples of non-normality in Section 6.10.1. For the album sales data, the distribution is very normal: the histogram is symmetrical and approximately bell-shaped. In the P-P plot the dots lie almost exactly along the diagonal, which we know indicates a normal distribution (see Section 6.10.1); hence this plot also suggests that the residuals are normally distributed.

9.12 Robust regression

Our model appears, in most senses, to be both accurate for the sample and generalizable to the population. The only slight glitch is some concern over whether image ratings violated the assumption of homoscedasticity. Therefore, we could conclude that in our sample advertising budget and airplay are equally important in predicting album sales. The image of the band is a significant predictor of album sales, but is less important than the other predictors (and probably needs verification because of possible heteroscedasticity). The assumptions seem to have been met, and so we can probably assume that this model would generalize to any album being released. You won't always (ever?) have such nice data: there will be times when you uncover problems that cast a dark shadow of evil over your model. It will invalidate significance tests,

confidence intervals and generalization of the model (use Chapter 6 to remind yourself of the implications of violating model assumptions).

Figure 9.24 Histogram and normal P-P plot for the residuals from our model



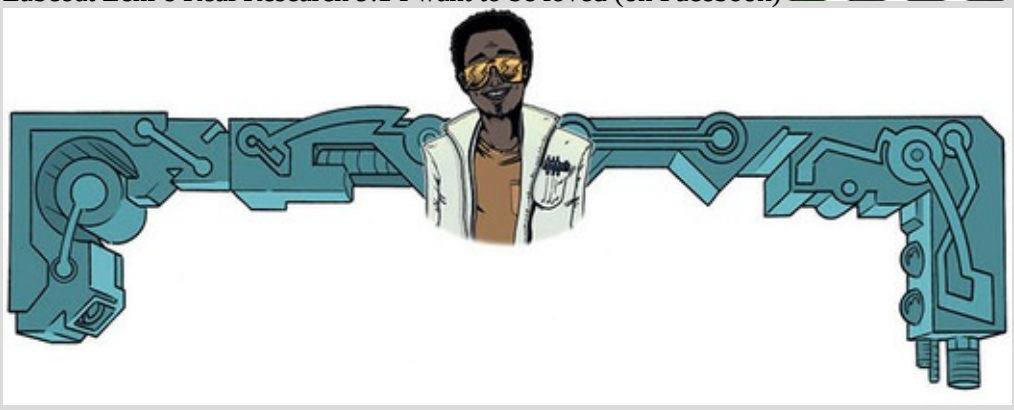
Cramming Sam's Tips Model assumptions



- Look at the graph of **ZRESID*** plotted against **ZPRED***. If it looks like a random array of dots then this is good. If the dots get more or less spread out over the graph (look like a funnel) then the assumption of homogeneity of variance is probably unrealistic. If the dots have a pattern to them (i.e., a curved shape) then the assumption of linearity is probably not true. If the dots seem to have a pattern and are more spread out at some points on the plot than others then this could reflect violations of both homogeneity of variance *and* linearity. Any of these scenarios puts the validity of your model into question. Repeat the above for all partial plots too.
- Look at the histogram and P-P plot. If the histogram looks like a normal distribution (and the P-P plot looks like a diagonal line), then all is well. If the histogram looks non-normal and the P-P plot looks like a wiggly snake curving around a diagonal line then things are less good. Be warned, though: distributions can look very non-normal in small samples even when they are normal.

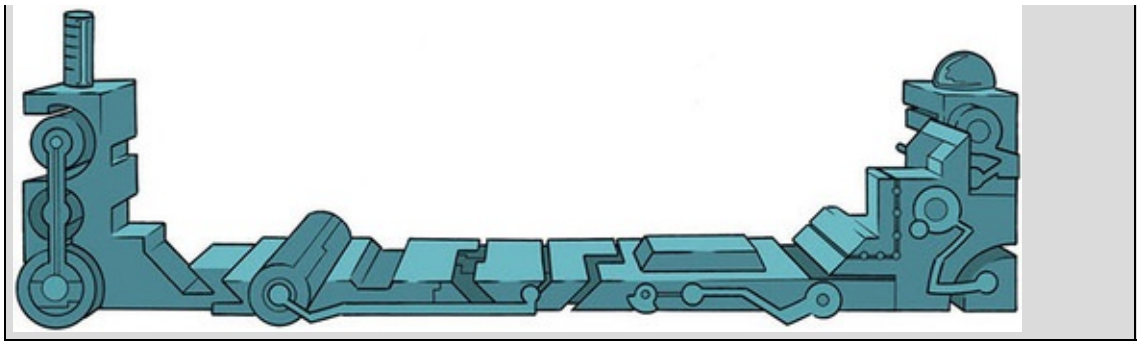


Labcoat Leni's Real Research 9.1 I want to be loved (on Facebook)



Ong, E. Y. L., *et al.* (2011). *Personality and Individual Differences*, *50*(2), 180–185.

Social media websites such as Facebook offer an unusual opportunity to carefully manage your self-presentation to others (i.e., you can appear rad when in fact you write statistics books, appear attractive when you have huge pustules all over your face, fashionable when you wear 1980s heavy metal band T-shirts, and so on). Ong *et al.* (2011) examined the relationship between narcissism and behaviour on Facebook in 275 adolescents. They measured the **Age, Gender** and **Grade** (at school), as well as extroversion and narcissism. They also measured how often (per week) these people updated their Facebook status (**FB_Status**), and also how they rated their own profile picture on each of four dimensions: coolness, glamour, fashionableness, and attractiveness. These ratings were summed as an indicator of how positively they perceived the profile picture they had selected for their page (**FB_Profile_TOT**). Ong *et al.* hypothesized that narcissism would predict the frequency of status updates and how positive a profile picture the person chose. To test this, they conducted two hierarchical regressions: one with **FB_Status** as the outcome and one with **FB_Profile_TOT** as the outcome. In both models they entered **Age, Gender** and **Grade** in the first block, then added extroversion (**NEO_FFI**) in a second block, and finally narcissism (**NPQC_R**) in a third block. Using **Ong *et al.* (2011).sav**, Labcoat Leni wants you to replicate the two hierarchical regressions and create a table of the results for each. Answers are on the companion website (or look at Table 2 in the original article).



Luckily, a lot of the problems can be overcome. If confidence intervals and significance tests of the model parameters are in doubt then use bootstrapping to generate confidence intervals and *p*-values. If homogeneity of variance is the issue then estimate the model with standard errors designed for heteroscedastic residuals (Hayes & Cai, 2007) – you can do this using the PROCESS tool described in Chapter 11. Finally, if the model parameters themselves are in doubt, estimate them using robust regression.

To get robust confidence intervals and significance tests of the model parameters re-estimate your model, selecting the same options as before but clicking Bootstrap... in the main dialog box (Figure 9.13) to access the dialog box explained in Section 6.12.3. To recap, select Perform bootstrapping to activate bootstrapping, and to get a 95% confidence interval click Percentile or Bias corrected accelerated (BCa). For this analysis, let's ask for a bias corrected (BCa) confidence interval. Bootstrapping won't work if you have set options to save diagnostics, so click Save... to open the dialog box in Figure 9.18 and *deselect everything*. Back in the main dialog box click OK to estimate the model.

Output 9.13

Bootstrap for Coefficients

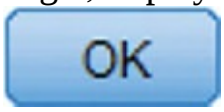
Model		B	Bootstrap[a] Bias	Std. Error	Sig. (2-tailed)	BCa 95% Confidence Interval Lower	Upper
1	(Constant)	134.14	.156	7.613	.001	119.470	150.048
	Advertising budget (thousands)	.096	.000	.008	.001	.081	.111
2	(Constant)	-26.61	-.028	15.733	.077	-54.589	2.715
	Advertising budget (thousands)	.085	.000	.007	.001	.071	.098
	No. of plays on radio	3.367	.005	.305	.001	2.773	3.972
	Band image rating (0-10)	11.086	-.019	2.223	.001	6.264	15.283

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

The output will contain a table of bootstrap confidence intervals for each predictor and their significance value (Output 9.13).[19] These tell us that advertising, $b = 0.09$ [0.07, 0.10], $p = 0.001$, airplay, $b = 3.37$ [2.77, 3.97], $p = 0.001$, and the band's image, $b = 11.09$ [6.26, 15.28], $p = 0.001$, all significantly predict album sales. These bootstrap confidence intervals and significance values do not rely on assumptions of normality or homoscedasticity, so they give us an accurate estimate of the population value of *b* for each predictor (assuming our sample is one of the 95% with confidence intervals that contain the population value).

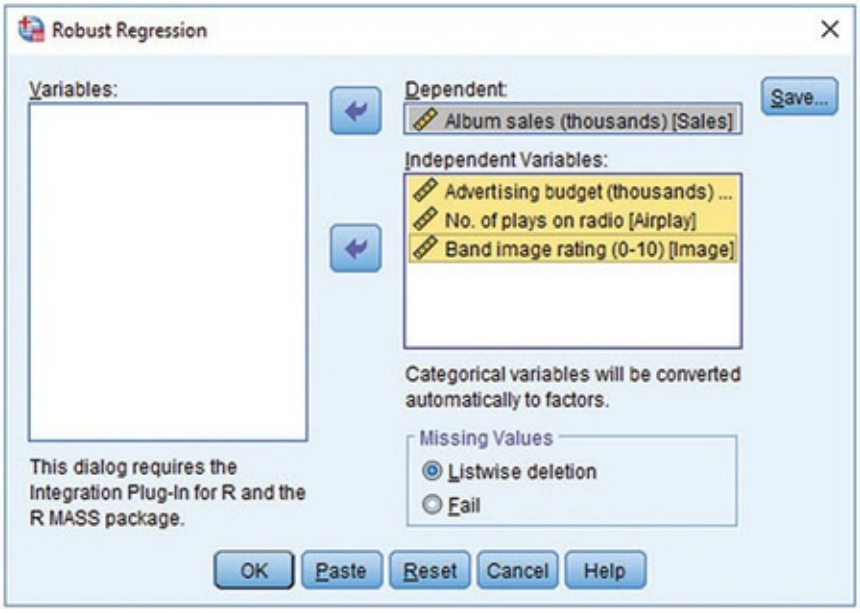
19 Remember that because of how bootstrapping works the values in your output will be different than mine, and different if you rerun the analysis.

To estimate the *b*s themselves using a robust method we can use the R plugin. If you have installed this plugin (Section 4.13.2) then you can access a dialog box (Figure 9.25) to run robust regression using R by selecting *Analyze*  *Regression* 

. If you haven't installed the plugin then this menu won't be there! Drag the outcome (album sales) to the box labelled *Dependent* and any predictors in the final model (in this case advertising budget, airplay and image rating) to the box labelled *Independent Variables*. Click 

to estimate the model.

Figure 9.25 Dialog box for robust regression



Output 9.14

Coefficients

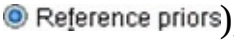
	Value	Std. Error	t value
(Intercept)	-28.858	17.569	-1.643
Adverts	.086	.007	12.219
Airplay	3.371	.281	11.984
Image	11.394	2.469	4.615

rlm(formula = Sales ~ Adverts+Airplay+Image, data = dta, na.action = na.exclude, method = "MM", model = FALSE)
Residual standard error: 45.37396
Degrees of freedom: 196

Output 9.14 shows the resulting robust *b*-values, their robust standard errors and *t*-statistics. Compare these with the non-robust versions in Output 9.8. The values are not much different (mainly because our original model didn't seem to violate its assumptions); for example, the *b* for image rating has changed from 11.09 (Output 9.8.) to 11.39 (Output 9.14), the associated standard error was 2.44 and the robust version is 2.47, and the associated *t*-statistic has changed from 4.55 to 4.62. Essentially our interpretation of the model won't have changed, but this is still a useful sensitivity analysis in that if robust estimates

are giving us basically the same results as non-robust estimates then we know that the non-robust estimates have not been unduly biased by properties of the data. So, this is always a useful double check, and if the robust estimates are hugely different from the original estimates then you can use and report the robust versions.

9.13 Bayesian regression 

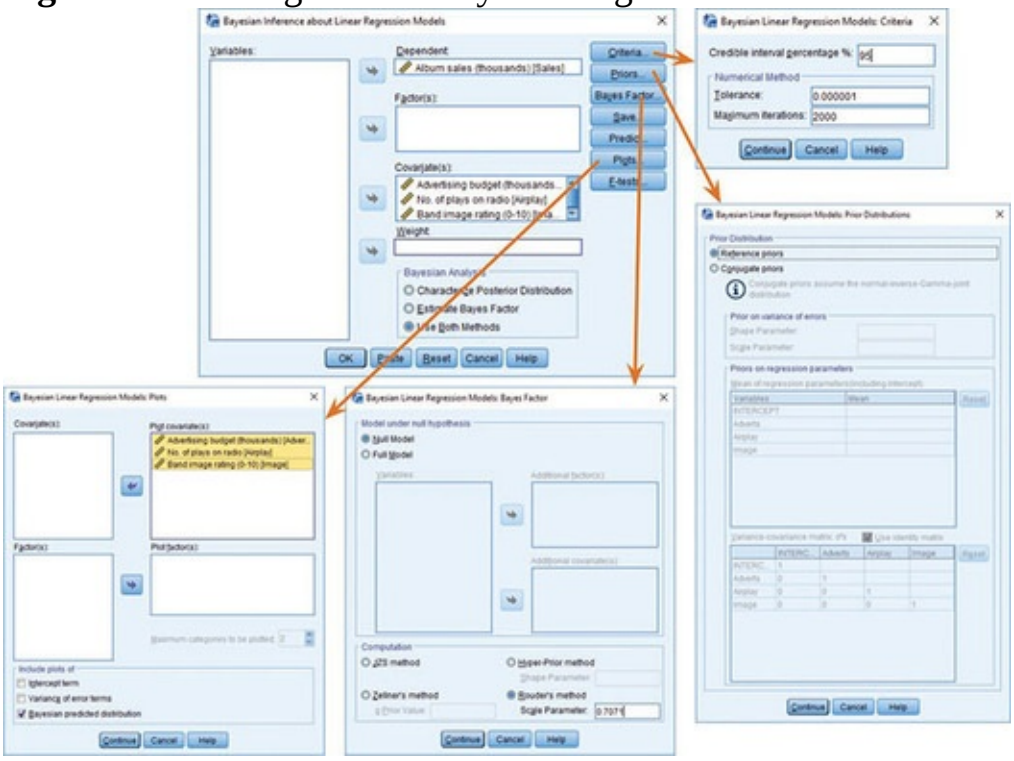
In Section 3.8.4 we looked at Bayesian approaches. To access a dialog box (Figure 9.26) to fit a Bayesian linear model select *Analyze*  *Bayesian Statistics*  *Linear Regression*. You can fit the model either using default priors (so called ), which set distributions that represent very diffuse prior beliefs, or conjugate priors, which allow you to specify more specific priors. One of the key strengths of Bayesian statistics (in my opinion) is that you can set evidence-based priors that you update with the data that you collect. However, this is not a trivial undertaking, and it requires a deeper understanding of the models being fit than we have covered. So, to help you to dip your toe in the water of Bayesian statistics we will stick to using the reference priors built into SPSS Statistics. The benefit of reference priors is that they enable you to get going with Bayesian models without drowning in a lot of quite technical material, but the cost is that you are building only very uninformative prior information into your models.[20]

20 Another downside of this convenience is that I find it hard to know what these priors actually represent (especially in the case of regression).

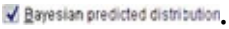
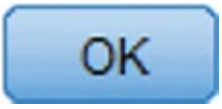
In the main dialog box (Figure 9.26) drag **Sales** to the box labelled *Dependent* (or click ) and drag **Adverts, Airplay** and **Image** to the

Covariate(s) box (or click ). If your model has categorical predictors (which we'll look at in the following chapter) drag them to the *Factor(s)* box. If you want to both compute Bayes factors and estimate the model parameters then select ◉ Use Both Methods.

Figure 9.26 Dialog box for Bayesian regression



If you want a credible interval other than 95% then click Criteria... and change the 95 to the value you want. Click Priors... to set your priors, although we'll stick with ◉ Reference priors. Click Bayes Factor... if you want to get a Bayes factor for your model. By default, the full model will be compared to the null model and there are four methods to compute them. I have selected ◉ JZS method (Jeffreys, 1961; Zellner & Siow, 1980). Click Priors... to inspect the prior and posterior distributions for each predictor. Drag all

predictors to the box labelled *Plot covariate(s)* (or click ) and select  . In the main dialog box click  to fit the model.

Output 9.15 (left) shows the Bayes factor for the full model compared to the null model, which I assume is the model including only the intercept. The right side of the output shows the parameter estimates based on Bayesian estimation. The Bayes factor is 1.066×10^{43} (that's what the E+43 means). In other words it is massive. In short, the probability of the data given the model including all three predictors is 1.07×10^{43} greater than the probability of the data given the model with only the intercept. We should shift our belief in the model (relative to the null model) by a factor of 1.07×10^{43}! This is very strong evidence for the model.

The Bayesian estimate of *b* can be found in the columns labelled *Posterior Mode* and *Posterior Mean*. In fact the columns contain identical values, but they won't always. The reason for the two columns is that we use the peak of the posterior distribution as our estimate and that peak can be defined by either the mode of the posterior or its mean. The values are 0.085 for advertising budget, 3.367 for airplay and 11.086 for image, compared to the values of 0.085, 3.37 and 11.09 (Output 9.8). from the non-Bayesian model. They are basically the same, which is not all that surprising because we started off with very diffuse priors (and so these priors will have had very little influence over the estimates – think back to Section 3.8). We can see this fact in Output 9.16, which shows the prior distribution for the *b* for advertising budget as a red line (in your output you will see similar plots for the other two predictors): the line is completely flat, representing a completely open and diffuse belief about the model parameters. The green line is the posterior distribution, which is quantified in Output 9.15 (right).

Output 9.15

Bayes Factor Model Summary[a,b]

Bayes Factor[c]	R	R Square	Adjusted R Square	Std. Error of the Estimate
1.066E+43	.815	.665	.660	47.09

a. Method: JZS

b. Model: (Intercept), Advertising budget (thousands), No. of plays on radio, Band image rating (0–10)

c. Bayes factor: Testing model versus null model (Intercept).

Bayesian Estimates of Coefficients[a,b,c]

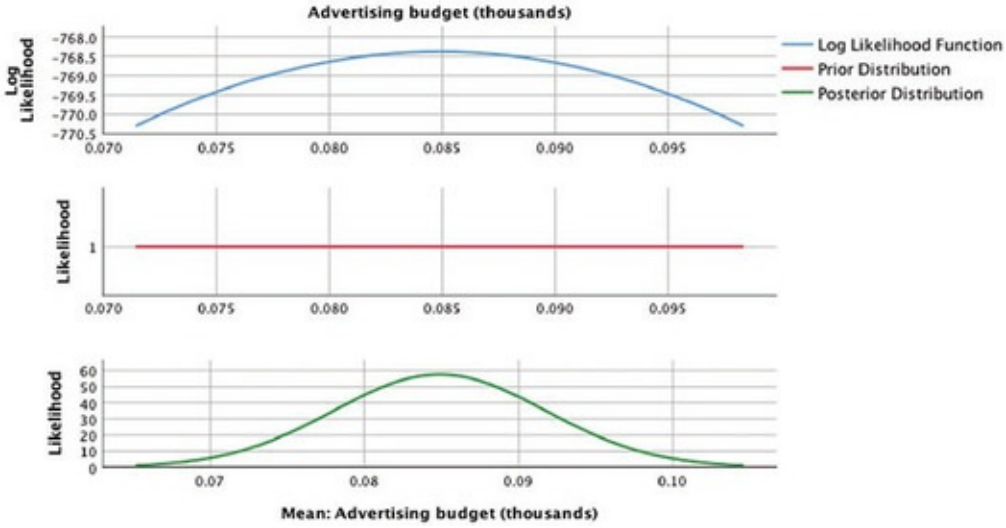
Parameter	Posterior Mode	Posterior Mean	Posterior Variance	95% Credible Interval Lower Bound	95% Credible Interval Upper Bound
(Intercept)	-26.61	-26.61	304.13	-60.830	7.604
Advertising budget (thousands)	.085	.085	.000	.071	.099
No. of plays on radio	3.367	3.367	.078	2.820	3.915
Band image rating (0–10)	11.086	11.086	6.004	6.279	15.894

a. Dependent Variable: Album sales (thousands)

b. Model: (Intercept), Advertising budget (thousands), No. of plays on radio, Band image rating (0–10)

c. Assume standard reference priors.

Output 9.16



Perhaps the most useful parts of Output 9.15 are the 95% credible intervals for the model parameters. Unlike confidence intervals, credible intervals contain the population value with a probability of 0.95 (95%). For advertising budget, therefore, there is a 95% probability that the population value of *b* lies between 0.071 and 0.099, for airplay the population value is plausibly between 2.820 and 3.915, and for image it plausibly lies between 6.279 and 15.894. These intervals are constructed assuming that an effect exists, so you cannot use them to test hypotheses, only to establish plausible population values of the *b*s in the model.

9.14 Reporting linear models

If your model has several predictors than you can't really beat a summary table as a concise way to report your model. As a bare minimum report the betas along with their standard errors and confidence interval (or credible interval if you've gone Bayesian). If you haven't gone Bayesian, report the significance value and perhaps the standardized beta. Include some general fit statistics about the model

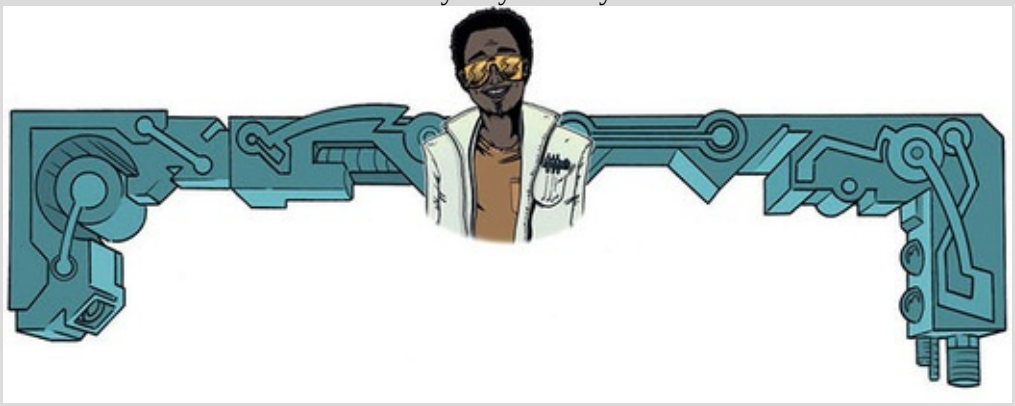
such as R^2 or the Bayes factor. Personally, I like to see the constant as well because then readers of your work can construct the full regression model if they need to. For hierarchical regression you should report these values at each stage of the hierarchy. For the example in this chapter we might produce a table like that in Table 9.2.

Table 9.2 Linear model of predictors of album sales. 95% bias corrected and accelerated confidence intervals reported in parentheses. Confidence intervals and standard errors based on 1000 bootstrap samples

	b	*SE B*	*β*	*p*
Step 1				
Constant	134.14 (120.11, 148.79)	7.95		0.001
Advertising Budget	0.10 (0.08, 0.11)	0.01	0.58	0.001
Step 2				
Constant	−26.61 (−55.40, 8.60)	16.30		0.097
Advertising Budget	0.09 (0.07, 0.10)	0.01	0.51	0.001
Plays on BBC Radio 1	3.37 (2.74, 4.02)	0.32	0.51	0.001
Image	11.09 (6.46, 15.01)	2.22	0.19	0.001

Note. $R^2 = 0.34$ for Step 1; $\Delta R^2 = 0.33$ for Step 2 (all *ps* < 0.001).

Labcoat Leni's Real Research 9.2 Why do you like your lecturers?



Chamorro-Premuzic, T., *et al.* (2008). *Personality and Individual Differences*, *44*, 965–976.

In the previous chapter we encountered a study by Chamorro-Premuzic *et al.* that linked students' personality traits with those they want to see in lecturers (see Labcoat Leni's Real

Research 8.1 for a full description). In that chapter we correlated these scores, but now Labcoat Leni wants you to carry out five multiple regression analyses: the outcome variables across the five models are the ratings of how much students want to see neuroticism, extroversion, openness to experience, agreeableness and conscientiousness. For each of these outcomes, force age and gender into the analysis in the first step of the hierarchy, then in the second block force in the five student personality traits (neuroticism, extroversion, openness to experience, agreeableness and conscientiousness). For each analysis create a table of the results. Answers are on the companion website (or look at Table 4 in the original article). The data are in the file **Chamorro-Premuzic.sav**.

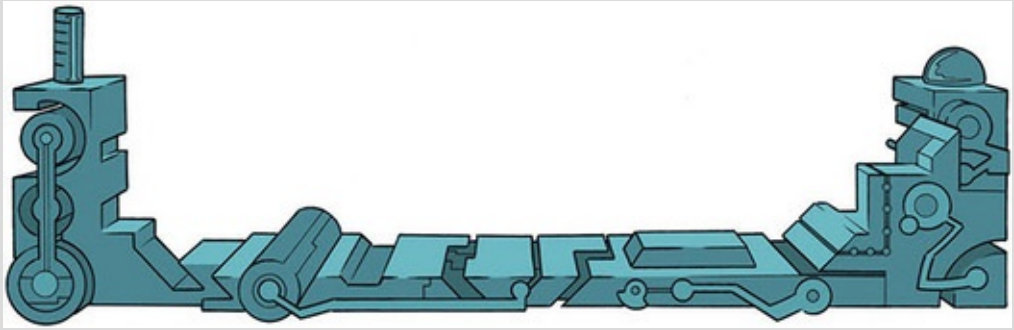
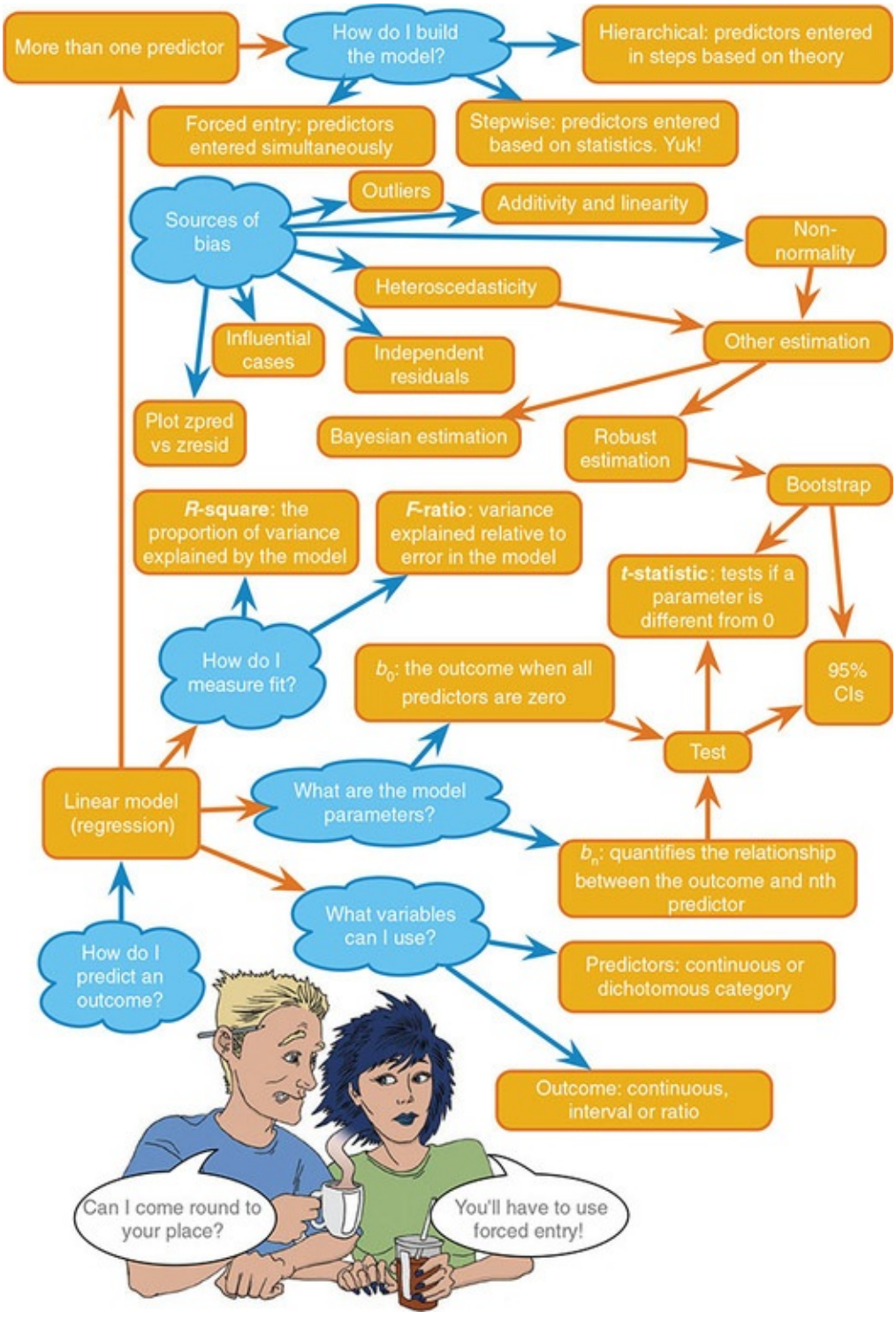


Figure 9.27 What Brian learnt from this chapter



Things to note are as follows: (1) I've rounded off to 2 decimal places throughout because this is a reasonable level of precision given the variables measured; (2) if you are following APA format (which I'm not), do not put zeros before the decimal point for the standardized betas, R^2 and *p*-values (because

these values shouldn't exceed 1); (3) I've reported exact *p*-values, which is good practice; (4) the R^2 for the initial model and the change in R^2 (denoted by ΔR^2) for each subsequent step of the model are reported below the table; and (5) in the title I have mentioned that confidence intervals and standard errors in the table are based on bootstrapping, which is important for readers to know.

9.15 Brian's attempt to woo Jane 

Jane put the fork down next to the jar and suppressed her reflex to gag. When she'd started at this university she'd had utter conviction in testing her flatworm theory. She would be her own single-case design. She knew that experimenting on herself would confound everything, but she wanted some evidence to firm up her beliefs. If it didn't work on her then she could move on, but if she found evidence for some effect then that was starting point for better research. She felt conflicted, though. Was it the experiments making her mind so unfocused, or was it the interest from campus guy? She hadn't come here looking for a relationship, she hadn't expected it, and it wasn't in the plan. Usually she was so good at ignoring other people, but his kindness was slowly corroding her shell. As she got up and replaced the jar on the shelf she told herself that the nonsense with campus guy had to stop. She needed to draw a line.

9.16 What next? 

This chapter is possibly the longest book chapter ever written, and if you feel like you aged several years while reading it then, well, you probably have (look around, there are cobwebs in the room, you have a long beard, and when you go outside you'll discover a second ice age has been and gone, leaving only you and a few woolly mammoths to populate the planet). However, on the plus side, you now know more or less everything you'll ever need to know about statistics. Seriously – you'll discover in the coming chapters that everything else we discuss is a variation of this chapter. So, although you may be near death, having spent your life reading this chapter (and I'm certainly near death having written it), you are officially a stats genius – well done!

We started the chapter by discovering that at 8 years old I could have really done with a linear model to tell me which variables are important in predicting talent competition success. Unfortunately I didn't have one, but I did have my dad (and he's better than a linear model). He correctly predicted the recipe for superstardom, but in doing so he made me hungry for more. I was starting to get a taste for the rock-idol lifestyle: I had friends, a fortune (well, two fake-gold-plated winner's medals), fast cars (a bike) and dodgy-looking 8-year-olds were giving me suitcases full of lemon sherbet to lick off mirrors. The only things needed to complete the job were a platinum-selling album and a heroin addiction. However, before I could get those my parents and teachers were about to impress reality upon my young mind …

9.17 Key terms that I've discovered

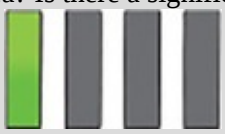
Adjusted predicted value
Adjusted R^2
Akaike information criterion (AIC)
Autocorrelation
b_i
β_i
Cook's distance
Covariance ratio (CVR)
Cross-validation
Deleted residual
DFBeta
DFFit
Durbin–Watson test
F-statistic
Generalization
Goodness of fit
Hat values
Heteroscedasticity
Hierarchical regression
Homoscedasticity
Independent errors
Leverage
Mahalanobis distance
Mean squares

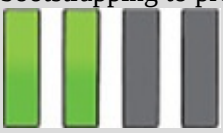
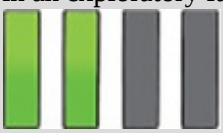
Model sum of squares
Multicollinearity
Multiple regression
Ordinary least squares (OLS)
Outcome variable
Perfect collinearity
Predicted Value
Predictor variable
Residual
Residual sum of squares
Shrinkage
Simple regression
Standardized DFBeta
Standardized DFFit
Standardized residuals
Stepwise regression
Studentized deleted residuals
Studentized residuals
Suppressor effects
t-statistic
Tolerance
Total sum of squares
Unstandardized residuals
Variance inflation factor (VIF)

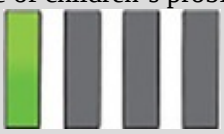
Smart Alex's tasks



- **Task 1**: In Chapter 4 (Task 7) we looked at data based on findings that the number of cups of tea drunk was related to cognitive functioning (Feng et al., 2010). Using a linear model that predicts cognitive functioning from tea drinking, what would cognitive

functioning be if someone drank 10 cups of tea? Is there a significant effect? (see Chapter 8, Task 9) (**Tea Makes You Brainy 716.sav**) 

- **Task 2**: Estimate a linear model for the **pubs.sav** data in Jane Superbrain Box 9.1 predicting **mortality** from the number of **pubs**. Try repeating the analysis but bootstrapping the confidence intervals. 
- **Task 3**: In Jane Superbrain Box 2.1 we encountered data (**HonestyLab.sav**) relating to people's ratings of dishonest acts and the likeableness of the perpetrator. Run a linear model with bootstrapping to predict ratings of dishonesty from the likeableness of the perpetrator. 
- **Task 4**: A fashion student was interested in factors that predicted the salaries of catwalk models. She collected data from 231 models (**Supermodel.sav**). For each model she asked them their salary per day (**salary**), their age (**age**), their length of experience as a model (**years**), and their industry status as a model as their percentile position rated by a panel of experts (**beauty**). Use a linear model to see which variables predict a model's salary. How valid is the model? 
- **Task 5**: A study was carried out to explore the relationship between **Aggression** and several potential predicting factors in 666 children who had an older sibling. Variables measured were **Parenting_Style** (high score = bad parenting practices), **Computer_Games** (high score = more time spent playing computer games), **Television** (high score = more time spent watching television), **Diet** (high score = the child has a good diet low in harmful additives), and **Sibling_Aggression** (high score = more aggression seen in their older sibling). Past research indicated that parenting style and sibling aggression were good predictors of the level of aggression in the younger child. All other variables were treated in an exploratory fashion. Analyse them with a linear model (**Child Aggression.sav**). 
- **Task 6**: Repeat the analysis in Labcoat Leni's Real Research 9.1 using bootstrapping for the confidence intervals. What are the confidence intervals for the regression parameters? 
- **Task 7**: Coldwell, Pike, & Dunn (2006) investigated whether household chaos predicted children's problem behaviour over and above parenting. From 118 families they recorded the age and gender of the youngest child (**child_age** and **child_gender**). They measured dimensions of the child's perceived relationship with their mum: (1) warmth/enjoyment (**child_warmth**), and (2) anger/hostility (**child_anger**). Higher scores indicate more warmth/enjoyment and anger/hostility respectively. They measured the mum's perceived

relationship with her child, resulting in dimensions of positivity (**mum_pos)** and negativity (**mum_neg**). Household chaos (**chaos**) was assessed. The outcome variable was the child's adjustment (**sdq**): the higher the score, the more problem behaviour the child was reported to display. Conduct a hierarchical linear model in three steps: (1) enter child age and gender; (2) add the variables measuring parent–child positivity, parent–child negativity, parent–child warmth, parent–child anger; (3) add chaos. Is household chaos predictive of children's problem behaviour over and above parenting? (**Coldwell *et al.* (2006).sav**). 

Answers & additional resources are available on the book's website at **https://edge.sagepub.com/field5e**



10 Comparing two means

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10.1 What will this chapter tell me?

We saw in the previous chapter that I had successfully conquered the holiday camps of Wales with my singing and guitar playing (and the Welsh know a thing or two about good singing). I had jumped on a snowboard called oblivion and thrown myself down the black run known as world domination. About 10 metres after starting this slippery descent I hit the lumpy patch of ice called ‘adults’. I was 9, life was fun, but every adult I encountered seemed obsessed with my future. ‘What do you want to be when you grow up?’ they would ask. Would I be a surgeon, a lawyer, a teacher? I was 9 and ‘grown up’ was a lifetime away. All I knew was that I was going to marry Clair Sparks (more on her in the next chapter) and be a rock legend who didn’t need to worry about such adult matters as having a job. It was a difficult question, but adults require answers and I wasn’t going to let them know that I didn’t care about ‘grown-up’ matters. Like all good scientists I drew upon past data: I hadn’t tried conducting brain surgery, neither did I have experience of sentencing psychopaths to prison for eating their husbands, nor had I taught anyone. I had, however, had a go at singing and playing guitar; therefore, I predicted I would be a rock star. Even at this early age I realized that not all adults would appreciate the raw talent that would see me parading across the lighted stage in front of tens of thousands of people. Some of them might not think that rock stardom was good career prospect. I needed to convince them. Adults tend to think money is important, so I decided I should demonstrate that rock stars earn more money than, say, a ‘respectable’ profession such as being a teacher. I could gather some teachers and rock stars, find out what their salaries were and compare them. Effectively I’d be ‘predicting’ salary from two categories: rock star or teacher. I could have done all that, but I didn’t know about statistics when I was 9. Happy days.

Figure 10.1 Practising for my career as a rock star by slaying the baying throng of Grove Primary School at the age of 10. (Note the girl with her hands covering her ears.)



10.2 Looking at differences

The last two chapters have focused on relationships between continuous variables, but sometimes researchers want to look at differences between groups of people or between people in different treatment conditions. Experimental research, for example, takes advantage of the fact that if we systematically manipulate what happens to people we can make causal inferences about the effects of those manipulations. The simplest form of experiment is one in which we split the sample into an experimental group and a control group that is identical to the experimental group in all respects except the one expected to have an impact on the outcome (see Field & Hole, 2003). For example, we might want to compare statistics knowledge before and after a lecture. One group, the experimental group, has to sit through the lecture, while the other group, the control group, gets to miss it and stay in bed. Here are some other examples of scenarios where we'd compare two conditions:

- Is the movie *Scream 2* scarier than the original, *Scream*? We could measure heart rates (which indicate anxiety) during both films and compare them.
- Is your work better when you do it listening to Andy's favourite music? You could get some people to write an essay (or book) listening to my favourite music (as listed in the Acknowledgements), and then write a different one in silence (this is a control) and compare the grades.
- Do diet pills work? We take two groups of people and randomly assign one group to a programme of diet pills and the other group to a programme of sugar pills (which they believe will help them lose weight). If the people who take the diet pills lose more weight than those on the sugar pills we can infer that the diet pills caused the weight loss.

Systematic manipulation of the independent (predictor) variable is a powerful

tool because it goes one step beyond merely observing variables.[1] This chapter is the first of many that look at these research scenarios. We start with the simplest scenario: when we have two groups, or, to be more specific, when we want to compare two means. We discovered in Chapter 1 that we can expose different entities to different experimental manipulations (a *between-groups* or *independent* design), or take a single group of entities and expose them to different experimental manipulations at different points in time (a *repeated-measures* or *within-subject* design). Researchers sometimes get tempted to compare artificially created groups by, for example, dividing people into groups based on a median score; avoid this temptation (Jane Superbrain Box 10.1).

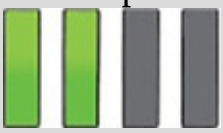
1 People sometimes get confused and think that certain statistical procedures allow causal inferences and others don't (see Jane Superbrain Box 1.4).

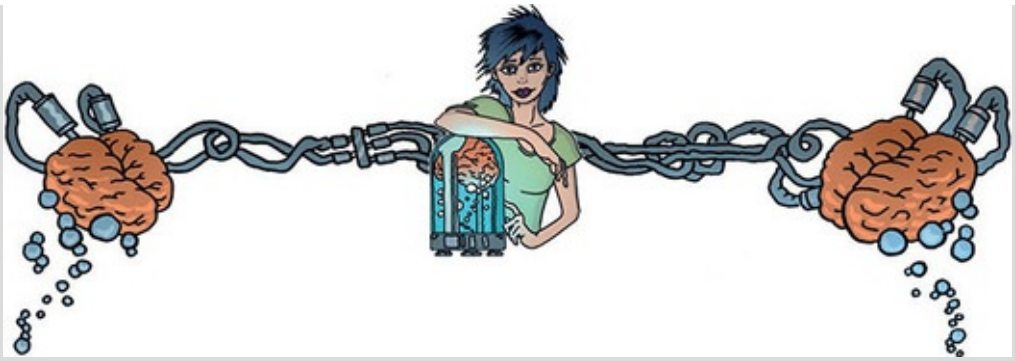
10.3 A mischievous example 

Two news stories caught my eye related to physics (Di Falco, Ploschner, & Krauss, 2010). In the first headline (November 2010) the *Daily Mirror* (a UK newspaper) reported 'Scientists make Harry Potter's invisible cloak'. I'm not really a Harry Potter aficionado,[2] so it wasn't his mention that caught my attention, but the idea of being able to don a cloak that would render me invisible and able to get up to mischief. That idea was very exciting indeed; where could I buy one? By February 2011 the same newspaper was reporting on a different piece of research (Chen et al., 2011) with a similarly exaggerated headline: 'Harry Potter-style "invisibility cloak" built by scientists'.

2 Perhaps I should be, given that a UK newspaper once tagged me 'the Harry Potter of the social sciences' (http://www.discoveringstatistics.com/docs/thes_170909.pdf). I wasn't sure whether this made me a heroic wizard battling against the evil forces of statistics, or an adult with a mental age of 11.

Jane Superbrain 10.1 Are median splits the devil's work?



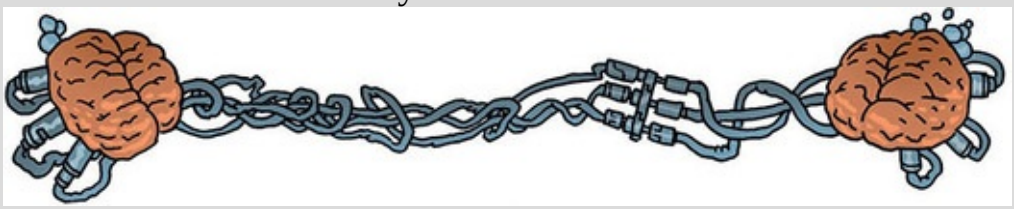


Sometimes scientists use a ‘median split’ of a predictor variable. For example, there is a stereotype that science fiction fans are recluses with no social skills. If you wanted to test this you might measure social skills and knowledge of the film *Star Wars*. You might then take the median score on *Star Wars* knowledge and classify anyone with a score above the median as a ‘*Star Wars* fan’, and those below as a ‘non-fan’. In doing this you ‘dichotomize’ a continuous variable. This practice is quite common, but there are several problems with median splits (MacCallum, Zhang, Preacher, & Rucker, 2002):

1. Imagine there are four people: Peter, Birgit, Jip and Kiki. We measure how much they know about *Star Wars* as a percentage and get Jip (100%), Kiki (60%), Peter (40%) and Birgit (0%). If we split these four people at the median (50%) then we’re saying that Jip and Kiki are the same (they get a score of 1 = fanatic) and Peter and Birgit are the same (they both get a score of 0 = not a fanatic). Median splits change the original information quite dramatically: Peter and Kiki are originally very similar but become opposite after the split, whereas Jip and Kiki are relatively dissimilar originally but become identical after the split.
2. Effect sizes get smaller. If you correlate two continuous variables then the effect size will be larger than if you correlate the same variables after one of them has been dichotomized. Effect sizes also get smaller in linear models.
3. There is an increased chance of finding spurious effects.

So, if your supervisor has just told you to do a median split, have a good think about whether it is the right thing to do and read up on the topic (I recommend DeCoster, Gallucci, & Iselin, 2011; DeCoster, Iselin, & Gallucci, 2009; MacCallum et al., 2002). According to MacCallum et al., one of the rare situations in which dichotomizing a continuous variable is justified is when there is a clear theoretical

rationale for distinct categories of people based on a meaningful break point (i.e., not the median); for example, phobic versus not phobic based on diagnosis by a trained clinician might be a legitimate dichotomization of anxiety.



Needless to say, scientists hadn't actually made Harry Potter's cloak of invisibility or anything close to it, but never let that get in the way of a headline. What Chen *et al.* had made wasn't so much a 'cloak' of invisibility as a 'calcite lump' of invisibility. This lump could hide small objects (centimetres and millimetres in scale): you could conceal my brain but little else. Nevertheless, with a suitably large piece of calcite in tow, I could theoretically hide my whole body (although people might get suspicious of the apparently autonomous block of calcite manoeuvring itself around the room on a trolley). Di Falco *et al.* had created a flexible material (Metaflex) with optical properties that meant that if you layered it up you might be able to create something around which light would bend. Not exactly a cloak in the clothing sense of the word, but easier to wear than, say, a slab of calcite.

Table 10.1 Data from **Invisibility.sav**

Participant	Cloak	Mischief
1	0	3
2	0	1
3	0	5
4	0	4
5	0	6
6	0	4
7	0	6
8	0	2
9	0	0
10	0	5
11	0	4
12	0	5
13	1	4
14	1	3
15	1	6
16	1	6
17	1	8
18	1	5
19	1	5
20	1	4
21	1	2
22	1	5
23	1	7
24	1	5

Although the newspapers overstated the case a little, these are exciting pieces of research that bring the possibility of a cloak of invisibility closer to a reality. I imagine a future in which we have some cloaks of invisibility to test out. Given my slightly mischievous streak, the future me is interested in the effect that wearing a cloak of invisibility has on the tendency for mischief. I take 24 participants and place them in an enclosed community. The community is riddled with hidden cameras so that we can record mischievous acts. Half of the participants are given cloaks of invisibility; they are told not to tell anyone else about their cloak and that they can wear it whenever they like. I measure how many mischievous acts they perform in a week. These data are in Table 10.1.



Enter these data into SPSS.

The file **Invisibility.sav** shows how you should have entered the data: the variable **Cloak** records whether a person was given a cloak (**Cloak** = 1) or not (**Cloak** = 0), and **Mischief** is how many mischievous acts were performed.



Produce some descriptive statistics for these data (using *Explore*).

Output 10.1 (your table will have more stuff in it – I edited mine down to save space) shows that people with invisibility cloaks performed more mischievous acts, $M = 5$, 95% CI [3.95, 6.05], $SD = 1.65$, than those with no cloak, $M = 3.75$, 95% CI [2.53, 4.97], $SD = 1.91$. Both groups' scores are normally distributed according to the K-S tests because the value of *Sig.* is greater than 0.05, but these tests will be highly underpowered because they are based on *N*s of 12 (Jane Superbrain Box 6.5).

Output 10.1

Descriptives

	Cloak of invisibility			Statistic	Std. Error
Mischievous Acts	No Cloak	Mean		3.75	.552
		95% Confidence Interval for Mean	Lower Bound	2.53	
			Upper Bound	4.97	
		5% Trimmed Mean		3.83	
		Median		4.00	
		Variance		3.659	
		Std. Deviation		1.913	
		Skewness		-.789	.637
		Kurtosis		-.229	1.232
	Cloak	Mean		5.00	.477
		95% Confidence Interval for Mean	Lower Bound	3.95	
			Upper Bound	6.05	
		5% Trimmed Mean		5.00	
		Median		5.00	
		Variance		2.727	
		Std. Deviation		1.651	
		Skewness		.000	.637
		Kurtosis		.161	1.232

Tests of Normality

		Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Cloak of invisibility	Statistic	df	Sig.	Statistic	df	Sig.
Mischievous Acts	No Cloak	.219	12	.118	.913	12	.231
	Cloak	.167	12	.200*	.973	12	.936

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

10.4 Categorical predictors in the linear model



If we want to compare differences between the means of two groups, all we are doing is predicting an outcome based on membership of two groups. For our invisibility example, we're predicting the number of mischievous acts from whether someone had a cloak of invisibility. This is a linear model with one dichotomous predictor. The *b* for the model reflects the differences between the mean levels of mischief in the two groups, and the resulting *t*-test will, therefore, tell us whether the difference between means is different from zero (because, remember, the *t*-test tests whether $b = 0$).

You might be thinking '*b*s show relationships, not differences between means – what is this fool going on about?'. You might be starting to mistrust me, or are stuffing the book in a box to post it back for a refund. I wouldn't blame you, because I used to think this too. To tame a land like the complex, thorny, weed-infested and Andy-eating tarantula-inhabited world of statistics you need an

epiphany. Mine was a paper by Cohen (1968) that showed me that when we compare means we are using a special case of the linear model. This revelation turned my statistical world into a beautiful meadow filled with bleating little lambs all jumping for joy at the wonder of life.

Recall from Chapter 2 that all statistical models are versions of the simple idea:

$$\text{outcome}_i = (\text{model}) + \text{error}_i \quad (10.1)$$

When using a linear model this general equation becomes equation (9.2) in which the model is defined by parameters: b_0 tells us the value of the outcome when the predictor is zero, and b_1 quantifies the relationship between the predictor (X_i) and outcome (Y_i). We've seen this equation lots of times, but let's tailor it to our example. The equation predicting the variable **Mischief** from the group to which a person belongs (the variable **Cloak**) is:

$$\begin{aligned} Y_i &= (b_0 + b_1 X_{1i}) + \varepsilon_i \\ \text{Mischief}_i &= (b_0 + b_1 \text{Cloak}_i) + \varepsilon_i \end{aligned} \quad (10.2)$$

Cloak is a nominal variable: people had a 'cloak' or 'no cloak'. We can't put words into a statistical model because it will burn a hole in the ozone layer. Instead, we convert this variable into numbers, just like we do when we enter nominal variables into IBM SPSS Statistics (see Section 4.6.5). When we enter nominal variables into SPSS it doesn't matter what numbers we choose, because SPSS converts them into sensible values behind the scenes. But the numbers we choose to represent our categories in a mathematical model are important: they change the meaning of the resulting *b*-values. There are different 'standard' ways to code variables (which we won't get into here), one of which is to use **dummy variables**. We'll look at these in Section 11.5.1, but the summary is that we code a baseline category with a 0, and other categories with a 1. In this example there are two categories. Our baseline category is no cloak (the control condition), and we assign these participants a 0 for the variable **Cloak**. The 'experimental' group contains those who were given a cloak, and we assign these participants a 1. This is the coding I've used in the SPSS file. Let's plug these numbers into the model and see what happens.

First, imagine someone is in the no cloak condition. Knowing that they are in that group, the best prediction we could make of the number of mischievous acts would be the group mean because this value is the summary statistic with the least squared error. So, the value of Y in the equation will be the group mean $\overline{X}_{\text{No Cloak}}$ (which is 3.75 in Output 10.1) and the value of the **Cloak** variable will be 0. If we ignore the error term, equation (10.2) becomes:

$$\begin{aligned}\text{Mischief}_i &= b_0 + b_1\text{Cloak}_i \\ \bar{X}_{\text{No Cloak}} &= b_0 + (b_1 \times 0) \\ b_0 &= \bar{X}_{\text{No Cloak}} \\ b_0 &= 3.75\end{aligned} \tag{10.3}$$

Note that b_0 (the intercept) is equal to the mean of the group coded as 0 (i.e., the no cloak group).

Now let's use the model to predict mischief in people who had an invisibility cloak. As before, the predicted value of the outcome would be the mean of the group to which the person belonged, because this is the summary statistic with the least squared error. The predicted value for someone in the cloak group is, therefore, the mean of the cloak group $\overline{X}_{\text{Cloak}}$, which is 5 in Output 10.1. The value of the **Cloak** variable is 1 (because this is the value we used to code group membership). Remember that b_0 is equal to the mean of the no cloak group $(\overline{X}_{\text{No Cloak}})$. If we place all of the values into equation (10.2) and rearrange it a bit we get:

$$\begin{aligned}\text{Mischief}_i &= b_0 + b_1\text{Cloak}_i \\ \bar{X}_{\text{Cloak}} &= b_0 + (b_1 \times 1) \\ \bar{X}_{\text{Cloak}} &= b_0 + b_1 \\ \bar{X}_{\text{Cloak}} &= \bar{X}_{\text{No Cloak}} + b_1 \\ b_1 &= \bar{X}_{\text{Cloak}} - \bar{X}_{\text{No Cloak}}\end{aligned} \tag{10.4}$$

which shows that b_1 represents the difference between the group means (in this case 5 – 3.75 = 1.25).

The take-home message is that we use the same linear model that we have used throughout the book to compare group means. In a model with a categorical predictor with two predictors, b_1 represents the difference between group means, and b_0 is equal to the mean of the group coded as 0. We have seen in the previous chapter that a *t*-statistic is used to ascertain whether a model parameter (b_1) is equal to 0; in this context, therefore, it would test whether the difference between group means is equal to 0.



To prove that I'm not making it up as I go along, fit a linear model to the data in **Invisibility.sav** with **Cloak** as the predictor and **Mischief** as the outcome using what you learnt in the previous chapter. **Cloak**

is coded using zeros and ones as described above.

If you do the self-test you should obtain the table in Output 10.2. First, notice that the value of the constant (b_0) is 3.75, the same as the mean of the base category (the no cloak group). Second, notice that the value of the regression coefficient b_1 is 1.25, which is the difference between the two group means (5 – 3.75 = 1.25). Finally, the *t*-statistic, which tests whether b_1 is significantly different from zero, is not significant because the significance value is greater than 0.05, which means that the difference between means (1.25) is not significantly different from 0. This section demonstrates that differences between means can be represented by linear models, which is a theme to which we'll return many times over the coming chapters.

Output 10.2

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	3.750	.516		7.270	.000
	Cloak of invisibility	1.250	.730	.343	1.713	.101

a. Dependent Variable: Mischievous Acts



10.5 The *t*-test

We have looked at how we can include a categorical predictor in a linear model to test for differences between two means. This approach is useful in showing you the joy that is the linear model, and to keep the thread of linear models running through the book. Historically, people think about comparing two means as a separate test, and SPSS keeps this historical convention with its menu structure. This is not as bonkers as it may seem, because the linear model framework gets complicated when we want to tackle repeated-measures designs. Therefore, when testing the difference between two means, researchers tend to apply the *t*-statistic but masquerading as something called the *t*-test (Student, 1908). In this section we'll look at the theoretical underpinnings of the test. There are two variants of this test:

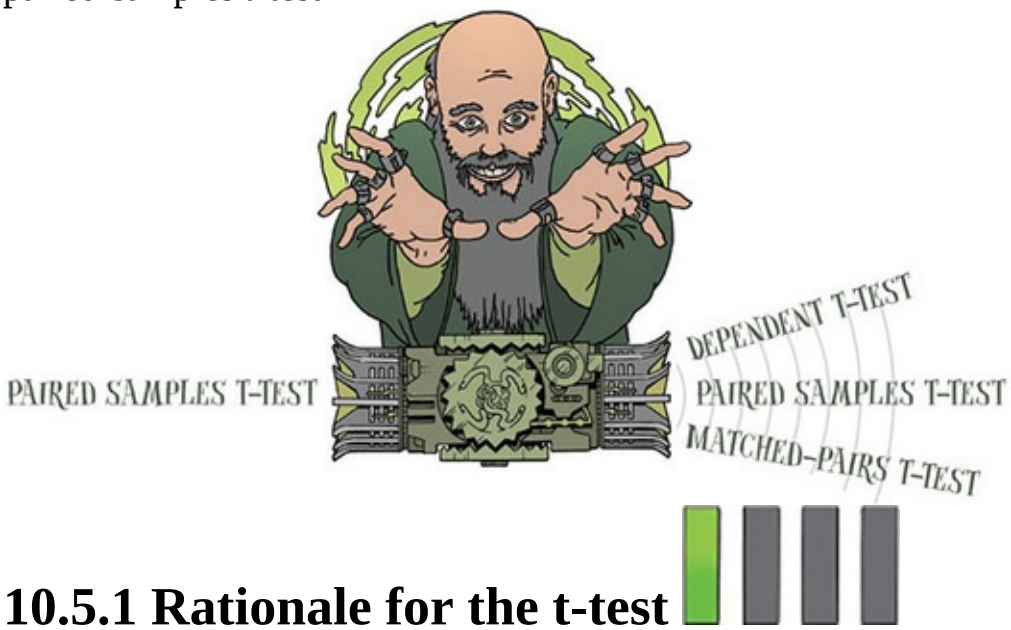


- **Independent *t*-test**: This test is used when you want to compare two means that come from conditions consisting of different entities (this is sometimes

called the *independent-measures* or *independent-means* *t*-test).

- **Paired-samples *t*-test**: This test, also known as the **dependent *t*-test**, is used when you want to compare two means that come from conditions consisting of the same or related entities (Figure 10.2).

Figure 10.2 Thanks to the Confusion machine there are lots of terms for the paired-samples *t*-test



10.5.1 Rationale for the t-test

Both *t*-tests have a similar rationale, which is based on what we learnt in Chapter 2 about hypothesis testing:

- Two samples of data are collected and the sample means calculated. These means might differ by either a little or a lot.
- If the samples come from the same population, then we expect their means to be roughly equal (see Section 2.7). Although it is possible for their means to differ because of sampling variation, we would expect large differences between sample means to occur very infrequently. Under the null hypothesis we assume that the experimental manipulation has no effect on the participant's behaviour: therefore, we expect means from two random samples to be very similar.
- We compare the difference between the sample means that we collected to the difference between the sample means that we would expect to obtain (in the long run) if there were no effect (i.e., if the null hypothesis were true). We use the standard error (see Section 2.7) as a gauge of the variability between sample means. If the standard error is small, then we expect most samples to have very similar means. When the standard error is large, large differences in sample means are more likely. If the difference between the

samples we have collected is larger than we would expect based on the standard error then one of two things has happened:

- There is no effect but sample means from our population fluctuate a lot and we happen to have collected two samples that produce very different means.
- The two samples come from different populations, which is why they have different means, and this difference is, therefore, indicative of a genuine difference between the samples. In other words, the null hypothesis is unlikely.
- The larger the observed difference between the sample means (relative to the standard error), the more likely it is that the second explanation is correct: that is, that the two sample means differ because of the different testing conditions imposed on each sample.

Most test statistics are a signal-to-noise ratio: the 'variance explained by the model' divided by the 'variance that the model can't explain' (reread Section 2.9.4). In other words, effect divided by error. When comparing two means the 'model' that we fit (the effect) is the difference between the two group means. Means vary from sample to sample (sampling variation), and we can use the standard error as a measure of how much means fluctuate (in other words, the error in the estimate of the mean) – see Chapter 2. Therefore, we can use the standard error of the differences between the two means as an estimate of the error in our model (or the error in the difference between means). Therefore, the *t*-statistic can be expressed as:

$$t = \frac{\text{observed difference between sample means} - \text{expected difference between population means (if null hypothesis is true)}}{\text{estimate of the standard error of the difference between two sample means}} \quad (10.5)$$

The top half of the equation is the 'model', which is that the difference between means is bigger than the expected difference under the null hypothesis, which in most cases will be 0. The bottom half is the 'error'. So, we're basically getting the test statistic by dividing the model (or effect) by the error in the model. The exact form that this equation takes depends on whether scores are independent (e.g., come from different entities) or related to each other (come from the same or related entities).

10.5.2 The paired-samples *t*-test equation explained



We'll start with the simpler scenario of when scores in the two conditions that you want to compare are related; for example, the same entities have been tested in the different conditions of your experiment, or perhaps you have data on a task from twins (you'd expect each person's score to be more similar to that of their twin than that of a stranger). If you choose not to think in terms of a linear model, then you can calculate the *t*-statistic using a numerical version of equation (10.5):



$$t = \frac{\bar{D} - \mu_D}{\sigma_{\bar{D}}} = \frac{\bar{D}}{\sigma_{\bar{D}}} \quad (10.6)$$

This equation compares the mean difference between our samples ($\overline{D}$) to the difference that we would expect to find between population means (μ_D), relative to the standard error of the differences ($\sigma_{\overline{D}}$). If the null hypothesis is true, then we expect no difference between the population means and $\mu_D = 0$ and it drops out of the equation.

Let's explore the logic of equation (10.6). Imagine you take a pair of samples from a population, calculate their means and then take the difference between them. We know from sampling theory (Section 2.7) that, on average, sample means will be very similar to the population mean; therefore, on average, most samples should have very similar means. Our pair of random samples should, therefore, have similar means, meaning that the difference between their means is zero, or close to zero. Imagine we repeated this process lots of times. We should find that most pairs of samples have differences between means that are close to zero, but sometimes one or both of the samples will have a mean very deviant from the population mean and we'd obtain a large difference between sample means. In short, sampling variation means that it's possible to get a difference between two sample means that is quite large, but it will happen relatively infrequently. If we plotted the frequency distribution of differences between means of pairs of samples we'd get the sampling distribution of differences between means. We might expect this distribution to be normal around zero, indicating that most pairs of samples have differences close to zero, and only very infrequently do we get large differences between sample means. The standard deviation of this sampling distribution is called the **standard error of differences**. Like any standard error (refresh your memory of Section 2.7 if

you need to), a small standard error suggests that the difference between means of most pairs of samples will be very close to the population mean (in this case 0 if the null is true) and that substantial differences are very rare. A large standard error tells us that the difference between means of most pairs of samples can be quite variable: although the difference between means of most pairs of samples will still be centred around 0 substantial differences from zero are more common (than when the standard error is small). As such, the standard error is a good indicator of the size of the difference between sample means that we can expect from sampling variation. In other words, it's a good baseline for what could reasonably happen if the conditions under which scores are collected are stable. The conditions under which scores are collected are not stable, though. In experiments, we systematically manipulate the conditions under which scores are collected. For example, to test whether looking like a human affects trust of robots, participants might have two interactions with a robot: in one the robot is concealed under clothes and realistic flesh, whereas in the other their natural titanium exoskeleton is visible. Each person's trust score in the first interaction could be different from the second; the question is whether this difference is the product of how the robot looked, or just what you'd get if you test the same person twice. The standard error helps us to gauge this by giving us a scale of likely variability between samples. If the standard error is small then we know that even a modest difference between scores in the two conditions would be unlikely from two random samples. If the standard error is large then a modest difference between scores is plausible from two random samples. As such, the standard error of differences provides a scale of measurement for how plausible it is that an observed difference between sample means could be the product of taking two random samples from the same population. That's what the bottom of equation (10.6) represents: it places the observed difference between sample means in the context of what's plausible for random samples.

The top half of equation (10.6) represents the size of the observed effect. $\overline{D}$ is the average difference between people's scores in two conditions. For each person, if we took their score in one condition and subtracted it from their score in the other, this would give us a difference score for each person; $\overline{D}$ is the mean of these difference scores. Going back to our robot example, if the appearance of the robot had no effect on people's trust, scores would be similar in the two conditions and we'd get an average difference of 0 (or close to it). If the robot's appearance matters, we'd expect scores to differ in the two conditions, and the resulting average difference would be different from 0.

So, $\overline{D}$ represents the effect size, and as I said earlier, we place this effect size

within the context of what's plausible for random samples by dividing by the standard error of differences. We know that the standard error can be estimated from the standard deviation divided by the square root of the sample size (equation (2.14) in Section 2.7). The standard error of differences ($\sigma_{\bar{D}}$) is likewise estimated from the standard deviation of differences within the sample (s_D) divided by the square root of the sample size (N). Replacing this term in equation (10.6) gives:

$$t = \frac{\bar{D}}{\sigma_{\bar{D}}} = \frac{\bar{D}}{s_D/\sqrt{N}} \quad (10.7)$$

Therefore, t is a signal-to-noise ratio or the systematic variance compared to the unsystematic variance. The top half of equation (10.7) is the signal or effect, whereas the bottom places that effect within the context of the natural variation between samples (the noise, or unsystematic variation). If the experimental manipulation creates difference between conditions, then we would expect the effect (the signal) to be greater than the unsystematic variation (the noise) and, at the very least, t will be greater than 1. We can compare the obtained value of t against the maximum value we would expect to get if the null hypothesis were true in a t-distribution with the same degrees of freedom (these values can be found in the Appendix). If the observed t exceeds the critical value for the predetermined alpha (usually 0.05), scientists tend to assume that this reflects an effect of their independent variable. We *can* compare the observed t to tabulated critical values, but we can also urinate in a bucket and throw it from our window into the street. It doesn't mean that we should, and seeing as it's not 1908 we won't do either of these things; instead we'll urinate in a computer and let a toilet compute an exact p-value for t. I think it's that way around. If the exact p-value for t is below the predetermined alpha value (usually 0.05), scientists take this to support the conclusion that the differences between scores are not due to sampling variation and that their manipulation (e.g., dressing up a robot as a human) has had a significant effect.

10.5.3 The independent *t*-test equation explained



When we want to compare scores that are independent (e.g., different entities have been tested in the different conditions of your experiment) we are in the same logical territory as when scores are related. The main difference is how we arrive at the values of interest. The equation for t based on independent scores is still a numerical version of equation (10.5). The main difference is that we're not

dealing with difference scores because there's no connection between scores in the two conditions that we want to compare.

When scores in two groups come from different participants, pairs of scores will differ not only because of the experimental manipulation reflected by those conditions, but also because of other sources of variance (individual differences between participants' motivation, IQ, etc.). These individual differences are eliminated when we use the same participants across conditions. Because the scores in the two conditions have no logical connection, we compare means on a *per condition* basis. We compute differences between the two sample means $(\bar{X}_1 - \bar{X}_2)$ and not between individual pairs of scores. The difference between sample means is compared to the difference we would expect to get between the means of the two populations from which the samples come ($\mu_1 - \mu_2$):

$$t = \frac{\left(\bar{X}_1 - \bar{X}_2\right) - \left(\mu_1 - \mu_2\right)}{\text{estimate of the standard error}} \quad (10.8)$$

If the null hypothesis is true then the samples have been drawn from populations that have the same mean. Therefore, under the null hypothesis $\mu_1 = \mu_2$, which means that $\mu_1 - \mu_2 = 0$, and so $\mu_1 - \mu_2$ drops out of the equation leaving us with:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\text{estimate of the standard error}} \quad (10.9)$$

Now, imagine we took several pairs of samples – each pair containing one sample from the two different populations – and compared the means of these samples. From what we have learnt about sampling distributions, we know that many samples from a population will have similar means. If the populations from which we're drawing have the same mean (which under the null hypothesis they do) then pairs of samples should also have the same means, and the difference between sample means should be zero, or close to it. We're now in very familiar territory to the paired-samples *t*-test because the sampling distribution of the differences between pairs of sample means would be normal with a mean equal to the difference between population means ($\mu_1 - \mu_2$), which under the null is zero. The sampling distribution would tell us by how much we can expect the means of two (or more) samples to differ (if the null were true). The standard deviation of this sampling distribution (the standard error) tells us how plausible differences between sample means are (under the null hypothesis). If the standard error is large then large differences between sample means can be expected; if it is small then only small differences between sample means are typical. As with related scores, it makes sense then to use the standard error to place the difference between sample means into the context of what's plausible given the null hypothesis. As such, equation (10.9) is conceptually the same as

equation (10.6); all that differs is how we arrive at the effect (top half) and the standard error (the bottom half). The standard error, in particular, is derived quite differently for independent samples.

I've already reminded you that the standard error can be estimated from the standard deviation and the sample size. It is, therefore, straightforward to estimate the standard error for the sampling distribution of each population by using the standard deviation (s) and size (N) for each sample:

$$SE \text{ of sampling distribution of population } 1 = \frac{s_1}{\sqrt{n_1}}$$
$$SE \text{ of sampling distribution of population } 2 = \frac{s_2}{\sqrt{n_2}} \quad (10.10)$$

These values don't tell us about the standard error for the sampling distribution of *differences* between means, though. To estimate that, we need to first convert these standard errors to variances by squaring them:[3]

3 Remember that a standard error is a standard deviation (it's just called a standard error because we're dealing with a sampling distribution), and the standard deviation is the square root of the variance.

$$\text{variance of sampling distribution of population } 1 = \left(\frac{s_1}{\sqrt{n_1}}\right)^2 = \frac{s_1^2}{n_1}$$
$$\text{variance of sampling distribution of population } 2 = \left(\frac{s_2}{\sqrt{n_2}}\right)^2 = \frac{s_2^2}{n_2} \quad (10.11)$$

Having converted to variances, we can take advantage of the **variance sum law**, which states that the variance of a difference between two independent variables is equal to the sum of their variances (see, for example, Howell, 2012). Put simply, the variance of the sampling distribution of differences between two sample means will be equal to the sum of the variances of the two populations from which the samples were taken. This law means that we can estimate the variance of the sampling distribution of differences by adding together the variances of the sampling distributions of the two populations:

$$\text{variance of sampling distribution of differences} = \frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \quad (10.12)$$

We convert this variance back to a standard error by taking the square root:

$$SE \text{ of sampling distribution of differences} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \quad (10.13)$$

If we pop this equation for the standard error of differences into equation (10.9) we get:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (10.14)$$

Equation (10.14) is true only when the sample sizes are equal, which in naturalistic studies may not be possible. To compare two groups that contain different numbers of participants we use a pooled variance estimate instead, which takes account of the difference in sample size by *weighting* the variance of each sample by a function of the size of sample on which it's based:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \quad (10.15)$$

This weighting makes sense because (as we saw in Chapter 1) large samples more closely approximate the population than small ones; therefore, they should carry more weight. In fact, rather than weighting by the sample size, we weight by the sample size minus 1 (the degrees of freedom).

The pooled variance estimate in equation (10.15) is a weighted average: each variance is multiplied (weighted) by its degrees of freedom, and then we divide by the sum of weights (or sum of the two degrees of freedom) to get an average. The resulting weighted average variance is plopped into the equation for *t*:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}} \quad (10.16)$$

One thing that might be apparent from equation (10.16) is that you don't actually need any raw data to compute *t;* you just need the group means, standard deviations and sample sizes (see SPSS Tip 10.1).

As with the paired-samples *t*, the Edwardians among you can compare the observed *t* to critical values in a table, but the rest of us flush a toilet and an exact *p*-value comes out that we use as information to decide whether the observed effect is indicative of something more theoretically interesting than sampling variation.

SPSS Tip 10.1 Computing *t* from means, *SD*s and *N*s 



Using syntax, you can compute an independent *t*-test from just the group means, standard deviations and sample sizes. Set up six variables in the data editor: **x1** (mean of group 1), **x2** (mean of group 2), **sd1** (standard deviation of group 1), **sd2** (standard deviation of group 2), **n1** (sample size of group 1) and **n2** (sample size of group 2). Type the values of each of these in the first row of the data editor. Open a syntax window and type the following:

```
COMPUTE df = n1+n2−2.
COMPUTE poolvar = (((n1−1)*(sd1**2))+((n2−1)*
(sd2**2)))/df.
COMPUTE t = (x1−x2)/sqrt(poolvar*((1/n1)+(1/n2))).
COMPUTE sig = 2*(1−(CDF.T(abs(t),df))).
Variable labels sig ’Significance (2-tailed)’.
EXECUTE.
```

The first line computes the degrees of freedom, the second the pooled variance, s_p^2, the third *t* and the fourth its two-tailed significance. All these values will be created in new columns in the data editor. The line beginning ‘Variable labels’ labels the significance variable so that we know that it is two-tailed. To display the results in the viewer window we could use this syntax:

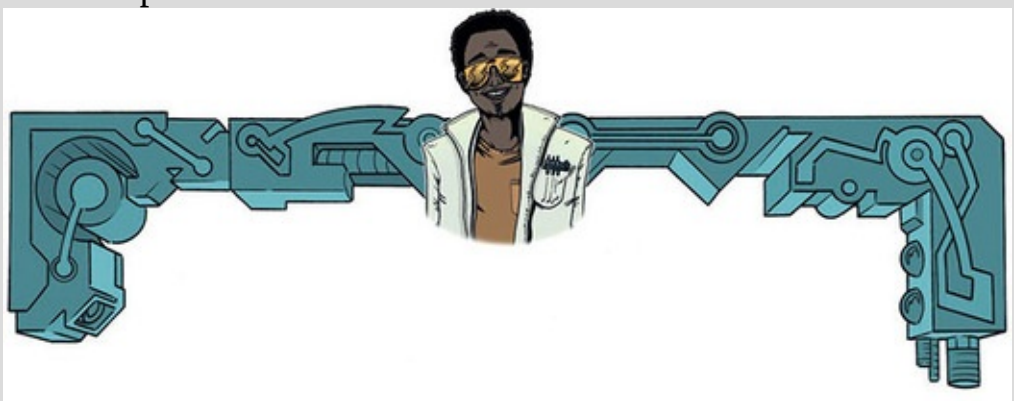
```
SUMMARIZE
/TABLES = x1 x2 df t sig
/FORMAT = VALIDLIST NOCASENUM TOTAL LIMIT =
100
/TITLE = ’T-test’
/MISSING = VARIABLE
/CELLS = NONE.
```

These commands produce a table of the variables **x1, x2, df, t** and

sig, so you'll see the means of the two groups, the degrees of freedom, the value of *t* and its two-tailed significance.
You can run lots of *t*-tests at the same time by putting different values for the means, SDs and sample sizes in different rows. If you do this, though, I suggest having a string variable called **Outcome** in the file in which you type what was being measured (or some other information so that you can identify to what the *t*-test relates). I've used these commands in a syntax file called **Independent t from means.sps**. My file is a bit more complicated because it also calculates Cohen's *d*. For an example of how to use it see Labcoat Leni's Real Research 10.1.



Labcoat Leni's Real Research 10.1 You don't have to be mad here, but it helps 



Board, B. J., & Fritzon, K. (2005). *Psychology, Crime & Law, 11*, 17–32.
In the UK you often see the 'humorous' slogan 'You don't have to be mad to work here, but it helps' stuck up in work places. Board and

Fritzon (2005) took this slogan a step further by measuring whether 39 senior business managers and chief executives from leading UK companies were mad (well, had personality disorders, PDs). They gave them the Minnesota Multiphasic Personality Inventory Scales for DSM III Personality Disorders (MMPI-PD), which measures 11 personality disorders: histrionic, narcissistic, antisocial, borderline, dependent, compulsive, passive-aggressive, paranoid, schizotypal, schizoid and avoidant. As a comparison group they chose 317 legally classified psychopaths from a high-security psychiatric hospital. The authors report the means and standard deviations for these two groups in Table 2 of their paper. Run the syntax file **Independent t from means.sps** on the data in **Board and Fritzon 2005.sav** to see whether managers score significantly higher on personality disorder questionnaires than legally classified psychopaths. Report these results. What do you conclude? Answers are on the companion website (or see Table 2 in the original article).

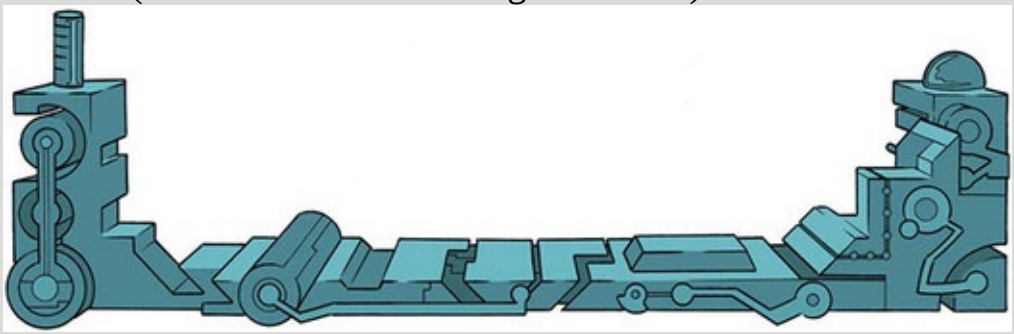
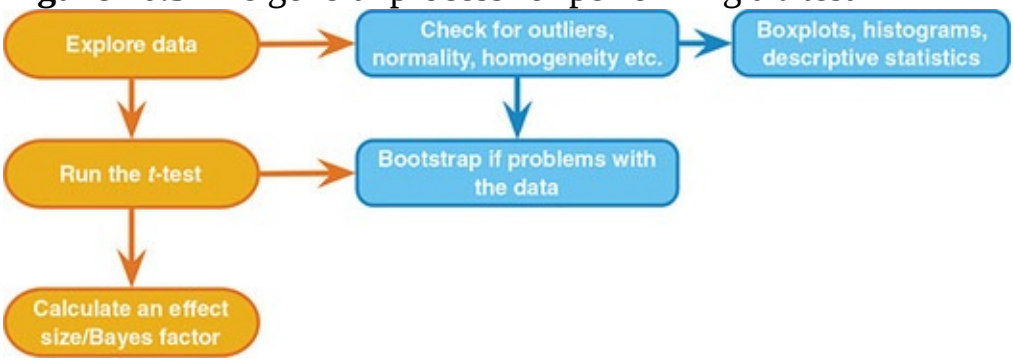


Figure 10.3 The general process for performing a *t*-test



10.6 Assumptions of the *t*-test

Both the independent *t*-test and the paired-samples *t*-test are *parametric tests* and

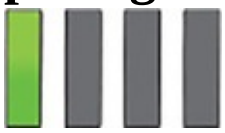
as such are prone to the sources of bias discussed in Chapter 6. For the paired-samples *t*-test the assumption of normality relates to the sampling distribution of the *differences* between scores, not the scores themselves (see Section 1.9.2). There are variants on these tests that overcome all of the potential problems, though.

10.7 Comparing two means: general procedure



I have probably bored most of you to the point of wanting to eat your own legs by now. Equations are tedious, and that is why computers were invented to help us minimize our contact with them. It's time to move on and do stuff. Figure 10.3 shows the general process for performing a *t*-test. As with fitting any model, we start by looking for the sources of bias identified in Chapter 6. Having satisfied ourselves that assumptions are met and outliers dealt with, we run the test. We can also consider using bootstrapping if any of the test assumptions were not met. Finally, we compute an effect size and Bayes factor.

10.8 Comparing two independent means using SPSS Statistics



Thinking back to our invisibility data (**Invisibility.sav**), we have 12 people who were given an invisibility cloak and 12 who were not (the groups are coded using the variable **Cloak**). Remember that the number of mischievous acts they performed was measured (**Mischief**). I have already described how the data are arranged (see Section 10.3), so we can move onto doing the test itself.

10.8.1 Exploring data and testing assumptions



We obtained descriptive statistics and looked at distributional assumptions in Section 10.3. We found evidence of normality in each group, and the mean

number of mischievous acts was higher for those with a cloak (M = 5) than for those without (M = 3.75). To look at homogeneity of variance (Section 6.11) SPSS will produce Levene's test when we run the *t*-test.

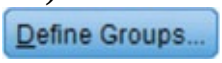


Produce an error bar chart of the **Invisibility.sav** data (**Cloak** will be on the *x*-axis and **Mischief** on the *y*-axis).

10.8.2 The independent *t*-test using SPSS Statistics



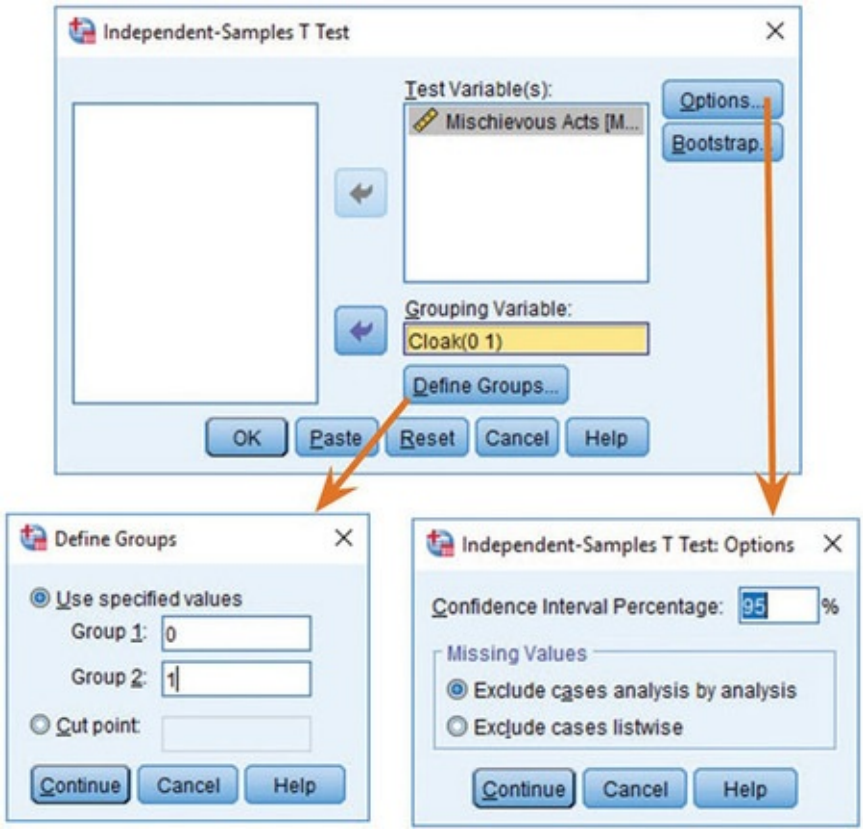
To run an independent *t*-test, we access the main dialog box by selecting *Analyze*  *Compare Means*  (see Figure 10.4). Once the dialog box is activated, select the outcome variable (**Mischief**) and drag it to the box labelled *Test Variable(s)* (or click ). If you want to carry out *t*-tests on several outcome variables at once then you can select several variables and transfer them to the *Test Variable(s)* box. However, remember that by doing lots of test you inflate the Type I error rate (see Section 2.9.7).

Next, we need to specify a predictor variable (the grouping variable). In this case, transfer **Cloak** to the box labelled *Grouping Variable*. The 

button will become active, so click it to activate the *Define Groups* dialog box. SPSS needs to know what numeric codes you assigned to your two groups, and there is a space for you to type the codes. In this example, we coded the no cloak group as 0 and our cloak group as 1, and so these are the values that we type into the boxes (as in Figure 10.4). Alternatively, you can specify a *Cut point* value whereby cases greater than or equal to that value are assigned to one group and values below the cut point to a second group. You can use this option to, for

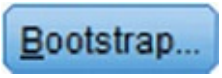
example, compare groups of participants based on something like a median split (see Jane Superbrain Box 10.1) – you would type the median value in the box labelled *Cut point*. When you have defined the groups, click Continue to return to the main dialog box.

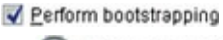
Figure 10.4 Dialog boxes for the independent-samples *t*-test



Clicking Options... activates a dialog box in which you can change the width of the confidence interval in the output. The default setting of a 95% confidence interval is fine, but there may be times where you want to be stricter (and specify something like a 99% confidence interval) even though you run a higher risk of failing to detect a genuine effect (a Type II error), or more lenient (e.g., a 90% confidence interval) which of course increases the chance of falsely accepting a hypothesis (a Type I error). You can also select how to deal with missing values (see SPSS Tip 6.1).

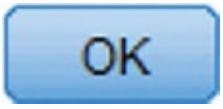
If you are worried about the assumption of normality, or simply want confidence intervals that don't rely on this assumption, then you can use bootstrapping

(Section 6.12.3). Select this option by clicking 

in the main dialog box to access the *Bootstrap* dialog box. We discussed this dialog box in Section 6.12.3; to recap, select 

to activate bootstrapping, and to get a 95% confidence interval click 

or 

. For this analysis, choose a bias corrected (BCa) confidence interval. Back in the main dialog box, click 

to run the analysis.

10.8.3 Output from the independent t-test 

The output from the independent *t*-test contains only three tables (two if you don't opt for bootstrapping). The first table (Output 10.3) provides summary statistics for the two experimental conditions (if you don't ask for bootstrapping this table will be a bit more straightforward). From this table, we can see that both groups had 12 participants (rows labelled *N*). The group that had no cloak performed, on average, 3.75 mischievous acts with a standard deviation of 1.913. What's more, the standard error of that group is 0.552. The bootstrap estimate of this standard error is a little smaller at 0.54, and the bootstrapped confidence interval for the mean ranges from 2.69 to 4.71. Those who were given an invisibility cloak performed, on average, 5 acts, with a standard deviation of 1.651, and a standard error of 0.477. The bootstrap estimate of this standard error is 0.47, and the confidence interval for the mean ranges from 4.20 to 5.79. Note that the confidence intervals for the two groups overlap, implying that they might be from the same population.

The second table of output (Output 10.4) contains the main test statistics. There are two rows containing values for the test statistics: one is labelled *Equal variances assumed*, while the other is labelled *Equal variances not assumed*. In Chapter 6, we saw that parametric tests assume that the variances in experimental groups are roughly equal. We also saw in Jane Superbrain Box 6.6 that there are adjustments that can be made in situations in which the variances are not equal. The rows of the table relate to whether these adjustments have been applied.

Output 10.3

Group Statistics

	Cloak of invisibility		Statistic	Bootstrap[a] Bias	Bootstrap[a] Std. Error	Bootstrap[a] BCa 95% Confidence Interval Lower	Bootstrap[a] BCa 95% Confidence Interval Upper
Mischievous Acts	No Cloak	N	12				
		Mean	3.75	.00	.54	2.69	4.71
		Std. Deviation	1.913	-.124	.367	1.206	2.258
		Std. Error Mean	.552				
	Cloak	N	12				
		Mean	5.00	.00	.47	4.20	5.79
		Std. Deviation	1.651	-.110	.329	1.131	1.958
		Std. Error Mean	.477				

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

Output 10.4

Independent Samples Test

		Levene's Test for Equality of Variances: F	Levene's Test for Equality of Variances: Sig.	t-test for Equality of Means: t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference: Lower	95% Confidence Interval of the Difference: Upper
Mischievous Acts	Equal variances assumed	.545	.468	-1.713	22	.101	-1.250	.730	-2.763	.263
	Equal variances not assumed			-1.713	21.541	.101	-1.250	.730	-2.765	.265

We saw in Section 6.11 that Levene's test tests whether variances are different in different groups. However, because the results of this test will depend on the sample size you have, and we can adjust the degrees of freedom of the *t*-test to compensate for the degree to which variances are unequal, there is a good argument for ignoring Levene's test and always reading results from the row labelled *Equal variances not assumed* (see Jane Superbrain Box 6.6).

We are told the mean difference ($\bar{X}_{\text{No Cloak}} - \bar{X}_{\text{Cloak}} = 3.75 - 5 = -1.25$) and the standard error of the sampling distribution of differences, which is calculated using the lower half of equation (10.14). The *t*-statistic is the mean difference divided by this standard error ($t = -1.25/0.730 = -1.71$). A *p*-value associated with this *t* is calculated based on a *t*-distribution with particular degrees of freedom. For the independent *t*-test, the degrees of freedom are calculated by adding the two sample sizes and then subtracting the number of samples ($df = N_1 + N_2 - 2 = 12 + 12 - 2 = 22$). This value is then reduced to compensate for any imbalance in group variances (in this case it becomes 21.54). The resulting (two-tailed) *p*-value is 0.107, which represents the probability of getting a *t* of –1.71 or smaller if the null hypothesis were true. Assuming our alpha is 0.05, we'd conclude that there was no significant difference between the means of these two samples because the observed *p* of 0.101 is greater than the criterion of 0.05. In terms of the experiment, we can infer that having a cloak of

invisibility did not significantly affect the amount of mischief a person got up to. Note that the value of *t* and the significance value are the same as when we ran the same test as a regression (see Output 10.2).[4]

4 The value of the *t*-statistic is the same but has a positive sign rather than negative. You'll remember from the discussion of the point-biserial correlation in Section 8.4.5 that when you correlate a dichotomous variable the direction of the correlation coefficient depends entirely upon which cases are assigned to which groups. Therefore, the direction of the *t*-statistic here is similarly influenced by which group we select to be the base category (the category coded as 0).

10.8.4 Robust tests of two independent means



Assuming you had reason to doubt the assumption of normality, you could interpret the bootstrap confidence intervals (Output 10.5). It's also a legitimate approach to always use bootstrapped confidence intervals and not think about normality one way or another. The table shows a robust re-estimate of the standard error of the mean difference (0.703 rather than 0.730, the value in Output 10.4).[5] The difference between means was −1.25, and the bootstrap confidence interval ranges from −2.653 to 0.111, which implies that the difference between means in the population could be negative, positive or even zero (because the interval ranges from a negative value to a positive one). In other words, if we assume that this is one of the 95% of intervals that will capture the population value, it's possible that the true difference between means is zero – no difference at all. The bootstrap confidence interval confirms our conclusion that having a cloak of invisibility seems not to affect acts of mischief.

5 Remember that the values for the standard error and confidence interval you get could differ from mine because of the way bootstrapping works.

Output 10.5

Bootstrap for Independent Samples Test

			Bootstrap[a]			
					BCa 95% Confidence Interval	
		Mean Difference	Bias	Std. Error	Lower	Upper
Mischievous Acts	Equal variances assumed	-1.250	.001	.703	-2.653	.111
	Equal variances not assumed	-1.250	.001	.703	-2.653	.111

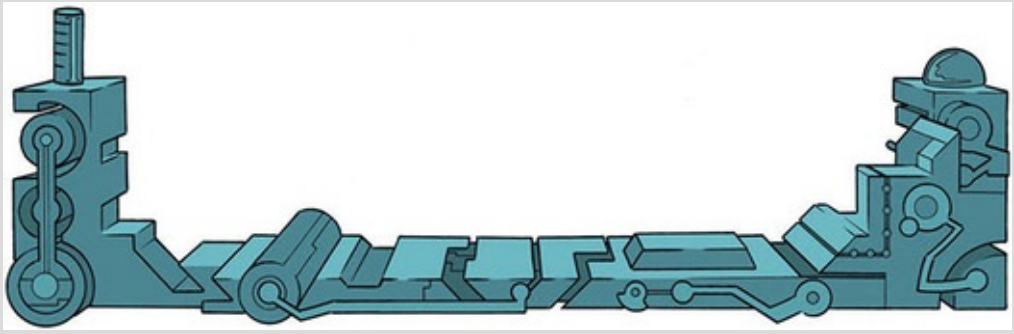
a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

Labcoat Leni's Real Research 10.2 Bladder control 



Tuk, M. A., *et al.* (2011). *Psychological Science*, *22*(5), 627–633.

Visceral factors that require us to engage in self-control (such as a filling bladder) can affect our inhibitory abilities in unrelated domains. In a fascinating study by Tuk, Trampe, and Warlop (2011) participants were given five cups of water: one group was asked to drink them all, whereas another was asked to take a sip from each. This manipulation led one group to have full bladders and the other group relatively empty (**Drink_Group**). Later on, these participants were given eight trials on which they had to choose between a small financial reward that they would receive soon (SS) or a large financial reward for which they would wait longer (LL). They counted how many trials participants chose the LL reward as an indicator of inhibitory control (**LL_Sum**). Do a *t*-test to see whether people with full bladders inhibited more than those without (**Tuk *et al.* (2011).sav**). Answers are on the companion website or see p. 629 of the article.



It's also possible to run a robust version of the *t*-test itself using R. To do this you need the Essentials for R plugin installed (Section 4.13.2) and you need to have installed the *WRS2* package (Section 4.13.3). The companion website contains a syntax file (**Robust independent t-test.sps**) for running a robust variant of the *t*-test based on Yuen (1974) which uses trimming and a bootstrap. The syntax in the file is as follows:

```
BEGIN PROGRAM R.
library(WRS2)
mySPSSdata = spssdata.GetDataFromSPSS(factorMode = "labels")
yuenbt(Mischief~Cloak, data = mySPSSdata)
END PROGRAM.
```

To obtain the robust test select and run these five lines of syntax, which are explained in SPSS Tip 10.2.

Having run the syntax, you'll find some rather uninspiring text output in the viewer (Output 10.6). This tells us that there is not a significant difference (because the confidence interval crosses zero and the *p*-value is greater than 0.05) in mischief scores across the two cloak groups, $Y_t = -1.36$ (−2.52, 0.52), $p = 0.167$.

Output 10.6

```
Test statistic: -1.3607 (df = NA), p-value = 0.16694
Trimmed mean difference: -1
95 percent confidence interval:
-2.5161 0.5161
```

SPSS Tip 10.2 Robust independent *t*-test 



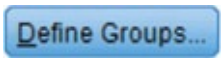
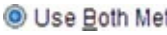
The first and lasts line of the syntax in **Robust independent t-test.sps** tell SPSS to begin and end communication with R, respectively. The three lines in between tell R what to do—these lines are R code, not SPSS syntax, and so do not end in full stops. The first loads the *WRS2* package that contains the function we want to use. The second grabs your data from the SPSS data editor and places it in a data frame called **mySPSSdata** (you can change this name if you like, but you don't need to). The final line runs the test using the function *yuenbt()*. Within this function we specify the data as being the data that we've just grabbed from SPSS, and we specify a formula for the test itself. This formula has the general form of *outcome~predictor*, in which *outcome* is the name of the outcome variable (from the data editor), and *predictor* is the name of the grouping variable. For these data the formula is **Mischief~Cloak** (which you could read as *predict mischief from cloak*). If you want to use this syntax on other data sets the only part of it that you need to edit is the formula by replacing **Mischief** and **Cloak** with the names of your variables.



10.8.5 Bayesian test of two independent means

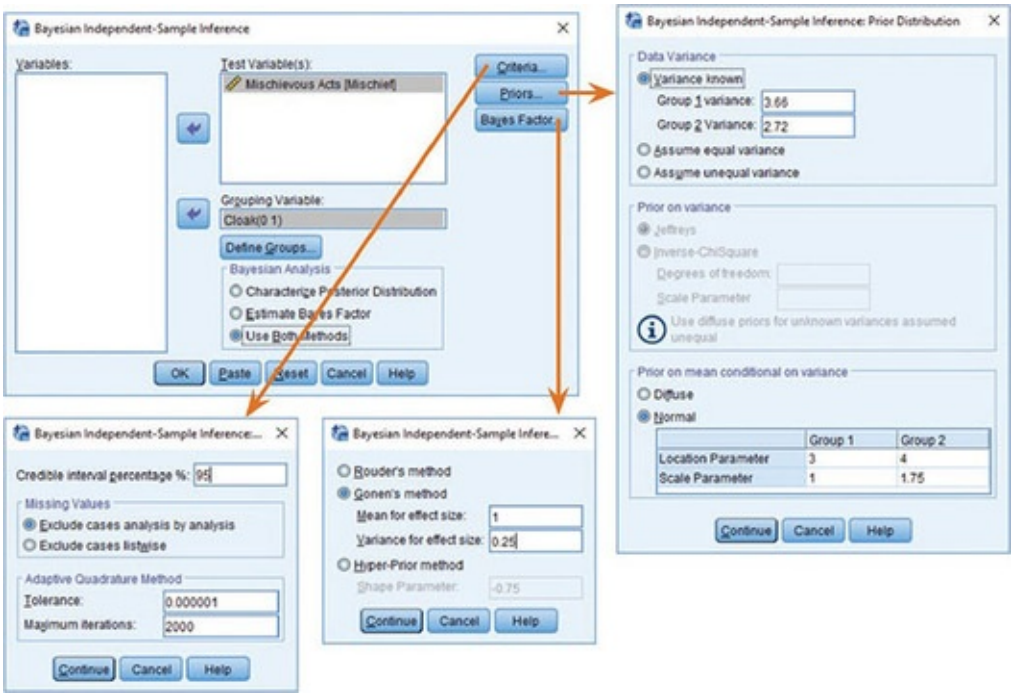


To access the dialog box to compute a Bayes factor and credible interval for

independent means select *Analyze*  *Bayesian Statistics*  *Independent Samples Normal* (Figure 10.5). Select the outcome variable (**Mischief**) and drag it into the box labelled *Test Variable(s)* (or click ), and select the group variable (**Cloak**) and drag it to the box labelled *Grouping Variable* (or click ). As with the regular *t*-test use the  button to tell SPSS that the first group (no cloak) was coded with 0 and group 2 (cloak) was coded with 1. To compute Bayes factors and estimate the model parameters select  .

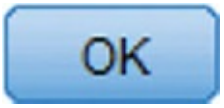
Imagine that prior to collecting data we believed that people without an invisibility cloak might be likely to perform anywhere between 0 and 6 mischievous acts. Obviously, we can't believe in a value less than 0 (you can't have a negative number of mischievous acts), but we're not prepared to believe in a value over 6. With a cloak of invisibility, we think the number of mischievous acts might fall anywhere between 0 and about 8. We could model these prior beliefs with a normal distribution. The graphs with red lines in Output 10.7 show these priors. The left graph is the prior for the no cloak group. As you can see it is a normal distribution centred at 3 and ranging from about 0 to 6. This equates to us believing that without an invisibility cloak, 3 acts of mischief are most probable, but we're prepared to accept values as low as 0 or as high as 6 (although we think these values are unlikely). The right graph is the prior for the cloak group, which is also a normal distribution but centred at 4 and ranging from about 0 to 8. This equates to us believing that, with an invisibility cloak, 4 acts of mischief are most probable, but we're prepared to accept (with low probability) values as low as 0 or as high as 8.

Figure 10.5 Dialog box for a Bayesian test of two independent means



To set up these priors click Priors... and first specify the variances of the two groups (remember that group 1 is the no cloak and group 2 is the cloak group). I've used the variances from the data, which I obtained by squaring the standard deviations in Output 10.3. Using these variances simplifies things (we don't need to set a prior for the variance) but as you get more experience you might want to estimate the variance too. At the bottom select Normal. For group 1 I have specified a distribution with mean (location) of 3 and standard deviation (scale) of 1. For group 2 I have specified a distribution with a mean of 4 and standard deviation of 1.75. To get an appropriate scale parameter, start by taking the range of plausible scores and divide by 6, then tweak it until the distribution looks correct. For example, for group 2 the range of plausible scores was 8, dividing by 6 gave me 1.33 but the resulting distribution was too narrow so I increased it slightly. If you want a credible interval other than 95% then click Criteria... and change the 95 to the value you want.

Click Bayes Factor... if you want to get a Bayes factor for the model of no difference between means (the null) relative to the model of a difference between means (the alternative). I have selected Gonen's method (Gönen, Johnson, Lu, & Westfall, 2005), because it allows us to incorporate some prior information rather than relying on defaults. Specifically, I have defined a prior

belief that the difference between means will be 1 (having an invisibility cloak would lead to one additional mischievous act compared to not having the cloak), and I've set this effect to have a variance of 0.25. Back in the main dialog box click 

to fit the model.

Output 10.7

Bayes Factor Independent Sample Test (Method = Gonen)[a]

	Mean Difference	Pooled Std. Error Difference	Bayes Factor[b]	t	df	Sig.(2-tailed)
Mischievous Acts	1.25	.730	.419	1.713	22	.101

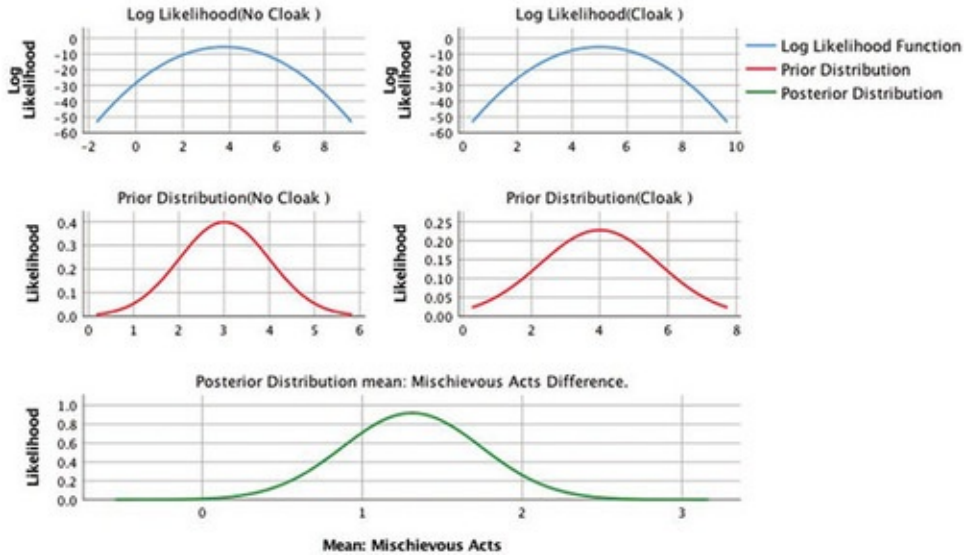
a. Assumes unequal variance between groups.

b. Bayes factor: Null versus alternative hypothesis.

Posterior Distribution Characterization for Independent Sample Mean[a]

	Posterior			95% Credible Interval	
	Mode	Mean	Variance	Lower Bound	Upper Bound
Mischievous Acts	1.31	1.31	.434	.02	2.60

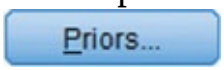
a. Prior for Variance: Diffuse. Prior for Mean: Normal.



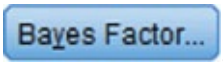
Output 10.7 shows that the Bayes factor is 0.419. SPSS reports the ratio of the null to the alternative hypothesis, so this value means that the data are 0.419 times as probable under the null hypothesis as under the alternative. If we divide 1 by the value (1/0.419 = 2.39) we flip the interpretation to say that the data are 2.39 times as probable under the alternative hypothesis as under the null. In other words, we should shift our belief towards the alternative hypothesis by a factor of 2.39. Although this effect is in the direction we hypothesised, it is not particularly strong evidence for the hypothesis that invisibility cloaks lead to

mischief.

Output 10.7 also shows our prior distributions for the group means (the graphs with the red lines). I have already described these. The Bayesian estimates of the difference between means (i.e., the *b*-value for group as a predictor of mischief) are in the columns labelled *Posterior Mode* and *Posterior Mean*. The 95% credible interval for this estimate ranged from 0.02 to 2.60. In other words, *assuming that the effect exists*, the population value of the effect will be between 0.02 to 2.60 with 95% probability. This tells us nothing about the null hypothesis (because it assumes the effect exists) but helps us to ascertain the likely population value if we're prepared to accept that the effect exists. So, we can say with 95% probability that having a cloak of invisibility will increase mischievous acts by anything as low as 0.02 (i.e., not at all) to 2.60 (although, of course, you can't have 0.6 of a mischievous act!).

If you want to see what happens when you use reference priors, then repeat this process but select Assume unequal variance when you click on 

and 

when you click 

. Doing so will fit a model with diffuse, uninformative priors.

10.8.6 Effect sizes for two independent means



The fact that the *t*-statistic is not statistically significant doesn't mean that our effect is unimportant, so it's worth quantifying it with an effect size (see Section 3.5). We can convert a *t*-value into an *r*-value using the following equation (e.g. Rosenthal, 1991; Rosnow & Rosenthal, 2005):

$$r = \sqrt{\frac{t^2}{t^2 + df}} \tag{10.17}$$

Using the value of *t* and the *df* from Output 10.4, we get:

$$r = \sqrt{\frac{(-1.713)^2}{(-1.713)^2 + 22}} = \sqrt{\frac{2.93}{24.93}} = 0.34 \tag{10.18}$$

Thinking back to our benchmarks for effect sizes, this represents a medium

effect. Therefore, even though the effect was non-significant, it represents a fairly substantial effect.

We could instead compute Cohen's *d* (Section 3.7.1), using the two means (5 and 3.75) and the standard deviation of the control group (no cloak):

$$\hat{d} = \frac{\bar{X}_{\text{Cloak}} - \bar{X}_{\text{No Cloak}}}{s_{\text{No Cloak}}} = \frac{5 - 3.75}{1.91} = 0.65 \quad (10.19)$$

This means that there is 0.65 standard deviations difference between the two groups in terms of their mischief making, which again is fairly substantial.

Oditi's Lantern *t*-tests



'I, Oditi, leader of the cult of undiscovered numerical truths, do not like differences. Everyone must conform to my cultish ideas, my view is the only view, and we must wage war on those who dare to have different views. Only by locating differences can we eliminate them and turn the world into mindless clones. Stare into my lantern to discover how to detect these differences. The more you stare, the more you will find yourself agreeing with everything I say …'



Cramming Sam's Tips The independent *t*-test



- The independent *t*-test compares two means, when those means have come from different groups of entities.
- You should probably ignore the column labelled *Levene's Test for Equality of Variance* and always look at the row in the table labelled *Equal variances not assumed*.
- Look at the column labelled *Sig*. If the value is less than 0.05 then the means of the two groups are significantly different.
- Look at the table labelled *Bootstrap for Independent Samples Test* to get a robust confidence interval for the difference between means.
- Look at the values of the means to see how the groups differ.
- A robust version of the test can be computed using syntax.
- A Bayes factor can be computed that quantifies the ratio of how probable the data are under the alternative hypothesis compared to the null.
- Calculate and report the effect size. Go on, you can do it! ☺



10.9 Comparing two related means using SPSS

Statistics

10.9.1 Entering data

Let's imagine that we had collected the cloak of invisibility data using a repeated-measures design: we might have recorded everyone's natural level of mischievous acts in a week, then given them an invisibility cloak and counted the number of mischievous acts in the following week.[6] The data will be identical to the previous example, not because I am too lazy to generate different scores, but because it allows me to illustrate various things.

6 In theory we'd counterbalance the weeks so that some people had the cloak and then it was taken away, whereas others had no cloak but were then given one. However, given that the research scenario relied on participants not knowing about the cloaks of invisibility it might be best just to have a baseline phase and then give everyone their cloak at the same time (unaware that others were getting cloaks too).



Enter the data in Table 10.1 into the data editor as though a repeated-measures design was used.

The data would now be arranged differently in the data editor. Instead of having a coding variable, and a single column with mischief scores in, we would arrange the data in two columns (one representing the **Cloak** condition and one representing the **No_Cloak** condition). The data are in **Invisibility RM.sav** if you had difficulty entering them yourself.

10.9.2 Exploring repeated-measures data 

We talked about the assumption of normality in Chapter 6. With the paired-samples *t*-test we're interested in the sampling distribution of the difference scores (not the raw scores). Therefore, if you want to test for normality before a paired-samples *t*-test then you should compute the differences between scores, and then check if this new variable is normally distributed as a proxy for the sampling distribution (or use a big sample or robust test and not worry about normality ☺). It is possible to have two measures that are highly non-normal and produce beautifully distributed differences.



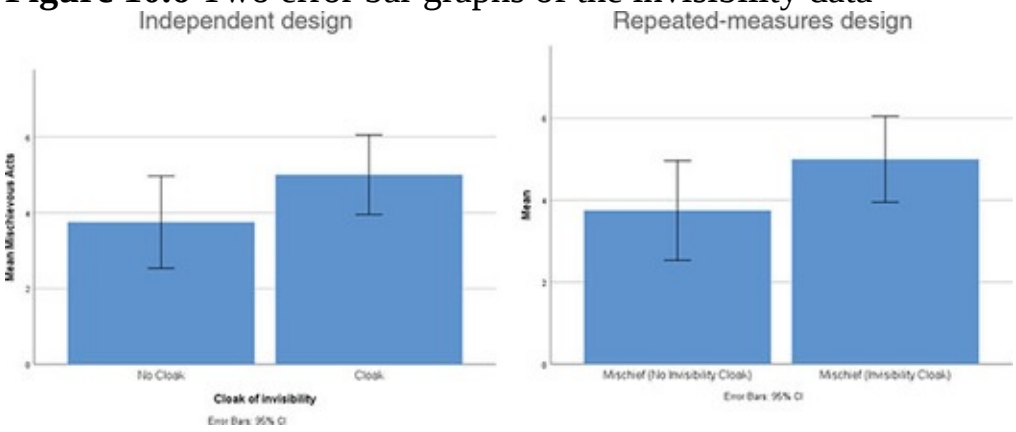
Using the **Invisibility RM.sav** data, compute the differences between the cloak and no cloak conditions and check the assumption of normality for these differences.

We saw in Chapter 5 that you can visualize group differences using error bars. However, there's a problem when we graph error bars from repeated-measures designs.



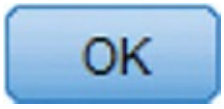
Produce an error bar chart of the **Invisibility RM.sav** data (**Cloak** on the *x*-axis and **Mischief** on the *y*-axis).

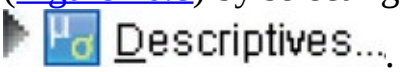
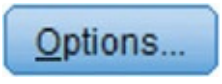
Figure 10.6 Two error bar graphs of the invisibility data



In one of the earlier self-tests I asked you to produce an error bar graph for the data when we treated it as an independent design. Compare that graph to the one you have just produced. Figure 10.6 shows both graphs; can you spot the difference? Hopefully you can't, because the graphs are identical (apart from axis labels). This is odd, isn't it? We discovered in Chapter 1 that repeated-measures designs eliminate some extraneous variables (such as age, IQ and so on), so why doesn't the graph of the repeated-measures design reflect the increased sensitivity of the design? It's because SPSS Statistics treats the data as though scores are independent, and consequently the error bars do not reflect the 'true' error around the means for repeated-measures designs. Let's correct these

error bars.

First, we need to calculate the average mischief for each participant. Select *Transform*  to access the *Compute Variable* dialog box (see Section 6.12.6). Enter the name **Mean** into the box labelled *Target Variable*. In the list labelled *Function group* select *Statistical*, and then in the list labelled *Functions and Special Variables* select *Mean*. Click  to transfer this command to the command area. The command will appear as *MEAN(?,?)*, and we need to replace the question marks with variable names by typing them or transferring them from the variables list. Replace the first question mark with the variable **No_Cloak** and the second one with the variable **Cloak**. The completed dialog box should look like Figure 10.7. Clicking  will create this new variable in the data editor.

The **grand mean** is the mean of all scores, and for the current data this value will be the mean of all 24 scores. The means we have just calculated are the average score for each participant; if we take the average of those mean scores, we will have the grand mean – phew, there were a lot of means in that sentence. We can use the *Descriptives* command (you could also use the *Explore* or *Frequencies* commands that we came across in Chapter 6, but as I've already covered those we'll try something different). Access the *Descriptives* dialog box (Figure 10.8) by selecting *Analyze*  *Descriptive Statistics*  . Select the variable **Mean** from the list and drag it to the box labelled *Variable(s)* (or click ). Clicking  activates a second dialog box in which we want to check only the option for the

mean (that's all we are interested in). The resulting output gives us the mean of the variable that we called **Mean**, which, remember, contains the average score for each participant across the cloak and no cloak conditions. This value (4.375) is the grand mean.

If you look at the variable labelled **Mean**, you'll notice that the values for each participant are different, which tells us that some people were, on average, more mischievous than others across the conditions. The fact that participants' mean mischief scores differ represents individual differences (it shows that some participants are generally more mischievous than others). According to Loftus and Masson (1994), these individual differences contaminate the error bars and need to be removed. In effect we want to equalize the means between participants (i.e., adjust the scores in each condition such that when we take the mean score across conditions, it is the same for all participants). To do this, we use the *compute* function to calculate an adjustment factor by subtracting each participant's mean score from the grand mean. Activate the *Compute Variable* dialog box, give the target variable a name (I suggest **Adjustment**) and use the command '4.375 − Mean'. This command will take the grand mean (4.375) and subtract from it each participant's average mischief level (see Figure 10.9).

Figure 10.7 Using the *compute* function to calculate the mean of two columns

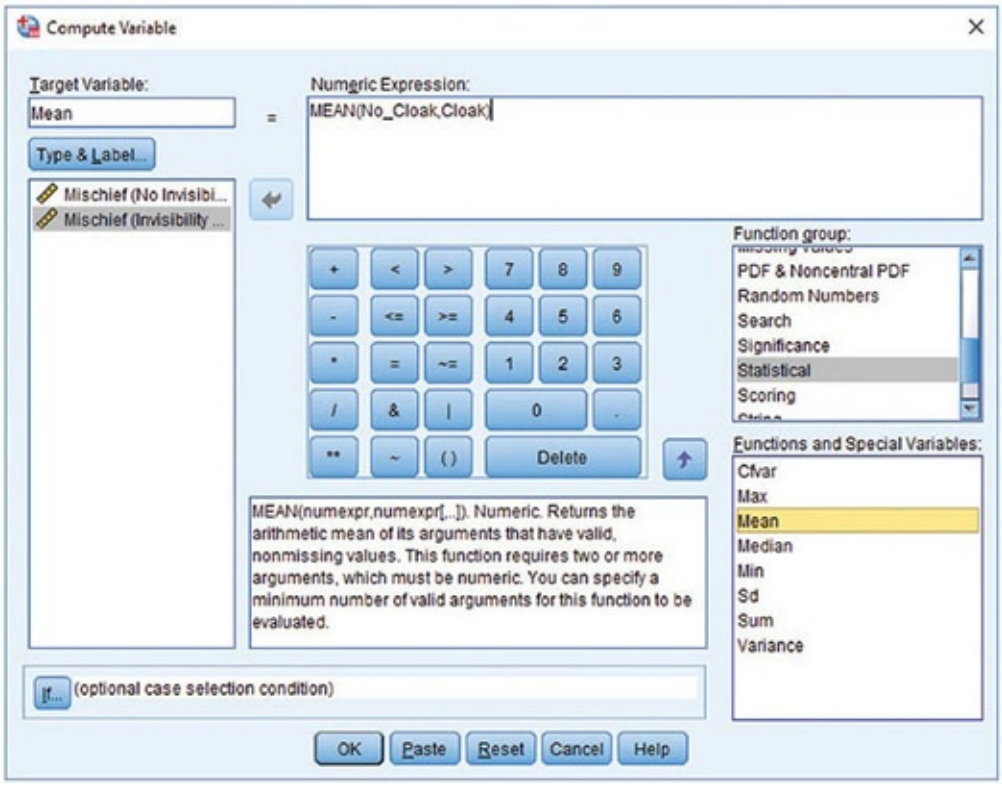


Figure 10.8 Dialog boxes and output for descriptive statistics

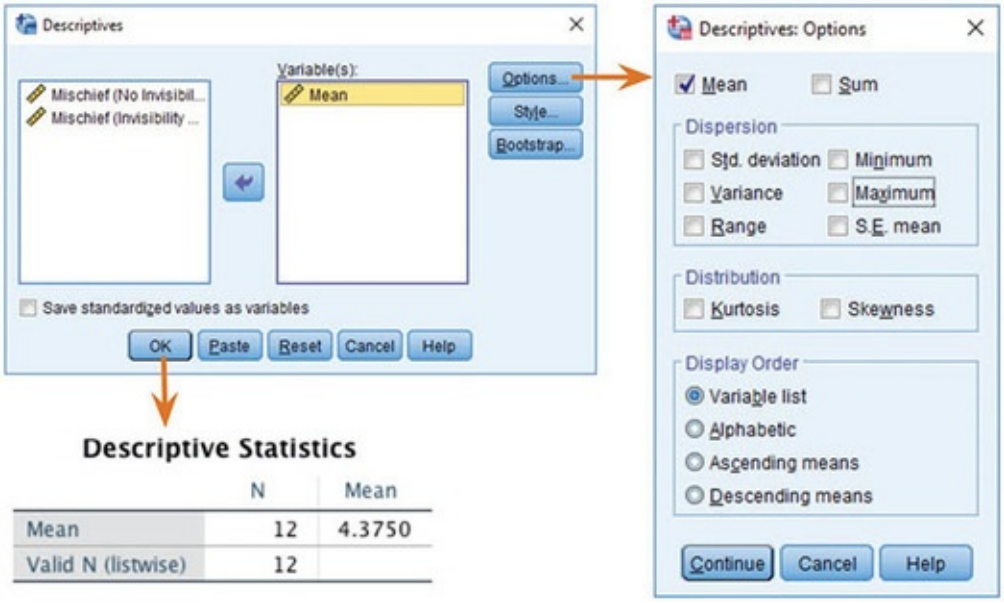
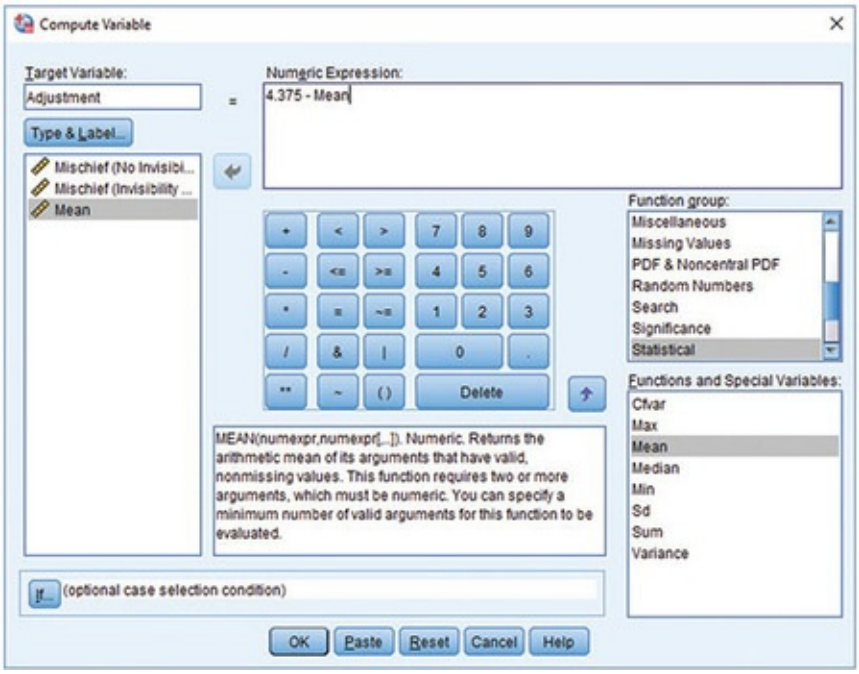
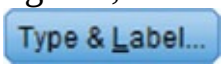


Figure 10.9 Calculating the adjustment factor



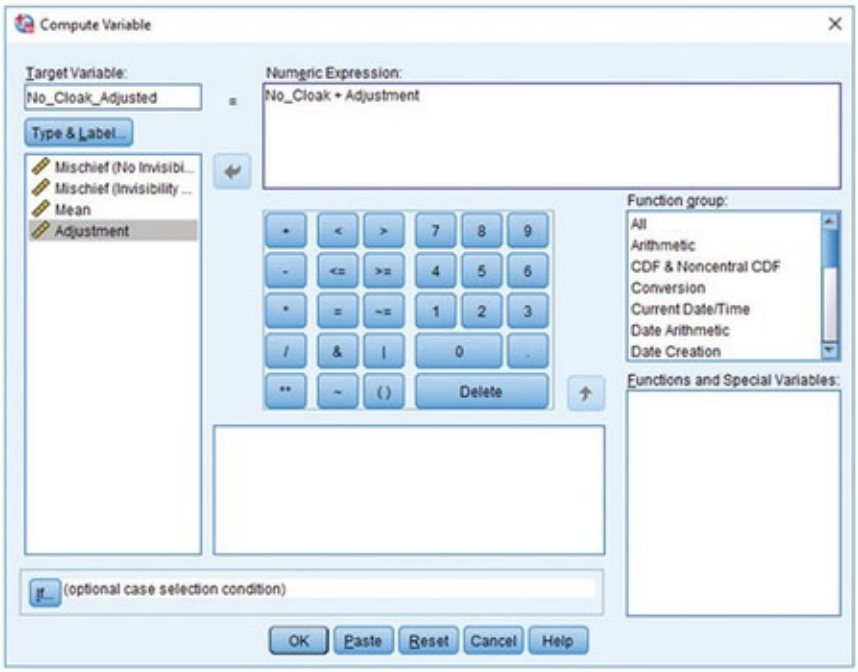
This process creates a new variable in the data editor called **Adjustment** which contains the difference between each participant's mean mischief levels and the mean mischief level across all participants. Some of the values are positive (participants who were less mischievous than average) and others negative (participants who were more mischievous than average). We can use these adjustment values to eliminate the between-subject differences in mischief.

First we'll adjust the scores in the **No_Cloak** condition again using the *Compute* command. Activate the *Compute* dialog box, and title the new variable **No_Cloak_Adjusted** (you can click 

to give this variable a label such as 'No Cloak Condition: Adjusted Values'). All we need to do is add each participant's score in the **No_Cloak** condition to their adjustment value. Select the variable **No_Cloak** and drag it to the command area (or click ), then click  and select the variable **Adjustment** and transfer it to the command area. The completed dialog box is in Figure 10.10. Do the same thing for the variable **Cloak**: create a variable called **Cloak_Adjusted** that contains the values of **Cloak** added to the value in the **Adjustment** column.

The variables **Cloak_Adjusted** and **No_Cloak_Adjusted** represent the mischief experienced in each condition, adjusted to eliminate between-subject differences. If you don't believe me, use the *Compute* command to create a variable **Mean2** that is the average of **Cloak_Adjusted** and **No_Cloak_Adjusted**. You should find that the value in this column is the same for every participant (it will be 4.375, the grand mean), demonstrating that the between-subject variability in means is gone.

Figure 10.10 Adjusting the values of **No_Cloak**





Create an error bar chart of the mean of the adjusted values that you have just made (**Cloak_Adjusted** and **No_Cloak_Adjusted**).

Compare the resulting error bar graph (Figure 10.11) to the graphs in Figure 10.6 —what difference do you notice? The first point to make is that the means in the two conditions have not changed. However, the error bars have got smaller. Also, whereas in Figure 10.6 the error bars overlap, in this new graph they do not. Therefore, when we plot the proper error bars for the repeated-measures data it shows the extra sensitivity that this design has: the differences between conditions appear to be significant (the error bars don't overlap), whereas when different participants are used, there does not appear to be a significant difference (error bars overlap a lot). Remember that the means in both situations are identical, but the sampling error is smaller in the repeated-measures design – I expand upon this point in Section 10.10.

10.9.3 The paired-samples t-test using SPSS Statistics



To conduct a paired-samples *t*-test, access the dialog box in Figure 10.12 by selecting *Analyze* *Compare Means* Paired-Samples T Test... . We need to select pairs of variables to be analyszed. In this case we have only one pair (**Cloak** vs. **No_Cloak**). To select a pair click the first variable that you want to select (in this case **No_Cloak**), then hold down the *Ctrl* key (*Cmd* on a Mac) and select the second (in this case **Cloak**). Transfer these variables to the box labelled *Paired Variables* by clicking . (You can select and transfer each variable individually, but selecting both variables as just described is quicker.) If you want to carry out several *t*-tests then you can select another pair of variables in the same way. Clicking Options... activates another dialog box that gives you the same options as for the independent *t*-test. Similarly, click Bootstrap... to access the bootstrap function (Section 6.12.3). As with the independent *t*-test, select Perform bootstrapping and Bias corrected accelerated (BCa). Back in the main dialog box, click OK to run the analysis.

Figure 10.11 Error bar graph of the adjusted values of the data in **Invisibility RM.sav**

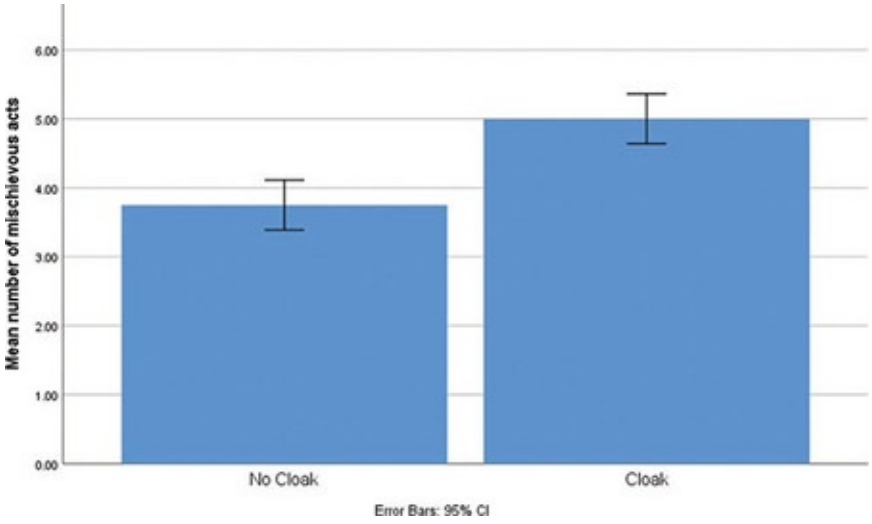
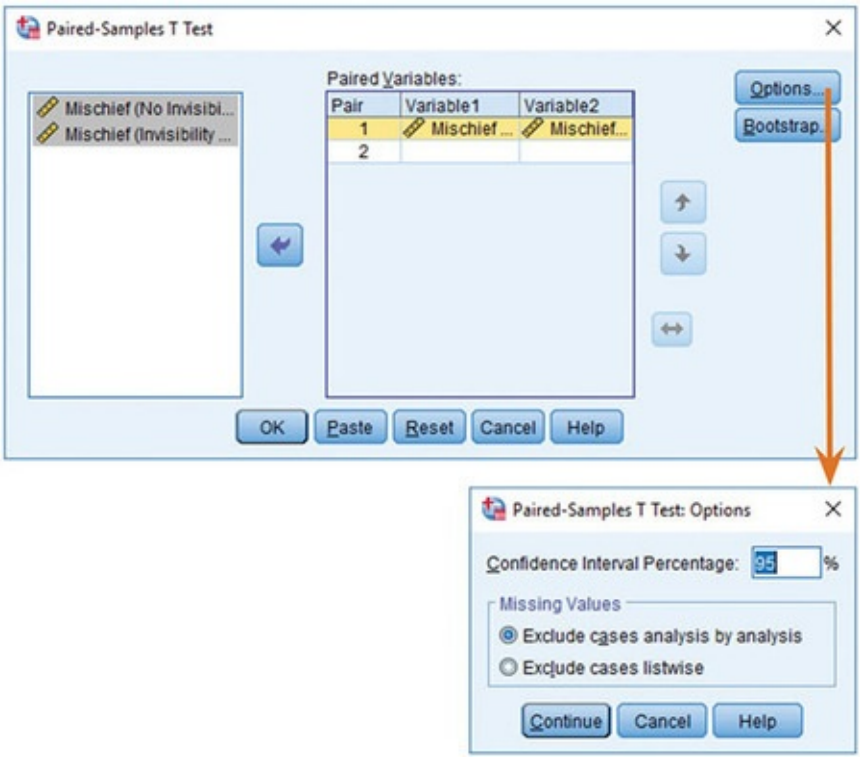


Figure 10.12 Main dialog box for paired-samples *t*-test



10.9.4 Output from the paired-samples t-test



The resulting output produces four tables (three if you don't select bootstrapping). Output 10.8 shows a table of summary statistics for the two experimental conditions (if you don't ask for bootstrapping this table will be a bit more straightforward). For each condition we are told the mean, the number of participants (*N*), the standard deviation and standard error. These values are the same as when we treated the data as an independent design and were described in Section 1.8.3.

Output 10.8 also shows the Pearson correlation between the two conditions. When repeated measures are used, scores in the experimental conditions will correlate to some degree (because the data in each condition come from the same entities you'd expect some constancy in their responses). The output contains the value of Pearson's *r* and the two-tailed significance value (see Chapter 8). For these data the experimental conditions yield a very large, highly significant,

correlation coefficient, $r = 0.806$, $p = 0.002$, with a bootstrap confidence interval that doesn't include zero, BCa 95% CI [0.417, 0.943].

Output 10.9 tabulates the main *t*-test results. The table contains the average difference score ($\overline{D}$ in equation (10.6)), which is 3.75 − 5 = −1.25, the standard deviation of the difference scores (1.138), and the standard error of difference scores (0.329, the lower half of equation (10.6)). The test statistic, *t*, is calculated using equation (10.6) by dividing the mean of differences by the standard error of differences ($t = -1.25/0.329 = -3.804$). For a repeated-measures design the degrees of freedom are the sample size minus 1 ($df = N - 1 = 11$). The *p*-value in the column labelled *Sig.* is the long-run probability that a value of *t* at least the size of the one obtained could occur if the population mean of difference scores was 0. This two-tailed probability is very low ($p = 0.003$); it tells us that there is only a 0.3% chance that a value of *t* of −3.804 or smaller could occur if the null hypothesis were true. Assuming we'd decided on an alpha of 0.05 before the experiment, we'd conclude that there was a significant difference between the means of these two samples, because 0.003 is less than 0.05. In terms of the experiment, we might conclude that having a cloak of invisibility significantly affected the amount of mischief a person got up to, $t(11) = -3.80$, $p = 0.003$. This result was predicted by the error bar chart in Figure 10.11.

Output 10.8

Paired Samples Statistics

				Bootstrap[a]			
						BCa 95% Confidence Interval	
			Statistic	Bias	Std. Error	Lower	Upper
Pair 1	Mischief (No Invisibility Cloak)	Mean	3.75	.02	.54	2.50	4.92
		N	12				
		Std. Deviation	1.913	-.129	.352	1.357	2.193
		Std. Error Mean	.552				
	Mischief (Invisibility Cloak)	Mean	5.00	.02	.47	3.83	6.17
		N	12				
		Std. Deviation	1.651	-.103	.310	1.115	1.946
		Std. Error Mean	.477				

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

Paired Samples Correlations

					Bootstrap for Correlation[a]		
						BCa 95% Confidence Interval	
		N	Correlation	Sig.	Bias	Std. Error	Lower	Upper
Pair 1	Mischief (No Invisibility Cloak) & Mischief (Invisibility Cloak)	12	.806	.002	-.029	.160	.417	.943

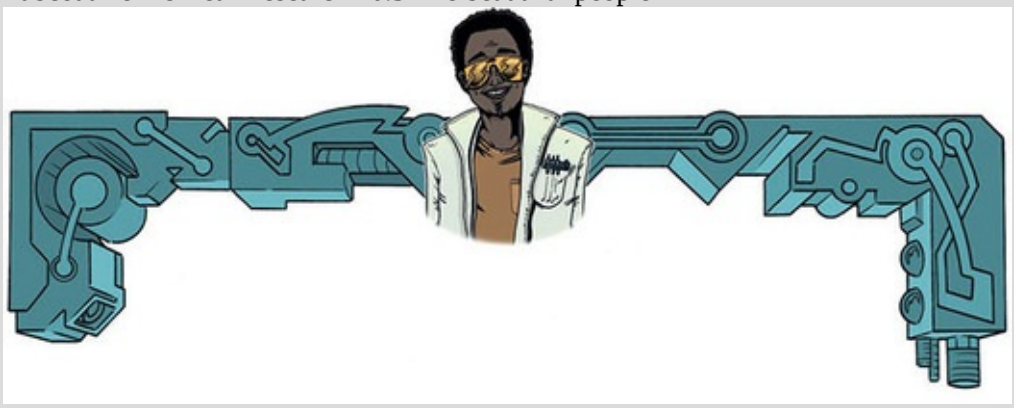
a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

Output 10.9

Paired Samples Test

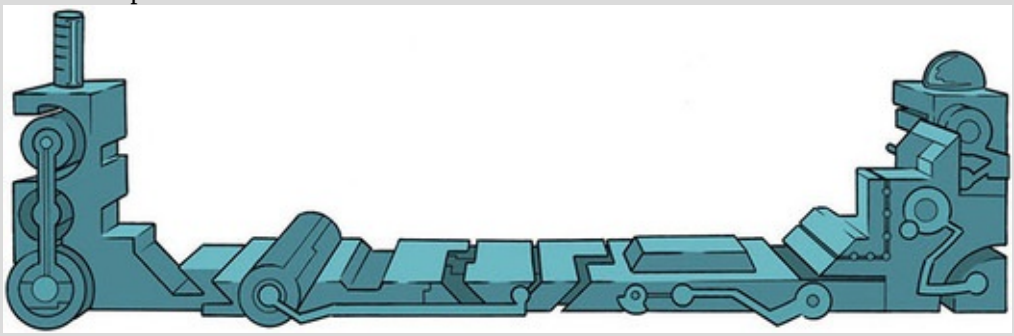
		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference Lower	Upper			
Pair 1	Mischief (No Invisibility Cloak) - Mischief (Invisibility Cloak)	-1.250	1.138	.329	-1.973	-.527	-3.804	11	.003

Labcoat Leni's Real Research 10.3 The beautiful people 



Gelman, A., & Weakliem, D. (2009). *American Scientist, 97*, 310–316.

Apparently there are more beautiful women in the world than there are handsome men. Satoshi Kanazawa explains this finding in terms of good-looking parents being moreer likely to have a baby daughter as their first child than a baby son. Perhaps more controversially, he suggests that, from an evolutionary perspective, beauty is a more valuable trait for women than for men (Kanazawa, 2007). In a playful and very informative paper, Andrew Gelman and David Weakliem discuss various statistical errors and misunderstandings, some of which have implications for Kanazawa's claims. The 'playful' part of the paper is that to illustrate their point they collected data on the 50 most beautiful celebrities (as listed by *People* magazine) of 1995–2000. They counted how many male and female children they had as of 2007. If Kanazawa is correct, these beautiful people would have produced more girls than boys. Do a *t*-test to find out whether they did. The data are in **Gelman & Weakliem (2009).sav**. Answers are on the companion website.



10.9.5 Robust tests of two dependent means 

As you've probably noticed, I'm keen on looking at the robust bootstrap confidence intervals (especially in a small sample like we have here). Output 10.10 shows these confidence intervals. Remember that confidence intervals are constructed such that in 95% of samples the intervals contain the true value of the mean difference. So, assuming that this sample's confidence interval is one of the 95 out of 100 that contain the population value, we can say that the true mean difference lies between −1.83 and −0.67. The importance of this interval is that it does not contain zero (both limits are negative), which tells us that the true value of the mean difference is unlikely to be zero. In other words, there is an effect in the population reflecting more mischievous acts performed when someone is given an invisibility cloak.

Output 10.10

Bootstrap for Paired Samples Test

			Bootstrap[a]				
						BCa 95% Confidence Interval	
		Mean	Bias	Std. Error	Sig. (2-tailed)	Lower	Upper
Pair 1	Mischief (No Invisibility Cloak) - Mischief (Invisibility Cloak)	-1.250	-.005	.311	.008	-1.833	-.667

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

It's also possible to run a robust version of the paired-samples *t*-test using R. As was the case for the robust independent *t*-test, we need the *Essentials for R* plugin and *WRS2* package (Section 4.13). The companion website contains a syntax file (**Robust paired-samples t-test.sps**) for running a robust variant of the paired-samples *t*-test based on Yuen (1974) and described by Wilcox (2017). The syntax in the file is as follows:

```
BEGIN PROGRAM R.
library(WRS2)
mySPSSdata = spssdata.GetDataFromSPSS()
yuend(mySPSSdata$No_Cloak, mySPSSdata$Cloak, tr = 0.2)
END PROGRAM.
```

Select and run these five lines of syntax (see SPSS Tip 10.3) and you'll find text output in the viewer (Output 10.11) which says that there was a significant difference (because the confidence interval does not cross zero and the *p*-value is

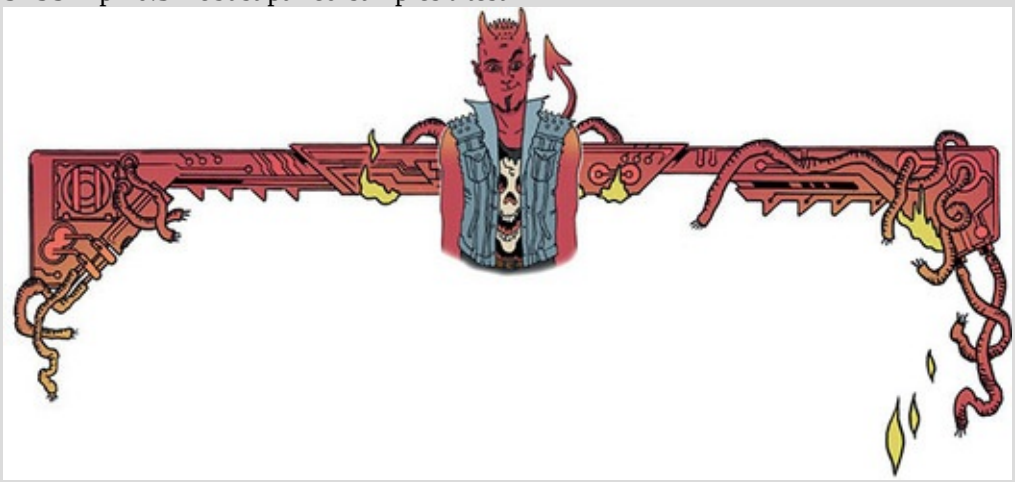
less than 0.05) in mischief scores across the two cloak groups, Y_t (7) = −2.70 [−1.87, −0.13], p = 0.031.

10.9.6 Bayesian test of two means from paired samples



To access the dialog box to compute a Bayes factor and credible interval for dependent means select *Analyze*  *Bayesian Statistics*  *Related Samples Normal* (Figure 10.13). Just like with the regular dependent *t*-test we start by selecting a pair of variables and transferring them to the *Paired Variables* list. We're expecting more mischievous acts in the cloak group than the no cloak group, therefore, if we specify our pair as **Cloak** (*Variable 1*) and **No_Cloak** (*Variable 2*) we'd predict a positive mean difference (when we subtract scores for the no cloak condition from those for the cloak condition the results should be, on average, positive if our prediction is correct). Conversely, if we specify our pair as **No_Cloak** (*Variable 1*) and **Cloak** (*Variable 2*) we'd predict a negative mean difference. In terms of setting up the prior distribution it's probably easier to think about a positive value for the mean difference, so let's specify **Cloak** as *Variable 1* and **No_Cloak** as *Variable 2*. To do this, select **Cloak** from the *Variables* list and click  , then select **No_Cloak** and click  . Select the *Variance Known* option and type 1.3 in the cell labelled *Variance Value*. The value of 1.3 is obtained by squaring the standard deviation of difference scores in Output 10.9 to convert it to the variance. Using the variance for the data simplifies things (we don't need to set a prior for the variance) but as you get more experience you might want to estimate the variance too. To compute Bayes factors and estimate the model parameters select  .

SPSS Tip 10.3 Robust paired-samples *t*-test 



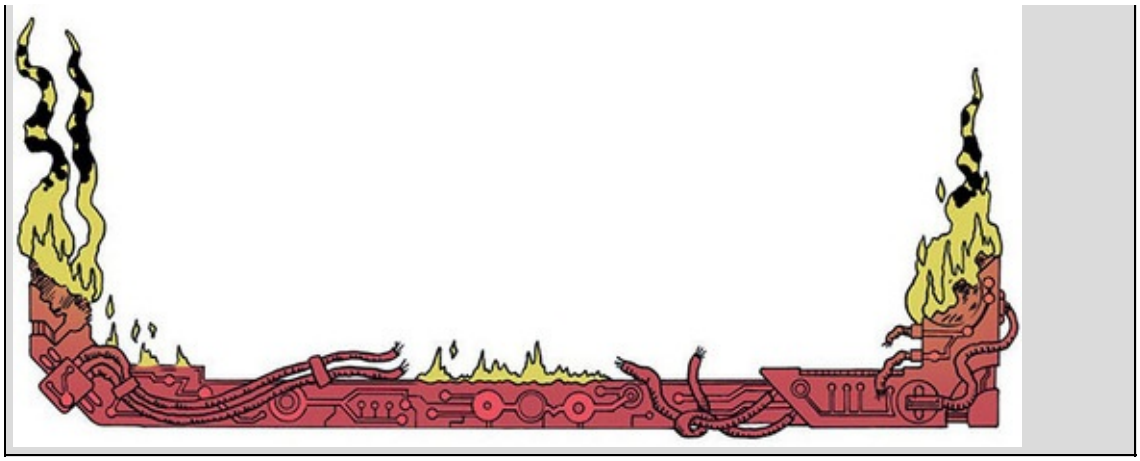
The syntax for a robust paired-samples *t*-test (**Robust paired-samples t-test.sps**) is the same as for the robust independent *t* (SPSS Tip 10.2) apart from the function itself, which is *yuend()*. This function takes the general form

yuend(scores in condition 1, scores in condition 2, tr = 0.2)

In our example we end up with:

yuend(mySPSSdata$No_Cloak, mySPSSdata$Cloak, tr = 0.2)

The scores in condition 1 are the scores in the no cloak condition. In the data editor for the paired-samples example this is the variable **No_Cloak**. However, I have written mySPSSdata$No_Cloak because R recognizes variables as belonging to data frames and mySPSSdata$No_Cloak means *the variable No_Cloak in the data frame mySPSSdata*. Remember that *mySPSSdata* contains the data pulled in from the SPSS data editor, so another way to translate this would be as *the variable No_Cloak from the data editor*. The scores in condition 2 are the scores in the cloak condition, which have similarly specified as mySPSSdata$Cloak. The *tr* relates to the amount of trimming to the data. By default 0.2 (20%) is trimmed, but you change this value to 0.1 for a 10% trimmed mean, or 0.05 for a 5% trimmed mean and so on. As with the robust independent *t*-test, if you want to use this test on your own data the only part of the syntax you need to edit is to replace the words ‘Mischief’ and ‘Cloak’ with the names of the variables representing your two conditions (remember to retain the *mySPSSdata$* before the variable name) and adjust the trim proportion if you want something other than a 20% trim.



Output 10.11

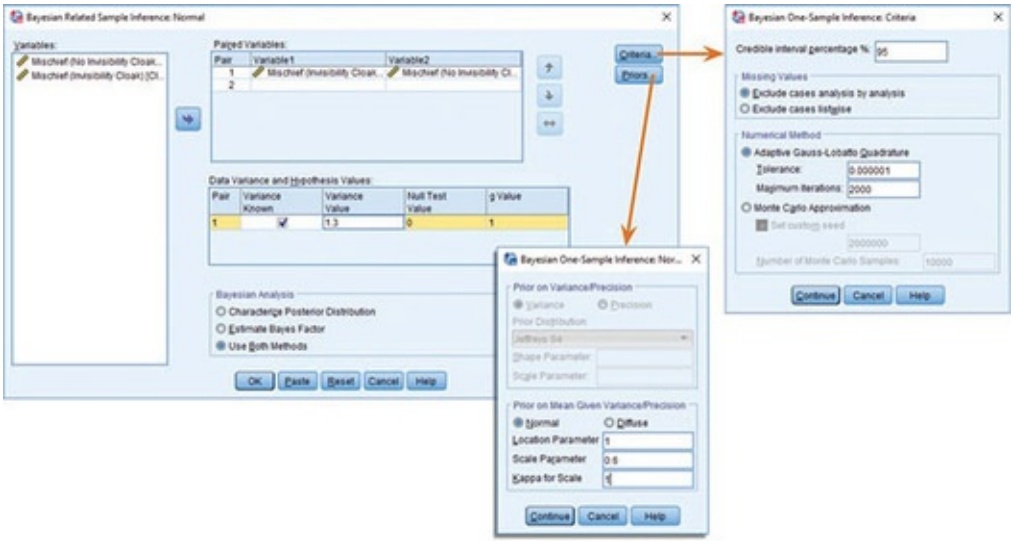
```
Test statistic: -2.7027 (df = 7), p-value = 0.03052

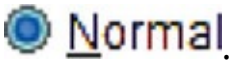
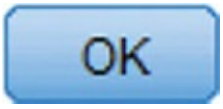
Trimmed mean difference: -1
95 percent confidence interval:
-1.8749 -0.1251

Explanatory measure of effect size: 0.4
```

Imagine that prior to collecting data we believed that the difference between the number of mischievous acts committed with and without an invisibility cloak would be 1 but you were prepared to believe that it could range from about -1 to 3. In other words, you believe the most probable outcome is that having an invisibility cloak would lead to 1 additional mischievous act compared to not having a cloak. However, you were prepared to accept that, at most, it might lead to 3 additional acts (but you think this possibility has a low probability – it is highly unlikely). Conversely, you're prepared to accept that you might be wrong and that having an invisibility cloak creates a sense of responsibility in the wearer that leads to fewer mischievous acts. At most, you be prepared to believe that a cloak could lead to 1 fewer act, but again you think this outcome is highly unlikely (you assign it low probability). The graph with the red line in Output 10.12 shows this prior distribution. It is a normal distribution centred at 1 (the most likely outcome in your prior opinion) and ranging from about -1 to 3. This equates to a prior belief that the difference between the number of mischievous acts committed when wearing an invisibility cloak compared to when not is most likely 1, but could be as low as -1 or as high as 3 (although we think these more extreme values are unlikely).

Figure 10.13 Dialog box for a Bayesian test of two related means



To set up these priors click  and select  . I have specified a distribution with mean (location) of 1 and standard deviation (scale) of 0.6. As before, to get this scale parameter I took the range of beliefs, 3−(−1) = 4, and divided by 6, 4/6 = 0.67. I then rounded down (although this won't have had an impact on the results). Kappa I set to be the SPSS default of 1. If you'd rather use a noninformative reference prior (i.e., prior to data collection you're prepared to believe any value for the mean difference) then select *Diffuse*. If you want a credible interval other than 95% then click  and change the 95 to the value you want. In the main dialog box click  to fit the model.

Output 10.12 shows that the Bayes factor is 0.005. SPSS reports the ratio of the null to the alternative hypothesis, so this value means that the data are 0.005 times as probable under the null hypothesis as under the alternative. We can flip the interpretation by dividing 1 by the value (1/0.005 = 200). This value tells us that the data are 200 times as probable under the alternative hypothesis as under the null. In other words, we should shift our belief towards the alternative hypothesis by a factor of 200. This is strong evidence for the hypothesis that invisibility cloaks lead to more mischief.

Output 10.12 also shows our prior distribution for the difference between group means (the graph with the red line). I have already described this graph. The Bayesian estimates of the difference between means are in the columns labelled

Posterior Mode and *Posterior Mean*. The 95% credible interval for this estimate ranged from 0.62 to 1.81. In other words, *assuming that the effect exists*, the population value of the effect will be between 0.62 to 1.81 with 95% probability. This tells us nothing about the null hypothesis (because it assumes the effect exists) but helps us to ascertain the likely population value if we're prepared to accept that the effect exists. So, we can say with 95% probability that having a cloak of invisibility will increase mischievous acts by anything as low as 0.62 to 1.81.

Output 10.12

Bayes Factor for Related-Sample T Test

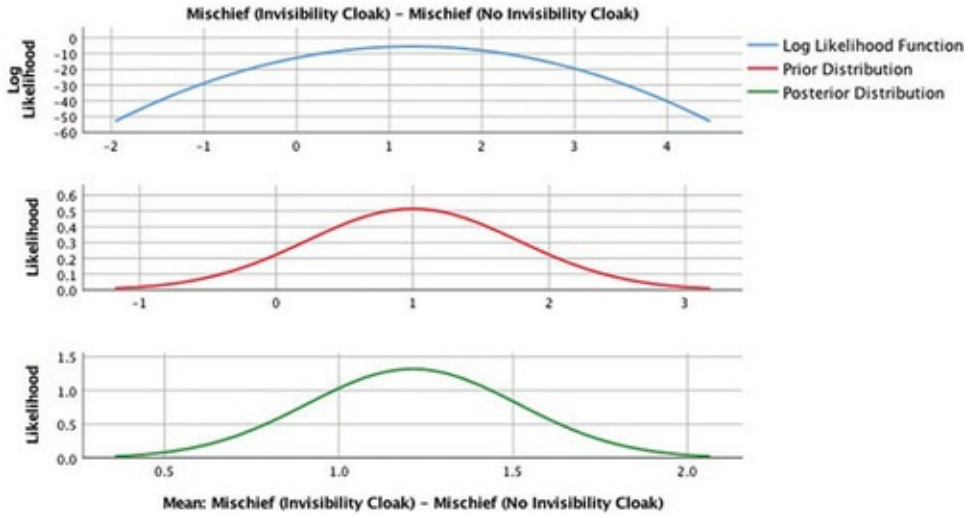
	N	Mean Difference	Std. Deviation	Std. Error Mean	Bayes Factor	t	df	Sig.(2-tailed)
Mischief (Invisibility Cloak) - Mischief (No Invisibility Cloak)	12	1.25	1.138	.329	.005	3.804	11	.003

Bayes factor: Null versus alternative hypothesis.

Posterior Distribution Characterization for Related-Sample Mean Difference

	N	Posterior Mode	Posterior Mean	Posterior Variance	95% Credible Interval Lower Bound	95% Credible Interval Upper Bound
Mischief (Invisibility Cloak) - Mischief (No Invisibility Cloak)	12	1.21	1.21	.092	.62	1.81

Prior on Variance: Diffuse. Prior on Mean: Normal.



10.9.7 Effect sizes for two related means

According to Rosenthal (1991), we can compute the effect size direct from the value of *t*, just as we did for the independent *t*-test. Using equation (10.17) and the values from Output 10.9, we get:

$$r = \sqrt{\frac{(-3.804)^2}{(-3.804)^2 + 11}} = \sqrt{\frac{14.47}{25.47}} = 0.75 \quad (10.20)$$

Therefore, as well as being statistically significant, this effect is very large. Notice that the effect is a lot bigger than when we treated the data as though they were from an independent design (r = 0.34), which is odd, given that we used exactly the same scores. This difference reflects the finding that using a t from a paired-samples t-test leads to an overestimation of the population effect size (Dunlap, Cortina, Vaslow, & Burke, 1996).

You could instead compute Cohen's d (Section 3.7.1) as we did in Section 10.8.6. Note the change in design does not affect the calculation: the effect size is still 0.65:

$$\hat{d} = \frac{\bar{X}_{\text{Cloak}} - \bar{X}_{\text{No Cloak}}}{s_{\text{No Cloak}}} = \frac{5 - 3.75}{1.91} = 0.65 \quad (10.21)$$

This consistency is a good thing because both studies did, in fact, show the same difference between means (because the scores used in the examples are identical). Some argue that you need to factor in the between scores in treatment conditions by dividing the estimate of d by the square root of 1 minus the correlation between the scores (which you can find in Output 10.8, r = 0.806). The corrected d is 1.48:

$$\hat{d}_D = \frac{\hat{d}}{\sqrt{1 - r}} = \frac{0.65}{\sqrt{1 - 0.806}} = \frac{0.65}{0.44} = 1.48 \quad (10.22)$$

Cramming Sam's Tips paired-samples t-test



- The paired-samples t-test compares two means, when those means have come from the same entities.
- Look at the column labelled *Sig*. If the value is less than 0.05 then the means of the two conditions are significantly different.

- Look at the values of the means to tell you how the conditions differ.
- Look at the table labelled *Bootstrap for Paired Samples Test* to get a robust confidence interval for the difference between means.
- A robust version of the test can be computed using syntax.
- A Bayes factor can be computed that quantifies the ratio of how probable the data are under the alternative hypothesis compared to the null.
- Calculate and report the effect size too.



which is more than double the original size! My issue with this 'correction' is precisely that the effect size now expresses information not just about the observed difference between means but the study design used to measure it. However, I include it in case you disagree with me.

10.10 Reporting comparisons between two means



As we have seen before, you usually state the finding to which the test relates and then report the test statistic, its degrees of freedom and its probability value. This applies whether we use a t-test or a robust test. Ideally report an estimate of the effect size and the Bayes factor too. If you used a robust test you should cite R (R Core Team, 2016) and the WRS2 package (Mair, Schoenbrodt, & Wilcox, 2017), because that's what was used to compute them.

10.10.1 Reporting *t*-tests 

For the data based on independent samples we could report this (to see from where the values come look at Output 10.4, Output 10.5, and Output 10.7):

✓ On average, participants given a cloak of invisibility engaged in more

acts of mischief (M = 5, SE = 0.48) than those not given a cloak (M = 3.75, SE = 0.55). This difference, −1.25, BCa 95% CI [−2.65, 0.11], was not significant, $t(21.54) = -1.71$, $p = 0.101$; however, it represented an effect of $d = 0.65$.

For the data based on paired samples you might report (see Outputs 10.9, 10.10 and 10.12):

> ✓ On average, participants given a cloak of invisibility engaged in more acts of mischief (M = 5, SE = 0.48) than those not given a cloak (M = 3.75, SE = 0.55). This difference, −1.25, BCa 95% CI [−1.83, −0.67], was significant, $t(11) = -3.80$, $p = 0.003$, and represented an effect of $d = 0.65$.

10.10.2 Reporting robust variants for the t-test



If you'd done the robust test, you could start with a general statement such as:

> ✓ R (R Core Team, 2016) was used to compute a robust variant of the *t*-test based on Yuen (1974) using the *WRS2* package (Mair et al., 2017).

Then for the data based on independent samples report this (see Output 10.6):

> ✓ On average, participants given a cloak of invisibility engaged in more acts of mischief (M = 5, SE = 0.48), than those not given a cloak (M = 3.75, SE = 0.55). This difference was not significant, $Y_t = -1.36$, 95% CI [−2.52, 0.52], $p = 0.167$.

Then for the data based on paired samples report this (see Output 10.11):

> ✓ On average, participants given a cloak of invisibility engaged in more acts of mischief (M = 5, SE = 0.48), than those not given a cloak (M = 3.75, SE = 0.55). This difference was significant, $Y_t(7) = -2.70$, 95% CI [−1.87, −0.13], $p = 0.031$.

10.10.3 Reporting Bayesian comparisons of means



For the data based on independent samples, report (see Output 10.7):

> ✓ On average, participants given a cloak of invisibility engaged in more acts of mischief (M = 5, SE = 0.48), than those not given a cloak (M = 3.75, SE = 0.55). The prior distributions for the group means were set as a normal distributions with a mean of 3 and standard deviation of 1 for the no cloak group and a mean of 4 and standard deviation of 1.75 for the cloak group. The Bayes factor was estimated using Gönen's method (Gönen et al., 2005) with a prior difference between means of 1 with a variance of 0.25. The Bayesian estimate of the true difference between means was 1.31, 95% credible interval [0.02, 2.60]. The associated Bayes factor, BF_{01} = 2.39, suggested that the data were weakly more probable under the alternative hypothesis than the null.

For the data based on paired samples (see Output 10.12):

> ✓ On average, participants given a cloak of invisibility engaged in more acts of mischief (M = 5, SE = 0.48), than those not given a cloak (M = 3.75, SE = 0.55). The prior distribution for the difference between means was set to be a normal distribution with mean of 1 and standard deviation of 0.6. Kappa was set at 1. The Bayesian estimate of the true difference between means was 1.21, 95% credible interval [0.62, 1.81]. The associated Bayes factor, BF_{01} = 200, suggested that the data were 200 times more probable under the alternative hypothesis than under the null.

10.11 Between groups or repeated measures?



The two examples in this chapter illustrate the difference between data collected using the same participants and data collected using different participants. The two examples use the same scores, but when analysed as though the data came from the same participants the result was a significant difference between means and large Bayes factor, and when analysed as though the data came from

different participants there was no significant difference and a small Bayes factor. This finding may puzzle you – after all, the numbers being analysed were identical, as reflected in the effect size (*d*) being the same. The explanation is that repeated-measures designs have relatively more power. When the same entities are used across conditions the unsystematic variance (error variance) is reduced dramatically, making it easier to detect the systematic variance. It is often assumed that the way in which you collect data is irrelevant, and in terms of the effect size it sort of is, but if you're interested in significance then it matters. Researchers have carried out studies using the same participants in experimental conditions, then replicated the study using different participants, and have found that the method of data collection interacts significantly with the results found (see Erlebacher, 1977).

10.12 Brian's attempt to woo Jane 

Brian wished he had a cloak of invisibility. Term was nearly over and Jane was still giving him the brush-off. The nicer he was to her, the more caustic she became. What was it with her? He felt embarrassed by his behaviour – it was a bit desperate. He wanted to be invisible for a while, or at least avoid Jane, and so he'd been hiding out in the library.

A week ago, he'd been struggling through his statistics assignment. He didn't get the whole Bayesian thing. Weeks of learning about *p*-values, and suddenly his lecturer was throwing this whole new approach into the mix. Why don't lecturers see how confusing that is? Mind you, with his big spacey eyes his lecturer looked like his brain might have crashed out on mushrooms in the '70s. Surely only a drugs casualty has a name like Oditi? Anyway, Brian was stuck, frustrated, and about to give up when he recognized Alex and Sam from his lectures.

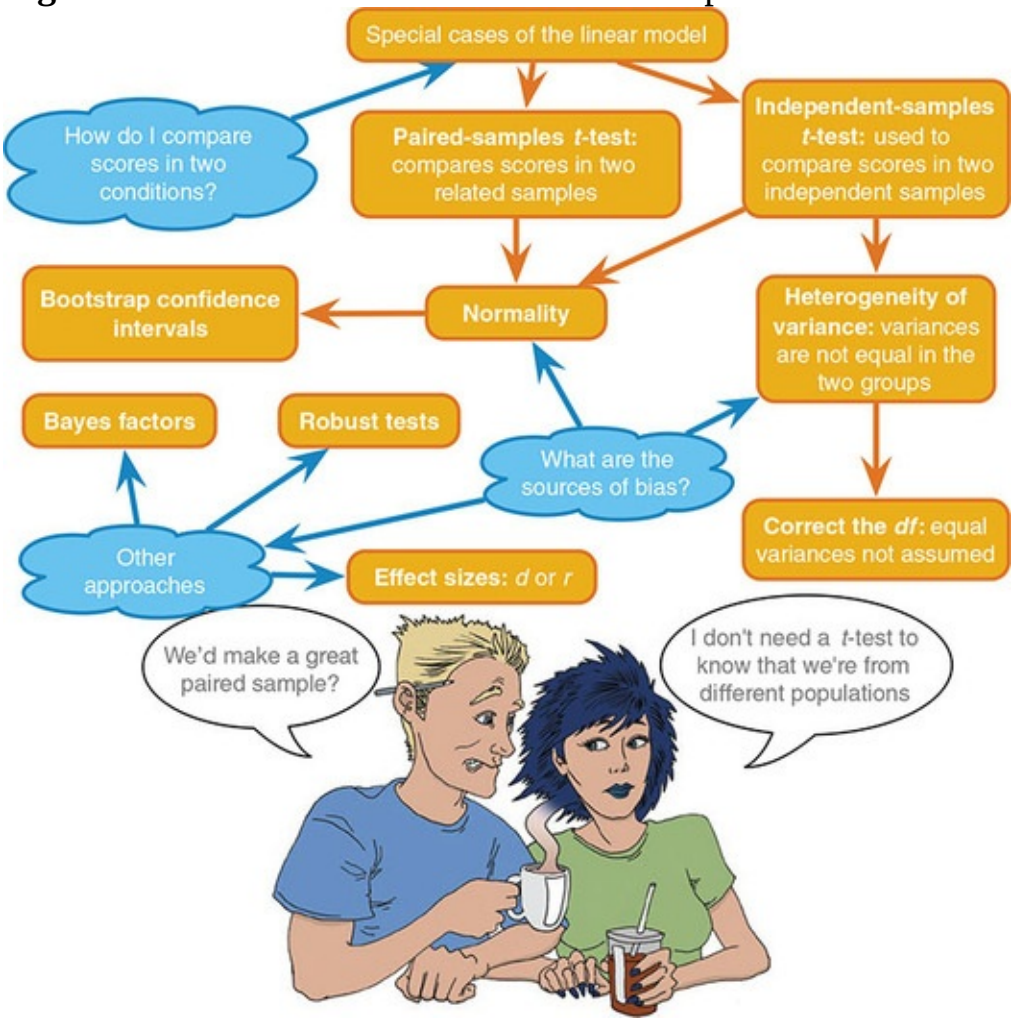
He went over and asked for help. They welcomed him. Alex was smart, that much was obvious, but her generosity and empathy in mentoring him and Sam was quite overwhelming. Alex was nervy around Brian and got a little tetchy whenever he mentioned Jane.

During his week of hiding out in the library he chatted to Alex more and more. She was big-hearted, funny and seemed to enjoy his company.

Today he was alone, though, and reflecting on what a good week it had been. He'd started to feel more human—like a regular guy with normal friends. He felt calm and relaxed for the first time since the term started. That was until a finger tapped his shoulder and he turned to face the knowing smile of Jane.

'Hi, stranger,' she said.

Figure 10.14 What Brian learnt from this chapter



10.13 What next?

I'd announced to my parents that my career of choice was a rock star. Obviously I hadn't presented them with a robust *t*-test showing how much more money I would earn compared to a university professor, but even if I had, I'm not sure it

would have mattered. My parents were quite happy for me to live this fantasy if I entertained the possibility that it might not work out and had a plan B. Preferably a plan B that was a little bit more sensible than being a rock star. At the age of 10, I think my plan B was probably to be a soccer star. One way or another I wanted my career to involve being a star, so if it wasn't rock, then soccer would do. However, we've seen already that I was at a genetic disadvantage when it came to soccer, but not so much when it came to rock stardom: my dad, after all, was quite musical. All I had to do was make it happen. The first step, I reasoned, was to build a fan base, and the best place to start a fan base is among your friends. With that in mind, I put on my little denim jacket with Iron Maiden patches sewn onto it, threw my guitar over my back and headed off down the rocky road of stardom. The first stop was my school.

10.14 Key terms that I've discovered

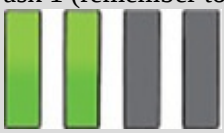
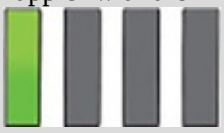
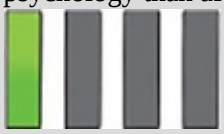
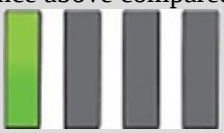
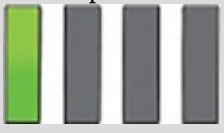
Dependent *t*-test
Dummy variables
Grand mean
Independent *t*-test
Paired-samples *t*-test
Standard error of differences
Variance sum law

Smart Alex's tasks

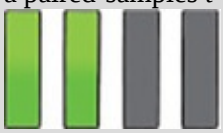
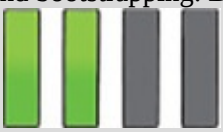


- **Task 1**: Is arachnophobia (fear of spiders) specific to real spiders or will pictures of spiders evoke similar levels of anxiety? Twelve arachnophobes were asked to play with a big hairy tarantula with big fangs and an evil look in its eight eyes and at a different point in time were shown only pictures of the same spider. The participants' anxiety was measured in each case. Do a *t*-test to see whether anxiety is higher for real spiders than

pictures (**Big Hairy Spider.sav**). 

- **Task 2**: Plot an error bar graph of the data in Task 1 (remember to adjust for the fact that the data are from a repeated measures design.) 
- **Task 3**: 'Pop psychology' books sometimes spout nonsense that is unsubstantiated by science. As part of my plan to rid the world of pop psychology I took 20 people in relationships and randomly assigned them to one of two groups. One group read the famous popular psychology book *Women are from Bras and Men are from Penis*, and the other read *Marie Claire*. The outcome variable was their relationship happiness after their assigned reading. Were people happier with their relationship after reading the pop psychology book? (**Penis.sav**). 
- **Task 4**: Twaddle and Sons, the publishers of *Women are from Bras and Men are from Penis*, were upset about my claims that their book was as useful as a paper umbrella. They ran their own experiment (N = 500) in which relationship happiness was measured after participants had read their book and after reading one of mine (Field & Hole, 2003). Participants read the books in counterbalanced order with a six-month delay. Was relationship happiness greater after reading their wonderful contribution to pop psychology than after reading my tedious tome about experiments? (**Field&Hole.sav**). 
- **Task 5**: In Chapter 4 (Task 6) we looked at data from people who had been forced to marry goats and dogs and measured their life satisfaction as well as how much they like animals (**Goat or Dog.sav**) Conduct a *t*-test to see whether life satisfaction depends upon the type of animal to which a person was married. 
- **Task 6**: What do you notice about the t-value and significance above compared to when you ran the analysis as a linear model in Chapter 8 Task 6? 
- **Task 7**: In Chapter 6 we looked at hygiene scores over three days of a rock music festival (**Download Festival.sav**). Do a paired-samples *t*-test to see whether hygiene scores on day 1 differed from those on day 3. 
- **Task 8**: Analyse the data in Chapter 7, Task 1 (whether men and dogs differ in their dog-like behaviours) using an independent *t*-test with bootstrapping. Do you reach the same

conclusions? (**MenLikeDogs.sav**). 

- **Task 9**: Analyse the data in Chapter 7, Task 2 (whether the type of music you hear influences goat sacrificing – **DarkLord.sav**), using a paired-samples *t*-test with bootstrapping. Do you reach the same conclusions? 
- **Task 10**: Thinking back to Labcoat Leni's Real Research 4.1, test whether the number of offers was significantly different in people listening to Bon Scott than in those listening to Brian Johnson, using an independent *t*-test and bootstrapping. Do your results differ from Oxoby (2008)? (**Oxoby (2008) Offers.sav**). 

Answers & additional resources are available on the book's website at **https://edge.sagepub.com/field5e**



11 Moderation, mediation and multicategory predictors

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11.1 What will this chapter tell me?

Having successfully slain audiences at holiday camps around Britain, my next step towards global domination was my primary school. I had learnt another Chuck Berry song ('Johnny B. Goode'), but broadened my repertoire to include songs by other artists (I have a feeling 'Over the edge' by Status Quo was one of them).[1] When the opportunity came to play at a school assembly I jumped at it. The headmaster tried to ban me,[2] but the show went on. It was a huge success (10-year-olds are very easily impressed). My classmates carried me around the playground on their shoulders. I was a hero. Around this time I had a childhood sweetheart called Clair Sparks. Actually, we had been sweethearts since before my newfound rock legend status. I don't think the guitar playing and singing impressed her much, but she rode a motorbike (really, a little child's one) which impressed *me* quite a lot; I was utterly convinced that we would one day get married and live happily ever after. I was utterly convinced, that is, until she ran off with Simon Hudson. Being 10, she probably literally did run off with him – across the playground. I remember telling my parents and them asking me how I felt about it. I told them I was being philosophical about it. I probably didn't know what philosophical meant at the age of 10, but I knew that it was the sort of thing you said if you were pretending not to be bothered about being dumped.

1 This would have been about 1982, so just before they abandoned hard rock in favour of a long series of increasingly cringeworthy publicity stunts. Nevertheless, up to the age of 10 they were my favourite band.

2 Seriously! I grew up in an age when a headmaster would try to ban a 10-year-old from playing guitar in assembly. The guy used to play hymns on an acoustic guitar, and I can assume only that he lost all perspective on the situation and decided that a 10-year-old blasting out some Quo in a squeaky little voice with an electric guitar was subversive or something.

If I hadn't been philosophical, I might have wanted to look at what had lowered Clair's relationship satisfaction. We've seen in previous chapters that we could predict things like relationship satisfaction using a linear model. Perhaps it's predicted from your partner's love of rock bands like Status Quo (I don't recall Clair liking that sort of thing). However, life is usually more complicated than this; for example, your partner's love of rock music probably depends on your own love of rock music. For example, if you both like rock music then your love of the same music might have an additive effect giving you huge relationship satisfaction (*moderation*), or perhaps the relationship between your partner's love of rock and your own relationship satisfaction can be explained by your own music tastes (*mediation*). In the previous chapter we also saw that a

dichotomous variable (e.g., rock fan or not) can be a predictor in a linear model, but what if you wanted to categorize musical taste into several categories (rock, hip-hop, R&B, etc.)? Surely you can't use multiple categories as a predictor variable? This chapter extends what we know about the linear model to these more complicated scenarios. First we look at two common linear models – moderation and mediation – before expanding what we already know about categorical predictors.

Figure 11.1 My 10th birthday. From left to right: my brother Paul (who still hides behind cakes rather than have his photo taken), Paul Spreckley, Alan Palsey, Clair Sparks and me



11.2 The *Process* tool

The best way to tackle moderation and mediation is with the *PROCESS* tool, which needs to be installed using the instructions in Section 4.13.1.

11.3 Moderation: interactions in the linear model

11.3.1 The conceptual model

So far we have looked at individual predictors in the linear model. It is possible for a statistical model to include the combined effect of two or more predictor variables on an outcome, which is known conceptually as **moderation**, and in statistical terms as an **interaction effect**. We'll start with the conceptual, using an example of whether violent video games make people antisocial. Video games are among the favourite online activities for young people: two-thirds of

5–16-year-olds have their own video games console, with 88% of boys aged 8–15 owning at least one games console (Ofcom, 2008). Although playing violent video games can enhance visuospatial acuity, visual memory, probabilistic inference, and mental rotation (Feng, Spence, & Pratt, 2007; Green & Bavelier, 2007; Green, Pouget, & Bavelier, 2010; Mishra, Zinni, Bavelier, & Hillyard, 2011), compared to games such as Tetris, these games have also been linked to increased aggression in youths (Anderson & Bushman, 2001). Another predictor of aggression and conduct problems is callous-unemotional traits such as lack of guilt, lack of empathy, and callous use of others for personal gain (Rowe, Costello, Angold, Copeland, & Maughan, 2010). Imagine that a scientist explored the relationship between playing violent video games (such as Grand Theft Auto, MadWorld and Manhunt) and aggression. She measured aggressive behaviour (**Aggress**), callous-unemotional traits (**CaUnTs**), and the number of hours per week they play video games (**Vid_Game**) in 442 youths (**Video Games.sav**).



Let's assume we're interested in the relationship between the hours spent playing these games (predictor) and aggression (outcome). The conceptual model of moderation in Figure 11.2 shows that a **moderator** variable is one that affects the relationship between two others. If callous-unemotional traits were a moderator then we're saying that the strength or direction of the relationship between game playing and aggression is affected by callous-unemotional traits.

Figure 11.2 Diagram of the *conceptual* moderation model

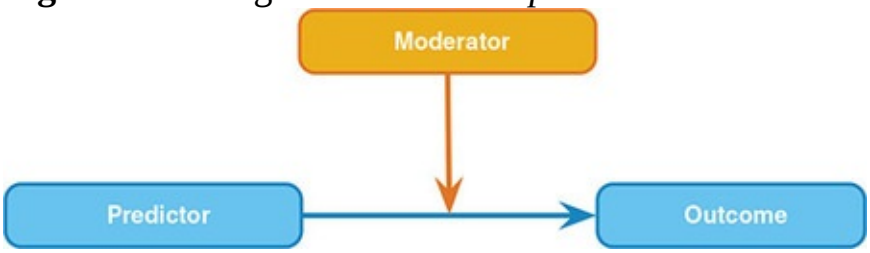
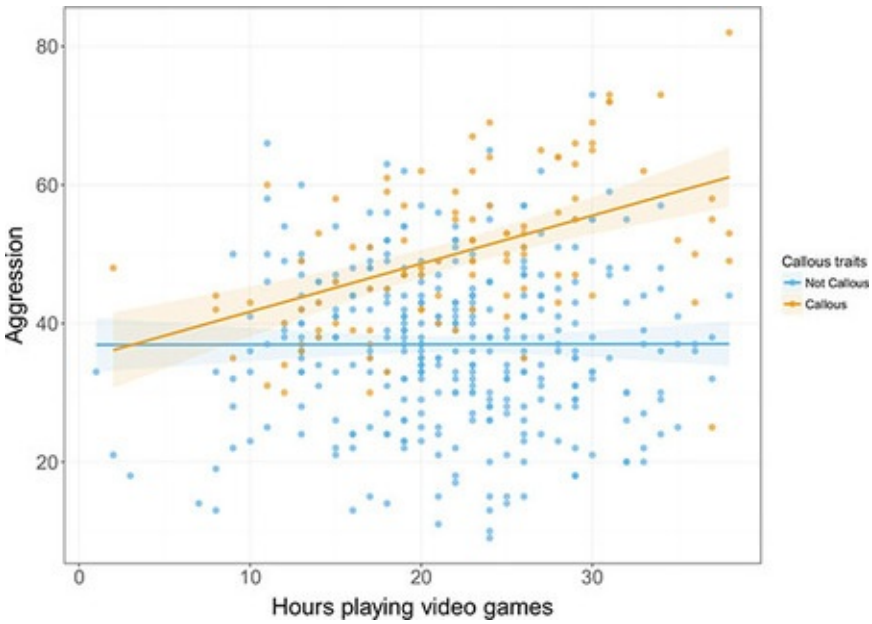


Figure 11.3 A categorical moderator (callous traits)



Suppose that we could classify people as having callous-unemotional traits or not. Our moderator variable would be categorical (callous or not callous). Figure 11.3 shows an example of moderation in this case: for people who are not callous there is no relationship between video games and aggression (the line is completely flat), but for people who are callous there is a positive relationship: as the time spent playing these games increases, so do aggression levels (the line slopes upwards). Therefore, callous-unemotional traits moderate the relationship between video games and aggression: there is a positive relationship for those with callous-unemotional traits but not for those without. This is a simple way to think about moderation, but it is not necessary that there is an effect in one group but not in the other, just that there is a difference in the relationship between video games and aggression in the two callousness groups. It could be that the effect is weakened or changes direction.

If we measure the moderator variable along a continuum it becomes trickier to visualize, but the basic interpretation stays the same. Figure 11.4 shows two graphs that display the relationships between the time spent playing video games, aggression and callous-unemotional traits (measured along a continuum rather than as two groups). We're still interested in how the relationship between video games and aggression changes as a function of callous-unemotional traits. We can explore this by comparing the slope of the regression plane for time spent gaming at low and high values of callous traits. To help you I have added arrows that show the relationship between video games and aggression. In Figure 11.4 (left) you can see that at the low end of the callous-unemotional traits scale, there is a slight positive relationship between playing video games and aggression (as time playing games increases so does aggression). At the high

end of the callous-unemotional traits scale, we see a very similar relationship between video games and aggression (the top and bottom of the regression plane slope at the same angle). The same is also true at the middle of the callous-unemotional traits scale. This is a case of no interaction or no moderation.

Figure 11.4 A continuous moderator (callous traits)

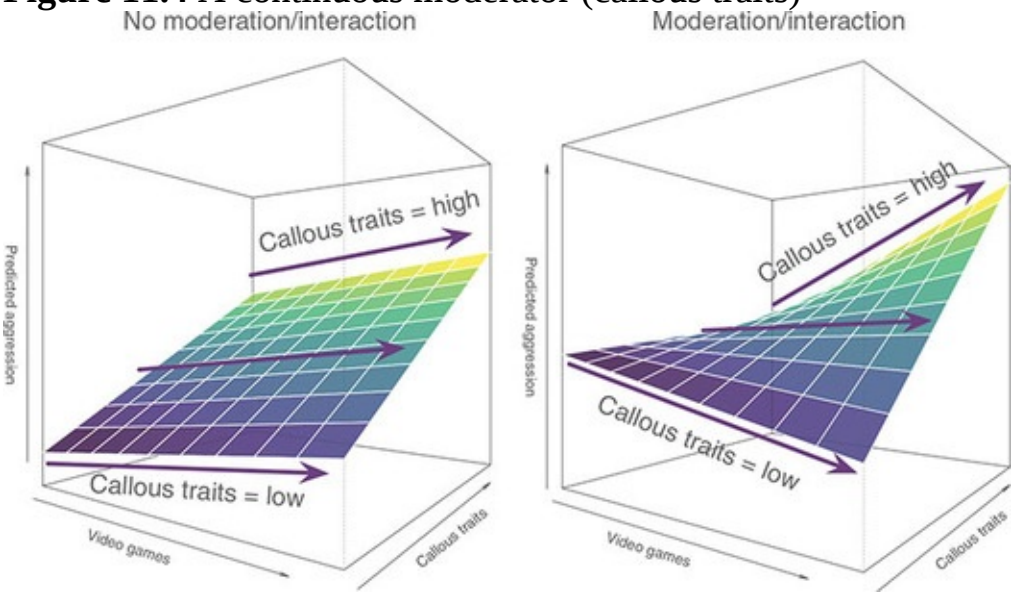


Figure 11.4 (right) shows moderation: at low values of callous-unemotional traits the plane slopes downwards, indicating a slightly negative relationship between playing video games and aggression, but at the high end of callous-unemotional traits the plane slopes upwards, indicating a strong positive relationship between gaming and aggression. At the midpoint of the callous-unemotional traits scale, the relationship between video games and aggression is relatively flat. So, as we move along the callous-unemotional traits variable, the relationship between gaming and aggression changes from slightly negative to neutral to strongly positive. We can say that the relationship between violent video games and aggression is moderated by callous-unemotional traits.

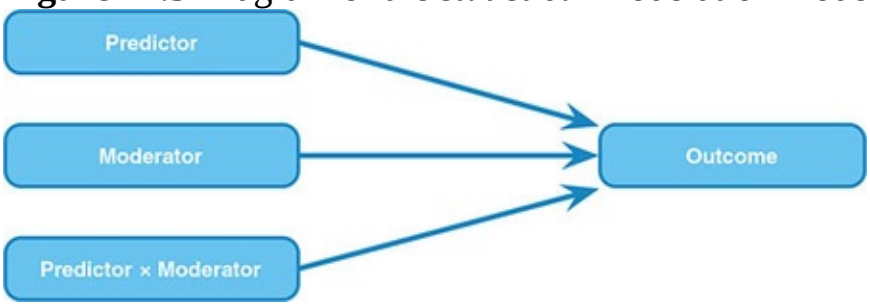
11.3.2 The statistical model

Now we know what moderation is conceptually, let's look at how we test for moderation within a statistical model. Figure 11.5 shows the statistical model: we predict the outcome from the predictor variable, the proposed moderator, and the interaction of the two. It is the interaction effect that tells us whether moderation has occurred,[3] but *we must include the predictor and moderator for the interaction term to be valid*. This point is very important. In our example, then, we'd be looking at fitting a linear model predicting aggression (the

outcome) from video game playing, callous-unemotional traits and their interaction.

3 The term ‘moderation’ implies that the moderator alters the relationship between the other two variables. A significant interaction effect does not justify making this causal assumption, it merely shows that two variables have a combined effect on the outcome. Therefore, although interactions are used to test for moderation, they are not the same thing (Hall & Sammons, 2014).

Figure 11.5 Diagram of the *statistical* moderation model



We’ve encountered the general form of the linear model many times already. The following equation is to refresh your memory (but if it is completely new information to you then read Chapter 9 before continuing):

$$\text{outcome}_i = (\text{model}) + \text{error}_i$$
$$Y_i = (b_0 + b_1 X_{1i} + b_2 X_{2i} + \cdots + b_n X_{ni}) + \varepsilon_i \quad (11.1)$$

By replacing the *X*s with the names of our predictor variables, and the *Y* with our outcome variable, the linear model becomes:

$$\text{Aggression}_i = (b_0 + b_1 \text{Gaming}_i + b_2 \text{Callous}_i) + \varepsilon_i \quad (11.2)$$

To test for moderation we need to consider the interaction between gaming and callous-unemotional traits. We have seen before that to add variables to a linear model we literally just add them in and assign them a parameter (*b*). Therefore, if we had two predictors labelled *A* and *B*, a model that tests for moderation would be expressed as:

$$Y_i = (b_0 + b_1 A_i + b_2 B_i + b_3 AB_i) + \varepsilon_i \quad (11.3)$$

If we replace the *A* and *B* with the names of the variables for this specific example we can express the model as:

$$\text{Aggression}_i = (b_0 + b_1 \text{Gaming}_i + b_2 \text{Callous}_i + b_3 \text{Interaction}_i) + \varepsilon_i \quad (11.4)$$

11.3.3 Centring variables

When an interaction term is included in the model the *b* parameters have a specific meaning: for the individual predictors they represent the regression of the outcome on that predictor when the other predictor is zero. So, in equation (11.4), b_1 represents the relationship between aggression and gaming when

callous traits are zero, and b_2 represents the relationship between aggression and callous traits when someone spends zero hours gaming per week. This interpretation isn't problematic because zero is a meaningful score for both predictors: it's plausible both that a child spends no hours playing video games, and that a child gets a score of 0 on the continuum of callous-unemotional traits. However, there are often situations where it makes no sense for a predictor to have a score of zero. Imagine that rather than measuring how much a child played violent video games we'd measured their heart rate while playing the games as an indicator of their physiological reactivity to them:

$$\text{Aggression}_i = (b_0 + b_1\text{Heart Rate}_i + b_2\text{Callous}_i + b_3\text{Interaction}_i) + \varepsilon_i \quad (11.5)$$

In this model b_2 is the regression of the aggression on callous traits when someone has a heart rate of zero while playing the games. This *b* makes no sense unless we're interested in knowing something about the relationship between callous traits and aggression in youths who die (and therefore have a heart rate of zero) while playing these games. It's fair to say that if video games killed the players, we'd have more to worry about than them developing aggression. The point is that the presence of the interaction term makes the *b*s for the main predictors uninterpretable in many situations.

For this reason, it is common to transform the predictors using **grand mean centring**. 'Centring' refers to the process of transforming a variable into deviations around a fixed point. This fixed point can be any value that you choose, but typically it's the grand mean. When we calculated *z*-scores in Chapter 1 we used grand mean centring because the first step was to take each score and subtract from it the mean of all scores. This is grand mean centring. Like *z*-scores, the subsequent scores are centred on zero, but unlike *z*-scores, we don't care about expressing the centred scores as standard deviations.[4] Therefore, grand mean centring for a given variable is achieved by taking each score and subtracting from it the mean of all scores (for that variable).

4 Remember that with *z*-scores we go a step further and divide the centred scores by the standard deviation of the original variable, which changes the units of measurements to standard deviations. You can do this if you want to compare *b*s across predictors within a model, but remember that *b* would represent the change in the outcome associated with a *1 standard deviation* change in the predictor.



Centring the predictors has no effect on the *b* for highest-order predictor, but will affect the *b*s for the lower-order predictors. *Order* refers to how many variables are involved: the gaming × callous traits interaction is a higher-order effect than gaming alone because it involves two variables rather than one. In our model (equation (11.4)), whether we centre the predictors has no effect on b_3 (the parameter for the interaction) but it changes the values of b_1 and b_2 (the parameters for gaming and callous traits). If we don't centre the gaming and callous variables, the *b*s represent the effect of the predictor when the other predictor is zero. However, if we centre the gaming and callous variables, the *b*s represent the effect of the predictor when the other predictor is at its mean value. For example, b_2 would represent the relationship between aggression and callous traits for someone who spends the average number of hours gaming per week. Centring is important when your model contains an interaction term because it makes the *b*s for lower-order effects interpretable. There are good reasons for not caring about the lower-order effects when the higher-order interaction involving those effects is significant; for example, if the gaming × callous traits interaction is significant, then it's not clear why we would be interested in the individual effects of gaming and callous traits. However, when the interaction is not significant, centring makes interpreting the main effects easier. With centred variables the *b*s for individual predictors have two interpretations: (1) they are the effect of that predictor at the mean value of the sample; and (2) they are the average effect of the predictor across the range of scores for the other predictors. To explain the second interpretation, imagine we took everyone who spent no hours gaming, estimated the linear model between aggression and callous traits and noted the *b*. We then took everyone who played games for 1 hour and did the same, and then took everyone who gamed for 2 hours per week and did the same. We continued doing this until we had estimated linear models for every different value of the hours spent gaming. We'd have a lot of *b*s, each one representing the relationship between callous traits and aggression for different amounts of gaming. If we took an average of these *b*s then we'd get the same value as the *b* for callous traits (centred) when we use it as a predictor with gaming (centred) and their interaction.

The *PROCESS* tool does the centring for us so we don't need to worry about

how it's done, but because centring is useful in other analyses Oliver Twisted has some additional material that shows you how to do it manually for this example.

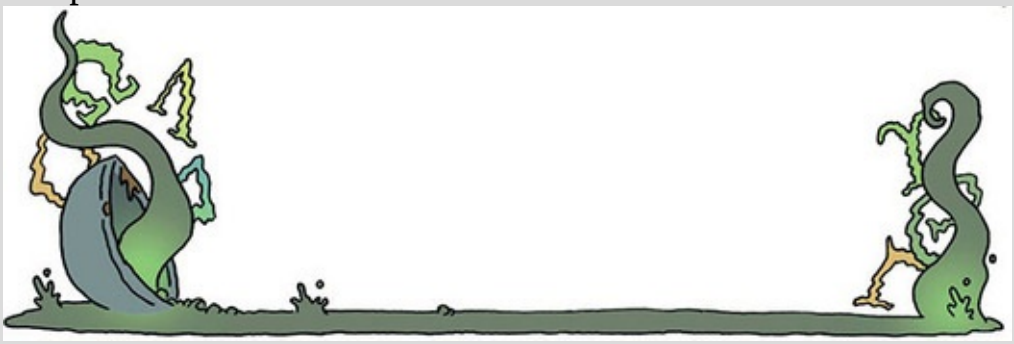
11.3.4 Creating interaction variables

Equation (11.4) contains a variable called 'Interaction', but the data file does not. You might ask how you enter a variable into the model that doesn't exist in the data. You create it, and it's easier than you might think. Mathematically speaking, when we look at the combined effect of two variables (an interaction) we are literally looking at the effect of the two variables multiplied together. The interaction variable in our example would be the scores on the time spent gaming multiplied by the scores for callous-unemotional traits. That's why interactions are denoted as *variable 1* × *variable 2*. The *PROCESS* tool creates the interaction variable for you, but the self-help task gives you practice at doing it manually (for future reference).

Oliver Twisted Please, Sir, can I have some more … centring?



'Recentgin,' babbles Oliver as he stumbles drunk out of Mrs Moonshine's alcohol emporium. 'I've had some recent gin.' I think you mean *centring* Oliver, not *recentgin*. If you want to know how to centre your variables using SPSS, then gaze drunkenly at the companion website.





Follow Oliver Twisted's instructions to create the centred variables **CUT_Centred** and **Vid_Centred**. Then use the *compute* command to create a new variable called **Interaction** in the **Video Games.sav** file, which is **CUT_Centred** multiplied by **Vid_Centred**.

11.3.5 Following up an interaction effect

If the moderation effect is significant, we need to interpret it. In our example, we're predicting that the moderator (callous traits) influences the relationship between playing violent video games and aggression. If the interaction of callous traits and time spent gaming is a significant predictor of aggression then we know that we have a significant moderation effect, but we don't know the nature of the effect. It could be that the time spent gaming always has a positive relationship with aggression, but that the relationship gets stronger the more a person has callous traits. Alternatively, perhaps the time spent gaming *reduces* aggression in people low on callous traits but *increases* aggression in those high on callous traits (i.e., the relationship reverses). To find out what is going on we need to do a **simple slopes analysis** (Aiken & West, 1991; Rogosa, 1981).

The idea behind simple slopes analysis is no different from what was illustrated in Figure 11.4. When describing that figure I talked about comparing the relationship between the predictor (time spent gaming) and outcome (aggression) at low and high levels of the moderator (callous traits). For example, in Figure 11.4 (right), we saw that time spent gaming and aggression had a slightly negative relationship at low levels of callous traits, but a positive relationship at high levels of callous traits. This is the essence of simple slopes analysis: we work out the model equations for the predictor and outcome at low, high and average levels of the moderator. The 'high' and 'low' levels can be anything you like, but *PROCESS* uses 1 standard deviation above and below the mean value of the moderator. Therefore, in our example, we would get the linear model for aggression predicted from hours spent gaming for the average value of callous traits, for 1 standard deviation above the mean value of callous traits and for 1 standard deviation below the mean value of callous traits. We compare these slopes in terms of both their significance and the value and direction of the *b* to see whether the relationship between hours spent gaming and aggression changes at different levels of callous traits.

A related approach is to look at how the relationship between the predictor and outcome changes at lots of different values of the moderator (not just at high, low and mean values). *PROCESS* implements such an approach based on Johnson and Neyman (1936). Essentially, it estimates the model including only the predictor and outcome at lots of different values of the moderator. For each model it computes the significance of the *b* for the predictor so you can see for which values of the moderator the relationship between the predictor and outcome is significant. It returns a 'zone of significance',[5] which consists of two values of the moderator. Typically, between these two values of the moderator the predictor does not significantly predict the outcome, whereas below the lower value and above the upper value of the moderator the predictor significantly predicts the outcome.

5 I must be careful not to confuse this with my wife, who is the Zoë of significance.

11.3.6 Moderation analysis using IBM SPSS Statistics



Given that moderation is demonstrated through a significant interaction between the predictor and moderator in a linear model, we could follow the general procedure in Chapter 9 (Figure 9.10). We would first centre the predictor and moderator, then create the interaction term as discussed already, then run a forced entry regression with the centred predictor, centred moderator and the interaction of the two centred variables as predictors. The advantage of this approach is that we can inspect sources of bias in the model.



Assuming you have done the previous self-test, fit a linear model predicting **Aggress** from **CUT_Centred, Vid_Centred** and **Interaction**.

Using the *PROCESS* tool has several advantages over using the normal regression menu in SPSS Statistics: (1) it centres predictors for us; (2) it computes the interaction term automatically; and (3) it produces simple slopes analysis. To access the dialog boxes in Figure 11.6 select *Analyze* 

Regression ▶ PROCESS, by Andrew F. Hayes (http://www.afhayes.com).[6]



Drag (or click) the outcome variable (**Aggress**) from the box labelled *Data File Variables* to the box labelled *Outcome Variable (Y)*. Similarly, drag the predictor variable (**Vid_Game**) to the box labelled *Independent Variable (X)*. Finally, drag the moderator variable (**CaUnTs**) to the box labelled *M Variable(s)*, which is where you specify moderators (you can have more than one).

[6] If the menu isn't there, see Section 4.13.1.

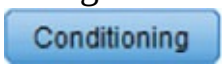
PROCESS can test 74 different types of model, and these models are listed in the drop-down box labelled *Model Number*.[7] The default model is 4 (mediation, which we'll look at next), so activate the drop-down list and select



, which is a simple moderation model. The rest of the options in this dialog box are for models other than simple moderation, so we'll ignore them.

[7] Details of the models are in the **templates.pdf** file that downloads with the *PROCESS* tool and in Hayes (2018).

Clicking 

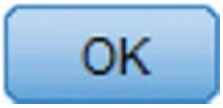
activates a dialog box containing four useful options for moderation. Selecting (1) *Mean center for products* centres the predictor and moderator for you; (2) *Heteroscedasticity-consistent SEs* removes the need to worry about the heteroscedasticity assumption; (3) *OLS/ML confidence intervals* produces confidence intervals for the model, and I've emphasized the importance of these many times; and (4) *Generate data for plotting* is helpful for interpreting and visualizing the simple slopes analysis. Talking of simple slopes, click 

for options related to this analysis. You can change whether you want simple slopes at ±1 standard deviation of the mean of the moderator (the default, which is fine) or at percentile points (*PROCESS* uses the 10th, 25th, 50th, 75th and 90th percentiles). Select the *Johnson-Neyman* method to get a zone of significance for the moderator.

Clicking 

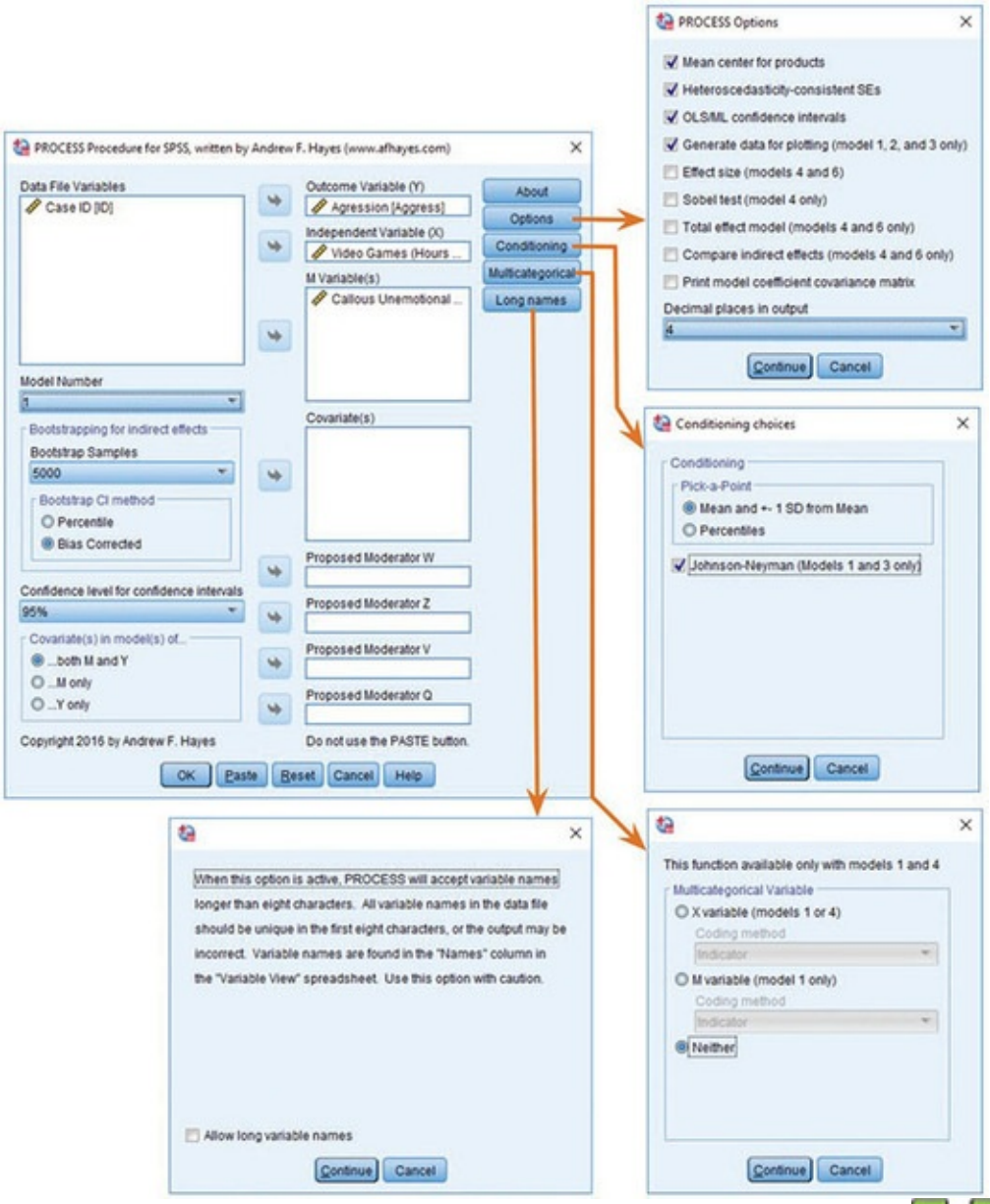
opens a dialog box where you can specify contrasts for the predictor (*X*) or moderator (*M*) if that variable is categorical with more than two categories. There is standard indicator or dummy coding (Section 11.5.1) as

well as some others (including Helmert coding, which we look at in Section 12.4.4). Clicking 

opens a dialog box with a check box for asking *PROCESS* to truncate your variable names. This option is here because *PROCESS* works only with variable names of 8 characters or less, so if you have variable names longer than this it won't work. One solution is to use this option to truncate the existing variable names, but you do need to be careful that you don't end up with multiple variables truncated to the same name (SPSS Tip 11.1). To be safe, I'd simply use variable names of 8 characters or less and leave this option unselected. Back in the main dialog box, click 

to run the analysis.

Figure 11.6 The dialog boxes for running moderation analysis



11.3.7 Output from moderation analysis

The output appears as text rather than nicely formatted in tables. Try not to let this formatting disturb you. If your output looks odd or contains warnings, or has a lot of zeros in it, it might be worth checking the variables that you input into *PROCESS* (SPSS Tip 11.1). Assuming everything has gone smoothly, you should see Output 11.1, which is the main moderation analysis. This output is pretty much the same as the table of coefficients that we saw in Chapter 9. We're told the *b*-value for each predictor, the associated standard errors (which have been adjusted for heteroscedasticity because we asked for them to be). Each *b* is

compared to zero using a *t*-test, which is computed from the beta divided by its standard error. The confidence interval for the *b* is also produced (because we asked for it). Moderation is shown up by a significant interaction effect, and that's what we've got here, $b = 0.027$, 95% CI [0.013, 0.041], $t = 3.71$, $p = 0.0002$, indicating that the relationship between the time spent gaming and aggression is moderated by callous traits.



Assuming you did the previous self-test, compare the table of coefficients that you got with those in Output 11.1.

To interpret the moderation effect we examine the simple slopes, which are shown in Output 11.2. The output shows us the results of three models: the model for time spent gaming as a predictor of aggression (1) when callous traits are low (to be precise, when the value of callous traits is −9.6177); (2) at the mean value of callous traits (because we centred callous traits its mean value is 0, as indicated in the output); and (3) when the value of callous traits is 9.6177 (i.e., high). We interpret these models as we would any other linear model by looking at the value of *b* (called *Effect* in the output), and its significance. We can interpret the three models as follows:

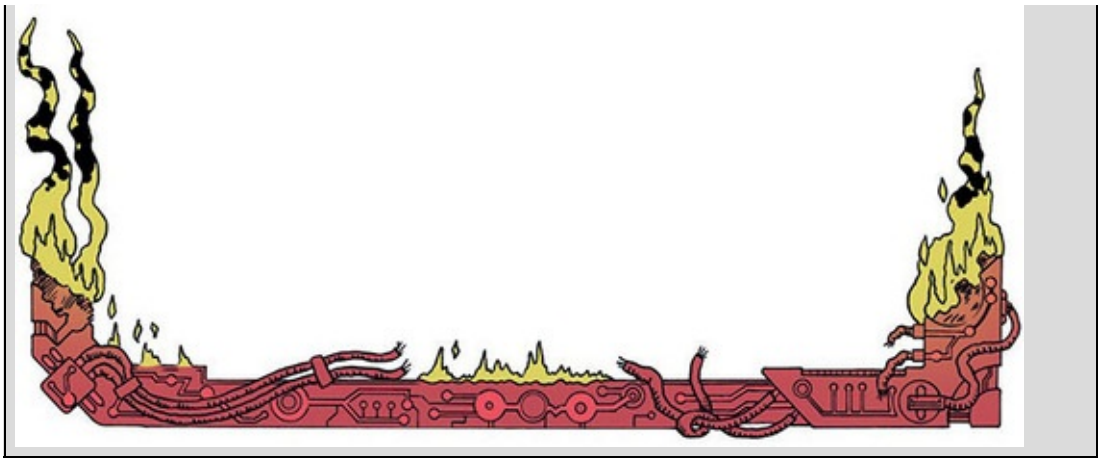
1. When callous traits are low, there is a non-significant negative relationship between time spent gaming and aggression, $b = -0.091$, 95% CI [−0.299, 0.117], $t = -0.86$, $p = 0.392$.
2. At the mean value of callous traits, there is a significant positive relationship between time spent gaming and aggression, $b = 0.170$, 95% CI [0.020, 0.319], $t = 2.23$, $p = 0.026$.
3. When callous traits are high, there is a significant positive relationship between time spent gaming and aggression, $b = 0.430$, 95% CI [0.231, 0.628], $t = 4.26$, $p < 0.001$.

SPSS Tip 11.1 Troubleshooting



Knowing the following about *PROCESS* might help to prevent weird stuff happening:

- If the variable names entered into *PROCESS* are longer than 8 characters, it blows up. Therefore, make sure your variable names are 8 characters or fewer, or select for *PROCESS* to truncate them by clicking Long names (Figure 11.6). Be careful, though, because if you have variables with similar long names *PROCESS* will get confused. For example, if you had two variables in the data editor called **NumberOfNephariousActs** and **NumberOfBlackSabbathAlbumsOwned** they would both be shortened to **numberof** (or possibly **number~1** and **number~2**). *PROCESS* will get confused about which variable is which. If your output looks weird, then check your variable names.
- Don't call any of your variables **xxx** (I'm not sure why you would) because that is a reserved variable name in *PROCESS*, so naming a variable **xxx** will confuse it.
- *PROCESS* is also confused by string variables, so only enter numeric variables.



These results tell us that the relationship between time spent playing violent video games and aggression only really emerges in people with average or greater levels of callous-unemotional traits.

Output 11.3 shows the results of the Johnson–Neyman method. First we're told that the boundaries of the zone of significance are −17.1002 and −0.7232. These are the values of the centred version of the callous-unemotional traits variable, and define regions within which the relationship between the time spent gaming and aggression is significant. The table underneath gives a detailed breakdown of these regions. Essentially it's doing a finer-grained simple slopes analysis: it takes different values of callous-unemotional traits and for each one computes the *b* (*Effect*) and its significance for the relationship between the time spent gaming and aggression. I have annotated the output to show the boundaries of the zone of significance. If you look at the column labelled *p* you can see that we start off with a significant negative relationship between time spent gaming and aggression, $b = -0.334$, 95% CI [−0.645, −0.022], $t = -2.10$, $p = 0.036$. As we move up to the next value of callous traits (−17.1002), the relationship between time spent gaming and aggression is still significant ($p = 0.0500$), but at the next value it becomes non-significant ($p = 0.058$). Therefore, the threshold for significance ends at −17.1002 (which we were told at the top of the output). As we increase the value of callous-unemotional traits the relationship between time spent gaming and aggression remains non-significant until the value of callous-unemotional traits is −0.723, at which point it crosses the threshold for significance again. For all subsequent values of callous-unemotional traits the relationship between time spent gaming and aggression is significant. Looking at the *b*-values themselves (in the column labelled *Effect*) we can also see that as callous-unemotional traits increases, the strength of relationship between time spent gaming and aggression goes from a small negative effect ($b = -0.334$) to a strong positive one ($b = 0.830$).

Output 11.1

```
**************************************************************

Model  = 1
     Y = Aggress
     X = Vid_Game
     M = CaUnTs

Sample size
         442

**************************************************************

Outcome: Aggress

Model Summary

         R       R-sq        MSE          F        df1        df2          p
     .6142      .3773    99.5266    90.5311     3.0000   438.0000      .0000

Model

              coeff         se          t          p       LLCI       ULCI
constant    39.9671      .4750    84.1365      .0000    39.0335    40.9007
CaUnTs        .7601      .0466    16.3042      .0000      .6685      .8517
Vid_Game      .1696      .0759     2.2343      .0260      .0204      .3188
int_1         .0271      .0073     3.7051      .0002      .0127      .0414


Interactions:

   int_1    Vid_Game    X    CaUnTs
```

Output 11.2

```
Conditional effect of X on Y at values of the moderator(s):

    CaUnTs     Effect         se          t          p       LLCI       ULCI
   -9.6177     -.0907      .1058     -.8568      .3920     -.2986      .1173
    .0000      .1696      .0759     2.2343      .0260      .0204      .3188
   9.6177      .4299      .1010     4.2562      .0000      .2314      .6284

Values for quantitative moderators are the mean and plus/minus one SD
from mean.

Values for dichotomous moderators are the two values of the moderator.
```

The final way to break down the interaction is to plot it. In Figure 11.6 we asked *PROCESS* to generate data for plotting, and these data are at the bottom of the output (see Figure 11.7). We're given values of the variable **Vid_Game** (−6.9622, 0, 6.9622) and of **CaUnTs** (−9.6177, 0, 9.6177). These values are not important in themselves, but they correspond to low, mean and high values of the variable. The final column contains the predicted values of the outcome (aggression) for these combinations of the predictors. For example, when **Vid_Game** and **CaUnTs** are both low (−6.9622 and −9.6177, respectively) the predicted value of aggression is 33.2879, when both variables are at their mean (0 and 0), the predicted value of aggression is 39.9671, and so on. To create the plot, put these values into the data editor. In a blank data editor window create coding variables that represent low, mean and high (use any codes you like). Then enter all combinations of these codes. For example, in Figure 11.7 I've created variables called **Games** and **CaUnTs** both of which are coding variables

(1 = low, 2 = mean, 3 = high), then entered the combinations of these codes that correspond to the *PROCESS* output (low–low, mean–low, high–low, etc.), and then typed in the corresponding predicted values from the *PROCESS* output. Hopefully you can see from Figure 11.7 how the output from *PROCESS* corresponds to the new data file. If you can't work out how to create the file yourself, use **Video Game Graph.sav**. Having transferred the output to a data file, we can draw a line graph using what we learnt in Chapter 5.



Draw a multiple line graph of **Aggress** (*y*-axis) against **Games** (*x*-axis) with different coloured lines for different values of **CaUnTs**.

Output 11.3

```
********************* JOHNSON-NEYMAN TECHNIQUE *************************

Moderator value(s) defining Johnson-Neyman significance region(s):
      Value     % below     % above
   -17.1002      1.3575     98.6425
    -.7232     48.8688     51.1312

Conditional effect of X on Y at values of the moderator (M)
     CaUnTs      Effect          se           t           p        LLCI        ULCI
   -18.5950     -.3336      .1587     -2.1027      .0361     -.6454     -.0218
   -17.1002     -.2931      .1492     -1.9654      .0500     -.5863      .0000
   -16.4450     -.2754      .1451     -1.8987      .0583     -.5605      .0097
   -14.2950     -.2172      .1319     -1.6467      .1003     -.4765      .0420
   -12.1450     -.1590      .1194     -1.3319      .1836     -.3937      .0756
    -9.9950     -.1009      .1077      -.9361      .3497     -.3126      .1109
    -7.8450     -.0427      .0972      -.4390      .6609     -.2338      .1484
    -5.6950      .0155      .0882       .1757      .8606     -.1579      .1889
    -3.5450      .0737      .0813       .9059      .3655     -.0862      .2336
    -1.3950      .1319      .0771      1.7111      .0878     -.0196      .2833
     -.7232      .1501      .0763      1.9654      .0500      .0000      .3001
      .7550      .1901      .0759      2.5053      .0126      .0410      .3392
     2.9050      .2482      .0779      3.1878      .0015      .0952      .4013
     5.0550      .3064      .0829      3.6980      .0002      .1436      .4693
     7.2050      .3646      .0903      4.0360      .0001      .1871      .5422
     9.3550      .4228      .0997      4.2386      .0000      .2267      .6188
    11.5050      .4810      .1106      4.3490      .0000      .2636      .6983
    13.6550      .5392      .1225      4.4013      .0000      .2984      .7799
    15.8050      .5973      .1352      4.4188      .0000      .3317      .8630
    17.9550      .6555      .1484      4.4160      .0000      .3638      .9473
    20.1050      .7137      .1621      4.4017      .0000      .3950     1.0324
    22.2550      .7719      .1762      4.3814      .0000      .4256     1.1181
    24.4050      .8301      .1905      4.3580      .0000      .4557     1.2044

************************************************************************
```

Significant

Not significant

Significant

The graph from the self-test (Figure 11.8) shows what we found from the simple slopes analysis. When callous traits are low (blue line) there is a non-significant negative relationship between time spent gaming and aggression; at the mean value of callous traits (red line) there is small positive relationship between time spent gaming and aggression; and this relationship gets even stronger at high levels of callous traits (green line).

Figure 11.7 Entering data for graphing simple slopes

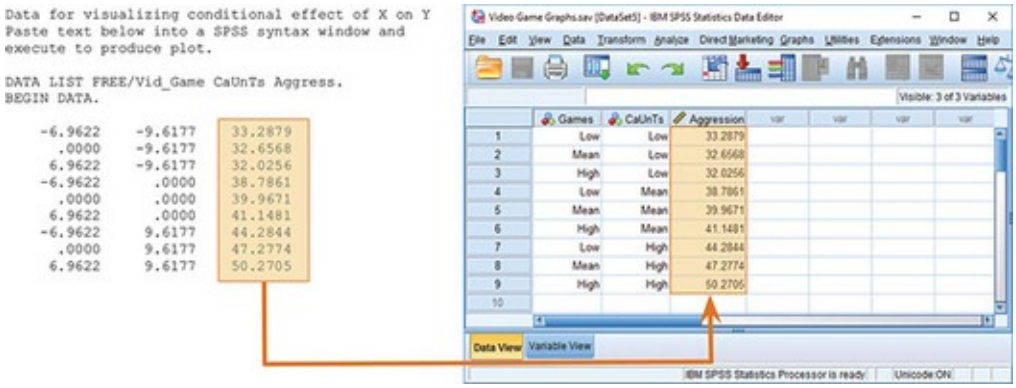
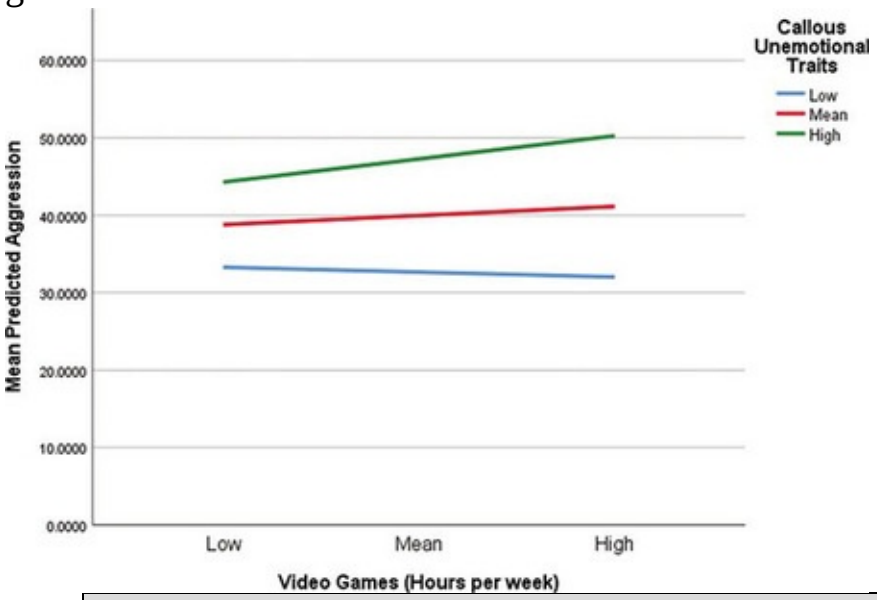


Figure 11.8 Simple slopes equations of the regression of aggression on video games at three levels of callous traits





Now draw a multiple line graph of **Aggress** (*y*-axis) against **CaUnTs** (*x*-axis) with different coloured lines for different values of **Games**.

11.3.8 Reporting Moderation Analysis

Moderation can be reported in the same way as described in Section 9.13. My personal preference would be to produce a table like Table 11.1.

Table 11.1 Linear model of predictors of aggression

	b	*SE B*	*t*	*p*
Constant	39.97 [39.03, 40.90]	0.475	84.13	$p < 0.001$
Callous Traits (centred)	0.76 [0.67, 0.85]	0.047	16.30	$p < 0.001$
Gaming (centred)	0.17 [0.02, 0.32]	0.076	2.23	$p = 0.026$
Callous Traits × Gaming	0.027 [0.01, 0.04]	0.007	3.71	$p < 0.001$

Note. $R^2 = 0.38$.

Note. $R^2 = 0.38$.

Cramming Sam's Tips Moderation



- Moderation occurs when the relationship between two variables changes as a function of a third variable. For example, the relationship between watching horror films and feeling scared at bedtime might increase as a function of how vivid an imagination a person has.
- Moderation is tested using a linear model in which the outcome (fear at bedtime) is predicted from a predictor (how many horror films are watched), the moderator (imagination) and the interaction of the predictor variables.
- Predictors should be centred before the analysis.
- The interaction of two variables is their scores multiplied together.
- If the interaction is significant then the moderation effect is also significant.
- If moderation is found, follow up the analysis with simple slopes analysis, which looks at the relationship between the predictor and outcome at low, mean and high levels of the moderator.



11.4 Mediation

11.4.1 The conceptual model

Whereas moderation alludes to the combined effect of two variables on an outcome, **mediation** refers to a situation when the relationship between a predictor variable and an outcome variable can be explained by their relationship to a third variable (the **mediator**). The top of Figure 11.9 shows a relationship between a predictor and an outcome (denoted by *c*). The bottom of the figure shows that these variables are also related to a third variable in specific ways: (1) the predictor also predicts the mediator through the path denoted by *a*; (2) the mediator predicts the outcome through the path denoted by *b*. The relationship between the predictor and outcome will probably be different when the mediator is also included in the model and so is denoted *c*'. The letters denoting each path (*a*, *b*, *c* and *c*') represent the unstandardized regression coefficient between the variables connected by the arrow; therefore, they symbolize the strength of relationship between variables. Mediation is said to have occurred if the strength of the relationship between the predictor and outcome is reduced by including the mediator (i.e., the *b*-value for *c*' is smaller than for *c*). Perfect mediation occurs when *c*' is zero: the relationship between the predictor and outcome is completely wiped out by including the mediator in the model.



This description is a bit abstract, so let's use an example. My wife and I often wonder what the important factors are in making a relationship last. For my part, I don't really understand why she'd want to be with a balding heavy rock fan with an oversized collection of vinyl and musical instruments and an unhealthy love of *Doctor Who* and numbers. It is important I gather as much information as possible about keeping her happy because the odds are stacked against me. For

her part, I have no idea why she wonders: her very existence makes me happy. Perhaps if you are in a relationship you have wondered how to make it last too. During our cyber-travels, Mrs Field and I have discovered that physical attractiveness (McNulty, Neff, & Karney, 2008), conscientiousness and neuroticism (good for us) predict marital satisfaction (Claxton, O'Rourke, Smith, & Delongis 2012). Pornography use probably doesn't: it is related to infidelity (Lambert, Negash, Stillman, Olmstead, & Fincham, 2012). Mediation is really all about the variables that explain relationships like these: it's unlikely that everyone who catches a glimpse of some porn suddenly rushes out of their house to have an affair – presumably it leads to some kind of emotional or cognitive change that undermines the love glue that holds us and our partners together. Lambert *et al.* tested this hypothesis. Figure 11.10 shows their mediator model: the initial relationship is that between pornography consumption (the predictor) and infidelity (the outcome), and they hypothesized that this relationship is mediated by commitment (the mediator). This model suggests that the relationship between pornography consumption and infidelity isn't a direct effect but operates though a reduction in relationship commitment. For this hypothesis to be true: (1) pornography consumption must predict infidelity in the first place (path *c*'); (2) pornography consumption must predict relationship commitment (path *a*); (3) relationship commitment must predict infidelity (path *b*); and (4) the relationship between pornography consumption and infidelity should be smaller when relationship commitment is included in the model than when it isn't. We can distinguish between the **direct effect** of pornography consumption on infidelity, which is the relationship between them controlling for relationship commitment, and the **indirect effect**, which is the effect of pornography consumption on infidelity through relationship commitment (Figure 11.10).

Figure 11.9 Diagram of a basic mediation model

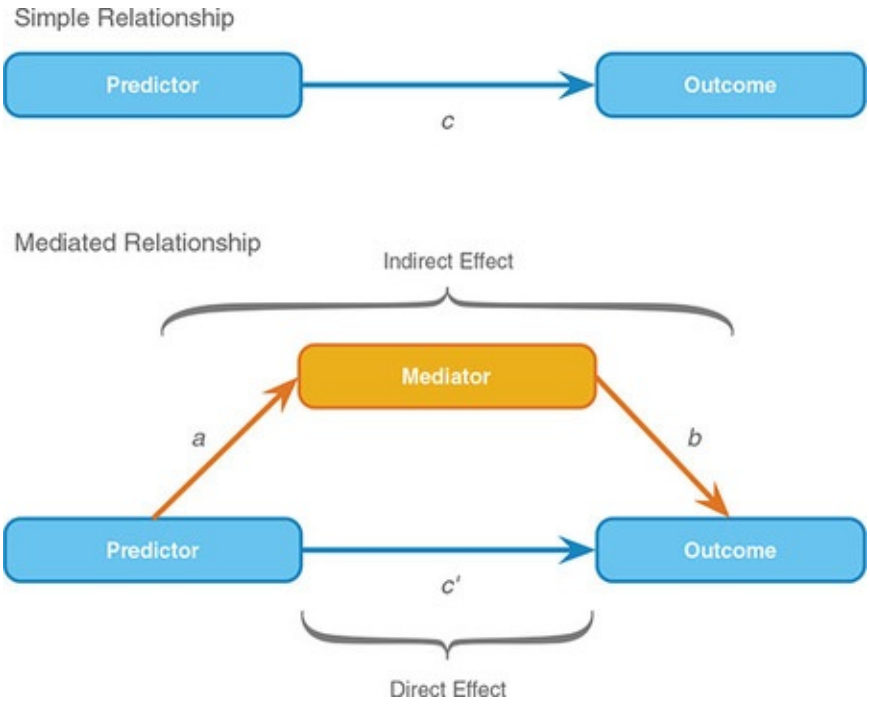
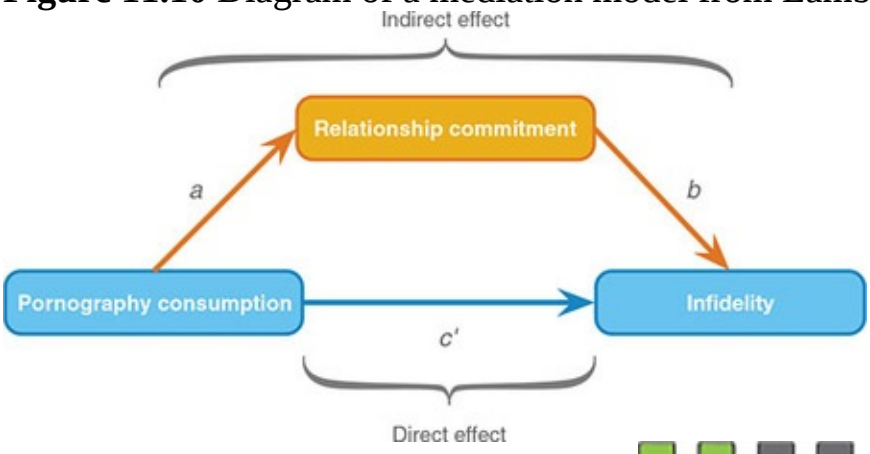


Figure 11.10 Diagram of a mediation model from Lambert *et al.* (2012)



11.4.2 The statistical model

Unlike moderation, the statistical model for mediation is basically the same as the conceptual model: it is characterized in Figure 11.9. Historically, this model was tested through a series of linear models that reflect the four conditions necessary to demonstrate mediation (Baron & Kenny, 1986). I have mentioned already that the letters denoting the paths in Figure 11.9 represent the unstandardized *b*-values for the relationships between variables denoted by the path. Therefore, to estimate any one of these paths, we want to know the unstandardized *b* for the two variables involved. For example, Baron and Kenny suggested that mediation is tested through three linear models (see also Judd &

Kenny, 1981):

1. A linear model predicting the outcome from the predictor variable. The *b*-value coefficient for the predictor gives us the value of *c* in Figure 11.9.
2. A linear model predicting the mediator from the predictor variable. The *b*-value for the predictor gives us the value of *a* in Figure 11.9.
3. A linear model predicting the outcome from both the predictor variable and the mediator. The *b*-value for the predictor gives us the value of *c*' in Figure 11.9, and the *b*-value for the mediator gives us the value of *b*.

These models test the four conditions of mediation: (1) the predictor variable must significantly predict the outcome variable in model 1; (2) the predictor variable must significantly predict the mediator in model 2; (3) the mediator must significantly predict the outcome variable in model 3; and (4) the predictor variable must predict the outcome variable less strongly in model 3 than in model 1.

In Lambert et al.'s (2012) study, all participants had been in a relationship for at least a year. The researchers measured pornography consumption on a scale from 0 (low) to 8 (high), but this variable, as you might expect, was skewed (most people had low scores) so they analysed log-transformed values (**LnPorn**). They also measured commitment to their current relationship (**Commit**) on a scale from 1 (low) to 5 (high). Infidelity was measured with questions asking whether the person had committed a physical act (**Phys_Inf**) that they or their partner would consider to be unfaithful (0 = no, 1 = one of them would consider the act unfaithful, 2 = both of them would consider it unfaithful),[8] and also using the number of people they had 'hooked up' with in the previous year (**Hook_Ups**), which would mean during a time period in which they were in their current relationship.[9] The actual data from Lambert et al.'s study are in the file **Lambert *et al.* (2012).sav**.

8 I've coded this variable differently from the original data to make interpretation of it more intuitive, but it doesn't affect the results.

9 A 'hook-up' was defined to participants as 'when two people get together for a physical encounter and don't necessarily expect anything further (e.g., no plan or intention to do it again)'.



Run the three models necessary to test mediation for Lambert et al.'s data: (1) a linear model predicting **Phys_Inf** from **LnPorn**; (2) a linear model predicting **Commit** from **LnPorn**; and (3) a linear

model predicting **Phys_Inf** from both **LnPorn** and **Commit**. Is there mediation?

Many people still use this approach to test mediation. I think it is very useful for illustrating the principles of mediation and for understanding what mediation means, but it has limitations. The main one is the fourth criterion: *the predictor variable must predict the outcome variable less strongly in model 3 than in model 1*. Although perfect mediation is shown when the relationship between the predictor and outcome is reduced to zero in model 3, in practice it rarely happens. Instead, you see a reduction in the relationship between the predictor and outcome, but it is not wiped out altogether. The question, therefore, arises of how much 'reduction' is sufficient to infer mediation.

Although Baron and Kenny advocated looking at the sizes of the *b*-values, in practice people tend to look for a change in significance; so, mediation would occur if the relationship between the predictor and outcome was significant ($p < 0.05$) when looked at in isolation (model 1) but not significant ($p > 0.05$) when the mediator is included too (model 3). This approach leads to all sorts of silliness because of the 'all-or-nothing' thinking that *p*-values encourage. You could have a situation in which the *b*-value for the relationship between the predictor and outcome changes very little in models with and without the mediator, but the *p*-values shift from one side of the threshold to the other (e.g., from $p = 0.049$ when the mediator isn't included to $p = 0.051$ when it is). Even though the *p*-values have changed from significant to not significant, the change is very small, and the size of the relationship between the predictor and outcome will not have changed very much at all. Conversely, you could have a situation where the *b* for the relationship between the predictor and the outcome reduces a lot when the mediator is included, but remains significant in both cases. For example, perhaps when looked at in isolation the relationship between the predictor and outcome is $b = 0.46$, $p < 0.001$, but when the mediator is included as a predictor as well it reduces to $b = 0.18$, $p = 0.042$. You'd conclude (based on significance) that mediation hadn't occurred despite the relationship between the predictor and outcome reducing to more than half its original value.

An alternative is to estimate the indirect effect and its significance. The indirect effect is illustrated in Figures 11.9 and 11.10; it is the combined effects of paths *a* and *b*, and the significance of this test can be assessed using the **Sobel test** (Sobel, 1982). If the Sobel test is significant it means that the predictor significantly affects the outcome variable via the mediator. In other words, there is significant mediation. This test works well in large samples, but you're better off computing confidence intervals for the indirect effect using bootstrap

methods (Section 6.12.3). Now that computers make it easy for us to estimate the indirect effect (i.e., the effect of mediation) and its confidence interval, this practice is becoming increasingly common and is preferable to Baron and Kenny's regressions and the Sobel test because it's harder to get sucked into the black-and-white thinking of significance testing (Section 3.2.2). People tend to apply Baron and Kenny's method in a way that is intrinsically bound to looking for 'significant' relationships, whereas estimating the indirect effect and its confidence interval allows us to simply report the degree of mediation observed in the data.

11.4.3 Effect sizes of mediation 

If we're going to look at the size of the indirect effect to judge the degree of mediation, then it's useful to have effect size measures to help us (see Section 3.5). Many effect size measures have been proposed and are discussed in detail elsewhere (MacKinnon, 2008; Preacher & Kelley, 2011). The simplest is to look at the *b*-value for the indirect effect and its confidence interval. Figure 11.9 shows us that the indirect effect is the combined effect of paths *a* and *b*. We have also seen that *a* and *b* are unstandardized model coefficients for the relationships between variables denoted by the path. To find the combined effect of these paths, we multiply these *b*-values:

$$\text{indirect effect} = ab \tag{11.6}$$

The resulting value is an unstandardized regression coefficient like any other, and consequently is expressed in the original units of measurement. As we have seen, it is sometimes useful to look at standardized *b*-values, because these can be compared across different studies using different outcome measures (see Chapter 9). MacKinnon (2008) suggested standardizing this measure by dividing by the standard deviation of the outcome variable:

$$\text{indirect effect (partially standardized)} = \frac{ab}{s_{\text{Outcome}}} \tag{11.7}$$

This standardizes the indirect effect with respect to the outcome variable, but not the predictor or mediator. As such, it is sometimes referred to as the partially standardized indirect effect. To fully standardize the indirect effect we would need to multiply the partially standardized measures by the standard deviation of the predictor variable (Preacher & Hayes, 2008b):

$$\text{indirect effect (standardized)} = \frac{ab}{s_{\text{Outcome}}} \times s_{\text{predictor}} \tag{11.8}$$

This measure is sometimes called the **index of mediation**. It is useful in that it can be compared across different mediation models that use different measures of the predictor, outcome and mediator. Reporting this measure would be

particularly helpful if anyone decides to include your research in a meta-analysis.

A different approach to estimating the size of the indirect effect is to look at the size of the indirect effect relative to either the total effect of the predictor or the direct effect of the predictor. For example, if we wanted the ratio of the indirect effect (*ab*) to the total effect (*c*) we could use the *b*-values from the various linear models displayed in Figure 11.9:

$$P_M = \frac{ab}{c} \tag{11.9}$$

Similarly, if we wanted to express the indirect effect as a ratio of the direct effect (*c'*), the models give us the values we need:

$$R_M = \frac{ab}{c'} \tag{11.10}$$

These ratio-based measures only really re-describe the original indirect effect. Both are very unstable in small samples, and MacKinnon (2008) advises against using P_M and R_M in samples smaller than 500 and 5000, respectively. Also, although it is tempting to think of P_M as a proportion (because it is the ratio of the indirect effect to the *total* effect) it is not: it can exceed 1 and even take negative values (Preacher & Kelley, 2011).

We used R^2 to assess the fit of a linear model. We can compute a form of R^2 for the indirect effect, which tells us the proportion of variance explained by the indirect effect. MacKinnon (2008) proposes several versions, but *PROCESS* computes the following:

$$R^2_M = R^2_{Y,M} - \left(R^2_{Y,MX} - R^2_{Y,X}\right) \tag{11.11}$$

This equation uses the proportion of variance in the outcome variables explained by the predictor ($R^2_{Y,X}$), the mediator ($R^2_{Y,M}$), and both ($R^2_{Y,MX}$). It can be interpreted as the variance in the outcome that is shared by the mediator and the predictor, but that cannot be attributed to either in isolation. Again, this measure is not bounded to fall between 0 and 1, and it's possible to get negative values (which usually indicate suppression effects rather than mediation).

Finally, Preacher and Kelley (2011) proposed a measure called kappa-squared (κ^2) that was available in versions of *PROCESS* prior to 2.16. Unfortunately, the maths behind the measure has subsequently been shown to be incorrect, leading to paradoxical (and undesirable) effects such as κ^2 *decreasing* as mediation effect *increases* (Wen & Fan, 2015). If you're using an older version of *PROCESS*, ignore this measure.

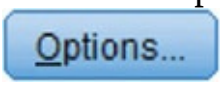
Probably the most useful of these measures are the unstandardized and standardized indirect effect. All the measures have accompanying confidence

intervals and are unaffected by sample sizes (although note my earlier comments about the variability of P_M and R_M in small samples). However, P_M, R_M and R^2_M cannot be interpreted easily because they aspire to being proportions but are not, and all of the measures are unbounded, which again makes interpretation tricky (Preacher & Kelley, 2011). For these reasons Wen and Fan (2015) argue against effect size measures for mediation, although for simple mediation models they concede that P_M can be useful provided that it is accompanied by the total effect (as important context for the size of the indirect effect).

11.4.4 Mediation using SPSS Statistics 

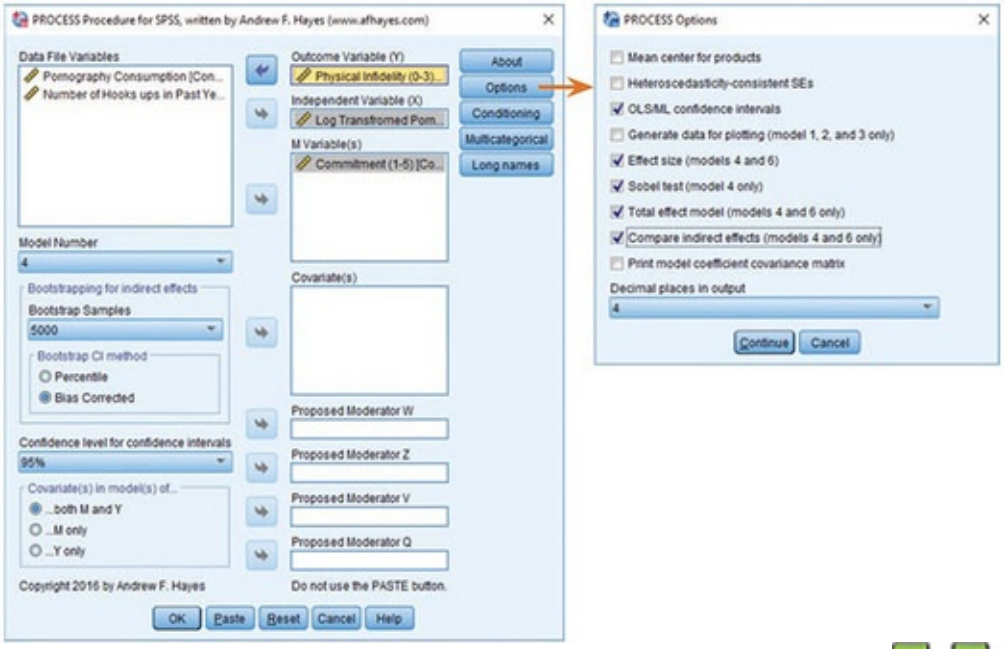
We can test Lambert's mediation model (Figure 11.10) using the *PROCESS* tool.

Access the dialog boxes in Figure 11.11 by selecting *Analyze*  *Regression* . Drag (or click ) the outcome variable (**Phys_Inf**) from the box labelled *Data File Variables* to the box labelled *Outcome Variable (Y)*, then drag the predictor variable (**LnPorn**) to the box labelled *Independent Variable (X)*. Finally, drag the mediator variable (**Commit**) to the box labelled *M Variable(s)*, where you specify any mediators (you can have more than one). Simple mediation is represented by model 4 (the default), therefore, make sure that  is selected in the drop-down list under *Model Number*. Unlike for moderation, there are other options in this dialog box that are useful. For example, to test the indirect effects we will use bootstrapping to generate a confidence interval around the indirect effect. By default *PROCESS* uses 5000 bootstrap samples, and will compute bias corrected and accelerated confidence intervals. These default options are fine, but be aware that you can ask for percentile bootstrap confidence intervals instead (see Section 6.12.3).

Click  to open another dialog box. Selecting (1) *Effect size* produces the estimates of the size of the indirect effect discussed in Section 11.4.3;[10] (2) *Sobel test* produces a significance test of the indirect effect devised

by Sobel; (3) *Total effect model* produces the direct effect of the predictor on the outcome (in this case the linear model of infidelity predicted from pornography consumption); and (4) *Compare indirect effects* will, when you have more than one mediator in the model, estimate the effect and confidence interval for the difference between the indirect effects resulting from these mediators. This final option is useful when you have more than one mediator to compare their relative importance in explaining the relationship between the predictor and outcome. However, we have only a single mediator, so we don't need to select this option (you can select it if you like, but it won't change the output produced). None of the options activated by clicking Conditioning apply to simple mediation models, so we can ignore this button. If our predictor variable (*X*) was categorical with more than two categories (which it's not), we could click Multicategorical to get *PROCESS* to automatically dummy-code (Section 11.5.1) it for us. Back in the main dialog box, click OK to run the analysis. 10 and κ^2 are produced only for models with a single mediator.

Figure 11.11 The dialog boxes for running mediation analysis



11.4.5 Output from mediation analysis

As with moderation, the output appears as text. Output 11.4 shows the first part of the output, which initially tells us the name of the outcome (*Y*), the predictor

(*X*) and the mediator (*M*) variables (SPSS Tip 11.1). This is useful for double-checking we have entered the variables in the correct place: the outcome is infidelity, the predictor is (log-transformed) pornography consumption, and the mediator is commitment. The next part of the output shows us the results of the linear model of commitment predicted from pornography consumption (i.e., path *a* in Figure 11.10). This output is interpreted just as we would interpret any linear model: pornography consumption significantly predicts relationship commitment, $b = -0.47$, 95% CI [−0.89, −0.05], $t = -2.21$, $p = 0.028$. The R^2 values tells us that pornography consumption explains 2% of the variance in relationship commitment, and the fact that the *b* is negative tells us that as consumption increases, commitment declines (and vice versa).

Oditi's Lantern Moderation and mediation



'I, Oditi, want you to join my cult of undiscovered numerical truths. I also want you to stare into my lantern to gain statistical enlightenment. It's possible that statistical knowledge mediates the relationship between staring into my lantern and joing my cult … or it could be mediated by neurological changes to your brain created by the subliminal messages in the videos. Stare into my lantern to find out about mediation and moderation.'



Output 11.4

```
**************************************************************
Model  = 4
     Y = Phys_Inf
     X = LnPorn
     M = Commit
Sample size
        239
**************************************************************
Outcome: Commit

Model Summary
          R   R-sq     MSE       F       df1         df2         p
      .1418   .0201   .5354  4.8633    1.0000   237.0000    .0284

Model
              coeff         se          t          p       LLCI       ULCI
  constant   4.2027      .0545    77.1777     .0000     4.0954     4.3100
    LnPorn   -.4697      .2130    -2.2053     .0284     -.8892     -.0501
```

Output 11.5 shows the results of the regression of infidelity predicted from both pornography consumption (i.e., path *c*′ in Figure 11.10) and commitment (i.e., path *b* in Figure 11.10). We can see that pornography consumption significantly predicts infidelity even with relationship commitment in the model, $b = 0.46$, 95% CI [−0.07, 0.84], $t = 2.35$, $p = 0.02$; relationship commitment also significantly predicts infidelity, $b = -0.27$, 95% CI [−0.39, −0.16], $t = -4.61$, $p < 0.001$. The R^2 value tells us that the model explains 11.4% of the variance in infidelity. The negative *b* for commitment tells us that as commitment increases, infidelity declines (and vice versa), but the positive *b* for consumptions indicates that as pornography consumption increases, infidelity increases also. These relationships are in the predicted direction.

Output 11.5

```
**************************************************************
Outcome: Phys_Inf

Model Summary
          R       R-sq        MSE          F        df1         df2          p
      .3383      .1144      .4379    15.2453     2.0000   236.0000      .0000

Model
              coeff         se          t          p       LLCI       ULCI
  constant   1.3704      .2518     5.4433     .0000      .8744     1.8663
    Commit   -.2710      .0587    -4.6128     .0000     -.3867     -.1552
    LnPorn    .4573      .1946     2.3505     .0196      .0740      .8407
```

Output 11.6 shows the total effect of pornography consumption on infidelity (outcome). You will get this bit of the output only if you selected *Total effect model* in Figure 11.11. The total effect is the effect of the predictor on the outcome when the mediator is not present in the model – in other words, path *c* in Figure 11.9. When relationship commitment is not in the model, pornography consumption significantly predicts infidelity, $b = 0.58$, 95% CI [0.19, 0.98], $t = 2.91$, $p = 0.004$. The R^2 value tells us that the model explains 3.46% of the variance in infidelity. As is the case when we include relationship commitment in the model, pornography consumption has a positive relationship with infidelity (as shown by the positive *b*-value).

Output 11.6

```
********************* TOTAL EFFECT MODEL *********************
Outcome: Phys_Inf

Model Summary
          R       R-sq        MSE          F        df1        df2          p
      .1859      .0346      .4754     8.4866     1.0000   237.0000      .0039

Model
              coeff         se          t          p       LLCI       ULCI
  constant      .2315      .0513     4.5123      .0000      .1304      .3326
    LnPorn      .5846      .2007     2.9132      .0039      .1893      .9800
```

Output 11.7 is the most important part of the output because it displays the results for the indirect effect of pornography consumption on infidelity (i.e., the effect via relationship commitment). First, we're told the effect of pornography consumption on infidelity in isolation (the total effect), and these values replicate the model in Output 11.6. Next, we're told the effect of pornography consumption on infidelity when relationship commitment is included as a predictor as well (the direct effect). These values replicate those in Output 11.5. The first bit of new information is the *Indirect effect of X on Y*, which in this case is the indirect effect of pornography consumption on infidelity. We're given an estimate of this effect ($b = 0.127$) as well as a bootstrapped standard error and confidence interval. As we have seen many times before, 95% confidence intervals contain the true value of a parameter in 95% of samples. People tend to assume that their sample isn't one of the 5% that does not contain the true value and use them to infer the population value of an effect. In this case, assuming our sample is one of the 95% that 'hits' the true value, we know that the true *b*-value for the indirect effect falls between 0.017 and 0.297.[11] This range does not include zero, and remember that $b = 0$ would mean 'no effect whatsoever'; therefore, the fact that the confidence interval does not contain zero supports the idea that relationship commitment really does mediate the relationship between pornography consumption and infidelity.

11 Remember that because of the nature of bootstrapping you will get slightly different values in your output.

The rest of Output 11.7 you will see only if you selected *Effect size* in Figure 11.11; it contains the effect size measures from Section 11.4.3. Rather than interpret them all, I'll note that for each one you get an estimate along with a confidence interval based on a bootstrapped standard error. As with the unstandardized indirect effect, if the confidence intervals don't contain zero then people assume that the true effect size is different from 'no effect'. In other words, there is mediation. All of the effect size measures have confidence intervals that don't include zero, so whatever one we look at we can assume that the indirect effect is probably greater than 'no effect'. Focusing on the most

useful of these effect sizes, the standardized *b* for the indirect effect, its value is *b* = 0.041, 95% BCa CI [0.005, 0.092].

Output 11.7

```
************** TOTAL, DIRECT, AND INDIRECT EFFECTS *****************

Total effect of X on Y
     Effect         SE          t          p       LLCI       ULCI
      .5846      .2007     2.9132     .0039     .1893     .9800
Direct effect of X on Y
     Effect         SE          t          p       LLCI       ULCI
      .4573      .1946     2.3505     .0196     .0740     .8407
Indirect effect of X on Y
                Effect    Boot SE  BootLLCI  BootULCI
     Commit      .1273      .0708     .0170     .2972
Partially standardized indirect effect of X on Y
                Effect    Boot SE  BootLLCI  BootULCI
     Commit      .1818      .0997     .0215     .4156
Completely standardized indirect effect of X on Y
                Effect    Boot SE  BootLLCI  BootULCI
     Commit      .0405      .0220     .0052     .0922
Ratio of indirect to total effect of X on Y
                Effect    Boot SE  BootLLCI  BootULCI
     Commit      .2177     8.6658     .0082    1.2609
Ratio of indirect to direct effect of X on Y
                Effect    Boot SE  BootLLCI  BootULCI
     Commit      .2783    13.7610    -.0392    4.1073
R-squared mediation effect size (R-sq_med)
                Effect    Boot SE  BootLLCI  BootULCI
     Commit      .0138      .0104     .0009     .0462
```

The final part of the output (Output 11.8) shows the results of the Sobel test. As I have mentioned before, it is better to interpret the bootstrap confidence intervals than formal tests of significance; however, if you selected *Sobel test* in Figure 11.11, this is what you will see. Again, we're given the size of the indirect effect (*b* = 0.127), the standard error, associated *z*-score (*z* = 1.95) and *p*-value (*p* = 0.051).[12] The *p*-value isn't quite under the not-at-all magic 0.05 threshold, so technically we'd conclude that there isn't a significant indirect effect, but this shows you how misleading these kind of tests can be: every single effect size had a confidence interval not containing zero, so there is compelling information that there is a small but meaningful mediation effect.

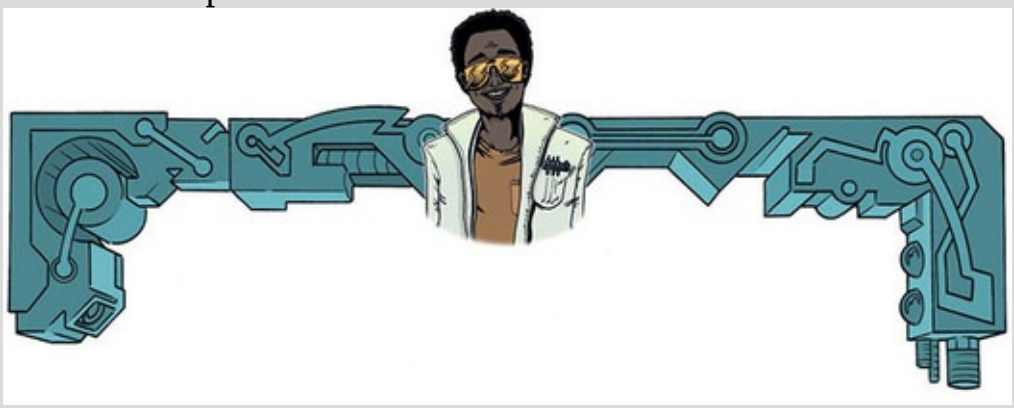
12 You might remember that in the linear model we calculate a test statistic (*t*) by dividing the regression coefficient by its standard error (as in equation (9.14)). We do the same here, except we get a *z* instead of a *t*: *z* = 0.1273/0.0652 = 1.9525.

Output 11.8

```
Normal theory tests for indirect effect
      Effect         se          Z          p
      .1273      .0652     1.9526     .0509
```

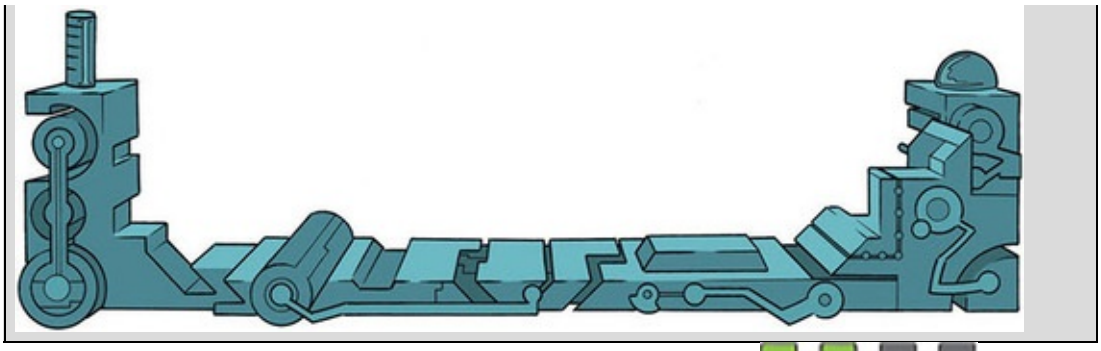
Labcoat Leni's Real Research 11.1 I heard that Jane has a boil and

kissed a tramp 



Massar, K., *et al.* (2012). *Personality and Individual Differences*, *52*, 106–109.

Everyone likes a good gossip from time to time, but apparently it has an evolutionary function. One school of thought is that gossip is used as a way to derogate sexual competitors – especially by questioning their appearance and sexual behaviour. For example, if you've got your eyes on a guy, but he has his eyes on Jane, then a good strategy is to spread gossip that Jane has a massive pus-oozing boil on her stomach and that she kissed a smelly vagrant called Aqualung. Apparently men rate gossiped-about women as less attractive, and they are more influenced by the gossip if it came from a woman with a high mate value (i.e., attractive and sexually desirable). Karlijn Massar and her colleagues hypothesized that if this theory is true then (1) younger women will gossip more because there is more mate competition at younger ages; and (2) this relationship will be mediated by the mate value of the person (because for those with high mate value gossiping for the purpose of sexual competition will be more effective). Eighty-three women aged from 20 to 50 (**Age**) completed questionnaire measures of their tendency to gossip (**Gossip**) and their sexual desirability (**Mate_Value**). Test Massar et al.'s mediation model using Baron and Kenny's method (as they did) but also using *PROCESS* to estimate the indirect effect (**Massar *et al.* (2011).sav**). Answers are on the companion website (or look at Figure 1 in the original article, which shows the parameters for the various regressions).



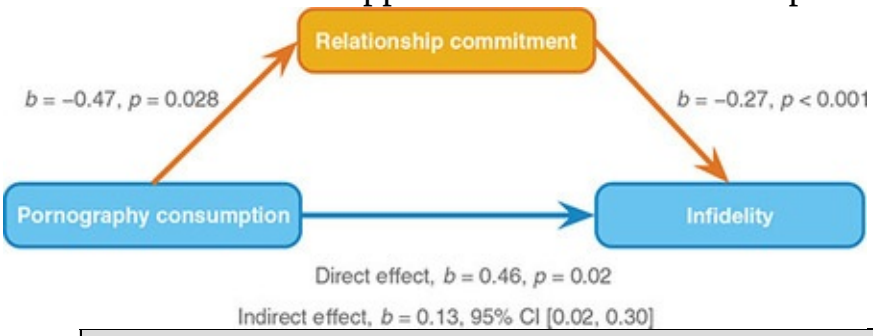
11.4.6 Reporting mediation analysis

Some people report only the indirect effect in mediation analysis, and possibly the Sobel test. However, I have repeatedly favoured using bootstrap confidence intervals, so you should report these:

> ✓ There was a significant indirect effect of pornography consumption on infidelity through relationship commitment, *b* = 0.127, 95% BCa CI [0.017, 0.297].

This is fine, but it can be quite useful to present a diagram of the mediation model, and indicate on it the regression coefficients, the indirect effect and its bootstrapped confidence intervals. For the current example, we might produce something like Figure 11.12.

Figure 11.12 Model of pornography consumption as a predictor of infidelity, mediated by relationship commitment. The confidence interval for the indirect effect is a BCa bootstrapped CI based on 5000 samples



Cramming Sam's Tips Mediation



- Mediation is when the strength of the relationship between a predictor variable and outcome variable is reduced by including another variable as a predictor. Essentially, mediation equates to the relationship between two variables being ‘explained’ by a third. For example, the relationship between watching horror films and feeling scared at bedtime might be explained by scary images appearing in your head.
- Mediation is tested by assessing the size of the *indirect effect* and its confidence interval. If the confidence interval contains zero then we tend to assume that a genuine mediation effect doesn’t exist. If the confidence interval doesn’t contain zero, then we tend to conclude that mediation has occurred.



11.5 Categorical predictors in regression

We saw in the previous chapter that categorical predictors with two categories can be included in a linear model: we simply code the categories with 0 and 1.[13] However, often you’ll collect data about groups of people in which there are more than two categories (e.g., ethnic group, gender, socio-economic status, diagnostic category). Given that we can include categorical predictors with two categories into a linear model (Section 10.4), and we can have several such predictors in a model, it follows that we can include a predictor with more than two categories by converting it to several variables each of which has two categories. There are several different ways to achieve this, one of the most common is dummy coding (a.k.a. indicator coding).[14]

13 We saw in Section 10.2.2 why we use 0 and 1, and I elaborate on this issue in

Section 12.2.1.
14 For some more detailed reading, see Hardy (1993).

11.5.1 Dummy coding 

Imagine a survey had a question about religiosity that resulted in many categories such as Muslim, Jewish, Hindu, Catholic, Buddhist, Protestant, Jedi,[15] other. These groups cannot be distinguished using a single variable coded with zeros and ones. If we wanted to include this variable in a linear model we would need to create **dummy variables**, which is a way of representing groups of people using only zeros and ones. To do it, we create several variables; in fact, the number of variables we need is one less than the number of groups we're recoding. There are eight basic steps:

15 Approximately 390,000 (almost 0.8%) people in England and Wales stated Jedi as their religion on the 2001 Census form, making it the fourth most popular religion. In my hometown of Brighton, 2.6% of the city claimed to be Jedi.

1. Count the number of groups you want to recode and subtract 1.
2. Create as many new variables as the value you calculated in step 1. These are your dummy variables.
3. Choose one of your groups as a baseline against which all other groups will be compared. Normally you'd pick a group that might be considered a control, or, if you don't have a specific hypothesis, the group that represents the majority of people (because it might be interesting to compare other groups against the majority).
4. Having chosen a baseline group, assign that group values of 0 for all dummy variables.
5. For your first dummy variable, assign the value 1 to the first group that you want to compare against the baseline group. Assign all other groups 0 for this variable.
6. For the second dummy variable assign the value 1 to the second group that you want to compare against the baseline group. Assign all other groups 0 for this variable.
7. Repeat this process until you run out of dummy variables.
8. Place all the dummy variables into the linear model in the same block.

Let's try this out using an example. In Chapter 6 we encountered a biologist who was worried about the potential health effects of music festivals. She originally collected data at a heavy metal festival (Download Festival), but was worried that her findings might not generalize beyond metal fans. Perhaps it's only metal fans who get smellier at festivals (at this point, as a metal fan, I would sacrifice

the biologist to Odin for her prejudices). To find out whether the type of music a person likes predicts whether hygiene decreases over the festival the biologist measured hygiene over the three days of the Glastonbury Music Festival, which has an eclectic clientele. Her hygiene measure ranged between 0 (you smell like you've bathed in sewage) and 4 (you smell like you've bathed in freshly baked bread). The data file (**GlastonburyFestival.sav**) contains the hygiene scores for each day of the festival and a variable called **change**, which is the change in hygiene from day 1 to day 3 of the festival. Not everyone could be followed up on day 3, so only a subset of the original sample has a change score. The biologist coded the festival-goer's musical affiliations into the categories 'indie kid' (people who mainly like alternative music), 'metaller' (people who like heavy metal), and 'crusty' (people who like hippy/folky/ambient stuff). Anyone not falling into these categories was labelled 'no musical affiliation'. These groups were coded 1, 2, 3 and 4 respectively in the variable **music**.

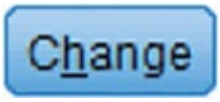
Table 11.2 Dummy coding for the Glastonbury Festival data

	Dummy Variable 1	Dummy Variable 2	Dummy Variable 3
No Affiliation	0	0	0
Indie Kid	0	0	1
Metaller	0	1	0
Crusty	1	0	0

With four groups we need three dummy variables (one less than the number of groups). The first step is to choose a baseline group. We're interested in comparing those who have different musical affiliations against those who don't, so our baseline category will be 'no musical affiliation'. We code this group with 0 for all dummy variables. For our first dummy variable, we could look at the 'crusty' group by assigning anyone who is a crusty a code of 1, and everyone else a code of 0. For our second dummy variable, we could look at the 'metaller' group by assigning a 1 to anyone who is a metaller, and a 0 to everyone else. Our final dummy variable codes the final category 'indie kid' by assigning 1 to anyone who is an indie kid, and a 0 to everyone else. Table 11.2 illustrates the resulting coding scheme. Note that each group has a code of 1 on only one of the dummy variables (except the base category, which is always coded as 0).

11.5.2 The recode function

To create these dummy variables select *Transform* ▸ Recode into Different Variables... to access the dialog box in Figure 11.13. Select the variable you want to recode (in this case

music) and drag it (or click ) to the box labelled *Numeric Variable → Output Variable*. To create a new variable we first type a name in the box labelled *Name* (let's call this first dummy variable **Crusty**). Give this variable a more descriptive name by typing something in the box labelled *Label* (I've labelled it 'No Affiliation vs. Crusty' which reflects what it represents). Click 

to transfer this new variable to the box labelled *Numeric Variable ↓ Output Variable* (this box should now say *music → Crusty*).
The variable **change** has missing values because the biologist couldn't get follow-up measures for everyone on day 3. If we recode the **music** variable, we will include all cases (including those for which we have missing values on the variable **change**). You may not care about this, but if you do you can set a 'do if' condition along the lines of 'if there is a value for the variable **change** then recode the variable **music'**, by clicking 

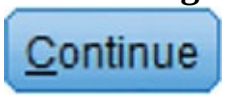
to access the dialog box in Figure 11.14. To set a condition that excludes cases for which the variable **change** has a missing value, select

. Click 

and then 

(or type '1−' in the command area). In the box labelled *Function group* select *Missing Values* and in *Functions and Special Variables* select *Missing*, and click  to transfer the command into the command area. The command will appear as MISSING(?). Drag the variable **change** across to replace the question mark (or delete it and type *change*). The

completed dialog box should look like Figure 11.14.
MISSING() returns 'true' (i.e., the value 1) for a case that has a system-missing or user-defined missing value for the specified variable; it returns 'false' (i.e., the value 0) if a case has a value. Hence, *MISSING(change)* returns a value of 1 for cases that have a missing value for the variable **change** and 0 for cases that do have values. By specifying *1–MISSING(change)* we reverse the command so that it returns 1 (true) for cases that have a value for the variable **change** and 0 (false) for system-missing or user-defined missing values. Therefore, this command says 'Do the following *recode* commands if the case has a value for the variable **change**'. If you don't have missing values you can skip this step.

Click 

to return to the main dialog box.

Figure 11.13 The *Recode* dialog box

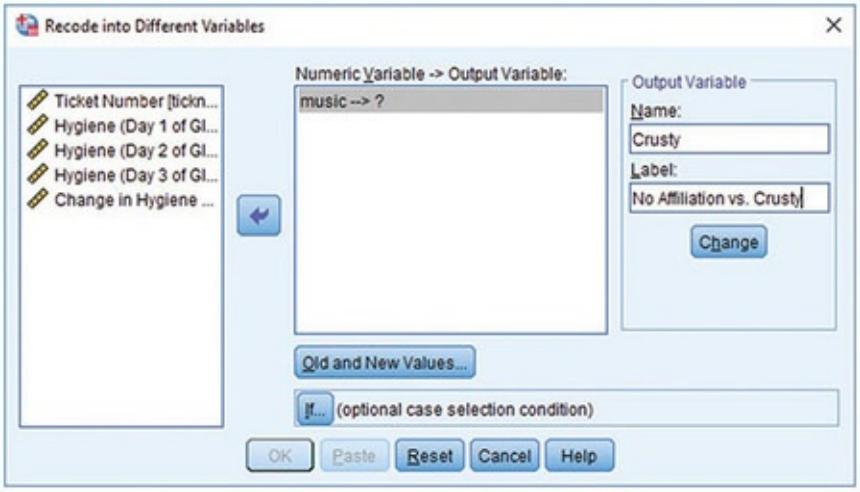
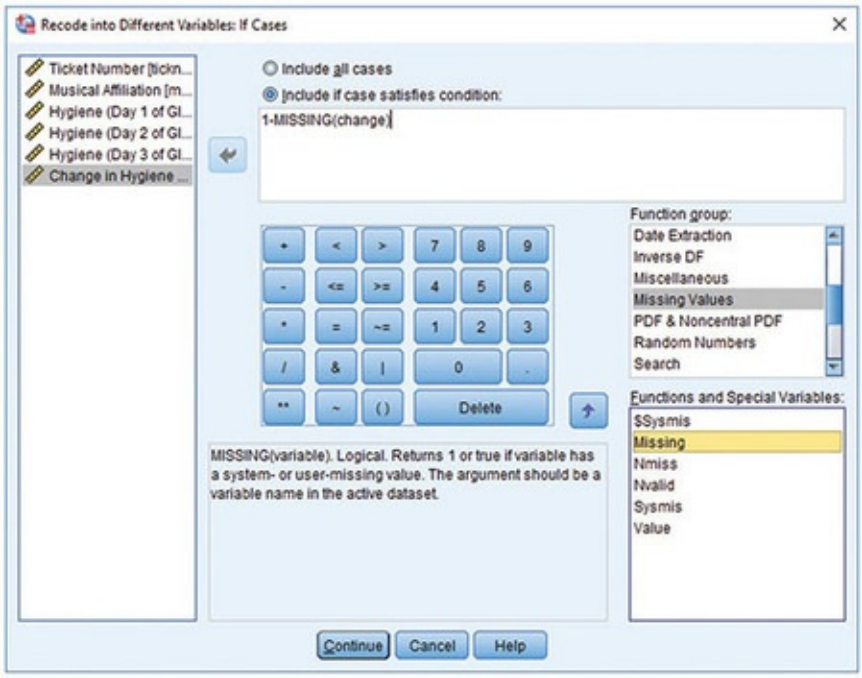
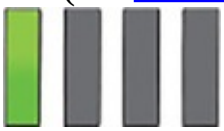
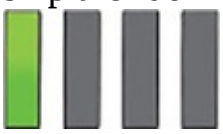


Figure 11.14 Setting an *if* command to include cases if the variable **change** is not a missing value



Now it's time to specify *how* to recode the values of the variable **music** into the values that we want for the new variable, **Crusty**. Click Old and New Values... to access the dialog box in Figure 11.15, which we'll use to recode values of the original variable into different values for the new variable. For our first dummy variable, we want anyone who was a crusty to get a code of 1 and everyone else to get a code of 0. In the original variable crusty was coded as 3, so in the section labelled *Old Value* select Value: and type '3' in the box underneath. We want this value to be 1 in the new variable, so in the section labelled *New Value* select Value: and type '1' in the empty box. Click Add and the box labelled *Old → New* should now include *3 → 1* as in Figure 11.15. The next thing to do is to recode the remaining groups in **music** to have a value of 0 for the first dummy variable. To do this select All other values,[16] and in the section labelled *New Value* select Value: and type '0' in the empty box type and click Add.The box labelled *Old → New* should now include *ELSE → 0* in the list. Click Continue to return to the main dialog box, and click OK to create the first dummy variable. This variable appears as a new

column in the data editor, and it will have a value of 1 for anyone originally classified as a crusty and a value of 0 for everyone else. It can be quicker to recode variables using syntax (see SPSS Tip 11.2).

16 Be careful about using  when you have missing values – remember that we set a 'do if', which means that we can use this option safe in the knowledge that missing values won't be recoded. An alternative method is to skip the 'do if' step and recode missing values specifically using the  option. It is a good idea to use the *frequencies* or *crosstabs* commands after a recode to check that you have caught all the missing values.



Try creating the remaining two dummy variables (call them **Metaller** and **Indie_Kid**) using the same principles.

11.5.3 Output for dummy variables 

Let's assume you've created the three dummy coding variables (if you're stuck there is a data file called **GlastonburyDummy.sav** (the 'Dummy' refers to the fact it has dummy variables in it – I'm not implying that if you need to use this file you're a dummy ☺). To put these dummy variables into a linear model you must enter them all into the model simultaneously (i.e. in the same block).



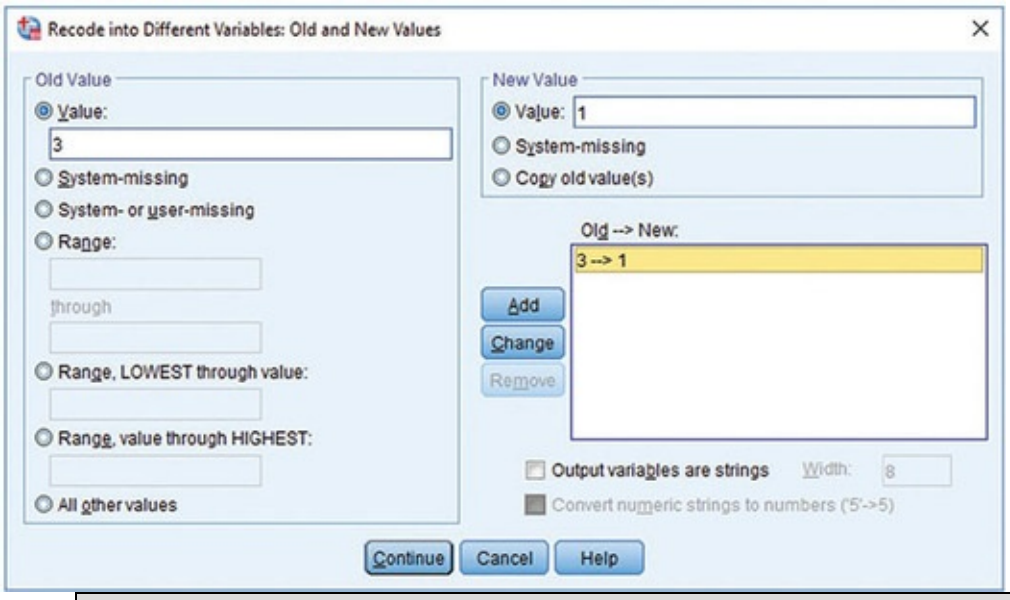
Use what you learnt in Chapter 9 to fit a linear model using the change scores as the outcome, and the three dummy variables as predictors.

Output 11.9 shows the model statistics: by entering the three dummy variables we can explain 7.6% of the variance in the change in hygiene scores. In other words, the musical affiliation of the person explains 7.6% of the variance in the change in hygiene. The F associated with the R^2 change and the model fit assess the same thing when the model has only one block, so these tells us that the

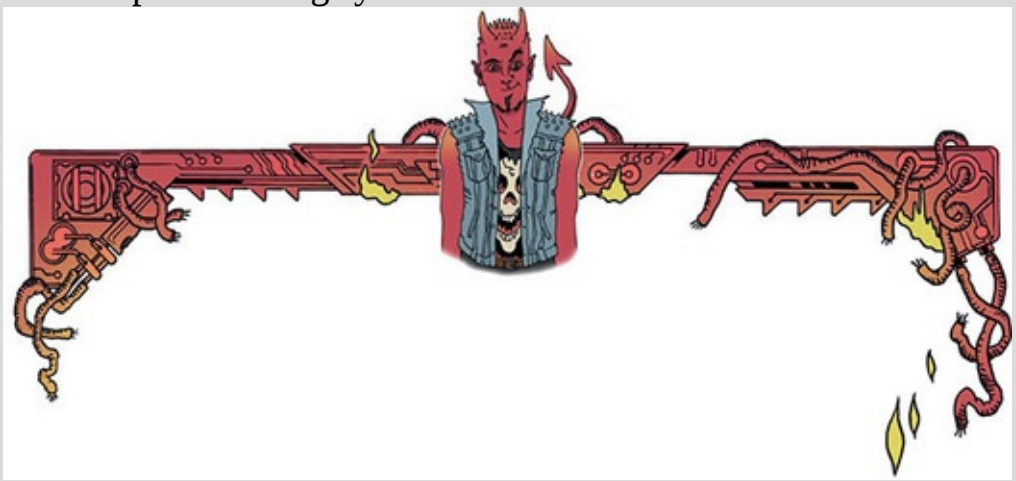
model is significantly better at predicting the change in hygiene scores than having no model (put another way, the 7.6% of explained variance is a significant amount).

Output 11.10 shows the *Coefficients* table for the dummy variables. Each dummy variable has a useful label (such as *No Affiliation vs. Crusty*) because I thought ahead and suggested typing useful labels when we created the variables (Figure 11.13); if we hadn't added labels the table would contain the less helpful variable names of *Crusty*, *Metaller* and *Indie_Kid*. The labels remind us of what each dummy variable represents.

Figure 11.15 Dialog boxes for the *Recode* function



SPSS Tip 11.2 Using syntax to recode



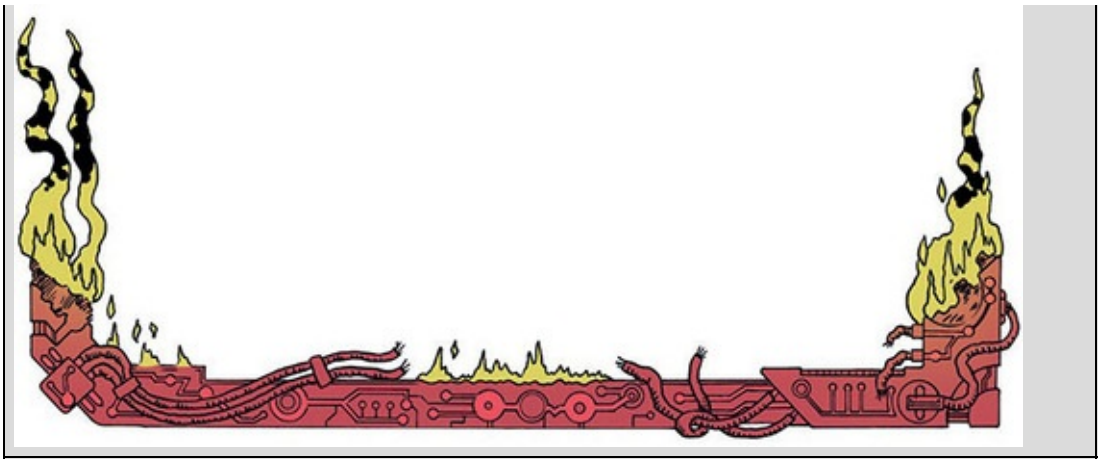
The syntax file, **RecodeGlastonburyData.sps**, creates all the

dummy variables we've discussed. Load this file or type these commands into a new syntax window (see Section 4.10):

```
DO IF(1-MISSING(change)).
RECODE music (3=1)(ELSE = 0) INTO Crusty.
RECODE music (2=1)(ELSE = 0) INTO Metaller.
RECODE music (1=1)(ELSE = 0) INTO Indie_Kid.
END IF.
VARIABLE LABELS Crusty 'No Affiliation vs. Crusty'.
VARIABLE LABELS Metaller 'No Affiliation vs. Metaller'.
VARIABLE LABELS Indie_Kid 'No Affiliation vs. Indie Kid'.
VARIABLE LEVEL Crusty Metaller Indie_Kid (Nominal).
FORMATS Crusty Metaller Indie_Kid (F1.0).
EXECUTE.
```

Each *recode* command does the equivalent of the dialog box in Figure 11.15. So, the three lines beginning *recode* create three new variables (**Crusty, Metaller** and **Indie_Kid**), based on the variable **music**. The first variable (**Crusty**) takes on values of 1 if **music** is 3 and 0 for all other values. The second is coded 1 if **music** is 2 and 0 for all other values, and so on for the third dummy variable. These *recode* commands are within an *if* statement (beginning *do if* and ending with *end if*), which means that they will only be executed if a certain condition is met. The condition we have set is *1 – MISSING(change)*, which is the same as in Figure 11.14 (see main text).

The *variable labels* command assigns the text in the quotations as labels for the variables **Crusty, Metaller**, and **Indie_Kid** respectively. The *variable level* command then sets these three variables to be 'nominal', and the *formats* command changes the variables to have a width of 1 and 0 decimal places (hence the 1.0). The *execute* command executes the commands above (without it nothing works). Note that every line ends with a full stop.



Output 11.9

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.276[a]	.076	.053	.68818	.076	3.270	3	119	.024

a. Predictors: (Constant), No Affiliation vs. Indie Kid, No Affiliation vs. Crusty, No Affiliation vs. Metaller

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4.646	3	1.549	3.270	.024[b]
	Residual	56.358	119	.474		
	Total	61.004	122			

a. Dependent Variable: Change in Hygiene Over The Festival

b. Predictors: (Constant), No Affiliation vs. Indie Kid, No Affiliation vs. Crusty, No Affiliation vs. Metaller

Remember that a *b*-value tells us the change in the outcome due to a unit change in the predictor. For our dummy variables, a unit change in the predictor is the change from 0 to 1. By including all three dummy variables in the model, zero represents the baseline category of no affiliation. For the first dummy variable 1 represents ‘crusty’ and so the change from 0 to 1 is the change from no affiliation to crusty. As such, the first dummy variable represents the difference in the change in hygiene scores for a crusty relative to someone with no musical affiliation. This difference is the difference between the two group means (see Section 10.4).

I’ve produced a table (Output 11.11) of the group means for each of the four groups and also the difference between the mean of each group and the mean of the no affiliation group. For example, the difference in the means of the no affiliation group and the crusty group is (−0.966) − (−0.554) = −0.412. The decrease in hygiene scores of the crusty group (−0.966) is larger in magnitude than for the no affiliation group (−0.554), showing that crusties’ hygiene decreases more over the festival than that of those with no musical affiliation. The difference between these two group means (−0.412) is the *unstandardized* beta in Output 11.10. This example demonstrates that *b*-values for dummy

variables tell us the difference in the mean of a particular group and the group that we chose as a baseline category.

Output 11.10

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			95.0% Confidence Interval for B	
		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound
1	(Constant)	-.554	.090		-6.134	.000	-.733	-.375
	No Affiliation vs. Crusty	-.412	.167	-.232	-2.464	.015	-.742	-.081
	No Affiliation vs. Metaller	.028	.160	.017	.177	.860	-.289	.346
	No Affiliation vs. Indie Kid	-.410	.205	-.185	-2.001	.048	-.816	-.004

a. Dependent Variable: Change in Hygiene Over The Festival

Bootstrap for Coefficients

Model		Bootstrap[a]					
						BCa 95% Confidence Interval	
		B	Bias	Std. Error	Sig. (2-tailed)	Lower	Upper
1	(Constant)	-.554	.005	.097	.001	-.736	-.349
	No Affiliation vs. Crusty	-.412	-.011	.179	.030	-.733	-.101
	No Affiliation vs. Metaller	.028	-.006	.149	.847	-.262	.293
	No Affiliation vs. Indie Kid	-.410	-.010	.201	.049	-.813	-.043

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

As with any linear model, the *b*-value has an associated *t*-statistic and *p*-value that tests whether it is significantly different from 0. For these dummy variables it is, therefore, testing whether the difference between group means is significantly different from 0. For our first dummy variable, the *t*-test is significant and the beta value is negative, so the change in hygiene scores decreases as a person changes from having no affiliation to being a crusty. In other words, hygiene decreased significantly more in crusties compared to those with no musical affiliation.

Our next dummy variable compares metallers to those who have no musical affiliation. The *b*-value (0.028 in Output 11.10) is the difference in the group means for the no affiliation group and the metaller group: (−0.526) − (−0.554) = 0.028. The *t*-test is not significant, which we could take to mean that the change in hygiene scores across the festival is similar in metallers to those with no affiliation.

The final dummy variable compares indie kids to those who have no musical affiliation. The *b*-value (−0.410 in Output 11.10) is the difference in the group means for the no affiliation group and the indie kid group: (−0.964) − (−0.554) = −0.410. The *t*-test is significant and the beta value has a negative value, so, as with the first dummy variable, we could say that the change in hygiene scores goes down as a person changes from having no affiliation to being an indie kid. In other words, hygiene decreased significantly more in indie kids than in those with no musical affiliation. We could report the results as in Table 11.3 (note that I've included the bootstrap confidence intervals). Overall the model shows that, compared to having no musical affiliation, crusties and indie kids get

significantly smellier across the three days of the festival, but metallers don't.

Output 11.11

OLAP Cubes

Change in Hygiene Over The Festival

Musical Affiliation	Mean
Indie Kid	-.9643
Metaller	-.5259
Crusty	-.9658
No Musical Affiliation	-.5543
Crusty - No Musical Affiliation	-.4115
Metaller - No Musical Affiliation	.0284
Indie Kid - No Musical Affiliation	-.4100
Total	-.6750

Table 11.3 Linear model of predictors of the change in hygiene scores. 95% bias corrected and accelerated confidence intervals reported in parentheses. Confidence intervals and standard errors based on 1000 bootstrap samples

	b	*SE B*	*β*	*p*
Constant	−0.55 (−0.74, −0.35)	0.10		$p = 0.001$
No Affiliation vs. Crusty	−0.41 (−0.73, −0.10)	0.18	−0.23	$p = 0.030$
No Affiliation vs. Metaller	0.03 (−0.26, 0.29)	0.15	0.02	$p = 0.847$
No Affiliation vs. Indie Kid	−0.41 (−0.81, −0.04)	0.20	−0.19	$p = 0.049$

Note. $R^2 = 0.08$ ($p = 0.024$).

Note. $R^2 = 0.08$ ($p = 0.024$).



11.6 Brian's attempt to woo Jane

Jane didn't like the library: it was full of people, and people made her uneasy. Why had she gone there to find campus guy? It made no sense to her, but as days passed without seeing him, she missed his nervy recitals of his statistics lectures. She'd watched from afar as he left his lectures and scuttled to the library. Was he avoiding her? The thought that he might be made her want to see him. She wanted to make amends for her barbed comments and yet when she found him, and he spat statistical theory at her like one of Pavlov's dogs, she cut him down again. Why did she do that? Why wouldn't she give him a millimetre? Had she spent too much time alone in the basement of the Pleiades building? Was she so broken that she needed to torture this guy with an elaborate rite of passage to make herself feel better about her past? Jane had realized as she'd looked at his deflated face in the library that he would reach a limit with her quips. She'd wanted to keep him at arm's length, but it was only now that she realized how long her emotional arms could be. If she wanted their chats to continue she

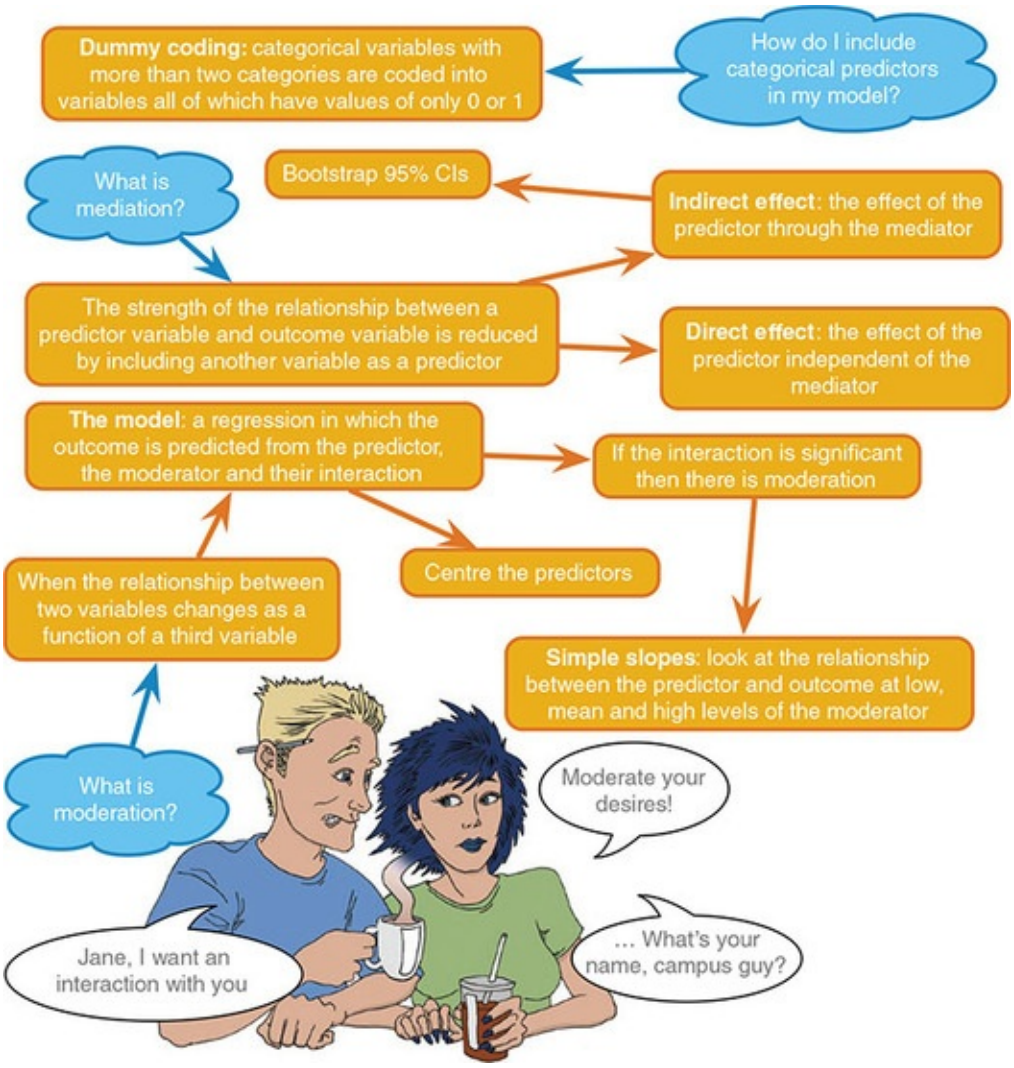
needed to give him some hope.

11.7 What next?



We started this chapter by looking at my relative failures as a human being compared to Simon Hudson. I then bleated on excitedly about moderation and mediation, which could explain why Clair Sparks chose Simon Hudson all those years ago. Perhaps she could see the writing on the wall. I was true to my word to my parents though: I *was* philosophical about it. I set my sights on a girl called Zoë during the obligatory lunchtime game of kiss chase (not the same Zoë whom I ended up marrying). I don't think she was all that keen, which was just as well because I was about to be dragged out of her life for ever. Not that I believe in these things, but if I did I would have believed that the scaly, warty, green, long-nailed hand of fate (I have always assumed that anything named as ominously as the hand of fate would need to look monstrous) had decided that I was far too young to be getting distracted by girls. Waggling its finger at me, it plucked me out of primary school and cast me into the flaming pit of hell, otherwise known as all-boys' grammar school. It's fair to say that my lunchtime primary school kiss chase games were the last I would see of girls for quite some time …

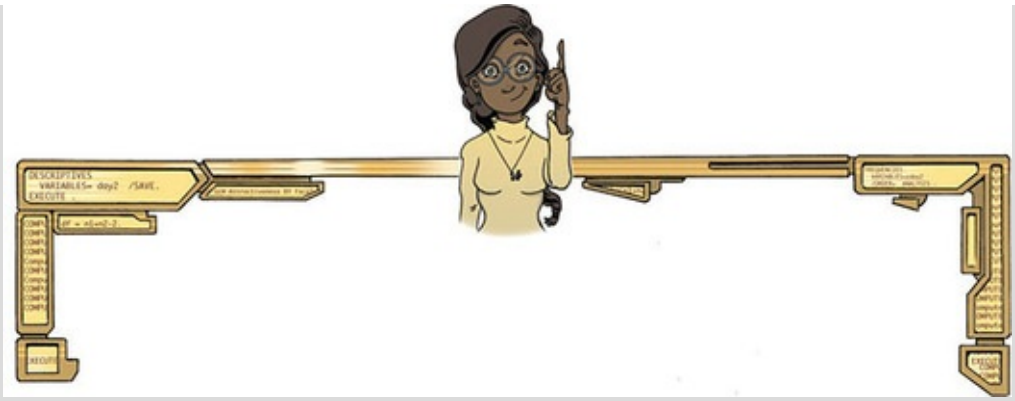
Figure 11.16 What Brian learnt from this chapter



11.8 Key terms that I've discovered

Direct effect
Dummy variables
Grand mean centring
Index of mediation
Indirect effect
Interaction effect
Mediation
Mediator
Moderation
Moderator
Simple slopes analysis
Sobel test

Smart Alex's tasks



- **Task 1**: McNulty *et al.* (2008) found a relationship between a person's **Attractiveness** and how much **Support** they give their partner among newlyweds. The data are in **McNulty *et al.* (2008).sav**. Is this relationship moderated by gender (i.e., whether the data were from the husband or wife)?[17]



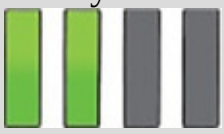
- **Task 2**: Produce the simple slopes graphs for Task 1.



- **Task 3**: McNulty *et al.* (2008) also found a relationship between a person's **Attractiveness** and their relationship **Satisfaction** among newlyweds. Using the same data as in Tasks 1 and 2, find out if this relationship is moderated by gender.

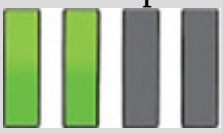


- **Task 4**: In this chapter we tested a mediation model of infidelity for Lambert et al.'s data using Baron and Kenny's regressions. Repeat this analysis but using **Hook_Ups** as the measure of infidelity.



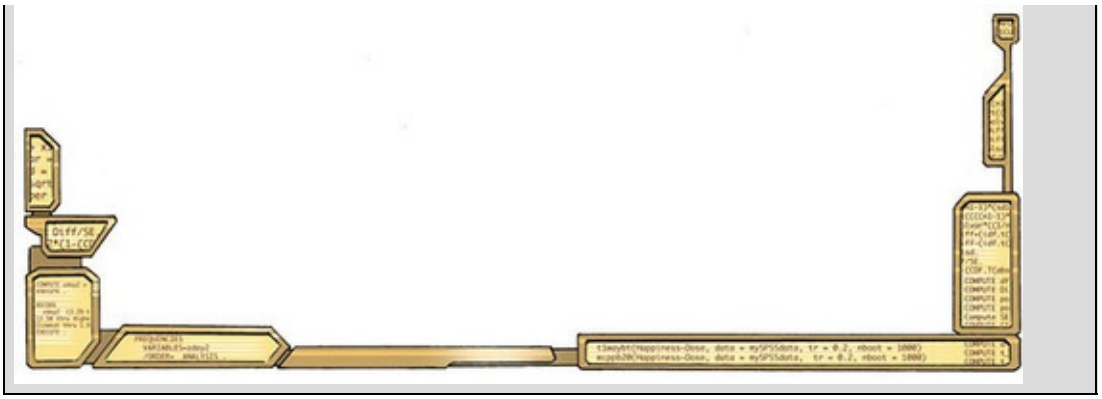
- **Task 5**: Repeat the analysis in Task 4 but using the ***PROCESS*** tool to estimate the indirect effect and its confidence interval.



- **Task 6**: In Chapter 4 (Task 6) we looked at data from people who had been forced to marry goats and dogs and measured their life satisfaction as well as how much they like animals **(Goat or Dog.sav)**. Fit a linear model predicting life satisfaction from the type of animal to which a person was married. Write out the final model. 
- **Task 7**: Repeat the analysis in Task 6 but include animal liking in the first block, and type of animal in the second block. Do your conclusions about the relationship between type of animal and life satisfaction change? 
- **Task 8**: Using the **GlastonburyDummy.sav** data, for which we have already fitted the model, comment on whether you think the model is reliable and generalizable. 
- **Task 9**: Tablets like the iPad are very popular. A company owner was interested in how to make his brand of tablets more desirable. He collected data on how cool people perceived a product's advertising to be (**Advert_Cool**), how cool they thought the product was (**Product_Cool**), and how desirable they found the product (**Desirability**). Test his theory that the relationship between cool advertising and product desirability is mediated by how cool people think the product is (**Tablets.sav**). Am I showing my age by using the word 'cool'? 

17 These are not the actual data from the study, but are simulated to mimic the findings in Table 1 of the original paper.

Answers & additional resources are available on the book's website at **https://edge.sagepub.com/field5e**



12 GLM 1: Comparing several independent means

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12.1 What will this chapter tell me?

There are pivotal moments in everyone's life, and one of mine was at the age of 11. Where I grew up in England there were three choices when leaving primary school and moving on to secondary school: (1) state school (where most people go); (2) grammar school (where clever people who pass an exam called the Eleven Plus go); and (3) private school (where rich people go). My parents were not rich and I am not clever, and consequently I failed my Eleven Plus, so private school and grammar school (where my older brother, 'the clever one', had gone) were out. There was no choice but for me to join my friends at the local state school. I could not have been happier: at age 11 I hadn't yet become a stats hermit and liked having friends. Imagine everyone's shock when my parents received a letter saying that some extra spaces had become available at the grammar school; although the local authority could scarcely believe it and had checked the Eleven Plus papers several million times to confirm their findings, I was next on their list. I could not have been unhappier. So, I waved goodbye to all my friends and trundled off to join my brother at Ilford County High School for Boys (a school at which the head teacher hit students with a cane if they were particularly bad and that, for some considerable time and with good reason, had 'H.M. Prison' painted in huge white letters on its roof). It was goodbye to normality, and hello to seven years of learning how not to function in society. I often wonder how my life would have turned out had I not gone to this school. In the parallel universes where the letter didn't arrive and the parallel Andy went to state school, or where my parents were rich and he went to private school, what became of him? If we wanted to compare these three situations we couldn't use the methods in Chapter 10 because there are more than two conditions.[1] However, this chapter tells us all about the statistical models that we use to analyse situations in which we want to compare more than two independent means. The model is typically called **analysis of variance** (or **ANOVA** to its friends) but, as we shall see, it is just a variant on the linear model. So, in effect, we're going to learn stuff that we already know from previous chapters. Hopefully that's reassuring.

1 This is the least of our problems – there's also the small issue of reinventing physics to access the parallel universes.

Figure 12.1 My brother Paul (left) and I (right) in our very fetching school uniforms (note the fear in my face)



12.2 Using a linear model to compare several means



We saw in Chapter 10 that if we include a predictor variable containing two categories into the linear model then the resulting *b* for that predictor compares the difference between the mean score for the two categories. We also saw in Chapter 11 that if we want to include a categorical predictor that contains more than two categories, this can be achieved by recoding that variable into several categorical predictors each of which has only two categories (dummy coding). When we do, the *b*s for predictors represent differences between means. Therefore, if we're interested in comparing more than two means we can use the linear model to do this. Remembering from Chapter 9 that we test the overall fit of a linear model with an *F*-statistic, we can do the same here: we first use an *F* to test whether we significantly predict the outcome variable by using group means (which tells us whether, overall, the group means are significantly different) and then use the specific model parameters (the *b*s) to tell us which means differ from which. It's not uncommon for researchers to think, and for people to be taught, that you compare means with 'ANOVA' and that this is somehow different from 'regression' (i.e. the linear model), which you apply to look for relationships between variables. This artificial division is unhelpful (in my view) and exists largely for weird historical reasons (Misconception Mutt 12.1). The 'ANOVA' to which some people allude is simply the *F*-statistic that we encountered as a test of the fit of a linear model, it's just that the linear model consists of group means. This chapter will develop what we discovered in Chapters 10 and 11 about using dummy variables in the linear model to compare means.

As a quick side note, there is a different way of teaching the use of the *F*-statistic to compare means known as the variance-ratio method. This approach is fine for simple designs, but becomes impossibly cumbersome in more complex

situations such as analysis of covariance, or when you have unequal sample sizes.[2] The linear model framework has various advantages. First, we're building on material that you have already learnt (this chapter is a natural progression from the bits of the book that you've hopefully already read). Second, the linear model extends very logically to the more complex situations (e.g., multiple predictors, unequal group sizes) without the need to get bogged down in mathematics. And third, IBM SPSS Statistics uses the linear model framework (known as the **general linear model**) for comparing means (on the whole).

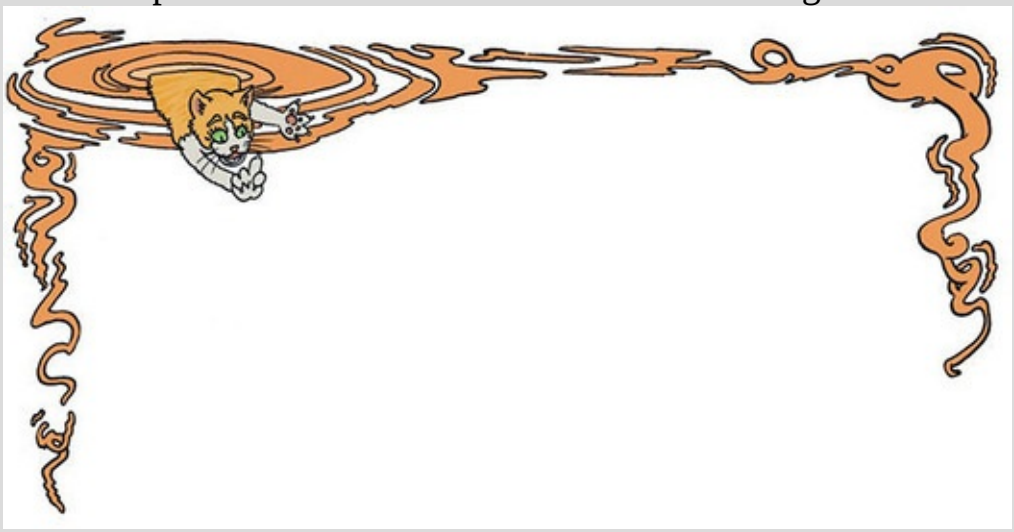
2 Having said this, it is well worth the effort of trying to obtain equal sample sizes in your different conditions because unbalanced designs do cause statistical complications (see Section 12.3).

Let's start with an example. You're about halfway through the book now, and there are a lot of equations in this chapter, so we probably need some puppy therapy. Puppy therapy is a form of animal-assisted therapy, in which puppy contact is introduced into the therapeutic process. Puppy rooms have been set up to de-stress students and staff at my own university (Sussex) in the UK along with universities in Bristol, Nottingham, Aberdeen and Lancaster. I've heard of similar things at Dalhousie and Simon Fraser in Canada and Tufts and Caldwell in the USA. My own contribution is to sometimes bring my adorable spaniel, Ramsey (Figure 12.2), into work to sit in my office and look cute at any students who break down in tears about some stats issue they have. He may pop up at strategic points in this chapter to help your mental state.

Despite this increase in puppies on campuses (which can only be a good thing) to reduce stress, the evidence base is pretty mixed. A review of animal-assisted therapy in childhood mental health found that of 24 studies, 8 found positive effects of animal-assisted therapy, 10 showed mixed findings, and 6 concluded that there was no effect (Hoagwood, Acri, Morrissey, & Peth-Pierce, 2017). Imagine we wanted to contribute to this literature by running a study in which we randomized people into three groups: (1) a control group (this could be a treatment as usual, a no treatment or ideally some kind of placebo group – for example, if our hypothesis was specifically about puppies we could give people in this group a cat disguised as a dog); (2) 15 minutes of puppy therapy (a low-dose group); and (3) 30 minutes of puppy contact (a high-dose group). The dependent variable was a measure of happiness ranging from 0 (as unhappy as I can possibly imagine being) to 10 (as happy as I can possibly imagine being). The design of this study mimics a very simple randomized controlled trial (as used in pharmacological, medical and psychological intervention trials) because people are randomized into a control group or groups containing the active intervention (in this case puppies, but in other cases a drug or a surgical

procedure). We'd predict that any form of puppy therapy should be better than the control (i.e. higher happiness scores) but also formulate a dose-response hypothesis that as exposure time increases (from 0 minutes to 15 and 30) happiness will increase too. The data are in Table 12.1 and the file **Puppies.sav**.

Misconception Mutt 12.1 ANOVA is the same as regression



The Misconception Mutt had been trying to train his owner. Every morning he tried doing a trick to see whether his owner would give him more grains for breakfast. He'd tried paw, rollover, and the classic spaniel move of just rolling around on his back with a crazy look in his eye and his tongue lolloping out of his mouth. If only there were some way he could compare the average amount of food he received after each trick. While his owner was asleep, the mutt had found his statistics textbook and rested his head on it. He found that written words drifted into his brain if he slept on things. Apparently he needed an ANOVA.

The next morning as he munched his grains, he muttered to himself about using an ANOVA to compare means. His dog flap clanked shut, making him jump on his heels to turn. There was nothing there. As he turned back to his bowl he noticed a large ginger cat helping himself to his food. The mutt growled.

'You know,' said the cat, 'ANOVA is the same thing as the linear model or regression.'

The growling continued.

'The reason why some people think of ANOVA and regression as separate statistical models is historical. Two distinct branches of methodology, correlational and experimental, developed in the social sciences. Researchers interested in controlled experiments adopted

ANOVA as their model of choice, whereas those looking for real-world relationships adopted multiple regression. With the divide in methodologies came a chasm between the statistical methods adopted by the two opposing camps (Cronbach, 1957, documents this divide in a lovely article). Never the twain shall meet, and all that.'
The mutt growled some more.
'Don't feel bad. For decades researchers have gone around oblivious to the fact that they were using the same model under different names. Students are often taught "regression" and "ANOVA" in different contexts as though they are different. But now, my canine friend, you know better. Nice grains, by the way.'
With that, the cat finished the last of the dog's food and darted out of the dog flap, leaving a ginger vapour trail.

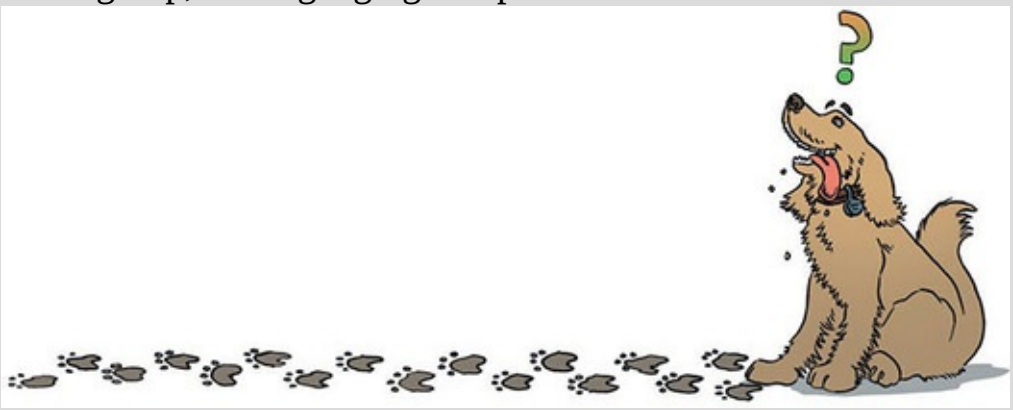


Figure 12.2 Some puppy therapy for you in the form of my dog, Ramsey

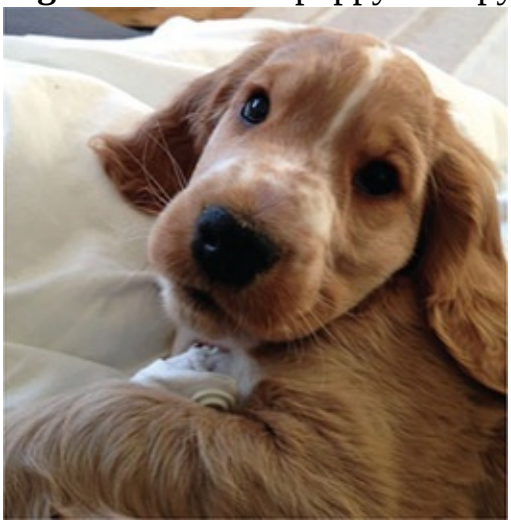


Table 12.1 Data in **Puppies.sav**

	Control	15 minutes	30 minutes
	3	5	7
	2	2	4
	1	4	5
	1	2	3
	4	3	6
$\bar{X}$	**2.20**	**3.20**	**5.00**
s	**1.30**	**1.30**	**1.58**
s^2	**1.70**	**1.70**	**2.50**

Grand mean = **3.467** Grand SD = **1.767**

Grand variance = **3.124**

If we want to predict happiness from group membership we can use the general equation that keeps popping up:

$$\text{outcome}_i = (\text{model}) + \text{error}_i \quad (12.1)$$

We've seen that with two groups we can use a linear model, by replacing the 'model' in equation (12.1) with one dummy variable that codes two groups (0 for one group and 1 for the other) and an associated *b*-value that would represent the difference between the group means (Section 10.4). We have three groups here, but we've also seen that this situation is easily incorporated into the linear model by including two dummy variables (each assigned a *b*-value), and that any number of groups can be included by extending the number of dummy variables to one less than the number of groups (Section 11.5).

We've also learnt already that when we use dummy variables we assign one group as the baseline category and assign that group a zero code on all dummy variables (remember in Section 11.5 that we chose the 'no musical affiliation' condition as a baseline). The baseline category should be the condition against which you intend to compare the other groups. In most well-designed experiments there will be a group of participants who act as a control for the other groups and, other things being equal, this will be your baseline category – although the group you choose will depend upon the particular hypotheses you want to test. In designs in which the group sizes are unequal it is important that the baseline category contains a large number of cases to ensure that the estimates of the *b*-values are reliable. In the puppy therapy example, we can take the control group (who received no puppy therapy) as the baseline category because we are interested in comparing both the 15-and 30-minute groups to this group. If the control group is the baseline category then the two dummy

variables need to represent the other two conditions: so, let's call one of them **Long** (30-minute dose) and the other **Short** (15 minutes) to reflect the length of the dose of puppies. Putting these dummy variables into the model as predictors gives:

$$\text{Happiness}_i = b_0 + b_1\text{Long}_i + b_2\text{Short}_i + \varepsilon_i \quad (12.2)$$

in which a person's happiness is predicted from knowing their group code (i.e., the numeric code for the **Long** and **Short** dummy variables) and the intercept (b_0) of the model. The dummy variables can be coded in several ways, but the simplest way is to use dummy coding (Section 11.5). The baseline category is coded as 0 for all dummy variables. If a participant received 30 minutes of puppy therapy then they are coded with a 1 for the **Long** dummy variable and 0 for **Short**. If a participant received 15 minutes of puppy therapy then they are coded with the value 1 for the **Short** dummy variable and coded with 0 for **Long**. Using this coding scheme, each group is uniquely expressed by the combined values for the two dummy variables (see Table 12.2).

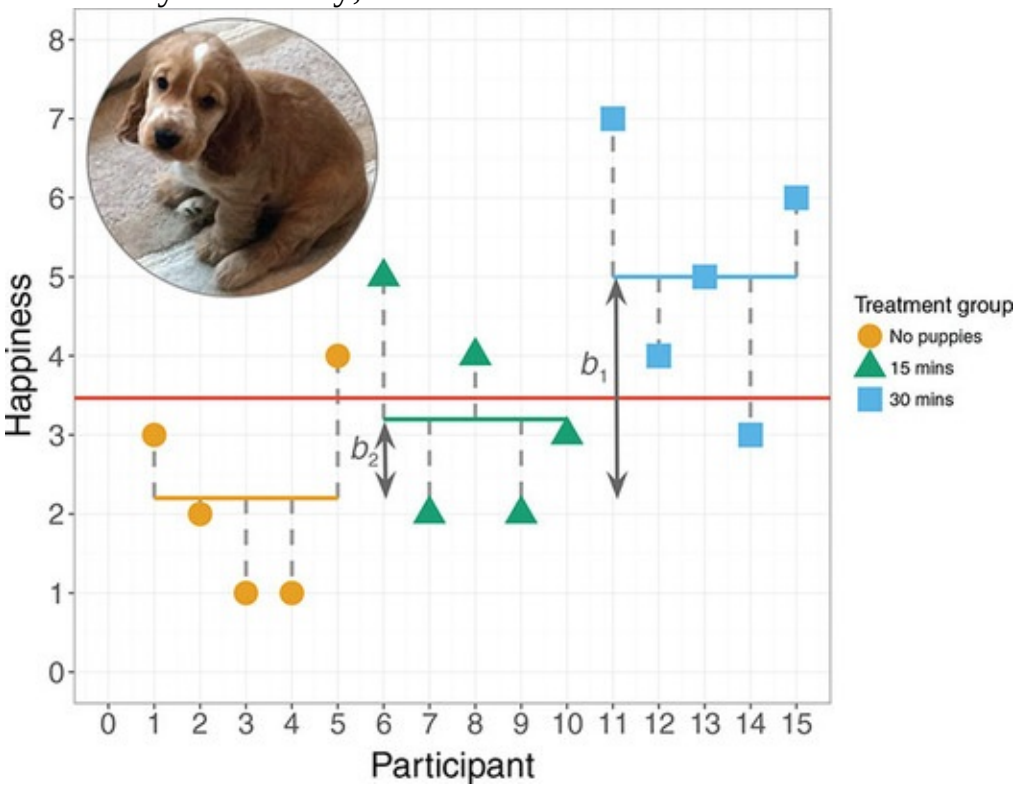
When we are predicting an outcome from group membership, predicted values from the model (the value of happiness in equation (12.2)) are the group means. This is illustrated in Figure 12.3, which splits the data by group membership. If we are trying to predict the happiness of a new person and we know to which group they have been assigned (but we don't have their score yet) then our best guess will be the group mean because, on average, we'll be correct. For example, if we know that someone is going to receive 30 minutes of puppy therapy and we want to predict their happiness, our best guess will be 5 because we know that, on average, people who have 30 minutes with a puppy rate their happiness as 5. If the group means are meaningfully different, then using the group means should be an effective way to predict scores (because we can successfully differentiate people's predicted happiness based on how much puppy time they received). We'll return to this point in the next section.

Table 12.2 Dummy coding for the three-group experimental design

Group	Dummy variable 1 (Long)	Dummy variable 2 (Short)
Control	0	0
15 minutes of puppy therapy	0	1
30 minutes of puppy therapy	1	0

Figure 12.3 The puppy therapy data. The coloured horizontal lines represent the mean happiness of each group. The shapes represent the happiness of individual participants (different shapes indicate different experimental groups). The red horizontal line is the average happiness of all participants. The puppy is my dog

Ramsey, as a pup. You should never needlessly apply puppies to your graphs … unless they're Ramsey, in which case it's fine



Let's first examine the model for the *control group*. Both the **Long** and **Short** dummy variables are coded 0 for people in the control group. Therefore, if we ignore the error term (ε_i), the model becomes:

$$\begin{aligned} \text{Happiness}_i &= b_0 + (b_1 \times 0) + (b_2 \times 0) \\ &= b_0 \\ \bar{X}_{\text{Control}} &= b_0 \end{aligned} \tag{12.3}$$

The 15-and 30-minute groups have dropped out of the model (because they are coded 0) and we're left with b_0. As we have just discovered, the predicted value of happiness will be the mean of the control group $\overline{X}_{\text{control}}$, so we can replace **Happiness** with this value. This leaves us with the epiphany that b_0 in the model is always the mean of the baseline category.

For someone in the *30-minute group*, the value of the dummy variable **Long** will be 1 and the value for **Short** will be 0. By replacing these values in equation (12.2) the model becomes:

$$\begin{aligned} \text{Happiness}_i &= b_0 + (b_1 \times 1) + (b_2 \times 0) \\ &= b_0 + b_1 \end{aligned} \tag{12.4}$$

which tells us that predicted happiness for someone in the 30-minute group is the sum of b_0 and the b for the dummy variable **Long** (b_1). We know already that b_0

is the mean of the control group $(\overline{X}_{\text{control}})$ and that the predicted value of **Happiness** for someone in the 30-minute group is the mean of that group $(\overline{X}_{30\text{ mins}})$. Therefore, we can replace b_0 with $\overline{X}_{\text{control}}$ and **Happiness** with $\overline{X}_{30\text{ mins}}$. The result is:

$$\begin{aligned}\text{Happiness}_i &= b_0 + b_1 \\ \bar{X}_{30\text{ mins}} &= \bar{X}_{\text{Control}} + b_1 \\ b_1 &= \bar{X}_{30\text{ mins}} - \bar{X}_{\text{Control}}\end{aligned} \tag{12.5}$$

which shows that the *b*-value for the dummy variable representing the 30-minute group is the difference between the means of that group and the control.

Finally, for someone assigned to the *15-minute group*, the dummy variable **Short** has a value of 1 and the dummy variable **Long** is 0. By replacing these values in equation (12.2) the model becomes:

$$\begin{aligned}\text{Happiness}_i &= b_0 + (b_1 \times 0) + (b_2 \times 1) \\ &= b_0 + b_2\end{aligned} \tag{12.6}$$

which tells us that predicted happiness for someone in the 15-minute group is the sum of b_0 and the *b* for the dummy variable **Short** (b_2). Again, we can replace b_0 with $\overline{X}_{\text{control}}$. The predicted value of **Happiness** for someone in the 15-minute group is the mean of that group so we can replace Happiness with $\overline{X}_{15\text{ mins}}$. The result is:

$$\begin{aligned}\text{Happiness}_i &= b_0 + b_2 \\ \bar{X}_{15\text{ mins}} &= \bar{X}_{\text{Control}} + b_2 \\ b_2 &= \bar{X}_{15\text{ mins}} - \bar{X}_{\text{Control}}\end{aligned} \tag{12.7}$$

which shows that the *b*-value for the dummy variable representing the 15-minute (**Short**) group is the difference between means for the 15-minute group and the control.

Using dummy coding is only one of many ways to code dummy variables. We shall see later in this chapter (Section 12.4.2) that an alternative is contrast coding, in which you code the dummy variables in such a way that the *b*-values represent differences between groups that you specifically hypothesized before collecting data.



To illustrate what is going on I have created a file called **Puppies**

Dummy.sav that contains the puppy therapy data along with the two dummy variables (**dummy1** and **dummy2**) we've just discussed (Table 10.2). Fit a linear model predicting happiness from **dummy1** and **dummy2**. If you're stuck, read Chapter 9 again.

Output 12.1 shows the model from the self-test. Quickly remind yourself of the group means from Table 12.1. The overall fit of the model has been tested with an F-statistic (i.e., ANOVA), which is significant, $F(2, 12) = 5.12$, $p = 0.025$. Given that our model represents the group means, this F tells us that using group means to predict happiness scores is significantly better than using the mean of all scores: in other words, the group means are significantly different.

Output 12.1

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	20.133	2	10.067	5.119	.025[b]
	Residual	23.600	12	1.967		
	Total	43.733	14			

a. Dependent Variable: Happiness (0–10)

b. Predictors: (Constant), Dummy 2: 15 mins vs. Control, Dummy 1: 30 mins vs. Control

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	2.200	.627		3.508	.004
	Dummy 1: 30 mins vs. Control	2.800	.887	.773	3.157	.008
	Dummy 2: 15 mins vs. Control	1.000	.887	.276	1.127	.282

a. Dependent Variable: Happiness (0–10)

The F-test is an overall test that doesn't identify differences between specific means. However, the model parameters (the b-values) do. As we just discovered, the constant (b_0) is equal to the mean of the base category (the control group), 2.2. The b-value for the first dummy variable (b_1) is equal to the difference between the means of the 30-minute group and the control group (5.0 − 2.2 = 2.8). Finally, the b-value for the second dummy variable (b_2) is equal to the difference between the means of the 15-minute group and the control group (3.2 − 2.2 = 1). This demonstrates what we saw in equations (12.3), (12.5) and (12.7). We can see from the significance values of the associated t-tests that the difference between the 30-minute group and the control group (b_1) is significant because $p = 0.008$, which is less than 0.05; however, the difference between the 15-minute and the control group is not ($p = 0.282$).

We can extend this three-group scenario to four groups (see Section 11.5 for an example). As before, we specify one category as a base category (a control

group) and assign this category a code of 0 for all dummy variables. The remaining three conditions will have a code of 1 for the dummy variable that describes that condition and a code of 0 for the other dummy variables (Table 12.3).

12.2.1 Logic of the F-statistic 

We learnt in Chapter 9 that the *F*-statistic (or *F*-ratio as it's also known) tests the overall fit of a linear model to a set of observed data. *F* is the ratio of how good the model is compared to how bad it is (its error). When the model is based on group means, our predictions from the model are those means. If the group means are the same then our ability to predict the observed data will be poor (*F* will be small), but if the means differ we will be able to better discriminate between cases from different groups (*F* will be large). So, in this context *F* basically tells us whether the group means are significantly different. Let me elaborate.

Table 12.3 Dummy coding for the four-group experimental design

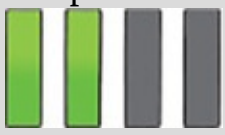
	Dummy variable 1	Dummy variable 2	Dummy variable 3
Group 1	1	0	0
Group 2	0	1	0
Group 3	0	0	1
Group 4 (base)	0	0	0

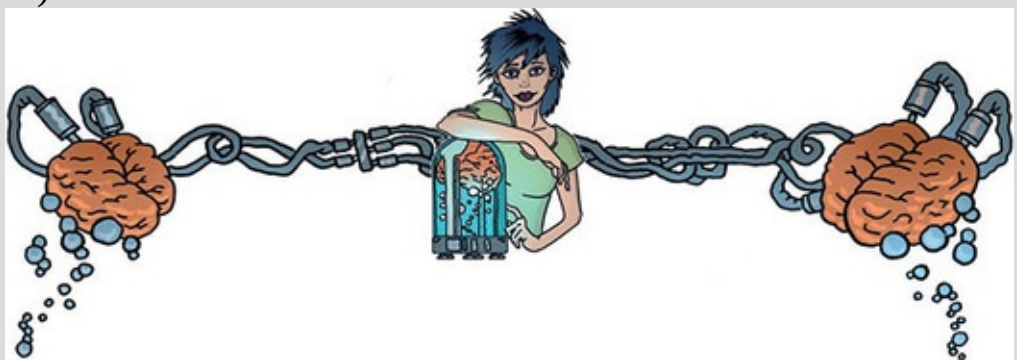
Figure 12.3 shows the puppy therapy data including the group means, the overall mean and the difference between each case and the group mean. We want to test the hypothesis that the means of three groups are different (so the null hypothesis is that the group means are the same). If the group means were all the same, then we would not expect the control group to differ from the 15-or 30-minute groups, and we would not expect the 15-minute group to differ from the 30-minute group. In this situation, the three coloured horizontal lines representing the group means in Figure 12.3 would be in the same vertical position (the exact position would be the grand mean – the red horizontal line in the figure). This is not the case in the figure: the coloured lines are in different vertical positions, showing that the group means are different. We have just found out that in the model, b_1 represents the difference between the control and 30-minute group means, and b_2 represents the difference between the 15-minute and control group means. These two distances are represented in Figure 12.3 by the vertical arrows. If the null hypothesis is true and all the groups have the same mean, then these *b* coefficients should be zero (because if the group means are

equal then the difference between them will be zero).
We can apply the same logic as for any linear model:

- The model that represents 'no effect' or 'no relationship between the predictor variable and the outcome' is one where the predicted value of the outcome is always the grand mean (the mean of the outcome variable).
- We can fit a different model to the data that represents our alternative hypotheses. We compare the fit of this model to the fit of the null model (i.e., using the grand mean).
- The intercept and one or more parameters (*b*) describe the model.
- The parameters determine the shape of the model that we have fitted; therefore, the bigger the coefficients, the greater the deviation between the model and the null model (grand mean).
- In experimental research the parameters (*b*) represent the differences between group means. The bigger the differences between group means, the greater the difference between the model and the null model (grand mean).
- If the differences between group means are large enough, then the resulting model will be a better fit to the data than the null model (grand mean).
- If this is the case we can infer that our model (i.e., predicting scores from the group means) is better than not using a model (i.e., predicting scores from the grand mean). Put another way, our group means are significantly different from the null (that all means are the same).

We use the *F*-statistic to compare the improvement in fit due to using the model (rather than the null, or grand mean, model) to the error that still remains. In other words, the *F*-statistic is the ratio of the explained to the unexplained variation. We calculate this variation using sums of squares (look back at Section 9.2.4 to refresh your memory), which might sound complicated, but isn't as bad as you think (see Jane Superbrain Box 12.1).

Jane Superbrain 12.1 ANOVA boils down to one equation (well, sort of) 

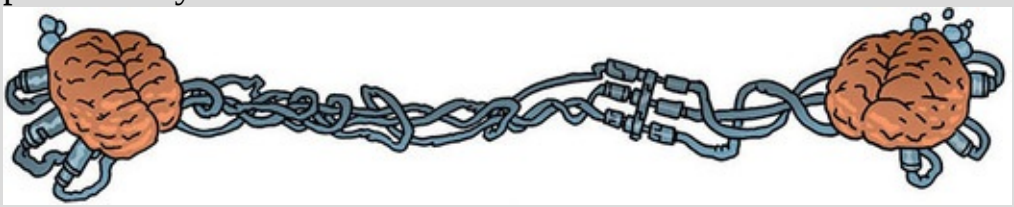


At every stage of the ANOVA we're assessing variation (or deviance) from a particular model (be that the null model, or the model representing our hypothesis). Back in Section 2.5.1 we saw that the extent to which a model deviates from the observed data can be expressed, in general, in the form of equation (2.11), which I repeat here as equation (12.8):

$$\text{Total error} = \sum_{i=1}^{n} (\text{observed}_i - \text{model}_i)^2 \quad (12.8)$$

So, when comparing means, as in the linear model generally, we use this equation to calculate the fit of the null model, and then the fit of the alternative model that represents our hypothesis. If the alternative model is any good then it should fit the data significantly better than the null model.

All of the sums of squares we look at in this chapter are variations on equation (12.8): all that changes is what we use as the model and observed data. As you read about the sums of squares, think back to equation (12.8) to remind yourself that the equations are just variants of looking at the difference between observed values and values predicted by a model.



12.2.2 Total sum of squares (SS_T)



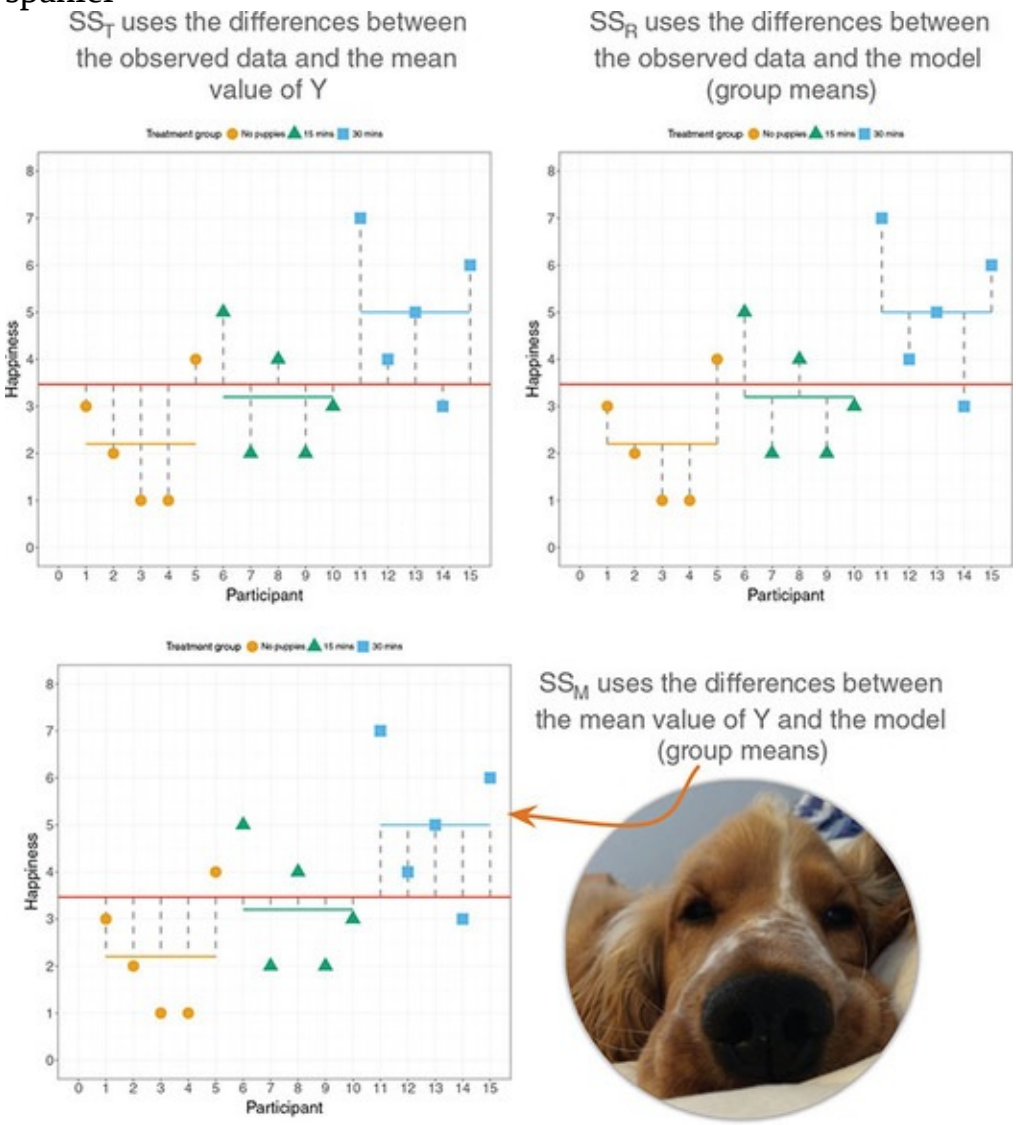
To find the total amount of variation within our data we calculate the difference between each observed data point and the grand mean. We square these differences and add them to give us the total sum of squares (SS_T):

$$SS_T = \sum_{i=1}^{N} (x_i - \bar{x}_{grand})^2 \quad (12.9)$$

The variance and the sums of squares are related such that variance, $s^2 = SS/(N - 1)$, where N is the number of observations (Section 2.5.1). Therefore, we can calculate the total sum of squares from the variance of all observations (the **grand variance**) by rearranging the relationship ($SS = s^2(N - 1)$). The grand variance is the variation between all scores, regardless of the group from which the scores come. Figure 12.4 shows the different sums of squares graphically

(note the similarity to Figure 9.5 which we looked at when we learnt about the linear model). The top left panel shows the total sum of squares: it is the sum of the squared distances between each point and the solid horizontal line (which represents the mean of all scores).

Figure 12.4 Graphical representation of the different sums of squares when comparing several means using a linear model. Also a picture of Ramsey as a puppy. Tufte would call him chartjunk, but I call him my adorable, crazy, spaniel



The grand variance for the puppy therapy data is given in Table 12.1, and there were 15 scores in all. Therefore, SS_T is 43.74:

$$\begin{aligned} SS_T &= s^2_{grand}(N-1) \\ &= 3.124(15-1) \\ &= 3.124 \times 14 \\ &= 43.74 \end{aligned} \tag{12.10}$$

Before we move on, have a look back at Jane Superbrain Box 2.2 to refresh your memory on degrees of freedom. When we estimate population values, the degrees of freedom are typically one less than the number of scores used to calculate the estimate. This is because to get the estimates we hold something constant in the population (e.g., to get the variance we hold the mean constant), which leaves all but one of the scores free to vary. For SS_T, we used the entire sample (i.e., 15 scores) to calculate the sums of squares and so the total degrees of freedom (df_T) are one less than the total sample size ($N - 1$). For the puppy therapy data, this value is 14.

12.2.3 Model sum of squares (SS_M) 

So far, we know that the total amount of variation within the outcome is 43.74 units. We now need to know how much of this variation the model can explain. Because our model predicts the outcome from the means of our treatment (puppy therapy) groups, the model sums of squares tell us how much of the total variation in the outcome can be explained by the fact that different scores come from entities in different treatment conditions.

The model sum of squares is calculated by taking the difference between the values predicted by the model and the grand mean (see Section 9.2.4, Figure 9.5). When making predictions from group membership, the values predicted by the model are the group means (the coloured horizontal lines in Figure 12.4). The bottom panel in Figure 12.4 shows the model sum of squared error: it is the sum of the squared distances between what the model predicts for each data point – the mean of the group to which the score belongs represented by the orange (control), green (15 mins) and blue (30 mins) horizontal lines – and the overall mean of the outcome (the red horizontal line). For example, the predicted value for the five participants in the control group (circles) is 2.2, for the participants (triangles) in the 15-minute condition is 3.2, and for the participants (squares) in the 30-minute condition is 5. The model sum of squares requires us to calculate the differences between each participant's predicted value and the grand mean. These differences are squared and added together (for reasons I've explained before). Given that the predicted value for participants in a group is the same value (the group mean), the easiest way to calculate SS_M is by using:

$$SS_M = \sum_{g=1}^{k} n_g \left(\bar{x}_g - \bar{x}_{grand}\right)^2 \quad (12.11)$$

This equation basically says:

- Calculate the difference between the mean of each group $(\bar{x}_g)$ and the grand mean $(\bar{x}_{grand})$.
- Square each of these differences.
- Multiply each result by the number of participants within that group (n_g).
- Add the values for each group together.

If we do this for the puppy therapy data, we get:

$$\begin{aligned} SS_M &= 5(2.200 - 3.467)^2 + 5(3.200 - 3.467)^2 + 5(5.000 - 3.467)^2 \\ &= 5(-1.267)^2 + 5(-0.267)^2 + 5(1.533)^2 \\ &= 8.025 + 0.355 + 11.755 \\ &= 20.135 \end{aligned} \quad (12.12)$$

For SS_M, the degrees of freedom (df_M) are one less than the number of 'things' used to calculate the SS. We used the three group means, so df_M is the number of groups minus one (which you'll see denoted as $k - 1$). So, in this example, we have three groups and the degrees of freedom are 2 (because the calculation of the sums of squares is based on the group means, two of which will be free to vary in the population if the third is held constant).

12.2.4 Residual sum of squares (SS_R)

We now know that there are 43.74 units of variation to be explained in our outcome variable, and that our model explains 20.14 of them (nearly half). The residual sum of squares (SS_R) tells us how much of the variation *cannot* be explained by the model. This value is the amount of variation created by things that we haven't measured such as measurement error and individual differences in things that might affect happiness. The simplest way to calculate SS_R is to subtract SS_M from SS_T ($SS_R = SS_T - SS_M$), but this provides little insight into what SS_R represents and, of course, if you've messed up the calculations of either SS_M or SS_T (or both!) then SS_R will be incorrect also.

We saw in Section 9.2.4 that the residual sum of squares is the difference between what the model predicts and what was observed. When using group membership to predict an outcome the values predicted by the model are the group means (the coloured horizontal lines in Figure 12.4). The top right panel

of Figure 12.4 shows the residual sum of squared error: it is the sum of the squared distances between each point and the horizontal line for the group to which the score belongs.

We already know that for a given participant, the model predicts the mean of the group to which that person belongs. Therefore, SS_R is calculated by looking at the difference between the score obtained by a person and the mean of the group to which the person belongs. In graphical terms the dashed vertical lines in Figure 12.3 represent this sum of squares. These distances between each data point and the group mean are squared and added together to give the residual sum of squares, SS_R:

$$SS_R = \sum_{g=1}^{k} \sum_{i=1}^{n} \left(x_{ig} - \bar{x}_g\right)^2 \tag{12.13}$$

Equation (12.13) says that the sum of squares for each group is the squared difference between each participant's score in a group (x_{ig}) and the group mean ($\bar{x}_g$), and the two sigma signs mean that we repeat this calculation for the first participant ($i = 1$) through to the last (n), in the first group ($g = 1$) through to the last (k). As such, we can also express SS_R as $SS_R = SS_{\text{group 1}} + SS_{\text{group 2}} + SS_{\text{group 3}} + \ldots + SS_{\text{group } k}$. We know that the variance is the sums of squares divided by $n - 1$, and we can express the residual sum of squares in terms of the variance like we did for the total sum of squares. The result is:

$$SS_R = \sum_{g=1}^{k} s_g^2 \left(n_g - 1\right) \tag{12.14}$$

which translates as 'multiply the variance for each group (s_g^2) by one less than the number of people in that group ($n_g - 1$), then add the results for each group together'. For the puppy therapy data, we get:

$$\begin{aligned} SS_R &= s^2_{\text{group 1}}\left(n_1 - 1\right) + s^2_{\text{group 2}}\left(n_2 - 1\right) + s^2_{\text{group 3}}\left(n_3 - 1\right) \\ &= 1.70(5-1) + 1.70(5-1) + 2.50(5-1) \\ &= (1.70 \times 4) + (1.70 \times 4) + (2.50 \times 4) \\ &= 6.8 + 6.8 + 10 \\ &= 23.60 \end{aligned} \tag{12.15}$$

The degrees of freedom for SS_R (df_R) are the total degrees of freedom minus the degrees of freedom for the model ($df_R = df_T - df_M = 14 - 2 = 12$). Put another way, this is $N - k$: the total sample size, N, minus the number of groups, k.

12.2.5 Mean squares 

SS_M tells us the *total* variation that the model (in this case the experimental manipulation) explains and SS_R tells us the *total* variation that is due to unmeasured factors. Because both values are sums, their size depends on the number of scores; for example, SS_M used the sum of three values (the group means), whereas SS_R and SS_T used the sum of 15 values. To eliminate this bias we calculate an average sum of squares (known as the *mean squares*, MS). Rather than dividing by the number of scores used for each SS, we divide by the degrees of freedom because we are trying to extrapolate to a population and so some parameters within that population will be held constant (remember we did this when calculating the variance – see Jane Superbrain Box 2.2). For the puppy therapy data we get a model mean squares (MS_M) of 10.067 and a residual mean squares (MS_R) of 1.96:

$$MS_M = \frac{SS_M}{df_M} = \frac{20.135}{2} = 10.067$$
$$MS_R = \frac{SS_R}{df_R} = \frac{23.60}{12} = 1.96 \quad (12.16)$$

MS_M represents the average amount of variation explained by the model (e.g., the systematic variation), whereas MS_R is a gauge of the average amount of variation explained by unmeasured variables (the unsystematic variation).

12.2.6 The F-statistic 

The *F*-statistic (a.k.a. the *F*-ratio) is a measure of the ratio of the variation explained by the model and the variation attributable to unsystematic factors. In other words, it is the ratio of how good the model is to how bad it is (how much error there is). It is calculated by dividing the model mean squares by the residual mean squares:

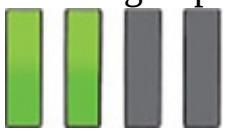
$$F = \frac{MS_M}{MS_R} \quad (12.17)$$

As with other test statistics that we have looked at (e.g., *t*) the *F*-statistic is a signal-to-noise ratio. In experimental research, it is the ratio of the experimental effect to the individual differences in performance. Because *F* is the ratio of systematic to unsystematic variance, if it is less than 1 it means that MS_R is greater than MS_M and that there is more unsystematic than systematic variance. In experimental research this means that the effect of natural variation is greater

than differences brought about by the experiment. In this scenario, we can, therefore, be sure that our experimental manipulation has been unsuccessful (because it has bought about less change than if we left our participants alone) and *F* will be non-significant. For the puppy therapy data, the *F*-statistic is:

$$F = \frac{MS_M}{MS_R} = \frac{10.067}{1.967} = 5.12 \quad (12.18)$$

meaning that the systematic variation is 5 times larger than the unsystematic variation; basically the experimental manipulation (puppy therapy groups) had some effect above and beyond the effect of individual differences in performance. Typically researchers are interested in whether this ratio is significant: in other words, what would the probability be of getting an *F* at least this big if the experimental manipulation, in reality, had no effect at all on happiness (i.e. the null hypothesis). When the time comes that society collapses and we revert to a simple bartering system and all technology has been jettisoned into space, we will compare the obtained value of *F* against an *F*-distribution with the same degrees of freedom (see the Appendix). If the observed value exceeds the critical value we will probably conclude that our independent variable has had a genuine effect (because an *F* at least as big as the one we have observed would be very unlikely if there were no effect in the population). With 2 and 12 degrees of freedom the critical values are 3.89 (p = 0.05) and 6.93 (p = 0.01). The observed value, 5.12, is, therefore, significant at the 0.05 level of significance but not significant at the 0.01 level. Until the dawn of this anti-technological future, we can get the exact *p* from a computer: if it is less than the alpha level that we set before the experiment (e.g., 0.05) then scientists typically conclude that the variable that they manipulated had a genuine effect (more generally, predicting the outcome from group membership improves prediction).

12.2.7 Interpreting F 

I've already mentioned that *F* assesses the overall fit of the model to the data. When the model is one that predicts an outcome from group means, *F* evaluates whether 'overall' there are differences between means; it does not provide specific information about which groups were affected. It is an *omnibus* test. In our puppy therapy example in which there are three groups, a significant *F* tells us that the means of these three samples are not equal (i.e., that $\overline{X}_1 = \overline{X}_2 = \overline{X}_3$ is *not* true). There are several ways in which the means can differ: (1) all three sample means could be significantly different

$(\overline{X}_1 \neq \overline{X}_2 \neq \overline{X}_3)$; (2) the means of groups 1 and 2 are similar to each other but different from that of group 3 $(\overline{X}_1 = \overline{X}_2 \neq \overline{X}_3)$; (3) the means of groups 2 and 3 are similar to each other but different from that of group 1 $(\overline{X}_1 \neq \overline{X}_2 = \overline{X}_3)$; or (4) the means of groups 1 and 3 are similar to each other but different from that of group 2 $(\overline{X}_1 = \overline{X}_3 \neq \overline{X}_2)$.



You might feel that *F* is a bit pointless because, given that you've gone to the trouble of running an experiment, you probably had more specific predictions than 'there's a difference, somewhere or other'. You might wonder why you don't fit lots of models each of which compares only two means at a time; after all, this would tell you specifically whether pairs of group means differ. The reason why we don't do this was explained in Section 2.9.7: every time you run a test on the same data you inflate the Type I error rate. We'll return to this point in Section 12.5 when we discover how to establish where the group differences lie. For now, though, the reason why the *F*-test is useful is that as a single test (albeit of a non-specific hypothesis) it controls the Type I error rate. Having established that overall group means differ (i.e., the outcome can be significantly predicted using the group means) we can use the parameters of the model (the *b*-values) to tell us where the differences lie.

12.3 Assumptions when comparing means 

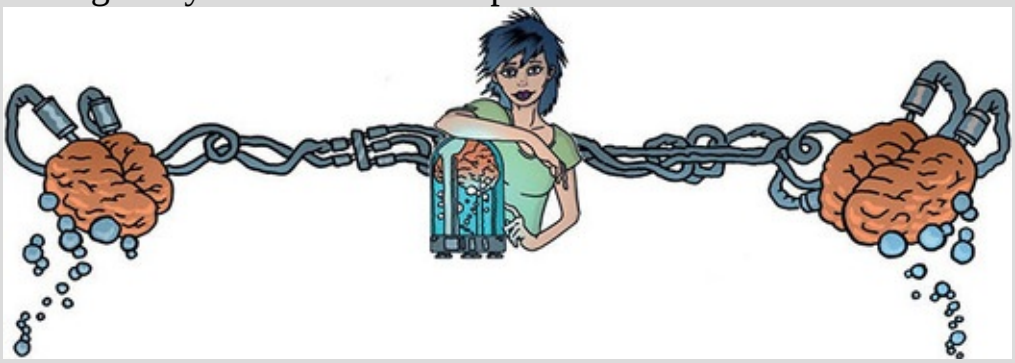
To compare means we use a linear model, so all of the potential sources of bias discussed in Chapter 6 apply. Normality is tested on scores *within groups*, not across the entire sample (see Jane Superbrain Box 6.1).

12.3.1 Homogeneity of variance 

As with any linear model, we assume that the variance of the outcome is steady as the predictor changes (in this context it means that variances in the groups are equal). When group sizes are unequal, violations of the assumption of

homogeneity of variance can have quite serious consequences. This assumption can be tested using Levene's test (see Section 6.11.2). A conventional approach to the assumption is that if Levene's test is significant (i.e., the *p*-value is less than 0.05) then we conclude that the variances are significantly different and try to rectify the situation. However, the *F*-statistic can be adjusted to correct for the degree of heterogeneity and so you may as well just use the corrected *F* because small deviations from homogeneity will result in very small corrections (see Jane Superbrain Box 6.6). Two such corrections are the **Brown–Forsythe *F*** (Brown & Forsythe, 1974), and **Welch's *F*** (Welch, 1951). If you're really bored, these two statistics are discussed in Jane Superbrain Box 12.2. You can also use a robust version of *F* that does not assume homogeneity (Section 12.8).

Jane Superbrain 12.2 What do I do in ANOVA when the homogeneity of variance assumption is broken?



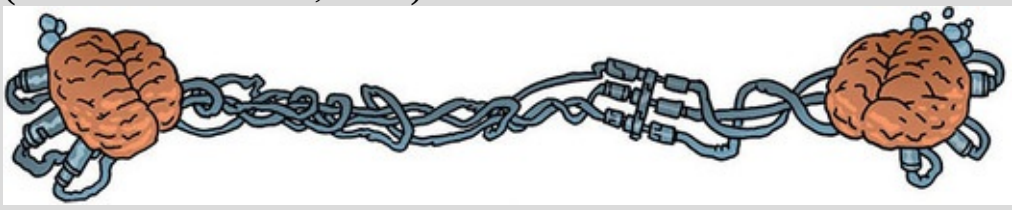
The Brown–Forsythe *F* is fairly easy to explain. When group sizes are unequal and the large groups have the biggest variance, *F* is conservative. If you think back to equation (12.14), this makes sense because to calculate SS_R variances are multiplied by their sample size (minus 1), so you get a large sample size cross-multiplied with a large variance, which will inflate the value of SS_R. *F* is proportionate to SS_M/SS_R, so if SS_R is big, then the *F*-statistic gets smaller (which is why it becomes conservative: its value is overly reduced). Brown and Forsythe get around this problem by weighting the group variances not by their sample size, but by the inverse of their sample sizes (they use n/N, which is the sample size as a proportion of the total sample size). This adjustment reduces the impact of large sample sizes with large variances:

$$F_{BF} = \frac{SS_M}{SS_{R_{BF}}} = \frac{SS_M}{\sum s_k^2\left(1 - \frac{n_k}{N}\right)}$$

For the puppy therapy data, SS_M is the same as before (20.135), so F becomes:

$$\begin{aligned} F_{BF} &= \frac{20.135}{s^2_{\text{group 1}}\left(1 - \frac{n_{\text{group 1}}}{N}\right) + s^2_{\text{group 2}}\left(1 - \frac{n_{\text{group 2}}}{N}\right) + s^2_{\text{group 3}}\left(1 - \frac{n_{\text{group 3}}}{N}\right)} \\ &= \frac{20.135}{1.7\left(1 - \frac{5}{15}\right) + 1.7\left(1 - \frac{5}{15}\right) + 2.5\left(1 - \frac{5}{15}\right)} \\ &= \frac{20.135}{3.933} \\ &= 5.119 \end{aligned}$$

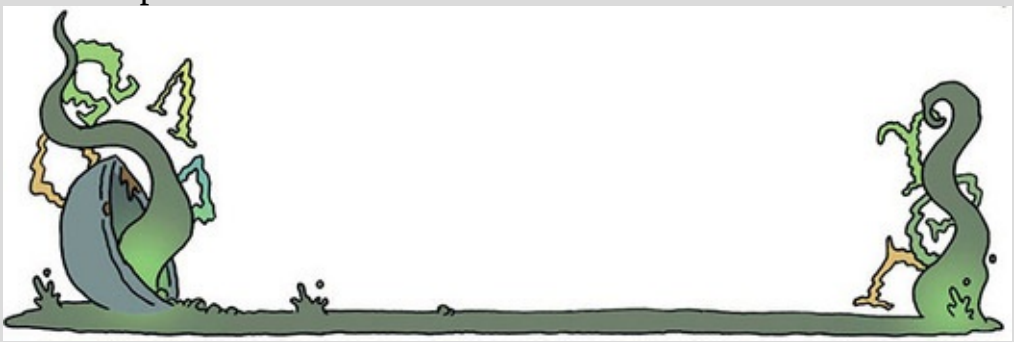
This statistic is evaluated using degrees of freedom for the model and error terms. For the model, df_M is the same as before (i.e., $k - 1 = 2$), but an adjustment is made to the residual degrees of freedom, df_R. Welch's (1951) F is an alternative adjustment that is more involved to explain – if you're interested, see Oliver Twisted. Both adjustments control the Type I error rate well (i.e., when there's no effect in the population you do get a non-significant F), but Welch's F has more power (i.e., is better at detecting an effect that exists) except when there is an extreme mean that has a large variance (Tomarken & Serlin, 1986).



Oliver Twisted Please, Sir, Can I have Some More … Welch's F?



‘You’re only telling us about the Brown–Forsythe *F* because you don’t understand Welch’s *F*,’ taunts Oliver. ‘Andy, Andy, brains all sandy ….’ Whatever, Oliver. Like the Brown–Forsythe *F*, Welch’s *F* adjusts *F* and the residual degrees of freedom to combat problems arising from violations of the homogeneity of variance assumption. There is a lengthy explanation about Welch’s *F* in the additional material available on the companion website. Oh, and Oliver, microchips are made of sand.



12.3.2 Is ANOVA robust?

People often say ‘ANOVA is a robust test’, which means that it doesn’t matter much if we break the assumptions, *F* will still be accurate. Remember from Chapter 6 that we mainly care about normality if we want to assess significance or construct confidence intervals. There are two issues to consider around the significance of *F*. First, does the *F* control the Type I error rate or is it significant even when there are no differences between means? Second, does *F* have enough power (i.e., is it able to detect differences when they are there)?



The myth that ‘ANOVA is robust’ is quite pervasive. It probably stems from

research done a long time ago that investigated a limited range of situations. Examples of this work suggested that *when group sizes are equal* the *F*-statistic can be quite robust to violations of normality (Lunney, 1970), notably skew (Donaldson, 1968), and that when variances are proportional to the means the power of *F* is unaffected by heterogeneity of variance (Budescu, 1982; Budescu & Appelbaum, 1981).

The trouble is that this research is over 35 years old, so let's update things (for a more detailed review see Field & Wilcox, 2017). Recent simulations show that differences in skewness, non-normality and heteroscedasticity interact in complicated ways that impact power (Wilcox, 2017). For example, in the absence of normality, violations of homoscedasticity will affect *F* even when group sizes are equal (Wilcox, 2010, 2012, 2016), and when means are equal the error rate (which should be 5%) can be as high as 18%. Wilcox (2016) suggests that *F* can be considered robust only if the group distributions are identical; for example, groups are skewed to an identical degree, which in practice is probably unlikely. Heavy-tailed distributions are particularly problematic: if you set up a situation with power of 0.9 to detect an effect in a normal distribution and contaminate that distribution with 10% of scores from a normal distribution with a bigger variance (so you get heavier tails), power drops to 0.28 (despite the fact that only 10% of scores have changed). Similarly, Cohen's *d* drops from 1 when distributions are normal to 0.28 (Wilcox, Carlson, Azen, & Clark, 2013). Also, heavy-tailed samples have implications for the central limit theorem, which says that in samples of 30 or more the sampling distribution should be normal (Section 6.6.1); for heavy-tailed distributions samples need to be much larger, up to 160 in some cases (Wilcox, 2010). To sum up, *F* is not robust, despite what your supervisor might tell you.

Violations of the assumption of independence are very serious indeed. Scariano and Davenport (1987) showed that if scores are made to correlate moderately (say, with a Pearson coefficient of 0.5), then when comparing three groups of 10 observations per group the Type I error rate is 0.74 (remember that we'd expect it to be 0.05). In other words, you think you'll make a Type I error 5% of the time but in fact you'll make one 74% of the time!

12.3.3 What to do when assumptions are violated



Violations of assumptions are nowhere near the headache they used to be. In Chapter 6 we discussed methods for correcting problems (e.g., the bias reduction methods in Section 6.12), but these can mostly be avoided. If you routinely

interpret Welch's *F* then you need never even think about homogeneity of variance, and you can bootstrap parameter estimates, which won't affect *F* itself, but at least you know that the model parameters are robust. There are also robust tests that use 20% trimmed means and a bootstrap, which we can implement in SPSS using R (see Section 4.13). There is also a school of thought that you should apply this robust test in all situations and don't bother thinking about assumptions. Finally, you can use the Kruskal–Wallis test from Chapter 7 (although personally I'd use the robust test). If you do apply the usual *F*-statistic then, at the very least, conduct a sensitivity analysis (i.e., apply a robust test to check that your conclusion doesn't change).

12.4 Planned contrasts (contrast coding) 

I've already alluded to the need to follow up a significant *F* by looking at the model parameters, which tell us about specific differences between means. In fact, in Section 12.2 we saw that if we dummy-code the groups that we want to compare then the *b* for each dummy variable compares the mean of the baseline group to the group coded with a 1 in the particular dummy variable. In the puppy therapy example we ended up with two *b*s, one comparing the mean of the 15-minute group to the control group and the other comparing the 30-minute group to the control (Figure 12.3). Each of these *b*s is tested with a *t*-statistic that tells us whether the *b* is significantly different from 0 (i.e. the means differ). The trouble is that with two dummy variables we end up with two *t*-tests, which inflates the familywise error rate (see Section 2.9.7). The other problem is that the dummy variables might not make all the comparisons that we want to make (e.g., the 15-and 30-minute groups are never compared).

There are a couple of solutions to these problems. The first is to use contrast coding rather than dummy coding. Contrast coding is a way of assigning weights to groups in dummy variables to carry out **planned contrasts** (also known as planned comparisons). Weights are assigned in such a way that the contrasts are independent, which means that the overall Type I error rate is controlled. A second option is to compare every group mean to all others (i.e., to conduct several overlapping tests using a *t*-statistic each time) but using a stricter acceptance criterion that keeps the familywise error rate at 0.05. These are known as using ***post hoc* tests** (see Section 12.5). Typically planned contrasts are done to test specific hypotheses, whereas *post hoc* tests are used when there were no specific hypotheses. Let's first look at contrast coding and planned contrasts.

12.4.1 Choosing which contrasts to do

In the puppy therapy example the primary hypothesis would be that any dose of puppy therapy should change happiness compared to the control group. A second hypothesis might be that a 30-minute session should increase happiness more than a 15-minute one. To do planned contrasts, these hypotheses must be derived *before* the data are collected. The *F*-statistic is based upon splitting the total variation into two component parts: the variation due to the model or experimental manipulation (SS_M) and the variation due to unsystematic factors (SS_R) (see Figure 12.5).

Planned contrasts extend this logic by breaking down the variation due to the model/experiment into component parts (see Figure 12.6). The exact contrasts will depend upon the hypotheses you want to test. Figure 12.6 shows a situation in which the model variance is broken down based on the two hypotheses that we already discussed. Contrast 1 looks at how much variation in happiness is created by the two puppy conditions compared to the no puppy control condition. Next, the variation explained by puppy therapy (in general) is broken down to see what proportion is accounted for by a 30-minute session relative to a 15-minute one (contrast 2).

Typically, students struggle with the notion of designing planned contrasts, but there are three rules that can help you to work out what to do.

1. If you have a control group, this is usually because you want to compare it against any other groups.
2. Each contrast must compare only two ‘chunks’ of variation.
3. Once a group has been singled out in a contrast it can’t be used in another contrast.

Figure 12.5 Partitioning variance for ANOVA

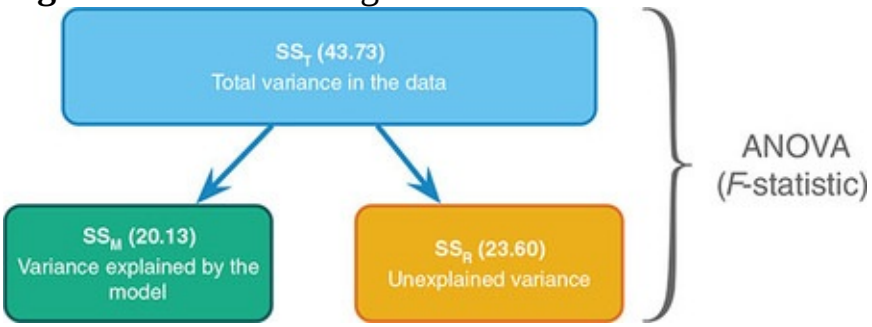
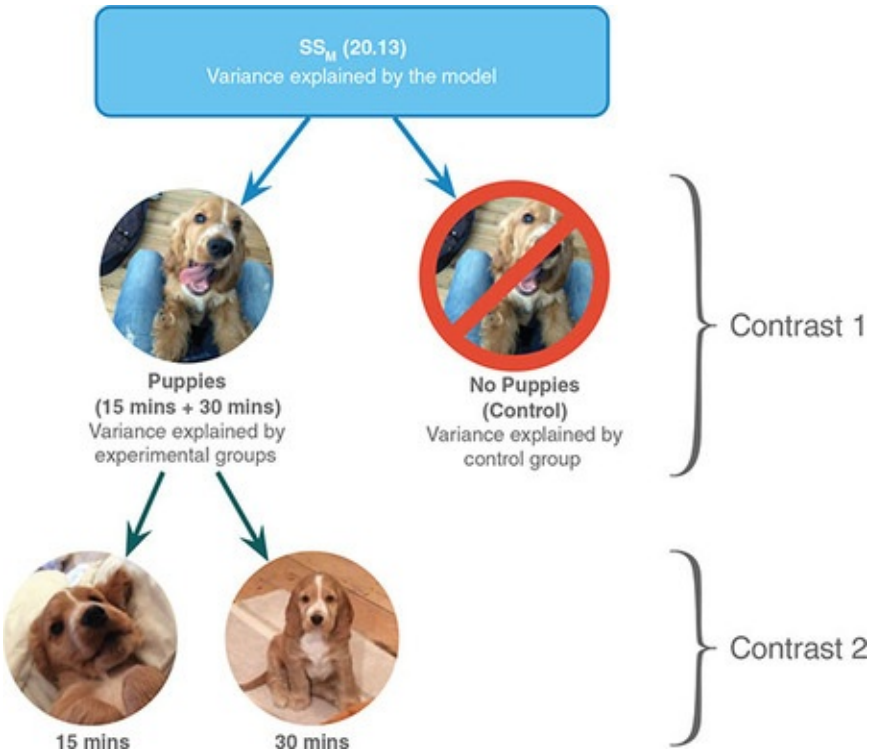


Figure 12.6 Partitioning of the model/experimental variance into component contrasts



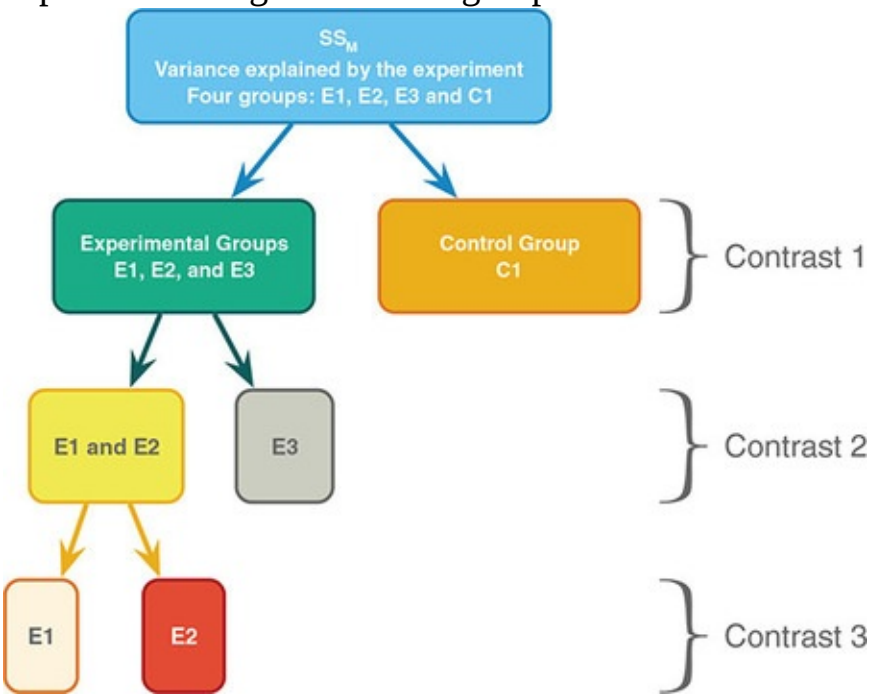
Let's look at these rules in reverse order. First, if a group is singled out in one contrast, then it should not reappear in another contrast. The important thing is that we are breaking down one chunk of variation into smaller independent chunks. This independence matters for controlling the Type I error rate. It's like slicing up a cake: you begin with a cake (the total sum of squares) and then cut it into two pieces (SS_M and SS_R), then you take the piece of cake that represents SS_M and slice it again into smaller pieces. Once you have cut off a piece of cake you cannot stick that piece back onto the original slice, and you cannot stick it onto other pieces of cake, but you can divide it into smaller pieces of cake. Likewise, once a slice of variance has been split from a larger chunk, it cannot be attached to any other pieces of variance, it can only be subdivided into smaller chunks of variance. All this talk of cake is making me hungry, but hopefully it illustrates a point. So, in Figure 12.6 contrast 1 compares the control group to the experimental groups and because the control group is singled out, it is not incorporated into contrast 2.

Second, each contrast must compare only two chunks of variance. This rule is so that we can interpret the contrast. The original *F* tells us that some of our means differ, but not which ones, and if we were to perform a contrast on more than two chunks of variance we would be no better off. By comparing only two chunks of variance we know that the result represents a significant difference (or not) between these two portions of variation. If you follow the independence of

contrasts rule (the cake slicing), and always compare only two pieces of variance, then you should end up with $k - 1$ contrasts (where k is the number of conditions you're comparing); in other words, one fewer contrasts than the number of conditions you have in your design.

The first rule in the list reminds us that we often use at least one control condition, and it (or they) usually exists because we predict that the experimental conditions will differ from it (or them). As such, when planning contrasts the chances are that your first contrast will be one that compares all the experimental groups with the control group (or groups). Once you have done this first comparison, any remaining contrasts will depend upon which groups you predict will differ (based on the theory you're testing).

Figure 12.7 Partitioning variance for planned contrasts in a four-group experiment using one control group



To further illustrate these principles Figures 12.7 and 12.8 show potential sets of contrasts for two different four-group experiments. In both examples there are three possible contrasts (one less than the number of groups) and every contrast compares only two chunks of variance. The first contrast is the same in both cases: the experimental groups are compared against the control group or groups. In Figure 12.7 there is only one control condition and this portion of variance is used only in the first contrast (because it cannot be broken down any further). In Figure 12.8 there were two control groups, and so the portion of variance due to the control conditions (contrast 1) can be broken down further to see whether the scores in the control groups differ from each other (contrast 3).

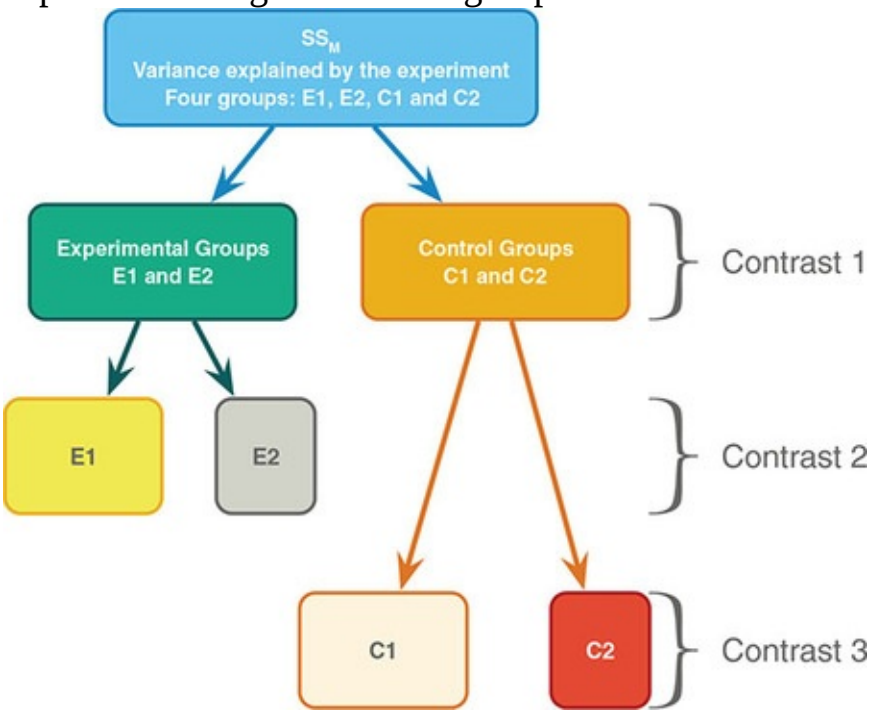
In Figure 12.7, the first contrast contains a chunk of variance that is due to the three experimental groups, and this chunk of variance is broken down by first looking at whether groups E1 and E2 differ from E3 (contrast 2). It is equally valid to use contrast 2 to compare groups E1 and E3 to E2, or to compare groups E2 and E3 to E1. The exact contrast that you choose depends upon your hypotheses. For contrast 2 in Figure 12.7 to be valid we need to have a good theoretical reason to expect group E3 to be different from the other two groups. The third contrast in Figure 12.7 depends on the comparison chosen for contrast 2. Contrast 2 necessarily had to involve comparing two experimental groups against a third, and the experimental groups chosen to be combined must be separated in the final contrast. As a final point about Figures 12.7 and 12.8, notice that once a group has been singled out in a contrast, it is never used in any subsequent one.



When we carry out a planned contrast, we compare ‘chunks’ of variance and these chunks often consist of several groups. When you design a contrast that compares several groups to one other group, you are comparing the means of the groups in one chunk with the mean of the group in the other chunk. As an example, for the puppy therapy data I suggested that an appropriate first contrast would be to compare the two dose groups with the control group. The means of the groups are 2.20 (control), 3.20 (15 minutes) and 5.00 (30 minutes), and so the first comparison, which compared the two experimental groups to the control, is comparing 2.20 (the mean of the control group) to the average of the other two groups ((3.20 + 5.00)/2 = 4.10). If this first contrast turns out to be significant, then we can conclude that 4.10 is significantly greater than 2.20, which in terms of the experiment tells us that the average of the experimental groups is significantly different from the average of the controls. You can probably see that logically this means that, if the standard errors are the same, the experimental group with the highest mean (the 30-minute group) will be significantly different from the mean of the control group. However, the experimental group with the lower mean (the 15-minute group) might not necessarily differ from the control group; we have to use the final contrast to make sense of the experimental conditions. For the puppy data the final contrast looked at whether the two experimental groups differ (i.e., is the mean of the 30-minute group significantly different from the mean of the 15-minute group?). If

this contrast turns out to be significant then we can conclude that having 30 minutes of puppy therapy significantly affected happiness compared to having 15 minutes. If the contrast is non-significant then we conclude that the dosage of puppy therapy made no significant difference to happiness. In this latter scenario it is likely that both doses affect happiness more than control, whereas the former case implies that having 15 minutes may be no different than having a control. However, the word *implies* is important here: it is possible that the 15-minute group might not differ from the control. To be completely sure you would need *post hoc* tests.

Figure 12.8 Partitioning variance for planned contrasts in a four-group experiment using two control groups



12.4.2 Defining contrasts using weights

Hopefully by now you have got some idea of how to plan which contrasts to do (i.e., if your brain hasn't exploded yet). The next issue is how to do them. To carry out contrasts we need to code our dummy variables in a way that results in *b*s that compare the 'chunks' that we set out in our contrasts. Remember that when we used dummy coding, we used values of 0 and 1 for the dummy variables and ended up with *b*-values that compared each group to a baseline group. We need to work out what values (instead of 0 and 1) to assign to each group to give us (in this example) two *b*-values, the first of which compares all puppy therapy scores to the control group, and the second of which compares 15

minutes of puppy therapy with 30 minutes (and ignores the control group). The values assigned to the dummy variables are known as **weights**.

This procedure is horribly confusing, but there are a few basic rules for assigning values to the dummy variables to obtain the contrasts you want. I will explain these simple rules before showing how the process works. Remember the previous section when you read through these rules, and remind yourself of what I mean by a 'chunk' of variation.

- **Rule 1**: Choose sensible contrasts. Remember that you want to compare only two chunks of variation and that if a group is singled out in one contrast, that group should be excluded from any subsequent contrasts.
- **Rule 2**: Groups coded with positive weights will be compared against groups coded with negative weights. So, assign one chunk of variation positive weights and the opposite chunk negative weights.
- **Rule 3**: If you add up the weights for a given contrast the result should be zero.
- **Rule 4**: If a group is not involved in a contrast, automatically assign it a weight of zero, which will eliminate it from the contrast.
- **Rule 5**: For a given contrast, the weights assigned to the group(s) in one chunk of variation should be equal to the number of groups in the opposite chunk of variation.

OK, let's follow these rules to derive the weights for the puppy therapy data. The first contrast we chose was to compare the two experimental groups against the control (Figure 12.9). The first chunk of variation contains the two experimental groups, and the second chunk contains only the control group. Rule 2 states that we should assign one chunk positive weights, and the other negative. It doesn't matter which way round we do this, but for convenience let's assign chunk 1 positive weights and chunk 2 negative weights, as in Figure 12.9. Using rule 5, the weight we assign to the groups in chunk 1 should be equivalent to the number of groups in chunk 2. There is only one group in chunk 2, and so we assign each group in chunk 1 a weight of 1. Likewise, we assign a weight to the group in chunk 2 that is equal to the number of groups in chunk 1. There are two groups in chunk 1 so we give the control group a weight of 2. Then we combine the sign of the weights with the magnitude to give us weights of −2 (control), +1 (15 minutes) and +1 (30 minutes), as in Figure 12.9. Rule 3 states that for a given contrast, the weights should add up to zero, and by following rules 2 and 5 this should be true (if you haven't followed the rules properly then it will become clear when you add the weights). Let's check by adding the weights: sum of weights = 1 + 1 − 2 = 0. Happy days.

The second contrast was to compare the two experimental groups, and so we

want to ignore the control group. Rule 4 tells us that we should automatically assign this group a weight of 0 (to eliminate it). We are left with two chunks of variation: chunk 1 contains the 15-minute group and chunk 2 contains the 30-minute group. By following rules 2 and 5 it should be obvious that one group is assigned a weight of +1 while the other is assigned a weight of −1 (Figure 12.10). If we add the weights for contrast 2 we find that they again add up to zero: sum of weights = 1 − 1 + 0 = 0. Even happier days.

The weights for each contrast are placed in the two dummy variables in the following equation:

$$\text{Happiness}_i = b_0 + b_1\text{Contrast}_{1i} + b_2\text{Contrast}_{2i} \quad (12.19)$$

Hence, the weights are used in a linear model in which b_1 represents contrast 1 (comparing the experimental groups to the control), b_2 represents contrast 2 (comparing the 30-minute group to the 15-minute group), and b_0 is the grand mean. Each group is specified now not by the 0 and 1 coding scheme that we initially used, but by the coding scheme for the two contrasts. Participants in the control group are identified by a code of −2 for contrast 1 and a code of 0 for contrast 2. Likewise, the 30-minute group is identified by a code of 1 for both variables, and the 15-minute group has a code of 1 for one contrast and a code of −1 for the other (Table 12.4).

Figure 12.9 Assigning weights for contrast 1

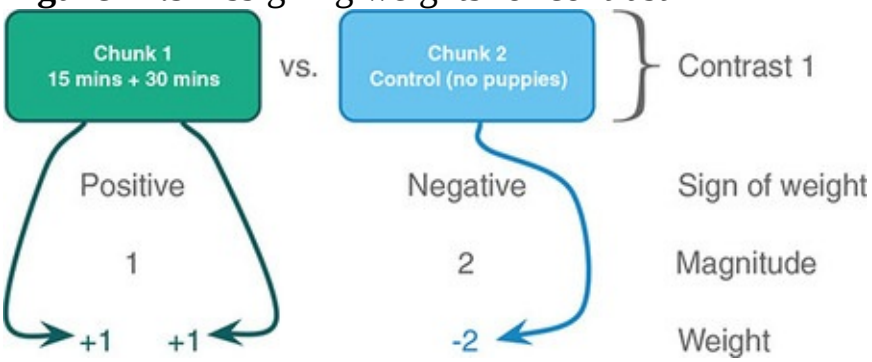


Figure 12.10 Assigning weights for contrast 2

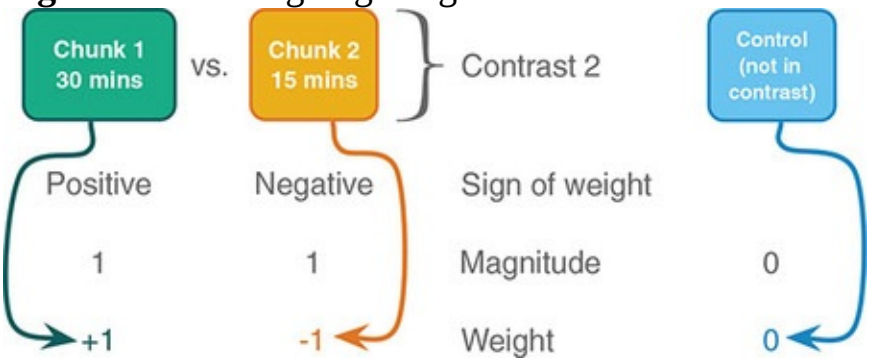


Table 12.4 Orthogonal contrasts for the puppy therapy data

Group	Dummy variable 1 ($Contrast_1$)	Dummy variable 2 ($Contrast_2$)	Product $Contrast_1 \times Contrast_2$
Control	−2	0	0
15 minutes	1	−1	−1
30 minutes	1	1	1
Total	0	0	0

It is important that the weights for a contrast sum to zero because it ensures that you are comparing two unique chunks of variation. Therefore, a *t*-statistic can be used (remember it assumes independence). A more important consideration is that when you multiply the weights for a particular group, these products should also add up to zero (see the final column of Table 12.4). If the products sum to zero then the contrasts are *independent* or **orthogonal**. When we used dummy coding and fit a linear model to the puppy therapy data, I commented that the familywise error rate for the *t*-statistics for the *b*-values would be inflated (see Section 2.9.7). This is because these contrasts are not independent (both *b*-values involve a comparison with the control group). However, if the contrasts *are* independent then the *t*-statistics for the *b*-values are also independent and the resulting *p*-values are uncorrelated. You might think that it is very difficult to ensure that the weights you choose for your contrasts conform to the requirements for independence but, provided you follow the rules I have laid out, you should always derive a set of *orthogonal* contrasts. You should double-check by looking at the sum of the multiplied weights, and if this total is not zero then go back to the rules and see where you have gone wrong.



Earlier on, I mentioned that when you contrast-code dummy variables in a linear model the *b*-values represent the differences between the means that the contrasts were designed to test. Let's have a look at how this works (this next part is not for the faint-hearted). When we do planned contrasts, the intercept b_0 is equal to the grand mean (i.e., the value predicted by the model when group membership is not known), which in this example (and because group sizes are equal) is:

$$b_0 = \text{grand mean} = \frac{\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}} + \bar{X}_{\text{Control}}}{3} \quad (12.20)$$

For a participant in the *control group*, we define their group membership using the values for the control group in Table 12.4 and the predicted value of

happiness is the mean of the control group. The model can be expressed as:

$$\text{Happiness}_i = b_0 + b_1\text{Contrast}_{1i} + b_2\text{Contrast}_{2i}$$

$$\bar{X}_{\text{Control}} = \left(\frac{\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}} + \bar{X}_{\text{Control}}}{3}\right) + (-2b_1) + (b_2 \times 0) \quad (12.21)$$

If we rearrange this equation and multiply everything by 3 (to get rid of the fraction) we get:

$$2b_1 = \left(\frac{\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}} + \bar{X}_{\text{Control}}}{3}\right) - \bar{X}_{\text{Control}} \quad (12.22)$$

$$6b_1 = \bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}} + \bar{X}_{\text{Control}} - 3\bar{X}_{\text{Control}}$$

$$= \bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}} - 2\bar{X}_{\text{Control}}$$

We then divide everything by 2 to reduce the equation to its simplest form:

$$3b_1 = \left(\frac{\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}}}{2}\right) - \bar{X}_{\text{Control}} \quad (12.23)$$

$$b_1 = \frac{1}{3}\left[\left(\frac{\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}}}{2}\right) - \bar{X}_{\text{Control}}\right]$$

We planned contrast 1 to look at the difference between the average of the two puppy groups and the control, and a final rearrangement of the equation shows how b_1 represents this difference:

$$3b_1 = \left(\frac{\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}}}{2}\right) - \bar{X}_{\text{Control}} \quad (12.24)$$

$$= \frac{5 + 3.2}{2} - 2.2$$

$$= 1.9$$

Rather than being the true value of the difference between experimental and control groups, b_1 is actually a third of this difference (b_1 = 1.9/3 = 0.633) – it is divided by the number of groups in the contrast. Nevertheless, it is proportionate to the difference it set out to test.

For someone in the *30-minute group*, the predicted value of happiness is the mean for the 30-minute group, and their group membership is coded using the values for the 30-minute group in Table 12.4. The resulting model is in equation (12.25).

$$\text{Happiness}_i = b_0 + b_1\text{Contrast}_{1i} + b_2\text{Contrast}_{2i}$$

$$\bar{X}_{30\,\text{mins}} = b_0 + (b_1 \times 1) + (b_2 \times 1) \quad (12.25)$$

$$b_2 = \bar{X}_{30\,\text{mins}} - b_1 - b_0$$

We know already what b_1 and b_0 represent, so we place these values into the equation and then multiply by 3 to get rid of some of the fractions:

$$
\begin{aligned}
b_2 &= \bar{X}_{30\,\text{mins}} - b_1 - b_0 \\
&= \bar{X}_{30\,\text{mins}} - \frac{1}{3}\left[\left(\frac{\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}}}{2}\right) - \bar{X}_{\text{Control}}\right] - \frac{\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}} + \bar{X}_{\text{Control}}}{3} \\
3b_2 &= 3\bar{X}_{30\,\text{mins}} - \left[\left(\frac{\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}}}{2}\right) - \bar{X}_{\text{Control}}\right] - \bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}} + \bar{X}_{\text{Control}}
\end{aligned}
\tag{12.26}
$$

If we multiply everything by 2 to get rid of the fraction, expand the brackets and collect terms we get:

$$
\begin{aligned}
6b_2 &= 6\bar{X}_{30\,\text{mins}} - \left(\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}} - 2\bar{X}_{\text{Control}}\right) - 2\left(\bar{X}_{30\,\text{mins}} + \bar{X}_{15\,\text{mins}} + \bar{X}_{\text{Control}}\right) \\
&= 6\bar{X}_{30\,\text{mins}} - \bar{X}_{30\,\text{mins}} - \bar{X}_{15\,\text{mins}} + 2\bar{X}_{\text{Control}} - 2\bar{X}_{30\,\text{mins}} - 2\bar{X}_{15\,\text{mins}} - 2\bar{X}_{\text{Control}} \\
&= 3\bar{X}_{30\,\text{mins}} - 3\bar{X}_{15\,\text{mins}}
\end{aligned}
\tag{12.27}
$$

Finally, let's divide the equation by 6 to find out what b_2 represents (remember that 3/6 = 1/2):

$$
b_2 = \frac{\bar{X}_{30\,\text{mins}} - \bar{X}_{15\,\text{mins}}}{2} \tag{12.28}
$$

We planned contrast 2 to look at the difference between the experimental groups:

$$
\bar{X}_{30\,\text{mins}} - \bar{X}_{15\,\text{mins}} = 5 - 3.2 = 1.8 \tag{12.29}
$$

and b_2 represents this difference (equation (12.28)). Again, rather than being the absolute value of the difference between the experimental groups, b_2 is this difference divided by the number of groups in the contrast (1.8/2 = 0.9), but the key thing is what it represents and it is proportional to the difference between experimental group means.



To illustrate these principles, I have created a file called **Puppies Contrast.sav** in which the puppy therapy data are coded using the contrast coding scheme used in this section. Fit a linear model using happiness as the outcome and **dummy1** and **dummy2** as the predictor variables (leave all default options).

Output 12.2 shows the result of the model from the self-test. The main ANOVA for the model is the same as when dummy coding was used (compare it to Output 12.1), showing that the model fit is the same (it should be because the model represents the group means and these have not changed); however, the *b*-values have changed because the values of our dummy variables have changed. The first thing to notice is that the intercept is the grand mean, 3.467 (see, I

wasn't telling lies). Second, the *b* for contrast 1 is one-third of the difference between the average of the experimental conditions and the control condition. Finally, the *b* for contrast 2 is half of the difference between the experimental groups (see above). The significance values of the *t*-statistics tell us that our puppy groups were significantly different from the control ($p = 0.029$) but that the 15-and 30-minutes of puppy therapy were not significantly different ($p = 0.065$).

Output 12.2

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	3.467	.362		9.574	.000
	Dummy 1: Dose vs. Control	.633	.256	.525	2.474	.029
	Dummy 2: 15 mins vs 30 mins	.900	.443	.430	2.029	.065

a. Dependent Variable: Happiness (0-10)

12.4.3 Non-orthogonal contrasts



Contrasts don't have to be orthogonal: non-orthogonal contrasts are contrasts that are related. The best way to get them is to disobey rule 1 in the previous section. Using my cake analogy again, non-orthogonal contrasts are where you slice up your cake and then try to stick slices of cake together again. Standard dummy coding (Section 12.2) is an example of non-orthogonal contrasts because the baseline group is used in each contrast. For the puppy therapy data another set of non-orthogonal contrasts might be to have the same initial contrast (comparing experimental groups against the control), but then to compare the 30-minute group to the control. This disobeys rule 1 because the control group is singled out in the first contrast but used again in the second contrast. The coding for this set of contrasts is shown in Table 12.5, and the last column makes clear that when you multiply and add the codings from the two contrasts the sum is not zero. This tells us that the contrasts are not orthogonal.



There is nothing intrinsically wrong with non-orthogonal contrasts, but you must be careful about how you interpret them because the contrasts are related and so the resulting test statistics and *p*-values will be correlated to some degree. Basically, the Type I error rate isn't controlled, so you should use a more

conservative probability level to accept a given contrast as statistically significant (see Section 12.5).

12.4.4 Built-in contrasts

Although under most circumstances you will design your own contrasts, there are also 'off-the-shelf' contrasts that you can choose. Table 12.6 shows the built-in contrasts in SPSS Statistics for procedures such as logistic regression (see Section 20.5.7), factorial designs and repeated-measures designs (see Chapters 14 and 15). The exact codings are not provided in Table 12.6, but I give examples of the contrasts done in a three-and four-group situation (where the groups are labelled 1, 2, 3 and 1, 2, 3, 4, respectively). When you code categorical variables in the data editor, SPSS Statistics treats the lowest-value code as group 1, the next highest code as group 2, and so on. Therefore, depending on which contrasts you want, you should code your grouping variable appropriately (and then use Table 12.6 as a guide to which contrasts you'll get). Some contrasts in Table 12.6 are orthogonal (i.e., Helmert and difference contrasts) while others are non-orthogonal (deviation, simple and repeated). You might also notice that simple contrasts are the same as those given by using the dummy variable coding described in Table 12.2.

Cramming Sam's Tips Planned contrasts



- If the *F* for the overall model is significant you need to find out which groups differ.
- When you have generated specific hypotheses before the experiment, use *planned contrasts*.
- Each contrast compares two 'chunks' of variance. (A chunk can contain one or more groups.)
- The first contrast will usually be experimental groups against control groups.
- The next contrast will be to take one of the chunks that contained more than one group (if there were any) and divide it

in to two chunks.

- You repeat this process: if there are any chunks in previous contrasts that contained more than one group that haven't already been broken down into smaller chunks, then create new contrasts that breaks them down into smaller chunks.
- Carry on creating contrasts until each group has appeared in a chunk on its own in one of your contrasts.
- The number of contrasts you end up with should be one less than the number of experimental conditions. If not, you've done it wrong.
- In each contrast assign a 'weight' to each group that is the value of the number of groups in the opposite chunk in that contrast.
- For a given contrast, randomly select one chunk, and for the groups in that chunk change their weights to be negative numbers.
- Breathe a sigh of relief.



12.4.5 Polynomial contrasts: trend analysis

One type of contrast deliberately omitted from Table 12.6 is the **polynomial contrast**. This contrast tests for trends in the data, and in its most basic form it looks for a linear trend (i.e., that the group means increase proportionately). However, there are other trends such as quadratic, cubic and quartic trends that can be examined. Figure 12.11 shows examples of the types of trend that can exist in data sets. The *linear* trend should be familiar to you and represents a simple proportionate change in the value of the dependent variable across ordered categories (the diagram shows a positive linear trend, but of course it could be negative). A **quadratic trend** is where there is a curve in the line (the curve can be subtler than in the figure and in the opposite direction). An example of this is a situation in which a drug enhances performance on a task at first but then as the dose increases the performance tails off or drops. To find a quadratic trend you need at least three groups because with two groups the means of the dependent variable can't be connected by anything other than a straight line. A

cubic trend is where there are two changes in the direction of the trend. So, for example, the mean of the dependent variable at first goes up across categories of the independent variable, then across the succeeding categories the means go down, but then across the last few categories the means rise again. To have two changes in the direction of the mean you must have at least four categories of the independent variable. The final trend that you are likely to come across is the **quartic trend**, and this trend has three changes of direction (so you need at least five categories of the independent variable).

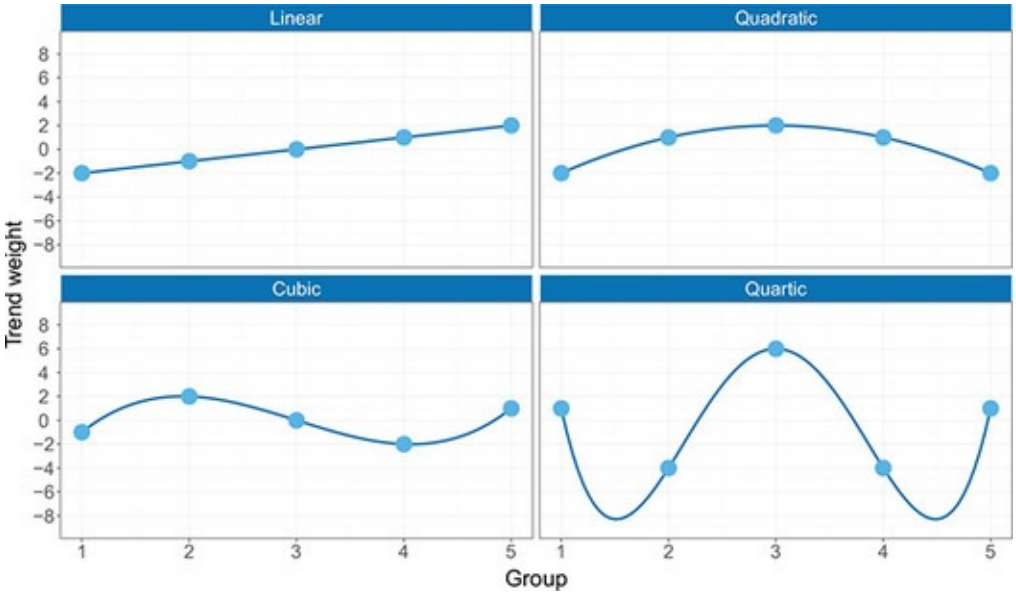
Table 12.5 Non-orthogonal contrasts for the puppy therapy data

Group	Dummy variable 1 (Contrast$_1$)	Dummy variable 2 (Contrast$_2$)	Product Contrast$_1$ × Contrast$_2$
Control	−2	−1	2
15 minutes	1	0	0
30 minutes	1	1	1
Total	0	0	3

Table 12.6 Standard contrasts available in SPSS

Name	Definition	Contrast	Three Groups	Four Groups
Deviation (first)	Compares the effect of each category (except the first) to the overall experimental effect	1	2 vs. (1, 2, 3)	2 vs. (1, 2, 3, 4)
		2	3 vs. (1, 2, 3)	3 vs. (1, 2, 3, 4)
		3		4 vs. (1, 2, 3, 4)
Deviation (last)	Compares the effect of each category (except the last) to the overall experimental effect	1	1 vs. (1, 2, 3)	1 vs. (1, 2, 3, 4)
		2	2 vs. (1, 2, 3)	2 vs. (1, 2, 3, 4)
		3		3 vs. (1, 2, 3, 4)
Simple (first)	Each category is compared to the first category	1	1 vs. 2	1 vs. 2
		2	1 vs. 3	1 vs. 3
		3		1 vs. 4
Simple (last)	Each category is compared to the last category	1	1 vs. 3	1 vs. 4
		2	2 vs. 3	2 vs. 4
		3		3 vs. 4
Repeated	Each category (except the first) is compared to the previous category	1	1 vs. 2	1 vs. 2
		2	2 vs. 3	2 vs. 3
		3		3 vs. 4
Helmert	Each category (except the last) is compared to the mean effect of all subsequent categories	1	1 vs. (2, 3)	1 vs. (2, 3, 4)
		2	2 vs. 3	2 vs. (3, 4)
		3		3 vs. 4
Difference (reverse Helmert)	Each category (except the first) is compared to the mean effect of all previous categories	1	3 vs. (2, 1)	4 vs. (3, 2, 1)
		2	2 vs. 1	3 vs. (2, 1)
		3		2 vs. 1

Figure 12.11 Examples of linear, quadratic, cubic and quartic trends across five groups



Polynomial trends should be examined in data sets in which it makes sense to order the categories of the independent variable (so, for example, if you have administered five doses of a drug it makes sense to examine the five doses in order of magnitude). For the puppy therapy data there are three groups, and so we can expect to find only a linear or quadratic trend (it would be pointless to test for any higher-order trends).

Each of these trends has a set of codes for the dummy variables in the model, so we are doing the same thing that we did for planned contrasts except that the codings have already been devised to represent the type of trend of interest. In fact, the graphs in Figure 12.11 have been constructed by plotting the code values for the five groups. If you add the codes for a given trend the sum will equal zero, and if you multiply the codes you will find that the sum of the products also equals zero. Hence, these contrasts are orthogonal.

12.5 *Post hoc* procedures

Often people have no specific *a priori* predictions about the data they have collected and instead they rummage around the data looking for any differences between means that they can find. It's a bit like *p*-hacking (Section 3.3), except that you adjust the *p* to make it harder to find significant differences. *Post hoc* tests consist of **pairwise comparisons** that are designed to compare all different combinations of the treatment groups. So, it is taking every pair of groups and performing a separate test on each. Now, this might seem like a particularly stupid thing to do in light of what I have already told you about the problems of inflated familywise error rates (Section 2.9.7). However, pairwise comparisons

control the familywise error by correcting the level of significance for each test such that the overall Type I error rate (α) across all comparisons remains at 0.05. There are several ways in which the familywise error rate can be controlled, and we have already discussed one of the most popular ones: the Bonferroni correction (Section 2.9.7).

There are other methods too (SPSS Statistics does about 18 different ones). Although I would love to go into the tedious detail of all 18 procedures, there really is little point. One reason is that there are excellent texts already available for those who wish to know (Klockars & Sax, 1986; Toothaker, 1993), but the main one is that to explain them I would have to learn about them first. I may be a nerd, but even I draw the line at reading up on 18 different *post hoc* tests. However, it *is* important that you know which *post hoc* tests perform best according to three important criteria. First, does the test control the Type I error rate? Second, does the test control the Type II error rate (i.e., does the test have good statistical power)? Third, is the test robust?

12.5.1 Type I and Type II error rates for post hoc tests



The Type I error rate and the statistical power of a test are linked. Therefore, there is always a trade-off: if a test is conservative (the probability of a Type I error is small) then it is likely to lack statistical power (the probability of a Type II error will be high). Therefore, it is important that multiple comparison procedures control the Type I error rate but without a substantial loss in power. If a test is too conservative then we are likely to reject differences between means that are, in reality, meaningful.

The *least-significant difference* (LSD) pairwise comparison makes no attempt to control the Type I error and is equivalent to performing multiple tests on the data. The only difference is that the LSD requires the overall ANOVA to be significant. The *Studentized Newman–Keuls* (SNK) procedure is also a very liberal test and lacks control over the familywise error rate. *Bonferroni's* and *Tukey's* tests both control the Type I error rate very well but are conservative (they lack statistical power). Of the two, Bonferroni has more power when the number of comparisons is small, whereas Tukey is more powerful when testing large numbers of means. Tukey generally has greater power than *Dunn* and *Scheffé*. The *Ryan, Einot, Gabriel and Welsch Q* procedure (REGWQ) has good power and tight control of the Type I error rate. In fact, when you want to test all pairs of means this procedure is probably the best. However, when group sizes are different this procedure should not be used.

12.5.2 Are post hoc procedures robust?

Most research on *post hoc* tests has looked at whether the test performs well when the group sizes are different (an unbalanced design), when the population variances are very different, and when data are not normally distributed. The good news is that most multiple comparison procedures perform relatively well under small deviations from normality. The bad news is that they perform badly when group sizes are unequal and when population variances are different. *Hochberg's GT2* and *Gabriel's* pairwise test procedure were designed to cope with situations in which sample sizes are different. Gabriel's procedure is generally more powerful but can become too liberal when the sample sizes are very different. Also, Hochberg's GT2 is very unreliable when the population variances are different, and so should be used only when you are sure that this is not the case. There are several multiple comparison procedures that have been specially designed for situations in which population variances differ. SPSS provides four options for this situation: *Tamhane's T2*, *Dunnett's T3*, *Games–Howell* and *Dunnett's C*. Tamhane's T2 is conservative and Dunnett's T3 and C keep very tight Type I error control. The Games–Howell procedure is the most powerful but can be liberal when sample sizes are small. However, Games–Howell is also accurate when sample sizes are unequal.

Cramming Sam's Tips *Post hoc* tests



- When you have no specific hypotheses before the experiment, follow up the model with *post hoc tests*.
- When you have equal sample sizes and group variances are similar use REGWQ or Tukey.
- If you want guaranteed control over the Type I error rate then use Bonferroni.
- If sample sizes are slightly different then use Gabriel's, but if sample sizes are very different use Hochberg's GT2.
- If there is any doubt that group variances are equal then use the

Games–Howell procedure.



12.5.3 Summary of *post hoc* procedures

The choice of comparison procedure will depend on the exact situation you have and whether it is more important for you to keep strict control over the familywise error rate or to have greater statistical power. However, some general guidelines can be drawn (Toothaker, 1993). When you have equal sample sizes and you are confident that your population variances are similar then use REGWQ or Tukey as both have good power and tight control over the Type I error rate. Bonferroni is generally conservative, but if you want guaranteed control over the Type I error rate then this is the test to use. If sample sizes are slightly different then use Gabriel's procedure because it has greater power, but if sample sizes are very different use Hochberg's GT2. If there is any doubt that the population variances are equal then use the Games–Howell procedure because this generally seems to offer the best performance. I recommend running the Games–Howell procedure in addition to any other tests you might select because of the uncertainty of knowing whether the population variances are equivalent.

Although these general guidelines provide a convention to follow, be aware of the other procedures available and when they might be useful to use (e.g., Dunnett's test is the only multiple comparison that allows you to test means against a control mean).

12.6 Comparing several means using SPSS Statistics

Because to compare means we simply fit a variant of the linear model, we could simply set up contrast dummy variables and run the analysis using the linear regression menus from Chapter 9. However, SPSS Statistics has a designated menu for situations where you're predicting an outcome from several group means (historically known as a one-way independent analysis of variance, ANOVA). Using this designated menu has the advantage that you don't have to

create dummy variables manually, as we shall see.

12.6.1 General procedure of one-way ANOVA

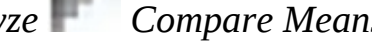


The data are in the file **Puppies.sav** (although it's good practice to enter them yourself). We enter the data into two columns (three if you want to include a participant ID). One column (**Dose**) specifies how much puppy therapy the participant was given, and is a variable that codes the group to which the individual belonged (I have coded 1 = control, 2 = 15 minutes and 3 = 30 minutes). The other column (**Happiness**) contains the person's happiness score. Because one-way **independent ANOVA** is a linear model with a different label attached, look back at the general procedure for linear models in Chapter 9. Figure 12.12 highlights the steps that are specific to this version of the linear model. As with any analysis, begin by graphing the data and looking for and correcting sources of bias.



Produce a line chart with error bars for the puppy therapy data.

Noting my earlier comments about homogeneity of variance and normality (Section 12.3), there is a case for always proceeding with a robust test or at the very least always using Welch's *F*.

To fit the model select *Analyze*  *Compare Means* 

to access the main dialog in Figure 12.13. There is an empty box in which to place one or more dependent variables and another to specify a categorical predictor or *factor*. 'Factor' is another term for independent variable or categorical predictor and not to be confused with a very different type of factor that we learn about in Chapter 18. For the puppy therapy data drag **Happiness** from the variables list into the box labelled *Dependent List* (or click ). Then drag the

grouping variable **Dose** to the box labelled *Factor* (*or* click ).

Figure 12.12 Overview of the general procedure for one-way ANOVA

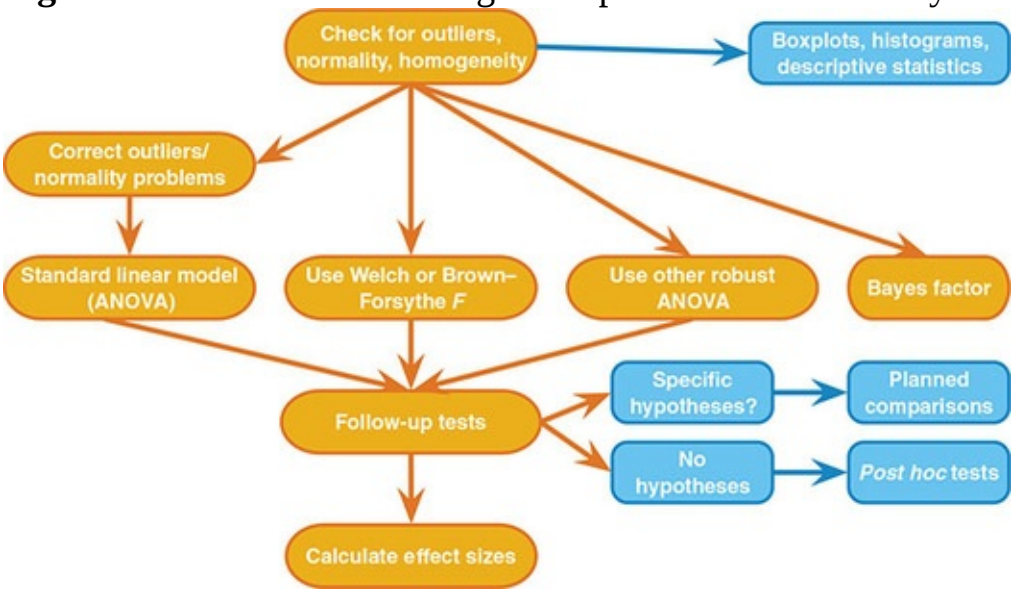
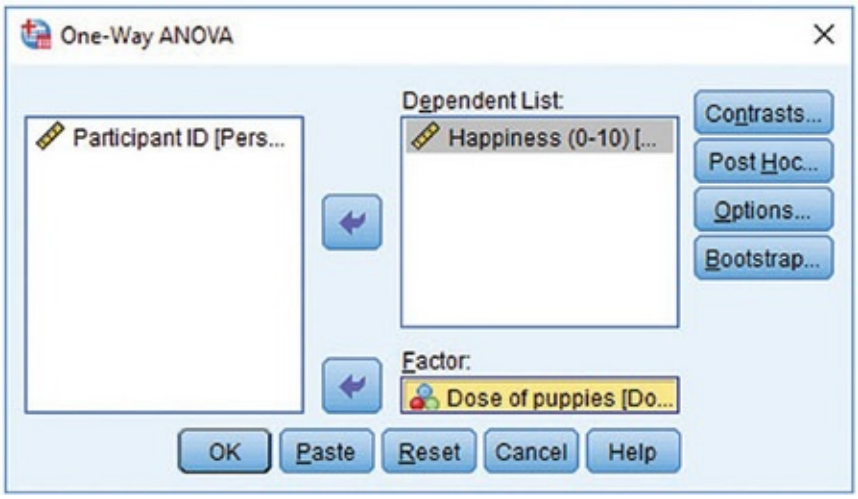


Figure 12.13 Main dialog box for one-way ANOVA



One thing that I dislike about SPSS Statistics is that in various procedures, such as this one, you are encouraged to carry out multiple tests (in this case by allowing you to specify several outcome variables at once). This is not a good thing because you lose control of the Type I error rate. If you had measured several dependent variables (e.g., you had measured not just happiness but other indicators of stress such as cortisol levels, non-verbal behaviour, and heart rate) it would be preferable to analyse these data using MANOVA (Chapter 17) rather than treating each outcome measure separately.

12.6.2 Planned contrasts using SPSS 

Click Reference priors to access a dialog box for specifying the contrast described in Section 12.4. The dialog box (Figure 12.14) has two sections. The first section is for specifying trend analyses. If you want to test for trends in the data then select Contrasts... . Once this option is active you can select the degree of polynomial you would like from the drop-down list (Polynomial). The puppy therapy data have only three groups and so the highest degree of trend there can be is quadratic (see Section 12.4.3). It is important from the point of view of trend analysis that we have coded the grouping variable in a meaningful order. We expect happiness to be smallest in the control group, to increase in the 15-minute group and then to increase again in the 30-minute group, so to detect a meaningful trend, these groups must be coded in ascending order. We have done this by coding the control group with the lowest value 1, the 15-minute group with the middle value 2 and the 30-minute group with the highest value of 3. If we coded the groups differently, this would influence both whether a trend is detected and, if one is detected, whether it has a meaningful interpretation. So, for the puppy therapy data select Polynomial and Quadratic . SPSS Statistics will test for the trend requested and all lower-order trends, so with *Quadratic* selected we'll get tests both for a linear and a quadratic trend.

The lower part of the dialog box in Figure 12.14 is for specifying weights for the planned contrasts that you have decided to do. We went through the process of generating weights for the contrasts we want in Section 12.4.2. The weights for contrast 1 were −2 (control group), +1 (15-minute group) and +1 (30-minute group). We will specify this contrast first. It is important to make sure that you enter the correct weight for each group; the first weight that you enter is the weight for the *first* group (i.e., the group coded with the lowest value in the data editor). For the puppy therapy data, the group coded with the lowest value was the control group (which had a code of 1) so we enter the weighting for this group first. Click in the box labelled *Coefficients*, type '−2' and click Add . Next, we input the weight for the second group, which for the puppy therapy data is the 15-minute group (because this group was coded in the data editor with the second-highest value). Click in the box labelled Coefficients, type '1' and click Add . Finally, we input the weight for the last group,

which for the puppy therapy data is the 30-minute group (because this group was coded with the highest value in the data editor). Click in the box labelled *Coefficients*, type '1' and click 

. The dialog box should now look like Figure 12.15 (left).

Figure 12.14 Dialog box for conducting planned contrasts

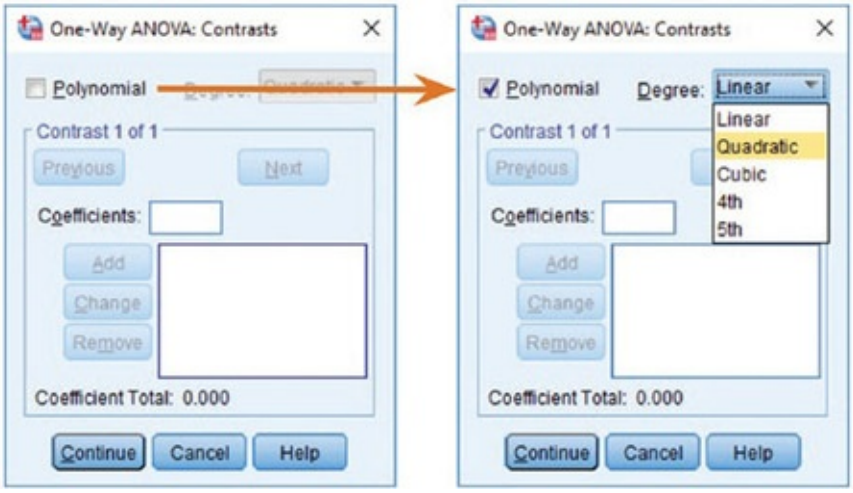
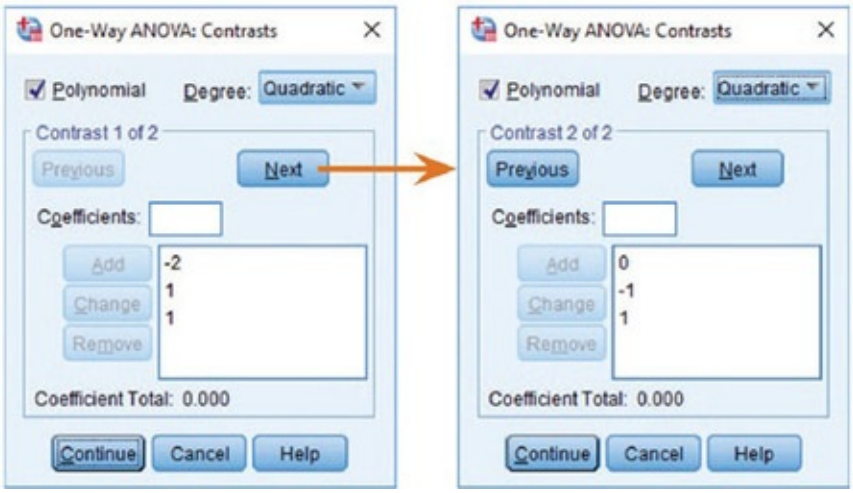
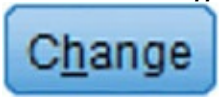


Figure 12.15 *Contrasts* dialog box completed for the two contrasts of the puppy therapy data



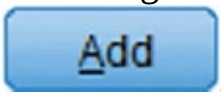
Once the weights are assigned you can change or remove any one of them by selecting the weight that you want to change. The weight will appear in the box labelled *Coefficients* where you can type a new weight and then click 

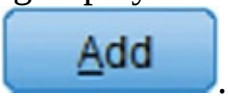
. Alternatively, you can click a weight and remove it completely by selecting 

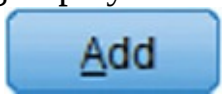
. Underneath the weights the sum of weights is displayed, which, as we saw in Section 12.4.2, should equal zero. If the

Coefficient Total is anything other than zero you should go back and check that the contrasts you have planned make sense and that you have followed the appropriate rules for assigning weights.

Once you have specified the first contrast, click 

. The weights that you have just entered will disappear and the dialog box will now read *Contrast 2 of 2*. We know from Section 12.4.2 that the weights for contrast 2 were: 0 (control group), −1 (15-minute group) and +1 (30-minute group). We specify this contrast as before. Remembering that the first weight we enter will be for the control group, we enter the value 0 as the first weight. Click in the box labelled *Coefficients*, then type '0' and click 

. Next, we input the weight for the 15-minute group by clicking in the box labelled *Coefficients*, typing '–1' and clicking 

. Finally, we input the weight for the 30-minute group by clicking in the box labelled *Coefficients*, typing '+1' and clicking 

. The dialog box should now look like Figure 12.15 (right). Notice that the weights add up to zero as they did for contrast 1. You must remember to input zero weights for any groups that are not in the contrast.

When the contrasts have been specified, click 

to return to the main dialog box.

12.6.3 Post hoc tests in SPSS 

In theory, if we have done planned contrasts we shouldn't need to do *post hoc* tests (because we have already tested the hypotheses of interest). Likewise, if we choose to conduct *post hoc* tests then we should not need to do planned contrasts (because we have no hypotheses to test). However, for the sake of space we will conduct some *post hoc* tests on the puppy therapy data as well as the contrasts we have just specified. Click 

in the main dialog box to access the dialog box in Figure 12.16.

In Section 12.5.3, I recommended various *post hoc* procedures for particular situations. For the puppy data there are equal sample sizes and so we need not use Gabriel's test. We should use Tukey's test and REGWQ and check the findings with the Games–Howell procedure. We have a specific hypothesis that

both the 30-and 15-minute groups should differ from the control group and so we could use Dunnett's test to examine these hypotheses. Once you have selected Dunnett, change the control category from Last to First so that the no puppies category (which, remember, we coded with the lowest value and so is the first group) is used as the comparison group. You can choose whether to conduct a two-tailed (2-sided) or a one-tailed test. If you choose a one-tailed test (which I advised against in Section 2.9.5) then you need to predict whether you believe that the mean of the first group (i.e., no puppies) will be less than each experimental group (< Control) or greater (< Control). These are all the *post hoc* tests that we need to specify (see Figure 12.16). Click Continue to return to the main dialog box.

Figure 12.16 Dialog box for specifying *post hoc* tests

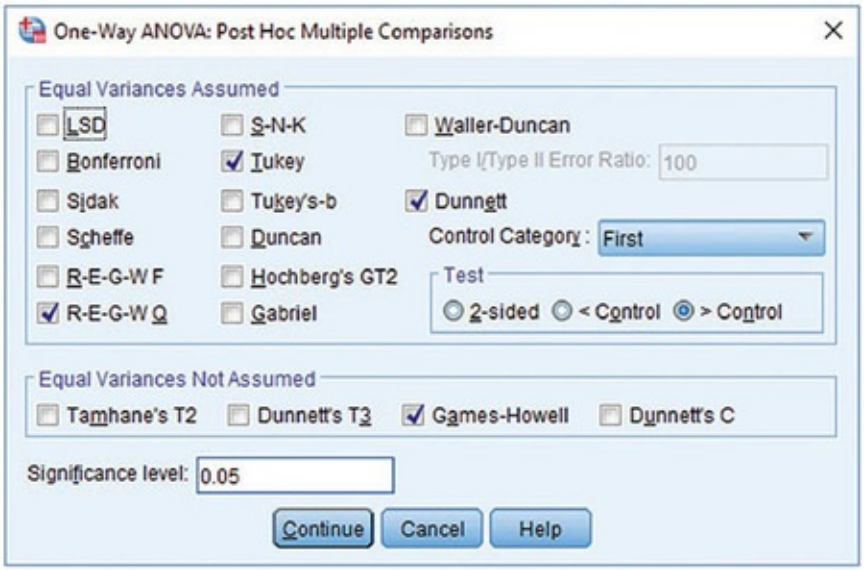
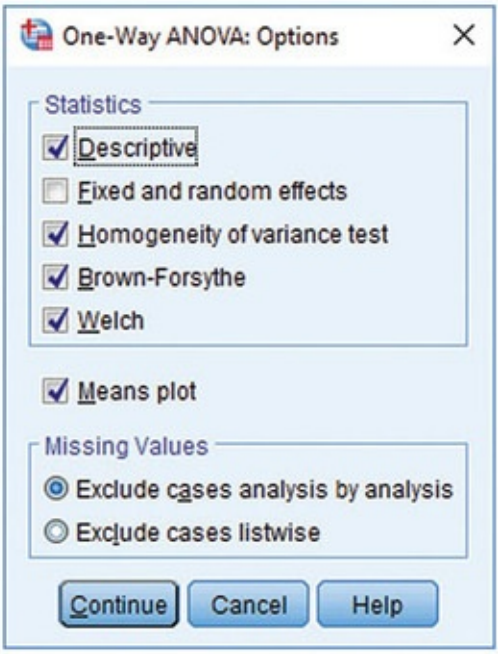
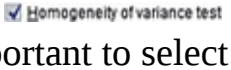


Figure 12.17 Options for one-way ANOVA

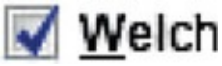


12.6.4 Options

Click 

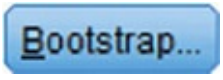
to access the dialog box in Figure 12.17. You can ask for descriptive statistics, which will produce a table of the means, standard deviations, standard errors, ranges and confidence intervals within each group. This information will help us to interpret the results. Select 

if you want Levene's test (Section 6.11.2), although it's more important to select 

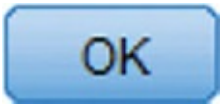
or 

so that you can interpret these if you're concerned about having unequal variances or, better still, use them by default. There is also an option to have a *Means plot* which produces a line graph of the group means. The resulting graph is a leprous tramp compared to what you can create using the Chart Builder and it's best to graph your data *before* the analysis, not during it. Finally, the options let us specify whether we want to exclude cases on a listwise basis or on a per analysis basis (see SPSS Tip 6.1 for an explanation). This option is useful only if you are fitting models to several outcome variables simultaneously (which hopefully you're not – ever).

12.6.5 Bootstrapping

Also in the main dialog box is the alluring 

button. We know that

bootstrapping is a good way to overcome bias, and this button glistens and tempts us with the promise of untold riches, like a diamond in a bull's rectum. However, if you use bootstrapping it'll be as disappointing as if you reached for that diamond only to discover that it's a piece of glass. You might, not unreasonably, think that if you select bootstrapping you'd get a nice bootstrap of the *F*-statistic. You won't. You will get bootstrap confidence intervals around the means (if you ask for descriptive statistics), contrasts and *post hoc* tests. All of which are useful, but the main test won't be bootstrapped. For this example, we have a very small data set so bootstrapping is going to go haywire, so we won't select it. Click 

in the main dialog box to run the analysis.

12.7 Output from one-way independent ANOVA



You should find that the output looks the same as what follows. If not, we should panic because one of us has done it wrong – hopefully not me, or a lot of trees have died for nothing. Figure 12.18 shows a line chart with error bars from the self-test earlier in the chapter (I have edited my graph; see if you can make yours look like mine). All the error bars overlap a fair bit, indicating that, at face value, there might not be between-group differences (see Section 2.9.9). The line that joins the means seems to indicate a linear trend in that, as the dose of puppy therapy increases, so does the mean level of happiness.

Oditi's Lantern One-way ANOVA

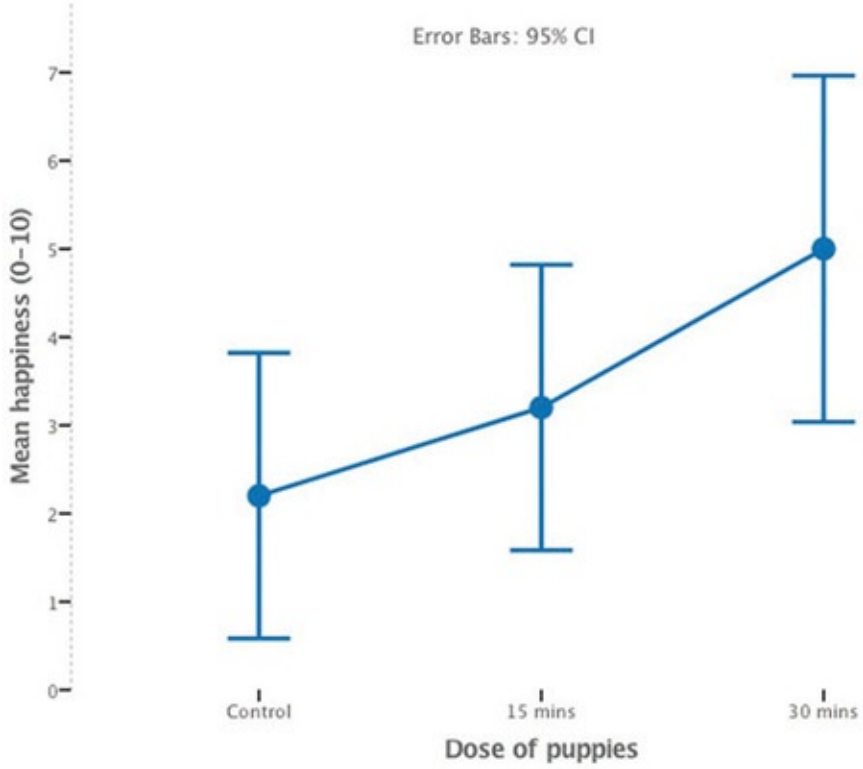


'I, Oditi, have made great progress in unearthing the hidden truth behind the numbers. This morning, one of my loyal followers reported to me that, based on an ANOVA he'd done, all dogs are controlled by cats who hide small remote controls up their rectums and manipulate them with their tongues. Everytime you see a cat

‘cleaning’ itself, there will be a dog nearby chasing its tail. Listen carefully and you can hear the cat laughing to itself. Be warned, cats are merely piloting the technology, and soon they will control us too, turning us into heated chairs and food vendors. We must find out more. Stare into my lantern so that you too can use ANOVA.’



Figure 12.18 Error bar (95% CI) chart of the puppy therapy data



12.7.1 Output for the main analysis

Output 12.3 shows the table of descriptive statistics for the puppy data. The means and standard deviations correspond to those shown in Table 12.1, which is reassuring. We are also given the standard error and confidence intervals for the mean. If this sample is one of the 95% that have confidence intervals containing the true value then the true value of the mean is between 0.58 and 3.82 for the control group. We will refer back to this table as we wade through the output.

Output 12.4 shows Levene's test (see Section 6.11.2) for those of you hell-bent on using it. For these data the variances are very similar (hence Levene's test is very non-significant, with *p* close to 1); in fact, if you look at Output 12.3 you'll see that the variances of the control and 15-minute groups are identical.

Output 12.5 shows the main ANOVA summary table. The table is divided into between-group effects (effects due to the model – the experimental effect) and within-group effects (this is the unsystematic variation in the data). The between-group effect is further broken down into a linear and quadratic component as requested in Figure 12.14. The between-group effect labelled *Combined* is the overall experimental effect, or, put another way, the improvement in the prediction of happiness scores resulting from using the group means. We are told the model sums of squares ($SS_M = 20.13$), which corresponds to the value calculated in Section 12.2.3. The degrees of freedom are 2 and the mean squares for the model corresponds to the value calculated in Section 12.2.5 (10.067).

The row labelled *Within Groups* gives details of the unsystematic variation within the data (the variation due to natural individual differences in happiness and different reactions to puppy therapy). The table tells us that the residual sum of squares (SS_R) is 23.60, as calculated in Section 12.2.4. The average amount of unsystematic variation, the mean squares (MS_R), is 1.967, as calculated in Section 12.2.5.

The test of whether the group means are the same is represented by the *F*-statistic for the combined between-group effect. This tells us whether predicting happiness from group means significantly improves the fit of the model. The value of *F* is 5.12, which we calculated in Section 12.2.6. The final column labelled *Sig*. tells us the probability of getting an *F* at least this big if there wasn't a difference between means in the population (see also SPSS Tip 12.1). In this case, there is a probability of 0.025 that an *F*-statistic of at least this size would occur if in reality the effect was zero. Assuming we set a cut-off point of 0.05 as a criterion for statistical significance before collecting data, most scientists would take the fact that 0.025 is less than the criterion of 0.05 as support for a significant effect of puppy therapy. At this stage we still do not know exactly what the effect of puppy therapy was (we don't know which groups differed). One interesting point is that we obtained a significant experimental effect yet the error bar plot suggested that no significant difference would be found. This contradiction illustrates how the error bar chart can act only as a rough guide to the data.

Output 12.3

Descriptives

Happiness (0–10)

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Control	5	2.20	1.304	.583	.58	3.82	1	4
15 mins	5	3.20	1.304	.583	1.58	4.82	2	5
30 mins	5	5.00	1.581	.707	3.04	6.96	3	7
Total	15	3.47	1.767	.456	2.49	4.45	1	7

Output 12.4

Test of Homogeneity of Variances

Happiness (0–10)

Levene Statistic	df1	df2	Sig.
.092	2	12	.913

Output 12.5

ANOVA

Happiness (0–10)

			Sum of Squares	df	Mean Square	F	Sig.
Between Groups	(Combined)		20.133	2	10.067	5.119	.025
	Linear Term	Contrast	19.600	1	19.600	9.966	.008
		Deviation	.533	1	.533	.271	.612
	Quadratic Term	Contrast	.533	1	.533	.271	.612
Within Groups			23.600	12	1.967		
Total			43.733	14			

Knowing that the overall effect of puppy therapy was significant, we can look at the trend analysis. First, let's look at the linear component. This contrast tests whether the means increase across groups in a linear way. For the linear trend the *F*-statistic is 9.97 and this value is significant at $p = 0.008$. Therefore, we can say that as the dose of puppy therapy increased from nothing to 15 minutes to 30 minutes, happiness increased proportionately. The quadratic trend tests whether the pattern of means is curvilinear (i.e., is represented by a curve that has one bend). The error bar graph of the data suggests that the means cannot be represented by a curve and the results for the quadratic trend bear this out. The *F*-statistic for the quadratic trend is non-significant (in fact, the value of *F* is less than 1, which immediately indicates that this contrast will not be significant).

Output 12.6

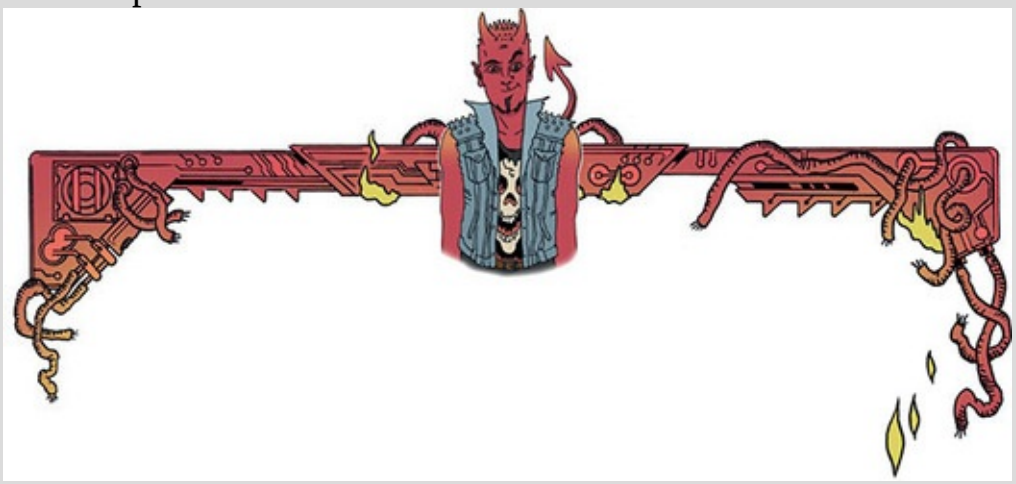
Robust Tests of Equality of Means

Happiness (0-10)

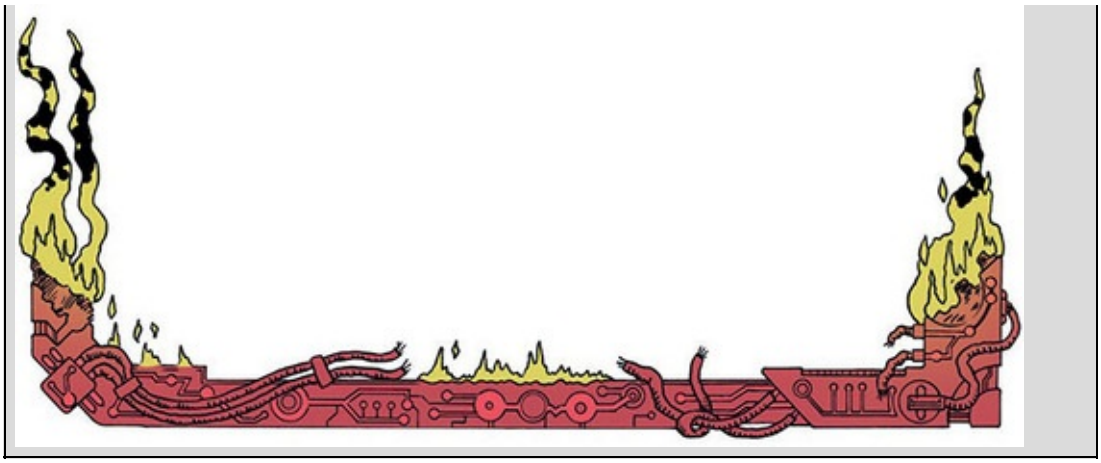
	Statistic[a]	df1	df2	Sig.
Welch	4.320	2	7.943	.054
Brown-Forsythe	5.119	2	11.574	.026

a. Asymptotically F distributed.

SPSS Tip 12.1 One and two-tailed tests in ANOVA 



A question I get asked a lot is 'is the significance of the ANOVA one-or two-tailed, and if it's two-tailed can I divide by 2 to get the one-tailed value?' Obviously I told you earlier not to do that sort of thing (see Section 2.9.5), but it's particularly daft in this context because to do a one-tailed test you have to be making a directional hypothesis (e.g., the mean for cats is greater than for dogs). When comparing more than two means (as you do with ANOVA) you can't make a directional hypothesis: you can predict only that the means will differ somehow. Therefore, it's invalid to halve the significance value of an *F*.



Output 12.6 shows the Welch and Brown–Forsythe *F*-statistics. If you're interested in how these values are calculated then look at Jane Superbrain Box 12.2, but to be honest it's just bloody confusing; you're much better off just looking at the values in Output 12.6 and trusting that they do what they're supposed to do (note that the error degrees of freedom have been adjusted and you should remember this when you report the values). Based on whether the observed *p* is less than 0.05, the Welch *F* yields a non-significant result ($p = 0.054$) whereas Brown–Forsythe is significant ($p = 0.026$). This is confusing, but only if you like to imbue the 0.05 threshold with magical powers and engage in black-and-white thinking of the sort that people who use *p*-values so often do.

12.7.2 Output for planned contrasts

In Section 12.6.2 we defined two planned contrasts: one to test whether the control group was different from the two groups which received puppy therapy, and one to see whether the two doses of puppy therapy made a difference to happiness. Output 12.7 shows the results of these. The first table displays the contrast coefficients that we entered in Section 12.6.2, and it is worth checking this table to make sure that the contrasts compare what they are supposed to. The second table gives the statistics for each contrast in their raw form, but also corrected for unequal variances. Some people use Levene's test as a decision rule: if it is significant, read the part of the table labelled *Does not assume equal variances*, and if it is not, use the part of the table labelled *Assume equal variances*. However, given the issues with Levene's test, it's probably more sensible to just routinely look at the corrected values. The table tells us the value of the contrast itself, which is the weighted sum of the group means. This value is obtained by taking each group mean, multiplying it by the weight for the contrast of interest, and then adding these values together.[3] The table also gives

the standard error of each contrast and a *t*-statistic. The *t*-statistic is derived by dividing the contrast value by the standard error ($t = 3.8/1.5362 = 2.47$) and then, in the bottom two rows of the table, adjusted for the degree of heterogeneity. The significance value of the contrast is given in the final column, and this value is two-tailed. For contrast 1, we can say that taking puppy therapy significantly increased happiness compared to the control group ($p = 0.031$), but contrast 2 tells us that 30 minutes of puppy therapy did not significantly affect happiness compared to 15 minutes ($p = 0.086$). Contrast 2 is almost significant, which again demonstrates how the NHST process can lead you to all-or-nothing thinking (Section 3.2.2).

[3] For the first contrast this value is

$$\sum \bar{X}W = (2.2 \times -2) + (3.2 \times 1) + (5 \times 1) = 3.8$$

.

Output 12.7

Contrast Coefficients

Contrast	Dose of puppies: Control	15 mins	30 mins
1	-2	1	1
2	0	-1	1

Contrast Tests

		Contrast	Value of Contrast	Std. Error	t	df	Sig. (2-tailed)
Happiness (0-10)	Assume equal variances	1	3.80	1.536	2.474	12	.029
		2	1.80	.887	2.029	12	.065
	Does not assume equal variances	1	3.80	1.483	2.562	8.740	.031
		2	1.80	.917	1.964	7.720	.086



12.7.3 Output for post hoc tests

If we had no specific hypotheses about the effect of puppy therapy on happiness then we would have selected *post hoc* tests to compare all group means to each other. Even though we wouldn't normally do contrasts and *post hoc* tests, to save space we did (Section 12.6.3) and Output 12.8 shows tables containing Tukey's test (known as Tukey's HSD),[4] the Games–Howell procedure and Dunnett's test. If we look at Tukey's test first (because we have no reason to doubt that the population variances are unequal) we see that for each pair of groups the difference between group means is displayed, the standard error of that difference, the significance level of that difference and a 95% confidence interval. The first row of Output 12.8 compares the control group to the 15-minute group and reveals a non-significant difference (*Sig.* of 0.516 is greater

than 0.05), and the second row compares the control group to the 30-minute group where there is a significant difference (*Sig.* of 0.021 is less than 0.05). It might seem odd that the planned contrast showed that any dose of puppy therapy produced a significant increase in happiness, yet the *post hoc* tests indicate that a 15-minute does not.

4 The HSD stands for 'honestly significant difference', which has a slightly dodgy ring to it, if you ask me.



Can you explain the contradiction between the planned contrasts and *post hoc* tests?

In Section 12.4.2, I explained that the first planned contrast would compare the experimental groups to the control group. Specifically, it would compare the average of the two group means of the experimental groups ((3.2 + 5.0)/2 = 4.1) to the mean of the control group (2.2). So, it was assessing whether the difference between these values (4.1 − 2.2 = 1.9) was significant. In the *post hoc* tests, when the 15-minute group is compared to the control it is testing whether the difference between the means of these two groups is significant. The difference in this case is only 1, compared to a difference of 1.9 for the planned contrast. This explanation illustrates how it is possible to have apparently contradictory results from planned contrasts and *post hoc* comparisons. More important, it illustrates the need to think carefully about what our planned contrasts test.

Output 12.8

Multiple Comparisons

Dependent Variable: Happiness (0-10)

	(I) Dose of puppies	(J) Dose of puppies	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
Tukey HSD	Control	15 mins	-1.000	.887	.516	-3.37	1.37
		30 mins	-2.800*	.887	.021	-5.17	-.43
	15 mins	Control	1.000	.887	.516	-1.37	3.37
		30 mins	-1.800	.887	.147	-4.17	.57
	30 mins	Control	2.800*	.887	.021	.43	5.17
		15 mins	1.800	.887	.147	-.57	4.17
Games-Howell	Control	15 mins	-1.000	.825	.479	-3.36	1.36
		30 mins	-2.800*	.917	.039	-5.44	-.16
	15 mins	Control	1.000	.825	.479	-1.36	3.36
		30 mins	-1.800	.917	.185	-4.44	.84
	30 mins	Control	2.800*	.917	.039	.16	5.44
		15 mins	1.800	.917	.185	-.84	4.44
Dunnett t (>control)[b]	15 mins	Control	1.000	.887	.227	-.87	
	30 mins	Control	2.800*	.887	.008	.93	

*. The mean difference is significant at the 0.05 level.

b. Dunnett t-tests treat one group as a control, and compare all other groups against it.

The third and fourth rows of Output 12.8 compare the 15-minute group to both

the control group and the 30-minute group. The test involving the 15-minute and 30-minute groups shows that these group means did not differ (because the *p* of 0.147 is greater than our alpha of 0.05). Rows 5 and 6 repeat comparisons already discussed.

The second block of the table describes the Games–Howell test, and a quick inspection reveals the same pattern of results: the only groups that differed significantly were the 30-minute and control groups. These results give us confidence in our conclusions from Tukey's test because even if the population variances are not equal (which seems unlikely given that the sample variances are very similar), then the profile of results holds true. Finally, Dunnett's test is described, and you'll hopefully remember that we asked the computer to compare both experimental groups against the control using a one-tailed hypothesis that the mean of the control group would be smaller than both experimental groups. Even as a one-tailed hypothesis, levels of happiness in the 15-minute group are equivalent to the control group. However, the 30-minute group has a significantly higher happiness than the control group.

Output 12.9 shows the results of Tukey's test and the REGWQ test. These tests display subsets of groups that have the same means with associated *p*-values for each subset. Tukey's test has created two subsets of groups with statistically similar means. The first subset contains the control and 15-minute groups (indicating that these two groups have similar means, $p = 0.516$), whereas the second subset contains the 30-and 15-minute groups (which also have similar means, $p = 0.147$). The REGWQ test agrees with the first subset, suggesting that the control and 15-minute groups have similar means ($p = 0.282$) and the second that the 15-and 30-minute groups have similar means ($p = 0.065$).

Output 12.9

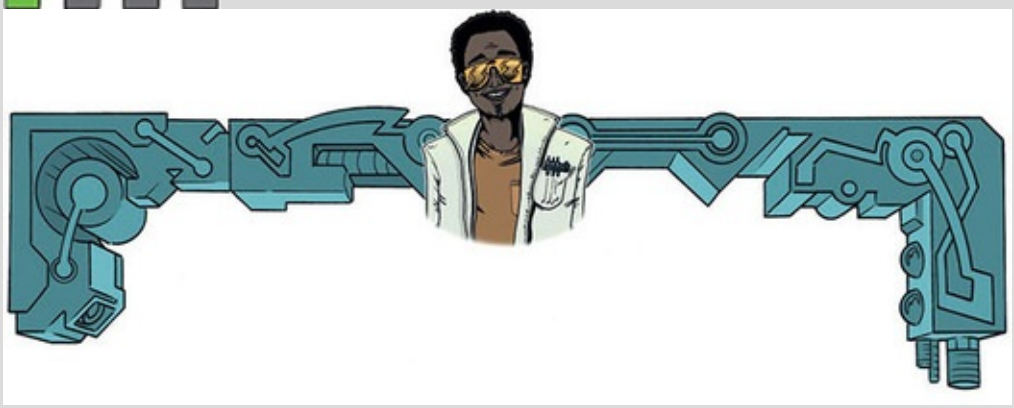
Happiness (0–10)

			Subset for alpha = 0.05	
	Dose of puppies	N	1	2
Tukey HSD[a]	Control	5	2.20	
	15 mins	5	3.20	3.20
	30 mins	5		5.00
	Sig.		.516	.147
Ryan-Einot-Gabriel-Welsch Range	Control	5	2.20	
	15 mins	5	3.20	3.20
	30 mins	5		5.00
	Sig.		.282	.065

Means for groups in homogeneous subsets are displayed.

a. Uses Harmonic Mean Sample Size = 5.000.

Labcoat Leni's Real Research 12.1Scraping the barrel?



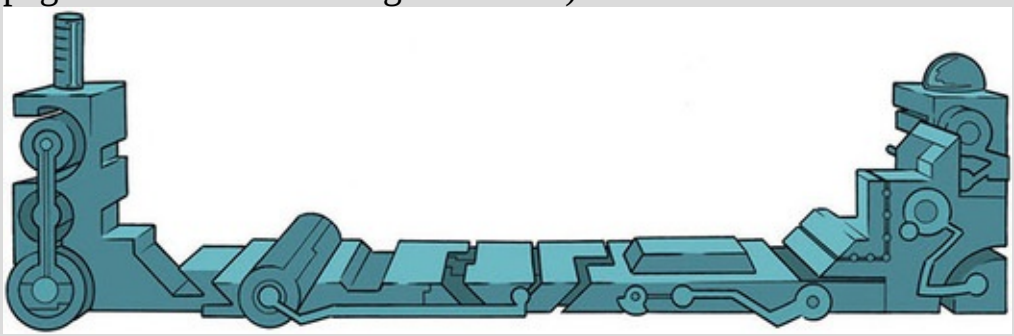
Gallup, G. G. J., *et al.* (2003). *Evolution and Human Behavior, 24*, 277–289.

Evolution has endowed us with many beautiful things (cats, dolphins, the Great Barrier Reef, etc.) all selected to fit their ecological niche. Given evolution's seemingly limitless capacity to produce beauty, it's something of a wonder how it managed to produce such a monstrosity as the human penis. One theory is sperm competition: the human penis has unusually large glans (the 'bell-end') compared to other primates, and this may have evolved so that the penis can displace seminal fluid from other males by 'scooping it out' during intercourse. Armed with various female masturbatory devices from Hollywood Exotic Novelties, an artificial vagina from California Exotic Novelties, and some water and cornstarch to make fake sperm, Gallup *et al.* (2003) put this theory to the test. They loaded the artificial vagina with 2.6 ml of fake sperm and inserted one of three female sex toys into it before withdrawing it: a control phallus that had no coronal ridge (i.e., no bell-end), a phallus with a minimal coronal ridge (small bell-end) and a phallus with a coronal ridge. They measured sperm displacement as a percentage using the following expression (included here because it is more interesting than all of the other equations in this book):

$$\frac{\text{weight of vagina with semen} - \text{weight of vagina following insertion and removal of phallus}}{\text{weight of vagina with semen} - \text{weight of empty vagina}} \times 100$$

100% means that all the sperm was displaced, and 0% means that none of the sperm was displaced. If the human penis evolved as a sperm displacement device then Gallup *et al.* predicted: (1) that having a bell-end would displace more sperm than not; and (2) that the phallus with the larger coronal ridge would displace more sperm

than the phallus with the minimal coronal ridge. The conditions are ordered (no ridge, minimal ridge, normal ridge), so we might also predict a linear trend. The data are in the file **Gallup et al.sav**. Draw an error bar graph of the means of the three conditions. Fit a model with planned contrasts to test the two hypotheses above. What did Gallup *et al.* find? Answers are on the companion website (or look at pages 280–281 in the original article).



Tukey's LSD uses the **harmonic mean** sample size, which is a weighted version of the mean that takes account of the relationship between variance and sample size. Although you don't need to know the intricacies of the harmonic mean, it is useful that the harmonic sample size is used because it reduces bias that might be introduced through having unequal sample sizes. However, as we have seen, these tests are still biased when sample sizes are unequal.

12.8 Robust comparisons of several means

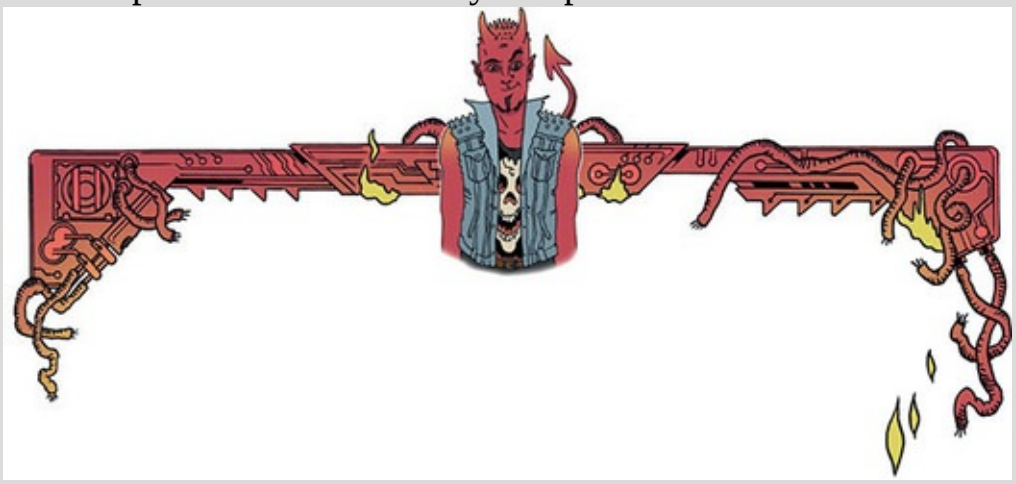
It's possible to run a robust test of several means using R. We need the *Essentials for R* plugin and *WRS2* package installed (Section 4.13). The companion website contains a syntax file (**t1waybt.sps**) for running a robust variant of one-way independent ANOVA (*t1waybt*) with *post hoc* tests (*mcppb20*) described by Wilcox (2017). These tests assume neither normality nor homogeneity of variance, so you can ignore these assumptions and plough ahead. The syntax in the file is as follows:

```
BEGIN PROGRAM R.
library(WRS2)
mySPSSdata = spssdata.GetDataFromSPSS(factorMode = "labels")
t1waybt(Happiness~Dose, data = mySPSSdata, tr = 0.2, nboot = 1000)
mcppb20(Happiness~Dose, data = mySPSSdata, tr = 0.2, nboot = 1000)
END PROGRAM.
```

Select and run these six lines of syntax (see SPSS Tip 10.3) and you'll find text output in the viewer (Output 12.10) that tells us that there was not a significant

difference (because the *p*-value is greater than 0.05) in happiness scores across the puppy therapy groups, $F_t = 3$, $p = 0.089$. The *post hoc* tests (which technically we should ignore because the overall test wasn't significant) show no significant difference between the control and 15-minute groups ($p = 0.381$), or between the 15-and 30-minute groups ($p = 0.081$), but there is a significant difference (that we should ignore because the overall test wasn't significant) between the control and 30-minute groups ($p = 0.010$).

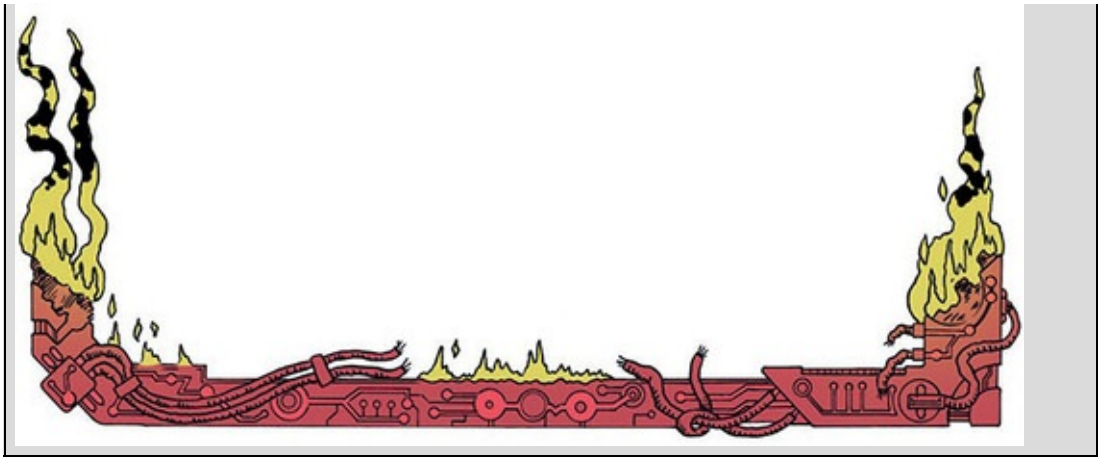
SPSS Tip 12.2 Robust one-way independent ANOVA 



To break down the syntax, we're using two functions (*t1waybt* and *mcppb20*) that both have the same form:

```
t1waybt(outcome~predictor, data = mySPSSdata, tr = 0.2, nboot = 1000)
mcppb20(outcome~predictor, data = mySPSSdata, tr = 0.2, nboot = 1000)
```

In our example we use our variable names to replace outcome with **Happiness** and predictor with **Dose**. The *tr* relates to the amount of trimming to the data (by default 0.2, or 20%, but you could change it to a different proportion). The *nboot* refers to the number of bootstrap samples, which I have set to 1000, which should be fine, but feel free to increase it.



Output 12.10

```
Effective number of bootstrap samples was 671.

Test statistic: 3
p-value: 0.08942
Variance explained 0.623
Effect size 0.789

mcppb20(formula = Happiness ~ Dose, data = mySPSSdata, tr = 0.2,
    nboot = 1000)

                    psihat ci.lower ci.upper p-value
Control vs. 15 mins     -1 -3.33333  1.33333   0.381
Control vs. 30 mins     -3 -5.00000 -0.33333   0.010
15 mins vs. 30 mins     -2 -4.00000  0.66667   0.081
```

Cramming Sam's Tips One-way independent ANOVA



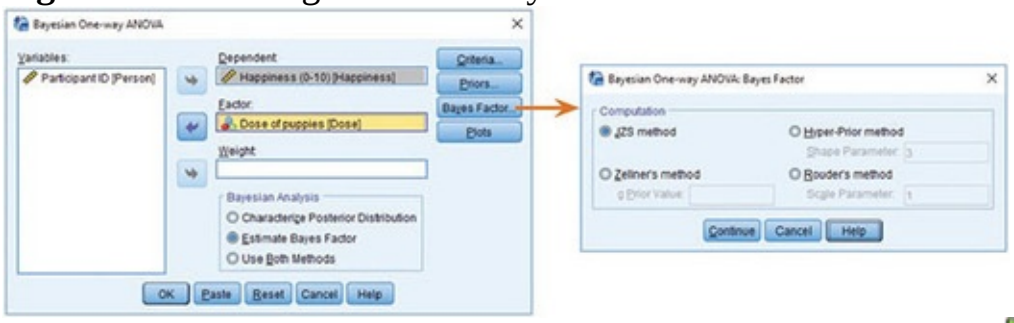
- One-way independent ANOVA compares several means, when those means have come from different groups of people; for example, if you have several experimental conditions and have used different participants in each condition. It is a special case of the linear model.
- When you have generated specific hypotheses before the experiment use *planned contrasts*, but if you don't have specific hypotheses use *post hoc* tests.
- There are lots of different *post hoc* tests: when you have equal sample sizes and homogeneity of variance is met, use REGWQ or Tukey's HSD. If sample sizes are slightly different then use

Gabriel's procedure, but if sample sizes are very different use Hochberg's GT2. If there is any doubt about homogeneity of variance use the Games–Howell procedure.

- You can test for homogeneity of variance using Levene's test, but consider using a *robust* test in all situations (the Welch or Browne–Forsythe *F*) or Wilcox's *t1way()* function.
- Locate the *p*-value (usually in a column labelled *Sig.*). If the value is less than 0.05 then scientists typically interpret this as the group means being significantly different.
- For contrasts and *post hoc* tests, again look to the columns labelled *Sig.* to discover if your comparisons are significant (they will be if the significance value is less than 0.05).



Figure 12.19 Dialog boxes for Bayesian ANOVA



12.9 Bayesian comparison of several means

Because we've been using a linear model to compare means, we can compute a Bayes factor (Section 3.8) just as we did in Section 9.13. You can instead access the dialog box in Figure 12.19 by selecting *Analyze* *Bayesian Statistics* *One-way ANOVA*. First, specify the model by dragging the outcome (**Happiness**) to the box labelled *Dependent* and the predictor (**Dose**) to the box labelled *Factor*. If we use estimation on a model with a categorical predictor we

will get Bayesian estimates for the group means (*not* the *b*s from the model), which are not particularly interesting. Therefore, I have selected to compute only Bayes factors (Estimate Bayes Factor). Click Bayes Factor... and select the default JZS method (Jeffreys, 1961; Zellner & Siow, 1980). In the main dialog box click OK to fit the model.

The Bayes factor here is essentially the same as the one we computed for the linear model (Section 9.13): it compares the full model (predicting **Happiness** from **Dose** and the intercept) to the null model (predicting **Happiness** from only the intercept). It, therefore, quantifies the overall effect of **Dose**. This is reflected by the fact that the Bayes factor appears as part of the table displaying the overall fit of the model (the *F*-statistic). Output 12.11 shows that the Bayes factor for including **Dose** as a predictor (compared to not including it) is 2.158. This means that the data are 2.158 times more likely under the alternative hypothesis (dose of puppy therapy has an effect) than under the null (dose of puppy therapy has no effect). This value is not strong evidence, but nevertheless suggests we should shift our belief about puppy therapy towards it being effective by a factor of about 2.

Output 12.11

ANOVA

Happiness (0-10)	Sum of Squares	df	Mean Square	F	Sig.	Bayes Factor[a]
Between Groups	20.133	2	10.067	5.119	.025	2.158
Within Groups	23.600	12	1.967			
Total	43.733	14				

a. Bayes factor: JZS

12.10 Calculating the effect size



SPSS Statistics doesn't provide an effect size, but we saw in equation (9.10) that we can compute R^2 for a linear model:

$$R^2 = \frac{SS_M}{SS_T} \quad (12.30)$$

We know these values from the output, so we can calculate R^2 using the between-group effect (SS_M), and the total amount of variance in the data (SS_T) – although for some bizarre reason it's usually called **eta squared, η^2**. It is then a simple matter to take the square root of this value to give us the effect size *r* of 0.68:

$$r^2 = \eta^2 = \frac{SS_M}{SS_T} = \frac{20.13}{43.73} = 0.46 \quad (12.31)$$

$$r = \sqrt{0.46} = 0.68$$

Therefore, the effect of puppy therapy on happiness is a substantive finding. However, this measure of effect size is slightly biased because it is based purely on sums of squares from the sample and no adjustment is made for the fact that we're trying to estimate the effect size in the population. Therefore, we often use a slightly more complex measure called **omega squared** (ω^2). This effect size estimate is still based on the sums of squares that we've met in this chapter: it uses the variance explained by the model, and the average error variance:

$$\omega^2 = \frac{SS_M - (df_M)MS_R}{SS_T + MS_R} \quad (12.32)$$

The df_M in the equation is the degrees of freedom for the effect, which you can get from the output. In this example we'd get:

$$\omega^2 = \frac{20.13 - (2)1.97}{43.73 + 1.97} = \frac{16.19}{45.70} = 0.35 \quad (12.33)$$

$$\omega = 0.60$$

This adjustment has led to a slightly lower estimate to using r, and in general ω is a more accurate measure. Think of ω as you would r (because it's basically an unbiased estimate of r). People normally report ω^2, and it has been suggested that values of 0.01, 0.06 and 0.14 represent small, medium and large effects, respectively (Kirk, 1996). Remember, though, that these are rough guidelines and that effect sizes need to be interpreted within the context of the research literature.

Most of the time it isn't that interesting to have effect sizes for the overall ANOVA because it's testing a general hypothesis. Instead, we really want effect sizes for the contrasts (because these compare only two things, so the effect size is easier to interpret). Planned contrasts are tested with the t-statistic and, therefore, we can use the same equation as in Section 10.9.5:

$$r_{Contrast} = \sqrt{\frac{t^2}{t^2 + df}} \quad (12.34)$$

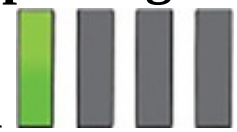
We know the value of t and the df from Output 10.7, and so we can compute r as follows:

$$r_{Contrast1} = \sqrt{\frac{2.474^2}{2.474^2 + 12}} = \sqrt{\frac{6.12}{18.12}} = 0.58 \quad (12.35)$$

As well as being statistically significant, this effect represents a substantive finding. For contrast 2 we get:

$$r_{Contrast2} = \sqrt{\frac{2.029^2}{2.029^2 + 12}} = \sqrt{\frac{4.12}{16.12}} = 0.51 \quad (12.36)$$

This too is a substantive finding.

12.11 Reporting results from one-way independent ANOVA 

We report the F-statistic and the degrees of freedom associated with it. Remember that F is the model mean squares divided by the residual mean squares for the residual, so the associated degrees of freedom are those for the effect of the model ($df_M = 2$) and those for the residuals of the model ($df_R = 12$). Also include an effect size estimate (have a go at calculating these to see whether you get the same values as me). Finally, report the exact p-value. Based on this advice we could report the overall effect as:

Notice that the degrees of freedom are in parentheses. I've advised you to always report Browne–Forsythe or Welch's F, so let's do that instead (note the adjusted residual degrees of freedom, and change in F and p):

The linear contrast can be reported in much the same way:

> ✓ There was a significant effect of puppy therapy on levels of happiness, $F(2, 12) = 5.12$, $p = 0.025$, $\omega = 0.60$.
>
> ✓ There was not a significant effect of puppy therapy on levels of happiness, $F(2, 7.94) = 4.32$, $p = 0.054$, $\omega = 0.60$.
>
> ✓ There was a significant effect of puppy therapy on levels of happiness, $F(2, 11.57) = 5.12$, $p = 0.026$, $\omega = 0.60$.
>
> ✓ There was a significant linear trend, $F(1, 12) = 9.97$, $p = 0.008$, $\omega = 0.62$, indicating that as the dose of puppy therapy increased, happiness increased proportionately.
>
> ✓ Planned contrasts revealed that having any dose of puppy therapy significantly increased happiness compared to having a control, $t(8.74) = 2.56$, $p = 0.031$, $r = 0.58$, but having 30 minutes did not significantly increase happiness compared to having 15 minutes, $t(7.72) = 1.96$, $p = 0.086$, $r = 0.51$.

The degrees of freedom have changed to reflect how F was calculated. We can do something similar for the planned contrasts:

12.12 Brian's attempt to woo Jane 

Brian had been astonished to see Jane in the library, and even more so when she dropped her guard enough to ask his name. He'd momentarily cursed his parents for calling him Brian, but Jane didn't seem to care. They'd even continued a bit

of an awkward conversation about music before Jane quickly excused herself and ran off. She'd looked pale, like she'd remembered something horrific. Brian wasn't used to seeing her like that: she was always so self-assured. He wanted to know what was up, but despite his attempt to follow her he lost sight of her as she darted across campus. The episode played on his mind: he didn't want her to feel bad. He had no way to contact her, though, other than by going to places where she hung out. He loitered on campus, but it was hopeless. He bumped into Alex. She seemed irked about his encounter with Jane. She said Jane was bad news. He pressed Alex for information, but she gave nothing away except that Jane often went to Blow Your Speakers, the record store in town, at the weekends. Brian knew it well: they had vinyl and coffee mornings on a Saturday. He'd always avoided it, fearing it would be full of pretentious hipsters. This Saturday he made an exception and sat among people in skinny trousers and strange hats, who sipped coffee, twiddled their moustaches and tried to choose more obscure cuts to play than the last person. He'd had enough and got up to leave as Jane walked in.

Jane was shocked to see Brian, but resisted the urge to flee. Brian smiled, rushed over and placed a reassuring hand on her shoulder. Jane flinched. Brian asked if she was alright and explained that he was worried he'd upset her. Jane was silent, unsure what to say, so Brian filled the silence with his statistics lecture. He needed to work on his social skills.

Figure 12.20 What Brian learnt from this chapter



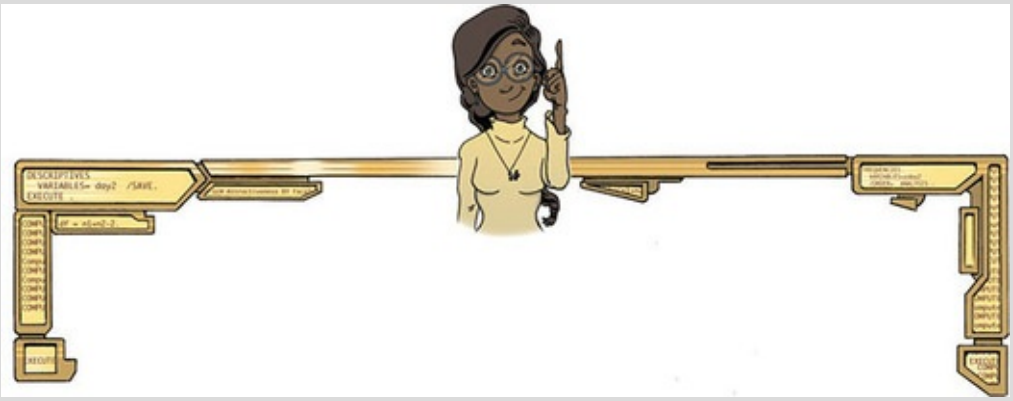
12.13 What next?

My life was changed by a letter popping through the letterbox one day saying that I could go to the local grammar school. When my parents told me, they were downbeat and not as celebratory as you might expect, but they knew how much I wanted to be with my friends. I had got used to my failure, and my initial reaction was to say that I wanted to go to the local school. I was unwavering in this view. Unwavering, that is, until my brother convinced me that being at the same school as him would be really cool. It's hard to measure how much I looked up to him, and still do, but the fact that I willingly subjected myself to a lifetime of social dysfunction just to be with him is a measure of sorts. As it turned out, being at school with him was not always cool – he was bullied for being a boffin (that's what they called nerds in the UK in the 1980s), and being the younger brother of a boffin made me a target. Luckily, unlike my brother, I was stupid and played football, which seemed to be good enough reasons for them to leave me alone. Most of the time.

12.14 Key terms that I've discovered

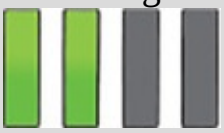
Analysis of variance (ANOVA)
Brown–Forsythe *F*
Cubic trend
Deviation contrast
Difference contrast (reverse Helmert contrast)
Eta squared, η^2
General linear model
Grand variance
Harmonic mean
Helmert contrast
Independent ANOVA
Omega squared
Orthogonal
Pairwise comparisons
Planned contrasts
Polynomial contrast
Post hoc tests
Quadratic trend
Quartic trend
Repeated contrast
Simple contrast
Weights
Welch's *F*

Smart Alex's tasks



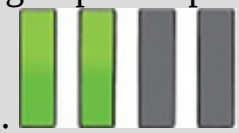
- **Task 1**: To test how different teaching methods affected students' knowledge I took three statistics modules where I taught the same material. For one module I wandered around with a large cane and beat anyone who asked daft questions or

got questions wrong (**punish**). In the second I encouraged students to discuss things that they found difficult and gave anyone working hard a nice sweet (**reward**). In the final course I neither punished nor rewarded students' efforts (**indifferent**). I measured the students' exam marks (**percentage**). The data are in the file **Teach.sav**. Fit a model with planned contrasts to test the hypotheses that: (1) reward results in better exam results than either punishment or indifference; and (2) indifference will lead to significantly better exam results than punishment.



- **Task 2**: Compute the effect sizes for Task 1 
- **Task 3**: Children wearing superhero costumes are more likely to harm themselves because of the unrealistic impression of invincibility that these costumes could create. For example, children have reported to hospital with severe injuries because of trying 'to initiate flight without having planned for landing strategies' (Davies, Surridge, Hole, & Munro-Davies, 2007). I can relate to the imagined power that a costume bestows upon you; indeed, I have been known to dress up as Fisher by donning a beard and glasses and trailing a goat around on a lead in the hope that it might make me more knowledgeable about statistics. Imagine we had data (**Superhero.sav**) about the severity of **injury** (on a scale from 0, no injury, to 100, death) for children reporting to the accident and emergency department at hospitals, and information on which superhero costume they were wearing (**hero**): Spiderman, Superman, the Hulk or a teenage mutant ninja turtle. Fit a model with planned contrasts to test the hypothesis that different costumes give rise to more severe injuries. 
- **Task 4**: In Chapter 7 (Section 7.6) there are some data looking at whether eating soya meals reduces your sperm count. Analyse these data with a linear model (ANOVA). What's the difference between what you find and what was found in Section 7.6.5?

Why do you think this difference has arisen? 

- **Task 5**: Mobile phones emit microwaves, and so holding one next to your brain for large parts of the day is a bit like sticking your brain in a microwave oven and pushing the 'cook until well done' button. If we wanted to test this experimentally, we could get six groups of people and strap a mobile phone on their heads, then by remote control turn the phones on for a certain amount of time each day. After six months, we measure the size of any tumour (in mm^3) close to the site of the phone antenna (just behind the ear). The six groups experienced 0, 1, 2, 3, 4 or 5 hours per day of phone microwaves for six months. Do tumours significantly increase with greater daily exposure? The data are in **Tumour.sav**. 
- **Task 6**: Using the Glastonbury data from Chapter 6 (**GlastonburyFestival.sav**), fit a model to see if the change in hygiene (**change**) is significant across people with different musical tastes (**music**). Do a simple contrast to compare each group against the no affiliation group. Compare the results to those described in Section 11.5. 
- **Task 7**: Labcoat Leni's Real Research 7.2 describes an experiment (Çetinkaya & Domjan, 2006) on quails with fetishes for terrycloth objects. There were two outcome variables (time spent near the terrycloth object and copulatory efficiency) that we didn't analyse. Read Labcoat Leni's Real Research 7.2 to get the full story, then fit a model with Bonferroni *post hoc* tests on the time spent near the terrycloth object. 
- **Task 8**: Repeat the analysis in Task 7 but using copulatory efficiency as the outcome. 
- **Task 9**: A sociologist wanted to compare murder rates

(**Murder**) each month in a year at three high-profile locations in London (**Street**). Fit a model with bootstrapping on the *post hoc* tests to see in which streets the most murders happened. The data are in **Murder.sav.**

Answers & additional resources are available on the book's website at **https://edge.sagepub.com/field5e**



13 GLM 2: Comparing means adjusted for other predictors (analysis of covariance)

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13.1 What will this chapter tell me?

My road to rock stardom had taken a bit of a knock with my unexpected entry to an all-boys' grammar school (rock bands and grammar schools really didn't go together). I needed to be inspired and I turned to the masters: Iron Maiden. I first heard Iron Maiden at the age of 11 when a friend lent me *Piece of Mind* on a cassette and told me to listen to 'The Trooper'. It was, to put it mildly, an epiphany. I became their smallest (I was 11) biggest fan and obsessed about them in the unhealthiest of ways. I bombarded the man who ran their fan club (a guy called Keith) with letters, and, bless him, he replied to them all. Eventually my stalking paid off and Keith arranged for me to go backstage when they played what was then (and to me always will be) the Hammersmith Odeon in London on 5 November 1986 (*Somewhere on Tour*, in case you're interested). Not only was it the first time I had seen them live, but I got to meet them too. It is difficult to convey how exciting and anxiety-provoking that night was. It was all quite overwhelming. I was so utterly awe-struck that I managed to say precisely nothing to any of the band (but I do have some good photos where my speechlessness is tangible; see Figure 13.1). Soon to become a theme in my life, a social situation had provoked me to make an utter fool of myself.[1] When it was over I was in no doubt that this was the best day of my life. In fact, I thought, I should just kill myself there and then because nothing would ever be as good.[2] This may be true, but I have subsequently had other very nice experiences, so who is to say that they were not better? I could compare experiences to see which one is the best, but there is an important confound: my age. At the age of 13, meeting Iron Maiden was bowel-weakeningly exciting, but adulthood (sadly) dulls your capacity for this kind of unqualified excitement. To really see which experience was best, I would have to take account of the variance in enjoyment that is attributable to my age at the time. Doing so will give me a purer measure of how much variance in my enjoyment is attributable to the event itself.

1 In my teens I met many bands I liked, and Iron Maiden were by far the nicest.

2 Apart from my wedding day, as it turned out.

This chapter extends the previous one to look at situations in which you want to compare groups means, but also adjust those means for another variable (or variables) that you expect to affect the outcome. This involves a linear model in which an outcome is predicted from dummy variables representing group membership but one or more other predictors (usually continuous variables) are included. These additional predictors are sometimes labelled covariates, and this configuration of the linear model is sometimes known as *analysis of covariance*.

Figure 13.1 Dave Murray (guitarist from Iron Maiden) and me backstage in

London in 1986 (my grimace reflects the utter terror I was feeling at meeting my hero)



13.2 What is ANCOVA?

In the previous chapter we saw how we can compare multiple group means with the linear model by using dummy variables to code group membership. In addition, in Chapter 9 we saw how the linear model can incorporate several continuous predictor variables. It should, therefore, be no surprise that the linear model to compare means can be extended to include one or more continuous variables that predict the outcome (or dependent variable). When the main focus of the model is to compare means (perhaps from different experimental groups) then these additional predictors in the model are sometimes referred to as **covariates**. Also, this form of the linear model is sometimes referred to as **analysis of covariance** (or **ANCOVA** for short).[3]

3 As we've discussed before, these labels for special cases of the linear model (such as one-way independent ANOVA in the previous chapter, and ANCOVA here) reflect historical divisions in methods (see Misconception Mutt 12.1). They are unhelpful because they create the impression that we're using distinct statistical models when we're not. I want you to focus on the general linear model that underpins these special cases, but I can't really avoid using the ANOVA/ANCOVA labels now and again so that when your supervisor tells you to do ANOVA/ANCOVA you can find the relevant part of the book!



In the previous chapter, we used an example about the effects of puppy therapy

on happiness. Let's think about things other than puppy therapy that might influence happiness. Well, the obvious one is how much you like dogs (a dog phobic is going to be about as happy after puppy therapy as I would be after tarantula therapy), but there are other things too such as individual differences in temperament. If these variables (the covariates) are measured, then it is possible to adjust for the influence they have on the outcome variable by including them in the linear model. From what we know of hierarchical regression (see Chapter 9) it should be clear that if we enter the covariate into the model first, and then enter the dummy variables representing the group means (e.g., the experimental manipulation), we can see what effect a predictor variable has, *adjusting for* the effect of the covariate. In essence, rather than predicting the outcome from group means, we predict it from group means that have been adjusted for the effect of covariate(s). There are two main reasons to include covariates in ANOVA:

- **To reduce within-group error variance**: When we predict an outcome from group means (e.g., when these represent the effect of an experiment), we compute an *F*-statistic by comparing the amount of variability in the outcome that the experiment can explain against the variability that it cannot explain. If we can attribute some of this 'unexplained' variance (SS_R) to other measured variables (covariates), then we reduce the error variance, allowing us to assess more sensitively the difference between group means (SS_M).
- **Elimination of confounds**: In any experiment, there may be unmeasured variables that confound the results (i.e., variables other than the experimental manipulation that affect the outcome variable). If any variables are known to influence the outcome variable being measured, then including them as covariates can remove these variables as potential explanations for the effect of interest.

13.3 ANCOVA and the general linear model 

The researchers who conducted the puppy therapy study in the previous chapter suddenly realized that a participant's love of dogs would affect whether puppy therapy would affect happiness. Therefore, they repeated the study on different participants, but included a self-report measure of love of puppies from 0 (I am a weird person who hates puppies, please be deeply suspicious of me) to 7 (puppies are the best thing ever, one day I might marry one). The data are in Table 13.1 and in the file **Puppy Love.sav**, which contains the variables **Dose** (1 = control, 2 = 15 minutes, 3 = 30 minutes), **Happiness** (the person's happiness

on a scale from 0 to 10), and **Puppy_love** (love of puppies from 0 to 7).



Use IBM SPSS Statistics to find the means and standard deviations of both happiness and love of puppies across all participants and within the three groups. (Answers are in Table 13.2.)

In the previous chapter, we characterized this experimental scenario as equation (12.2), and knowing what we know about the linear model (Chapter 9) you can hopefully see that that equation can be extended to include the covariate as follows:

$$\text{Happiness}_i = b_0 + b_1\text{Long}_i + b_2\text{Short}_i + b_3\text{Covariate}_i + \varepsilon_i$$
$$\text{Happiness}_i = b_0 + b_1\text{Long}_i + b_2\text{Short}_i + b_3\text{Puppy_love}_i + \varepsilon_i \quad (13.1)$$

We can compare the means of different groups using a linear model (see Section 12.2) in which groups are coded as the dummy variables **Long** and **Short**: **Long** takes the value of 1 only for the 30-minute group, **Short** takes a value of 1 only for the 15-minute group, and in all other situations they have a value of 0. We can add a covariate as a predictor to the model to test the difference between group means *adjusted for the covariate*. Let's look at a practical example.



Add two dummy variables to the file **Puppy Love.sav** that compare the 15-minute group to the control (**Dummy 1**) and the 30-minute group to the control (**Dummy 2**) – see Section 12.2 for help. If you get stuck use **Puppy Love Dummy.sav.**



Fit a hierarchical regression with **Happiness** as the outcome. In the first block enter love of puppies (**Puppy_love**) as a predictor, and then in a second block enter both dummy variables (forced entry) – see Section 9.10 for help.

Table 13.1 Data from **Puppy Love.sav**

Dose	Participant's happiness	Love of puppies
Control	3	4
	2	1
	5	5
	2	1
	2	2
	2	2
	7	7
	2	4
	4	5
15 minutes	7	5
	5	3
	3	1
	4	2
	4	2
	7	6
	5	4
	4	2
30 minutes	9	1
	2	3
	6	5
	3	4
	4	3
	4	3
	4	2
	6	0
	4	1
	6	3
	2	0
	8	1
	5	0

Table 13.2 Means (and standard deviations) from **Puppy Love.sav**

Dose	Participant's happiness	Love of puppies
Control	3.22 (1.79)	3.44 (2.07)
15 minutes	4.88 (1.46)	3.12 (1.73)
30 minutes	4.85 (2.12)	2.00 (1.63)
Total	**4.37 (1.96)**	**2.73 (1.86)**

The summary of the model resulting from the self-test (Output 13.1) shows us the goodness of fit of the model first when only the covariate is used in the model, and second when both the covariate and the dummy variables are used. The difference between the values of R^2 (0.288 − 0.061 = 0.227) represents the individual contribution of puppy therapy to predicting happiness. Puppy therapy accounted for 22.7% of the variation in happiness, whereas love of puppies

accounted for only 6.1%. This additional information provides some insight into the substantive importance of puppy therapy. The next table is the ANOVA table, which is also divided into two sections. The top half represents the effect of the covariate alone, whereas the bottom half represents the whole model (i.e., covariate and puppy therapy included). Notice at the bottom of the ANOVA table (the bit for model 2) that the entire model (love of puppies and the dummy variables) accounts for 31.92 units of variance (SS_M), there are 110.97 units in total (SS_T) and the unexplained variance (SS_R) is 79.05.



The interesting bit is the table of model coefficients (Output 13.2). The top half shows the effect when only the covariate is in the model, and the bottom half contains the whole model. The *b*-values for the dummy variables represent the difference between the means of the 15-minute group and the control group (**Dummy 1**) and the 30-minute group and the control group (**Dummy 2**) – see Section 12.2 for an explanation of why. The means of the 15-and 30-minute groups were 4.88 and 4.85 respectively, and the mean of the control group was 3.22. Therefore, the *b*-values for the two dummy variables should be roughly the same (4.88 − 3.22 = 1.66 for **Dummy 1** and 4.85 − 3.22 = 1.63 for **Dummy 2**). The astute among you might notice that the *b*-values in Output 13.2 are not only very different from each other (which shouldn't be the case because the 15-and 30-minute groups means are virtually the same), but also different from the values I've just calculated. Does this mean I've been lying to you for the past 50 pages about what the beta values represent? I'm evil, but I'm not *that* evil. The reason for this apparent anomaly is that with a covariate present, the *b*-values represent the differences between the means of each group and the control *adjusted for the covariate(s)*. In this case, they represent the difference in the means of the puppy therapy groups adjusted for the love of puppies.

Output 13.1

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.246[a]	.061	.027	1.929
2	.536[b]	.288	.205	1.744

a. Predictors: (Constant), Love of puppies (0–7)

b. Predictors: (Constant), Love of puppies (0–7), Dummy 1: (control vs. 15 mins), Dummy 2: (control vs. 30 mins)

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	6.734	1	6.734	1.809	.189[b]
	Residual	104.232	28	3.723		
	Total	110.967	29			
2	Regression	31.920	3	10.640	3.500	.030[c]
	Residual	79.047	26	3.040		
	Total	110.967	29			

a. Dependent Variable: Happiness (0–10)

b. Predictors: (Constant), Love of puppies (0–7)

c. Predictors: (Constant), Love of puppies (0–7), Dummy 1: (control vs. 15 mins), Dummy 2: (control vs. 30 mins)

These **adjusted means** come directly from the model. If we replace the *b*-values in equation (13.1) with the values in Output 13.2, our model becomes:

$$\text{Happiness}_i = 1.789 + 2.225\text{Long}_i + 1.786\text{Short}_i + 0.416\text{Puppy_love}_i \quad (13.2)$$

Remember that **Long** and **Short** are dummy variables such that **Long** takes the value of 1 only for the 30-minute group, and **Short** takes a value of 1 only for the 15-minute group; in all other situations they have a value of 0. To get the adjusted means, we use this equation, but rather than replacing the covariate with an individual's score, we replace it with the mean value of the covariate from Table 13.2 (2.73) because we're interested in the predicted value for each group at the mean level of the covariate. For the control group, the dummy variables are both coded as 0, so we replace **Long** and **Short** in the model with 0. The adjusted mean will, therefore, be 2.925:

$$\begin{aligned}\overline{\text{Happiness}}_{\text{Control}} &= 1.789 + (2.225 \times 0) + (1.786 \times 0) + \left(0.416 \times \bar{X}_{\text{Puppy_love}}\right) \\ &= 1.789 + (0.416 \times 2.73) \\ &= 2.925\end{aligned} \quad (13.3)$$

For the 15-minute group, the dummy variable **Short** is 1 and **Long** is 0, so the adjusted mean is 4.71:

$$\begin{aligned}\overline{\text{Happiness}}_{15\text{ mins}} &= 1.789 + (2.225 \times 0) + (1.786 \times 1) + \left(0.416 \times \bar{X}_{\text{Puppy_love}}\right) \\ &= 1.789 + 1.786 + (0.416 \times 2.73) \\ &= 4.71\end{aligned} \tag{13.4}$$

For the 30-minute group, the dummy variable **Short** is 0 and **Long** is 1, so the adjusted mean is 5.15:

$$\begin{aligned}\overline{\text{Happiness}}_{30\text{ mins}} &= 1.789 + (2.225 \times 1) + (1.786 \times 0) + \left(0.416 \times \bar{X}_{\text{Puppy_love}}\right) \\ &= 1.789 + 2.225 + (0.416 \times 2.73) \\ &= 5.15\end{aligned} \tag{13.5}$$

We can now see that the *b*-values for the two dummy variables represent the differences between these *adjusted* means (4.71 − 2.93 = 1.78 for **Dummy 1** and 5.15 − 2.93 = 2.22 for **Dummy 2**). These adjusted means are the average amount of happiness for each group at the mean level of love of puppies. Some people think of this kind of model (i.e., ANCOVA) as 'controlling' for the covariate, because it compares the predicted group means at the average value of the covariate, so the groups are being compared at a level of the covariate that is the same for each group. However, as we shall see, the 'controlling for the covariate' analogy is not a good one.

Output 13.2

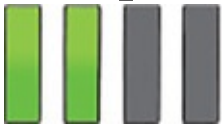
Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	3.657	.634		5.764	.000
	Love of puppies (0-7)	.260	.193	.246	1.345	.189
2	(Constant)	1.789	.867		2.063	.049
	Love of puppies (0-7)	.416	.187	.395	2.227	.035
	Dummy 1: (control vs. 15 mins)	1.786	.849	.411	2.102	.045
	Dummy 2: (control vs. 30 mins)	2.225	.803	.573	2.771	.010

a. Dependent Variable: Happiness (0-10)

13.4 Assumptions and issues in ANCOVA 

Including covariates doesn't change the fact we're using the general linear model, so all the sources of potential bias (and counteractive measures) discussed in Chapter 6 apply. There are two additional considerations: (1) independence of the covariate and treatment effect; and (2) homogeneity of regression slopes.

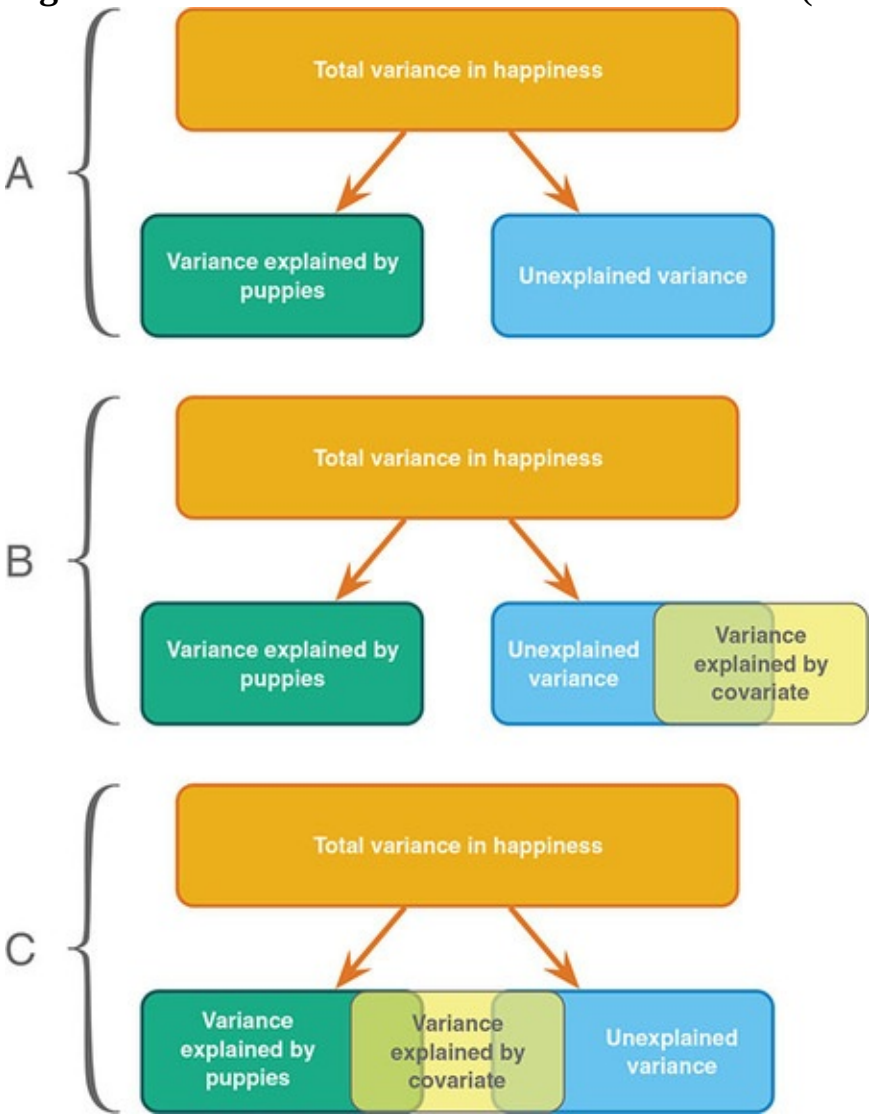
13.4.1 Independence of the covariate and treatment effect 

I said in the previous section that covariates can be used to reduce within-group error variance if the covariate explains some of this error variance, which will be the case if the covariate is independent of the experimental effect (group means). Figure 13.2 shows three different scenarios. Part A shows a basic model that compares group means (it is like Figure 12.5). The variance in the outcome (in our example happiness) can be partitioned into two parts that represent the experimental or treatment effect (in this case the administration of puppy therapy) and the error or unexplained variance (i.e., factors that affect happiness that we haven't measured). Part B shows the ideal scenario when including a covariate, which is that the covariate shares its variance only with the bit of happiness that is currently unexplained. In other words, it is completely independent of the treatment effect (it does not overlap with the effect of puppy therapy at all). Some argue that this scenario is the only one in which ANCOVA is appropriate (Wildt & Ahtola, 1978). Part C shows a situation in which the effect of the covariate overlaps with the experimental effect. In other words, the experimental effect is confounded with the effect of the covariate. In this situation, the covariate will reduce (statistically speaking) the experimental effect because it explains some of the variance that would otherwise be attributable to the experiment. When the covariate and the experimental effect (independent variable) are not independent, the treatment effect is obscured, spurious treatment effects can arise, and at the very least the interpretation of the ANCOVA is seriously compromised (Wildt & Ahtola, 1978).

The problem of the covariate and treatment sharing variance is common and is ignored or misunderstood by many people (Miller & Chapman, 2001). Miller and Chapman are not the only people to point this out, but their paper is very readable and they cite many examples of people misapplying ANCOVA. Their main point is that when treatment groups differ on the covariate, putting the covariate into the analysis will not 'control for' or 'balance out' those differences (Lord, 1967, 1969). This situation arises mostly when participants are not randomly assigned to experimental treatment conditions. For example, anxiety and depression are closely correlated (anxious people tend to be depressed), so if you wanted to compare an anxious group of people against a non-anxious group on some task, the chances are that the anxious group would also be more depressed than the non-anxious group. You might think that by adding depression as a covariate into the analysis you can look at the 'pure' effect of anxiety, but you can't. This situation matches part C of Figure 13.2 because the effect of the covariate (depression) would contain some of the variance from the effect of anxiety. Statistically speaking, all that we know is that anxiety and depression share variance; we cannot separate this shared

variance into ‘anxiety variance’ and ‘depression variance’, it will always be ‘shared’. Another common example is if you happen to find that your experimental groups differ in their ages. Placing age into the analysis as a covariate will not solve this problem – it is still confounded with the experimental manipulation. The use of covariates cannot solve this problem (see Jane Superbrain Box 13.1).

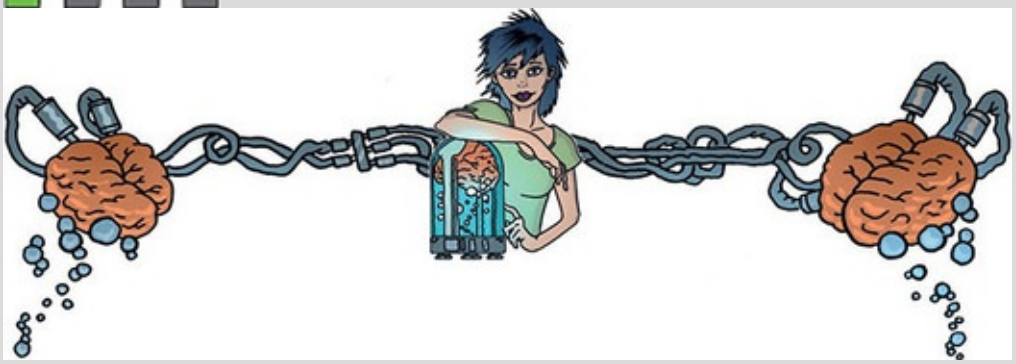
Figure 13.2 The role of the covariate in ANCOVA (see text for details)



This problem can be avoided by randomizing participants to experimental groups, or by matching experimental groups on the covariate (in our anxiety example, you could try to find participants for the low-anxiety group who score high on depression). We can see whether this problem is likely to be an issue by checking whether experimental groups differ on the covariate before fitting the model. To use our anxiety example again, we could test whether our high-and low-anxiety groups differ on levels of depression. If the groups do not

significantly differ then we might consider it reasonable to use depression as a covariate.

Jane Superbrain 13.1 An interpretational or statistical requirement?

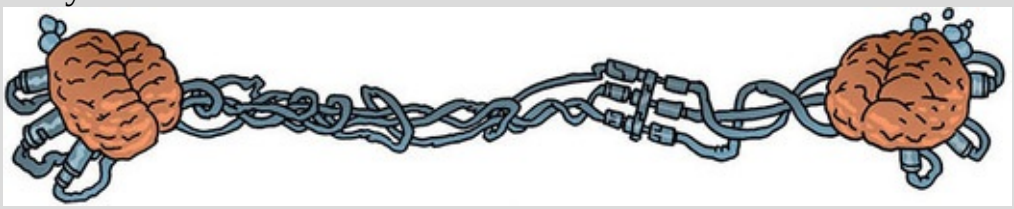


The treatment effect and covariate are simply predictor variables in a general linear model, yet despite several hundred pages discussing linear models, I haven't before mentioned that predictors should be completely independent. I've said that they shouldn't overlap too much (e.g., collinearity) but that's quite different from saying that they shouldn't overlap at all. If, in general, we don't care about predictors being independent in linear models, why should we care now? The short answer is we don't – there is no *statistical* requirement for the treatment variable and covariate to be independent.

However, there are situations in which ANCOVA can be biased when the covariate is not independent of the treatment variable. One situation, common in medical research, has been discussed a lot: an outcome (e.g., hypertension) is measured at baseline, and after a treatment intervention (with participants assigned to a treatment or control group). This design can be analysed using an ANCOVA in which treatment effects on post-intervention hypertension are analysed while covarying baseline levels of hypertension. In this scenario the independence of treatment and covariate variables means that baseline levels of hypertension are equal in the different treatment groups. According to Senn (2006), the idea that ANCOVA is biased unless treatment groups are equal on the covariate applies only when there is *temporal additivity*. To use our hypertension example, temporal additivity is the assumption that both treatment groups would experience the same change in hypertension over time

if the treatment had no effect. In other words, had we left the two groups alone, their hypertension would change by exactly the same amount. Given that the groups have different overall levels of hypertension to begin with, this assumption might not be reasonable, which undermines the argument for requiring group equality in baseline measures.

To sum up, the independence of the covariate and treatment makes interpretation more straightforward but is not a statistical requirement. ANCOVA can be unbiased when groups differ on levels of the covariate, but as Miller and Chapman point out, it creates an interpretational problem that ANCOVA cannot magic away.

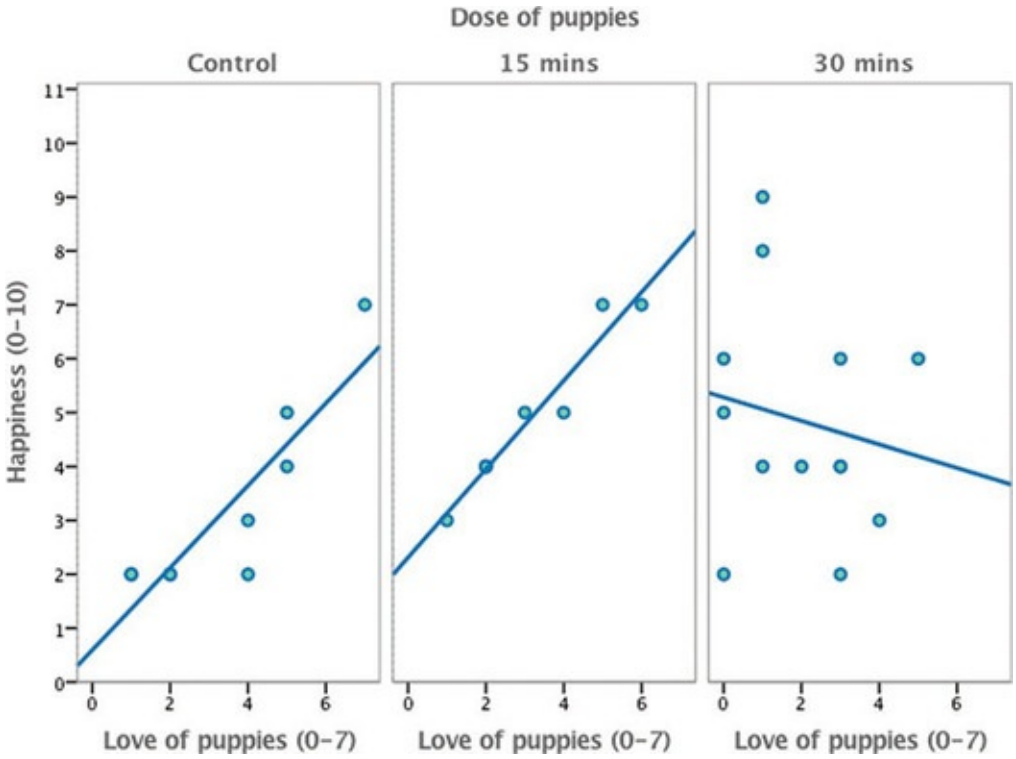




13.4.2 Homogeneity of regression slopes

When a covariate is used we look at its overall relationship with the outcome variable: we ignore the group to which a person belongs. We assume that this relationship between covariate and outcome variable holds true for all groups of participants, which is known as the assumption of **homogeneity of regression slopes**. Think of the assumption like this: imagine a scatterplot for each group of participants with the covariate on one axis, the outcome on the other, and a regression line summarizing their relationship. If the assumption is met then the regression lines should look similar (i.e., the values of *b* in each group should be equal).

Figure 13.3 Scatterplot and regression lines of happiness against love of puppies for each of the experimental conditions

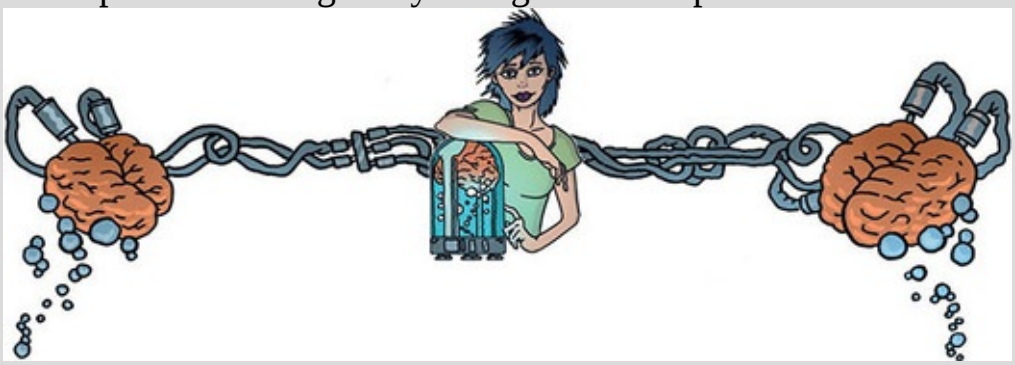


Let's make this concept a bit more concrete. Remember that the main example in this chapter looks at whether different doses of puppy therapy affect happiness when including love of puppies as a covariate. The *homogeneity of regression slopes* assumption means that the relationship between the outcome (dependent variable) and the covariate is the same in each of our treatment groups. Figure 13.3 shows a scatterplot with regression line that summarizes this relationship (i.e., the relationship between love of puppies, the covariate, and the outcome, participant's happiness) for the three experimental conditions (shown in different panels). There is a positive relationship (the regression line slopes upwards from left to right) between love of puppies and participant's happiness in both the control (left panel) and 15-minute conditions (middle panel). In fact, the slopes of the lines for these two groups are very similar, showing that the relationship between happiness and love of puppies is very similar in these two groups. This situation is an example of *homogeneity* of regression slopes. However, in the 30-minute condition (right panel) there is a slightly negative relationship between happiness and love of puppies. The slope of this line differs from the slopes in the other two groups, suggesting *heterogeneity of regression slopes* (because the relationship between happiness and love of puppies is different in the 30-minute group compared to the other two groups).

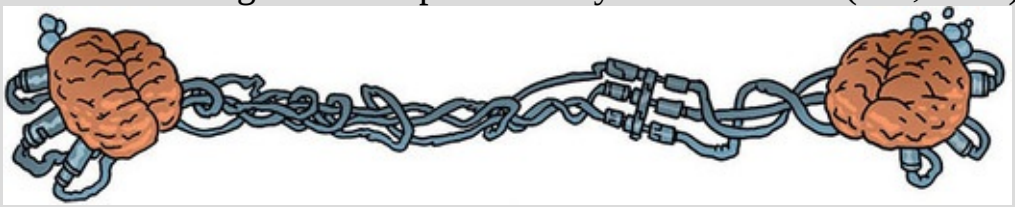
Although in a traditional ANCOVA heterogeneity of regression slopes is a bad thing (Jane Superbrain Box 13.2), there are situations where you might expect regression slopes to differ across groups and that variability may be interesting.

For example, when research is conducted across different locations, you might expect the effects to vary across those locations. Imagine you had a new treatment for backache, and you recruit several physiotherapists to try it out in different hospitals. The effect of the treatment is likely to differ across these hospitals (because therapists will differ in expertise, the patients they see will have different problems and so on). As such, heterogeneity of regression slopes is not a bad thing *per se*. If you have violated the assumption of homogeneity of regression slopes, or if the variability in regression slopes is an interesting hypothesis in itself, then you can explicitly model this variation using multilevel linear models (see Chapter 21).

Jane Superbrain 13.2 What are the consequences of violating the assumption of homogeneity of regression slopes?



When the assumption of homogeneity of regression slopes is met the resulting *F*-statistic can be assumed to have the corresponding *F*-distribution; however, when the assumption is not met it can't, meaning that the resulting *F*-statistic is being evaluated against a distribution different than the one that it actually has. Consequently, the Type I error rate of the test is inflated and the power to detect effects is not maximized (Hollingsworth, 1980). This is especially true when group sizes are unequal (Hamilton, 1977) and when the standardized regression slopes differ by more than 0.4 (Wu, 1984).



13.4.3 What to do when assumptions are violated

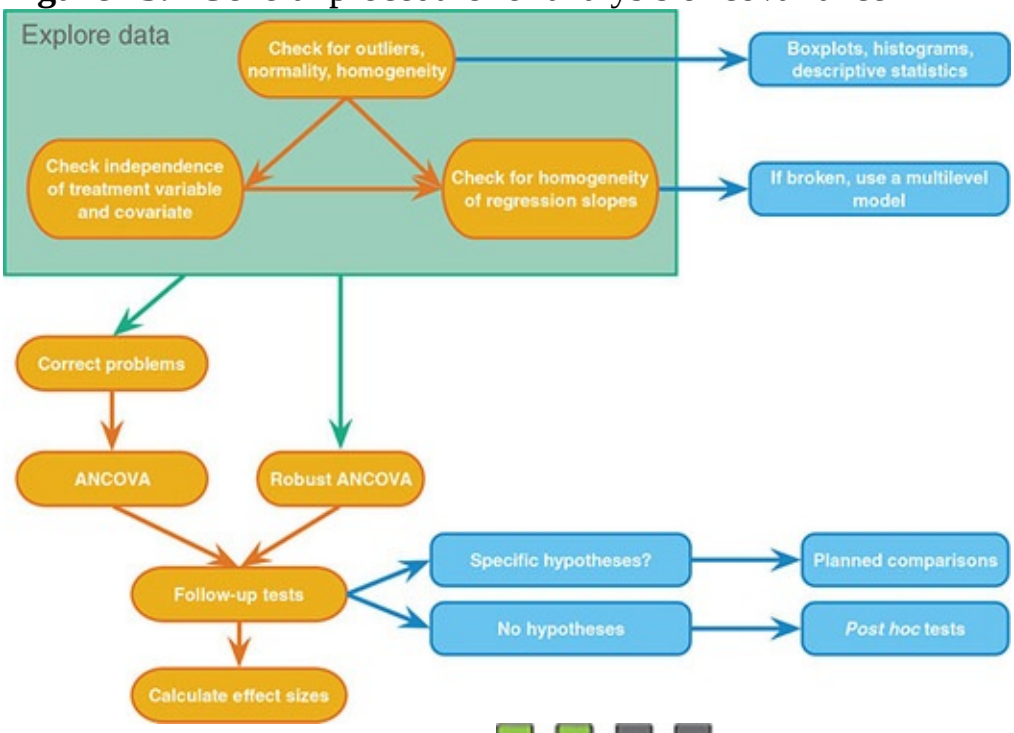
A bootstrap for the model parameters and *post hoc* tests can be used so that these, at least, are robust (see Chapter 6). The bootstrap won't help for the *F*-tests though. There is a robust variant of ANCOVA that can be implemented using R, and we'll discuss this in Section 13.8.

13.5 Conducting ANCOVA using SPSS Statistics

13.5.1 General procedure

The general procedure is much the same as for any linear model, so remind yourself of the steps for fitting a linear model (Chapter 9). Figure 13.4 shows a simpler overview of the process that highlights some of the specific issues for ANCOVA-style models. As with any analysis, begin by graphing the data and looking for and correcting sources of bias.

Figure 13.4 General procedure for analysis of covariance



13.5.2 Inputting data

We have already looked at the data (Table 13.1) and the data file (**Puppy Love.sav**). To remind you, the data file is set out like Table 13.1 and contains three columns: a coding variable called **Dose** (1 = control, 2 = 15 minutes, 3 = 30 minutes), a variable called **Happiness** containing the scores for the person's happiness, and a variable called **Puppy_love** containing the scores for love of puppies from 0 to 7. The 30 rows correspond to each person's scores on these three variables.

13.5.3 Testing the independence of the treatment variable and covariate



In Section 13.4.1, I mentioned that if the covariate and group means (independent variable) are independent then the interpretation of ANCOVA models is a lot more straightforward. In this case, the covariate is love of puppies, so we'd want to check that the mean level of love of puppies is roughly equal across the three puppy therapy groups by fitting a linear model with **Puppy_love** as the outcome and **Dose** as the predictor.



Fit a model to test whether love of puppies (our covariate) is independent of the dose of puppy therapy (our independent variable).

Output 13.3 shows that the main effect of dose is not significant, $F(2, 27) = 1.98$, $p = 0.16$, which shows that the average level of love of puppies was roughly the same in the three puppy therapy groups. In other words, the means for love of puppies in Table 13.2 are not significantly different across the control, 15-and 30-minute groups. This result is good news for using love of puppies as a covariate in the model.

Output 13.3

ANOVA

Love of puppies (0-7)

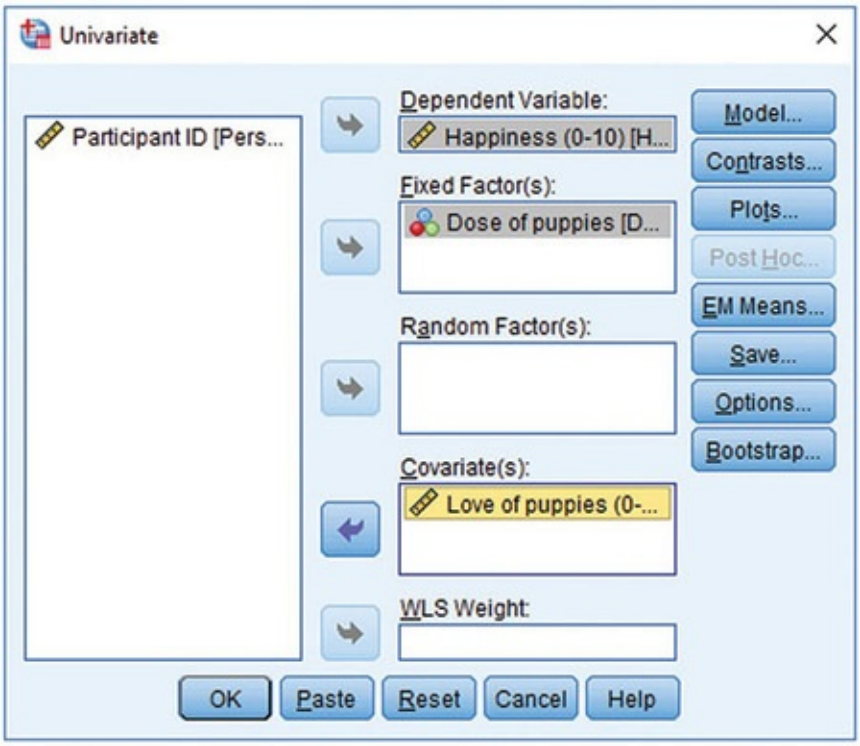
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	12.769	2	6.385	1.979	.158
Within Groups	87.097	27	3.226		
Total	99.867	29			

13.5.4 The main analysis



Most of the *General Linear Model* (GLM) procedures in SPSS Statistics contain the facility to include one or more covariates. For designs that don't involve repeated measures it is easiest to include covariates by selecting *Analyze*  *General Linear Model* to activate the dialog box in Figure 13.5. Drag the variable **Happiness** into the box labelled *Dependent Variable* (or click ), drag **Dose** into the box labelled *Fixed Factor(s)* and drag **Puppy_love** into the box labelled *Covariate(s)*.

Figure 13.5 Main dialog box for GLM univariate



13.5.5 Contrasts 

There are various dialog boxes that can be accessed from the main dialog box. If a covariate is selected, the *post hoc* tests are disabled because the tests that we used in the previous chapter are not designed for models that include covariates. However, comparisons can be done by clicking  to access the

Contrasts dialog box in Figure 13.6. You cannot enter codes to specify user-defined contrasts (but see SPSS Tip 13.1); instead you can select one of the standard contrasts that we met in Table 12.6. In this example, there was a control condition (coded as the first group), so a sensible set of contrasts would be simple contrasts comparing each experimental group to the control (this results in the same contrasts as dummy coding). Click the drop-down list (None) and select a type of contrast (in this case *Simple*) from this list. For simple contrasts you need to specify the reference category (i.e., the category against which all other groups are compared). By default the last category is used, which for our data is the 30-minute group. We need to change the reference category to be the control group, which is the first category (assuming that you coded control as 1). We make this change by selecting First. Having selected a contrast, click Change to register the selection. Figure 13.6 shows the completed dialog box. Click Continue to return to the main dialog box.

13.5.6 Other options

You can get a limited range of *post hoc* tests by clicking EM Means... to access the *Estimated Marginal Means* dialog box (see Figure 13.7). To specify *post hoc* tests, drag the independent variable (in this case **Dose**) from the box labelled *Estimated Marginal Means: Factor(s) and Factor Interactions* to the box labelled *Display Means for* (or click Save...). Once a variable has been transferred, you'll be able to select Compare main effects to activate the drop-down list (LSD(none)) of *post hoc* tests. The default is to perform a Tukey LSD *post hoc* test which makes no adjustment for multiple tests (and which I don't recommend). The other options are a Bonferroni *post hoc* test (recommended) and a **Šidák correction**, which is like the Bonferroni correction but is less conservative and so should be selected if you are concerned about the loss of power associated with Bonferroni. For this example we'll use the Šidák correction just for variety (we have used Bonferroni in previous examples). As well as producing *post hoc* tests for the **Dose** variable, the options we've selected will create a table of estimated marginal means for this variable: these are the

group means adjusted for the effect of the covariate. Click Continue.

Clicking Options... opens a dialog box containing the options described in Jane Superbrain Box 13.3. The most useful are (in my opinion) descriptive statistics, parameter estimates, residual plot and HC4 robust standard errors (see Figure 13.7).

Figure 13.6 Options for standard contrasts in GLM univariate

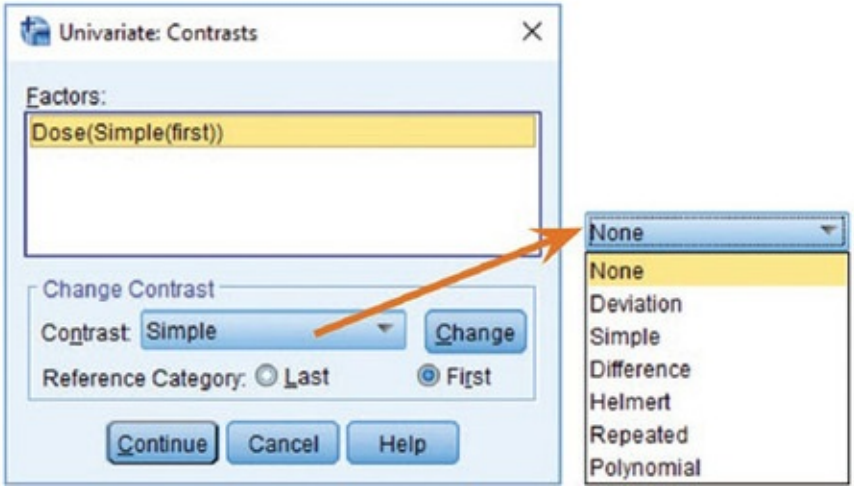
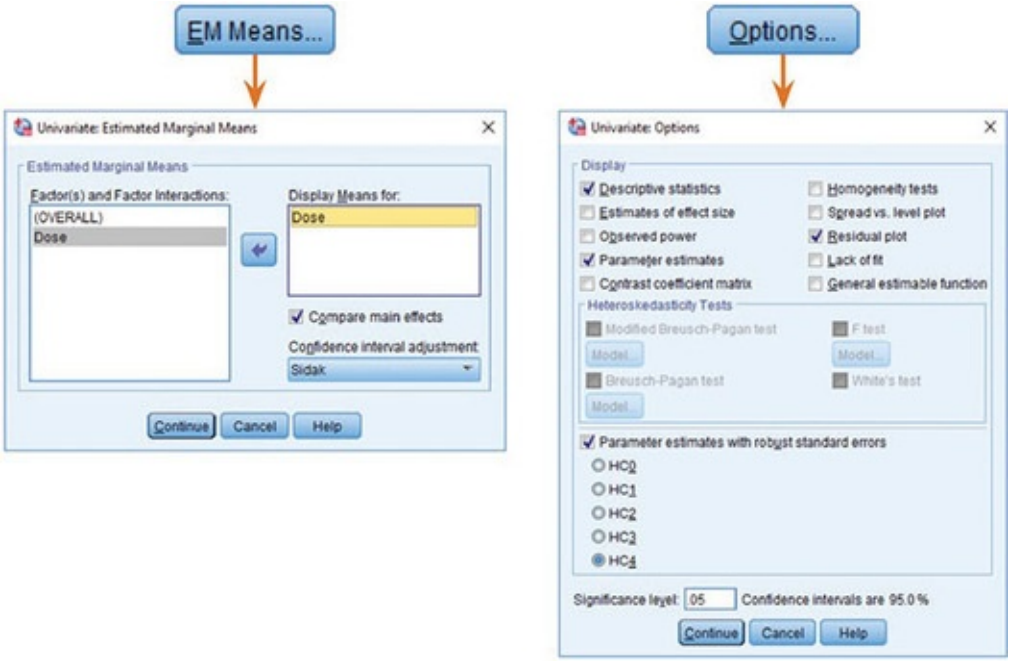
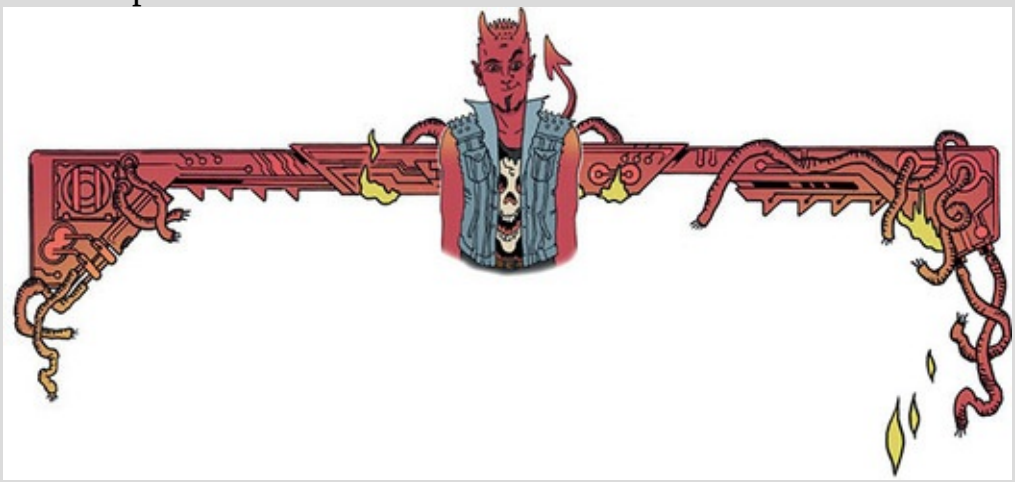


Figure 13.7 *Estimated marginal means* and *Options* dialog boxes for GLM univariate



SPSS Tip 13.1 Planned contrasts for ANCOVA



There is no option for specifying planned contrasts like we used in the previous chapter (see Section 12.6.2). However, these contrasts can be done if we fit the model using the regression menu. Imagine you chose some planned contrasts as in Chapter 12, in which the first contrast compared the control group to all doses of puppy therapy, and the second contrast then compared the 30-and 15-minute groups (see Section 12.4). We saw in Sections 12.4 and 12.6.2 that we specify these contrasts with codes. For the first contrast we discovered that an appropriate set of codes was −2 for the control group and then 1 for both the 30-and 15-minute groups. For the second contrast the codes were 0 for the control group, −1 for the 15-minute group and 1 for the 30-minute group (see Table 12.4). To do these contrasts when a covariate is included in the model, enter these values as two dummy variables. In other words, add a column called **Dummy1** in which every person in the control group has a value of −2 and all other participants have a value of 1. Add a second column called **Dummy2**, in which everyone in the control group has the value 0, everyone in the 15-minute group has the value −1 and those in the 30-minute group have a value of 1. The file **Puppy Love Contrast.sav** includes these dummy variables.

Output 13.4

Coefficients[a]

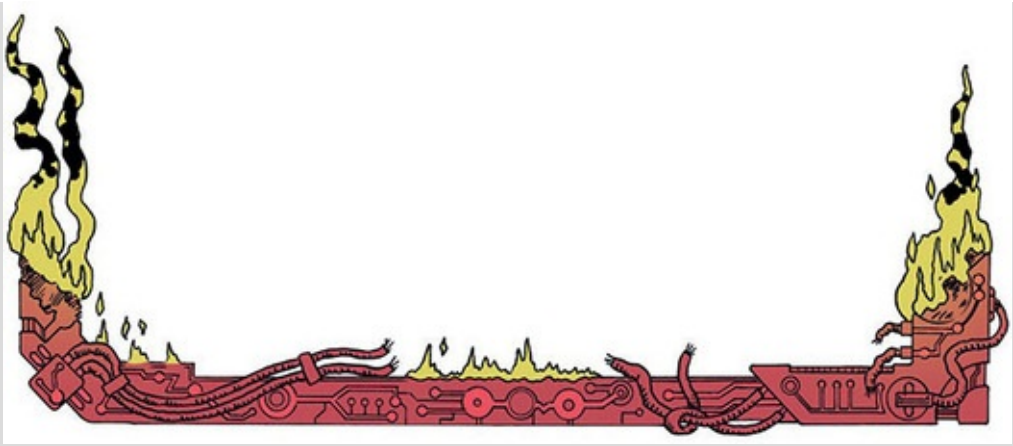
Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	3.657	.634		5.764	.000
	Love of puppies (0-7)	.260	.193	.246	1.345	.189
2	(Constant)	3.126	.625		5.002	.000
	Love of puppies (0-7)	.416	.187	.395	2.227	.035
	Dummy 1: puppies vs. control	.668	.240	.478	2.785	.010
	Dummy 2: 15 mins vs. 30 mins	.220	.406	.094	.541	.593

a. Dependent Variable: Happiness (0-10)

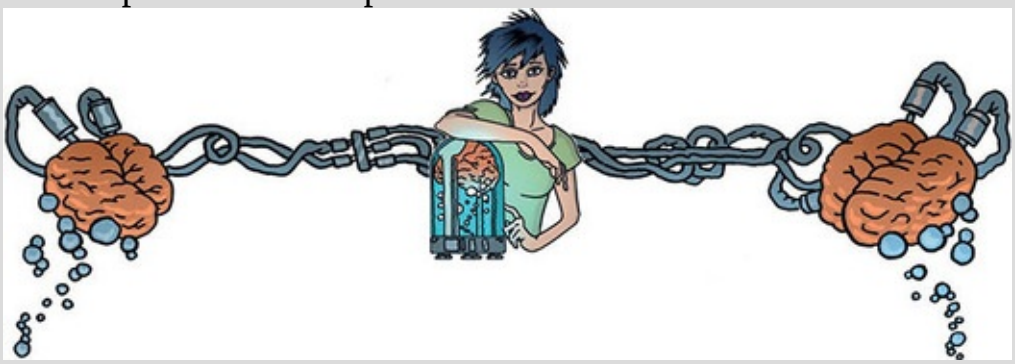
Run the analysis as described in Section 13.3. The model summary and ANOVA table for the model will be identical to Output 13.1 (because we've done the same thing as before; the only difference is how the model variance is subsequently broken down with the contrasts). The *b*-values for the dummy variables will be different than before because we've specified different contrasts. Output 13.4 shows the model parameters. The first dummy variable compares the control group with the 15-and 30-minute groups. As such, it compares the adjusted mean of the control group (2.93) with the average of the adjusted means for the 15-and 30-minute groups ((4.71+5.15)/2 = 4.93). The *b*-value for the first dummy variable should reflect the difference between these values: 4.93 − 2.93 = 2. We discovered in a rather complex and boring bit of Section 12.4.2 that this value gets divided by the number of groups within the contrast (i.e., 3) and so will be 2/3 = 0.67 (as in Output 13.4).[4] The associated *t*-statistic is significant ($p = 0.010$), indicating that the control group was significantly different from the combined adjusted mean of the puppy therapy groups.

4 The output contains the value 0.668 rather than 0.67. This difference is because we've rounded values to 2 decimal places in our caculations whereas SPSS retains all decimal places in its calculations.

The second dummy variable compares the 15-and 30-minute groups, and so the *b*-value should reflect the difference between the adjusted means of these groups: 5.15 − 4.71 = 0.44. In Section 12.4.2 we discovered that this value gets divided by the number of groups within the contrast (i.e., 2) and so will be 0.44/2 = 0.22 (as in Output 13.4). The associated *t*-statistic is not significant ($p = 0.593$), indicating that the 30-minute group did not produce significantly higher happiness than the 15-minute group after adjusting for love of puppies.



Jane Superbrain 13.3 Options for ANCOVA

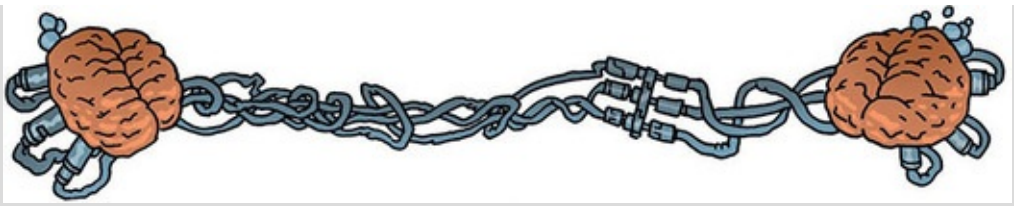


The remaining options in this dialog box are as follows:

- *Descriptive statistics*: This option produces a table of means and standard deviations for each group.
- *Estimates of effect size*: This option produces the value of partial eta squared (partial η^2) – see Section 13.10.
- *Observed power*: This option provides an estimate of the probability that the statistical test could detect the difference between the observed group means (see Section 2.9.7). This measure is pointless because if the *F*-test is significant then the probability that the effect was detected will, of course, be high. Likewise, if group differences were small, the observed power would be low. Do power calculations before the experiment is conducted, not after (see Section 2.9.8).
- *Parameter estimates*: This option produces a table of model parameters (*b*-values) and their tests of significance for the variables in the model (see Section 13.6.2).
- *Contrast coefficient matrix*: This option produces matrices of

the coding values used for any contrasts in the analysis, which is useful for checking which groups are being compared in which contrast.

- *Homogeneity tests*: This option produces Levene's test of the homogeneity of variance assumption (see Section 9.3). You'll have seen by now that I think this test needs to be taken with a pinch of salt.
- *Spread vs. level plot*: This option produces a chart that plots the mean of each group of a factor (x-axis) against the standard deviation of that group (y-axis). This plot is useful to check that there is no relationship between the mean and standard deviation. If a relationship exists then the data may need to be stabilized using a logarithmic transformation (see Chapter 6).
- *Residual plot*: This option produces a matrix scatterplot of all combinations of pairs of the following variables: observed values of the outcome, predicted values from the model, standardized residuals from the model. These plots can be used to assess the assumption of homoscedasticity. In particular, the plot of the standardized residuals against the predicted values from the model can be interpreted in a similar way to the zpred vs. zresid plot that we have discussed before.
- *Heteroskedasticity tests*: There are four tests for heteroscedasticity that you can select (two variants of the Breusch-Pagan test, White's test and an F-test). For the same reasons that I don't recommend Levene's test, I also don't recommend these (that is, because they are significance tests your decisions based on them will be confounded by your sample size).
- *Parameter estimates with robust standard errors*: This produces one of 5 methods (HC0 to HC4) to estimate standard errors (and, therefore, confidence intervals) for the model parameters that are robust to heteroscedasticity. These methods are described clearly in Hayes and Cai (2007). In short, HC3 has been shown to outperform HC0 to HC2 (Long & Ervin, 2000) but HC4 outperforms HC3 in some circumstances (Cribari-Neto, 2004). Basically choose HC3 or HC4.



Oditi's Lantern ANCOVA



'I, Oditi, have discovered that covariates give us greater control. I like control, especially controlling people's minds and making them worship me, erm, I mean controlling people's minds for the benevolent purpose of helping them to seek truth and personal enlightenment. As long as they are personally enlightened to worship me. In any case, stare into my lantern to discover more about using covariates and ANCOVA.'



13.5.7 Bootstrapping and plots

There are other options available from the main dialog box. For example, if you have several independent variables you can plot them against each other (which is useful for interpreting interaction effects – see Section 14.7). There's also the Bootstrap... button, which you can use to activate bootstrapping. Selecting this option will bootstrap confidence intervals around the estimated marginal means, parameter estimates (*b*-values) and *post hoc* tests, but not the main *F*-statistic.

Select the options described in Section 6.12.3 and click OK in the

main dialog box to run the analysis.

13.6 Interpreting ANCOVA

13.6.1 What happens when the covariate is excluded?

Output 13.5 shows (for illustrative purposes) the ANOVA table for these data when the covariate is not included. It is clear from the significance value, which is greater than 0.05, that puppy therapy seems to have no significant effect on happiness. Note that the total amount of variation in happiness (SS_T) was 110.97 (*Corrected Total*), of which the therapy condition accounted for 16.84 units (SS_M), while 94.12 were unexplained (SS_R).

Output 13.5

Tests of Between-Subjects Effects

Dependent Variable: Happiness (0-10)

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	16.844[a]	2	8.422	2.416	.108
Intercept	535.184	1	535.184	153.522	.000
Dose	16.844	2	8.422	2.416	.108
Error	94.123	27	3.486		
Total	683.000	30			
Corrected Total	110.967	29			

a. R Squared = .152 (Adjusted R Squared = .089)

Output 13.6

Tests of Between-Subjects Effects

Dependent Variable: Happiness (0-10)

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	31.920[a]	3	10.640	3.500	.030
Intercept	76.069	1	76.069	25.020	.000
Puppy_love	15.076	1	15.076	4.959	.035
Dose	25.185	2	12.593	4.142	.027
Error	79.047	26	3.040		
Total	683.000	30			
Corrected Total	110.967	29			

a. R Squared = .288 (Adjusted R Squared = .205)



Fit the model without the covariate to see whether the three groups

differ in their levels of happiness.

13.6.2 The main analysis

The format of the ANOVA table in Output 13.6 is largely the same as without the covariate, except that there is an additional row of information about the covariate (**Puppy_love**). Looking first at the significance values, the covariate significantly predicts the dependent variable (p = 0.035, which is less than 0.05). Therefore, the person's happiness is significantly influenced by their love of puppies. What's more interesting is that when the effect of love of puppies is removed, the effect of puppy therapy is significant (p = 0.027, which is less than 0.05). The amount of variation accounted for by puppy therapy has increased to 25.19 units and the unexplained variance (SS_R) has been reduced to 79.05 units. Notice that SS_T has not changed; all that has changed is how that total variation is partitioned.[5]

5 I am often asked what the *Corrected Model* represents in this table. It is the fit of the model overall (i.e., the model containing the intercept, **Puppy_love** and **Dose**). Note that the SS of 31.92, *df* of 3, F of 3.5 and p of 0.03 are identical to the values in Output 13.1 (model 2), which tested the overall fit of this model when we ran the analysis as a regression.



This example illustrates how covariates can help us to exert stricter experimental control by taking account of confounding variables to give us a 'purer' measure of effect of the experimental manipulation. Looking back at the group means from Table 13.1, you might think that the significant F-statistic reflects a difference between the control group and the two experimental groups – because the 15-and 30-minute groups have very similar means (4.88 and 4.85) whereas the control group mean is much lower at 3.22. However, we can't use these group means to interpret the effect because they have not been adjusted for the effect of the covariate. These original means tell us nothing about the group differences reflected by the significant F. Output 13.7 gives the adjusted values of the group means (which we calculated in Section 13.3), and we use these values for interpretation (this is why we selected *Display Means for* in Section 13.5.6). From these adjusted means you can see that happiness increased across

the three doses.

Output 13.7

Estimates

Dependent Variable: Happiness (0–10)

Dose of puppies	Mean	Std. Error	95% Confidence Interval		Bootstrap for Mean[gn]			
							BCa 95% Confidence Interval	
			Lower Bound	Upper Bound	Bias	Std. Error	Lower	Upper
Control	2.926[a]	.596	1.701	4.152	.030	.446	2.111	4.125
15 mins	4.712[a]	.621	3.436	5.988	.033[go]	.392[go]	3.988[go]	5.620[go]
30 mins	5.151[a]	.503	4.118	6.184	.041	.651	3.923	6.771

a. Covariates appearing in the model are evaluated at the following values: Love of puppies (0–7) = 2.73.

gn. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

go. Based on 999 samples

Output 13.8 shows the parameter estimates selected in the *Options* dialog box and their bootstrapped confidence intervals and *p*-values (bottom table). These estimates result from **Dose** being coded using two dummy coding variables. The dummy variables are coded with the last category (the category coded with the highest value in the data editor, in this case the 30-minute group) as the reference category. This reference category (labelled Dose=3 in the output) is coded with a 0 for both dummy variables (see Section 12.2 for a reminder of how dummy coding works). Dose=2, therefore, represents the difference between the group coded as 2 (15 minutes) and the reference category (30 minutes), and Dose=1 represents the difference between the group coded as 1 (control) and the reference category (30 minutes). The *b*-values represent the differences between the adjusted means in Output 13.7 and the significances of the *t*-tests tell us whether these adjusted group means differ significantly. The *b* for Dose=1 in Output 13.8 is the difference between the adjusted means for the control group and the 30-minute group, 2.926 − 5.151 = −2.225, and the *b* for Dose=2 is the difference between the adjusted means for the 15-minute group and the 30-minute group, 4.712 − 5.151 = −0.439.

The degrees of freedom for the *t*-test of the *b*-values are $N - k - 1$ (see Section 9.2.5), in which N is the total sample size (in this case 30) and k is the number of predictors (in this case 3, the two dummy variables and the covariate). For these data, $df = 30 - 3 - 1 = 26$. Based on the bootstrapped significance and confidence intervals (remember you'll get different values than me because of how bootstrapping works), we could conclude that the 30-minute group differs significantly from the control group, $p = 0.021$ (Dose=1 in the table), but not from the 15-minute group, $p = 0.558$, (Dose=2 in the table).

The final thing to note is the value of *b* for the covariate (0.416), which is the same as in Output 13.2 (when we ran the analysis through the regression menu). This value tells us that if love of puppies increases by one unit, then the person's happiness should increase by just under half a unit (although there is nothing to suggest a causal link between the two); because the coefficient is positive we

know that as love of puppies increases so does happiness. A negative coefficient would mean the opposite: as one increases, the other decreases.

Output 13.9 repeats the parameter estimates from Output 13.8 but with standard errors, *p*-values and confidence intervals robust to heteroscedasticity (the HC4 estimates that we asked for). We can interpret the effects for **Dose** in the same way as for the regular and Bootstrap *p*-values and confidence intervals. For the effect of puppy love, the HC4 robust confidence interval and *p*-value supports the conclusion from the non-robust model: the *p*-value is 0.038, which is less than 0.05, and the confidence interval does not contain zero (0.025, 0.807). However, the bootstrap confidence interval (Output 13.8) contradicts this conclusion because it contains zero (−0.023, 0.698) and has a $p = 0.052$ (again, we're reminded of how daft it is to have a threshold that yields such opposing conclusions from such small differences in a value).

Output 13.8

Parameter Estimates

Dependent Variable: Happiness (0-10)

Parameter	B	Std. Error	t	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Intercept	4.014	.611	6.568	.000	2.758	5.270
Puppy_love	.416	.187	2.227	.035	.032	.800
[Dose=1]	-2.225	.803	-2.771	.010	-3.875	-.575
[Dose=2]	-.439	.811	-.541	.593	-2.107	1.228
[Dose=3]	0[a]	.	.	.	.	.

a. This parameter is set to zero because it is redundant.

Bootstrap for Parameter Estimates

Dependent Variable: Happiness (0-10)

Parameter	B	Bootstrap[a]				
		Bias	Std. Error	Sig. (2-tailed)	BCa 95% Confidence Interval	
					Lower	Upper
Intercept	4.014	.091[b]	.843[b]	.003[b]	1.969[b]	5.949[b]
Puppy_love	.416	-.029[b]	.202[b]	.052[b]	-.023[b]	.698[b]
[Dose=1]	-2.225	-.011[b]	.760[b]	.021[b]	-3.753[b]	-.823[b]
[Dose=2]	-.439	-.008[b]	.745[b]	.558[b]	-1.937[b]	.935[b]
[Dose=3]	0	0[b]	0[b]		.[b]	.[b]

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

b. Based on 999 samples

Output 13.9

Parameter Estimates with Robust Standard Errors

Dependent Variable: Happiness (0–10)

Parameter	B	Robust Std. Error[a]	t	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Intercept	4.014	.805	4.989	.000	2.360	5.668
Puppy_love	.416	.190	2.187	.038	.025	.807
[Dose=1]	-2.225	.690	-3.226	.003	-3.642	-.807
[Dose=2]	-.439	.695	-.632	.533	-1.868	.990
[Dose=3]	0[b]	.	.	.	.	.

a. HC4 method

b. This parameter is set to zero because it is redundant.

Output 13.10

Contrast Results (K Matrix)

Dose of puppies Simple Contrast[a]			Dependent Variable Happiness (0–10)
Level 2 vs. Level 1	Contrast Estimate		1.786
	Hypothesized Value		0
	Difference (Estimate – Hypothesized)		1.786
	Std. Error		.849
	Sig.		.045
	95% Confidence Interval for Difference	Lower Bound	.040
		Upper Bound	3.532
Level 3 vs. Level 1	Contrast Estimate		2.225
	Hypothesized Value		0
	Difference (Estimate – Hypothesized)		2.225
	Std. Error		.803
	Sig.		.010
	95% Confidence Interval for Difference	Lower Bound	.575
		Upper Bound	3.875

a. Reference category = 1

Output 13.11

Pairwise Comparisons

Dependent Variable: Happiness (0-10)

(I) Dose of puppies	(J) Dose of puppies	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b] Lower Bound	Upper Bound
Control	15 mins	-1.786	.849	.130	-3.953	.381
	30 mins	-2.225*	.803	.030	-4.273	-.177
15 mins	Control	1.786	.849	.130	-.381	3.953
	30 mins	-.439	.811	.932	-2.509	1.631
30 mins	Control	2.225*	.803	.030	.177	4.273
	15 mins	.439	.811	.932	-1.631	2.509

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Sidak.

Bootstrap for Pairwise Comparisons

Dependent Variable: Happiness (0-10)

(I) Dose of puppies	(J) Dose of puppies	Mean Difference (I-J)	Bootstrap[a] Bias	Std. Error	Sig. (2-tailed)	BCa 95% Confidence Interval Lower	Upper
Control	15 mins	-1.786	-.003[b]	.535[b]	.003[b]	-2.778[b]	-.765[b]
	30 mins	-2.225	-.011	.760	.021	-3.752	-.832
15 mins	Control	1.786	.003[b]	.535[b]	.003[b]	.663[b]	2.879[b]
	30 mins	-.439	-.008[b]	.745[b]	.558[b]	-1.937[b]	.935[b]
30 mins	Control	2.225	.011	.760	.021	.686	3.923
	15 mins	.439	.008[b]	.745[b]	.558[b]	-.938[b]	1.945[b]

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

b. Based on 999 samples

13.6.3 Contrasts



Output 13.10 shows the result of the contrast analysis specified in Figure 13.6 and compares level 2 (15 minutes) against level 1 (control) as a first comparison, and level 3 (30 minutes) against level 1 (control) as a second comparison. The group differences are displayed: a difference value, standard error, significance value and 95% confidence interval. These results show that both the 15-minute group (contrast 1, $p = 0.045$) and 30-minute group (contrast 2, $p = 0.010$) had significantly different happiness compared to the control group (note that contrast 2 is identical to the parameter for Dose=1 in the previous section). Output 13.11 shows the results of the Šidák corrected *post hoc* comparisons that were requested in Section 13.5.6. The bottom table shows the bootstrapped significance and confidence intervals for these tests and because these will be robust we'll interpret this table (again, remember your values will differ because of how bootstrapping works). There is a significant difference between the control group and both the 15- ($p = 0.003$) and 30-minute ($p = 0.021$) groups. The 30-and 15-minute groups did not significantly differ ($p = 0.558$). It is interesting that the significant difference between the 15-minute and control groups when bootstrapped ($p = 0.003$) is not present for the normal *post hoc* tests ($p = 0.130$). This anomaly could reflect properties of the data that have biased

the non-robust version of the *post hoc* test.

13.6.4 Interpreting the covariate

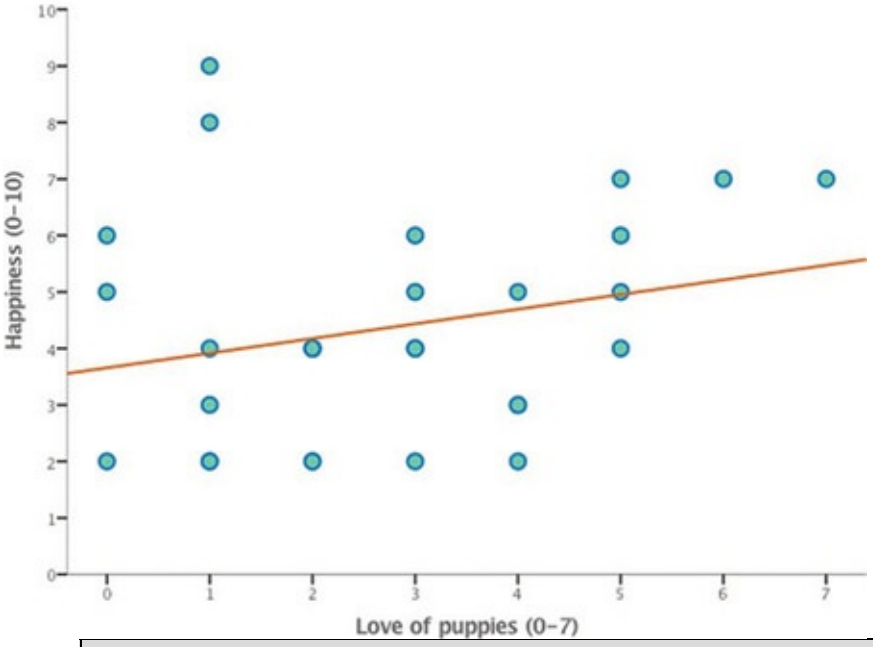
I've already mentioned that the parameter estimates (Output 13.8) tell us how to interpret the covariate: the sign of the *b*-value tells us the direction of the relationship between the covariate and outcome variable. For these data the *b*-value was positive, indicating that as the love of puppies increases, so does the participant's happiness. Another way to discover the same thing is to draw a scatterplot of the covariate against the outcome.

SELF TEST

Produce a scatterplot of love of puppies (horizontal axis) against happiness (vertical axis).

Figure 13.8 confirms that the effect of the covariate is that as love of puppies increases, so does the participant's happiness (as shown by the slope of the line).

Figure 13.8 Scatterplot of happiness against love of puppies



Labcoat Leni's Real Research 13.1 Space invaders



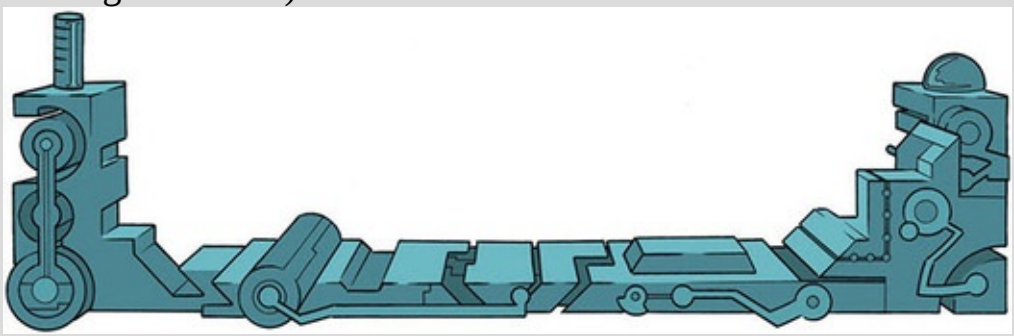
Muris, P., *et al.* (2008). *Child Psychiatry and Human Development, 39*(4), 469–480.

Anxious people tend to interpret ambiguous information in a negative way. For example, being highly anxious myself, if I overheard a student saying 'Andy Field's lectures are really *different*,' I would assume that 'different' meant rubbish, but it could also mean 'refreshing' or 'innovative'. Muris, Huijding, Mayer, and Hameetman (2008) addressed how these interpretational biases develop in children. Children imagined that they were astronauts who had discovered a new planet. They were given scenarios about their time on the planet (e.g., 'On the street, you encounter a spaceman. He has a toy handgun and he fires at you …') and the child had to decide whether a positive ('You laugh: it is a water pistol and the weather is fine anyway') or negative ('Oops, this hurts! The pistol produces a red beam which burns your skin!') outcome occurred. After each response the child was told whether their choice was correct. Half of the children were *always* told that the negative interpretation was correct, and the reminder were told that the positive interpretation was correct.

Over 30 scenarios children were trained to interpret their experiences on the planet as negative or positive. Muris *et al.* then measured interpretational biases in everyday life to see whether the training had created a bias to interpret things negatively. In doing so, they could ascertain whether children might learn interpretational biases through feedback (e.g., from parents).

The data from this study are in the file **Muris et al (2008).sav**. The independent variable is **Training** (positive or negative) and the outcome is the child's interpretational bias score (**Interpretational_Bias**) – a high score reflects a tendency to interpret situations negatively. It is important to adjust for the **Age**

and **Gender** of the child and also their natural anxiety level (which they measured with a standard questionnaire of child anxiety called the **SCARED**) because these things affect interpretational biases also. Labcoat Leni wants you to fit a model to see whether **Training** significantly affected children's **Interpretational_Bias** using **Age, Gender** and **SCARED** as covariates. What can you conclude? Answers are on the companion website (or look at pages 475–476 in the original article).



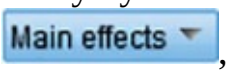
13.7 Testing the assumption of homogeneity of regression slopes

Remember that the assumption of homogeneity of regression slopes means that the relationship between the covariate and outcome variable (in this case **Puppy_love** and **Happiness**) should be similar at different levels of the predictor variable (in this case in the three **Dose** groups). Figure 13.3 shows that the relationship between **Puppy_love** and **Happiness** looks comparable in the 15-minute and control groups, but seems different in the 30-minute group.

To test the assumption of homogeneity of regression slopes we need to refit the model but customize it to include the interaction between the covariate and categorical predictor. Access the main dialog box as before and place the variables in the same boxes as before (the finished dialog box should look like Figure 13.5). To customize the model, click 

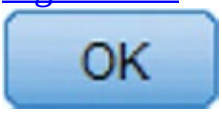
to access the dialog box in Figure 13.9 and select 

. The variables specified in the main dialog box are listed on the left-hand side. We need a model that includes the interaction between the covariate and grouping variable. To test this interaction term it's important to also include the main effects otherwise variance in the outcome (happiness) may be attributed to the interaction term that would otherwise be attributed to the main effects. To begin with, then, select **Dose** and

Puppy_love (you can select both simultaneously by holding down *Ctrl*, or *Cmd* on a Mac), change the drop-down menu to 

, and click  to transfer the main effects of **Dose** and **Puppy_love** to the box labelled *Model*. Next specify the interaction term by selecting **Dose** and **Puppy_love** simultaneously (as just described), change the drop-down menu to 

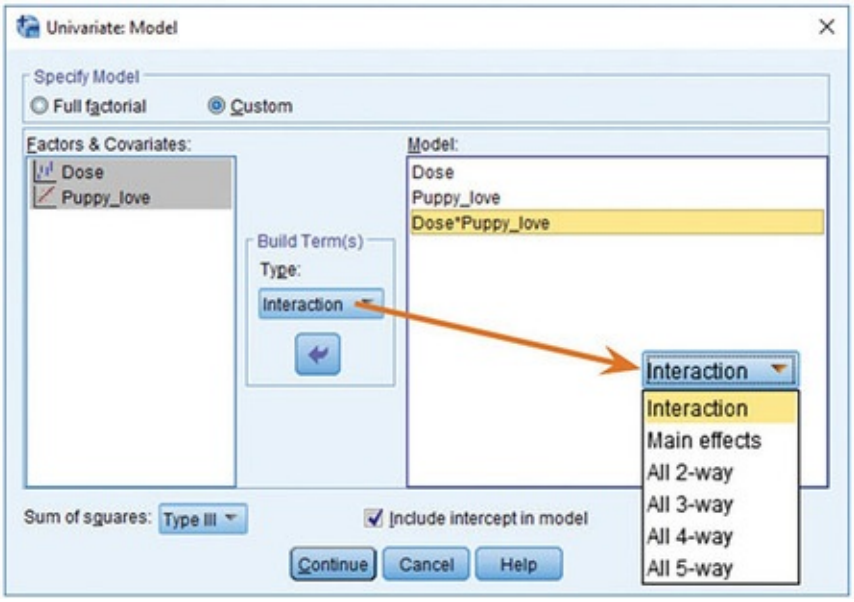
and click  to transfer the interaction of **Dose** and **Puppy_love** to the box labelled *Model*. The finished dialog box should look like Figure 13.9. Click 

to return to the main dialog box and 

to run the analysis.

Output 13.11 shows the main summary table for the model including the interaction term. The effects of the dose of puppy therapy and love of puppies are still significant, but so is the covariate by outcome interaction (**Dose** × **Puppy_love**), implying that the assumption of homogeneity of regression slopes is not realistic (p = 0.028). Although this finding is not surprising given the pattern of relationships shown in Figure 13.3, it raises concerns about the main analysis.

Figure 13.9 *Model* dialog box for GLM univariate



Output 13.12

Tests of Between-Subjects Effects

Dependent Variable: Happiness (0-10)

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	52.346[a]	5	10.469	4.286	.006
Intercept	53.542	1	53.542	21.921	.000
Dose	36.558	2	18.279	7.484	.003
Puppy_love	17.182	1	17.182	7.035	.014
Dose * Puppy_love	20.427	2	10.213	4.181	.028
Error	58.621	24	2.443		
Total	683.000	30			
Corrected Total	110.967	29			

a. R Squared = .472 (Adjusted R Squared = .362)

Cramming Sam's Tips Covariates



- When the linear model is used to compare several means adjusted for the effect of one or more other variables (called *covariates*) it can be referred to as analysis of covariance (ANCOVA).
- Before the analysis check that the covariate(s) are independent

of any independent variables by seeing whether those independent variables predict the covariate (i.e., the covariate should not differ across groups).
- In the table labelled *Tests of Between-Subjects Effects*, assuming you're using an alpha of 0.05, look to see if the value in the column labelled *Sig.* is below 0.05 for both the covariate and the independent variable. If it is for the covariate then this variable has a significant relationship to the outcome variable; if it is for the independent variable then the means (adjusted for the effect of the covariate) are significantly different across categories of this variable.
- If you have generated specific hypotheses before the experiment use planned contrasts; if not, use *post hoc* tests.
- For parameters and *post hoc* tests, look at the columns labelled *Sig.* to discover if your comparisons are significant (they will be if the significance value is less than 0.05). Use bootstrapping to get robust versions of these tests.
- In addition to the assumptions in Chapter 6, test for *homogeneity of regression slopes* by customizing the model to look at the independent variable × covariate interaction.

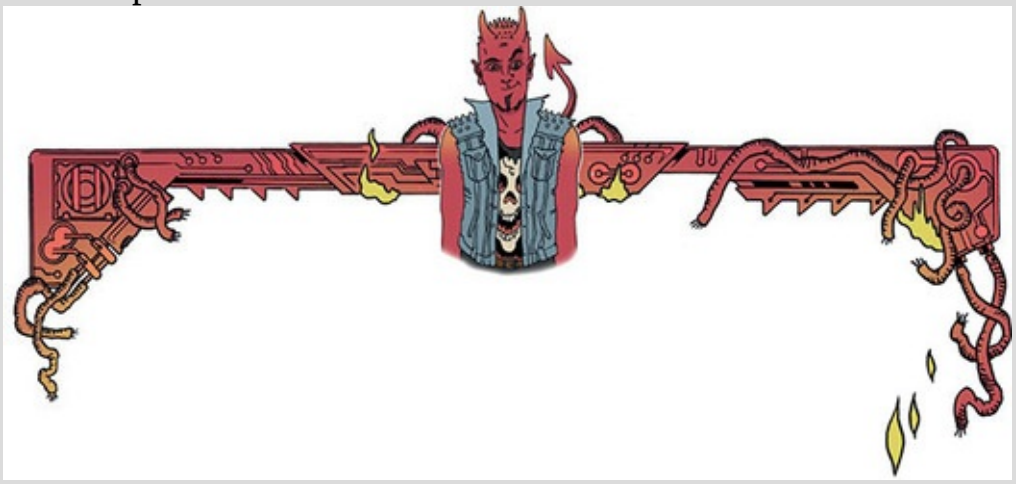


13.8 Robust ANCOVA

We have already looked at robust confidence intervals and *p*-values for the model parameters that were computed using bootstrapping and heteroscedasticity robust standard errors (Section 13.6.2). In addition, the companion website contains a syntax file (**robustANCOVA.sps**) for running a robust variant of ANCOVA (*ancboot*) that works on trimmed means and is described by Wilcox (2017). We need the *Essentials for R* plugin and *WRS2* package installed (Section 4.13). This test is limited to the situation where the independent variable (the categorical predictor) has two categories and there is one covariate. But it does enable you to ignore assumptions and get on with your life. Because this syntax only works when you have two groups, I have provided a data file called **PuppiesTwoGroup.sav**, which contains the example data for

this chapter but excluding the 15-minute condition, so it compares the control (no puppies) with the 30-minute group (**Dose**), and has the scores for the love of puppies covariate too (**Puppy_love**). The syntax to run the robust test is as follows: BEGIN PROGRAM R.
library(WRS2)
mySPSSdata = spssdata.GetDataFromSPSS(factorMode = "labels")
ancboot(Happiness ~ Dose + Puppy_love, data = mySPSSdata, tr = 0.2, nboot = 1000) END PROGRAM.
Select and run these five lines of syntax (see SPSS Tip 10.3). As Output 13.13 shows, the test works by identifying values of the covariate for which the relationship between the covariate and outcome are comparable in the two groups. In this example it identifies five values of **Puppy_love** (2, 3, 5, 6, and 8) for which the relationship between love of puppies and happiness is comparable. At each of these design points, we're told the number of cases for the two groups (*n1* and *n2*) that have a value of the covariate (**Puppy_love**) close to these design points (not exactly *x*, but close to it). Based on these two samples, trimmed means (20% by default) are computed and the difference between them tested. This difference is stored in the column *Diff* along with the boundaries of the associated 95% bootstrap confidence interval (corrected to control for doing five tests) in the next two columns. The test statistic comparing the difference is in the column *statistic*, with its *p*-value in the final column. Output 13.12 shows no significant differences between trimmed means for any of the design points (all *p*-values are greater than 0.05).

SPSS Tip 13.2 Robust ANCOVA



The function *ancboot* takes the form:

ancboot(outcome ~ categorical predictor + covariate, data =

```
        mySPSSdata, tr = 0.2, nboot = 1000)
```

In our example we would replace outcome with **Happiness,** categorical predictor with **Dose** and covariate with **Puppy_love**. *tr* relates to the amount of trimming to the data (by default 0.2, or 20%, so change it if you want to use a different proportion). *nboot* refers to the number of bootstrap samples which I have set to a reasonable 1000, but feel free to increase it.



Output 13.13

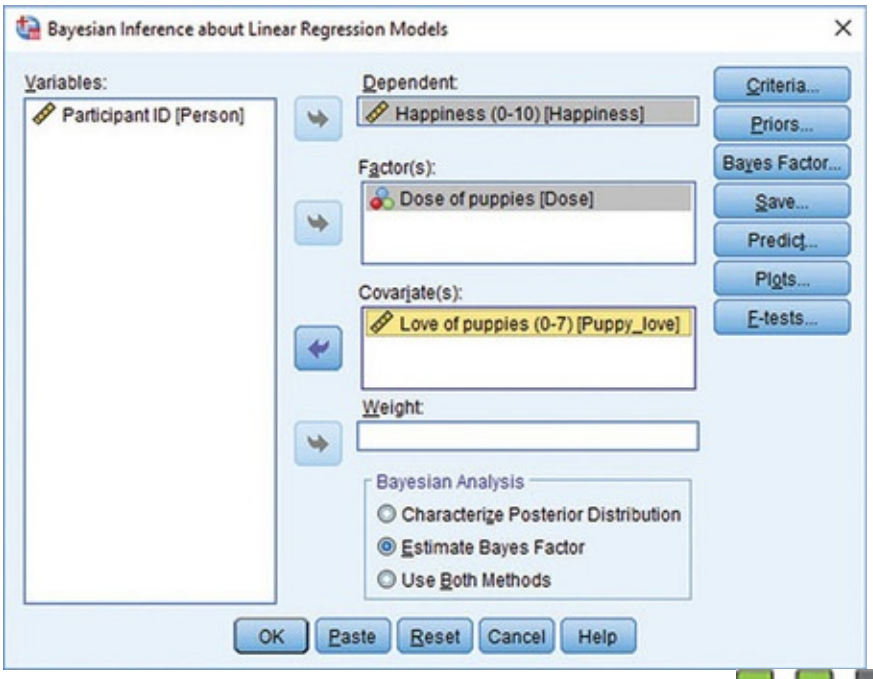
	n1	n2	diff	lower CI	upper CI	statistic	p-value
Puppy_love = 2	22	13	-1.0873	-3.1547	0.9801	-1.6952	0.098
Puppy_love = 3	27	15	-1.0719	-2.8097	0.6659	-1.9881	0.058
Puppy_love = 5	30	22	-0.6508	-2.4220	1.1204	-1.1843	0.250
Puppy_love = 6	23	21	-0.9846	-3.3281	1.3589	-1.3542	0.207
Puppy_love = 8	12	13	-1.5278	-4.3223	1.2667	-1.7622	0.119



13.9 Bayesian analysis with covariates

Because the model we have fitted is a linear model with a categorical predictor and a continuous predictor, you can use what you learned in Section 9.13 to run a Bayesian regression. You would need to manually create dummy variables (as in the file **Puppy Love Dummy.sav**) and drag these to the box labelled *Factor(s)* and drag **Puppy_Love** to the box labelled *Covariate(s)* see Figure 13.10. You would interpret in the same way as the model we fitted in Section 9.13.

Figure 13.10



13.10 Calculating the effect size

In the previous chapter we used eta squared, η^2, as an effect size measure when comparing means (Section 12.10). When we include a covariate too we have more than one effect and we could calculate eta squared for each effect. We can also use an effect size measure called **partial eta squared (partial η2)**. This differs from eta squared in that it looks not at the proportion of total variance that a variable explains, but at the proportion of variance that a variable explains that *is not explained by other variables in the analysis*. Let's look at this with our example. Suppose we want to know the effect size of the dose of puppy therapy. Partial eta squared is the proportion of variance in happiness that the dose of puppy therapy shares that is not attributed to love of puppies (the covariate). If you think about the variance that the covariate cannot explain, there are two sources: it cannot explain the variance attributable to the dose of puppy therapy, $SS_{\text{puppy therapy}}$, and it cannot explain the error variability, SS_R. Therefore, we use these two sources of variance instead of the total variability, SS_T, in the calculation. The difference between eta squared and partial eta squared is illustrated by comparing the following two equations:

$$\eta^2 = \frac{SS_{\text{Effect}}}{SS_{\text{Total}}} \quad (13.6)$$

$$\text{Partial } \eta^2 = \frac{SS_{\text{Effect}}}{SS_{\text{Effect}} + SS_{\text{Residual}}} \quad (13.7)$$

SPSS Statistics will produce partial eta squared for us (see Jane Superbrain Box

13.3), but to illustrate its calculation look at equation (13.8), where we use the sums of squares in Output 13.6 for the effect of dose (25.19), the covariate (15.08) and the error (79.05):

$$\text{Partial } \eta^2_{\text{Dose}} = \frac{SS_{\text{Dose}}}{SS_{\text{Dose}} + SS_{\text{Residual}}} = \frac{25.19}{25.19 + 79.05} = \frac{25.19}{104.24} = 0.24$$
$$\text{Partial } \eta^2_{\text{Puppy_love}} = \frac{SS_{\text{Puppy_love}}}{SS_{\text{Puppy_love}} + SS_{\text{Residual}}} = \frac{15.08}{15.08 + 79.05} = \frac{15.08}{94.13} = 0.16 \quad (13.8)$$

These values show that **Dose** explained a bigger proportion of the variance not attributable to other variables than **Puppy_love**.



Rerun the analysis but select ☑ Estimates of effect size in Figure 13.7. Do the values of partial eta squared match the ones we have just calculated?

You can also use omega squared (ω^2). However, as we saw in Section 12.8, this measure can be calculated only when we have equal numbers of participants in each group (which is not the case in this example). So, we're a bit stumped! Not all is lost, though, because, as I've said many times already, the overall effect size is not nearly as interesting as the effect size for more focused comparisons. These are easy to calculate because we selected to see the model parameters (see Output 13.8) and so we have *t*-statistics for the covariate and comparisons between the 15-and 30-minute groups and the control and 30-minute group. These *t*-statistics have 26 degrees of freedom (see Section 13.6.1). We can use the same equation as in Section 10.9.5:[6]

6 Strictly speaking, we should use a slightly more elaborate procedure when groups are unequal. It's a bit beyond the scope of this book, but Rosnow, Rosenthal, and Rubin (2000) give a very clear account.

$$r_{\text{Contrast}} = \sqrt{\frac{t^2}{t^2 + df}} \quad (13.9)$$

Therefore, we get (using *t* from Output 13.8) values of 0.40 for the covariate, and 0.48 and 0.11 respectively for the comparison of the 30-minute group and control, and the 15-and 30-minute groups:

$$r_{\text{Covariate}} = \sqrt{\frac{2.23^2}{2.23^2 + 26}} = \sqrt{\frac{4.97}{30.97}} = 0.40$$
$$r_{30\text{ mins vs. control}} = \sqrt{\frac{(-2.77)^2}{(-2.77)^2 + 26}} = \sqrt{\frac{7.67}{33.67}} = 0.48 \quad (13.10)$$
$$r_{30\text{ vs. }15\text{ mins}} = \sqrt{\frac{(-0.54)^2}{(-0.54)^2 + 26}} = \sqrt{\frac{0.29}{26.29}} = 0.11$$

For the effect of the covariate and the difference between the 30-minute and control groups the effects are not only statistically significant but also substantive in size. The difference between the 30-and 15-minute groups was a fairly small effect.

13.11 Reporting results 

When using covariates you can report the model in much the same way as any other. For the covariate and the experimental effect give details of the *F*-statistic and the degrees of freedom from which it was calculated. In both cases, the *F*-statistic was derived from dividing the mean squares for the effect by the mean squares for the residual. Therefore, the degrees of freedom used to assess the *F*-statistic are the degrees of freedom for the effect of the model (df_M = 1 for the covariate and 2 for the experimental effect) and the degrees of freedom for the residuals of the model (df_R = 26 for both the covariate and the experimental effect) – see Output 13.6. The correct way to report the main findings would be:

> ✓ The covariate, love of puppies, was significantly related to the participant's happiness, $F(1, 26) = 4.96$, $p = 0.035$, $r = 0.40$. There was also a significant effect of puppy therapy on levels of happiness after controlling for the effect of love of puppies, $F(2, 26) = 4.14$, $p = 0.027$, partial $\eta^2 = 0.24$.

We can also report some contrasts (see Output 13.8):

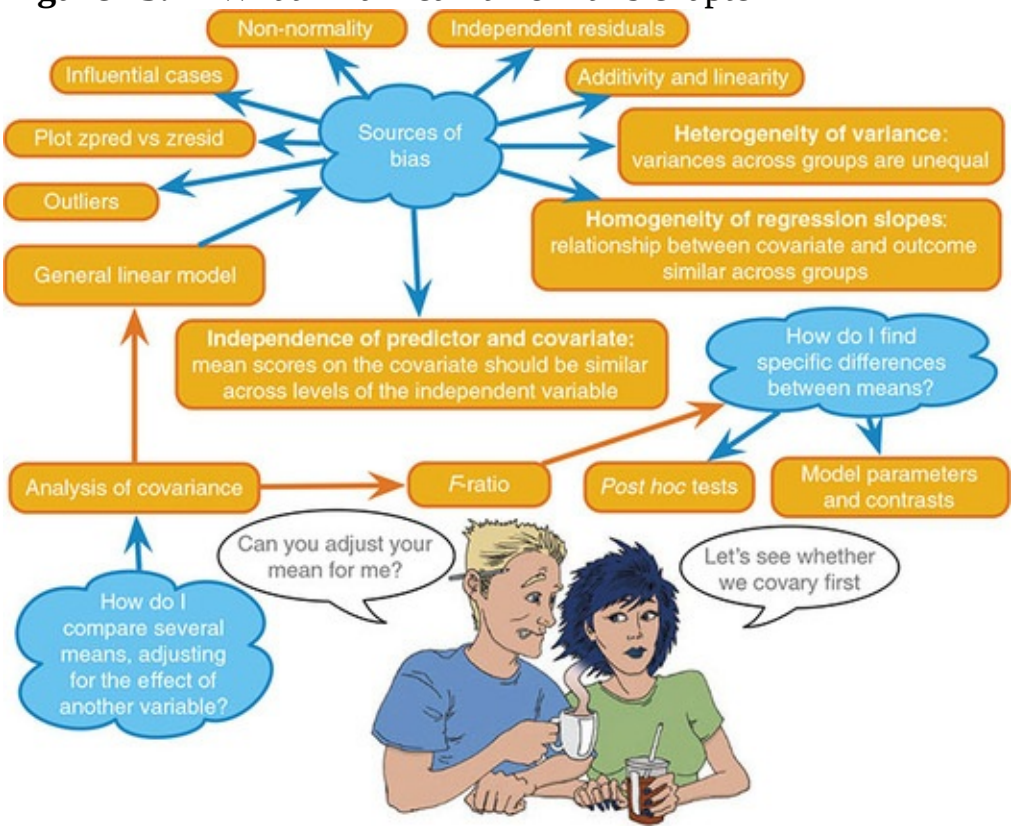
> ✓ Planned contrasts revealed that having 30 minutes of puppy therapy significantly increased happiness compared to having a control, $t(26) = -2.77$, $p = 0.01$, $r = 0.48$, but not compared to having 15 minutes, $t(26) = -0.54$, $p = 0.59$, $r = 0.11$.

13.12 Brian's attempt to woo Jane 

The encounter in Blow Your Speakers had been beyond weird. Jane felt terrible. This Brian guy was so nice to her, and she'd just told him where to go – again! It

had been easy to dismiss Brian at first, he'd seemed like a loser, a waste of her time. But there was more to him than that: he'd been working hard to learn statistics, and he'd made impressive progress. She liked how awkward he was around her, and how he always defaulted to talking stats. It was endearing. It could derail her research, though, and he could never know about that. She was a monster, and if he found out the truth it would be another let-down. Best to keep her distance.

Figure 13.11 What Brian learnt from this chapter



The phone rang. It was her brother, Jake. She loved and admired Jake like no one else. Until he left home, he'd been her sanity in the madhouse that they grew up in. Their parents, both highly successful academics, were at home only long enough to pile the pressure on them both to succeed. Jane reacted by spending her youth in books, in a futile pursuit of their attention. Every set of straight As was met with 'these are just a step towards the exams that really matter, you'll need to up your game'. She was tired of trying to impress them. Jake was her opposite – he'd realized early on that he could never win. He let the pressure roll off him, and left home as soon as he could. But he always looked out for Jane.

'Mum is in hospital,' he said as the blood drained from Jane's legs.

'I don't care,' she replied, but she did. She also wanted to see Brian, because he was the closest thing she had to a friend in this town.

13.13 What next? 

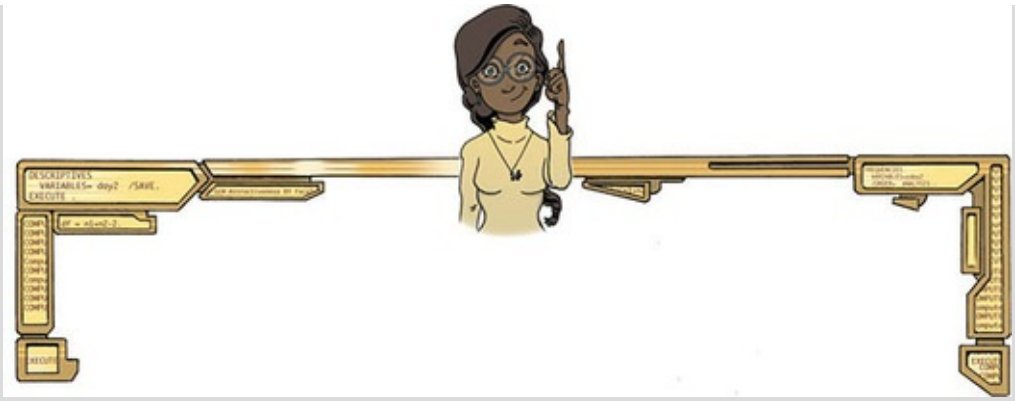
At the age of 13 I met my heroes, Iron Maiden, and very nice they were too. I've met them a couple of times since (not because they're my best buddies or anything exciting like that, but over the years the fan club put on various events where you were allowed to stand next to them and gibber like a fool while they humoured you politely). You'll notice that the photo at the start of this chapter is signed by Dave Murray. This wasn't possible because I had my own darkroom installed backstage at the Hammersmith Odeon in which I could quickly process photographs, or because I had access to time travel (sadly), but because I took the photo with me when I met him in 2000. I recounted the tale of how terrified I had been about meeting him in 1986. If he thought I was some strange stalker he certainly didn't let on. Uncharacteristic of most people who've sold millions of albums, they're lovely blokes.

Anyway, having seen Iron Maiden in their glory, I was inspired. They still inspire me: I still rate them as the best live band I've ever seen (and I've seen them over 35 times, so I ought to know). Although I had briefly been deflected from my destiny by the shock of grammar school, I was back on track. I *had* to form a band. There was just one issue: no one else played a musical instrument. The solution was easy: through several months of covert subliminal persuasion I convinced my two best friends (both called Mark, oddly enough) that they wanted nothing more than to start learning the drums and bass guitar. A power trio was in the making.

13.14 Key terms that I've discovered

Adjusted mean
Analysis of covariance (ANCOVA)
Covariate
Homogeneity of regression slopes
Partial eta squared (partial η^2)
Šidák correction

Smart Alex's tasks



- **Task 1**: A few years back I was stalked. You'd think they could have found someone a bit more interesting to stalk, but apparently times were hard. It could have been a lot worse, but it wasn't particularly pleasant. I imagined a world in which a psychologist tried two different therapies on different groups of stalkers (25 stalkers in each group – this variable is called **Group**). To the first group he gave cruel-to-be-kind therapy (every time the stalkers followed him around, or sent him a letter, the psychologist attacked them with a cattle prod). The second therapy was psychodyshamic therapy, in which stalkers were hypnotized and regressed into their childhood to discuss their penis (or lack of penis), their father's penis, their dog's penis, the seventh penis of a seventh penis and any other penis that sprang to mind. The psychologist measured the number of hours stalking in one week both before (**stalk1**) and after (**stalk2**) treatment (**Stalker.sav**). Analyse the effect of therapy on stalking behaviour after therapy, covarying for the amount of stalking behaviour before therapy. 
- **Task 2**: Compute effect sizes for Task 1 and report the results. 
- **Task 3**: A marketing manager tested the benefit of soft drinks for curing hangovers. He took 15 people and got them drunk. The next morning as they awoke, dehydrated and feeling as though they'd licked a camel's sandy feet clean with their tongue, he gave five of them water to drink, five of them Lucozade (a very nice glucose-based UK drink) and the

remaining five a leading brand of cola (this variable is called **drink**). He measured how well they felt (on a scale from 0 = I feel like death to 10 = I feel really full of beans and healthy) two hours later (this variable is called **well**). He measured how **drunk** the person got the night before on a scale of 0 = as sober as a nun to 10 = flapping about like a haddock out of water on the floor in a puddle of their own vomit (**HangoverCure.sav**). Fit a model to see whether people felt better after different drinks when covarying for how drunk they were the night before. 

- **Task 4**: Compute effect sizes for Task 3 and report the results. 
- **Task 5**: The highlight of the elephant calendar is the annual elephant soccer event in Nepal (google search it). A heated argument burns between the African and Asian elephants. In 2010, the president of the Asian Elephant Football Association, an elephant named Boji, claimed that Asian elephants were more talented than their African counterparts. The head of the African Elephant Soccer Association, an elephant called Tunc, issued a press statement that read 'I make it a matter of personal pride never to take seriously any remark made by something that looks like an enormous scrotum'. I was called in to settle things. I collected data from the two types of elephants (**elephant**) over a season and recorded how many goals each elephant scored (**goals**) and how many years of experience the elephant had (**experience**). Analyse the effect of the type of elephant on goal scoring, covarying for the amount of football experience the elephant has (**Elephant Football.sav**). 
- **Task 6**: In Chapter 4 (Task 6) we looked at data from people who had been forced to marry goats and dogs and measured their life satisfaction and also how much they like animals (**Goat or Dog.sav**). Fit a model predicting life satisfaction from

the type of animal to which a person was married and their animal liking score (covariate).

- **Task 7**: Compare your results for Task 6 to those for the corresponding task in Chapter 11. What differences do you notice and why?
- **Task 8**: In Chapter 10 we compared the number of mischievous acts (**mischief2**) in people who had invisibility cloaks to those without (**cloak**). Imagine we also had information about the baseline number of mischievous acts in these participants (**mischief1**). Fit a model to see whether people with invisibility cloaks get up to more mischief than those without when factoring in their baseline level of mischief (**Invisibility Baseline.sav**).

Answers & additional resources are available on the book's website at **https://edge.sagepub.com/field5e**



14 GLM 3: Factorial designs

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14.1 What will this chapter tell me?

After persuading my two friends (Mark and Mark) to learn the bass and drums, I took the rather odd decision to *stop* playing the guitar. I didn't stop, as such, but I focused on singing instead. In retrospect, this was a bad decision because I am *not* a good singer. Mind you, I'm not a good guitarist either. The upshot was that a classmate, Malcolm, ended up as our guitarist. I can't recall how or why we ended up in this configuration, but we called ourselves Andromeda, we learnt several Queen and Iron Maiden songs and we were truly awful. I have some recordings somewhere to prove just what a cacophony of tuneless drivel we produced, but the chances of them appearing on the companion website are slim at best. Suffice it to say, you'd be hard pushed to recognize *which* Iron Maiden and Queen songs we were trying to play. The fact that we were only 14 or 15 at the time cannot even begin to mitigate the depths of ineptitude to which we sank. On the plus side, we garnered a reputation for being too loud in school assembly and we did a successful tour of London's East End (well, our friends' houses). It's common for bands to tire of cover versions and to get lofty ambitions to write their own tunes. I wrote one called 'Escape From Inside' about the film *The Fly* that contained the rhyming couplet 'I am a fly, I want to die' – the great lyricists of the time quaked in their boots at the young new talent on the scene. The only thing we did that resembled the activities of a 'proper' band was to split up due to 'musical differences': Malcolm wanted to write 15-part symphonies about a boy's journey to worship electricity pylons and discover a mythical beast called the cuteasaurus, whereas I wanted to write songs about flies and dying (preferably both). When we could not agree on a musical direction the split became inevitable. Had I had the power of statistics in my hands back then, rather than split up we could have tested empirically the best musical direction for the band. Suppose Malcolm and I had each written a 15-part symphony and a 3-minute song about a fly. We could have played these songs to some very lucky people and measured their screams of agony. From these data, we could have ascertained the best musical direction to gain popularity. We have two variables that predict screams: whether Malcolm or I wrote the song (songwriter), and whether the song was a 15-part symphony or a song about a fly (song type). This design is called a factorial design, and this chapter looks at how the linear model extends to incorporate multiple categorical predictors.

Figure 14.1 Andromeda coming to a living room near you in 1988 (I'm the one wearing the *Anthrax* T-shirt) M



14.2 Factorial designs

In the previous two chapters we have used the linear model to test for differences between group means when those groups have belonged to a single predictor variable (in experimental designs one independent variable has been manipulated). This chapter extends the linear model to situations in which there are two categorical predictors (independent variables).

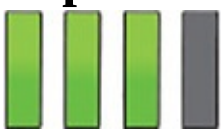


Predictor variables (independent variables) often get lonely and want to have friends. Scientists are obliging individuals and often put a second (or third) independent variable into their designs to keep the others company. When an experiment has two or more independent variables it is known as a *factorial design* (because, as we have seen, independent variables are sometimes referred to as *factors*). There are several types of factorial design:

- **Independent factorial design**: There are several independent variables or predictors and each has been measured using different entities (between groups). We discuss this design in this chapter.
- **Repeated-measures (related) factorial design**: Several independent variables or predictors have been measured, but the same entities have been used in all conditions (see Chapter 15).
- **Mixed design**: Several independent variables or predictors have been measured; some have been measured with different entities, whereas others used the same entities (see Chapter 16).

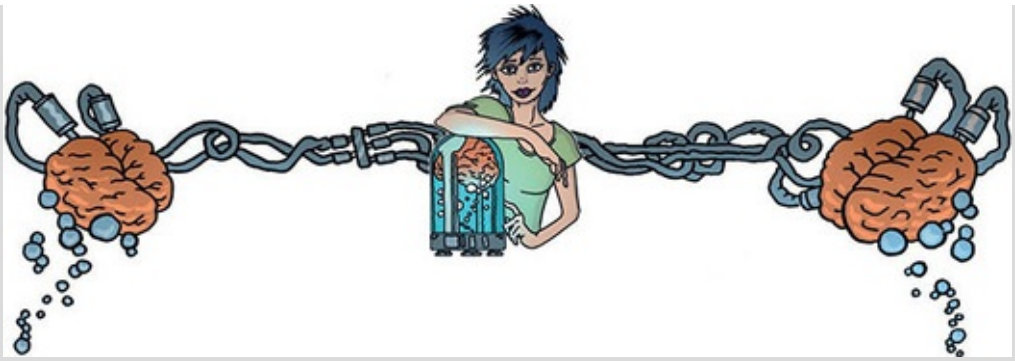
As you might imagine, analysing factorial designs can get quite complicated. Fortunately, we can offset this complexity by realizing that we can still fit a

linear model to the design, which at least grounds everything in a model with which (I hope) by now are familiar. Remembering that for weird historical reasons (see Misconception Mutt 12.1) people often label this form of the linear model as 'ANOVA', people often refer to the linear model with two or more categorical predictors that represent experimental independent variables as **factorial ANOVA**. There are also a family of more specific labels that reflect the experimental design that is being analysed (see Jane Superbrain Box 14.1). However, I think our lives would be simpler if we abandoned these labels because they deflect attention from the fact that the underlying model is the same.

14.3 Independent factorial designs and the linear model 

Throughout this chapter we'll use an example of an experimental design with two independent variables (a two-way independent design — Jane Superbrain Box 14.1). The study tested the prediction that subjective perceptions of physical attractiveness become inaccurate after drinking alcohol (the well-known **beer-goggles effect**). The example is based on real research by Chen, Wang, Yang, and Chen (2014) who looked at whether the beer-goggles effect was influenced by the attractiveness of the faces being rated. The logic is that alcohol consumption has been shown to reduce accuracy in symmetry judgements, and symmetric faces have been shown to be rated as more attractive. If the beer-goggles effect is driven by alcohol impairing symmetry judgements then you'd expect a stronger effect for unattractive (asymmetric) faces (because alcohol will affect the perception of asymmetry) than attractive (symmetric) ones. The data we'll analyse are fictional, but the results mimic the findings of this research paper.

Jane Superbrain 14.1 Naming ANOVAs 



Statistical models of experimental designs can appear confusing because there appears to be an endless list of them. It can create the false impression that the models are completely distinct, rather than the truth that they are variations on a common model (the linear model). To add to the confusion, scientists refer to these models as 'ANOVA' rather than the linear model, because the *F*-statistic that tests the fit of the model (Section 9.2.4) partitions variance, and so is called 'analysis of variance'.

The names people use have two things in common: (1) they involve some quantity of independent (predictor) variables; and (2) they label whether these variables were measured using the same or different entities. If the same entities are tested multiple times the term *repeated measures* is applied, and if different entities take part in the treatment conditions the term *independent* is used. With two or more independent variables, it's possible that some were measured with the same entities and others with different entities; this is referred to as a *mixed* design. In general, people name models that compare means as:

- a [number of independent variables]-way of [how these variables were measured] ANOVA.

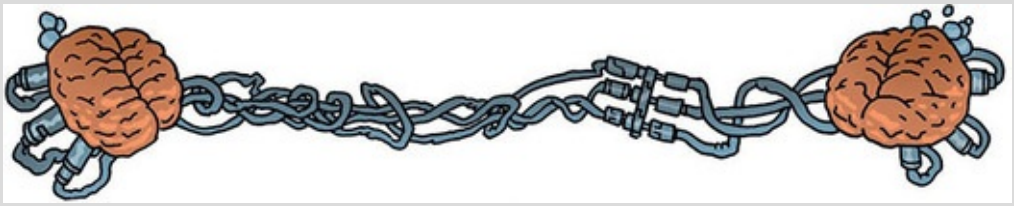
By remembering this you can decipher the name of any ANOVA model. Look at these examples and work out how many independent variables were used and how they were measured:

- One-way independent ANOVA
- Two-way repeated-measures ANOVA
- Two-way mixed ANOVA
- Three-way independent ANOVA

The answers you should get are:

- One independent variable measured using different entities.
- Two independent variables both measured using the same entities.

- Two independent variables: one measured using different entities and the other measured using the same entities.
- Three independent variables all of which are measured using different entities.



An anthropologist was interested in the effects of facial attractiveness on the beer-goggles effect. She randomly selected 48 participants. Participants were randomly subdivided into three groups of 16: (1) a placebo group drank 500 ml of alcohol-free beer; (2) a low-dose group drank 500 ml of average strength beer (4% ABV); and (3) a high-dose group drank 500 ml of strong beer (7% ABV). Within each group, half (n = 8) rated the attractiveness of 50 photos of unattractive faces on a scale from 0 (pass me a paper bag) to 10 (pass me their phone number) and the remaining half rated 50 photos of attractive faces.[1] The outcome for each participant was their median rating across the 50 photos (Table 14.1 and **Goggles.sav**).

1 These photographs were from a larger pool of 500 that had been pre-rated by a different sample. The 50 photos with the highest and lowest ratings were used.

Table 14.1 Data for the beer-goggles effect

Alcohol	Placebo		Low dose		High dose	
FaceType	Attractive	Unattractive	Attractive	Unattractive	Attractive	Unattractive
	6	2	7	3	5	5
	7	4	6	5	6	6
	6	3	8	7	7	8
	7	3	7	5	5	6
	6	4	6	4	7	7
	5	6	7	4	6	8
	8	5	6	5	5	7
	6	1	5	6	8	6
Total	51	28	52	39	49	53
Mean	6.375	3.500	6.500	4.875	6.125	6.625
Variance	0.839	2.571	0.857	1.554	1.268	1.125
	Grand mean = 5.667					
	Grand variance = 2.525					

To keep things simple, imagine for now that we have only two levels of the alcohol variable (placebo and high dose). As such, we have two predictor variables, each with two levels. We've seen many times that the general linear model takes the following general form:

$$Y_i = b_0 + b_1X_{1i} + b_2X_{2i} + \cdots + b_nX_{ni} + \varepsilon_i \quad (14.1)$$

When we first looked at using the linear model to compare means (see equation (12.2)) we used an example of the effect of three doses of puppy therapy (none, 15 and 30 minutes) on happiness. We saw that the linear model became:

$$\text{Happiness}_i = b_0 + b_1\text{Long}_i + b_2\text{Short}_i + \varepsilon_i \quad (14.2)$$

in which the **Long** and **Short** predictor variables were dummy variables that coded to which of the groups a participant belongs using values of 0 and 1. In our current example, we also have two variables that represent categories: **FaceType** (unattractive or attractive) and **Alcohol** (placebo and high dose). Just like we did for puppy therapy, we can code participant's category membership on these variables with zeros and ones; for example, we might code type of face as unattractive = 0, attractive = 1, and alcohol group as 0 = placebo, 1 = high dose. We can copy the puppy therapy model (equation (14.2)) but replace the predictors with our two independent variables:

$$\text{Attractiveness}_i = b_0 + b_1\text{FaceType}_i + b_2\text{Alcohol}_i + \varepsilon_i \quad (14.3)$$

Table 14.2 Coding scheme for factorial ANOVA

Type of faces rated	Alcohol	Dummy (FaceType)	Dummy (Alcohol)	Interaction	Mean
Unattractive	Placebo	0	0	0	3.500
Unattractive	High dose	0	1	0	6.625
Attractive	Placebo	1	0	0	6.375
Attractive	High dose	1	1	1	6.125

However, this model does not consider the interaction between type of face and alcohol. To include this term, we extend the model to become:

$$\begin{aligned}\text{Attractiveness}_i &= b_0 + b_1A_i + b_2B_i + b_3AB_i + \varepsilon_i \\ &= b_0 + b_1\text{FaceType}_i + b_2\text{Alcohol}_i + b_3\text{Interaction}_i + \varepsilon_i\end{aligned} \quad (14.4)$$

You might wonder how we code the interaction term, but we saw how to do this in Section 11.3. The interaction term represents the combined effect of **Alcohol** and **FaceType**, and it is obtained by multiplying the variables involved. This multiplication is the reason why interaction terms are written as type of face × alcohol. Table 14.2 shows the resulting predictor variables for the model (the group means for the combinations of type of face and alcohol are included because they'll come in useful in due course). Note that the interaction variable is the type of face dummy variable multiplied by the alcohol dummy variable. For example, someone receiving a high dose of alcohol and rating unattractive faces would have a value of 0 for the type of face variable, 1 for the alcohol variable and 0 for the interaction variable.

To see what the *b*-values in equation (14.4) represent we can insert values of our predictors and see what happens. Let's start with participants in the placebo group rating unattractive faces. In this case, the values of type of face, alcohol and the interaction are all 0. The predicted value of the outcome, as we have seen in Chapter 12, will be the group mean (3.500). Our model becomes:

$$
\begin{aligned}
\text{Attractiveness}_i &= b_0 + b_1\text{FaceType}_i + b_2\text{Alcohol}_i + b_3\text{Interaction}_i + \varepsilon_i \\
\bar{X}_{\text{Unattractive, Placebo}} &= b_0 + (b_1 \times 0) + (b_2 \times 0) + (b_3 \times 0) \\
b_0 &= \bar{X}_{\text{Unattractive, Placebo}} \\
&= 3.500
\end{aligned}
\tag{14.5}
$$

and you can see that the constant b_0 represents the mean of the group for which all variables are coded as 0. As such it's the mean value of the baseline category (in this case participants in the placebo group rating unattractive faces).

Now, let's see what happens when we look at participants in the placebo group rating attractive faces. The outcome is the mean rating of attractive faces after a placebo drink, the type of face variable is 1 and the alcohol and interaction variables are 0. Remember that b_0 is the mean rating of unattractive faces after a placebo drink. The model becomes:

$$
\begin{aligned}
\bar{X}_{\text{Attractive, Placebo}} &= b_0 + (b_1 \times 1) + (b_2 \times 0) + (b_3 \times 0) \\
&= b_0 + b_1 \\
&= \bar{X}_{\text{Unattractive, Placebo}} + b_1 \\
b_1 &= \bar{X}_{\text{Attractive, Placebo}} - \bar{X}_{\text{Unattractive, Placebo}} \\
&= 6.375 - 3.500 \\
&= 2.875
\end{aligned}
\tag{14.6}
$$

which shows that b_1 represents the difference between ratings of unattractive and attractive faces in people who drank a placebo. More generally we can say it's the effect of type of face for the baseline category of alcohol (the category coded with 0, in this case the placebo).

Let's look at people who had a high dose of alcohol and rated unattractive faces. The outcome is the mean rating of unattractive faces after a high dose of alcohol, the type of face variable is 0, the alcohol variable is 1 and the interaction variable is 0. We can replace b_0 with the mean rating of unattractive faces after a placebo drink. The model becomes:

$$
\begin{aligned}
\bar{X}_{\text{Unattractive, High dose}} &= b_0 + (b_1 \times 0) + (b_2 \times 1) + (b_3 \times 0) \\
&= b_0 + b_2 \\
&= \bar{X}_{\text{Unattractive, Placebo}} + b_2 \\
b_2 &= \bar{X}_{\text{Unattractive, High dose}} - \bar{X}_{\text{Unattractive, Placebo}} \\
&= 6.625 - 3.500 \\
&= 3.125
\end{aligned}
\tag{14.7}
$$

which shows that b_2 in the model represents the difference between ratings of unattractive faces after a high dose of alcohol compared to a placebo drink. Put more generally, it's the effect of alcohol in the baseline category of type of face (i.e., the category coded with a 0, in this case unattractive).

Finally, we can look at ratings of attractive faces after a high dose of alcohol. The predicted outcome is the mean rating of attractive faces by people after a high dose of alcohol, and the type of face, alcohol and interaction variables are all 1. We can replace b_0, b_1, and b_2, with what we now know they represent from equations (14.5)–(14.7). The model becomes:

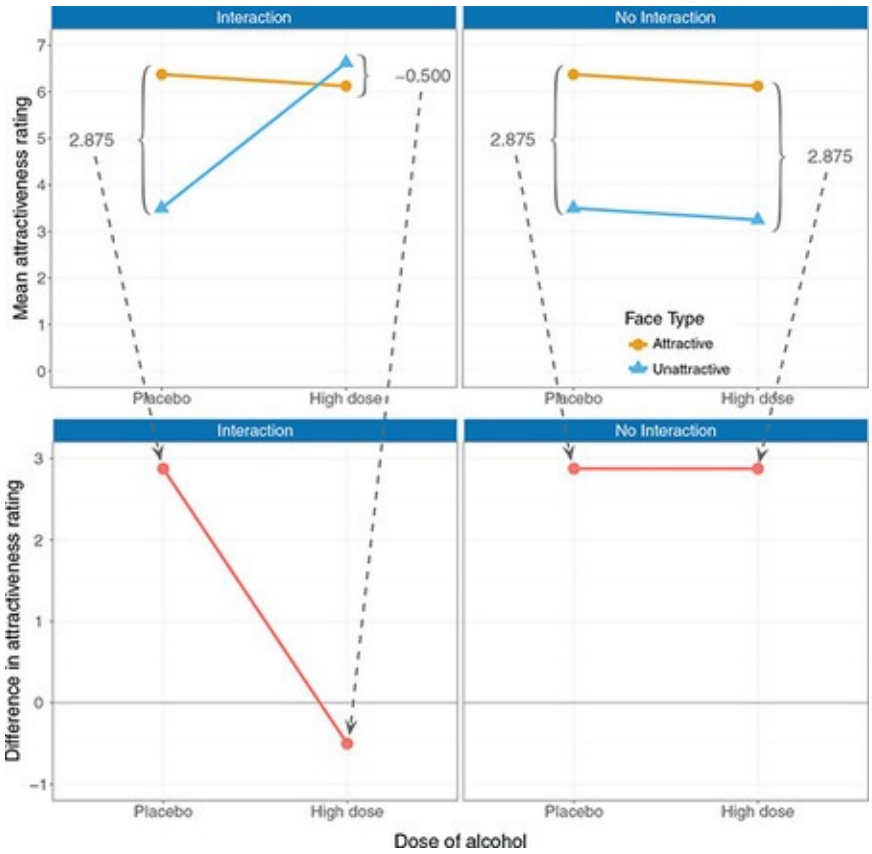
$$\begin{aligned}
\bar{X}_{\text{Attractive, High dose}} &= b_0 + (b_1 \times 1) + (b_2 \times 1) + (b_3 \times 1) \\
&= b_0 + b_1 + b_2 + b_3 \\
&= \bar{X}_{\text{Unattractive, Placebo}} + \left(\bar{X}_{\text{Attractive, Placebo}} - \bar{X}_{\text{Unattractive, Placebo}}\right) \\
&\quad + \left(\bar{X}_{\text{Unattractive, High dose}} - \bar{X}_{\text{Unattractive, Placebo}}\right) + b_3 \\
&= \bar{X}_{\text{Attractive, Placebo}} + \bar{X}_{\text{Unattractive, High dose}} - \bar{X}_{\text{Unattractive, Placebo}} + b_3 \qquad (14.8) \\
b_3 &= \bar{X}_{\text{Unattractive, Placebo}} - \bar{X}_{\text{Attractive, Placebo}} + \bar{X}_{\text{Attractive, High dose}} \\
&\quad - \bar{X}_{\text{Unattractive, High dose}} \\
&= 3.500 - 6.375 + 6.125 - 6.625 \\
&= -3.375
\end{aligned}$$

which is scary, but we'll break it down. It shows that b_3 compares the difference between ratings of unattractive and attractive faces in the placebo group to the same difference in the high-dose group. Put more generally, it compares the effect of type of face after a placebo drink to the effect of type of face after a high dose of alcohol.[2]

2 If you rearrange the terms in the equation you'll see that you can also phrase the interaction the opposite way around: it represents the effect of alcohol on ratings of attractiveness for attractive faces compared to unattractive ones.

This explanation makes more sense if you think in terms of an interaction graph. Figure 14.2 (top left) shows the interaction graph for these data. The difference between ratings of unattractive and attractive faces in the two placebo groups is the distance between the lines on the graph for the placebo group (the difference between group means, which is 2.875). If we look at the same difference for the high-dose group, we find that the difference between ratings of unattractive and attractive faces is −0.500. If we plotted these two 'difference' values as a new graph we'd get a line connecting 2.875 to −0.500 (see Figure 14.2, bottom left). The slope of this line reflects the difference between the effect of type of face after a placebo compared to after a high dose of alcohol. We know that *b*-values represent gradients of lines and b_3 is the gradient of the line that connects the difference scores (this is (−0.500) − 2.875 = −3.375).

Figure 14.2 Breaking down what an interaction represents



The right-hand side of Figure 14.2 illustrates what happens if there isn't an interaction effect by showing the same data as on the left except that the mean rating of unattractive pictures after a high dose of alcohol has been changed to 3.25. If we calculate the difference between ratings of unattractive and attractive faces after a placebo drink we get the same as before: 2.875. However, if we calculate the difference between ratings of unattractive and attractive faces after a high dose of alcohol we now also get 2.875. If we again plot these differences on a new graph, we find a completely flat line. So, when there's no interaction, the line connecting the effect of type of face after a placebo drink and after a high dose is flat and the resulting b_3 would be 0 (remember that a zero gradient means a flat line). If we calculate this difference, this is what we get: 2.875 − 2.875 = 0.



The file **GogglesRegression.sav** contains the dummy variables used in this example. Just to prove that this works, use this file to fit a linear model predicting attractiveness ratings from **FaceType, Alcohol** and the interaction variable.

Output 14.1

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	3.500	.426		8.219	.000
	Attractivenss of facial stimuli	2.875	.602	.851	4.774	.000
	Alcohol consumption	3.125	.602	.925	5.189	.000
	Interaction	-3.375	.852	-.866	-3.963	.000

a. Dependent Variable: Median attractiveness rating

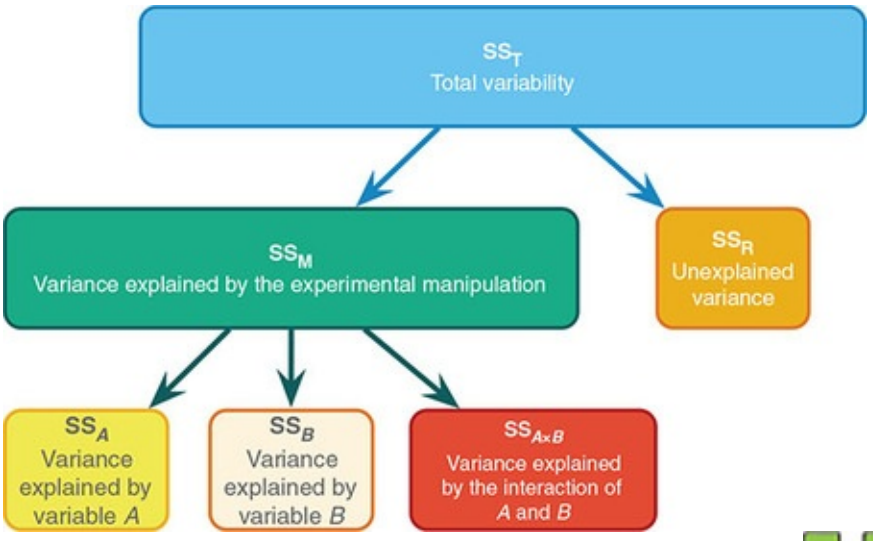
Output 14.1 shows the resulting table of coefficients from the self-test. Note that the *b*-value for the type of face matches equation (14.6), the *b* for the alcohol group matches equation (14.7) and the *b* for the interaction matches equation (14.8). All of which I hope convinces you that we can use a linear model to analyse designs that incorporate multiple categorical predictors.



14.3.1 Behind the scenes of factorial designs

Now that we have a good conceptual understanding of factorial designs as an extension of the linear model, we will turn our attention to the specific calculations that go on behind the scenes. The reason for doing this is that it should help you to understand what the output of the analysis means. Calculating the *F*-statistic with two categorical predictors is very similar to when we had only one: we still find the total sum of squared errors (SS_T) and break this variance down into variance that can be explained by the model/experiment (SS_M) and variance that cannot be explained (SS_R). The main difference is that with factorial designs, the variance explained by the model/experiment is made up of not one predictor (experimental manipulation) but two. Therefore, the model sum of squares gets further subdivided into variance explained by the first predictor/independent variable (SS_A), variance explained by the second predictor/independent variable (SS_B) and variance explained by the interaction of these two predictors ($SS_{A \times B}$) – see Figure 14.3.

Figure 14.3 Breaking down the variance in a two-way factorial design



14.3.2 Total sums of squares (SS_T)

We start off in the same way as we did for a one-way ANOVA. That is, we calculate how much variability there is between scores when we ignore the experimental condition from which they came. Remember from one-way ANOVA (equation (12.9)) that SS_T is calculated using the following equation:

$$\begin{aligned} SS_T &= \sum_{i=1}^{N} \left(x_i - \bar{x}_{grand}\right)^2 \\ &= s^2_{grand}\left(N-1\right) \end{aligned} \tag{14.9}$$

The grand variance is the variance of all scores when we ignore the group to which they belong. So we treat the data as one big group, calculate the variance of all scores, and find it's 2.525 as in Table 14.1 (try this on your calculator if you don't trust me). We used 48 scores to generate this value, and so N is 48. As such the total sum of squares is 118.675 (equation (14.10)). The degrees of freedom for this SS will be $N - 1$, or 47.

$$\begin{aligned} SS_T &= s^2_{grand}\left(N-1\right) \\ &= 2.525\left(48-1\right) \\ &= 118.675 \end{aligned} \tag{14.10}$$

14.3.3 The model sum of squares, SS_M

The next step is to work out the model sum of squares, which is then further broken down into the variance attributable to the first independent variable (SS_A), the variance attributable to the second independent variable (SS_B) and

variance attributable to the interaction of these two variables ($SS_{A \times B}$). The model sum of squares (as we've seen a few times) is the difference between what the model predicts and the overall mean of the outcome variable. We've also seen that with predictors that represent group membership 'what the model predicts' is the group means. Therefore, we work out the model sum of squares by looking at the difference between each group mean and the overall mean (see Section 12.2.3). If we combine all levels of the two independent variables (three alcohol groups that rated unattractive faces and another three that rated attractive faces) we have six groups. We can apply the equation for the model sum of squares that we've used before (equation (12.11)):

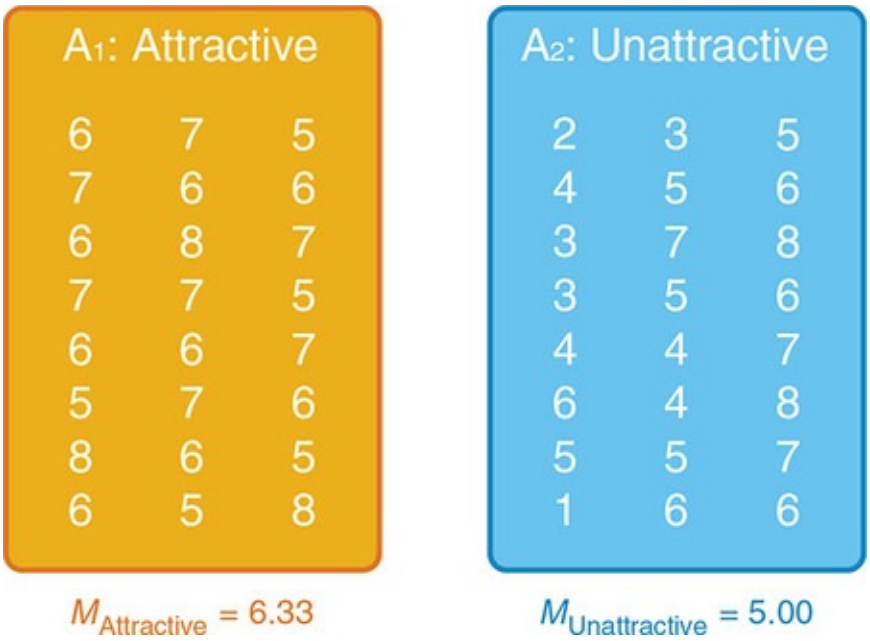
$$SS_M = \sum_{g=1}^{k} n_g \left(\bar{x}_g - \bar{x}_{grand}\right)^2 \quad (14.11)$$

The grand mean is the mean of all scores and is 5.667 (Table 14.1) and *n* is the number of scores in each group (8 in this case). Therefore, the model sum of squares is 61.17:

$$\begin{aligned} SS_M &= 8(6.375-5.667)^2 + 8(3.500-5.667)^2 + 8(6.500-5.667)^2 \\ &\quad + 8(4.875-5.667)^2 + 8(6.125-5.667)^2 + 8(6.625-5.667)^2 \\ &= 8(0.708)^2 + 8(-2.167)^2 + 8(0.833)^2 + 8(-0.792)^2 + 8(0.458)^2 + 8(0.958)^2 \\ &= 4.010 + 37.567 + 5.551 + 5.018 + 1.678 + 7.342 \\ &= 61.17 \end{aligned} \quad (14.12)$$

The degrees of freedom are the number of groups, *k*, minus 1; we had six groups and so *df* = 5. At this stage we know that the model (our experimental manipulations) can explain 61.17 units of variance out of the total of 118.675 units. The next stage is to break down this model sum of squares to separate the variance in attractiveness ratings explained by each of our independent variables.

Figure 14.4 The main effect of type of face



14.3.4 The main effect of type of face, SS_A

To work out the variance accounted for by the first predictor/independent variable (in this case, type of face) we group attractiveness ratings according to which type of face was being rated. So, basically we ignore the dose of alcohol that was drunk, and we place all the ratings of unattractive faces into one group and all of the ratings of attractive ones into another. The data will look like Figure 14.4 (note that the first box contains the three attractive columns from Table 14.1 and the second box contains the unattractive columns).

We then apply the same equation for the model sum of squares that we just used (compare equation (14.11)):

$$SS_A = \sum_{g=1}^{k} n_g \left(\bar{x}_g - \bar{x}_{\text{grand}}\right)^2 \tag{14.13}$$

The grand mean is the mean of all scores (5.667, as above) and *n* is the number of scores in each group (i.e., the number of participants who rated unattractive and attractive faces; 24 in both cases). The means of the two groups have been calculated in Figure 14.4. The resulting model sum of squares for the main effect of the type of face is 21.32:

$$\begin{aligned} SS_{\text{FaceType}} &= 24(6.33 - 5.667)^2 + 24(5.00 - 5.667)^2 \\ &= 24(0.666)^2 + 24(-0.667)^2 \\ &= 10.645 + 10.677 \\ &= 21.32 \end{aligned} \tag{14.14}$$

The degrees of freedom for this SS are the number of groups used, *k*, minus 1.

We used two groups (unattractive and attractive) and so *df* = 1. To sum up, the main effect of type of face compares the mean of all ratings of unattractive faces to the corresponding mean for attractive faces (regardless of how much alcohol was consumed).

Figure 14.5 The main effect of alcohol



14.3.5. The main effect of alcohol, SS_B

To work out the variance accounted for by the second independent variable (dose of alcohol) we group the attractiveness ratings according to the dose of alcohol. In other words, we ignore the type of face the participant was rating, and we place all of the scores after the placebo drink in one group, the ratings after a low dose in another group and the ratings after a high dose in a third group. The data will look like Figure 14.5. We apply the same equation as for the main effect of type of face:

$$\mathrm{SS}_B = \sum_{g=1}^{k} n_g \left(\bar{x}_g - \bar{x}_{\text{grand}}\right)^2 \tag{14.15}$$

The grand mean is the mean of all scores (5.667 as before), *n* is the number of scores in each group (i.e., the number of scores in each alcohol condition, in this case 16), and the group means are given in Figure 14.5. The resulting sum of squares is 16.53:

$$\begin{aligned}\mathrm{SS}_{\text{Alcohol}} &= 16(4.938-5.667)^2 + 16(5.688-5.667)^2 + 16(6.375-5.667)^2 \\ &= 16(-0.729)^2 + 16(0.021)^2 + 16(0.708)^2 \\ &= 8.503 + 0.007 + 8.020 \\ &= 16.53\end{aligned} \tag{14.16}$$

The degrees of freedom are the number of groups used minus 1 (see Section 12.2.3). We used three groups and so $df = 2$. To sum up, the main effect of alcohol compares the means of the placebo, low dose and high dose (regardless of whether the ratings were of unattractive or attractive faces).

14.3.6 The interaction effect, $SS_{A \times B}$



The final stage is to calculate how much variance is explained by the interaction of the two variables. The simplest way to do this is to remember that the SS_M is made up of three components (SS_A, SS_B and $SS_{A \times B}$). Therefore, given that we know SS_A and SS_B, we can calculate the interaction term by subtraction:

$$SS_{A\times B} = SS_M - SS_A - SS_B \tag{14.17}$$

For these data, the value is 23.32:

$$\begin{aligned} SS_{A\times B} &= SS_M - SS_A - SS_B \\ &= 61.17 - 21.32 - 16.53 \\ &= 23.32 \end{aligned} \tag{14.18}$$

The degrees of freedom can be calculated in the same way, but are also the product of the degrees of freedom for the main effects – either method works:

$$\begin{aligned} df_{A\times B} &= df_M - df_A - df_B \\ &= 5 - 1 - 2 = 2 \end{aligned} \qquad \begin{aligned} df_{A\times B} &= df_A \times df_B \\ &= 1 \times 2 = 2 \end{aligned} \tag{14.19}$$

14.3.7 The residual sum of squares, SS_R



The residual sum of squares is calculated in the same way as in Section 12.2.4. As ever, it represents errors in prediction from the model, but in experimental designs this also reflects individual differences in performance or variance that can't be explained by factors that were systematically manipulated. The value is calculated by taking the squared error between each data point and its corresponding group mean. An alternative way to express this is (see equation (12.14)):

$$\begin{aligned} SS_R &= \sum_{g=1}^{k} s_g^2 (n_g - 1) \\ &= s^2_{\text{group}\,1}(n_1 - 1) + s^2_{\text{group}\,2}(n_2 - 1) + \cdots + s^2_{\text{group}\,n}(n_n - 1) \end{aligned} \tag{14.20}$$

We use the individual variances of each group from Table 14.1 and multiply them by one less than the number of people within the group (n), in this case $n = 8$. The resulting residual sum of squares is 57.50:

$$
\begin{aligned}
\mathrm{SS}_{\mathrm{R}} &= s^2_{\text{group 1}}(n_1 - 1) + s^2_{\text{group 2}}(n_2 - 1) + \cdots + s^2_{\text{group 6}}(n_6 - 1) \\
&= 0.839(8-1) + 2.571(8-1) + 0.857(8-1) + 1.554(8-1) \\
&\quad + 1.268(8-1) + 1.125(8-1) \\
&= (0.839 \times 7) + (2.571 \times 7) + (0.857 \times 7) + (1.554 \times 7) \\
&\quad + (1.268 \times 7) + (1.125 \times 7) \\
&= 5.873 + 17.997 + 5.999 + 10.878 + 8.876 + 7.875 \\
&= 57.50
\end{aligned}
\tag{14.21}
$$

The degrees of freedom for each group will be one less than the number of scores per group (i.e., 7). We add the degrees of freedom for each group to get a total of 6 × 7 = 42.

14.3.8 The F-statistics 

Each effect in a factorial design has its own *F*-statistic. In a two-way design this means we compute *F* for the two main effects and the interaction. To calculate these we first calculate the mean squares for each effect by taking the sum of squares and dividing by the respective degrees of freedom (think back to Section 12.2.5). We also need the mean squares for the residual term. So, for this example we'd have four mean squares:

$$
\begin{aligned}
\mathrm{MS}_A &= \frac{\mathrm{SS}_A}{df_A} = \frac{21.32}{1} = 21.32 \\
\mathrm{MS}_B &= \frac{\mathrm{SS}_B}{df_B} = \frac{16.53}{2} = 8.27 \\
\mathrm{MS}_{A\times B} &= \frac{\mathrm{SS}_{A\times B}}{df_{A\times B}} = \frac{23.32}{2} = 11.66 \\
\mathrm{MS}_{\mathrm{R}} &= \frac{\mathrm{SS}_{\mathrm{R}}}{df_{\mathrm{R}}} = \frac{57.50}{42} = 1.37
\end{aligned}
\tag{14.22}
$$

The *F*-statistic for each effect is then calculated, as we've seen before, by dividing its mean squares by the residual mean squares:

$$
\begin{aligned}
F_A &= \frac{\mathrm{MS}_A}{\mathrm{MS}_{\mathrm{R}}} = \frac{21.32}{1.37} = 15.56 \\
F_B &= \frac{\mathrm{MS}_B}{\mathrm{MS}_{\mathrm{R}}} = \frac{8.27}{1.37} = 6.04 \\
F_{A\times B} &= \frac{\mathrm{MS}_{A\times B}}{\mathrm{MS}_{\mathrm{R}}} = \frac{11.66}{1.37} = 8.51
\end{aligned}
\tag{14.23}
$$

IBM SPSS Statistics will compute an exact *p*-value for each of these *F*-statistics to tell us the probability (in the long run) of getting an *F* at least as big as the one we have if there were no effect in the population. One important issue here is that these *F*s do not control the familywise Type I error (Misconception Mutt

14.1).
To sum up, when you have a factorial design, the computations behind the linear model are basically the same as when you have only one categorical predictor, except that the model sum of squares is partitioned into three parts: the effect of each of the independent variables and the effect of how these variables interact.



14.4 Model assumptions in factorial designs

When using the linear model to analyse a factorial design the sources of potential bias (and counteractive measures) discussed in Chapter 6 apply. If you have violated the assumption of homogeneity of variance there are corrections based on the Welch procedure that was described a couple of chapters back. However, this is quite technical, SPSS Statistics doesn't do it, and if you have anything more complicated than a 2 × 2 design then, really, it would be less painful to cover your body in paper cuts and then bathe in chilli sauce (see Algina & Olejnik, 1984). One practical solution is to bootstrap the *post hoc* tests so that these will be robust. You can also ask for confidence intervals and *p*-values for parameter estimates that are robust to heteroscedasticity (refer back to Section 13.5.6). This won't help for the *F*-statistics. There is a robust test based on trimmed means that can be done using the R extension (see Section 14.8).



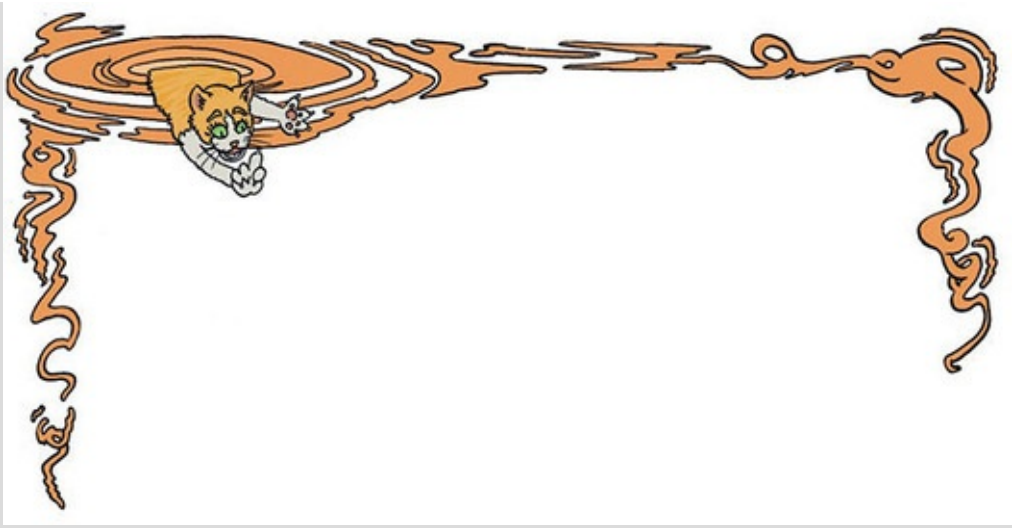
14.5 Factorial designs using SPSS Statistics

14.5.1 General procedure for factorial designs



Using the linear model to test differences between means on several predictors/independent variables requires the same steps as for one predictor/independent variable, so refer back to Figure 12.4 as a guide.

Misconception Mutt 14.1 *F*-statistics and Type I errors



The Misconception Mutt was very tired. His owner had been walking him early before lectures and he needed some sleep. This morning's game of fetch was killing him, and yet his little legs seemed incapable of not speeding his body off towards the stick that his owner seemed to lose so carelessly, persistently and forcefully. As he sprinted, his mind was wandering to *F*-statistics in factorial designs. He thought they were a neat way to control Type I errors. As he picked up the stick, it winked at him. Was it the sleep deprivation? He wasn't sure but then its bark turned a ginger hue, and he noticed fur shooting out from it. Soon legs began to poke out, and a tail. Before he knew it he was carrying a very heavy ginger cat in his mouth. This was all he needed.

'Did you notice how the *F*-statistics in factorial designs are computed?' the cat said.

Determined not to be outwitted by a cat, the mutt replied, 'Yes, the mean squares for the effect divided by the residual mean squares.'

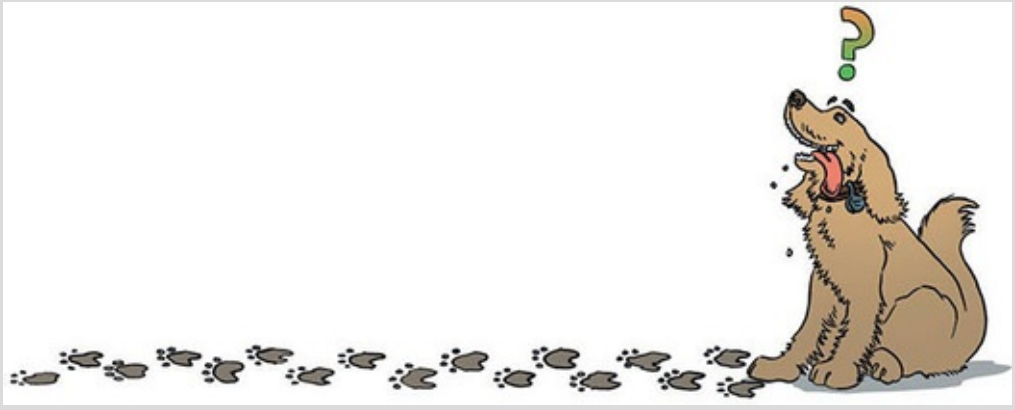
'Exactly,' the cat grinned. 'They all use the residual mean squares. This means that they are not independent.' The cat waited as though expecting applause. The mutt was confused, so the cat continued.

'The *F*-statistics are not independent, which means that they do not control the Type I error rate. In the example from your lecture there are three *F*s. The probability that at least one of them is a Type I error is not 0.05, but is 0.143 because the *F*s are not independent. If you had even more predictors and interactions the probability of at least one Type 1 error would be even greater.'

'What can be done?' the mutt asked.

'You could use a more stringent criterion for accepting an *F* as

significant. With three effects you might use 0.05/3 = 0.017, which is effectively a Bonferroni correction. Or simply ignore any effects that don't test your substantive hypothesis. In this case interpret the interaction and ignore the main effects (even if they are significant).' With that the legs, fur and grin retracted and the mutt dropped the stick at his owner's feet.



14.5.2 Entering the data and accessing the main dialog box

To enter the data we need two different coding variables to represent type of face and alcohol consumption. Create a variable called **FaceType** in the data editor. We have had a lot of experience with coding variables, so you should be able to define value labels to represent the two types of faces: I've used the codes unattractive = 0 and attractive = 1. Having created the variable, enter a code of 0 or 1 in the **FaceType** column indicating to which group the participant was assigned. Create a second variable called **Alcohol** and assign value labels of placebo = 0, low dose = 1 and high dose = 2. In the data editor, enter 0, 1 or 2 into the alcohol column to represent the amount of alcohol consumed by the participant. Remember that if you turn on the *value labels* option you will see text in the data editor rather than the numerical codes. The coding scheme I have suggested is summarized in Table 14.3. Once you have created the two coding variables, create a third called **Attractiveness** and use the *labels* option to give it the fuller name of *Median attractiveness rating*. Enter the scores in Table 14.1 into this column, taking care to ensure that each score is associated with the correct combination of type of face and alcohol.

Table 14.3 Coding two independent variables

FaceType	Alcohol	The participant ...
0	0	Rated unattractive faces after a placebo drink
0	1	Rated unattractive faces after a low dose of alcohol
0	2	Rated unattractive faces after a high dose of alcohol
1	0	Rated attractive faces after a placebo drink
1	1	Rated attractive faces after a low dose of alcohol
1	2	Rated attractive faces after a high dose of alcohol

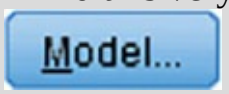


Use the Chart Builder to plot an error bar graph of the attractiveness ratings with alcohol consumption on the *x*-axis and different coloured lines to represent whether the faces being rated were unattractive or attractive.

To fit a linear model for independent factorial designs select *Analyze* *General Linear Model* Univariate... to access the dialog box in Figure 14.6. Drag the outcome variable **Attractiveness** from the variables list on the left-hand side and to the space labelled *Dependent Variable* (or click). Drag **Alcohol** and **FaceType** (to select them simultaneously hold down *Ctrl*, or *Cmd* on a Mac, while clicking each one) from the variables list to the *Fixed Factor(s)* box (or click). There are various other spaces that are available for conducting more complex analyses such as factorial ANCOVA, which extends the model at the beginning of this chapter to include a covariate (as in the previous chapter).

Oliver Twisted Please, Sir, Can I … customize my model?



‘My friend told me that there are different types of sums of squares,’ complains Oliver with an air of impressive authority. ‘Why haven’t you told us about them? Is it because you have a microbe for a brain?’ No, Oliver, it’s because everyone but you will find this very tedious. If you want to find out more about what the 

button does, and the different types of sums of squares that can be used, then the companion website will tell you.

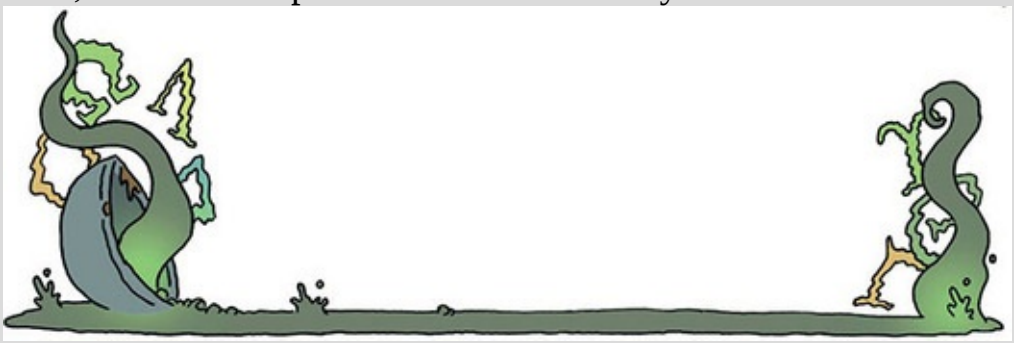
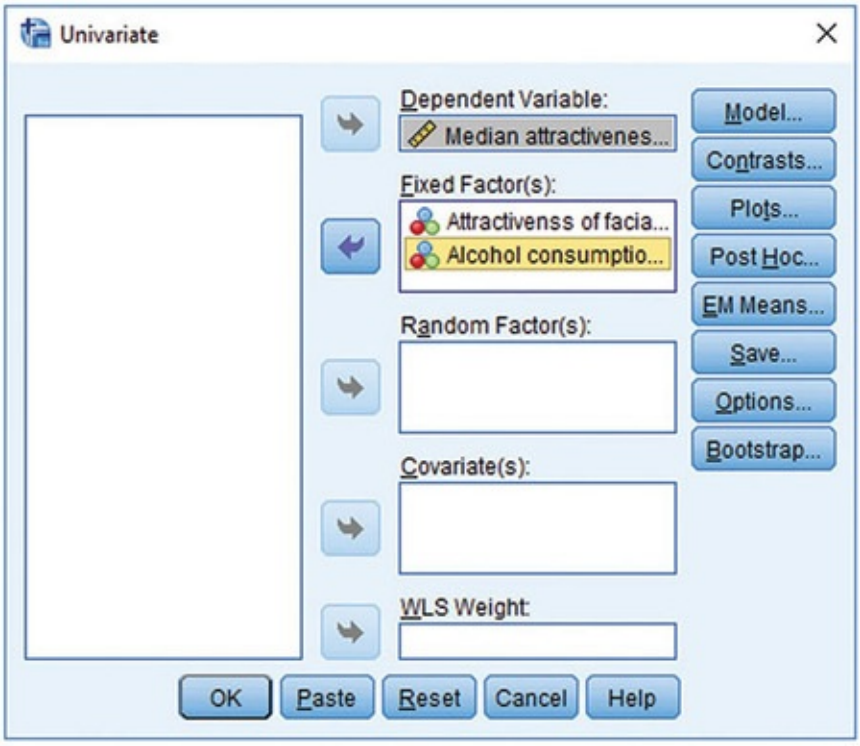
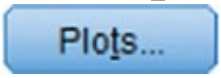


Figure 14.6 Main dialog box for independent factorial designs



14.5.3 Graphing interactions

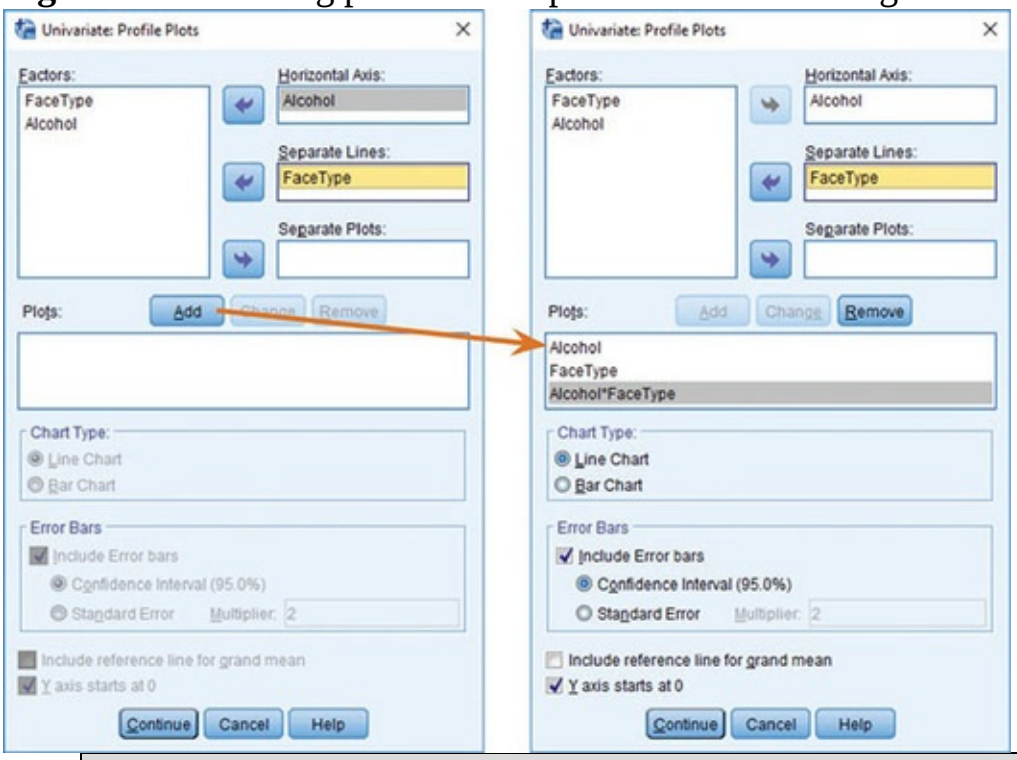
Click 

to access the dialog box in Figure 14.7, which allows you to specify line graphs of your data. These graphs can be useful for interpreting interaction effects, but they do tend to be scaled so as to create the impression of massive differences between means, so, seriously, plot your data properly *before* the main analysis. If you want to use this plotting function to create an **interaction graph** to show the combined effect of type of face and alcohol consumption, drag **Alcohol** from the variables list to the space labelled *Horizontal Axis* (or click ) and drag **FaceType** to the space labelled *Separate Lines*. Alternatively, you could plot the variables the other way round – it doesn't matter, so use your discretion as to which way produces the graph that helps make sense of the interaction. When you have moved the two variables to the appropriate boxes, click 

to add the plot to the list at the bottom of the box. You can choose either a bar or line chart (I have chosen

a line chart), and you should ask for error bars displaying confidence intervals. Using the default scaling, you can end up with a chart showing apparently massive group differences that turn out to be tiny when you inspect the *y*-axis. To avoid this crushing disappointment it is generally (although not always) a good idea to select *Y axis starts at 0* to scale the *y*-axis from zero. In Figure 14.7 I have also specified graphs for the two main effects by dragging them (in turn) to the space labelled *Horizontal Axis* and clicking Add. Click Continue to return to the main dialog box.

Figure 14.7 Defining plots for independent factorial designs



Oliver Twisted Please, Sir, can I have some more … contrasts?



‘I don’t want to use standard contrasts,’ sulks Oliver as he stamps his feet on the floor. ‘They smell of rotting cabbage.’ Actually, Oliver, I

think the stench of rotting cabbage is because you stood your Dickensian self under Mr. Mullycents' window when he emptied his toilet bucket into the street. Nevertheless, I do get asked a fair bit about how to do contrasts with syntax and because I'm a complete masochist I've prepared a fairly detailed guide in the additional material for this chapter. These contrasts are useful to follow up a significant interaction effect.

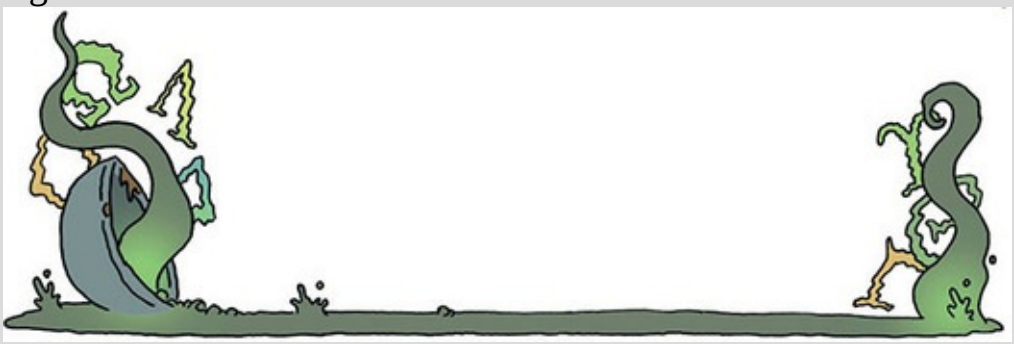
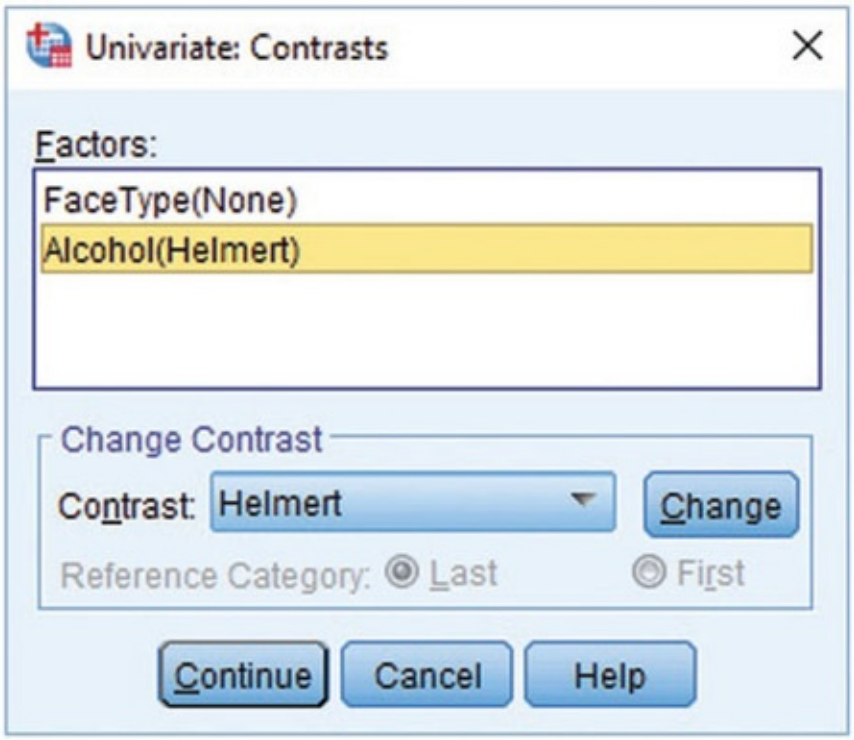


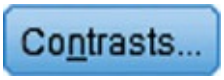
Figure 14.8 Defining contrasts for independent factorial designs



14.5.4 Contrasts

We saw in Chapter 12 that contrasts help us to break down main effects and tell us where the differences between groups lie. With one independent variable we could enter codes to define the contrasts we want. However, with two

independent variables no such facility exists (although it can be done using syntax – see Oliver Twisted). Instead we are restricted to several standard contrasts that were described in Table 12.6.
Standard contrasts will be fine for this example. The type of face had only two levels, so we don't need contrasts for that main effect because it compares only two means. However, the effect of alcohol had three levels: placebo, low dose and high dose. We could select a simple contrast and use the first category as a reference. Doing so would compare the low-dose group to the placebo group, and then compare the high-dose group to the placebo group. As such, the alcohol groups would get compared to the placebo group. We could also select a *repeated* contrast, which would compare the low-dose group to the placebo group, and then the high-dose group to the low-dose group (i.e., it moves through the groups comparing each group to the one before). Again, this might be useful. We could also do a **Helmert contrast**, which compares each category against all subsequent categories. In this case it would compare the placebo group to the remaining categories (i.e., all of the groups that had some alcohol) and then would move on to the low-dose group and compare it to the high. Any of these contrasts would be fine, but they give us contrasts only for the main effects. In reality, most of the time we want contrasts for our interaction term, and they can be obtained only through syntax (it looks like you might have to look at Oliver Twisted after all!).

To get contrasts for the main effect of alcohol click 

in the main dialog box. We've used the *Contrasts* dialog box before (Section 13.5.5), so refer back to that section to help you select a Helmert contrast for the **Alcohol** variable (Figure 14.8). Click 

to return to the main dialog box.

14.5.5 Post hoc tests 

Click 

to access *post hoc* tests (Figure 14.9). The variable **FaceType** has only two levels and so we don't need *post hoc* tests (because any significant effects must reflect the difference between unattractive and attractive faces). However, there were three levels of **Alcohol** (placebo, low dose and high dose) so if we didn't have prior hypotheses to test we might want *post hoc* tests (remember that normally you would conduct contrasts *or post hoc* tests, not both). Drag the variable **Alcohol** from the box labelled *Factors* to the box labelled *Post Hoc Tests for:*. For my recommendations for *post hoc* procedures

see Section 12.5 or select the ones in Figure 14.9. Click Continue to return to the main dialog box.

Figure 14.9 Dialog box for *post hoc* tests

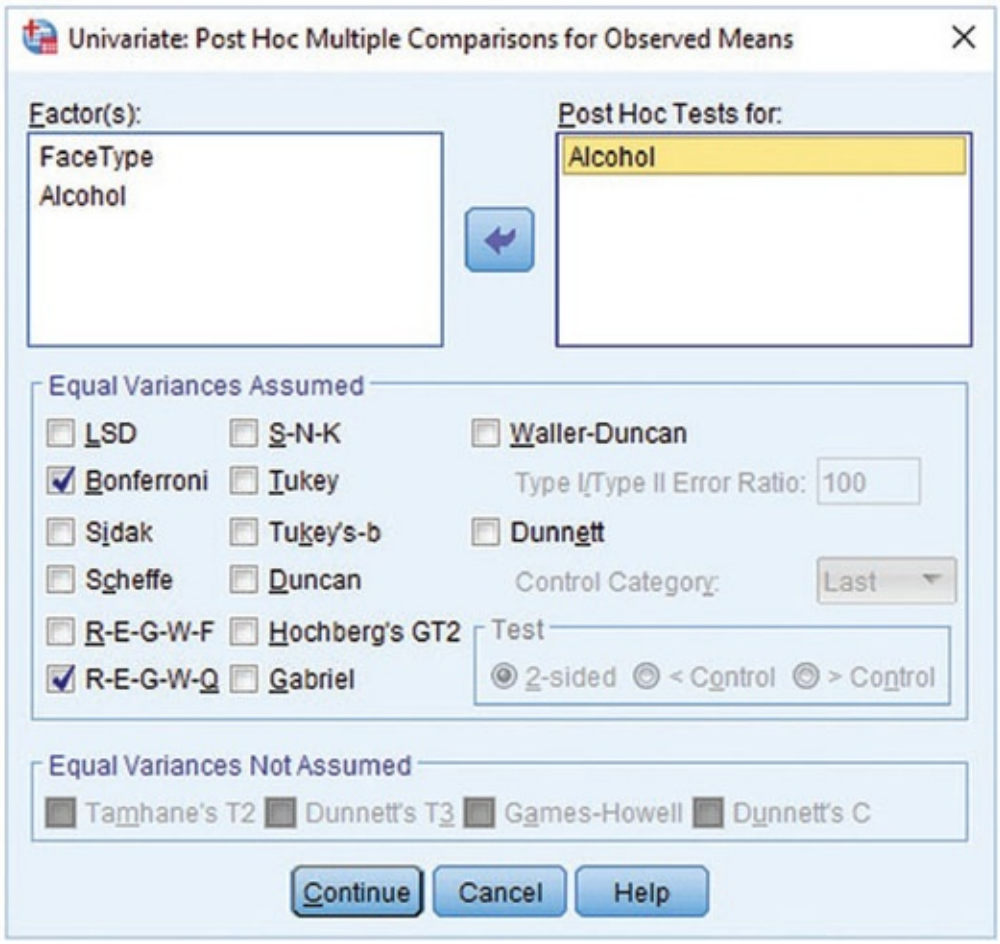
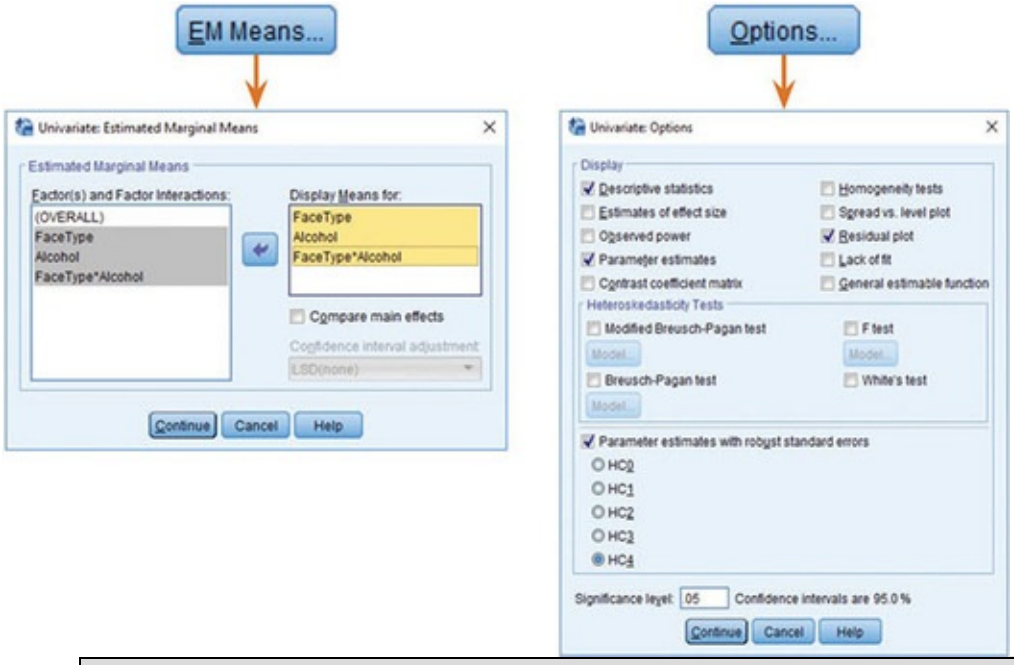


Figure 14.10 *Estimated marginal means* and *Options* dialog boxes for GLM univariate



Oditi's Lantern Factorial ANOVA



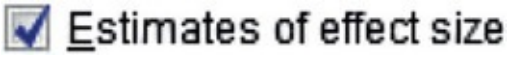
'I, Oditi, enjoy interactions immensely. I want to interact with all of my followers, invite them around to my large desert ranch and let them sup on my tasty mint tea. I grow mint in my special mushroom patch, which gives it a unique flavour, and sometimes makes people obey my every command. I have learnt that interactions like these are powerful tools to understand the secrets of global domina … erm, I mean "life" and how to breed cute bunny rabbits of love. Stare into my lantern and discover more about factorial designs.'



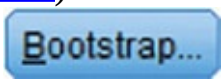
14.5.6 Bootstrapping and other options 

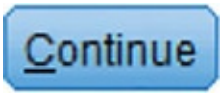
Click 

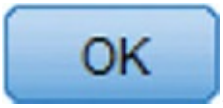
and 

to activate the same dialog boxes that we saw in the previous chapter (Figure 13.7; the options are explained in Jane Superbrain Box 13.3). The main thing is to get estimated marginal means by transferring all effects into the box labelled *Display Means for* (Figure 14.10). Some people select *Homogeneity tests* to produce Levene's test (Section 6.11.2), but I'm not a fan. You can select 

to obtain partial eta squared (see Section 13.10).

The main dialog box contains the 

button, which can be used to specify bootstrapped confidence intervals for the estimated marginal means, descriptive statistics and *post hoc* tests, but not the main *F*-statistic. This option is mainly useful if you plan to look at *post hoc* tests, which we will, so select the options described in Section 6.12.3. Once these options have been selected click 

to return to the main dialog box, then click 

.

14.6 Output from factorial designs 

If you're a Levene's test kind of person (which I'm not) and selected that option then Output 14.2 will appear in the viewer (see Jane Superbrain Box 6.6). With eight participants in each group this test will be horrifically underpowered so the fact that the result is non-significant ($p = 0.625$) could mean that the variance in attractiveness ratings is roughly equal across the combinations of type of face and alcohol, or it could be that we don't have enough power to detect differences in the variance across groups.

14.6.1 The main effect of type of face 

Output 14.3 tells us whether any of the predictors/independent variables had a significant effect on attractiveness ratings. Note that the sums of squares, mean squares and *F*-statistics match (to within rounding error) the values we calculated in Sections 14.3.2–14.3.8. The main effect of type of face is significant because the *p* associated with the *F*-statistic is given as 0.000, which is less than 0.05. This effect means that overall *when we ignore how much*

alcohol had been drunk the type of face being rated significantly affected attractiveness ratings. We can visualize this effect by plotting the average attractiveness rating at each level of type of face (ignoring the dose of alcohol completely).

Output 14.2

Levene's Test of Equality of Error Variances[a]

Dependent Variable: Median attractiveness rating

F	df1	df2	Sig.
.702	5	42	.625

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Design: Intercept + FaceType + Alcohol + FaceType * Alcohol

Output 14.3

Tests of Between-Subjects Effects

Dependent Variable: Median attractiveness rating

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	61.167[a]	5	12.233	8.936	.000
Intercept	1541.333	1	1541.333	1125.843	.000
FaceType	21.333	1	21.333	15.583	.000
Alcohol	16.542	2	8.271	6.041	.005
FaceType * Alcohol	23.292	2	11.646	8.507	.001
Error	57.500	42	1.369		
Total	1660.000	48			
Corrected Total	118.667	47			

a. R Squared = .515 (Adjusted R Squared = .458)

Figure 14.11 plots the means and shows that the significant main effect reflects the fact that average attractiveness ratings were higher for the photos of attractive faces than unattractive ones. Of course, this result is not at all surprising because the attractive faces were pre-selected to be more attractive than the unattractive faces. This result is a useful manipulation check though: our participants, other things being equal, found the attractive faces more attractive than the unattractive ones.

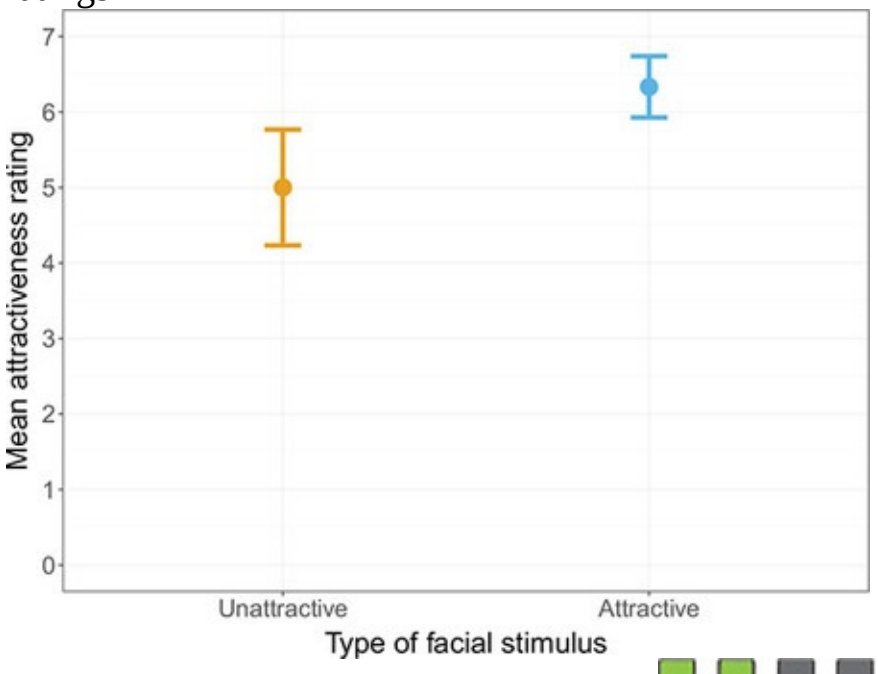
14.6.2 The main effect of alcohol



Output 14.3 also shows a significant main effect of alcohol on attractiveness ratings (because the significance value of 0.005 is less than 0.05). This result means that *when we ignore whether the participant rated unattractive or*

attractive faces the amount of alcohol influenced their attractiveness ratings. The best way to understand this effect is to plot the average attractiveness rating at each level of alcohol (ignoring the type of face completely) – we calculated these means in Section 14.3.5. Figure 14.12 shows such a graph, the mean attractiveness ratings increase quite linearly as more alcohol is drunk. This significant main effect is *likely* to reflect this trend. Looking at the error bars (95% confidence intervals), there is a lot of overlap between the placebo and low-dose groups (implying that these groups have similar average ratings) but the overlap between the placebo and high-dose groups is less, and quite possibly within what you'd expect from a significant difference (see Section 2.9.9). The confidence intervals for the low-and high-dose groups also overlap a lot, suggesting that these groups do not differ. Therefore, we might speculate based on the confidence intervals that this main effect reflects a difference between the placebo and high-dose groups but that no other groups differ. It could also reflect the linear increase in ratings as the dose of alcohol increases.

Figure 14.11 Graph to show the main effect of type of face on attractiveness ratings



14.6.3 The interaction effect

Finally, Output 14.3 tells us about the interaction between the effect of type of face and the effect of alcohol. The *F*-statistic is highly significant (because the observed *p*-value of 0.001 is less than 0.05). This effect means that the effect of alcohol on attractiveness ratings was different when rating unattractive faces

compared to when rating attractive faces. The output will include the (probably badly scaled) plot that we asked for in Figure 14.7. Figure 14.13 is a nicer version of this graph that shows the estimated marginal means from Output 14.4. We can use this graph to get a handle on the interaction effect. Focus first on the blue line, which is flat and shows very little difference in average attractiveness ratings across the alcohol conditions. This line shows that *when rating attractive faces, alcohol has very little effect*. Now look at the orange line, which slopes upwards, showing that when rating unattractive faces ratings increase with the dose of alcohol. This line shows that *when rating unattractive faces, alcohol has an effect*. The significant interaction reflects the differing effect of alcohol when rating attractive and unattractive faces; that is, that alcohol has an effect on ratings of unattractive faces but not for attractive ones. This example illustrates an important point about interaction effects. We concluded earlier that alcohol significantly affected attractiveness ratings (the **Alcohol** main effect) but the interaction effect qualifies this conclusion by showing that this is true only when rating unattractive faces (ratings of attractive faces seem unaffected by alcohol). The take-home message is that *you should not interpret a main effect in the presence of a significant interaction involving that main effect.*



Figure 14.12 Graph showing the main effect of alcohol on attractiveness ratings

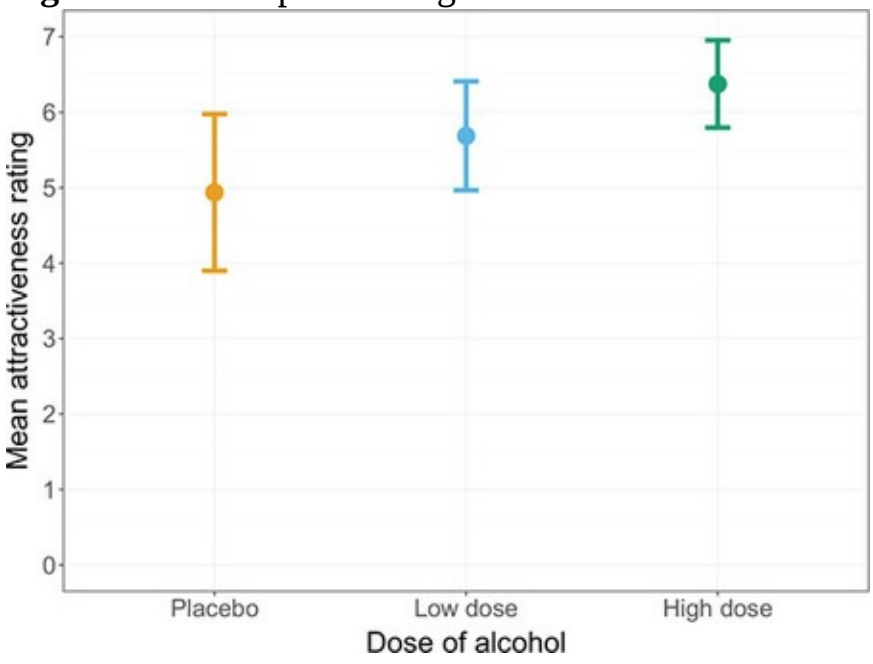
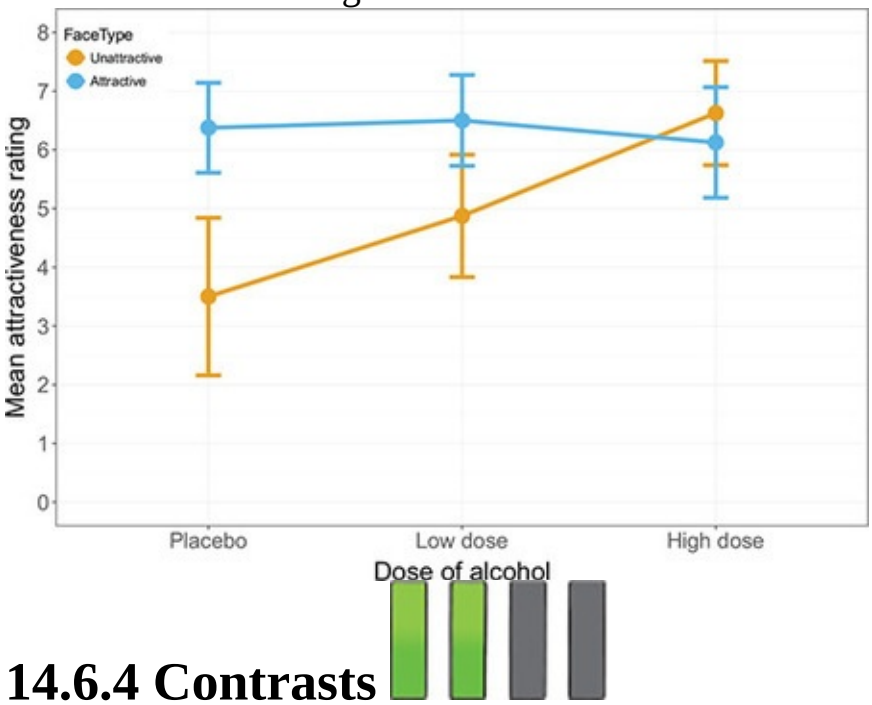


Figure 14.13 Graph of the interaction of type of face and alcohol consumption on attractiveness ratings



14.6.4 Contrasts

Output 14.5 shows the Helmert contrasts on the effect of alcohol. Ordinarily we wouldn't look at these because the interaction involving alcohol was significant, but for the sake of explanation I will. The top of the table shows the contrast for *Level 1 vs. Later*, which in this case means the placebo group compared to the two alcohol groups. This contrast tests whether the mean of the placebo group (4.938) is different from the mean of the low-and high-dose groups combined ((5.688 + 6.375)/2 = 6.032). This is a difference of −1.094 (4.938 − 6.032), which both the *Contrast Estimate* and the *Difference* in the table tell us. The value of *Sig.* (0.004) tells us that this difference is significant because it is smaller than the criterion of 0.05. The confidence interval for this difference also doesn't cross zero so, assuming this sample is one of the 95 out of 100 that produce a confidence interval containing the true value of the difference, the real difference between the placebo and alcohol groups is not zero (between −1.817 and −0.371, to be precise). We could conclude that the effect of alcohol is that any amount of alcohol increases the attractiveness ratings of pictures compared to when a placebo was drunk. However, we need to look at the remaining contrast to qualify this statement.

Output 14.4

3. Attractivenss of facial stimuli * Alcohol consumption

Dependent Variable: Median attractiveness rating

Attractivenss of facial stimuli	Alcohol consumption	Mean	Std. Error	95% Confidence Interval		Bootstrap for Mean[a]			
				Lower Bound	Upper Bound	Bias	Std. Error	BCa 95% Confidence Interval Lower	Upper
Unattractive	Placebo	3.500	.414	2.665	4.335	-.018	.570	2.429	4.571
	Low dose	4.875	.414	4.040	5.710	.006	.433	4.000	5.800
	High dose	6.625	.414	5.790	7.460	.002	.363	6.000	7.333
Attractive	Placebo	6.375	.414	5.540	7.210	-.002	.312	5.826	7.000
	Low dose	6.500	.414	5.665	7.335	.015	.316	5.833	7.200
	High dose	6.125	.414	5.290	6.960	.007	.404	5.375	7.000

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

The bottom of the table shows the contrast for *Level 2 vs. Level 3*, which in this case means the low-dose group compared to the high. This contrast compares the mean of the low-dose group (5.688) to the mean of the high (6.375). This is a difference of −0.687 (5.688 − 6.375),[3] which both the *Contrast Estimate* and the *Difference* in the table tell us. This difference is not significant (because *Sig.* is 0.104, which is greater than 0.05). The confidence interval for this difference also crosses zero so, assuming this sample is one of the 95 out of 100 that produced confidence intervals that contain the true value of the difference, the real difference is between −1.522 and 0.0147 and could be zero. This contrast tells us that having high dose of alcohol doesn't significantly affect attractiveness ratings compared to having a low dose.

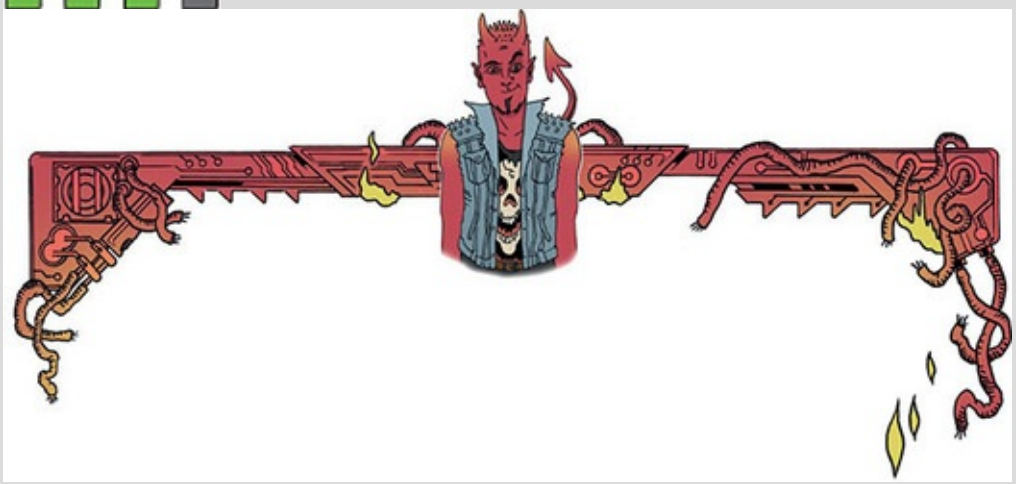
3 Because I've rounded the means to 3 decimal places, the value here differs from Output 14.5 very slightly.

Output 14.5

Contrast Results (K Matrix)

Alcohol consumption	Helmert Contrast		Dependent Variable: Median attractiveness rating
Level 1 vs. Later	Contrast Estimate		-1.094
	Hypothesized Value		0
	Difference (Estimate - Hypothesized)		-1.094
	Std. Error		.358
	Sig.		.004
	95% Confidence Interval for Difference	Lower Bound	-1.817
		Upper Bound	-.371
Level 2 vs. Level 3	Contrast Estimate		-.688
	Hypothesized Value		0
	Difference (Estimate - Hypothesized)		-.688
	Std. Error		.414
	Sig.		.104
	95% Confidence Interval for Difference	Lower Bound	-1.522
		Upper Bound	.147

SPSS Tip 14.1 Simple effects analysis using SPSS Statistics



The syntax for a simple effects analysis of the beer-goggles data is in the file **GogglesSimpleEffects.sps**. The commands are:

```
GLM Attractiveness by FaceType Alcohol
/EMMEANS = TABLES(FaceType*Alcohol)
COMPARE(FaceType).
```

The first line specifies the model using the *GLM* command, followed by the outcome variable (**Attractiveness**), the *BY* command and then a list of predictor/independent variables (**FaceType** and **Alcohol**). The line beginning */EMMEANS* specifies the simple effects. *COMPARE(FaceType)* specifies to look at the effect of type of face at each level of alcohol. Run the syntax (make sure you have **Goggles.sav** loaded) and you'll get the same output as in the chapter, but with an extra table containing the simple effects (Output 14.6). There was a significant difference in ratings of unattractive and attractive faces in the placebo group, $p < 0.001$, and in the low-dose group, $p = 0.008$, but not in the high-dose group, $p = 0.398$. These results confirm our speculation based on Figure 14.13 (see main text).

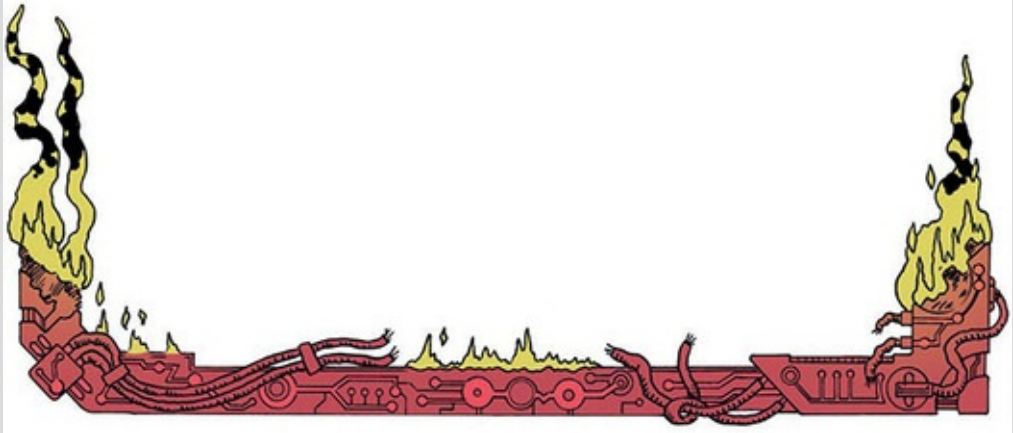
Output 14.6

Univariate Tests

Dependent Variable: Median attractiveness rating

Alcohol consumption		Sum of Squares	df	Mean Square	F	Sig.
Placebo	Contrast	33.063	1	33.063	24.150	.000
	Error	57.500	42	1.369		
Low dose	Contrast	10.563	1	10.563	7.715	.008
	Error	57.500	42	1.369		
High dose	Contrast	1.000	1	1.000	.730	.398
	Error	57.500	42	1.369		

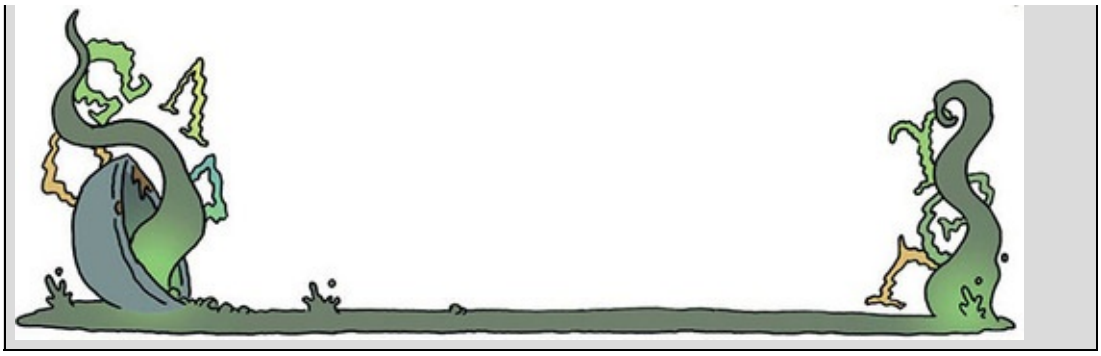
Each F tests the simple effects of Attractivenss of facial stimuli within each level combination of the other effects shown. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.



Oliver Twisted Please, Sir, can I have some more … simple effects?



‘I want to impress my friends by doing a simple effects analysis by hand,’ boasts Oliver. You don’t really need to know how simple effects analyses are calculated to run them, Oliver, but seeing as you asked, it is explained on the companion website.



Output 14.7

Multiple Comparisons

Dependent Variable: Median attractiveness rating

	(I) Alcohol consumption	(J) Alcohol consumption	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
Bonferroni	Placebo	Low dose	-.75	.414	.231	-1.78	.28
		High dose	-1.44*	.414	.004	-2.47	-.41
	Low dose	Placebo	.75	.414	.231	-.28	1.78
		High dose	-.69	.414	.312	-1.72	.34
	High dose	Placebo	1.44*	.414	.004	.41	2.47
		Low dose	.69	.414	.312	-.34	1.72

Based on observed means.
The error term is Mean Square(Error) = 1.392.

*. The mean difference is significant at the .05 level.

Bootstrap for Multiple Comparisons

Dependent Variable: Median attractiveness rating

	(I) Alcohol consumption	(J) Alcohol consumption	Mean Difference (I-J)	Bootstrap[a] Bias	Bootstrap[a] Std. Error	Bootstrap[a] BCa 95% Confidence Interval Lower	Bootstrap[a] BCa 95% Confidence Interval Upper
Bonferroni	Placebo	Low dose	-.75	-.02	.59	-1.94	.40
		High dose	-1.44	-.01	.57	-2.54	-.41
	Low dose	Placebo	.75	.02	.59	-.40	1.94
		High dose	-.69	.01	.42	-1.49	.14
	High dose	Placebo	1.44	.01	.57	.31	2.60
		Low dose	.69	-.01	.42	-.12	1.45

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

Output 14.8

Median attractiveness rating

	Alcohol consumption	N	Subset 1	Subset 2
Ryan-Einot-Gabriel-Welsch Range[a]	Placebo	16	4.94	
	Low dose	16	5.69	5.69
	High dose	16		6.38
	Sig.		.077	.104

Means for groups in homogeneous subsets are displayed.
Based on observed means.
The error term is Mean Square(Error) = 1.369.

a. Alpha = .05.

Cramming Sam's Tips Factorial ANOVA



- Two-way independent designs compare several means when there are two independent variables and different entities have been used in all experimental conditions. For example, if you wanted to know whether different teaching methods worked better for different topics, you could take students from four courses (Psychology, Geography, Management, and Statistics) and assign them to either lecture-based or book-based teaching. The two variables are topic and method of teaching. The outcome might be the end-of-year mark (as a percentage).
- In the table labelled *Tests of Between-Subjects Effects*, look at the column labelled *Sig.* for all main effects and interactions; if the value is less than 0.05 then the effect is significant using the conventional criterion.
- To interpret a significant interaction, plot an interaction graph and conduct simple effects analysis.
- You don't need to interpret main effects if an interaction effect involving that variable is significant.
- If significant main effects are not qualified by an interaction then consult *post hoc* tests to see which groups differ: significance is shown by values smaller than 0.05 in the columns labelled *Sig.*, and bootstrap confidence intervals that do not contain zero.
- Test the same assumptions as for any linear model (see Chapter 6).



14.6.5 Simple effects analysis 

A particularly effective way to break down interactions is **simple effects analysis**, which looks at the effect of one independent variable at individual levels of the other independent variable. For example, we could do a simple effects analysis looking at the effect of type of face at each level of alcohol. This would mean taking the average attractiveness rating of unattractive faces and comparing it to that for attractive faces after a placebo drink, then making the same comparison after a low dose of alcohol, and then finally for a high dose. Another way of looking at this is to say we would look at the distance between the blue and orange dots in Figure 14.13 at each dose of alcohol: based on the graph, we might expect to find a difference in ratings after a placebo and possibly the low dose (the blue and orange dots are very far apart) but not after a high dose (the dots are in practically the same location).

An alternative is to quantify the effect of alcohol (the pattern of means across the placebo, low dose and high dose) separately for unattractive and attractive faces. This analysis would look at whether the means portrayed by the blue line in Figure 14.13 differ, and then separately do the same for the orange line. Simple effects analyses can't be run through dialog boxes – you need to use syntax (see SPSS Tip 14.1).

14.6.6 Post hoc analysis 

The Bonferroni *post hoc* tests (Output 14.7) break down the main effect of alcohol and can be interpreted as if **Alcohol** were the only predictor in the model (i.e., the reported effects for alcohol are collapsed with regard to the type of face). The tests show (both by the significance and whether the bootstrap confidence intervals cross zero) that when participants had a high dose of alcohol their ratings of faces were significantly higher than those who had a placebo drink ($p = 0.004$) but not than those who had a low dose of alcohol ($p = 0.312$), and that ratings were not significantly different between those who had a low dose of alcohol and those in the placebo group ($p = 0.231$).

The REGWQ test (Output 14.8) confirms that the means of the placebo and low-dose conditions were equivalent ($p = 0.077$), as were the means of the low-and high-dose groups ($p = 0.104$). I note again that we wouldn't normally interpret *post hoc* tests when there is a significant interaction involving that main effect (as there is here).

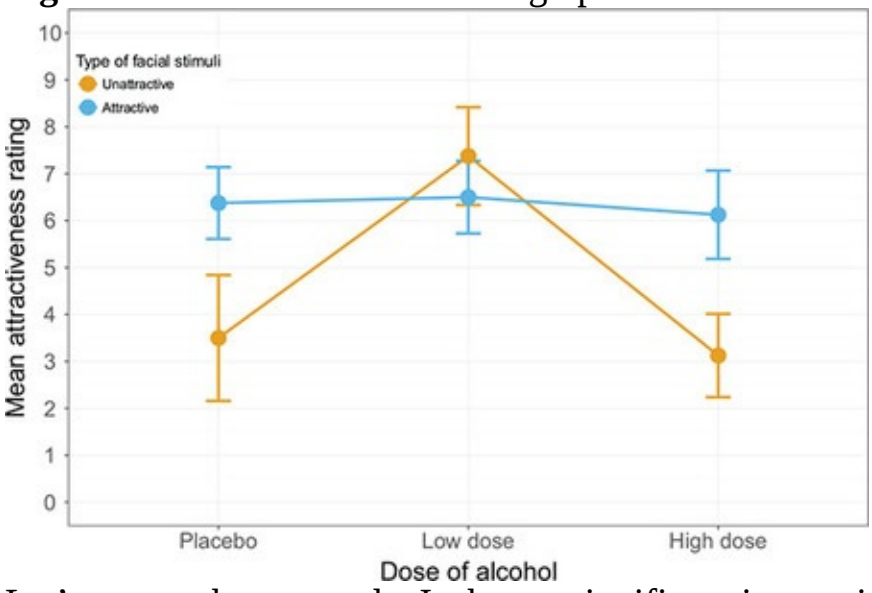
14.7 Interpreting interaction graphs

We can sum up the findings of the study as follows. Alcohol has an effect on ratings of the attractiveness of pictures of faces, but only when those pictures depict unattractive faces. When the faces are attractive, alcohol has no significant effect on judgements of attraction. This pattern of results was evident from the interaction graph (Figure 14.13).

Let's have a look at other examples of interaction graphs to practise interpreting them. Imagine we got the profile of results in Figure 14.14. Do you think the interaction effect would still have been significant?

This profile of data probably would also give rise to a significant interaction term because the ratings of attractive and unattractive faces are different in the placebo and high-dose group, but relatively similar in the low-dose group. Visually, the blue line shows a different pattern than the orange one. This situation reflects a world in which the beer-goggles effect kicks in at low doses of alcohol, but after a higher dose reality kicks back in again. Theoretically more difficult to explain, but nevertheless this profile of results reflects an interaction because the difference between ratings of attractive and unattractive faces varies depending on how much alcohol has been drunk.

Figure 14.14 Another interaction graph



Let's try another example. Is there a significant interaction in Figure 14.15?

The data in Figure 14.15 are unlikely to reflect a significant interaction because the effect of alcohol is the same for unattractive and attractive photos: ratings of attractiveness increase as the dose of alcohol increases, and this is true for both attractive and unattractive faces. Visually, the blue line shows the same pattern

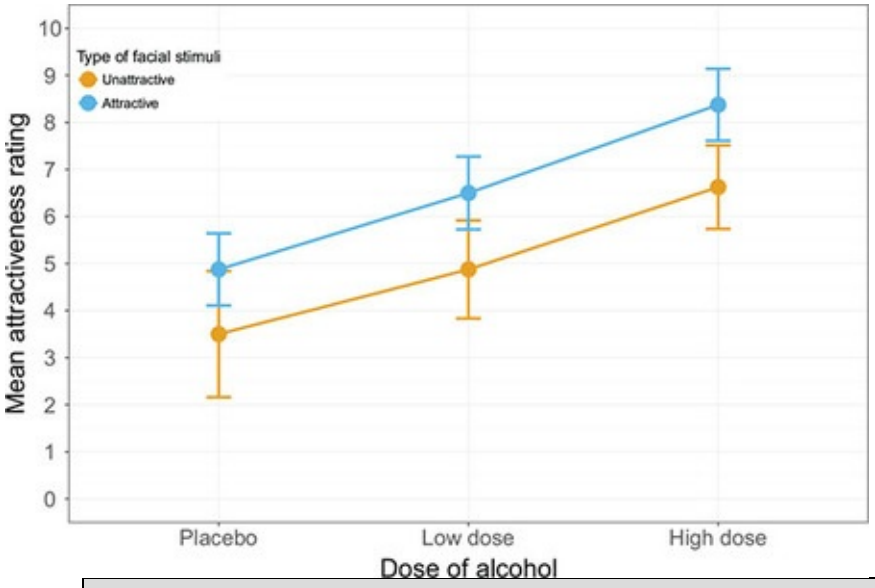
as the orange one.

Two general points that we can make from these examples are:

1. Non-parallel lines on an interaction graph indicate some degree of interaction, but how strong and whether the interaction is significant depends on how non-parallel the lines are.
2. Lines on an interaction graph that cross are very non-parallel, which hints at a possible significant interaction. However, crossing lines don't *always* reflect a significant interaction.

Sometimes people use bar charts to show interactions, so let's look at those in Figure 14.16. Panels (a) and (b) display the data from the example used in this chapter (have a go at plotting them for yourself). The data are presented in two different ways: panel (a) shows the data when levels of alcohol are placed along the *x*-axis and different-coloured bars show the means for unattractive and attractive faces, and panel (b) shows the opposite scenario where type of face is plotted on the *x*-axis and different colours distinguish the dose of alcohol. Both graphs show the significant interaction effect from the example. You're looking for the differences in heights of coloured bars to vary at different points along the *x*-axis. For example, in panel (a) you'd look at the difference between the light and dark blue bars for placebo, and then look at, say, the high dose and ask, 'Is the difference between the bars the same as for the placebo?' In this case the difference in height between the dark-and light-blue bars for the placebo is bigger than the same difference for the high-dose group, hence an interaction. Panel (b) shows the same thing, but plotted the other way around. Again look at the pattern of responses. First look at the unattractive faces and see that the pattern is that attractiveness ratings increase as the dose of alcohol goes up (the bars increase in height). Now look at the attractive faces. Is the pattern the same? No, the bars are all the same height, showing that ratings don't change as a function of alcohol. The interaction effect is shown up by the fact that for the attractive faces the bars follow a different pattern than the unattractive faces.

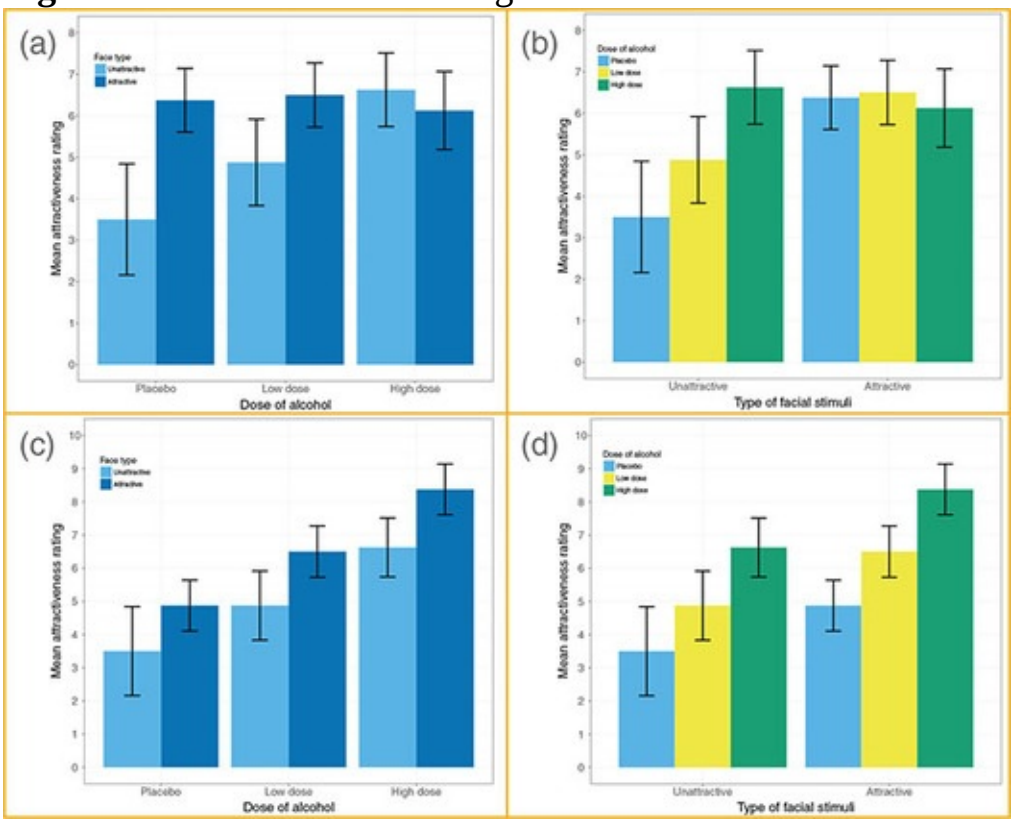
Figure 14.15 A 'lack of' interaction graph





What about panels (c) and (d): do you think there is an interaction?

Figure 14.16 Bar charts showing interactions between two variables



Panels (c) and (d) display the same data in two different ways, albeit different data than the example in the chapter. In the placebo group in panel (c) the dark

bar is a little bit bigger than the light one; moving on to the low-dose group, the dark bar is also a little bit taller than the light bar; and finally, for the high-dose data the dark bar is again higher than the light one. In all conditions the same pattern is shown – the dark-blue bar is a similar amount taller than the light-blue one (i.e., the amount by which attractive faces are rated more attractive than unattractive faces is similar in all three alcohol groups) – therefore, there is no interaction. Panel (d) shows a similar result. For unattractive faces the pattern is that attractiveness ratings increase as more alcohol is drunk (the bars increase in height), and for the attractive faces the same pattern emerges, with ratings increasing as more alcohol is consumed. This again indicates no interaction: the change in attractiveness ratings due to alcohol is similar for unattractive and attractive faces.

14.8 Robust models of factorial designs 

If you requested robust standard errors in Figure 14.10 then you'll find a table of parameter estimates for the model in the output with confidence intervals and *p*-values robust to heteroscedasticity (Output 14.9). These are a little hard to unpick. SPSS Statistics automatically uses the last category as the reference category. The parameter estimate for face type, because it had only two categories, gives us a robust estimate of the effect on ratings of attractive faces compared to unattractive ones. For the main effect of alcohol the first estimate *[Alcohol = 0]* compares the placebo to the high dose and *[Alcohol = 1]* compares the low dose to the high dose. Both parameters are not significant. For the interaction term, the first parameter, *[FaceType = 0]*[Alcohol = 0]* suggests that the difference between ratings of attractive and unattractive faces in the placebo was significantly different to the same difference in the high dose group ($p < 0.001$). The second parameter, *[FaceType = 0]*[Alcohol = 1]* suggests that the difference between ratings of attractive and unattractive faces in the low dose group was significantly different to the same difference in the high dose group ($p = 0.009$). These conclusions are based on robust confidence intervals and *p*-values.

It's possible to run a robust test for two-and three-way factorial designs using R (make sure the *Essentials for R* plugin and *WRS2* package are installed – see Section 4.13). The companion website contains the syntax file **robust2way.sps** for running a robust test of trimmed means for two independent variables/predictors (*t2way*) and associated robust *post hoc* tests (*mcp2atm*) described by Wilcox (2017).[4] The syntax in the file is as follows:

4 You'll also find the file robust3way.sps, which includes the function *t3way*

that performs the same test for designs with three independent variables and predictors, but you'll need to edit this file to include the variable names from your data.

```
BEGIN PROGRAM R.
library(WRS2)
mySPSSdata = spssdata.GetDataFromSPSS(factorMode = "labels")
t2way(Attractiveness ~ FaceType*Alcohol, data = mySPSSdata, tr = 0.2)
mcp2atm(Attractiveness ~ FaceType*Alcohol, data = mySPSSdata, tr =
0.2)
END PROGRAM.
```

Select and run these six lines of syntax (see SPSS Tip 14.2) to get the output in Output 14.10. These results confirm what the linear model showed: significant main effects of alcohol, $F_t = 10.31$, $p = 0.019$, and face type, $F_t = 14.57$, $p = 0.001$, and the interaction, $F_t = 16.60$, $p = 0.003$. In the *post hoc* tests, *Alcohol1* is the difference between placebo and low dose, *Alcohol2* is the difference between placebo and high dose, and *Alcohol3* is the difference between low and high dose. We can see from the *p*-values and confidence intervals that attractiveness ratings were significantly different between the placebo and high-dose groups, $\hat{\Psi} = -2.83$, $p = 0.005$, but not between the placebo and low-dose groups, $\hat{\Psi} = -1.5$, $p = 0.080$, or between the low-and high-dose groups, $\hat{\Psi} = -1.33$, $p = 0.106$. Most interesting are the interaction terms, which look at the difference between ratings of attractive and unattractive faces across the group comparisons we've just looked at. These effects tell us that the difference in ratings between attractive and unattractive faces in the high-dose group was significantly different from the corresponding difference in the placebo group, $\hat{\Psi} = -3.5$, $p = 0.001$, and the low-dose group, $\hat{\Psi} = -2.33$, $p = 0.008$. However, the difference in ratings between attractive and unattractive faces in the low-dose group was not significantly different from the corresponding difference in the placebo group, $\hat{\Psi} = -1.17$, $p = 0.163$. In a nutshell, the interaction seems to be driven by effects of a high dose of alcohol compared to the other conditions.

Output 14.9

Parameter Estimates with Robust Standard Errors

Dependent Variable: Median attractiveness rating

Parameter	B	Robust Std. Error[a]	t	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Intercept	6.125	.398	15.386	.000	5.322	6.928
[FaceType=0]	.500	.547	.914	.366	-.604	1.604
[FaceType=1]	0[b]	.	.	.	.	.
[Alcohol=0]	.250	.513	.487	.629	-.786	1.286
[Alcohol=1]	.375	.515	.728	.471	-.665	1.415
[Alcohol=2]	0[b]	.	.	.	.	.
[FaceType=0] * [Alcohol=0]	-3.375	.852	-3.963	.000	-5.094	-1.656
[FaceType=0] * [Alcohol=1]	-2.125	.775	-2.742	.009	-3.689	-.561
[FaceType=0] * [Alcohol=2]	0[b]	.	.	.	.	.
[FaceType=1] * [Alcohol=0]	0[b]	.	.	.	.	.
[FaceType=1] * [Alcohol=1]	0[b]	.	.	.	.	.
[FaceType=1] * [Alcohol=2]	0[b]	.	.	.	.	.

a. HC4 method

b. This parameter is set to zero because it is redundant.

Output 14.10

```
t2way(formula = Attractiveness ~ FaceType * Alcohol, data = mySPSSdata
  tr = 0.2)
                  value p.value
FaceType        14.5730   0.001
Alcohol         10.3117   0.019
FaceType:Alcohol 16.6038   0.003

Call:
mcp2atm(formula = Attractiveness ~ FaceType * Alcohol, data = mySPSSdata,
  tr = 0.2)
                    psihat    ci.lower    ci.upper      p-value
FaceType1         -3.83333    -5.90979    -1.75688      0.00088
Alcohol1          -1.50000    -3.63960     0.63960      0.07956
Alcohol2          -2.83333    -5.17113    -0.49554      0.00543
Alcohol3          -1.33333    -3.38426     0.71759      0.10569
FaceType1:Alcohol1 -1.16667   -3.30627     0.97294      0.16314
FaceType1:Alcohol2 -3.50000   -5.83780    -1.16220      0.00110
FaceType1:Alcohol3 -2.33333   -4.38426    -0.28241      0.00802
```

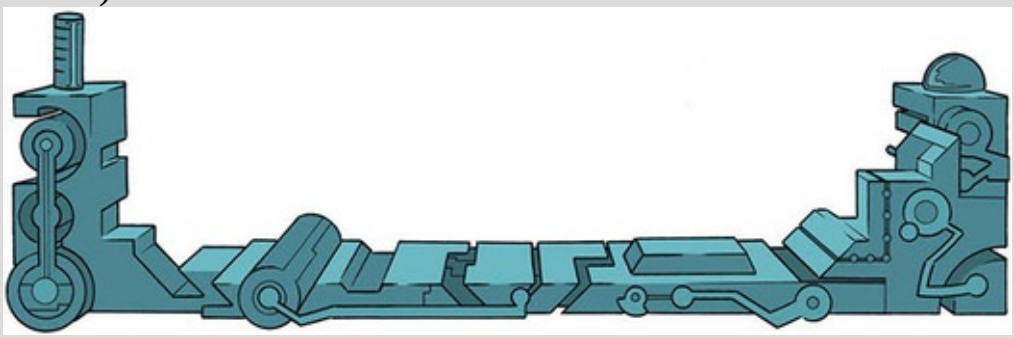
Labcoat Leni's Real Research 14.1 Going out on the pierce



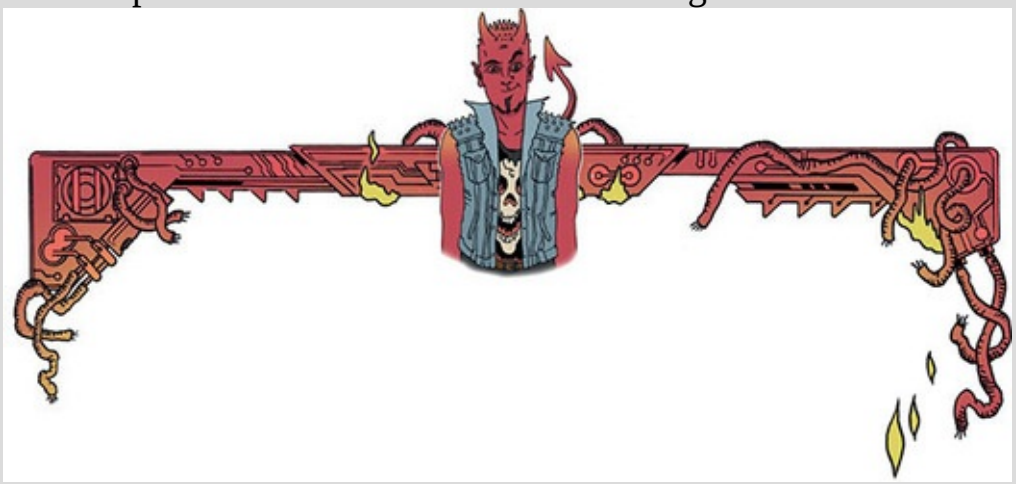
Guéguen, N. (2012). *Alcoholism: Clinical and Experimental Research*, *36*(7), 1253–1256.

Tattoos and body piercings have become very popular since I was young. I have often contemplated having Ronald Fisher's face tattooed over my own so that people will think I'm a genius. But I digress. Research has shown that people who have tattoos and piercings are more likely to engage in risky behaviour. Nicolas Guéguen (2012) measured the level of intoxication (mass of alcohol per litre of breath exhaled, **Alcohol**) in 1965 French youths as they left bars. This measure was an indicator of risky behaviour. Each youth was also classified as having tattoos, piercings, both or neither (**Group**), and their gender was noted (**Gender**). The data are in the

file **Gueguen (2012).sav**. Was the level of risk (i.e., alcohol) greater in groups who had tattoos and piercings? Did this effect interact with gender? Draw an error bar chart of the data too. Answers are on the companion website (or look at pages 1254–1255 in the original article).



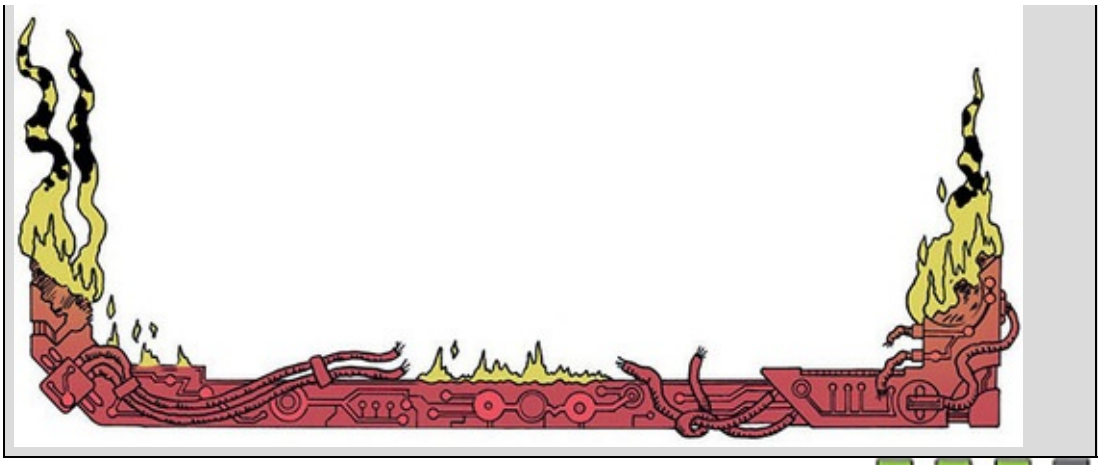
SPSS Tip 14.2 Robust tests for factorial designs 



To break down the syntax, we're using two functions (*t2way* and *mcp2atm*) that both have the same form:

```
t2way(outcome~predictor1*predictor2, data = mySPSSdata, tr = 0.2)
mcp2atm(outcome~predictor1*predictor2, data = mySPSSdata, tr = 0.2)
```

In our example we use our variable names to replace outcome with **Attractiveness** and predictors with **FaceType** and **Alcohol**. So the formula in the function becomes *Attractiveness ~ FaceType*Alcohol*. *tr* specifies the amount of trimming (by default 0.2, or 20%, but you could change it to a different proportion).



14.9 Bayesian models of factorial designs

You can't easily compute Bayes factors (Section 3.8.4) for factorial designs. Although we're using a linear model, so we could follow a similar process to that in Section 9.13 and specify **FaceType** and **Alcohol** as *Factor(s)* the problem is that the model will include only the main effects, and not the interaction term. This is a problem because the interaction term is typically where our interests lie. The way around this would be to manually dummy code **FaceType** and **Alcohol** (remembering that this variable would become two dummy variables), and then manually create the interaction term by multiplying the **FaceType** dummy variable by each of the **Alcohol** dummy variables to create two dummy variables for the interaction term. The file **goggles_dummy.sav** contains these dummy variables and the coding is illustrated in Table 14.4. **FaceType** is coded with 0s for unattractive faces and 1s for attractive faces. **Alcohol** is split into two dummy variables (columns 4 and 5). The first (**Alc_high_pla**) codes the high dose against placebo and the second (**Alc_low_pla**) codes the low dose against placebo. This coding scheme should be familiar from Chapter 12. The final two columns code the interaction term. The first (**Int_high_pla**) are the codes from columns 3 and 4 multiplied (i.e., **FaceType** × **Alc_high_pla**) and represent the difference between ratings for unattractive faces compared to attractive ones in the placebo group relative to the same difference in the high dose group. The second (**Int_low_pla**) are the codes from columns 3 and 5 multiplied (i.e., **FaceType** × **Alc_low_pla**) and represent the difference between ratings for unattractive faces compared to attractive ones in the placebo group relative to the same difference in the low dose group.

To get Bayes factors and estimates with the default reference priors follow the process in Section 9.13 and drag all five dummy variables (**FaceType,**

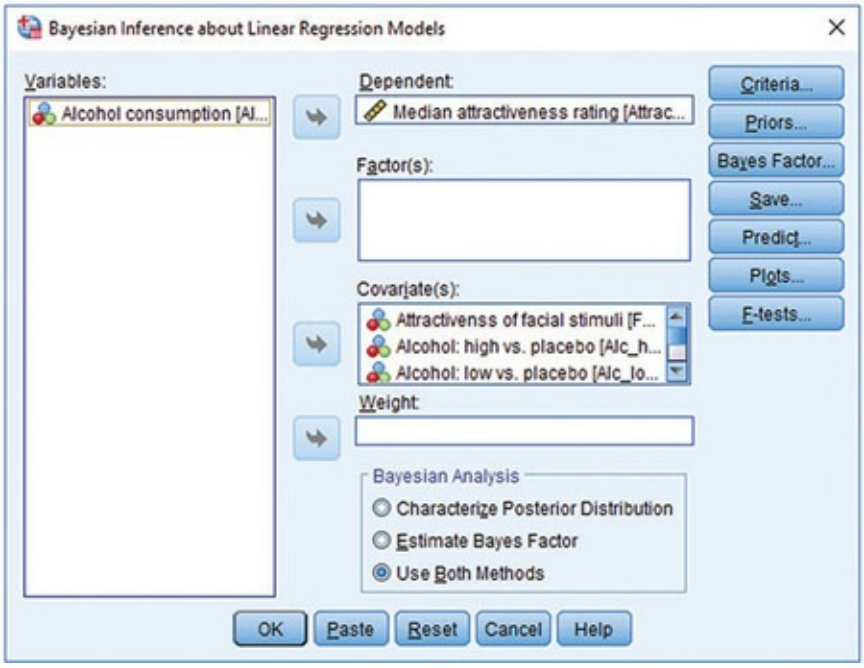
Alc_high_pla, Alc_low_pla, Int_high_pla, Int_low_pla) to the box labelled Covariate(s). (If you drag them to box labelled Factor(s) then SPSS Statistics will recode them, so by using the Covariate(s) box the output will match the dummy coding that we used in the data editor.) Select and leave the default options alone.

Output 14.11 shows the Bayes factor for the model including all of the predictors (all 5 dummy variables) against the null model (which I assume includes only the intercept). This Bayes factor is 1195.598. This is very strong evidence for the model. It means that the probability of the data given the model including all five predictors is about 1196 greater than the probability of the data given the model with only the intercept. We should shift our belief in the model (relative to the null model) by a factor of 1196!

Table 14.4 Dummy coding scheme for the beer goggles data

Type of faces rated	Alcohol	FaceType	**Alc_high_pla** (High vs Placebo)	**Alc_low_pla** (Low vs Placebo)	**Int_high_pla** (High vs Placebo)	**Int_low_pla** (Low vs Placebo)
Unattractive	Placebo	0	0	0	0	0
Unattractive	Low dose	0	0	1	0	0
Unattractive	High dose	0	1	0	0	0
Attractive	Placebo	1	0	0	0	0
Attractive	Low dose	1	0	1	0	1
Attractive	High dose	1	1	0	1	0

Figure 14.17 Dialog box for Bayesian analysis of factorial designs



Output 14.11

Bayes Factor Model Summary[a,b]

Bayes Factor[c]	R	R Square	Adjusted R Square	Std. Error of the Estimate
1195.598	.718	.515	.458	1.17

a. Method: JZS

b. Model: (Intercept), Attractivenss of facial stimuli, Alcohol: high vs. placebo, Alcohol: low vs. placebo, Interaction: high vs placebo, att vs unatt, Interaction: low vs placebo, att vs unatt

c. Bayes factor: Testing model versus null model (Intercept).

Bayesian Estimates of Coefficients[a,b,c]

		Posterior		95% Credible Interval	
Parameter	Mode	Mean	Variance	Lower Bound	Upper Bound
(Intercept)	3.500	3.500	.180	2.665	4.335
Attractivenss of facial stimuli	2.875	2.875	.359	1.694	4.056
Alcohol: high vs. placebo	3.125	3.125	.359	1.944	4.306
Alcohol: low vs. placebo	1.375	1.375	.359	.194	2.556
Interaction: high vs placebo, att vs unatt	-3.375	-3.375	.719	-5.045	-1.705
Interaction: low vs placebo, att vs unatt	-1.250	-1.250	.719	-2.920	.420

a. Dependent Variable: Median attractiveness rating

b. Model: (Intercept), Attractivenss of facial stimuli, Alcohol: high vs. placebo, Alcohol: low vs. placebo, Interaction: high vs placebo, att vs unatt, Interaction: low vs placebo, att vs unatt

c. Assume standard reference priors.

The Bayesian estimate of the *b*s for the dummy variables can be found in the columns labelled *Posterior Mode* and *Posterior Mean*. The 95% credible intervals for the model parameters contain the population value with a probability of 0.95 (95%). For the difference between ratings for unattractive faces compared to attractive ones in the placebo group relative to the same difference in the high dose group (i.e., the first interaction term), the relative difference has a 95% probability of being between about −5 and −1.7. For the difference between ratings for unattractive faces compared to attractive ones in the placebo group relative to the same difference in the low dose group (i.e., the second interaction term), the relative difference has a 95% probability of being between about −2.9 and 0.4. These credible intervals assume that the effect exists but in the later case it suggests that the effect could plausibly be positive, negative and very small indeed.

14.10 Calculating effect sizes 

SPSS can produce partial eta squared, η^2 (Section 13.8). However, we've seen before that omega squared (ω^2) is less biased. The calculation of omega squared becomes somewhat more cumbersome in factorial designs ('somewhat' being an understatement). Howell (2012), as ever, does a wonderful job of explaining it all (and has a nice table summarizing the various components for a variety of situations). Condensing all of this information, I'll say only that we first compute a variance component for each of the effects (the two main effects and the

interaction term) and the error, and then use these to calculate effect sizes for each. If we call the first main effect *A*, the second main effect *B* and the interaction effect $A \times B$, then the variance components for each of these are calculated from the mean squares of each effect and the sample sizes on which they're based:

$$\hat{\sigma}^2_{\alpha} = \frac{(a-1)(\mathrm{MS}_A - \mathrm{MS}_\mathrm{R})}{nab}$$
$$\hat{\sigma}^2_{\beta} = \frac{(b-1)(\mathrm{MS}_B - \mathrm{MS}_\mathrm{R})}{nab} \qquad (14.24)$$
$$\hat{\sigma}^2_{\alpha\beta} = \frac{(a-1)(b-1)(\mathrm{MS}_{A\times B} - \mathrm{MS}_\mathrm{R})}{nab}$$

In these equations, *a* is the number of levels of the first independent variable, *b* is the number of levels of the second independent variable, and *n* is the number of people per condition.

Let's calculate these for our data. Output 14.3 contains the mean squares for each effect and for the error term. The first predictor, alcohol, had three levels (hence $a = 3$) and a mean squares of 8.271. Type of face had two levels (hence $b = 2$) and a mean squares of 21.333. The interaction had a mean squares of 11.646. The number of people in each group was 8 and the residual mean squares were 1.369. Therefore, our equations become:

$$\hat{\sigma}^2_{\alpha} = \frac{(3-1)(8.271-1.369)}{8\times 3\times 2} = 0.288$$
$$\hat{\sigma}^2_{\beta} = \frac{(2-1)(21.333-1.369)}{8\times 3\times 2} = 0.416 \qquad (14.25)$$
$$\hat{\sigma}^2_{\alpha\beta} = \frac{(3-1)(2-1)(11.646-1.369)}{8\times 3\times 2} = 0.428$$

We estimate the total variability by adding the estimates in equation (14.25) to the residual mean squares:

$$\begin{aligned}\hat{\sigma}^2_{\text{total}} &= \hat{\sigma}^2_{\alpha} + \hat{\sigma}^2_{\beta} + \hat{\sigma}^2_{\alpha\beta} + \mathrm{MS}_\mathrm{R} \\ &= 0.288 + 0.416 + 0.428 + 1.369 \\ &= 2.501\end{aligned} \qquad (14.26)$$

The effect size is the variance estimate for the effect in which you're interested divided by the total variance estimate from equation (14.26):

$$\omega^2_{\text{effect}} = \frac{\hat{\sigma}^2_{\text{effect}}}{\hat{\sigma}^2_{\text{total}}} \qquad (14.27)$$

For the main effect of alcohol we get 0.115:

$$\omega^2_{\text{alcohol}} = \frac{\hat{\sigma}^2_{\text{alcohol}}}{\hat{\sigma}^2_{\text{total}}} = \frac{0.288}{2.501} = 0.115 \qquad (14.28)$$

For the main effect of type of face we get 0.166:

$$\omega^2_{\text{type of face}} = \frac{\hat{\sigma}^2_{\text{type of face}}}{\hat{\sigma}^2_{\text{total}}} = \frac{0.416}{2.501} = 0.166 \quad (14.29)$$

For the interaction of type of face and alcohol we get 0.171:

$$\omega^2_{\text{alcohol} \times \text{type of face}} = \frac{\hat{\sigma}^2_{\text{alcohol} \times \text{type of face}}}{\hat{\sigma}^2_{\text{total}}} = \frac{0.428}{2.501} = 0.171 \quad (14.30)$$

To make these values comparable to *r* we can take the square root, which gives us effect sizes of 0.34 for alcohol and 0.41 for both type of face and the interaction term.

It's also possible to calculate effect sizes for our simple effects analysis (Section 14.6.5). These effects have 1 degree of freedom for the model (which means they're comparing only two things), and in these situations *F* can be converted to *r* using the following equation (which just uses the *F*-statistic and the residual degrees of freedom):[5]

5 If your *F* compares more than two things then a different equation is needed (see Rosenthal, Rosnow, & Rubin, 2000: 44), but I think effect sizes for situations in which only two things are being compared are most useful because they have a clear interpretation.

$$r = \sqrt{\frac{F\left(1, df_R\right)}{F\left(1, df_R\right) + df_R}} \quad (14.31)$$

Looking at Output 14.6, we got *F*s of 24.150, 7.715 and 0.730 for the effects of type of face at placebo, low dose and high dose, respectively. For each of these, the degrees of freedom were 1 for the model and 42 for the residual. Therefore, we get the following effect sizes:

$$r_{\text{Type of face (placebo)}} = \sqrt{\frac{24.15}{24.15 + 42}} = 0.604$$

$$r_{\text{Type of face (low dose)}} = \sqrt{\frac{7.715}{7.715 + 42}} = 0.394 \quad (14.32)$$

$$r_{\text{Type of face (high dose)}} = \sqrt{\frac{0.730}{0.730 + 42}} = 0.131$$

The effect of type of face is large at both placebo and low dose, but becomes small at a high dose of alcohol.

14.11 Reporting the results of factorial designs



We report the details of the *F*-statistic and the degrees of freedom for each effect. For the effects of alcohol and the alcohol × type of face interaction, the model degrees of freedom were 2 ($df_M = 2$), but for the effect of type of face the

degrees of freedom were 1 ($df_M = 1$). For all effects, the degrees of freedom for the residuals were 42 ($df_R = 42$). We can, therefore, report the three effects as follows:

✓ There was a significant main effect of the amount of alcohol consumed on ratings of the attractiveness of faces, $F(2, 42) = 6.04$, $p = 0.005$, $\omega^2 = 0.12$. Bonferroni *post hoc* tests revealed that the attractiveness ratings were significantly higher after a high dose than after a placebo drink ($p = 0.004$). The attractiveness ratings were not significantly different after a low dose compared to a placebo ($p = 0.231$), or a high dose compared to a low dose ($p = 0.312$).

✓ Attractive faces were rated significantly higher than unattractive faces, $F(1, 42) = 15.58$, $p < 0.001$, $\omega^2 = 0.17$.

✓ There was a significant interaction between the amount of alcohol consumed and the type of face of the person rated on attractiveness, $F(2, 42) = 8.51$, $p = 0.001$, $\omega^2 = 0.17$. This effect indicates that ratings of unattractive and attractive faces were affected differently by alcohol. Simple effects analysis revealed that ratings of attractive faces were significantly higher than unattractive faces in the placebo group, $F(1, 42) = 24.15$, $p < 0.001$, and in the low-dose group, $F(1, 42) = 7.72$, $p = 0.008$, but not in the high dose group, $F(1, 42) = 0.73$, $p = 0.398$.

Labcoat Leni's Real Research 14.2 Don't forget your toothbrush?

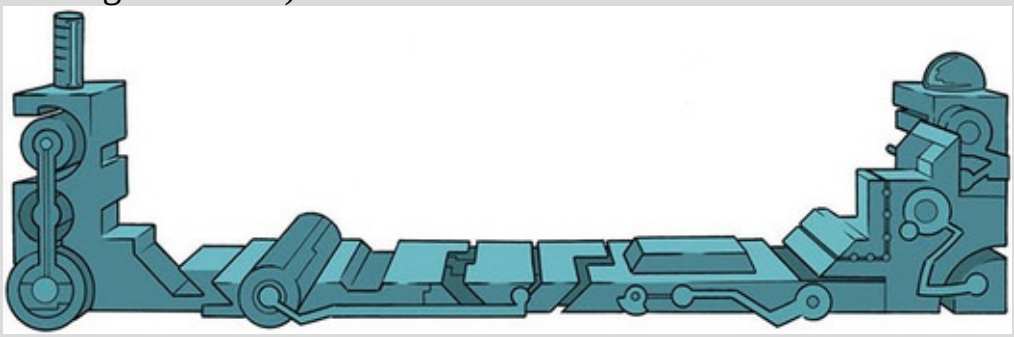


Davey, G. C. L., *et al.* (2003). *Journal of Behavior Therapy & Experimental Psychiatry, 34*, 141–160.

Many of us have experienced that feeling after we have left the house of wondering whether we remembered to lock the door, close the window, or remove the bodies from the fridge in case the police turn

up. However, some people with obsessive compulsive disorder (OCD) check things so excessively that they might, for example, take hours to leave the house. One theory is that this checking behaviour is caused by the mood you are in (positive or negative) interacting with the rules you use to decide when to stop a task (do you continue until you feel like stopping, or until you have done the task as best as you can?). Davey, Startup, Zara, MacDonald, and Field (2003) tested this hypothesis by getting people to think of as many things as they could that they should check before going on holiday (**Checks**) after putting them into a negative, positive or neutral mood (**Mood**). Within each mood group, half of the participants were instructed to generate as many items as they could, whereas the remainder were asked to generate items for as long as they felt like continuing the task (**Stop_Rule**).

Plot an error bar chart and then conduct the appropriate analysis to test Davey et al.'s hypotheses that (1) people in negative moods who use an 'as many as can' stop rule would generate more items than those using a 'feel like continuing' stop rule; (2) people in a positive mood would generate more items when using a 'feel like continuing' stop rule compared to an 'as many as can' stop rule; (3) in neutral moods, the stop rule used won't have an effect (**Davey(2003).sav**). Answers are on the companion website (or look at pages 148–149 in the original article).

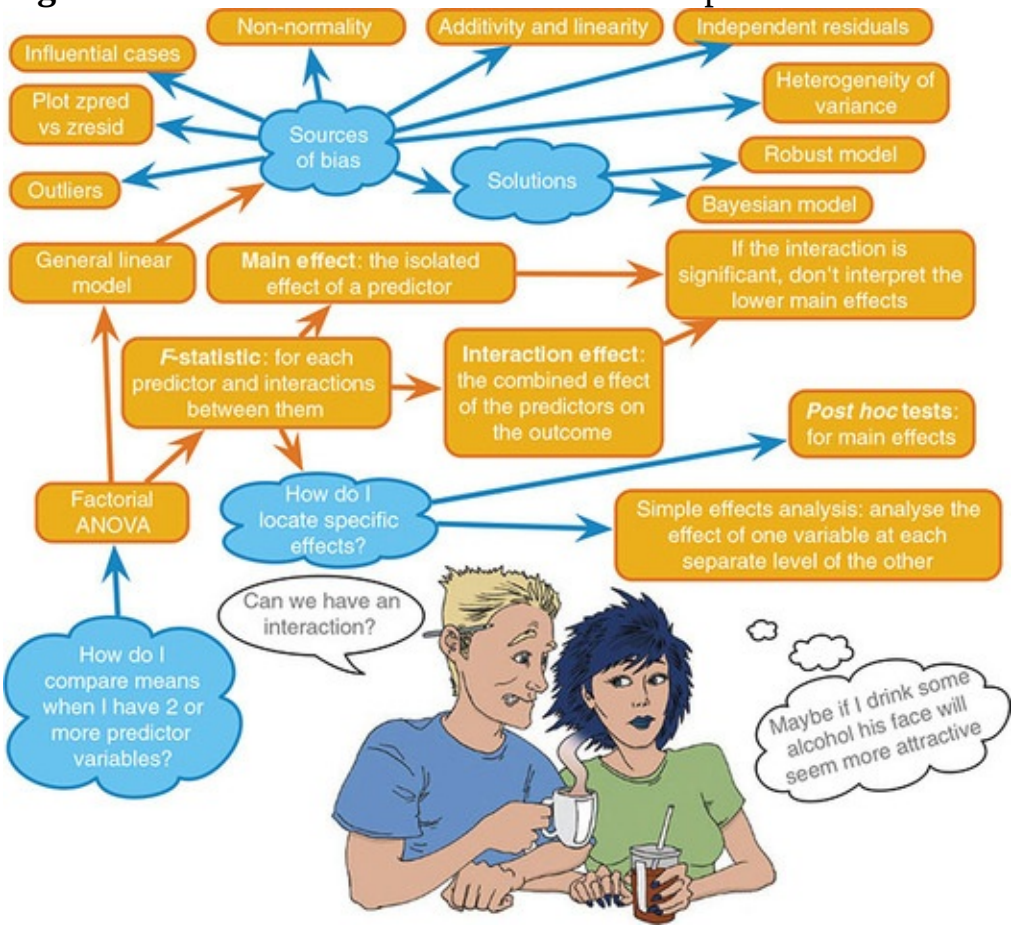


14.12 Brian's attempt to woo Jane

After walking the streets alone with her thoughts, Jane arrived at Brian's apartment. It was going to seem weird that she was there: she barely knew his name, they'd never exchanged numbers, let alone addresses, but these things were easy enough to find out if you knew how to hack. She felt anxious. At the third attempt she mustered the courage to press the buzzer. No answer. She

buzzed again. And again.

Figure 14.18 What Brian learnt from this chapter



'Hi!' said the puzzled voice through the intercom.

'It's Jane. I need to talk.'

A minute later Brian opened the door, linked arms with her and walked. She had expected to be invited in, but this was cool too. They walked the neighbourhood in the moonlight. She needed to talk, but she didn't: their lack of intimacy muted her. Brian filled the silence, as he always did, with his latest statistical knowledge. When he ran out of things to impress her with he ended, as he usually did, with a question. She had no clever put-down this time, no enthusiasm for flirtatious banter, and no one else to talk to. She told him about her mum, about her ambivalence towards her parents, about her upbringing, her brother. It was unfiltered and unguarded and made her feel vulnerable. Brian listened and told her to go to her mother.

14.13 What next?



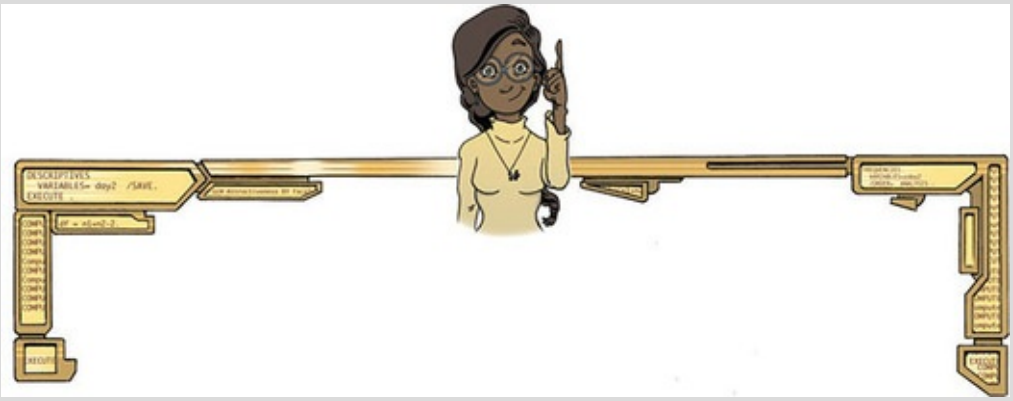
No sooner had I started my first band than it disintegrated. I went with drummer

Mark to sing in a band called the Outlanders, who were much better musically but were not, if the truth be told, metal enough for me. They also sacked me after a very short period of time for not being able to sing like Bono (an insult at the time, but in retrospect …). So, that was two failed bands in very quick succession. You'd have thought that the message that perhaps singing wasn't the thing for me might have started to sink in. It hadn't and didn't for quite some time (still hasn't entirely). I needed a new plan, and one was hatched one evening while walking along a cliff top in Cornwall with drummer Mark. Fortunately, it wasn't a plan that involved throwing myself off into the sea …

14.14 Key terms that I've discovered

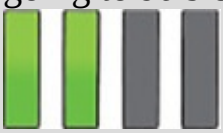
Beer-goggles effect
Factorial ANOVA
Independent factorial design
Interaction graph
Mixed design
Related factorial design
Simple effects analysis

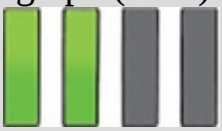
Smart Alex's tasks



- **Task 1**: I've wondered whether musical taste changes as you get older: my parents, for example, after years of listening to relatively cool music when I was a kid, hit their mid-forties and developed a worrying obsession with country and western. This possibility worries me immensely, because if the future is listening to Garth Brooks and thinking 'oh boy, did I underestimate Garth's immense talent when I was in my twenties', then it is bleak indeed. To test the idea I took two groups (**age**): young people (which I arbitrarily decided was under 40 years of age) and older people (above 40 years of age). I split each of these groups of 45 into three smaller groups of 15

and assigned them to listen to Fugazi, ABBA or Barf Grooks[6] (**music**). Each person rated the music (**liking**) on a scale ranging from +100 (this is sick) through 0 (indifference) to −100 (I'm going to *be* sick). Fit a model to test my idea (**Fugazi.sav**).



- **Task 2**: Compute omega squared for the effects in Task 1 and report the results of the analysis. 
- **Task 3**: In Chapter 5 we used some data that related to male and female arousal levels when watching *The Notebook* or a documentary about notebooks (**Notebook.sav**). Fit a model to test whether men and women differ in their reactions to different types of films. 
- **Task 4**: Compute omega squared for the effects in Task 3 and report the results of the analysis. 
- **Task 5**: In Chapter 4 we used some data that related to learning in men and women when either reinforcement or punishment was used in teaching (**Method Of Teaching.sav**). Analyse these data to see whether men and women's learning differs according to the teaching method used. 
- **Task 6**: At the start of this chapter I described a way of empirically researching whether I wrote better songs than my old bandmate Malcolm, and whether this depended on the type of song (a symphony or song about flies). The outcome variable was the number of screams elicited by audience members during the songs. Draw an error bar graph (lines) and analyse these data (**Escape From Inside.sav**). 
- **Task 7**: Compute omega squared for the effects in Task 6 and

report the results of the analysis. 

- **Task 8**: Using SPSS Tip 14.1, change the syntax in **GogglesSimpleEffects.sps** to look at the effect of alcohol at different levels of type of face. 
- **Task 9**: There are reports of increases in injuries related to playing Nintendo Wii (http://ow.ly/ceWPj). These injuries were attributed mainly to muscle and tendon strains. A researcher hypothesized that a stretching warm-up before playing Wii would help lower injuries, and that athletes would be less susceptible to injuries because their regular activity makes them more flexible. She took 60 athletes and 60 non-athletes (**athlete**); half of them played Wii and half watched others playing as a control (**wii**), and within these groups half did a 5-minute stretch routine before playing/watching whereas the other half did not (**stretch**). The outcome was a pain score out of 10 (where 0 is no pain, and 10 is severe pain) after playing for 4 hours (**injury**). Fit a model to test whether athletes are less prone to injury, and whether the prevention programme worked (**Wii.sav**). 
- **Task 10**: A researcher was interested in what factors contributed to injuries resulting from game console use. She tested 40 participants who were randomly assigned to either an active or static game played on either a Wii or Xbox Kinect. At the end of the session their physical condition was evaluated on an injury severity scale. The data are in the file **Wii vs Xbox Injuries.sav** which contains the variables **Game** (0 = static, 1 = active), **Console** (0 = Wii, 1 = Xbox), and **InjurySeverity** (a score ranging from 0 (no injury) to 20 (severe injury)). Fit a model to see whether injury severity is significantly predicted from the type of game, the type of console and their interaction.


6 A less well-known country musician not to be confused with anyone who has a similar name and produces music that makes you want to barf.

Answers & additional resources are available on the book's website at **https://edge.sagepub.com/field5e**



15 GLM 4: Repeated-measures designs

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15.1 What will this chapter tell me?

At the age of 15, I was on holiday with my friend Mark (the drummer) in Cornwall. I had a pretty decent mullet by this stage (nowadays I wish I had enough hair to grow a mullet) and a respectable collection of heavy metal T-shirts from going to various gigs. We were walking along the clifftops one evening at dusk, reminiscing about Andromeda. We decided that the only thing we hadn't enjoyed about that band was Malcolm and that maybe we should re-form it with a different guitarist.[1] As I wondered who we could get to play guitar, Mark pointed out the blindingly obvious: I played guitar. So, when we got home Scansion was born.[2] As the singer, guitarist and songwriter I set about writing some songs. Lyrically speaking, flies were so 'last year', so I moved on to exploring existential angst (I have never really stopped). Lots of songs about the futility of existence, social leprosy, incapacitating fear of death. That sort of thing. Obviously we failed totally in our musical ambitions, otherwise I wouldn't be writing this book, but we did get reviewed in the music magazine *Kerrang!* They called us 'twee' in a live review, which is not really what you want to read as a metal band, but were slightly more flattering about one of our demos (only slightly though). Our career high point was playing at the famous Marquee Club in London (which, like most of the venues of my youth, has closed now, but not because of us playing there).[3] This was the biggest gig of our career, and we needed to play like we never had before. As it turned out, we did: I ran on stage and fell over, de-tuning my guitar beyond recognition and breaking the zip on my trousers in the process. I spent the whole gig out of tune and spread-eagle to prevent my trousers falling down. Like I said, I'd never played like *that* before. We used to get quite obsessed with comparing how we played at different gigs. I didn't know about statistics then (happy days), but if I had I would have realized that we could rate ourselves and compare the mean ratings for different gigs; because we would always be the ones rating the gigs, this would be a repeated-measures design. That's what this chapter is about; hopefully it won't make our trousers fall down.

1 Although this is what we thought at the time, in retrospect I feel bad because Malcolm was never anything other than a very nice, positive, enthusiastic guy. To be honest, at that age (and, some would argue, beyond) I could be a bit of a dolt.

2 Scansion is a term for the rhythm of poetry. We got the name by searching through a dictionary until we found a word that we liked. Originally we didn't think it was 'metal' enough, and we decided that any self-respecting heavy metal band needed to have a big spiky 'X' in their name. So, for the first couple of

years we spelt it ‘Scanxion’. Like I said, I could be a bit of a dolt back then.
3 In its day the Marquee Club started the careers of people like Jimi Hendrix, The Who, Iron Maiden and Led Zeppelin. Google search it if you need to procrastinate.

Figure 15.1 Scansion in the early days; I used to stare a lot (L-R: me, Mark and Mark)



15.2 Introduction to repeated-measures designs



So far in this book, when comparing means, we’ve concentrated on situations in which different entities contribute to different means. I have tended to focus on examples with different *people* taking part in different experimental conditions, but it could be different plants, companies, plots of land, viral strains, goats or even duck-billed platypuses (or whatever the plural is). I’ve completely ignored situations in which the same people (plants, goats, hamsters, seven-eyed green galactic leaders from space, or whatever) contribute to the different means. I’ve put it off long enough, and now I’m going to take you through the models we fit to repeated-measures data.

Figure 15.2 The Confusion machine has created different ways to refer to repeated-measures designs





What is a repeated-measures design? (Clue: it is described in Chapter 1.)

'Repeated measures' is a term used when the same entities participate in all conditions of an experiment or provide data at multiple time points. For example, you might test the effects of alcohol on enjoyment of a party. Some people can drink a lot of alcohol without really feeling the consequences, whereas others, like myself, have only to sniff a pint of lager and they start flapping around on the floor waving their arms and legs around shouting 'Look at me, I'm Andy, King of the lost world of the Haddocks.' To control for these individual differences in tolerance to alcohol you can test the same people in all conditions of the experiment: participants could be given a questionnaire assessing their enjoyment of the party after they had consumed 1 pint, 2 pints, 3 pints and 4 pints of lager. There are lots of different ways to refer to this sort of design (Figure 15.2).

Table 15.1 Data for the bushtucker example

Celebrity	Stick Insect	Kangaroo Testicle	Fish Eye	Witchetty Grub	Mean	s^2
1	8	7	1	6	5.50	9.67
2	9	5	2	5	5.25	8.25
3	6	2	3	8	4.75	7.58
4	5	3	1	9	4.50	11.67
5	8	4	5	8	6.25	4.25
6	7	5	6	7	6.25	0.92
7	10	2	7	2	5.25	15.58
8	12	6	8	1	6.75	20.92
Mean	**8.13**	**4.25**	**4.13**	**5.75**		

15.3 A grubby example

I'm a Celebrity, Get Me Out of Here! is a TV show in which celebrities (more like ex-celebrities), in a pitiful attempt to salvage their careers (or to have careers in the first place), go and live in the jungle in Australia for a few weeks. During the show, they are subjected to various humiliating and degrading tasks to win food for their campmates. The tasks frequently involve creepy-crawlies; for example, a contestant might be locked in a coffin full of rats, forced to put their head in a bowl of large spiders, or have eels and cockroaches poured onto them. It's cruel, voyeuristic, gratuitous, car-crash TV, and I love it. As a vegetarian, for

me the bushtucker trial in which the celebrities eat various animal-related things is particularly noxious. Some examples are eating live stick insects or witchetty grubs, and chewing on fish eyes, kangaroo testicles and (surprisingly elastic) kangaroo penises. Seeing a fish eye exploding in someone's mouth is a mental scar that's hard to shake off.

I've often wondered (perhaps a little too much) which of the bushtucker foods is the most revolting. Imagine that I answered this question by getting eight celebrities and forcing them to eat four different animals (the aforementioned stick insect, kangaroo testicle, fish eye and witchetty grub) in counterbalanced order. On each occasion I measured the time it took the celebrity to retch, in seconds. This is a repeated-measures design because every celebrity eats every food. The predictor/independent variable is the type of food eaten and the outcome/dependent variable is the time taken to retch. Table 15.1 shows the data: the eight rows show the different celebrities, and the columns indicate their time to retch after eating each animal. Because they'll come in handy later, the mean time to retch (and variance) for each celebrity and the mean time to retch for each animal are shown. The overall variance in retching times will, in part, be caused by the fact that the foods differ in their palatability (the manipulation) and, in part, by differences in the celebrities' constitutions (individual differences).

15.4 Repeated-measures and the linear model



Up until now we have considered everything as a variation of the general linear model. Repeated-measures designs can also be conceptualized in this way. First, imagine we wanted to conceptualize the experiment I've just described as a linear model in which different people ate the different animals. To keep things simple, let's imagine there were just two animals eaten: the stick insect and testicle. Thinking back to Chapters 10 and 12, we could write the model as:

$$\begin{aligned} Y_i &= b_0 + b_1 X_{1i} + \varepsilon_i \\ \text{Retch}_i &= b_0 + b_1 \text{Food}_i + \varepsilon_i \end{aligned} \tag{15.1}$$

We predict retching time from the food that was eaten, plus the average retching time when food is equal to zero (b_0).

This model doesn't account for the fact that the same people took part in all conditions. First off, in an independent design we have one observation for the outcome from each person, so we predict the outcome for the individual (Y_i) based on the value of the predictor of that person (X_i), but with repeated

measures the person has several values of the predictor so we predict the outcome from both the individual (i) and the specific value of the predictor that's of interest (g).[4] So we might write a simple model as:

[4] When using this model for repeated-measures designs people typically use subscripts i to represent different conditions and j to represent individuals, so my subscripts are a bit weird, but I think it's less confusing than referring to an earlier model (equation (15.2)) in which i represents individuals but then having i represent something else in equation (15.3).

$$\text{Retch}_{gi} = b_0 + b_1\text{Food}_{gi} + \varepsilon_{gi} \quad (15.2)$$

which acknowledges that we predict the retch time (Y_{gi}) for food g within person i from the specific food eaten (X_{gi}), and the error in prediction, X_{gi}, to which both the individual and the food eaten contribute. All that's changed is the subscripts in the model, which acknowledge that levels of the treatment condition (g) occur within individuals (i).

Equation (15.2) is a very simple model, and we might, for example, want to factor in the possibility that individuals will vary in their constitution. We can do this by adding a variance term to the intercept. Remember that the intercept represents the time to wretch when the predictor is 0, so if we allow this parameter to vary across individuals, we're effectively modelling the possibility that different people will have different natural retching latencies. This is known as a random intercept model, and we'll look at these in detail in Chapter 21. This model is written as:

$$\begin{aligned}\text{Retch}_{gi} &= b_{0i} + b_1 X_{gi} + \varepsilon_{gi} \\ b_{0i} &= b_0 + u_{0i}\end{aligned} \quad (15.3)$$

All that's happened is the intercept has had an i added to the subscript to reflect that it is specific to an individual, and underneath we define the intercept as being made up of the group-level intercept (b_0) plus the deviation of the individual's intercept from the group-level intercept (u_{0i}). Put simply, u_{0i} reflects individual differences in retching. The first line of equation (15.3) becomes a model for an individual, and the group-level effects are incorporated into the second line.

We might also want to factor in the possibility that the effect of different foods varies across individuals. We can do this by adding a variance term to the slope. Remember that the slope represents the effect that the different foods have on retching. By allowing this parameter to vary across individuals, we model the possibility that the effect of the food on retching will be different in different participants. This is known as a random slope model (again, see Chapter 21 for a fuller discussion). We write this model as.

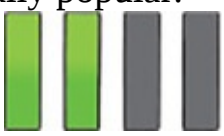
$$\begin{aligned} \text{Retch}_{gi} &= b_{0i} + b_{1i}X_{gi} + \varepsilon_{gi} \\ b_{0i} &= b_0 + u_{0i} \\ b_{1i} &= b_1 + u_{1i} \end{aligned} \quad (15.4)$$

Compare it to equation (15.3): the main change is that the slope (b_1) has had an *i* added to the subscript to reflect that it is specific to an individual, and underneath we define it as being made up of the group-level slope (b_1) plus the deviation of the individual's slope from the group-level slope (u_{1i}). As before, u_{1i} reflects individual differences in the effect of food on retching. As in equation (15.3), the top of equation (15.4) is a model for the individual, and the group-level effects are incorporated into the second and third lines.

We are getting ahead of ourselves a bit, but I just want to show that repeated measures can be incorporated into the general linear model. As we'll see in Chapter 21, these models get considerably more complex, but the take-home point is simply that we're again dealing with a linear model.

15.5 The ANOVA approach to repeated-measures designs 

Although you can conceptualize repeated measures as a linear model, there are other ways too. The way that people typically handle repeated measures in IBM SPSS Statistics is to use a **repeated-measures ANOVA** approach. Very crudely, this is a linear model like I have just described, but with some very restrictive constraints. In a nutshell, we have seen that the standard linear model assumes that residuals are independent (not related to each other – see Section 12.3), but as you can see from the models above, this assumption isn't true for repeated-measures designs: the residuals are affected by both between-participant factors (which should be independent) and within-participant factors (which won't be independent). There are (broadly) two solutions. One is to model this within-participant variability, which is what the models in the previous section do. The other is to apply additional assumptions that allow a simpler, less flexible model to be fit. The later approach is the one that is historically popular.

15.5.1 The assumption of sphericity 

The assumption that permits us to use a simpler model to analyse repeated-measures data is **sphericity**, which, trust me, is a pain in the butt to pronounce when you're giving statistics lectures at 9 a.m. on a Monday. Put crudely, sphericity is about assuming that the relationship between scores in pairs of

treatment conditions is similar (i.e., the level of dependence between means is roughly equal).

The assumption of sphericity (denoted by ε and sometimes referred to as *circularity*) can be likened to the assumption of homogeneity of variance in between-group designs. It is a form of **compound symmetry**, which holds true when both the variances across conditions are equal (this is the same as the homogeneity of variance assumption in between-group designs) and the covariances between pairs of conditions are equal. So, we assume that the variation within conditions is similar and that no two conditions are any more dependent than any other two. Sphericity is a more general, less restrictive form of compound symmetry and refers to the equality of variances of the *differences* between treatment levels (Misconception Mutt 15.1).



The best way to explain what this means is by checking the assumption of sphericity by hand, which incidentally only a complete lunatic would do. Table 15.2 shows data from an experiment with three conditions and five people who have contributed scores to each condition. That's the first three columns. The second three show the differences between pairs of scores in all combinations of the treatment levels. I have calculated the variance of these differences in the bottom row. Sphericity is met when these variances are roughly equal. For these data, sphericity will hold when:

Table 15.2 Hypothetical data to illustrate the calculation of the variance of the differences between conditions

Condition *A*	Condition *B*	Condition *C*	*A* – *B*	*A* – *C*	*B* – *C*
10	12	8	−2	2	4
15	15	12	0	3	3
25	30	20	−5	5	10
35	30	28	5	7	2
30	27	20	3	10	7
		Variance:	**15.7**	**10.3**	**10.7**

$$\text{variance}_{A-B} \approx \text{variance}_{A-C} \approx \text{variance}_{B-C} \quad (15.5)$$

Table 15.2 shows that there is some deviation from sphericity in these data because the variance of the differences between conditions *A* and *B* (15.7) is greater than the variance of the differences between both *A* and *C* (10.3) and *B* and *C* (10.7). However, these data have *local circularity* (or local sphericity) because two of the variances of differences are very similar, which means that

sphericity can be assumed for any multiple comparisons involving these conditions (for a discussion of local circularity, see Rouanet & Lépine, 1970). For the data in Table 15.2 the biggest difference in variances is between *A* and *B* (15.7) and *A* and *C* (10.3), but how do we know whether this difference is large enough to be a problem?

15.5.2 Assessing the severity of departures from sphericity 

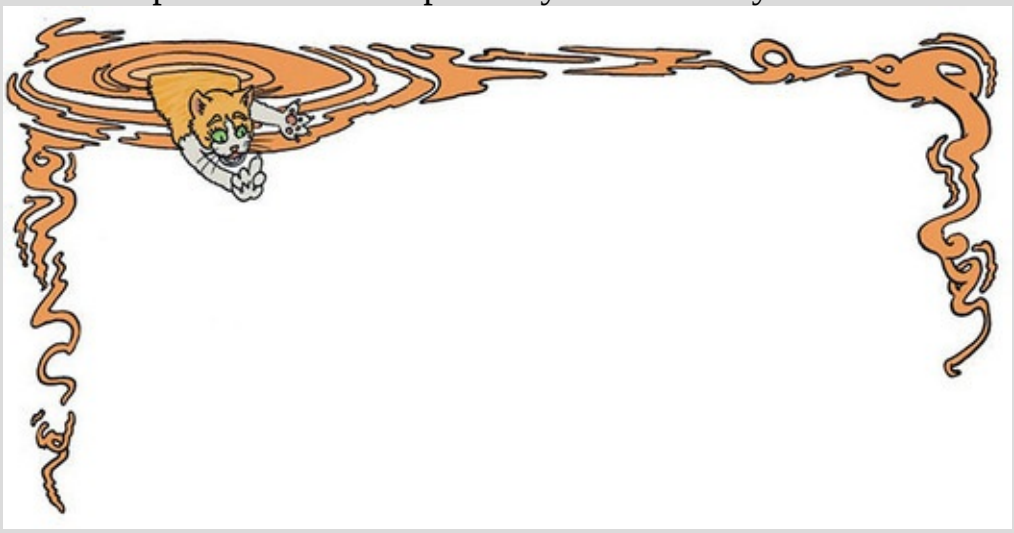
Mauchly's test assesses the hypothesis that the variances of the differences between conditions are equal. If Mauchly's test statistic is significant (i.e., has a probability value less than 0.05) it implies that there are significant differences between the variances of differences and, therefore, sphericity is not met. If Mauchly's test statistic is non-significant (i.e., $p > 0.05$) then the implication is that the variances of differences are roughly equal and sphericity is met. However, like any significance test, Mauchly's test depends upon sample size and is probably best ignored (Misconception Mutt 15.1). Instead we can estimate the degree of sphericity using the **Greenhouse–Geisser estimate,** $\hat{\varepsilon}$ (Greenhouse & Geisser, 1959), or the **Huynh–Feldt estimate** (Huynh & Feldt, 1976), $\tilde{\varepsilon}$. The Greenhouse–Geisser estimate varies between $1/(k-1)$ (where k is the number of repeated-measures conditions) and 1. For example, in a situation in which there are five conditions the lower limit of $\hat{\varepsilon}$ will be $1/(5-1)$, or 0.25 (known as the **lower-bound estimate** of sphericity). You can live a long, happy life oblivious to the precise calculation of these estimates (interested readers should consult Girden, 1992) so we won't go there. We need know only that these estimates are used to correct for departures from sphericity and so are considerably more use than Mauchly's test (Misconception Mutt 15.1).

15.5.3 What's the effect of violating the assumption of sphericity? 

Rouanet and Lépine (1970) provided a detailed account of the validity of the *F*-statistic under violations of sphericity (see also Mendoza, Toothaker, & Crain, 1976). I summarized (Field, 1998) their findings in a very obscure newsletter that no one can ever access (see Oliver Twisted). The bottom line is that sphericity creates a loss of power and an *F*-statistic that doesn't have the distribution that it's supposed to have (i.e., an *F*-distribution). Lack of sphericity

also causes some amusing complications for *post hoc* tests. If you don't want to worry about what these complications are then when sphericity is violated, the Bonferroni method is the most robust in terms of power and control of the Type I error rate. When sphericity is definitely not violated, Tukey's test can be used (Jane Superbrain Box 15.1).

Misconception Mutt 15.1 Sphericity and Mauchly's test



The Misconception Mutt was spinning around in circles. In the lecture he'd been taken to, the lecturer kept rambling on about circularity, which had reminded the mutt how much fun it was to chase his tail. A ginger mist arose like a tornado from the epicentre of his spinning body, forming into a feline cloud above him. The cloud raised its eyebrows at him and muttered 'Stupid dogs!' The mutt stopped to throw the cat a look of disgust.

'What *are* you doing?' said the cat disapprovingly.

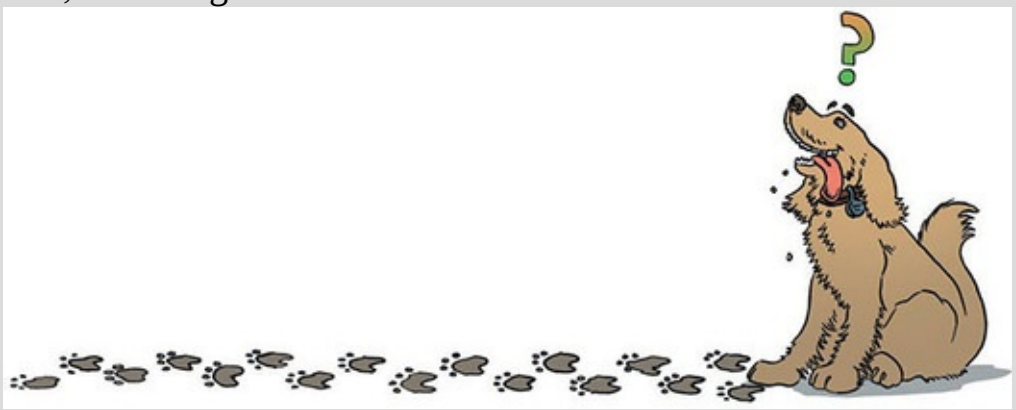
'I learnt about circularity today,' the dog wheezed, 'about how you need to look for sphericity when you compare two related means, and you use Mauchly's test, and if it's significant then sphericity can't be assumed but if it's non-significant then you can assume it …'

The cat's paw reached towards the mutt's lips and gently closed them. The dog looked puzzled. 'Shhh …', said the cat, throwing him a patronizing look.

'Where to begin?' the cat whined. 'First, sphericity is not relevant if you're comparing only two means. The assumption is that the variances of difference scores between pairs of treatment levels are equal, and with only two conditions you have only one set of difference scores, and only one variance. Y*ou need at least three conditions for sphericity to be an issue.* Second, Mauchly's test is,

essentially, pointless. In large samples a significant Mauchly's test could mean a small departure from sphericity that we have a lot of power to detect, and in small samples a non-significant Mauchly's could mean a large departure from sphericity that we didn't have the power to detect. Mauchly's test can't be interpreted without the context of the sample size. Also, if sphericity is violated then you can apply a correction proportionate to the extent of the violation (for example, the Greenhouse–Geisser
correction) so you may as well ignore Mauchly's test and always apply the correction. When the data are spherical no correction is made, for small violations a small correction is made, and for large violations a large correction is made.'

The mutt started spinning furiously in the opposite direction, chasing his tail as though his life depended on it. As he spun he created a reverse vortex that sucked the cat back into the ether. 'That'll learn him,' he thought to himself.



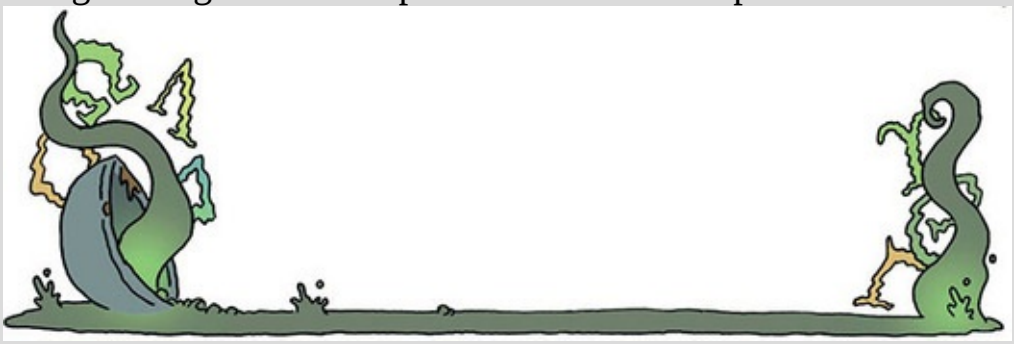
Oliver Twisted Please, Sir, can I have some more … sphericity?



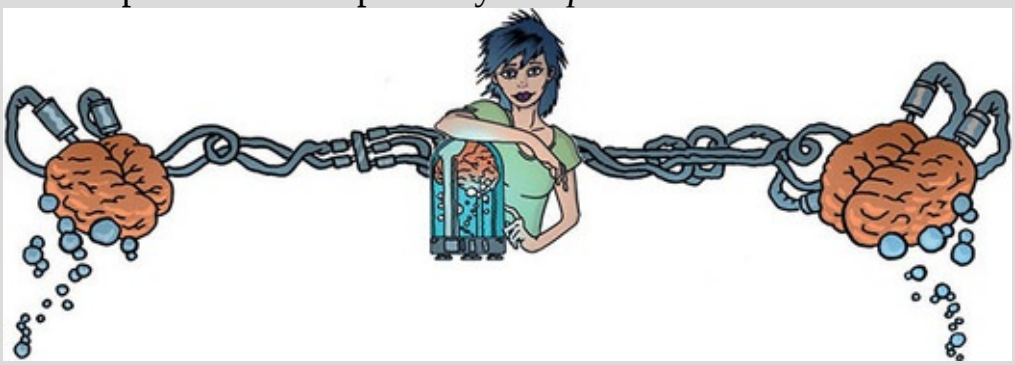
'Balls!' says Oliver. 'Balls are spherical, and I like them. Maybe I'd like sphericity too if only you could explain it to me in more detail.'

Be careful what you wish for, Oliver. In my youth I wrote an article called 'A bluffer's guide to sphericity', which I used to cite in this book, roughly on this page. Occasionally people ask me for it, so I

thought I might as well reproduce it on the companion website.

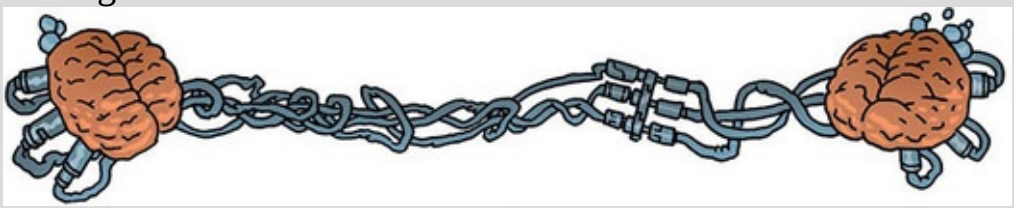


Jane Superbrain 15.1 Sphericity and *post hoc* tests 



The violation of sphericity has implications for multiple comparisons. There is a more detailed summary of these online (see Oliver Twisted), but here are a few take home-messages. Boik (1981) recommends against using the *F*-statistic for repeated-measures contrasts because even very small departures from sphericity produce large biases. Maxwell (1980) compared the power and alpha levels for five *post hoc* tests: three variants of the Tukey procedure, the Bonferroni method, and a multivariate approach (the Roy–Bose simultaneous confidence interval). The multivariate approach was always 'too conservative for practical use' (p. 277), and this was most extreme when *n* (the number of participants) is small relative to *k* (the number of conditions). All variants of Tukey's test inflated the Type I error rate unacceptably with increasing departures from sphericity. The Bonferroni method was extremely robust (although *slightly* conservative) and controlled Type I error rates well. In terms of power (the Type II error rate), Tukey's wholly significant difference was the most powerful under non-sphericity in tiny samples ($n = 8$), but this advantage was severely reduced in even slightly larger samples ($n = 15$). Keselman and Keselman (1988)

extended Maxwell's work to unbalanced designs and concluded that Bonferroni was more powerful than the multivariate approach as the number of repeated treatment levels increases. A simple take-home message is that the Bonferroni method has much to recommend it.



15.5.4 What do you do if you violate sphericity?



You might think that if your data violate the sphericity assumption then you need to have a nervous breakdown, or book in to see a counsellor or something. You *can* do that, but a less costly option (emotionally and financially) is to adjust the degrees of freedom of any *F*-statistics affected. You can estimate sphericity in various ways (see above), resulting in a value that is 1 when your data are spherical and less than 1 when they are not. You multiply the degrees of freedom for an affected *F* by this estimate. The result is that when you have sphericity the degrees of freedom don't change (because you multiply them by 1) but when you don't the degrees of freedom get smaller (because you multiply them by a value less than 1). The greater the violation of sphericity, the smaller the estimate gets, and the smaller the degrees of freedom become. Smaller degrees of freedom make the *p*-value associated with the *F*-statistic less significant. By adjusting the degrees of freedom by the extent to which the data are not spherical, we also make the *F*-statistic more conservative. In doing so, the Type I error rate is controlled.



As I mentioned above, the degrees of freedom are adjusted using either the Greenhouse–Geisser or Huynh–Feldt estimates of sphericity. When the Greenhouse–Geisser estimate is greater than 0.75 the correction is too conservative (Huynh & Feldt, 1976), and this can also be true when the sphericity estimate is as high as 0.90 (Collier, Baker, Mandeville, & Hayes, 1967). However, the Huynh–Feldt estimate tends to overestimate sphericity

(Maxwell & Delaney, 1990). Many authors recommend that when estimates of sphericity are greater than 0.75 the Huynh–Feldt estimate should be used, but when the Greenhouse–Geisser estimate of sphericity is less than 0.75 or nothing is known about sphericity the Greenhouse–Geisser correction should be used (Barcikowski & Robey, 1984; Girden, 1992; Huynh & Feldt, 1976). Stevens (2002) suggests taking an average of the two estimates and adjusting the *df* by this average. We will see how these values are used in due course.

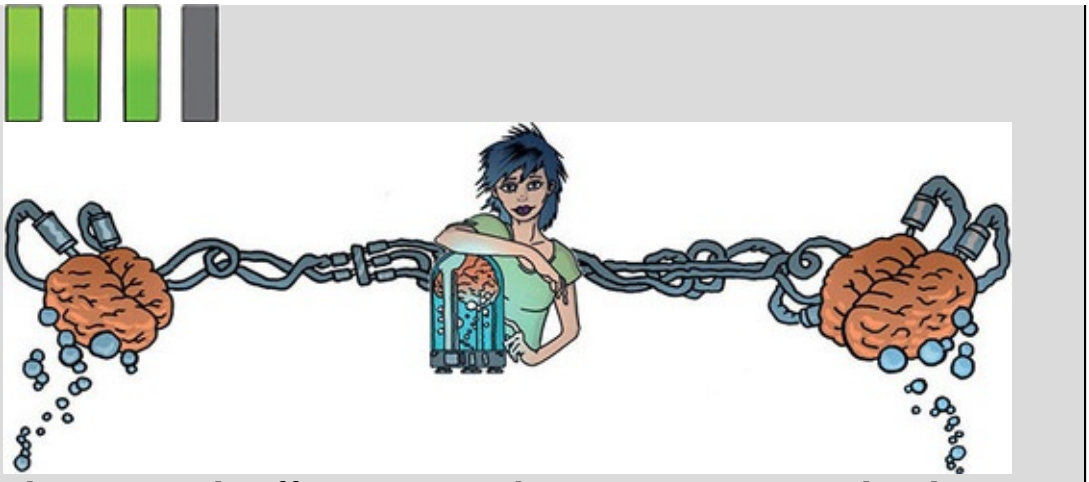
Another option when you have data that violate sphericity is to fit the kind of model described in Section 15.4, which is known as a multilevel model (see Chapter 21 for more detail). A third option is to use multivariate test statistics (MANOVA), because they do not assume sphericity (see O'Brien & Kaiser, 1985). SPSS Statistics produces multivariate test statistics (see also Chapter 17). However, there may be trade-offs in power between these univariate and multivariate tests (see Jane Superbrain Box 15.2).

15.6 The *F*-statistic for repeated-measures designs



In a repeated-measures design the effect of the experiment (the independent variable) is shown up in the within-participant variance (rather than in the between-group variance). Remember that in independent designs (Section 12.2) the within-participant variance is the residual sum of squares (SS_R); it is the variance created by individual differences in performance. When we carry out our experimental manipulation on the same entities, the within-participant variance will be made up of not just individual differences in performance but also the effect of our manipulation. Therefore, the main difference with a repeated-measures design is that we look for the experimental effect (the model sum of squares) within the individual rather than within the group. Figure 15.3 illustrates how the variance is partitioned in a repeated-measures design. The important thing to note is that the types of variances are the same as in independent designs: we have a total sum of squares (SS_T), a model sum of squares (SS_M) and a residual sum of squares (SS_R). The *only* difference is from where those sums of squares come: in repeated-measures designs the model and residual sums of squares are both part of the within-participant variance. Let's have a look at an example.

Jane Superbrain 15.2 Power in ANOVA and MANOVA



There is a trade-off in test power between univariate and multivariate approaches. Univariate techniques are relatively powerless to detect small reliable changes between highly correlated conditions when other less correlated conditions are also present (Davidson, 1972). As the degree of violation of compound symmetry increases, the power of multivariate tests also increases, whereas for univariate tests it decreases (Mendoza, Toothaker, & Nicewander, 1974). However, multivariate approaches should probably not be used if n is less than $k + 10$, where k is the number repeated-measures levels (Maxwell & Delaney, 1990). To sum up, if you have a large violation of sphericity (ε ε 0.7) and your sample size is greater than $k + 10$ then multivariate procedures are more powerful, but with small sample sizes or when sphericity holds (ε ε0.7) use the univariate approach (Stevens, 2002). It is also worth noting that the power of MANOVA varies as a function of the correlations between dependent variables (see Jane Superbrain Box 17.1) and so the relationship between treatment conditions must be considered.

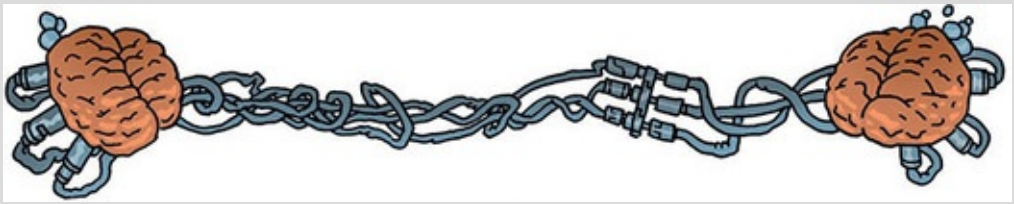
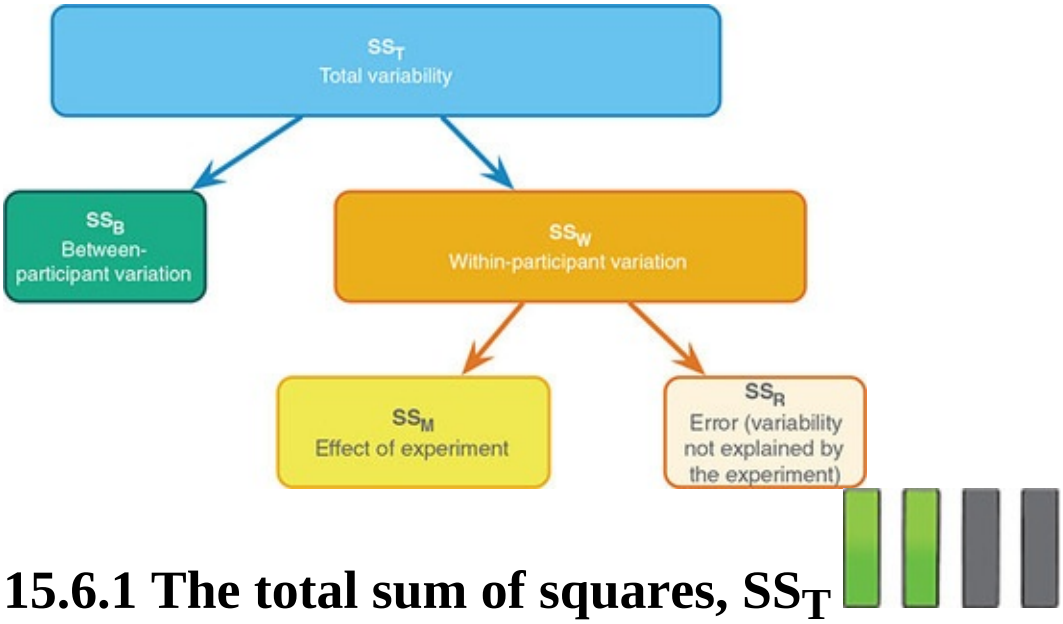


Figure 15.3 Partitioning variance for repeated-measures designs



15.6.1 The total sum of squares, SS_T

Remember that for one-way independent designs SS_T is calculated as:

$$SS_T = s^2_{grand}(N-1) \tag{15.6}$$

In repeated-measures designs the total sum of squares is calculated in the same way. The grand variance in the equation is the variance of all scores when we ignore the group to which they belong. So, if we treated the data as one big group (as in Figure 15.4), the variance of these scores is 8.19 (try this on your calculator), and this is the grand variance. We used 32 scores to generate this value, so N is 32. Our sum of squares becomes:

$$\begin{aligned} SS_T &= s^2_{grand}(N-1) \\ &= 8.19(32-1) \\ &= 253.89 \end{aligned} \tag{15.7}$$

The degrees of freedom for this sum of squares, as for an independent design, are $N - 1$, or 31.

Figure 15.4 Treating the data as a single group

8	7	1	6
9	5	2	5
6	2	3	8
5	3	1	9
8	4	5	8
7	5	6	7
10	2	7	2
12	6	8	1

Grand mean = 5.56

Grand variance = 8.19

15.6.2 The within-participant sum of squares, SS_W



The crucial difference from an independent design is that in a repeated-measures design there is a within-participant variance component, which represents individual differences within participants. In independent designs these individual differences were quantified with the residual sum of squares (SS_R) using the following equations:

$$SS_R = \sum_{g=1}^{k}\sum_{i=1}^{n}\left(x_{ig} - \bar{x}_g\right)^2 = \sum_{g=1}^{k} s_g^2\left(n_g - 1\right) \quad (15.8)$$

In independent designs, because there are different participants within each condition, we calculated SS_R within each condition and added these values to get a total:

$$SS_R = s^2_{\text{group}\,1}\left(n_1 - 1\right) + s^2_{\text{group}\,2}\left(n_2 - 1\right) + s^2_{\text{group}\,3}\left(n_3 - 1\right) + \ldots + s^2_{\text{group}\,n}\left(n_n - 1\right) \quad (15.9)$$

In a repeated-measures design, because we've subjected entities to more than one experimental condition, we're interested in the variation not within a condition but *within an entity*. Therefore, we use the same equation but adapt it to look within participants rather than groups. If we call this sum of squares SS_W (for within-participant SS), we adapt equation (15.9) to give:

$$SS_W = s^2_{\text{entity}\,1}\left(n_1 - 1\right) + s^2_{\text{entity}\,2}\left(n_2 - 1\right) + s^2_{\text{entity}\,3}\left(n_3 - 1\right) + \ldots + s^2_{\text{entity}\,n}\left(n_n - 1\right) \quad (15.10)$$

This equation translates as looking at the variation in each individual's scores and then adding these variances for all the entities in the study. The *n*s represent the number of scores within the person (i.e., the number of experimental conditions, or in this case the number of foods). All the variances we need are in Table 15.1, so we can calculate SS_W as:

$$\begin{aligned} SS_W &= s^2_{\text{celebrity 1}}(n_1 - 1) + s^2_{\text{celebrity 2}}(n_2 - 1) + \ldots + s^2_{\text{celebrity } n}(n_n - 1) \\ &= 9.67(4-1) + 8.25(4-1) + 7.58(4-1) + 11.67(4-1) + 4.25(4-1) \\ &\quad + 0.92(4-1) + 15.58(4-1) + 20.92(4-1) \\ &= 29 + 24.75 + 22.75 + 35 + 12.75 + 2.75 + 46.75 + 62.75 \\ &= 236.50 \end{aligned} \tag{15.11}$$

The degrees of freedom for each entity are $n - 1$ (i.e., the number of conditions minus 1). To get the total degrees of freedom we add the *df*s for all participants. With eight participants (celebrities) and four conditions (i.e., $n = 4$), there are 3 degrees of freedom for each celebrity and $8 \times 3 = 24$ degrees of freedom in total.

15.6.3 The model sum of squares, SS_M

So far, we know that the total amount of variation within the retching scores is 253.58 units. We also know that 236.50 of those units are explained by the variance created by individuals' (celebrities') performances under different conditions. Some of this variation is the result of our experimental manipulation and some of this variation is due to unmeasured factors. The next step is to work out how much variance is explained by our manipulation (different foods) and how much is not.

In independent designs, we worked out how much variation could be explained by our experiment (the model SS) by looking at the means for each group and comparing these to the overall mean. We measured the variance resulting from the differences between group means and the overall mean (see equation (12.11)). We do the same thing in a repeated-measures design:

$$SS_M = \sum_{g=1}^{k} n_g \left(\bar{x}_g - \bar{x}_{\text{grand}}\right)^2 \tag{15.12}$$

Using the means from the bushtucker data (see Table 15.1), we calculate SS_M as:

$$\begin{aligned} SS_M &= 8(8.13 - 5.56)^2 + 8(4.25 - 5.56)^2 + 8(4.13 - 5.56)^2 + 8(5.75 - 5.56)^2 \\ &= 8(2.57)^2 + 8(-1.31)^2 + 8(-1.44)^2 + 8(0.196)^2 \\ &= 83.13 \end{aligned} \tag{15.13}$$

With independent designs the model degrees of freedom are the number of conditions (k) minus 1. The same is true here: there were four conditions (foods),

so the degrees of freedom will be 3.

15.6.4 The residual sum of squares, SS_R 

We now know that there are 253.58 units of variation to be explained in our data, and that the variation across our conditions accounts for 236.50 units. Of these 236.50 units, our experimental manipulation can explain 83.13 units. The final sum of squares is the residual sum of squares (SS_R), which tells us how much of the variation cannot be explained by the model. This value is the amount of variation caused by extraneous factors outside experimental control. Knowing SS_W and SS_M already, the simplest way to calculate SS_R is to subtract SS_M from SS_W:

$$\begin{aligned} SS_R &= SS_W - SS_M \\ &= 236.50 - 83.13 \\ &= 153.37 \end{aligned} \tag{15.14}$$

The degrees of freedom are calculated in a similar way:

$$\begin{aligned} df_R &= df_W - df_M \\ &= 24 - 3 \\ &= 21 \end{aligned} \tag{15.15}$$

15.6.5 The mean squares 

SS_M tells us how much variation the model (e.g., the experimental manipulation) explains and SS_R tells us how much variation is due to extraneous factors. Both these values are totals and depend on how many scores have contributed to them, so to make them comparable we convert to the mean (or average) sum of squares (MS) by dividing by the degrees of freedom:

$$\begin{aligned} MS_M &= \frac{SS_M}{df_M} = \frac{83.13}{3} = 27.71 \\ MS_R &= \frac{SS_R}{df_R} = \frac{153.37}{21} = 7.30 \end{aligned} \tag{15.16}$$

MS_M represents the average variation explained by the model (e.g., the average systematic variation), whereas MS_R is a gauge of the average variation explained by extraneous variables (the average unsystematic variation).

15.6.6 The F-statistic 

The *F*-statistic is the ratio of the variation explained by the model and the

variation explained by unsystematic factors. As for independent designs, it is the model mean squares divided by the residual mean squares:

$$F = \frac{MS_M}{MS_R} \quad (15.17)$$

So, as with independent designs, the *F*-statistic is the ratio of systematic to unsystematic variation: it is the experimental effect on performance relative to the effect of unmeasured factors.

For the bushtucker data, the *F*-statistic is 3.79:

$$F = \frac{MS_M}{MS_R} = \frac{27.71}{7.30} = 3.79 \quad (15.18)$$

This value is greater than 1, which indicates that the experimental manipulation had some effect above and beyond the effect of unmeasured factors. This value can be compared against a critical value based on its degrees of freedom (which are df_M and df_R, which are 3 and 21 in this case), but more generally, it's possible to compute an exact *p*-value that is the probability of getting an *F* at least as big as the one we have observed if the null hypothesis is true.

15.6.7 The between-participant sum of squares



We've sort of forgotten about the between-participant variation in Figure 15.3 because we didn't need it to calculate the *F*-statistic. I will briefly mention what it represents. The easiest way to calculate this term is by subtraction:

$$SS_T = SS_B + SS_W$$
$$SS_B = SS_T - SS_W \quad (15.19)$$

We have already calculated SS_W and SS_T, so by replacing the values of these terms, we get:

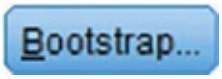
$$\begin{aligned} SS_B &= SS_T - SS_W \\ &= 253.89 - 236.50 \\ &= 17.39 \end{aligned} \quad (15.20)$$

SS_B represents individual differences between cases. In this example different celebrities will have different tolerances for eating these foods. This variation is illustrated by the different means for the celebrities in Table 15.1. For example, celebrity 4 (M = 4.50) was, on average, more than 2 seconds quicker to retch than participant 8 (M = 6.75). Celebrity 8 had, on average, a stronger constitution than celebrity 4. The between-participant sum of squares reflects these differences between individuals. In this case only 17.39 units of variation in retching latencies is down to individual differences between celebrities.

15.7 Assumptions in repeated-measures designs



Repeated-measures designs still use the linear model so all of the sources of potential bias (and counteractive measures) discussed in Chapter 6 apply. Using the ANOVA approach, the assumption of independence is replaced by assumptions about the relationships between differences scores (sphericity). In the multilevel approach, sphericity isn't required, and you have a lot more flexibility to model different types of assumptions about residuals (see Section 21.4.2.)

If assumptions are not met, there is a robust variant of a one-way repeated-measures ANOVA that we'll cover in Section 15.10 – and you have the option of Friedman's ANOVA (see Chapter 7) although I'd use that as a last resort. The 

button is noticeable by its absence in repeated-measures designs, which is a sad story. If you have a factorial design using repeated measures then you're stuffed.

15.8 One-way repeated-measures designs using SPSS



The general procedure for a one-way repeated-measures design (using ANOVA) is much the same as any other linear model, so, remind yourself of the general procedure in Chapter 9. Figure 15.5 shows a simpler overview that highlights some of the specific issues when using repeated measures.

15.8.1 The main analysis 

The data for the bushtucker example (**Bushtucker.sav**) can be entered into the data editor in the same format as Table 15.1 (although you don't need to include the columns labelled *Mean* or s^2). If you're entering the data by hand, create a variable called **stick** and use the labels dialog box to give this variable a full title of 'Stick Insect'. In the next column, create a variable called **testicle**, with a full title of 'Kangaroo Testicle', and so on for variables called **eye** ('Fish Eye') and **witchetty** ('Witchetty Grub').

To compare means from a one-way repeated-measures design select *Analyze*  *General Linear Model* 

. We use the *Define Factor(s)* dialog

box to name our within-subject (repeated-measures) variable, which in this example is the type of animal eaten, so replace the word *factor1* with the word *Animal* (Figure 15.6).[5] Next, we specify how many levels the variable had (i.e., how many conditions participants took part in). There were four different animals eaten by each celebrity, so type '4' into the box labelled *Number of Levels*. Click 

to register this variable in the list of repeated-measures variables, where it appears as *Animal(4)* as in Figure 15.6. If your design has several repeated-measures variables you can add more factors to the list (see the next example). When you have finished creating repeated-measures factors, click 

to go to the main dialog box.

[5] The name cannot have spaces in it.

The main dialog box (Figure 15.7) has a space labelled *Within-Subjects Variables* that contains a list of four question marks followed by a number. These question marks are placeholders for the variables representing the four levels of the independent variable and need to be replaced with the variables corresponding to each level. The order of levels is not important for this example, so we can select all four variables in the data editor (click the stick insect variable then click the witchetty grub variable while holding the *Shift* key) and drag them to the box labelled *Within-Subjects Variables* (or click 
). The finished dialog box is also shown in Figure 15.7.

Figure 15.5 The process for analysing repeated-measures designs

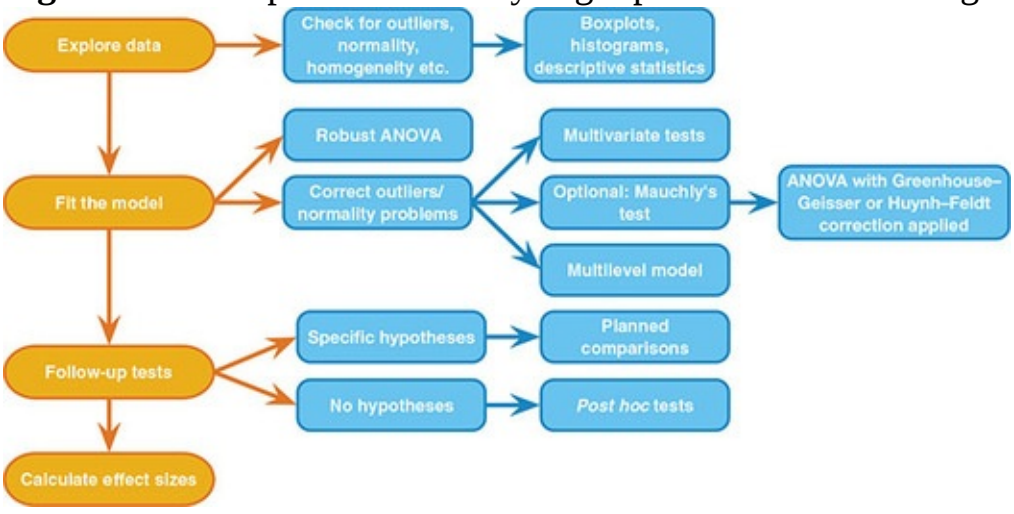


Figure 15.6 The *Define Factor(s)* dialog box for repeated-measures ANOVA

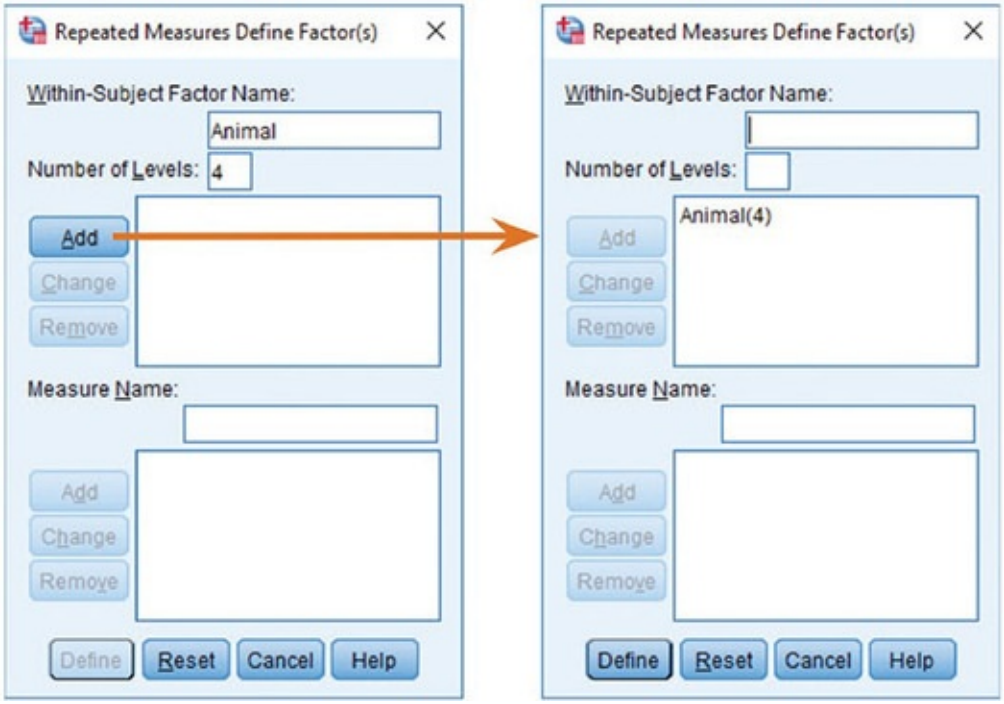


Figure 15.7 The main dialog box for repeated-measures designs (before and after completion)

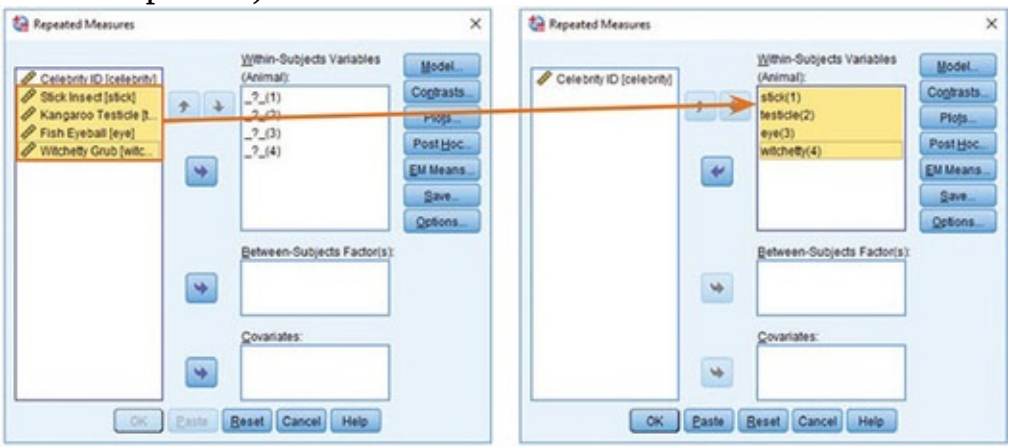
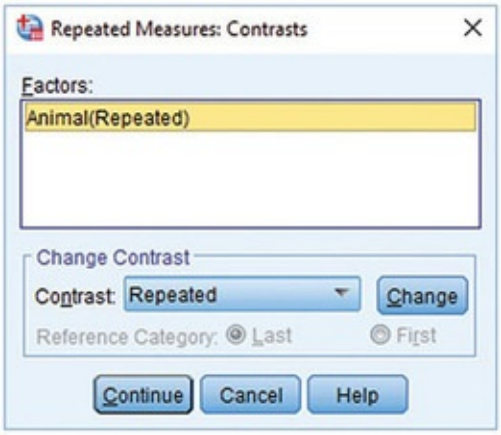


Figure 15.8 Repeated-measures contrasts



15.8.2 Defining contrasts for repeated measures



Click Contrasts... to access the dialog box in Figure 15.8, which is used to specify one of the many standard contrasts that we have come across before (see Section 12.4.4). The default is a polynomial contrast, but you can change it by selecting a variable in the box labelled *Factors*, clicking Polynomial, selecting a contrast from the list, and clicking Change. If you choose to conduct a simple contrast then you can specify whether to compare groups against the first or last category. The first category would be the one entered as (1) in the main dialog box and, for these data, the last category would be the one entered as (4). Therefore, the order in which you enter variables in the main dialog box is important for the contrasts you choose.

There is no obvious contrast for this example (the simple contrast is not useful because we have no control category), so let's use the *repeated* contrast, just because we haven't looked at it before. A repeated contrast compares each category against the previous one, which can be useful in repeated-measures designs in which the levels of the independent variable have a meaningful order – for example, if you've measured the outcome variable at successive points in time, or administered increasing doses of a drug. When you have selected this contrast, click Continue.

15.8.3 Custom contrasts



In Section 12.4 we looked at planned contrasts. Two of our animals are eaten alive (the witchetty grub and stick insect) whereas the others are dead body parts (the testicle and eye). We might predict that eating live animals is more disgusting than eating body parts and test this with a set of planned contrasts. Contrast 1 would compare live animals to dead ones, contrast 2 would then compare the two live animals (and ignore the testicle and eye), and contrast 3 would compare the eye and testicle while ignoring the stick insect and witchetty grub. The partitioning of variance is like that depicted in Figure 12.8.



Devise some contrast codes for the contrasts described in the text.

Table 15.3 Contrast codes for the bushtucker data

	Stick insect	Witchetty grub	Kangaroo testicle	Fish eye
Contrast 1 (live vs. dead)	0.5	0.5	−0.5	−0.5
Contrast 2 (stick vs. witchetty)	1	−1	0	0
Contrast 3 (testicle vs. eye)	0	0	1	−1

The resulting contrast codes are in Table 15.3. If you follow the 'rules' in Section 12.4 you'll end up with codes of 2, 2, −2, −2 for contrast 1. I have divided these values by the number of groups (4) to get codes of 0.5, 0.5, −0.5, −0.5. I've done this so that the value for the contrast is the actual difference between means of the live and dead animals (rather than a multiple of it). Noting the group means in Table 15.1, my first contrast will yield a value of 2.75:

$$\frac{\bar{X}_{\text{Stick}} + \bar{X}_{\text{Witchetty}}}{2} - \frac{\bar{X}_{\text{Testicle}} + \bar{X}_{\text{Eye}}}{2} = \frac{8.13 + 5.75}{2} - \frac{4.25 + 4.13}{2} = 2.75 \quad (15.21)$$

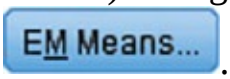
whereas using codes of 2 and −2 will yield a value 4 times this value (11). It doesn't affect the significance value at all, so in this sense you can use what codes you like, but it is handy to have the contrast value equal the actual difference between means because we can more easily interpret the confidence interval for the contrast (it will be the confidence interval for the difference between means, rather than the confidence interval for four times the difference). To operationalize these contrasts you have to use the following syntax (**BushtuckerContrast.sps**):

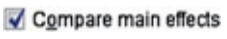
```
GLM stick testicle eye witchetty
/WSFACTOR=Animal 4 Polynomial
/WSDESIGN=Animal
/MMATRIX =
'Live vs. dead' stick 0.5 witchetty 0.5 testicle -0.5 eye -0.5;
'Stick vs. witchetty' stick 1 witchetty -1 testicle 0 eye 0;
```

’Testicle vs. eye’ stick 0 witchetty 0 testicle 1 eye -1.

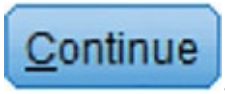
The first three lines specify a basic repeated-measures model. The contrasts are defined within the */MMATRIX* subcommand. I specify each contrast on a separate line (not essential, but it makes the syntax easier to read). I start with a name for the contrast in straight quotes, for example, ’Live vs. dead’ names the first contrast *Live vs. dead*, which describes what the contrast tests. Then I list the variables that make up the levels of the predictor variable using the variable names from the data editor (*stick*, *testicle*, *eye* and *witchetty*) and after each one type its contrast code. Note that each contrast ends with a semicolon (which tells SPSS that the contrast specification is finished) except the last one, which ends with a period to tell SPSS that the entire *GLM* command is complete. Executing this syntax produces a table that is explained in Section 15.9.4.

15.8.4 Post hoc tests and additional options 

Lack of sphericity creates entertaining complications for *post hoc* tests (see Jane Superbrain Box 15.1). When sphericity is definitely not violated, Tukey’s test can be used, but if sphericity can’t be assumed then the Games–Howell procedure is preferable. Because of these sphericity-related complications the standard *post hoc* tests for independent designs are not available for repeated-measures variables (the *post hoc* test dialog box does not list repeated-measured factors). The good news is that you can do some *post hoc* procedures by clicking 

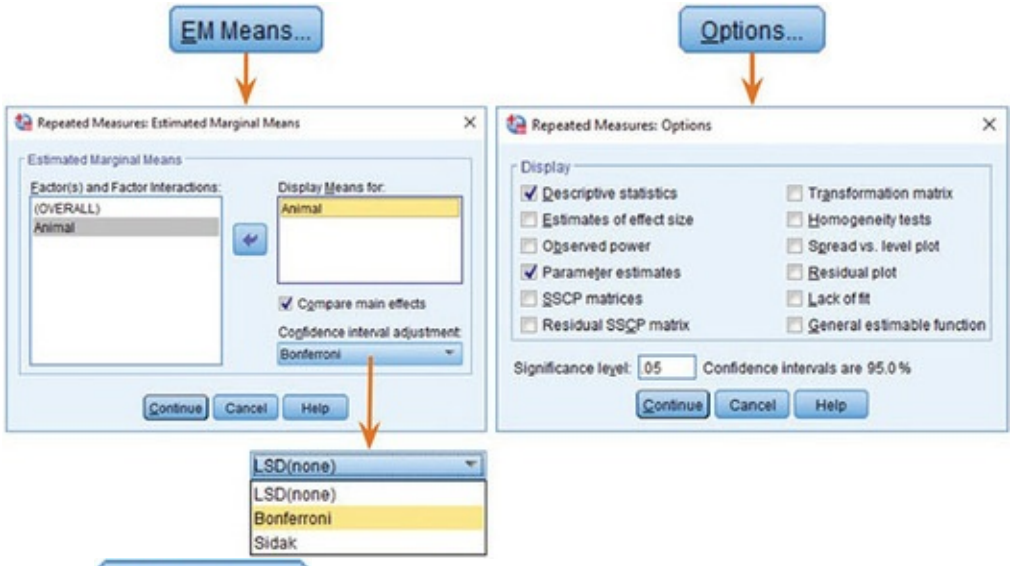
. To specify *post hoc* tests, drag the repeated-measures variable (in this case **Animal**) from the box labelled *Estimated Marginal Means: Factor(s) and Factor Interactions* to the box labelled *Display Means for* (or click 
). Then select 

to activate the 

drop-down menu (Figure 15.9). The default is to have no adjustment for multiple tests (Tukey LSD), which I don’t recommend, but you can choose a Bonferroni or Šidák correction (recommended for the reasons already mentioned). I’ve selected Bonferroni. Click 

.

Figure 15.9 The *Options* dialog box



Click Options... if you'd like to see things like descriptive statistics, a transformation matrix (which provides the coding values for any contrast selected in the *Contrasts* dialog box in Figure 15.8), and you can print the hypothesis, error and residual sum of squares and cross-products matrices (see Chapter 17) – Figure 15.9. If you have a between-group factor as well (mixed designs – see the next chapter) and you're a Levene's test kind of person then there is an option for homogeneity of variance tests. You can also change the level of significance at which to test any *post hoc* tests (you can change from the 0.05 level if you want to do a manual correction for multiple tests). Click Continue to return to the main dialog box, and OK to run the analysis.

15.9 Output for one-way repeated-measures designs



15.9.1 Descriptive statistics 

Output 15.1 shows two tables: the first (left) tells us the variables that represent each level of the predictor variable, which is a useful check that we entered the variables in the order that we intended to. The second table (right) shows that, on average, the time taken to retch was longest after eating the stick insect, and quickest after eating a testicle or eyeball. These means are useful for interpreting the main analysis.

Output 15.1

Within-Subjects Factors

Measure: MEASURE_1

Animal	Dependent Variable
1	stick
2	testicle
3	eye
4	witchetty

Descriptive Statistics

	Mean	Std. Deviation	N
Stick Insect	8.13	2.232	8
Kangaroo Testicle	4.25	1.832	8
Fish Eyeball	4.13	2.748	8
Witchetty Grub	5.75	2.915	8

Output 15.2

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser	Epsilon[b] Huynh-Feldt	Epsilon[b] Lower-bound
Animal	.136	11.406	5	.047	.533	.666	.333

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: Animal

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

15.9.2 Correcting for sphericity



Output 15.2 shows Mauchly's test for the bushtucker data. The significance value (0.047) is less than the critical value of 0.05, which implies that the assumption of sphericity has been violated, but I suggested you ignore this test and routinely apply a correction for whatever deviation from sphericity is present in the data (Misconception Mutt 15.1). The more informative part of the table contains the Greenhouse–Geisser ($\hat{\varepsilon}$ = 0.533) and Huynh–Feldt ($\hat{\varepsilon}$ = 0.666) estimates of sphericity.[6] If the data are perfectly spherical then these estimates will be 1. Therefore, both estimates indicate a departure from sphericity, so we may as well correct for it regardless of what Mauchly's test says. Jane Superbrain Box 15.3 explains how these estimates are used to correct the degrees of freedom for the *F*-statistic.

6 The lowest possible value of the Greenhouse–Geisser estimate is $1/(k - 1)$, which with four conditions is $1/(4 - 1) = 0.33$. This is given as the lower-bound estimate in Output 15.2.

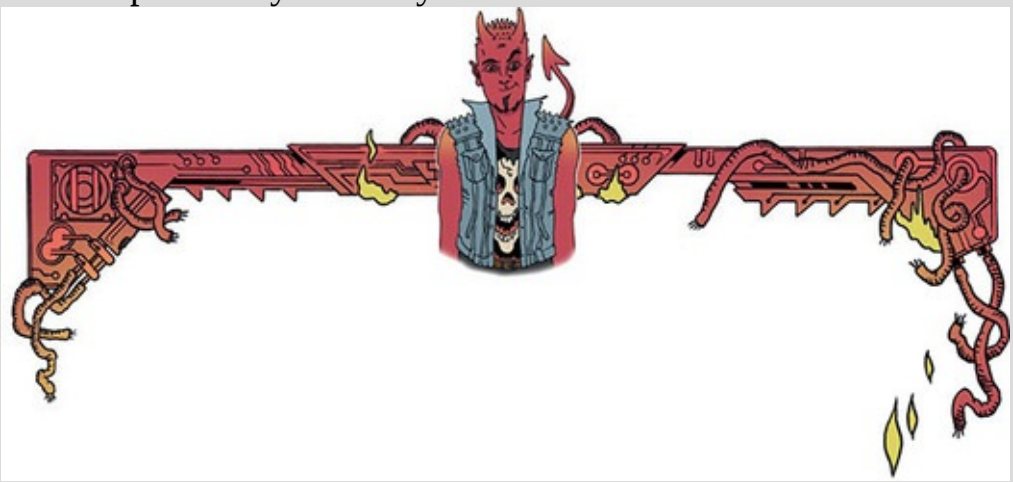
15.9.3 The F-statistic



Output 15.4 shows the summary information for the *F*-statistic that tests whether

we can significantly predict retching times from the group means (i.e., are the means significantly different?). Note that the values of the sum of squares for the repeated-measures effect of **Animal**, the model sum of squares (SS_M), the residual sum of squares (SS_R), the mean squares and *F*-statistic are the same as we calculated in Sections 15.6.3–15.6.6. The *p*-value associated with the *F*-statistic is 0.026, which is significant because it is less than the criterion value of 0.05. This result implies that there was a significant difference in the capacity of the four animals to induce retching when eaten. Remember though, the *F* does not tell us which animals differed from which.

SPSS Tip 15.1 My Mauchly's test looks weird



Sometimes the significance for Mauchly's test shows a dot and no significance value, as in Output 15.3. Naturally, you fear that SPSS has gone crazy and is going to break into your bedroom at night and tattoo the equation for the Greenhouse–Geisser estimate on your face. Fear not, the reason for the dot is that you need at least three conditions for sphericity to be an issue (see Misconception Mutt 15.1). Therefore, if the repeated-measures variable has only two levels then sphericity is met, the estimates of sphericity are 1 (perfect sphericity), the chi-square is zero and has no degrees of freedom and so a *p*-value can't be computed. It would be a lot easier if SPSS just printed in big letters 'Hooray! Hooray! Sphericity has gone away!' We can dream.

Output 15.3

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser	Huynh-Feldt	Lower-bound
Animal	1.000	.000	0	.	1.000	1.000	1.000

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: Animal

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Although the result seems plausible, we have learnt that departures from sphericity make the *F*-test inaccurate, and Output 15.2 shows that these data are not spherical. As well as showing the *F*-statistic and associated degrees of freedom when sphericity is assumed, Output 15.4 shows the results adjusted using the three estimates of sphericity in Output 15.2 (Greenhouse–Geisser, Huynh–Feldt, and the lower-bound value). These estimates are used to correct the degrees of freedom, which has the effect of increasing *p* (Jane Superbrain Box 15.3).

The adjustments result in the observed *F* being non-significant when using the Greenhouse–Geisser correction (because $p > 0.05$) but significant using the Huynh–Feldt correction (because the probability value of 0.048 is just below the criterion value of 0.05). I noted earlier that the Greenhouse–Geisser correction is probably too strict and that the Huynh–Feldt correction is probably not strict enough, and we see this here because one of them takes the significance value above the conventional 0.05 threshold whereas the other doesn't (see Jane Superbrain Box 15.4). This leaves us with the puzzling dilemma of whether to accept this *F*-statistic as significant.

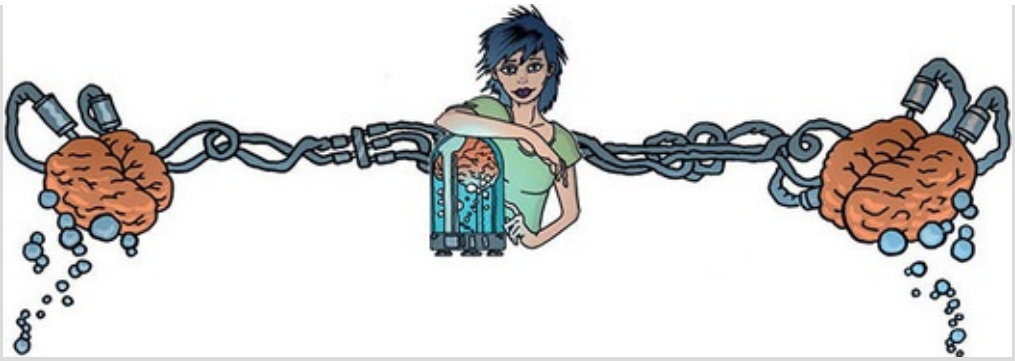
Output 15.4

Tests of Within-Subjects Effects

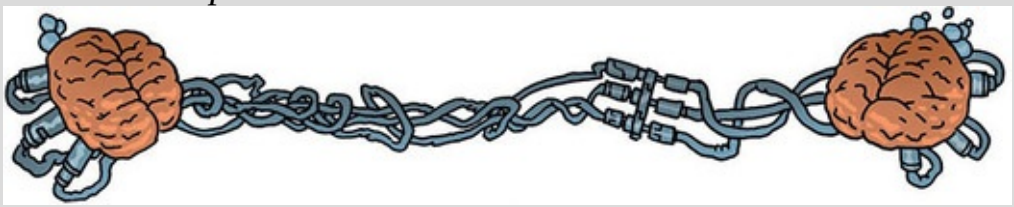
Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
Animal	Sphericity Assumed	83.125	3	27.708	3.794	.026
	Greenhouse-Geisser	83.125	1.599	52.001	3.794	.063
	Huynh-Feldt	83.125	1.997	41.619	3.794	.048
	Lower-bound	83.125	1.000	83.125	3.794	.092
Error(Animal)	Sphericity Assumed	153.375	21	7.304		
	Greenhouse-Geisser	153.375	11.190	13.707		
	Huynh-Feldt	153.375	13.981	10.970		
	Lower-bound	153.375	7.000	21.911		

Jane Superbrain 15.3 Adjusting for sphericity 



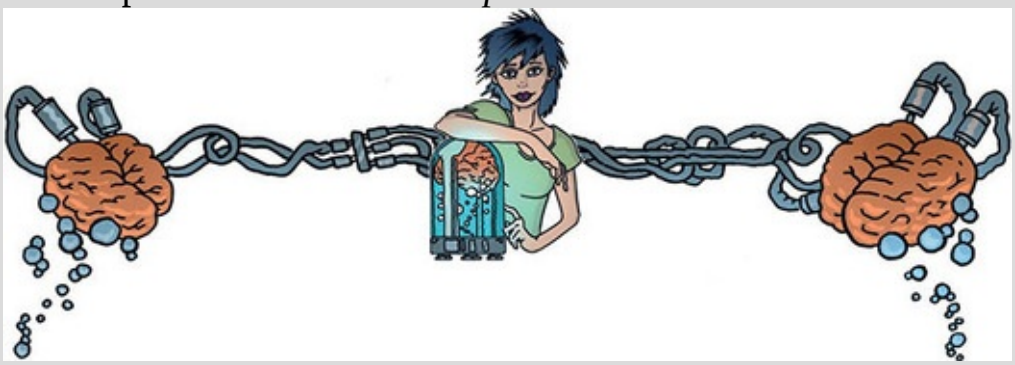
To adjust for the extent to which the data are not spherical, the degrees of freedom are multiplied by the estimates of sphericity in Output 15.2. For example, the Greenhouse–Geisser estimate of sphericity was 0.533 (Output 15.2). The original degrees of freedom for the model sum of squares were 3 and for the residual sum of squares they were 21. These values are adjusted by multiplying by the estimate of sphericity (0.533) resulting in 3 × 0.533 = 1.599 for the model and 21 × 0.533 = 11.19 for the residual. The Huynh–Feldt correction is applied in the same way (see Oliver Twisted on sphericity). The effect of reducing the degrees of freedom is to change the shape of the *F*-distribution that is used to obtain *p*, so the *F*-statistic is unchanged, but its *p* is based on a distribution with 1.599 and 11.19 degrees of freedom instead of 3 and 21. This increases the *p*-value associated with *F*.



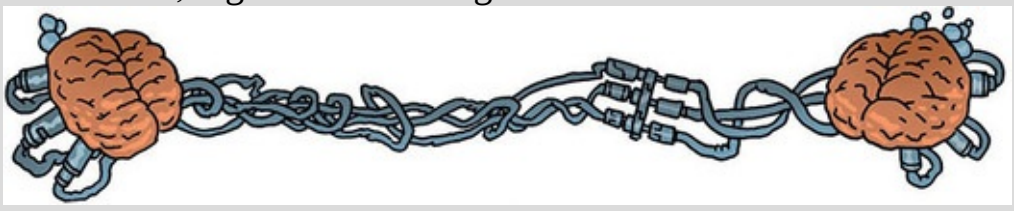
A recommendation mentioned earlier is to use the Greenhouse–Geisser estimate unless it's greater than 0.75 (which it's not in this case). Some people also suggest taking an average of the two estimates (Stevens, 2002). In practical terms, rather than averaging the estimates, correcting the degrees of freedom manually and trying with an abacus or two to generate exact *p*-values, we could average the two *p*-values instead, which yields $p = (0.063 + 0.048)/2 = 0.056$. In both cases the conclusion would be that there was no significant difference between means. Another option is to use multivariate test statistics (MANOVA), which do not assume sphericity (see O'Brien & Kaiser, 1985). Output 15.5 shows the multivariate test statistics (details of these test statistics can be found in Section 17.4), all of which are significant (because *p* is 0.002, which is less than the criterion value of 0.05). Based on multivariate tests, we'd conclude that

there are significant differences between the times taken to retch after eating different animals. It's easy to see how the decision rule applied to *p*-values can lead to results that don't replicate, conclusions that have been influenced by researcher degrees of freedom, and a lot of noise in the scientific literature.

Jane Superbrain 15.4 Pointless *p*



In Section 3.2.2 I discussed significance testing leading to all-or-nothing thinking. These data illustrate this point beautifully: the two sphericity corrections lead to significance values just above (0.063) or just below (0.048) the 0.05 criterion. These significance values differ by only 0.015 but lead to completely opposite conclusions. The decision about 'significance' becomes rather arbitrary: choose one correction for a result that is 'significant' but choose another for non-significance. The conclusion of the study will largely be a function of researcher degrees of freedom (Section 3.3.1). The means and the effect size are unaffected by sphericity corrections, and so whether the *p* falls slightly above or slightly below 0.05 side-tracks us from the more important question of how big the effect was. We might be well advised to look at an effect size to see whether the effect is substantive, regardless of its significance.



Output 15.5

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
Animal	Pillai's Trace	.942	26.955[b]	3.000	5.000	.002
	Wilks' Lambda	.058	26.955[b]	3.000	5.000	.002
	Hotelling's Trace	16.173	26.955[b]	3.000	5.000	.002
	Roy's Largest Root	16.173	26.955[b]	3.000	5.000	.002

a. Design: Intercept
Within Subjects Design: Animal

b. Exact statistic



15.9.4 Contrasts

Output 15.6 shows the transformation matrix requested in the options. Thinking back to contrast coding (Chapter 12), a code of 0 means that the group is not included in a contrast. Therefore, contrast 1 (labelled *Level 1 vs. Level 2*) ignores the fish eyeball and witchetty grub. Also remember that groups with a negative weight are compared to groups with a positive weight. For this first contrast, this means that the stick insect is compared against the kangaroo testicle. Using the same logic, contrast 2 (labelled *Level 2 vs. Level 3*) ignores the stick insect and witchetty grub and compares the kangaroo testicle with the fish eye.



What does contrast 3 (*Level 3 vs. Level 4*) compare?

Contrast 3 compares the fish eyeball with the witchetty grub. This pattern is consistent with a repeated contrast in that all groups except the first are compared to the preceding category.

Output 15.7 lists each contrast and its *F*-statistic, which compares the two chunks of variation within the contrast. We can conclude that celebrities took significantly longer to retch after eating the stick insect compared to the kangaroo testicle, $p = 0.002$ (*Level 1 vs. Level 2*), but that the time to retch was roughly the same after eating the kangaroo testicle and the fish eyeball, $p = 0.920$ (*Level 2 vs. Level 3*) and after eating a fish eyeball compared to eating a witchetty grub, $p = 0.402$ (*Level 3 vs. Level 4*). It's worth remembering that, by some criteria, our main effect of the type of animal eaten was not significant, and if this is the case then we really shouldn't look at these contrasts. However, given the multivariate tests, there is some justification in looking at these contrasts.

For those of you brave enough to attempt the custom contrasts in Section 15.8.3, Output 15.8 shows the tables of contrasts. The top table is probably most useful

because it includes the confidence interval for the contrast, but the bottom table reports the test statistics and degrees of freedom (note that the *p*-values are identical in the two tables). We can conclude from these tables that retching times were significantly longer after eating live animals than dead ones, $F(1, 7) = 18.41$, $p = 0.004$, but there was no significant difference between the stick insect and witchetty grub, $F(1, 7) = 1.76$, $p = 0.227$, or between the fish eye and kangaroo testicle, $F(1, 7) = 0.011$, $p = 0.920$. The confidence intervals tell us (assuming this sample is one of the 95% that produce intervals containing the population value) that the difference in retching times after eating live animals compared to body parts is likely to fall between 1.24 and 4.27 seconds, the difference between stick insect and witchetty grub lies between −1.86 and 6.61 seconds, and the difference between kangaroo testicle and fish eye lies between −2.72 and 2.97 seconds.

Output 15.6

Animal[a]

Measure: MEASURE_1

	Animal		
Dependent Variable	Level 1 vs. Level 2	Level 2 vs. Level 3	Level 3 vs. Level 4
Stick Insect	1	0	0
Kangaroo Testicle	-1	1	0
Fish Eyeball	0	-1	1
Witchetty Grub	0	0	-1

a. The contrasts for the within subjects factors are:
Animal: Repeated contrast

Output 15.7

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	Animal	Type III Sum of Squares	df	Mean Square	F	Sig.
Animal	Level 1 vs. Level 2	120.125	1	120.125	22.803	.002
	Level 2 vs. Level 3	.125	1	.125	.011	.920
	Level 3 vs. Level 4	21.125	1	21.125	.796	.402
Error(Animal)	Level 1 vs. Level 2	36.875	7	5.268		
	Level 2 vs. Level 3	80.875	7	11.554		
	Level 3 vs. Level 4	185.875	7	26.554		

Output 15.8

Contrast Results (K Matrix)

Contrast[a]			Transformed Variable: Live vs. dead	Stick vs. witchetty	Testicle vs. eye
L1	Contrast Estimate		2.750	2.375	.125
	Hypothesized Value		0	0	0
	Difference (Estimate – Hypothesized)		2.750	2.375	.125
	Std. Error		.641	1.792	1.202
	Sig.		.004	.227	.920
	95% Confidence Interval for Difference	Lower Bound	1.235	-1.863	-2.717
		Upper Bound	4.265	6.613	2.967

a. Estimable Function for Intercept

Univariate Test Results

Source	Transformed Variable	Sum of Squares	df	Mean Square	F	Sig.
Contrast	Live vs. dead	60.500	1	60.500	18.413	.004
	Stick vs. witchetty	45.125	1	45.125	1.756	.227
	Testicle vs. eye	.125	1	.125	.011	.920
Error	Live vs. dead	23.000	7	3.286		
	Stick vs. witchetty	179.875	7	25.696		
	Testicle vs. eye	80.875	7	11.554		



15.9.5 Post hoc tests

If you selected *post hoc* tests for the repeated-measures variable (see Section 15.8.3), then Output 15.9 is produced. The table should have a familiar feel from other *post hoc* tests we've looked at: it shows the difference between group means and their confidence intervals, the standard error, and the significance value. Based on the significance values and the means (in Output 15.1) we can conclude that the time to retch was significantly longer after eating a stick insect compared to a kangaroo testicle ($p = 0.012$) and a fish eye ($p = 0.006$), but not compared to a witchetty grub ($p = 1$). The time to retch after eating a kangaroo testicle was not significantly different compared to after eating a fish eye or witchetty grub (both ps = 1). Finally, the time to retch was not significantly different after eating a fish eyeball compared to a witchetty grub ($p = 1$). Again, it's worth noting that we wouldn't interpret these effects if we decide that the main effect of the type of animal eaten wasn't significant.

Output 15.9

Pairwise Comparisons

Measure: MEASURE_1

(I) Animal	(J) Animal	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b] Lower Bound	Upper Bound
1	2	3.875*	.811	.012	.925	6.825
	3	4.000*	.732	.006	1.339	6.661
	4	2.375	1.792	1.000	-4.141	8.891
2	1	-3.875*	.811	.012	-6.825	-.925
	3	.125	1.202	1.000	-4.244	4.494
	4	-1.500	1.336	1.000	-6.359	3.359
3	1	-4.000*	.732	.006	-6.661	-1.339
	2	-.125	1.202	1.000	-4.494	4.244
	4	-1.625	1.822	1.000	-8.249	4.999
4	1	-2.375	1.792	1.000	-8.891	4.141
	2	1.500	1.336	1.000	-3.359	6.359
	3	1.625	1.822	1.000	-4.999	8.249

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Cramming Sam's Tips One-way repeated-measures designs



- One-way repeated-measures designs compares several means, when those means come from the same entities; for example, if you measured people's statistical ability each month over a year-long course.
- When you have three or more repeated-measures conditions there is an additional assumption: *sphericity*.
- You can test for sphericity using *Mauchly's test*, but it is better to always adjust for the departure from sphericity in the data.
- The table labelled *Tests of Within-Subjects Effects* shows the main *F*-statistic. Other things being equal, always read the row labelled *Greenhouse–Geisser* (or *Huynh–Feldt*, but you'll have to read this chapter to find out the relative merits of the two procedures). If the value in the column labelled *Sig.* is less than 0.05 then the means of the conditions are significantly different.
- For contrasts and *post hoc* tests, again look to the columns labelled *Sig.* to discover if your comparisons are significant (i.e.,

the value is less than 0.05).



15.10 Robust tests of one-way repeated-measures designs

There is a robust test of several dependent means with *post hoc* tests that can be run using the syntax file **rmanova.sps**. As with other robust tests in this book, this syntax requires the *Essentials for R* plugin and *WRS2* package (Section 4.13). The syntax uses the *rmanova* and *rmmcp* functions described by Wilcox (2017). These tests assume neither normality nor homogeneity of variance, so you can ignore these assumptions and plough ahead. The syntax is a little more complicated than in previous chapters because the data need to be restructured, so I have written a bunch of stuff to try to make that process as painless as possible. There's a bunch of stuff that looks to see whether certain packages are installed and if they aren't it installs them, after which the core syntax is (see SPSS Tip 15.2 for an explanation):

```
BEGIN PROGRAM R.
mySPSSdata = spssdata.GetDataFromSPSS(factorMode = ''labels'')
ID<-''celebrity''
rmFactor<-c(''stick'', ''testicle'', ''eye'', ''witchetty'')
df<-melt(mySPSSdata, id.vars = ID, measure.vars = rmFactor)
names(df)[names(df) == ID] <- ''id''
rmanova(df$value, df$variable, df$id, tr = 0.2)
rmmcp(df$value, df$variable, df$id, tr = 0.2)
END PROGRAM.
```

Running this syntax (see SPSS Tip 15.2) should produce Output 15.10, which tells us that there was not a significant difference between means, F_t (2.31, 11.55) = 2.75, p = 0.100. The *post hoc* tests (which technically we should ignore because the overall test wasn't significant) show no significant difference between any groups (note that the column labelled *sig* shows FALSE for all comparisons, the *p*-values are all greater than the critical values (*p.crit*) and the confidence intervals all contain zero).

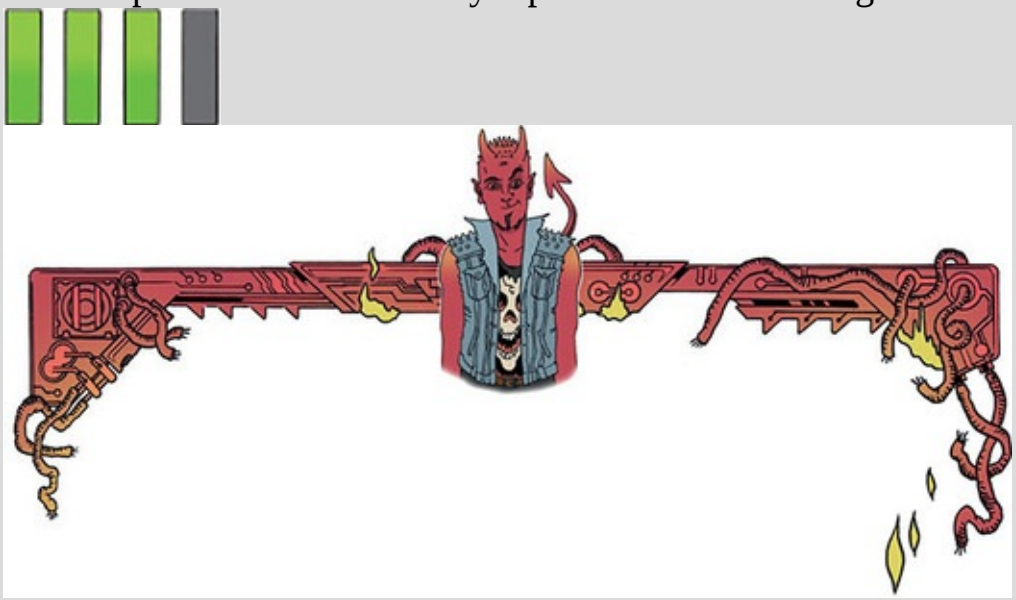
Output 15.10

```
Test statistic: 2.7528
Degrees of Freedom 1: 2.31
Degrees of Freedom 2: 11.55
p-value: 0.1002

Call:
rmmcp(y = df$value, groups = df$variable, blocks = df$id, tr = 0.2)

                         psihat ci.lower ci.upper p.value  p.crit   sig
stick vs. testicle      3.66667 -0.48300  7.81633 0.01360 0.01020 FALSE
stick vs. eye           4.00000 -0.35728  8.35728 0.01172 0.00851 FALSE
stick vs. witchetty     2.00000 -8.09920 12.09920 0.44148 0.01690 FALSE
testicle vs. eye        0.00000 -5.38802  5.38802 1.00000 0.05000 FALSE
testicle vs. witchetty -1.83333 -9.23266  5.56599 0.34371 0.01270 FALSE
eye vs. witchetty      -2.00000-12.54827  8.54827 0.46001 0.02500 FALSE
```

SPSS Tip 15.2 Robust one-way repeated-measures designs



I'll break down the syntax line by line. First we grab the data from SPSS and put it in a data frame called *mySPSSdata*:

mySPSSdata = spssdata.GetDataFromSPSS(factorMode = ''labels'')

Next, we define the name of the variable that represents the participant IDs. If you use this file on your own data, replace the word *celebrity* with the (case-sensitive) name of your own ID variable. Make sure the name is in straight quotes and don't edit anything else:

ID<-"celebrity"

Next, we list the variables that represent the levels of the repeated-measures predictor. If you use this file on your own data replace the words *stick, testicle, etc.* with the (case-sensitive) names from your own data file. Make sure the names are in straight quotes and don't edit anything else:

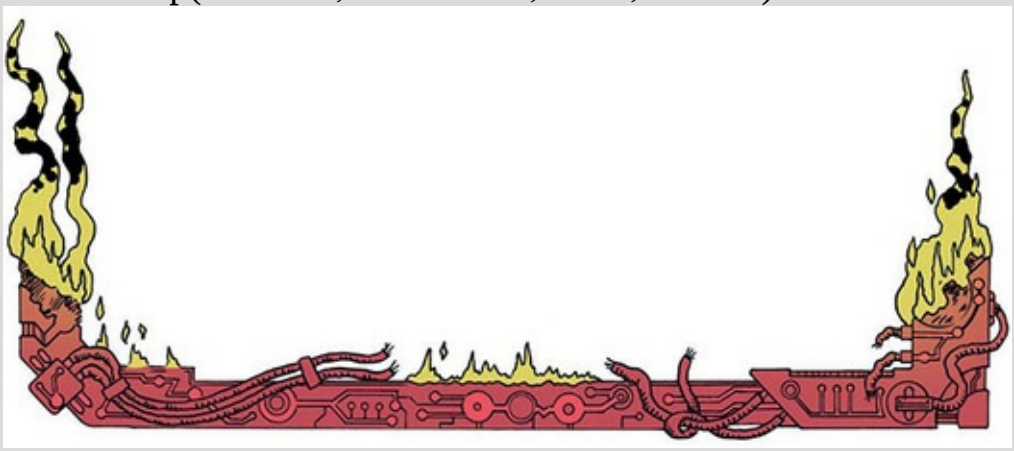
rmFactor<-c(''stick'', ''testicle'', ''eye'', ''witchetty'')

The next two lines convert your dataframe from wide format to long (Section 4.6.1) and then rename the participant ID variable to be *id* (which is done to save you needing to edit the final two commands). If you've set up *ID* and *rmFactor* correctly in the two lines above everything should go smoothly:

```
df<-melt(mySPSSdata, id.vars = ID, measure.vars = rmFactor)
names(df)[names(df) == ID] <- "id">
```

The final two lines run the robust test (*rmanova*) and *post hoc* tests (*rmmcp*). Because I've used generic variable names that were set up before we got to this point, you don't need to edit these functions unless you want to change the trim (*tr = 0.2*) to something other than the default of 20%:

```
rmanova(df$value, df$variable, df$id, tr = 0.2)
rmmcp(df$value, df$variable, df$id, tr = 0.2)
```



15.11 Effect sizes for one-way repeated-measures designs 

As with independent designs, the best measure of the overall effect size is omega squared (ω^2). However, just to make life even more complicated, the equations we've previously used for omega squared can't be used for repeated-measures designs: if you do use them the effect size will be overestimated. For the sake of simplicity some people do use the same equation, but I'm going to hit simplicity in the face with Stingy the particularly poison-ridden jellyfish, and embrace complexity like I do my children.

In repeated-measures designs, the equation for omega squared is (hang onto your hat):

$$\omega^2 = \frac{\left[\frac{k-1}{nk}(MS_M - MS_R)\right]}{MS_R + \frac{MS_B - MS_R}{k} + \left[\frac{k-1}{nk}(MS_M - MS_R)\right]} \quad (15.22)$$

I know what you're thinking and, no, I'm not having a laugh. Far from it. The equation isn't too bad if you break it down. First, there are some mean squares that we've come across before: the mean square for the model (MS_M) and the residual mean square (MS_R), both of which can be obtained from Output 15.4. There's also *k*, the number of conditions, which for these data would be 4 (there were four animals), and *n*, the number of people who took part (in this case 8 celebrities).

The main problem is MS_B. At the beginning of Section 15.3 (Figure 15.3) I mentioned that the total variation is broken down into a within-participant variation and a between-participant variation (SS_B), which we can calculate using equation (15.19). SPSS doesn't give us SS_W, but we know that it's made up of SS_M and SS_R, which we are given. By substituting these terms and rearranging the equation we get:

$$SS_T = SS_B + SS_M + SS_R \quad (15.23)$$
$$SS_B = SS_T - SS_M - SS_R$$

SPSS, which is clearly trying to hinder us at every step, also doesn't give us SS_T, and I'm afraid (unless I've missed something in the output) you're going to have to calculate it by hand (see Section 15.6.1). For these data it's 17.39 (equation (15.20)). The next step is to convert this value to a mean squares by dividing by the degrees of freedom, which in this case are the number of people in the sample minus 1:

$$MS_B = \frac{SS_B}{df_B} = \frac{SS_B}{N-1} = \frac{17.38}{8-1} = 2.48 \quad (15.24)$$

Having done all this and probably died of boredom in the process, we now resurrect our corpses with renewed vigour for the effect size equation:

$$\omega^2 = \frac{\left[\frac{4-1}{8\times 4}(27.71 - 7.30)\right]}{7.30 + \frac{2.48 - 7.30}{4} + \left[\frac{4-1}{8\times 4}(27.71 - 7.30)\right]} \quad (15.25)$$
$$= \frac{1.91}{8.01}$$
$$= 0.24$$

I hope you agree it was worth the effort.

I've mentioned at various other points that it's more useful to have effect size measures for focused comparisons anyway (rather than an overall *F*-statistic),

and so an easier approach is to calculate effect sizes for the contrasts in Output 15.7. We can use the familiar equation to convert the *F*-values (because they all have 1 degree of freedom for the model) to *r*:

$$r = \sqrt{\frac{F(1, df_R)}{F(1, df_R) + df_R}} \quad (15.26)$$

For the three contrasts we get the following values:

$$r_{\text{stick insect vs. kangaroo testicle}} = \sqrt{\frac{22.80}{22.80 + 7}} = 0.87$$

$$r_{\text{kangaroo testicle vs. fish eyeball}} = \sqrt{\frac{0.01}{0.01 + 7}} = 0.04 \quad (15.27)$$

$$r_{\text{fish eyeball vs. witchetty grub}} = \sqrt{\frac{0.80}{0.80 + 7}} = 0.32$$

The difference between the stick insect and the testicle was large, between the fish eye and witchetty grub reasonable, but between the testicle and eyeball the effect was tiny.

15.12 Reporting one-way repeated-measures designs



Reporting repeated-measures designs is much the same as independent designs, except that we need to pay attention to reporting the corrected degrees of freedom. The degrees of freedom used to assess the *F*-statistic are the degrees of freedom for the effect of the model (df_M = 1.60) and the degrees of freedom for the residuals of the model (df_R = 11.19). Therefore, we could report the main finding as:

> ✓ The Greenhouse–Geisser estimate of the departure from sphericity was ε = 0.53. The time to retch was not significantly affected by the type of animal eaten, $F(1.60, 11.19) = 3.79$, $p = 0.063$, $\omega^2 = 0.24$.

For Huynh–Feldt corrected values:

> ✓ The Huynh–Feldt estimate of the departure from sphericity was $\varepsilon = 0.67$. The time to retch was significantly affected by the type of animal eaten, $F(2, 13.98) = 3.79$, $p = 0.048$, $\omega^2 = 0.24$.

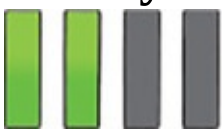
We could report multivariate tests. There are four different test statistics, but in most situations you should report Pillai's trace, *V* (see Chapter 17) and the associated *F* and its degrees of freedom (all from Output 15.6):

> ✓ The Greenhouse–Geisser estimate of sphericity showed a substantial deviation (ε = 0.53), therefore multivariate tests are reported. The time to retch was significantly affected by the type of animal eaten, $V = 0.94$, $F(3,$

5) = 26.96, $p = 0.002$, $\omega^2 = 0.24$.

Finally, robust tests:

✓ The Greenhouse–Geisser estimate of sphericity showed a substantial deviation ($\varepsilon = 0.53$). Robust tests of 20% trimmed means implemented with the *WRS2* package in R (Mair et al., 2015) showed that the time to retch was not significantly affected by the type of animal eaten, F_t (2.31, 11.55) = 2.75, $p = 0.100$, $\omega^2 = 0.24$.

15.13 A boozy example: a factorial repeated-measures design 

We have seen that between-group designs can be extended to incorporate multiple predictor/independent variables. The same is true for repeated-measures designs. There is evidence that attitudes towards stimuli can be changed using positive and negative imagery (e.g., Hofmann, De Houwer, Perugini, Baeyens, & Crombez, 2010; Stuart, Shimp, & Engle, 1987). As part of an initiative to stop binge drinking in teenagers, the government funded scientists to look at whether negative imagery could be used to make teenagers' attitudes towards alcohol more negative. The scientists compared the effects of negative imagery against positive and neutral imagery for different types of drinks. Table 15.4 illustrates the experimental design and contains the data for this example (each row represents a single participant).

Table 15.4 Data from **Attitude.sav**

Drink	Beer			Wine			Water		
Image	+ve	−ve	Neut	+ve	−ve	Neut	+ve	−ve	Neut
Male	1	6	5	38	−5	4	10	−14	−2
	43	30	8	20	−12	4	9	−10	−13
	15	15	12	20	−15	6	6	−16	1
	40	30	19	28	−4	0	20	−10	2
	8	12	8	11	−2	6	27	5	−5
	17	17	15	17	−6	6	9	−6	−13
	30	21	21	15	−2	16	19	−20	3
	34	23	28	27	−7	7	12	−12	2
	34	20	26	24	−10	12	12	−9	4
	26	27	27	23	−15	14	21	−6	0
Female	1	−19	−10	28	−13	13	33	−2	9
	7	−18	6	26	−16	19	23	−17	5
	22	−8	4	34	−23	14	21	−19	0
	30	−6	3	32	−22	21	17	−11	4
	40	−6	0	24	−9	19	15	−10	2
	15	−9	4	29	−18	7	13	−17	8
	20	−17	9	30	−17	12	16	−4	10
	9	−12	−5	24	−15	18	17	−4	8
	14	−11	7	34	−14	20	19	−1	12
	15	−6	13	23	−15	15	29	−1	10

Participants viewed a total of nine videos over three sessions. In one session, they saw three videos: (1) a brand of beer (Strange Brew) presented alongside negative imagery (a bunch of inanimate dead bodies in a trendy bar with the slogan ‘Strange Brew: who needs a liver?’); (2) a brand of wine (Liquid Fire) presented within positive imagery (a bunch of sexy hipster types in a trendy bar with the slogan ‘Liquid Fire: your life would be so much better if you were a sexy hipster type’); and (3) a brand of water (Backwater) presented with neutral imagery (some completely average people in a trendy bar accompanied by the slogan ‘Backwater: it will make no difference to your life one way or another’). In a second session (a week later), the participants saw the same three brands, but this time Strange Brew was accompanied by the positive imagery, Liquid Fire by the neutral image and Backwater by the negative. In a third session, the participants saw Strange Brew accompanied by the neutral image, Liquid Fire by the negative image and Backwater by the positive. After each advert participants rated the drinks from −100 (dislike very much) through 0 (neutral) to 100 (like very much). The order of adverts was randomized, as was the order in which people participated in the three sessions. This design is quite complex. There are two predictor/independent variables: the type of drink (beer, wine or water) and the type of imagery used (positive, negative or neutral). These two variables

completely cross over, producing nine experimental conditions.

15.14 Factorial repeated-measures designs using SPSS Statistics

When entering the data in Table 15.4 remember that each row represents a single participant. If a person participates in all conditions (in this case each person sees all types of drink presented with all types of imagery) then each condition is represented by a column. So, we need to create nine variables in the data editor with the names and value labels in Table 15.5.

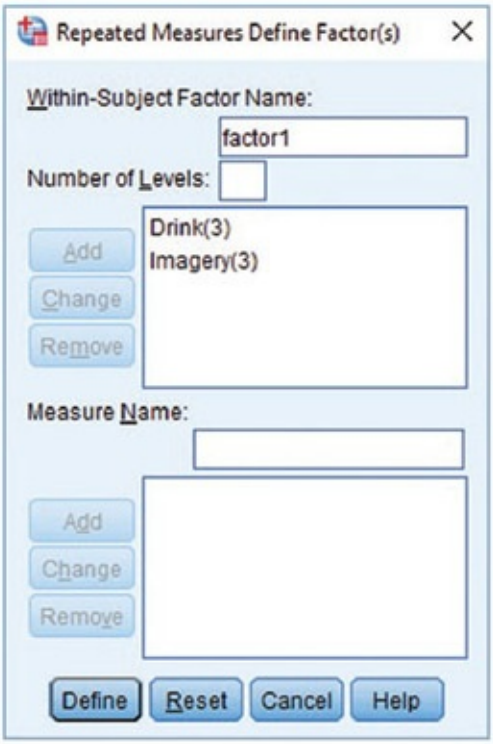


Once these variables have been created, enter the data as in Table 15.4. If you have problems entering the data then use the file **Attitude.sav**.

Table 15.5 Variable names and labels for the **Attitude.sav** data

Variable name	Variable label
beerpos	Beer + sexy hipsters
beerneg	Beer + dead bodies
beerneut	Beer + average people
winepos	Wine + sexy hipsters
wineneg	Wine + dead bodies
wineneut	Wine + average people
waterpos	Water + sexy hipsters
waterneg	Water + dead bodies
waterneut	Water + average people

Figure 15.10 The *Define Factor(s)* dialog box for factorial repeated-measures designs



Select *Analyze* *General Linear Model* Repeated Measures... to access the dialog boxes for a repeated-measures model. First we define the repeated-measures predictors. In this case there are two: **Drink** (beer, wine or water) and **Imagery** (positive, negative and neutral). Replace the word *factor1* with the word 'Drink', type '3' into the box labelled *Number of Levels*, and click Add. This variable appears in the list of variables as *Drink(3)*, which means we have defined a predictor called Drink that has three levels. We repeat this process for the second predictor by typing 'Imagery' into the space labelled *Within-Subject Factor Name*, typing '3' into the space labelled *Number of Levels*, and clicking Add. The variable will appear in the list as *Imagery(3)* (Figure 15.10). Click Define to go to the main dialog box.

The main dialog box is the same as for designs with one predictor variable, except that there are nine question marks (Figure 15.11). At the top of the *Within-Subjects Variables* box, SPSS lists the variables that we defined: **Drink** and **Imagery**. Underneath is a series of question marks followed by bracketed numbers. The numbers in brackets represent the levels of the predictor variables

(independent variables):

- *?*(1,1) ⇒ variable representing 1st level of drink and 1st level of imagery
- *?*(1,2) ⇒ variable representing 1st level of drink and 2nd level of imagery
- *?*(1,3) ⇒ variable representing 1st level of drink and 3rd level of imagery
- *?*(2,1) ⇒ variable representing 2nd level of drink and 1st level of imagery
- *?*(2,2) ⇒ variable representing 2nd level of drink and 2nd level of imagery
- *?*(2,3) ⇒ variable representing 2nd level of drink and 3rd level of imagery
- *?*(3,1) ⇒ variable representing 3rd level of drink and 1st level of imagery
- *?*(3,2) ⇒ variable representing 3rd level of drink and 2nd level of imagery
- *?*(3,3) ⇒ variable representing 3rd level of drink and 3rd level of imagery

Because we have two predictors there are two numbers in the brackets: the first refers to levels of the first predictor listed above the box (in this case **Drink**) and the second refers to levels of the second predictor listed above the box (in this case **Imagery**). We need to replace these question marks with variable names from the data editor (which are listed on the left-hand side of the dialog box). At this stage we need to think about which conditions to assign to which level of each variable. For example, if we entered **beerpos** into the list first, then SPSS would treat beer as the first level of **Drink** and positive imagery as the first level of the **Imagery** variable. However, if we entered **wineneg** into the list first, we're specifying wine as the first level of **Drink**, and negative imagery as the first level of **Imagery**. For this reason, we need to think about what contrasts we might want *before* assigning variables in this dialog box.

Figure 15.11 The main dialog box for factorial repeated-measures designs before completion